

# IBM System Storage SAN Volume Controller Best Practices and Performance Guidelines

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**Storage**





IBM Redbooks

**IBM System Storage SAN Volume Controller Best  
Practices and Performance Guidelines**

April 2021

**Note:** Before using this information and the product it supports, read the information in “Notices” on page xi.

**First Edition (April 2021)**

This edition applies to IBM Spectrum Virtualize Version 8.4.

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# Preface

This IBM® Redbooks® publication describes several of the preferred practices and describes the performance gains that can be achieved by implementing the IBM SAN Volume Controller powered by IBM Spectrum® Virtualize V8.4. These practices are based on field experience.

This book highlights configuration guidelines and preferred practices for the storage area network (SAN) topology, clustered system, back-end storage, storage pools, and managed disks, volumes, Remote Copy services, and hosts. Then, it provides performance guidelines for IBM SAN Volume Controller, back-end storage, and applications.

It explains how you can optimize disk performance with the IBM System Storage Easy Tier® function. It also provides preferred practices for monitoring, maintaining, and troubleshooting IBM SAN Volume Controller.

This book is intended for experienced storage, SAN, and IBM SAN Volume Controller administrators and technicians. Understanding this book requires advanced knowledge of the IBM SAN Volume Controller, IBM FlashSystem, and SAN environments.

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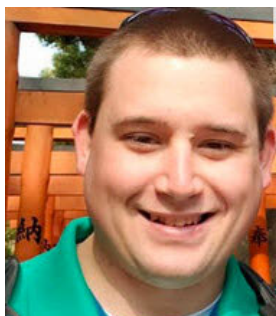
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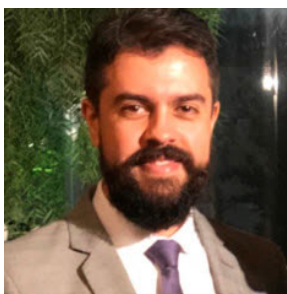
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Thanks to the following for their contributions that made this book possible:

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**IBM Hursley, UK**

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# IBM SAN Volume Controller introduction

This chapter introduces the IBM SAN Volume Controller storage subsystem range that is supported with the new Spectrum Virtualize software v8.4. It describes all of the relevant models, their key features, benefits, and technology.

The software that runs on the IBM SAN Volume Controller products is called IBM Spectrum Virtualize. The name of the underlying hardware platform remains intact.

The IBM Spectrum Virtualize software is a proven offering that was available for years in IBM SAN Volume Controller family of storage solutions. It provides an ideal way to manage and protect huge volumes of data from mobile and social applications, enables rapid and flexible cloud services deployments, and delivers the performance and scalability that is needed to gain insights from the latest analytics technologies.

This chapter includes the following topics:

- ▶ 1.1, “Benefits of using IBM Spectrum Virtualize” on page 2
- ▶ 1.2, “IBM SAN Volume Controller supported product range” on page 4
- ▶ 1.3, “IBM SAN Volume Controller product range” on page 7
- ▶ 1.4, “Advanced functions for data reduction” on page 14
- ▶ 1.5, “Advanced software features” on page 15

## 1.1 Benefits of using IBM Spectrum Virtualize

The storage virtualization functions of IBM Spectrum Virtualize are a powerful tool in the hands of storage administrators. However, for an organization to fully realize the benefits of storage virtualization, its implementation must be the result of a process that begins with identifying the organization's goals. For a storage virtualization project to be a success, the organization must identify what it wants to achieve before it starts to think how to implement the solution.

Today, organizations are searching for affordable and efficient ways to store, use, protect, and manage their data. Also, a storage environment is required to have an easy to manage interface and be sufficiently flexible to support a many applications, servers, and mobility requirements. Although business demands change quickly, some recurring client concerns drive adoption of storage virtualization, including the following example:

- ▶ Growing data center costs
- ▶ Inability of IT organizations to respond quickly to business demands
- ▶ Poor asset usage
- ▶ Poor availability and resultant unsatisfactory (for the clients) or challenging (for the providers) service levels
- ▶ Lack of skilled staff for storage administration

Storage virtualization is one of the foundations of building a flexible and reliable infrastructure solution that enables companies to better align to their technological needs. Storage virtualization enables an organization to achieve affordability and manageability by implementing storage pools across several physically separate disk systems (which might be from different vendors).

Storage can then be deployed from these pools, and migrated transparently between pools without interruption to the attached host systems and their applications. Storage virtualization provides a single set of tools for advanced functions, such as instant copy and remote mirroring solutions, which enables faster and seamless deployment of storage regardless of the underlying hardware.

Because the storage virtualization represented by IBM Spectrum Virtualize is a software-enabled function, it offers more features that are typically not available on a regular pure storage subsystem, including (but not limited to) the following features:

- ▶ Data compression
- ▶ Software and hardware encryption
- ▶ IBM Easy Tier for workload balancing
- ▶ Thin provisioning
- ▶ Mirroring and copy services
- ▶ Interface to Cloud Service Providers

Figure 1-1 shows these features.

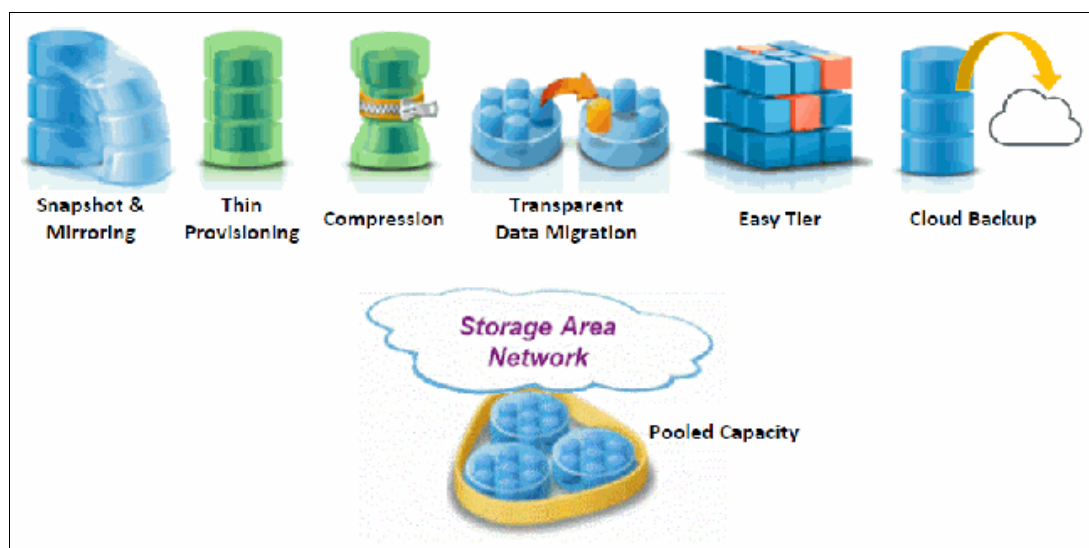


Figure 1-1 IBM Spectrum Storage virtualization

IBM SAN Volume Controller systems that are running IBM Spectrum Virtualize software v8.4 reduce the number of separate environments that must be managed down to a single system.

After the initial configuration of the back-end storage subsystems, all of the daily storage management operations are performed by way of a single graphical user interface. At the same time, administrators gain access to the rich function set that is provided by IBM Spectrum Virtualize, even features that are natively available on the back-end storage systems.

Spectrum Virtualize also provides a robust command-line interface with access to many bash utilities, such as cut, grep, and sed. RESTful API and Ansible support for enterprise integration and automation are also available.

## 1.2 IBM SAN Volume Controller supported product range

Figure 1-2 shows the current IBM SAN Volume Controller models that are supported by Spectrum Virtualize software v8.4.

Earlier models of IBM SAN Volume Controller (2145-CG8) are not supported because of hardware restrictions and thus not covered in this IBM Redbooks publication.

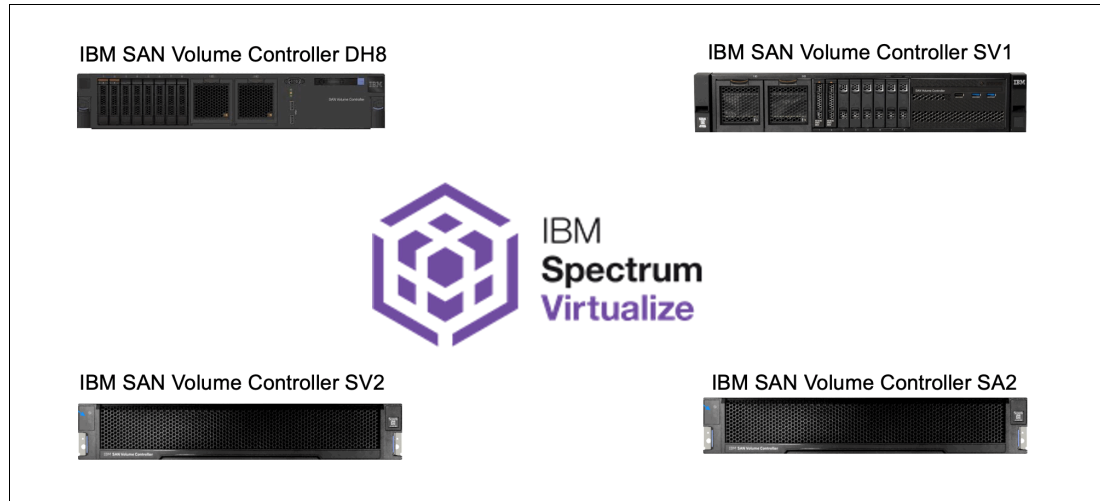


Figure 1-2 IBM SAN Volume Controller current models

### 1.2.1 New in Spectrum Virtualize v8.4

IBM Spectrum Virtualize 8.4 provides more features and updates to the IBM Spectrum Virtualize family of products, of which IBM SAN Volume Controller is part.

Version 8.4 features the following major software changes:

- ▶ Data Reduction Pool (DRP) improvements:
  - Data Reduction Child Pool: Allows for more flexibility, such as multi-tenancy.
  - FlashCopy® with redirect-on-write support: Uses DRP's internal deduplication referencing capabilities to reduce overhead by creating references instead of copying the data. Redirect-on-write (RoW) is an alternative to the copy-on-write (CoW) capabilities.

**Note:** At the time of this writing, this capability might be used only for volumes with supported deduplication without mirroring relationships and within the same pool and I/O group. The mode selection (RoW/CoW) is automatic based on these conditions.

- Compressimator always on: Allows the systems to sample each volume at regular intervals, and provides the ability to display the compressibility of the data in the GUI and IBM Storage Insights at any time.
- RAID Reconstruct Read: Increases reliability and availability by reducing chances of DRP going offline because of fixable array issues. It also uses RAID capabilities, and DRP asks for a specific data block reconstruction when a potential corruption is detected.

- ▶ Distributed RAID 1 (DRAID 1) support: Provides the ability to extend distributed RAID advantages to smaller pools of drives. This feature improves performance over traditional RAID 1 implementations, which allows a better use of flash technology. These distributed arrays can support as few as two drives with no rebuild area, and 3 - 16 drives with a single rebuild area.

**Note:** This function is not applicable to IBM SAN Volume Controller products because its primary use is with the new NVMe SCM type drives in the IBM FlashSystem products. It is included here for a complete summary of the v8.4 new features and functions.

- ▶ Expansion of mirrored vDisks (also known as *volumes*): Allows the vDisks capacity to be expanded or reduced online, without requiring an offline format and sync. This ability improves the availability of the volume for use because the new capacity is available immediately.
- ▶ Three-site replication with HyperSwap® support: Provides improved availability for data in three-site implementations. This feature expands on the Disaster Recovery (DR) capabilities that are inherent in this topology.

**Important:** Three-site replication by using Metro Mirror was supported on version 8.3.1 only in limited installations by way of RPQ process. With 8.4.0, this implementation is generally available.

- ▶ Host attachment support with FC-NVMe in HyperSwap systems.
- ▶ DNS support for LDAP and NTP with full DNS length (that is, 256 characters).
- ▶ Updates to maximum configuration limits: Doubles FlashCopy mapping from 5,000 to 10,000 and increases HyperSwap volumes limit from 1,250 to 2,000.
- ▶ Password and login changes on the Spectrum Virtualize v8.4 GUI to meet today's extra regulatory compliance with password and ID expiry and security enhancements.
- ▶ Support for internal proxy servers (also known as customer web proxy) by using IBM Call Home with cloud services and log upload features.

For more information about these new features, see [IBM Documentation](#).

## 1.2.2 Supported products

The following IBM SAN Volume Controller products are supported to run the Spectrum Virtualize software v8.4 software. Here, we list the IBM Storwize and IBM SAN Volume Controller series name and then the hardware machine type and model for extra clarity:

- ▶ 2145-DH8, 2147-DH8
- ▶ 2145-SV1, 2147-SV1
- ▶ 2145-SA2, 2147-SA2
- ▶ 2145-SV2, 2147-SV2

### 1.2.3 IBM SAN Volume Controller high-level features

This IBM Redbooks publication describes and focuses on the best practices and options to gain the optimum performance from the product, including the set of software-defined storage features.

It also describes data reduction techniques, including deduplication, compression, dynamic tiering, thin provisioning, snapshots, cloning, replication, data copy services, enhanced stretch cluster and IBM HyperSwap for high availability.

**Note:** The detailed technical explanations, and theory of operations, of these features are not covered in this publication. For more information about these areas, see the following publications:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8467
- ▶ *IBM SAN Volume Controller 2145-DH8 Introduction and Implementation*, SG24-8229

All IBM SAN Volume Controller products, running Spectrum Virtualize software, feature two types of enclosures: control engine enclosures and expansion enclosures:

- ▶ A *control enclosure* or *storage engine* manages your storage systems, communicates with the host, and manages interfaces.

Each control enclosure or storage engine is a standard 2U high, 19-inch rack mounted unit.

- ▶ An *expansion enclosure* enables you to increase the available capacity of the IBM SAN Volume Controller cluster that communicates with the control enclosure by way of a pair of 12 Gbps SAS connections. These expansion enclosures can house many flash (SSD) SAS type drives or hard disk drives (HDD), depending on which model of expansion enclosure is ordered.

The expansion enclosures generally have three types:

- Figure 1-3 shows the large form factor (LFF) expansion enclosure, which can hold 12 3.5-inch drives and is 2U high.



Figure 1-3 2145 LFF expansion enclosure

- Figure 1-4 shows the small form factor (SFF) expansion enclosure, which can hold 24 2.5-inch drives and is 2U high.



Figure 1-4 2145 SFF expansion enclosure

- Figure 1-5 shows the large form factor high density (LFF HD) expansion enclosure, which can hold 92 3.5-inch drives (or 92 2.5-inch drives in carriers) and is 5U high.

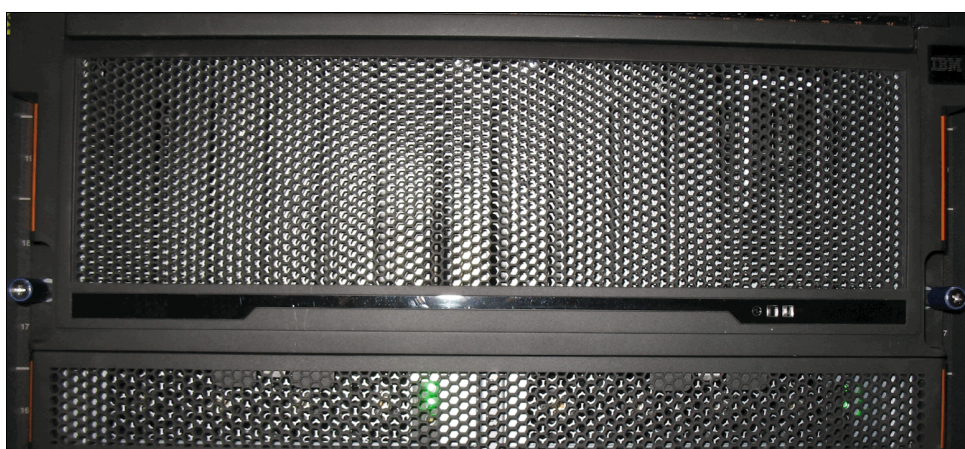


Figure 1-5 2145 LFF HD expansion enclosure

The type and models of expansion enclosures that can attach to the relevant control enclosure or storage engine is model dependent and is described next.

**Note:** IBM SAN Volume Controller models SV2 and SA2 do not support any type of SAS expansion enclosures.

## 1.3 IBM SAN Volume Controller product range

Next, we discuss the various IBM SAN Volume Controller products that are supported in IBM Spectrum Virtualize software v8.4, with some more in-depth information about each product, its capabilities, features, and functions. Also supplied for each product range are links to more information about their configuration limits and restrictions, so that the customer can research any information or values that are needed to give optimum performance and adhere to best practices.

The IBM SAN Volume Controller is available in the following machines types. The only difference between them is the warranty period and both machines are functionally the same:

- ▶ 2145 has 1-year warranty
- ▶ 2147 has 3-year warranty



The 2147 also includes Enterprise Class Support (ESC), which gives more benefits the normal warranty terms.

For more information about the ECS program is available in [IBM Documentation](#).

The SAN Volume Controller Machine Type 2145 storage engines can be clustered with IBM SAN Volume Controller Machine Type 2147 storage engines only if the extra IBM Support services that upgrades IBM SAN Volume Controller Machine Type 2145 to the equivalent IBM SAN Volume Controller Machine Type 2147 Support Terms and Conditions is purchased.

For more information about supported environments, devices, and configurations, see [IBM System Storage Interoperation Center](#).

## IBM SAN Volume Controller Model DH8

IBM SAN Volume Controller is a combined hardware and software storage virtualization system with a single point of control for storage resources. The IBM SAN Volume Controller includes many functions that are traditionally deployed separately in disk systems. By including these functions in a virtualization system, IBM SAN Volume Controller standardizes functions across virtualized storage for greater flexibility and potentially lower costs.

IBM SAN Volume Controller improves business application availability and delivers greater resource usage so you can get the most from your storage resources, and achieve a simpler, more scalable, cost-efficient IT infrastructure.

Figure 1-6 shows the front view of the IBM SAN Volume Controller storage engine model DH8.



Figure 1-6 IBM SAN Volume Controller storage engine model DH8

The IBM SAN Volume Controller Storage Engine Model DH8 offers the following benefits:

- ▶ One or two Intel Xenon E5 v2 Series eight-core processors, each with 32 GB memory
- ▶ 16 Gb FC, 8 Gb FC, 10 Gb Ethernet, and 1 Gb Ethernet I/O ports for FC, iSCSI, and FCoE connectivity
- ▶ Hardware-assisted compression acceleration (optional feature)
- ▶ 12 Gb SAS expansion enclosure attachment for internal flash storage (optional feature)
- ▶ 2U, 19-inch rack mount enclosure

**Note:** The 2145 / 2147 Model DH8 are now End of Marketing (EOM) since December 2016, and are no longer available to purchase from IBM. They are included in this publication for completeness because they support running the Spectrum Virtualize software v8.4.

The 2145 Model DH8 includes three 1 Gb Ethernet ports standard for iSCSI connectivity. It can be configured with up to four I/O adapter features that provide up to 16 16 Gb FC ports, up to 16 8 Gb FC ports, or up to four 10 Gb Ethernet (iSCSI) ports.



For more information, see the [Technical Description - Adapter Cards](#) section of the announcement letter for supported configurations.

Compression workloads can benefit from Model DH8 configurations with two eight-core processors with 64 GB of memory (total system memory). Compression workloads can also benefit from the hardware-assisted acceleration that is offered by the addition of up to two compression accelerator cards.

IBM SAN Volume Controller Storage Engines can be clustered to help deliver greater performance, bandwidth, and scalability. An IBM SAN Volume Controller clustered system can contain up to four node pairs or I/O groups.

IBM SAN Volume Controller Storage Engine Model DH8 can also support expansion enclosures with the following models:

- ▶ The IBM 2145 SAN Volume Controller LFF Expansion Enclosure Model 12F  
Holds up to 12 3.5-inch SAS drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SAN Volume Controller SFF Expansion Enclosure Model 24F  
Holds up to 24 2.5-inch SAS internal flash (solid-state) drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SAN Volume Controller HD LFF Expansion Enclosure Model 92F  
Holds up to 92 3.5-inch SAS internal flash (solid-state) drive capacity drives in a 5U, 19-inch rack mount enclosure.

Table 1-1 lists the IBM SAN Volume Controller storage engine model DH8 host, expansion drive capacity, and functions.

*Table 1-1 IBM SAN Volume Controller DH8 host, drive capacity and functions summary*

| Feature/Function         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Host/SAS interfaces      | <ul style="list-style-type: none"> <li>▶ Four-port 16 Gb FC adapter card with shortwave SFP transceivers for 16 Gb FC connectivity</li> <li>▶ Two-port 16 Gb FC adapter card with shortwave SFP transceivers for 16 Gb FC connectivity</li> <li>▶ Four-port 8 Gb FC adapter card with shortwave SFP transceivers for 8 Gb FC connectivity</li> <li>▶ Four-port 10 Gb Ethernet adapter card with SFP+ transceivers for 10 Gb iSCSI/FCoE connectivity</li> <li>▶ Four-port 12 Gb SAS expansion enclosure attachment card</li> </ul> |
| SAS expansion enclosures | <ul style="list-style-type: none"> <li>▶ Model 12F/24F 2U 12 or 24 drives</li> <li>▶ Model 92F 5U 92 drives</li> <li>▶ NL-SAS disk drives: 4 TB, 6 TB, and 8 TB 7,200 rpm</li> <li>▶ SAS disk drives: <ul style="list-style-type: none"> <li>– 300 GB, 600 GB, and 900 GB 15,000 rpm</li> <li>– 900 GB, 1.2 TB, and 1.8 TB 10,000 rpm</li> </ul> </li> <li>▶ Flash (solid-state) drives: <ul style="list-style-type: none"> <li>– 400 GB, 800 GB, 1.6 TB, 1.92 TB, 3.2 TB, 3.84 TB, 7.68 TB, and 15.36 TB</li> </ul> </li> </ul>  |
| RAID levels              | DRAID 5 (CLI-only) and 6, TRAIID 5 and 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Feature/Function                            | Description                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advanced features included with each system | <ul style="list-style-type: none"> <li>▶ Virtualization of expansion and external storage</li> <li>▶ Data migration</li> <li>▶ Data Reduction Pools with thin provisioning</li> <li>▶ UNMAP</li> <li>▶ Compression and deduplication</li> <li>▶ Metro Mirror (synchronous) and Global Mirror (asynchronous)</li> </ul> |
| Other available advanced features           | <ul style="list-style-type: none"> <li>▶ Remote mirroring</li> <li>▶ Easy Tier compression</li> <li>▶ External virtualization</li> <li>▶ Encryption</li> <li>▶ IBM FlashCopy</li> <li>▶ IBM Spectrum Control</li> <li>▶ IBM Spectrum Protect Snapshot</li> </ul>                                                       |

For more information about the V8.4.0.x configuration limits and restrictions for IBM System Storage SAN Volume Controller, see [this IBM Support web page](#).

## IBM SAN Volume Controller Model SV1

The IBM SAN Volume Controller (2145-SV1) is the hardware component of IBM SAN Volume Controller family, and is a combined hardware and software storage virtualization system. The IBM SAN Volume Controller includes many functions traditionally deployed separately in disk systems. By including these functions in a virtualization system, IBM SAN Volume Controller standardizes functions across virtualized storage for greater flexibility and potentially lower costs.

The IBM 2145 IBM SAN Volume Controller Storage Engine Model SV1 features the following specifications:

- ▶ Two Intel Xeon E5 v4 Series eight-core processors
- ▶ 64 GB of memory (options for 256 GB of memory)
- ▶ 2U, 19-inch rack mount enclosure.
- ▶ 10 Gb iSCSI connectivity is standard (options for 16 Gb FC, 10 Gb iSCSI, and 25 Gb iSCSI connectivity)

Figure 1-7 shows the front view of the IBM SAN Volume Controller storage engine model SV1.



Figure 1-7 IBM SAN Volume Controller storage engine model SV1

IBM SAN Volume Controller Storage Engine Model SV1 can also support expansion enclosures with the following models:

- ▶ The IBM 2145 SAN Volume Controller LFF Expansion Enclosure Model 12F, which holds up to 12 3.5-inch SAS drives in a 2U, 19-inch rack mount enclosure
- ▶ The IBM 2145 SAN Volume Controller SFF Expansion Enclosure Model 24F, which holds up to 24 2.5-inch SAS internal flash (solid state) drives in a 2U, 19-inch rack mount enclosure
- ▶ The IBM 2145 SAN Volume Controller HD LFF Expansion Enclosure Model 92F, which holds up to 92 3.5-inch SAS internal flash (solid state) or HDD capacity drives in a 5U, 19-inch rack mount enclosure

Table 1-2 lists the IBM SAN Volume Controller storage engine model SV1 host, expansion drive capacity, and functions summary.

*Table 1-2 IBM SAN Volume Controller model SV1 host, expansion drive capacity, and functions*

| Feature/Function                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Host/SAS interfaces                         | <ul style="list-style-type: none"> <li>▶ Two-port 25 Gb Ethernet adapter card with SFP28 transceivers for 25 Gb iSCSI connectivity</li> <li>▶ Four-port 16 Gb FC adapter card with shortwave SFP transceivers for 16 Gb FC connectivity</li> <li>▶ Four-port 10 Gb Ethernet adapter card with SFP+ transceivers for 10 Gb iSCSI/FCoE connectivity</li> <li>▶ Four-port 12 Gb SAS expansion enclosure attachment card</li> <li>▶ 16 Gb FC longwave SFP transceivers</li> </ul> |
| SAS expansion enclosures                    | <ul style="list-style-type: none"> <li>▶ Model 12F/24F 2U 12 or 24 drives</li> <li>▶ Model 92F 5U 92 drives</li> <li>▶ NL-SAS disk drives: 4 TB, 6 TB, and 8 TB 7,200 rpm</li> <li>▶ SAS disk drives: <ul style="list-style-type: none"> <li>– 300 GB, 600 GB, and 900 GB 15,000 rpm</li> <li>– 900 GB, 1.2 TB, and 1.8 TB 10,000 rpm</li> </ul> </li> <li>▶ Flash (solid state) drives: 400 GB, 800 GB, 1.6 TB, 1.92 TB, 3.2 TB, 3.84 TB, 7.68 TB, and 15.36 TB</li> </ul>   |
| RAID levels                                 | DRAID 5 (CLI-only) and 6, TRAIID 5 and 6                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Advanced features included with each system | <ul style="list-style-type: none"> <li>▶ Virtualization of expansion and external storage</li> <li>▶ Data migration</li> <li>▶ Data Reduction Pools with thin provisioning</li> <li>▶ UNMAP</li> <li>▶ Compression and deduplication</li> <li>▶ Metro Mirror (synchronous) and Global Mirror (asynchronous)</li> </ul>                                                                                                                                                        |
| Additional available advanced features      | <ul style="list-style-type: none"> <li>▶ Remote mirroring</li> <li>▶ Easy Tier compression</li> <li>▶ External virtualization</li> <li>▶ Encryption</li> <li>▶ FlashCopy</li> <li>▶ IBM Spectrum Control</li> <li>▶ IBM Spectrum Protect Snapshot</li> </ul>                                                                                                                                                                                                                  |

Model SV1 storage engines can be added to IBM SAN Volume Controller clustered systems that include previous generation storage engine models. All nodes within a clustered system must use the same version of IBM SAN Volume Controller software. An IBM SAN Volume Controller clustered system can contain up to four node pairs.

For more information about the V8.4.0.x configuration limits and restrictions for IBM System Storage SAN Volume Controller, see this [IBM Support web page](#).

## IBM SAN Volume Controller Model SV2 and SA2

IBM SAN Volume Controller, a combined hardware and software storage virtualization system with a single point of control for storage resources, delivers a single system to manage and provision heterogeneous storage systems. IBM SAN Volume Controller storage engines enable clients to update their storage technology without taking the system offline, which helps to lower total cost of ownership of their storage infrastructure.

The engines are available in two models:

- ▶ IBM SAN Volume Controller Entry Storage Engine Model SA2 with two Intel Cascade Lake eight-core processors running at 2.1 GHz
- ▶ IBM SAN Volume Controller Storage Engine Model SV2 with two Intel Cascade Lake 16-core processors running at 2.30 GHz

Both models include the following features:

- ▶ 128 GB of base memory
- ▶ Four 10 Gb Ethernet ports standard for iSCSI connectivity and service technician use
- ▶ Support for up to three I/O adapter cards for 16 or 32 Gb Fibre Channel (FC) and 25 Gb iSCSI/ RoCE/iWARP over Ethernet connectivity
- ▶ Two integrated AC power supplies
- ▶ Integrated battery backup

**Note:** IBM SAN Volume Controller SV2 and SA2 do not support any type of SAS expansion enclosures.

Figure 1-8 shows the front view of the IBM SAN Volume Controller storage engine models SV2 and SA2. The front view of these two machine models are identical.

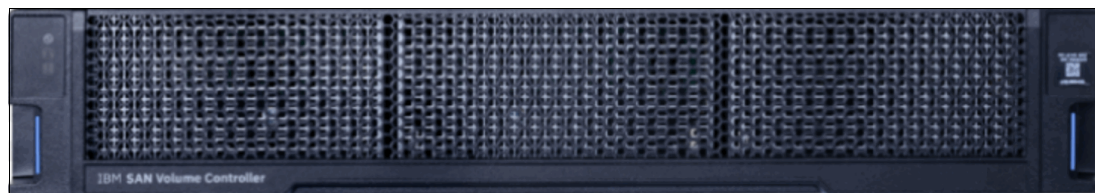


Figure 1-8 IBM SAN Volume Controller models SV2 and SA2

Table 1-3 lists the IBM SAN Volume Controller storage engine model SV2 and SA2 host connections and functions.

*Table 1-3 IBM SAN Volume Controller model SV2 and SA2 host connections and functions summary*

| Feature/Function                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Host/SAS interfaces                         | <ul style="list-style-type: none"> <li>▶ Four 10 Gb Ethernet ports standard for iSCSI connectivity and service technician processes</li> <li>▶ Three I/O adapter cards: <ul style="list-style-type: none"> <li>– 16 or 32 Gb Fibre Channel (FC) and</li> <li>– 25 Gb iSCSI/ RoCE/iWARP over Ethernet connectivity</li> <li>– 16 Gb FC longwave SFP transceivers</li> </ul> </li> </ul> <p><b>Note:</b> A minimum of one Fibre Channel or one Ethernet adapter is required.</p> |
| RAID levels                                 | DRAID 5 (CLI-only) and 6, TRAIID 5 and 6                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Advanced features included with each system | <ul style="list-style-type: none"> <li>▶ Virtualization of expansion and external storage</li> <li>▶ Data migration</li> <li>▶ Data Reduction Pools with thin provisioning</li> <li>▶ UNMAP</li> <li>▶ Compression and deduplication</li> <li>▶ Metro Mirror (synchronous) and Global Mirror (asynchronous)</li> <li>▶ Compression acceleration is built into the SA2 and SV2 hardware</li> </ul>                                                                              |
| Additional available advanced features      | <ul style="list-style-type: none"> <li>▶ Remote mirroring</li> <li>▶ Easy Tier compression</li> <li>▶ External virtualization</li> <li>▶ Encryption</li> <li>▶ FlashCopy</li> <li>▶ IBM Spectrum Control</li> <li>▶ IBM Spectrum Protect Snapshot</li> </ul>                                                                                                                                                                                                                   |

IBM SAN Volume Controller storage engines can be clustered to help deliver greater performance, bandwidth, scalability, and availability. An IBM SAN Volume Controller clustered system can contain up to four node pairs or I/O groups, for a total of eight nodes. These storage engines can be added to IBM SAN Volume Controller clustered systems that include previous generation storage engine models; that is, DH8 and SV1.

## Hot-spare nodes

The loss of a node for unplanned reasons, such as hardware failure, or planned outages, such as upgrades, can result in loss of redundancy or degraded system performance. To reduce this possibility, a *hot-spare node* is kept powered on and visible on the system. The hot-spare node is a feature that can be purchased separately. For more information, contact your Business Partner or local IBM Sales representative.

A hot-spare node features active system ports, but no host I/O ports, and is not part of any I/O group. If a node fails or is upgraded, this spare node joins the system and assumes the place of the failed node, restoring redundancy. Only host connection on Fibre Channel ports that support node port virtualization (NPIV) can be used for hot-spare nodes.

The hot-spare node uses the same N\_Port ID Virtualization (NPIV) worldwide port names (WWPNs) for its Fibre Channel ports as the failed node; therefore, host operations are not disrupted. The hot-spare node retains its node identifier when it was the spare.

During an upgrade, the spare node is added to the system when a node is removed. As each node in a system shuts down for the upgrade, it is replaced by the hot-spare node.

In addition, up to four hot-spare nodes can be configured to deliver even higher availability for the solution.

For more information about V8.4.0.x configuration limits and restrictions for IBM System Storage SAN Volume Controller, see this [IBM Support web page](#).

## **1.4 Advanced functions for data reduction**

The IBM SAN Volume Controller range can function as a feature-rich, software-defined storage layer that virtualizes and extends the functions of all managed storage. These functions include data reduction, dynamic tiering, copy services, and high-availability configurations. In this capacity, it acts as the virtualization layer between the host and other external storage systems, providing flexibility and extending functionality to the virtualized external storage capacity.

### **1.4.1 Data Reduction Pools**

Data Reduction Pools (DRPs) represent a significant enhancement to the storage pool concept. The virtualization layer is primarily a simple layer that runs the task of lookups between virtual and physical extents.

With the introduction of data reduction technology, compression, and deduplication, it became more of a requirement to have an uncomplicated way to stay “thin”. DRPs enable you to automatically de-allocate (not to be confused with deduplicate) and reclaim the capacity of thin-provisioned volumes that contain deleted data.

### **1.4.2 Deduplication**

Deduplication can be configured with thin-provisioned and compressed volumes in data reduction pools for added capacity savings. The deduplication process identifies unique chunks of data, or byte patterns, and stores a signature of the chunk for reference when writing new data chunks. If the new chunk’s signature matches an existing signature, the new chunk is replaced with a small reference that points to the stored chunk. The same byte pattern can occur many times, resulting in the amount of data that must be stored being greatly reduced.

### **1.4.3 Thin provisioning**

In a shared storage environment, thin provisioning is a method for optimizing the use of available storage. It relies on allocation of blocks of data on demand versus the traditional method of allocating all of the blocks up front.

This methodology eliminates almost all white space, which helps avoid the poor usage rates (often as low as 10%) that occur in the traditional storage allocation method. Traditionally, large pools of storage capacity are allocated to individual servers, but remain unused (not written to).

## 1.4.4 Thin-provisioned flash copies

Thin-provisioned IBM FlashCopy (or snapshot function in the GUI) uses disk space only when updates are made to the source or target data, and not for the entire capacity of a volume copy.

## 1.5 Advanced software features

The IBM SAN Volume Controller storage engine includes the following advanced software features:

- ▶ Data migration
- ▶ Copy services:
  - Metro Mirror
  - Global Mirror
  - Global Mirror with Change Volumes
  - FlashCopy
  - Remote Mirroring
- ▶ EasyTier
- ▶ External virtualization
- ▶ IBM HyperSwap
- ▶ Licensing

### 1.5.1 Data migration

The IBM SAN Volume Controller range provides online volume migration while applications are running, which is possibly the greatest single benefit for storage virtualization. This capability enables data to be migrated on and between the underlying storage subsystems without any effect on the servers and applications.

In fact, this migration is performed without the knowledge of the servers and applications that it even occurred. The IBM SAN Volume Controller range delivers these functions in a homogeneous way on a scalable and highly available platform over any attached storage and to any attached server

### 1.5.2 Copy services

Advanced copy services are a class of functionality within storage arrays and storage devices that enable various forms of block-level data duplication locally or remotely. By using advanced copy services, you can make mirror images of part or all of your data eventually between distant sites.

Copy services functions are implemented within an IBM SAN Volume Controller (FlashCopy and Image Mode Migration), or between one IBM SAN Volume Controller and another IBM SAN Volume Controller, or any other member of the IBM Spectrum Virtualize family, in the different modes:

- ▶ *Metro Mirror* is the IBM branded term for synchronous Remote Copy function.
- ▶ *Global Mirror* is the IBM branded term for the asynchronous Remote Copy function.

- ▶ *Global Mirror with Change Volumes* is the IBM branded term for the asynchronous Remote Copy of a locally and remotely created FlashCopy.

Remote replication can be implemented by using both Fibre Channel and Internet Protocol (IP) network methodologies.

For more information, see Chapter 6, “Copy services” on page 219.

## FlashCopy

FlashCopy is the IBM branded name for point-in-time copy, which is sometimes called time-zero (T0) copy. This function makes a copy of the blocks on a source volume and can duplicate them on 1 - 256 target volumes.

## Remote mirroring

The three remote mirroring modes are implemented at the volume layer within the IBM SAN Volume Controller storage engine. They are collectively referred to as Remote Copy capabilities. In general, the purpose of these functions is to maintain two copies of data.

Often, but not necessarily, the two copies are separated by distance. The Remote Copy can be maintained in one of two modes: synchronous or asynchronous, with a third asynchronous variant:

- ▶ Metro Mirror
- ▶ Global Mirror
- ▶ Global Mirror with Change Volumes

## 1.5.3 Easy Tier

Easy Tier is a performance function that automatically migrates or moves extents of a volume to or from one storage tier to another storage tier. The IBM SAN Volume Controller storage engine supports the following types of Easy Tier storage tiers:

- ▶ Storage Class Memory

This tier exists when the pool contains drives that use persistent memory technologies that improve endurance and speed of current flash storage device technologies.

- ▶ Tier 0 flash

This exists when the pool contains high performance flash drives.

- ▶ Tier 1 flash

This exists when the pool contains tier 1 flash drives. Tier 1 flash drives typically offer larger capacities, but slightly lower performance and write endurance characteristics.

- ▶ Enterprise tier

This tier exists when the pool contains enterprise-class MDisk, which are disk drives that are optimized for performance.

- ▶ Nearline tier

This tier exists when the pool contains nearline-class MDisk, which are disk drives that are optimized for capacity.



Consider the following points about Easy Tier:

- ▶ Easy Tier monitors the host volume I/O activity as extents are read, and migrates the most active extents to higher performing tiers.
- ▶ The monitoring function of Easy Tier is continual but, in general, extents are migrated over a 24-hour period. As extent activity cools, Easy Tier moves extents to slower performing tiers.
- ▶ Easy Tier creates a migration plan that organizes its activity to decide how to move extents. This plan can also be used to predict how extents will be migrated.

## 1.5.4 External virtualization

The IBM SAN Volume Controller range includes data virtualization technology to help insulate hosts, hypervisors, and applications from physical storage. This technology enables them to run without disruption, even when changes are made to the underlying storage infrastructure. The IBM SAN Volume Controller storage engine functions benefit all virtualized storage.

For example, Easy Tier and DRPs with compression help improve performance and increase effective capacity, where high-performance thin provisioning helps automate provisioning. These benefits can help extend the useful life of storage assets, and reduce costs. Because these functions are integrated into the IBM SAN Volume Controller storage engine, they also can operate smoothly together, which reduces management effort.

## 1.5.5 Enhanced stretch cluster

In a stretched system configuration, each site is defined as an independent failure domain. If one site experiences a failure, the other site can continue to operate without disruption. You must also configure a third site to host a quorum device that provides an automatic tie-break in the event of a potential link failure between the two main sites. The main site can be in the same room or across rooms in the data center, buildings on the same campus, or buildings in different cities. Different kinds of sites protect against different types of failures.

A stretched system is designed to continue operation after the loss of one failure domain however it cannot guarantee that it can operate after the failure of two failure domains. If the enhanced stretched system function is configured, you can enable a manual override for this situation. You can also use Metro Mirror or Global Mirror on a second system for extended DR with either an enhanced stretched system or a conventional stretched system. You configure and manage Metro Mirror or Global Mirror partnerships that include a stretched system in the same way as other Remote Copy relationships. The system supports SAN routing technology, which includes FCIP links, for inter-system connections that use Metro Mirror or Global Mirror.

The two partner systems cannot be in the same production site. However, they can be colocated with the storage system that provides the active quorum disk for the stretched system.

For more information about the Enhanced Stretch Cluster function, see [IBM Documentation](#).

## 1.5.6 IBM HyperSwap

HyperSwap capability enables each volume to be presented by two IBM SAN Volume Controller storage engine I/O groups. The configuration tolerates combinations of node and site failures by using host multipathing drivers that are based on the one that is available for the IBM SAN Volume Controller storage engine.

The IBM SAN Volume Controller storage engine provides GUI management of the HyperSwap function.

For more information about the HyperSwap function, see [IBM Documentation](#).

## 1.5.7 Licensing

The base license that is provided with the system includes the use of its basic functions. However, extra licenses can be purchased to expand the capabilities of the system. Administrators are responsible for purchasing extra licenses and configuring the systems within the license agreement, which includes configuring the settings of each licensed function on the system.

For more information about the licensing on the IBM SAN Volume Controller storage engine, see the chapter “Licensing and Features” in *Implementing the IBM FlashSystem with IBM Spectrum Virtualize V8.4*, SG24-8467.



## Storage area network

The storage area network (SAN) is one of the most important aspects when implementing and configuring an IBM Spectrum Virtualize cluster. Because of their unique behavior and the interaction with other storage, specific SAN design and zoning recommendations differ from classic storage practices.

This chapter does not describe how to design and build a flawless SAN from the beginning. Rather, it provides guidance to connect IBM Spectrum Virtualize in a SAN to achieve a stable, redundant, resilient, scalable, and performance-likely environment. However, you can consider the principles that are presented here when building your SAN.

This chapter includes the following sections:

- ▶ 2.1, “SAN topology general guidelines” on page 20
- ▶ 2.2, “SAN topology-specific guidelines” on page 22
- ▶ 2.3, “SAN Volume Controller ports” on page 29
- ▶ 2.4, “Zoning” on page 33
- ▶ 2.5, “Distance extension for Remote Copy services” on page 60
- ▶ 2.6, “Tape and disk traffic that share the SAN” on page 66
- ▶ 2.7, “Switch interoperability” on page 66

## 2.1 SAN topology general guidelines

The SAN topology requirements for IBM SAN Volume Controller do not differ too much from any other SAN. Remember that a well-sized and designed SAN enables you to build a redundant and failure-proof environment, and minimizing performance issues and bottlenecks. Therefore, before installing any of the products covered by this book, ensure that your environment follows an actual SAN design and architecture, with vendor-recommended SAN devices and code levels.

For more information about SAN design and preferred practices, see [SAN Fabric Administration Best Practices Guide](#).

A topology is described in terms of how the switches are interconnected. There are several different SAN topologies, such as core-edge, edge-core-edge, or full mesh. Each topology has its utility, scalability, and also its cost, so one topology will be a better fit for some SAN demands than others. Independent of the environment demands, there are a few best practices that must be followed to keep your SAN working correctly, performing well, redundant, and resilient.

### 2.1.1 SAN performance and scalability

Regardless of the storage and the environment, planning and sizing of the SAN makes a difference when growing your environment and when troubleshooting problems.

Because most SAN installations continue to grow over the years, the main SAN industry-lead companies design their products in a way to support a certain growth. Keep in mind that your SAN must be designed to accommodate both short-term and medium-term growth.

From the performance standpoint, the following topics must be evaluated and considered:

- ▶ Host-to-storage fan-in fan-out ratios
- ▶ Host to inter-switch link (ISL) oversubscription ratio
- ▶ Edge switch to core switch oversubscription ratio
- ▶ Storage to ISL oversubscription ratio
- ▶ Size of the trunks
- ▶ Monitor for slow drain device issues

From the scalability standpoint, ensure that your SAN will support the new storage and host traffic. Make sure that the chosen topology will also support a growth not only in performance, but also in port density.

If new ports need to be added to the SAN, you might need to drastically modify the SAN to accommodate a larger-than-expected number of hosts or storage. Sometimes these changes increase the number of hops on the SAN, and so cause performance and ISL congestion issues. For additional information, see 2.1.2, “ISL considerations” on page 21.

Consider the use of SAN director-class switches. They reduce the number of switches in a SAN and provide the best scalability available. Most of the SAN equipment vendors provide high port density switching devices. With MDS 9718 Multilayer Director, Cisco offers the industry’s highest port density single chassis with up to 768 16/32 Gb ports. The Brocade UltraScale Inter-Chassis Links (ICL) technology enables you to create multichassis configurations with up to 4608 16/32 Gb ports.

Therefore, if possible, plan for the maximum size configuration that you expect your IBM SAN Volume Controller (SAN Volume Controller or IBM SAN Volume Controller) installation to reach. Planning for the maximum size does not mean that you must purchase all of the SAN hardware initially. It requires you to design only the SAN to reach the expected maximum size.

## 2.1.2 ISL considerations

ISLs are responsible for interconnecting the SAN switches, creating SAN flexibility and scalability. For this reason, they can be considered as the core of a SAN topology. Consequently, they are sometimes the main cause of issues that can affect a SAN. For this reason it is important to take extra caution when planning and sizing the ISL in your SAN.

Regardless of your SAN size, topology, or the size of your SAN Volume Controller installation, consider the following practices to your SAN Inter-switch link design:

- ▶ Beware of the ISL oversubscription ratio

The standard recommendation is up to 7:1 (seven hosts using a single ISL). However, it can vary according to your SAN behavior. Most successful SAN designs are planned with an oversubscription ratio of 7:1 and some extra ports are reserved to support a 3:1 ratio. However, high-performance SANs start at a 3:1 ratio.

Exceeding the standard 7:1 oversubscription ratio requires you to implement fabric bandwidth threshold alerts. If your ISLs exceed 70%, schedule fabric changes to distribute the load further.

- ▶ Avoid unnecessary ISL traffic

Connect all SAN Volume Controller node ports in a clustered system to the same SAN switches or Directors as all of the storage devices with which the clustered system of SAN Volume Controller is expected to communicate. Conversely, storage traffic and internode traffic must never cross an ISL (except during migration scenarios).

Keep high-bandwidth use servers and I/O Intensive application on the same SAN switches as the SAN Volume Controller host ports. Placing these servers on a separate switch can cause unexpected ISL congestion problems. Also, placing a high-bandwidth server on an edge switch wastes ISL capacity.

- ▶ Properly size the ISLs on your SAN. They must have adequate bandwidth and buffer credits to avoid traffic or frames congestion. A congested inter-switch link can affect the overall fabric performance.
- ▶ Always deploy redundant ISLs on your SAN. Using an extra ISL avoids congestion if an ISL fails because of certain issues, such as a SAN switch line card or port blade failure.
- ▶ Use the link aggregation features, such as Brocade Trunking or Cisco Port Channel, to obtain better performance and resiliency.
- ▶ Avoid exceeding two hops between the SAN Volume Controller and the hosts. More than two hops are supported. However, when ISLs are not sized properly, more than two hops can lead to ISL performance issues and buffer credit starvation (SAN congestion).

When sizing over two hops, consider that all of the ISLs that go to the switch where the SAN Volume Controller is connected also handle the traffic that is coming from the switches on the edges, as shown in Figure 2-1.

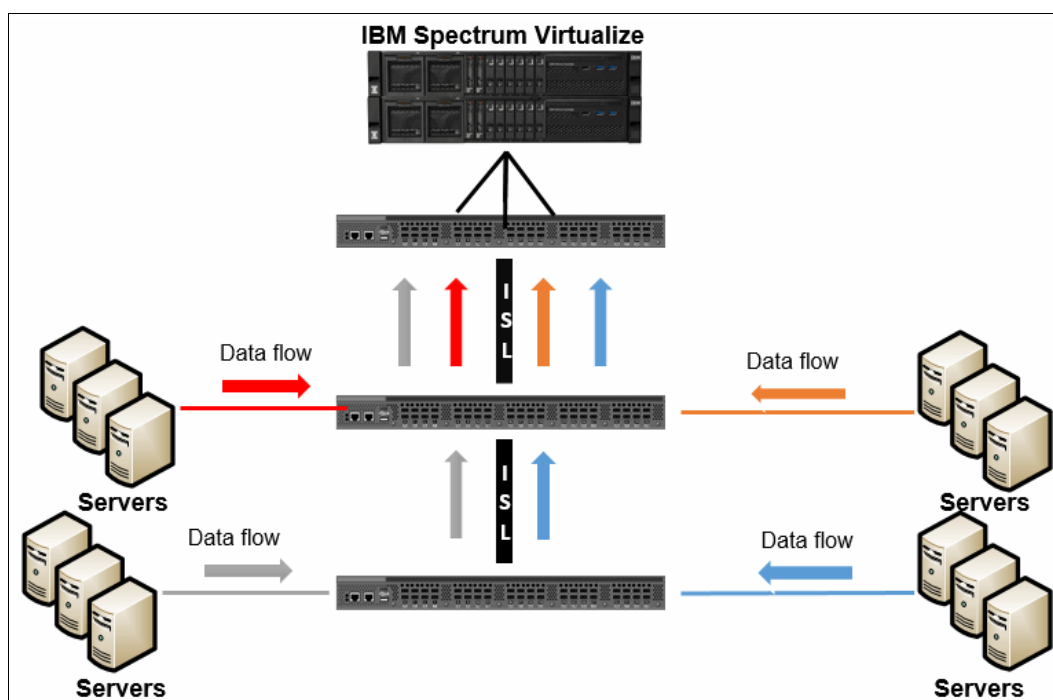


Figure 2-1 ISL data flow

Consider the following points:

- ▶ If possible, use SAN directors to avoid many ISL connections. Problems that are related to oversubscription or congestion are much less likely to occur within SAN director fabrics.
- ▶ When interconnecting SAN directors through ISL, spread the ISL cables across different directors blades. In a situation where an entire blade fails, the ISL will still be redundant through the links connected to other blades.
- ▶ Plan for the peak load, not for the average load.

## 2.2 SAN topology-specific guidelines

Some preferred practices, as mentioned in 2.1, “SAN topology general guidelines” on page 20, apply to all SANs. However, there are specific preferred practices requirements to each SAN topology available. The following topic shows the difference between the different kinds of topology and highlights the specific considerations for each of them.

This section covers the following topologies:

- ▶ Single switch fabric
- ▶ Core-edge fabric
- ▶ Edge-core-edge
- ▶ Full mesh

## 2.2.1 Single switch SAN Volume Controller SANs

The most basic SAN Volume Controller topology consists of a single switch per SAN fabric. This switch can range from a 24-port 1U switch for a small installation of a few hosts and storage devices, to a director with hundreds of ports. This configuration is a low-cost design solution that has the advantage of simplicity and is a sufficient architecture for small-to-medium SAN Volume Controller installations.

One of the advantages of a single switch SAN is that when all servers and storages are connected to the same switches, no hop exists.

**Note:** To meet redundancy and resiliency requirements, a single switch solution needs at least two SAN switches and directors, with one per different fabric.

The preferred practice is to use a multislot director-class single switch over setting up a core-edge fabric that is made up solely of lower-end switches, as described in 2.1.1, “SAN performance and scalability” on page 20.

The single switch topology, as shown in Figure 2-2, has only two switches; therefore, the SAN Volume Controller ports must be equally distributed on both fabrics.

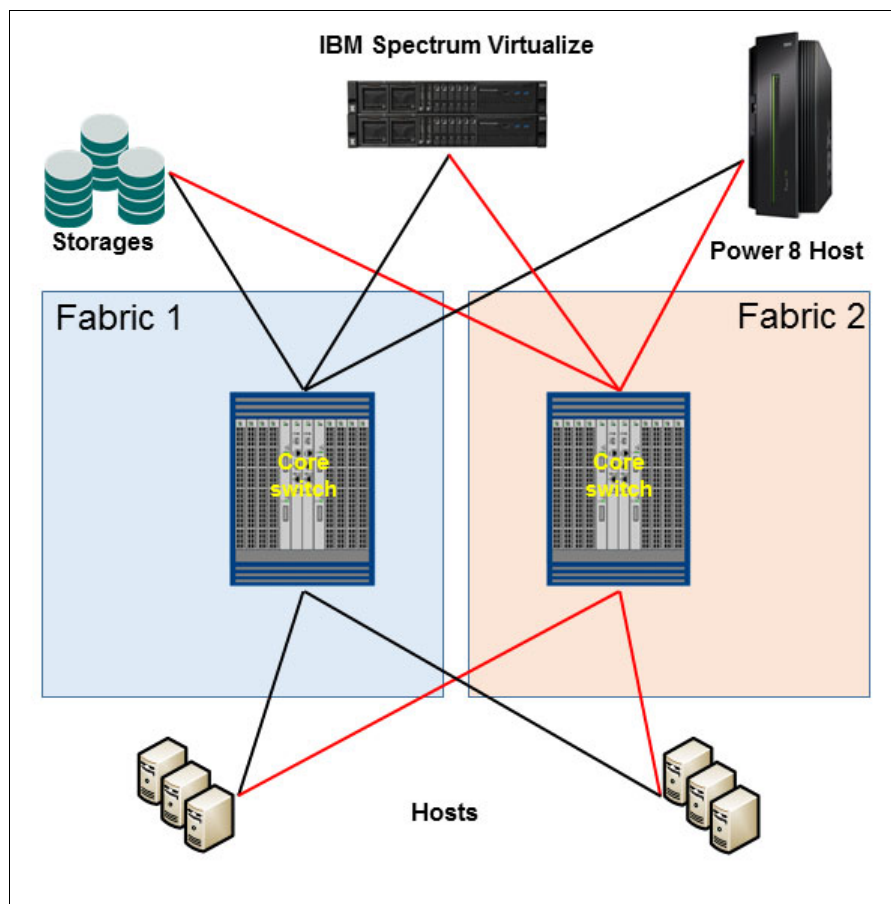


Figure 2-2 Single switch SAN

## 2.2.2 Basic core-edge topology

The core-edge topology (as shown in Figure 2-3) is easily recognized by most SAN architects. This topology consists of a switch in the center (usually, a director-class switch), which is surrounded by other switches. The *core switch* contains all SAN Volume Controller ports, storage ports, and high-bandwidth hosts. It is connected by using ISLs to the edge switches. The edge switches can be of any size from 24 port switches up to multi-slot directors.

When the SAN Volume Controller nodes and servers are connected to different switches, the hop count for this topology is one.

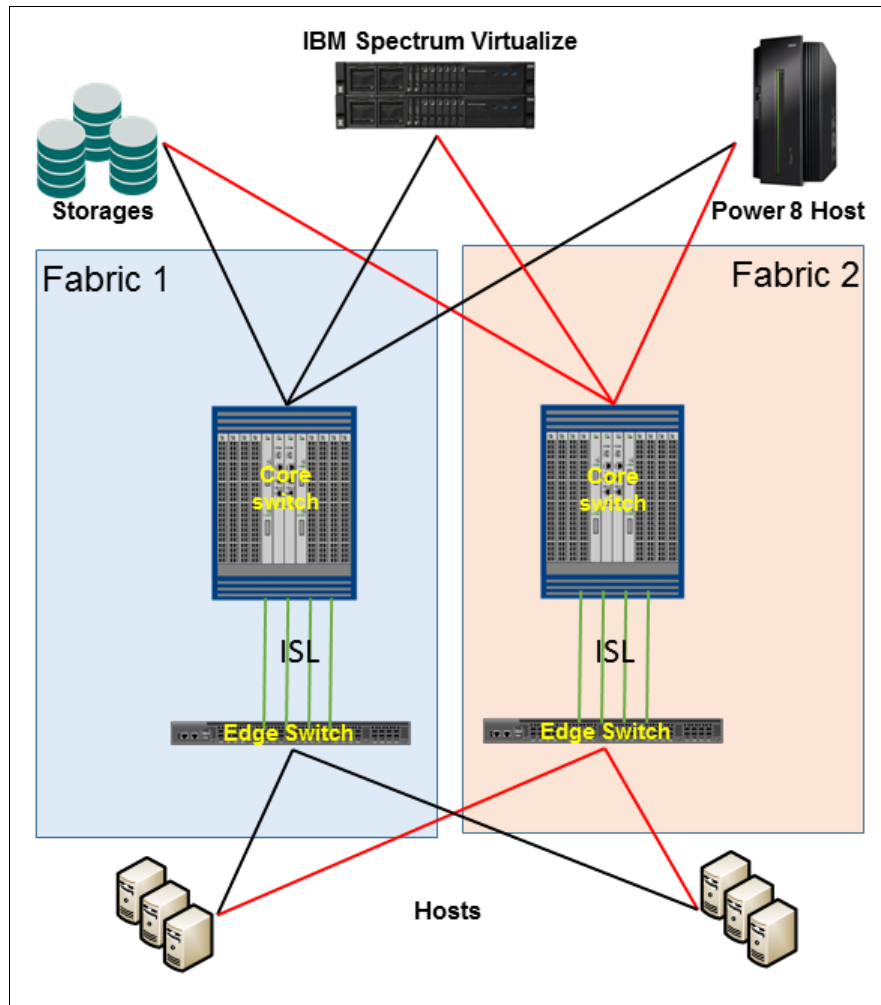


Figure 2-3 Core-edge topology

## 2.2.3 Edge-core-edge topology

Edge-core-edge is the most scalable topology, it is used for installations where a core-edge fabric made up of multislot director-class SAN switches is insufficient. This design is useful for large, multiclustered system installations. Similar to a regular core-edge, the edge switches can be of any size, and multiple ISLs must be installed per switch.



Figure 2-4 shows an edge-core-edge topology with two different edges, one of which is exclusive for the storage, SAN Volume Controller, and high-bandwidth servers. The other pair is exclusively for servers.

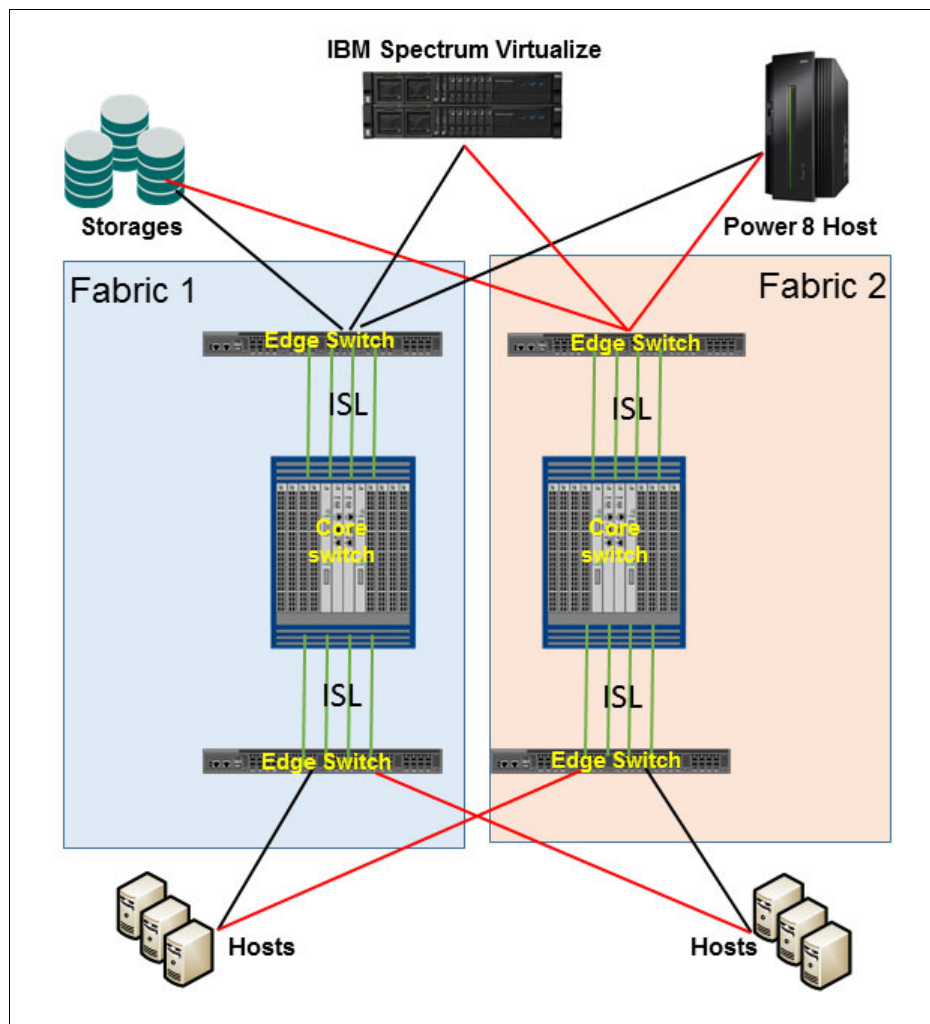


Figure 2-4 Edge-core-edge topology

Edge-core-edge fabrics allow better isolation between tiers. For additional information, see 2.2.6, “Device placement” on page 27.

## 2.2.4 Full mesh topology

In a full mesh topology, all switches are interconnected to all other switches on the same fabric. So the server and storage placement is not a concern after the number of hops is no more than one hop. Figure 2-5 shows a full mesh topology.

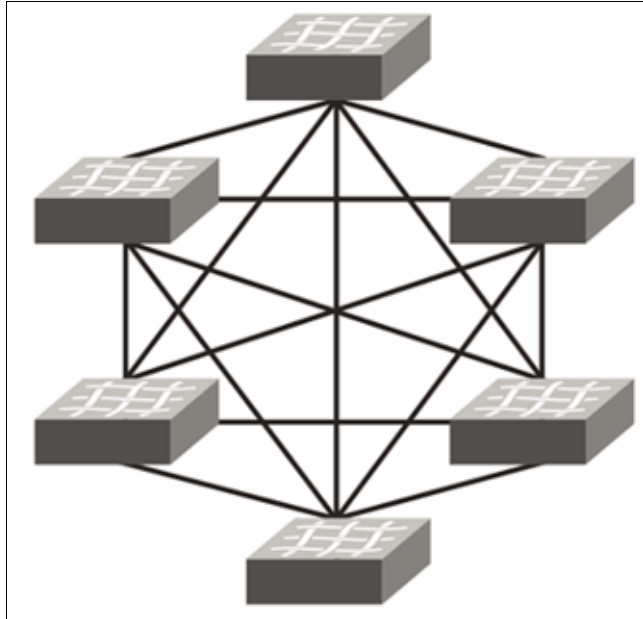


Figure 2-5 Full mesh topology

## 2.2.5 IBM Spectrum Virtualize as a SAN bridge

IBM SAN Volume Controller nodes now have a maximum of 16 ports. In addition to the increased throughput capacity, this number of ports enables new possibilities and allows different kinds of topologies and migration scenarios.

One of these topologies is the use of a SAN Volume Controller as a bridge between two isolated SANs. This configuration is useful for storage migration or sharing resources between SAN environments without merging them. Another use is if your devices with different SAN requirements are used in your installation.

Figure 2-6 shows an example of an IBM Spectrum Virtualize as a SAN bridge.

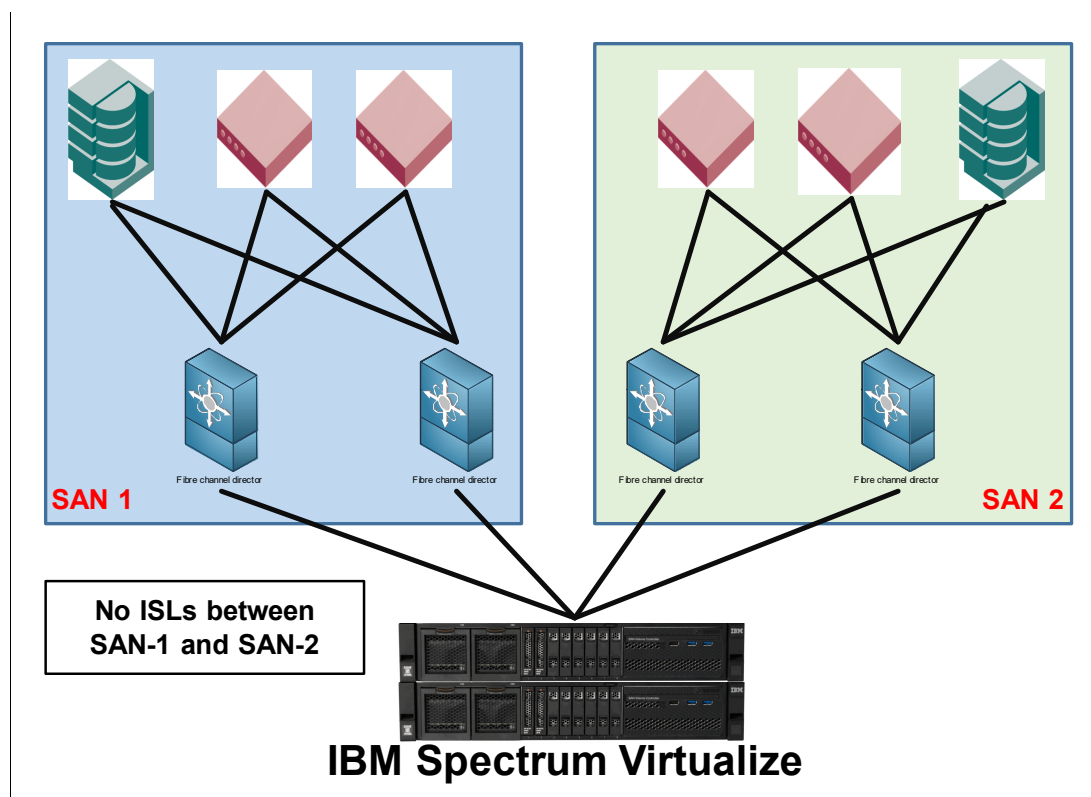


Figure 2-6 IBM Spectrum Virtualize as SAN bridge

Notice in Figure 2-6 that both SANs (Blue and Green) are isolated and no communication exists through ISLs. When connected to both fabrics, the SAN Volume Controller can virtualize storage from either fabrics. They also can provide disks from storage on the Green SAN (right), for example, to hosts on blue SAN (left).

## 2.2.6 Device placement

With the growth of virtualization, it is not usual to experience frame congestion on the fabric. Device placement seeks to balance the traffic across the fabric to ensure that the traffic is flowing in a specific way to avoid congestion and performance issues. The ways to balance the traffic consist of isolating traffic by using zoning, virtual switches, or traffic isolation zoning.

Keeping the traffic local to the fabric is a strategy to minimize the traffic between switches (and ISLs) by keeping storages and hosts attached to the same SAN switch, as shown in Figure 2-7.

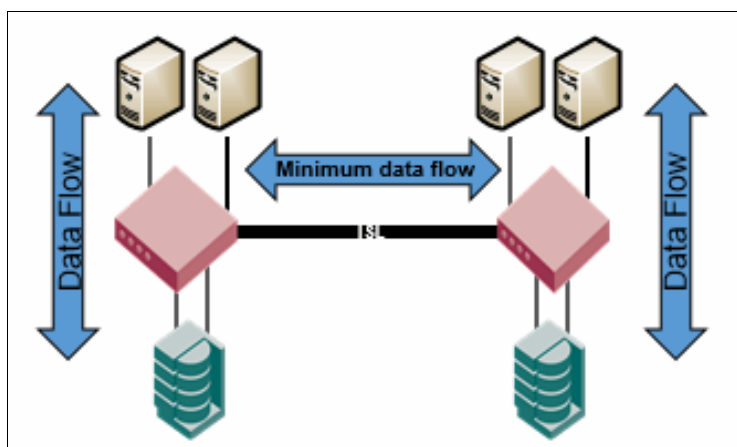


Figure 2-7 Storage and hosts attached to the same SAN switch

This solution can fit perfectly in small and medium SANs. However, it is not as scalable as other topologies available. As described in 2.2, “SAN topology-specific guidelines” on page 22, the most scalable SAN topology is the edge-core-edge.

In addition to scalability, this topology provides different resources to isolate the traffic and reduce possible SAN bottlenecks. Figure 2-8 shows an example of traffic segregation on the SAN by using edge-core-edge topology.

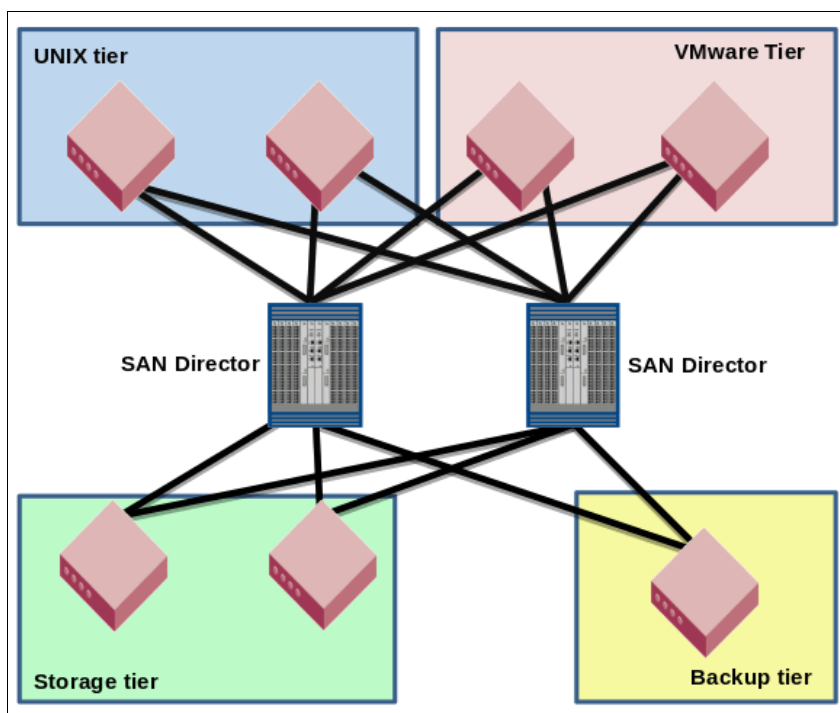


Figure 2-8 Edge-core-edge segregation

Even when sharing the same core switches, it is possible to use virtual switches (see “SAN partitioning” on page 29) to isolate one tier from the other. This configuration helps avoid traffic congestion that is caused by slow drain devices that are connected to the backup tier switch.

## SAN partitioning

SAN partitioning is a hardware-level feature that allows SAN switches to share hardware resources by partitioning its hardware into different and isolated virtual switches. Both Brocade and Cisco provide SAN partitioning features called, respectively, *Virtual Fabric* and *Virtual SAN* (VSAN).

Hardware-level fabric isolation is accomplished through the concept of switch virtualization, which allows you to partition physical switch ports into one or more “virtual switches.” Virtual switches are then connected to form virtual fabrics.

From a device perspective, SAN partitioning is completely transparent and so the same guidelines and practices that apply to physical switches apply also to the virtual ones.

Although the main purposes of SAN partitioning are port consolidation and environment isolation, this feature is also instrumental in the design of a business continuity solution based on IBM Spectrum Virtualize.

For a description of the IBM Spectrum Virtualize business continuity solutions, see Chapter 7, “Business continuity” on page 329.

## 2.3 SAN Volume Controller ports

IBM SAN Volume Controller hardware significantly increased port connectivity options. Models 2145-DH8 and 2145-SV1 deliver up to 16x 16 Gb FC ports per node, as listed in Table 2-1.

Table 2-1 IBM SAN Volume Controller connectivity

| Feature                        | 2145-DH8                                       | 2145-SV1                 |
|--------------------------------|------------------------------------------------|--------------------------|
| Fibre Channel Host Bus Adapter | 4x Quad 8 Gb<br>4x Dual 16 Gb<br>4x Quad 16 Gb | 4x Quad 16 Gb            |
| Ethernet I/O                   | 4x Quad 10 Gb iSCSI/FCoE                       | 4x Quad 10 Gb iSCSI/FCoE |
| Built in ports                 | 4x 1 Gb                                        | 4x 10 Gb                 |
| SAS expansion ports            | 4x 12 Gb SAS                                   | 4x 12 Gb SAS             |

This new port density expands the connectivity options and provides new ways to connect the IBM SAN Volume Controller to the SAN. This sections describes some preferred practices and use cases that show how to connect a SAN volume controller on the SAN to use this increased capacity.

### 2.3.1 Slots and ports identification

The SAN Volume Controller can have up to four quad Fibre Channel (FC) host bus adapter (HBA) cards (16 FC ports) per node. Figure 2-9 shows the port location in the rear view of the 2145-SV1 node.

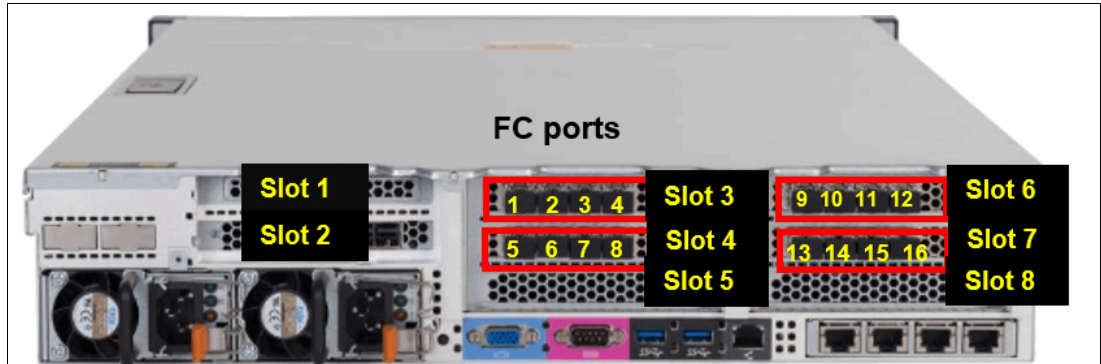


Figure 2-9 SAN Volume Controller 2145-SV1 rear port view

Figure 2-10 shows the port locations for the SV2/SA2 nodes.

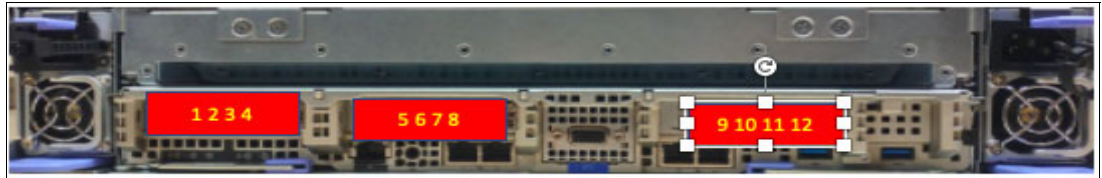


Figure 2-10 SV2/SA2 Node Layout

For maximum redundancy and resiliency, spread the ports across different fabrics. Because the port count varies according to the number of cards that is included in the solution, try to keep the port count equal on each fabric.

### 2.3.2 Port naming and distribution

In the field, fabric naming conventions vary. However, it is common to find fabrics that are named, for example, PROD\_SAN\_1 and PROD\_SAN\_2, or PROD\_SAN\_A and PROD\_SAN\_B. This type of naming convention is used to simplify the SAN Volume Controller, after their denomination followed by 1 and 2 or A and B, which specifies that the devices connected to those fabrics contains the redundant paths of the same servers and SAN devices.

To simplify the SAN connection identification and troubleshooting, keep all odd ports on the odd fabrics, or “A” fabrics and the even ports on the even fabric or “B” fabrics, as shown in Figure 2-11.

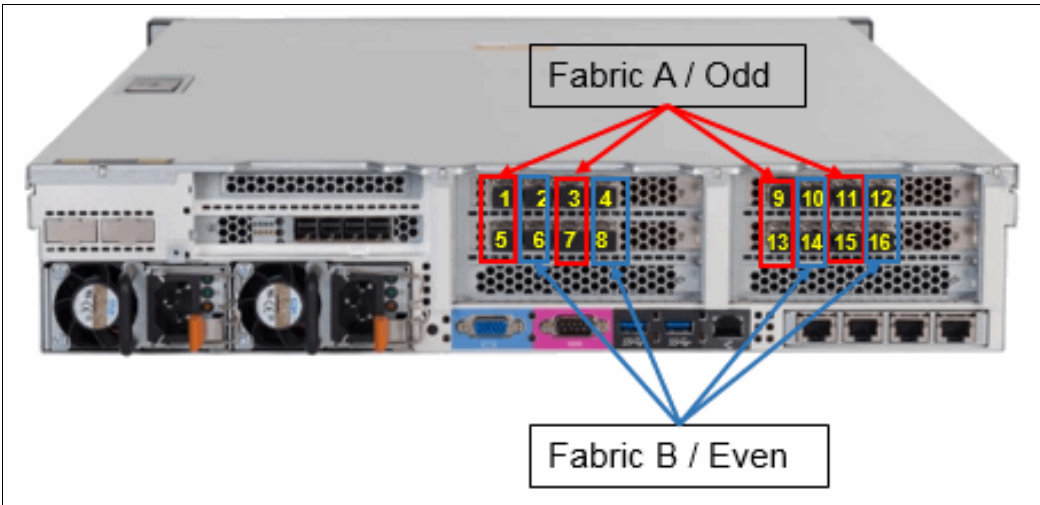


Figure 2-11 SAN Volume controller model 2145-SV1 Port distribution

SAN Volume Controller clusters follow the same arrangement, odd ports to odd or “A” Fabric, and even ports that are attached to even fabrics or “B” fabric.

As a preferred practice, assign specific uses to specific SAN volume controller ports. This technique helps optimize the port utilization by aligning the internal allocation of hardware CPU cores and software I/O threads to those ports.

Figure 2-12 shows the specific port use guidelines for IBM SAN Volume Controller and Spectrum Virtualize products.

|                                                                                                                                                                                                                                                                                                                                                                                | 4 port                  | 8 port                     | 12 port                      | 16 port                     | SAN Fabric |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|------------------------------|-----------------------------|------------|
| Adapter 1 Port 1                                                                                                                                                                                                                                                                                                                                                               | Host+Storage            | Host+Storage               | Host+Storage                 | Host+Storage                | A          |
| Adapter 1 Port 2                                                                                                                                                                                                                                                                                                                                                               | Host+Storage            | Host+Storage               | Host+Storage                 | Host+Storage                | B          |
| Adapter 1 Port 3                                                                                                                                                                                                                                                                                                                                                               | Intracuster+Replication | Intracuster                | Intracuster                  | Intracuster                 | A          |
| Adapter 1 Port 4                                                                                                                                                                                                                                                                                                                                                               | Intracuster+Replication | Intracuster                | Intracuster                  | Intracuster                 | B          |
| Adapter 2 Port 1                                                                                                                                                                                                                                                                                                                                                               |                         | Host+Storage               | Host+Storage                 | Host+Storage                | A          |
| Adapter 2 Port 2                                                                                                                                                                                                                                                                                                                                                               |                         | Host+Storage               | Host+Storage                 | Host+Storage                | B          |
| Adapter 2 Port 3                                                                                                                                                                                                                                                                                                                                                               |                         | Intracuster or Replication | Replication or Host+Storage  | Replication or Host+Storage | A          |
| Adapter 2 Port 4                                                                                                                                                                                                                                                                                                                                                               |                         | Intracuster or Replication | Replication or Host+Storage  | Replication or Host+Storage | B          |
| Adapter 3 Port 1                                                                                                                                                                                                                                                                                                                                                               |                         |                            | Host+Storage                 | Host+Storage                | A          |
| Adapter 3 Port 2                                                                                                                                                                                                                                                                                                                                                               |                         |                            | Host+Storage                 | Host+Storage                | B          |
| Adapter 3 Port 3                                                                                                                                                                                                                                                                                                                                                               |                         |                            | Intracuster                  | Intracuster                 | A          |
| Adapter 3 Port 4                                                                                                                                                                                                                                                                                                                                                               |                         |                            | Intracuster                  | Intracuster                 | B          |
| Adapter 4 Port 1                                                                                                                                                                                                                                                                                                                                                               |                         |                            |                              | Host+Storage                | A          |
| Adapter 4 Port 2                                                                                                                                                                                                                                                                                                                                                               |                         |                            |                              | Host+Storage                | B          |
| Adapter 4 Port 3                                                                                                                                                                                                                                                                                                                                                               |                         |                            |                              | Replication or Host+Storage | A          |
| Adapter 4 Port 4                                                                                                                                                                                                                                                                                                                                                               |                         |                            |                              | Replication or Host+Storage | B          |
| localfcportmask                                                                                                                                                                                                                                                                                                                                                                | 1100                    | 11001100 OR 00001100       | 110000001100 OR 110000001100 | 0000110000001100            |            |
| remotefcportmask                                                                                                                                                                                                                                                                                                                                                               | 1100                    | 00000000 OR 11000000       | 000000000000 OR 000011000000 | 1100000011000000            |            |
| Host refers to host objects defined in the system.<br>Storage refers to controller objects defined in the system.<br>Replication refers to nodes which are part of a different cluster.<br>Intracuster refers to nodes within the same cluster.<br>The "+" indicates that both types are should to be used<br>The word "or" indicates that one of the options must be selected |                         |                            |                              |                             |            |

Figure 2-12 Port use guidelines for IBM SAN Volume Controller and Spectrum Virtualize products

Because of port availability on IBM SAN Volume Controller clusters, and the increased bandwidth with the 16 Gb ports, it is possible to segregate the port assignment between hosts and storage, thus isolating their traffic.

Host and storage ports have different traffic behavior, so keeping host and storage ports together produces maximum port performance and utilization by benefiting from its full duplex bandwidth. For this reason, sharing host and storage traffic in the same ports is generally the preferred practice.

However, traffic segregation can also provide some benefits in terms of troubleshooting and host zoning management. For example, consider SAN congestion conditions that are the result of a slow draining device. Because of this issue, all slower hosts (that is, hosts with port speeds that are more than 2x less the IBM SAN Volume Controller node port speeds) should be isolated to their own IBM SAN Volume Controller node ports with no other hosts or controllers zoned to these ports where possible.

In this case, segregating the ports simplifies the identification of the device causing the problem. At the same time, it limits the effects of the congestion to the hosts or backend ports only. Furthermore, dedicating ports for host traffic reduces the possible combinations of host zoning and simplifies SAN management. It is advised to implement the port traffic segregation with configurations with a suitable number of ports (that is, 12 ports or more) only.

**Note:** Consider the following points:

- ▶ On IBM SAN Volume Controller clusters with 12 ports or more per node, if you use advanced copy services (such as volume mirroring, flash copies, or remote copies), it is recommended that you use four ports per node for inter-node communication. The IBM SAN Volume Controller uses the internode ports for all of these functions and the use of these features greatly increases the data rates that are sent across these ports. If you are implementing a new cluster and plan to use any copy services, because of this you should plan on having 12 ports per cluster node.
- ▶ Use port masking to assign specific uses to the SAN Volume Controller ports. For more information, see Chapter 8, “Hosts” on page 339.

## Buffer credits

SAN Volume Controller ports feature a predefined number of buffer credits. The amount of buffer credits determines the available throughput over distances:

- ▶ All 8 Gbps adapters have 41 credits available per port, saturating links at up to 10 km (6.2 miles) at 8 Gbps
- ▶ 2-port 16 Gbps (DH8 only nodes) adapters have 80 credits available per port, saturating links at up to 10 km (6.2 miles) at 16 Gbps
- ▶ 4-port 16 Gbps adapters have 40 credits available per port, saturating links at up to 5 km at (3.1 miles) 16 Gbps

**Switch port buffer credit:** For stretched cluster and IBM HyperSwap configurations that do not use ISLs for the internode communication, it is advised to set the switch port buffer credits to match the IBM Spectrum Virtualize port.

## Port designation and CPU cores utilization

The ports assignment/designation recommendation is based on the relationship between a single port to a CPU and core.



Figure 2-13 shows the Port to CPU core mapping for a 2145-SV1 node.

| Uncompressed-Physical port to CPU Core |    |    |    |    |    |    |    |    |
|----------------------------------------|----|----|----|----|----|----|----|----|
| CPU1 CORE                              | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| Port                                   | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  |
| CPU2 Core                              | 8  | 9  | 10 | 11 | 12 | 13 | 14 | XX |
| Port                                   | 7  | 6  | 5  | 4  | 3  | 2  | 1  | XX |
|                                        |    |    |    |    | 16 |    |    |    |

| Compressed-Physical port to CPU Core |    |    |   |    |    |    |   |   |
|--------------------------------------|----|----|---|----|----|----|---|---|
| CPU1 CORE                            | 0  | 1  | 2 | 3  | 4  | 5  | 6 | 7 |
| Port                                 | 15 | 14 | 1 | 12 | 11 | 10 | 9 | 8 |
|                                      | 7  | 6  | 3 | 4  | 3  | 2  | 1 |   |
|                                      |    |    | 5 |    | 16 |    |   |   |

Slot 3

Slot 4

Slot 6

Slot 7

Figure 2-13 Port to CPU core mapping for SV1 nodes

**N\_Port ID Virtualization feature:** On N\_Port ID Virtualization (NPIV)-enabled systems, the port to CPU core assignment for the virtual WWPNN is the same as the physical WWPNN.

## 2.4 Zoning

Because of the nature of storage virtualization and cluster scalability, the IBM SAN Volume Controller zoning differs from traditional storage devices. Zoning an IBM SAN Volume Controller cluster into a SAN fabric requires planning and following specific guidelines.

**Important:** Errors that are caused by improper SAN Volume Controller zoning are often difficult to isolate and the steps to fix them can affect the SAN environment. Therefore, create your zoning configuration carefully.

The initial configuration for SAN Volume Controller requires the following different zones:

- ▶ Internode and intra-cluster zones
- ▶ Replication zones (if replication is used)
- ▶ Back-end storage to SAN Volume Controller zoning
- ▶ Host to SAN Volume Controller zoning

For each zoning type, different guidelines must be followed, which are described in 2.4.1, “Types of zoning” on page 34.

**Note:** Although internode and intra-cluster zone is not necessary for non-clustered IBM Storwize family systems, it is generally preferred to use one of these zones.

## 2.4.1 Types of zoning

Modern SAN switches feature two types of zoning: Port zoning, and worldwide port name (WWPN) zoning. The preferred method is to use only WWPN zoning. A common misconception is that WWPN zoning provides poorer security than port zoning, which is not the case. Modern SAN switches enforce the zoning configuration directly in the switch hardware. Also, you can use port binding functions to enforce a WWPN to be connected to a specific SAN switch port.

When switch-port based zoning is used, the ability to allow only specific hosts to connect to an IBM SAN Volume Controller cluster is lost.

Consider an NPV device, such as an Access Gateway that is connected to a fabric. If 14 hosts are attached to that NPV Device, and switch port-based zoning is used to zone the switch port for the NPV device to IBM SAN Volume Controller node ports, all 14 hosts can potentially connect to the IBM SAN Volume Controller cluster, even if the IBM SAN Volume Controller is virtualizing storage for only 4 or 5 of those hosts.

However, the problem is exacerbated when the IBM SAN Volume Controller NPIV feature is used in transitional mode. In this mode, a host can connect to the physical and virtual WWPNs on the cluster. With switch port zoning, this configuration doubles the connection count for each host attached to the IBM SAN Volume Controller cluster. This issue can affect the function of path failover on the hosts by resulting in too many paths, and the IBM SAN Volume Controller Cluster can exceed the maximum host connection count on a large fabric.

If you have the NPIV feature enabled on your IBM SAN Volume Controller, you must use WWPN-based zoning.

**Zoning types:** Avoid the use of a zoning configuration that includes a mix of port and WWPN zoning.

A preferred practice for traditional zone design calls for *single initiator* zoning. That is, a zone can consist of many target devices, but only one initiator because target devices often wait for an initiator device to connect to them, while initiators actively attempt to connect to each device to which they are zoned. The single initiator approach removes the possibility of a misbehaving initiator affecting other initiators.

The drawback to single initiator zoning is that on a large SAN that features many zones, the SAN administrator’s job can be more difficult, and the number of zones on a large SAN can exceed the zone database size limits.

Cisco and Brocade developed features that can reduce the number of zones by allowing the SAN administrator to control which devices in a zone can communicate with other devices in the zone. The features are called Cisco Smart Zoning and Brocade Peer Zoning, which are supported with IBM Spectrum Virtualize systems.

A brief overview of these features is provided next.

**Note:** Brocade Traffic Isolation (TI) zoning is deprecated in FOS v9.0. You can still use TI zoning if you have existing zones, but you must keep at least one switch running a pre-9.0 version of FOS in the fabric to be able to make changes to the TI zones.

## Cisco Smart Zoning

Cisco Smart Zoning is a feature that, when enabled, restricts the initiators in a zone to communicate only with target devices in the same zone. For our cluster example, this feature allows a SAN administrator to zone all of the host ports for a VMware cluster in the same zone with the storage ports to which all of the hosts need access. Smart Zoning configures the access control lists in the fabric routing table to allow only the initiator (host) ports to communicate with target ports.

For more information about Smart Zoning, see [this web page](#).

For more information about implementation, see this [IBM Support web page](#).

## Brocade Peer Zoning

Brocade Peer Zoning is a feature that provides a similar functionality of restricting what devices can see other devices within the same zone. However, Peer Zoning is implemented such that some devices in the zone are designated as principal devices. The non-principal devices can only communicate with the principal device, not with each other. As with Cisco, the communication is enforced in the fabric routing table.

For more information, see Peer Zoning in *Modernizing Your IT Infrastructure with IBM b-type Gen 6 Storage Networking and IBM Spectrum Storage Products*, SG24-8415.

**Note:** Use Smart and Peer zoning for the host zoning only. Use traditional zoning for intracluster, back-end, and intercluster zoning.

## Simple zone for small environments

As an option for small environments, IBM Spectrum Virtualize-based systems support a simple set of zoning rules that enable a small set of host zones to be created for different environments.

For systems with fewer than 64 hosts that are attached, zones that contain host HBAs must contain no more than 40 initiators, including the ports that acts as initiators, such as the IBM Spectrum Virtualize based system ports that are target + initiator. Therefore, a valid zone can be 32 host ports plus 8 IBM Spectrum Virtualize based system ports. Include only one port from each node in the I/O groups that are associated with this host.

**Note:** Do not place more than one HBA port from the same host in the same zone. Also, do not place dissimilar hosts in the same zone. Dissimilar hosts are hosts that are running different operating systems or are different hardware products.

## 2.4.2 Prezoning tips and shortcuts

In this section, we describe several tips and shortcuts that are available for SAN Volume Controller zoning.

### Naming convention and zoning scheme

When you create and maintaining a SAN Volume Controller zoning configuration, you must have a defined naming convention and zoning scheme. If you do not define a naming convention and zoning scheme, your zoning configuration can be difficult to understand and maintain.

Remember that environments have different requirements, which means that the level of detailing in the zoning scheme varies among environments of various sizes. Therefore, ensure that you have an easily understandable scheme with an appropriate level of detail. Then, make sure that you use it consistently and adhere to it whenever you change the environment.

For more information about SAN Volume Controller naming convention, see 10.12.1, “Naming conventions” on page 452.

### Aliases

Use zoning aliases when you create your SAN Volume Controller zones if they are available on your specific type of SAN switch. Zoning aliases makes your zoning easier to configure and understand, and causes fewer possibilities for errors.

One approach is to include multiple members in one alias because zoning aliases can normally contain multiple members (similar to zones). This approach can help avoid some common issues that are related to zoning and make it easier to maintain the port balance in a SAN.

Create the following zone aliases:

- ▶ One zone alias for each SAN Volume Controller port
- ▶ Zone an alias group for each storage subsystem port pair (the SAN Volume controller must reach the same storage ports on both I/O group nodes)

You can omit host aliases in smaller environments, as we did in the lab environment that was used for this publication. Figure 2-14 on page 37 shows some alias examples.

## 2.4.3 SAN Volume Controller internode communications zones

Internode (or intra-cluster) communication is critical to the stable operation of the cluster. The ports that carry internode traffic are used for mirroring write cache and metadata exchange between nodes and canisters.

In FlashSystem systems, internode communication primarily occurs through the internal PCI connectivity between the two canisters of a control enclosure. However, for the clustered FlashSystem systems, the internode communication requirements are similar to the SAN Volume Controller requirements.

To establish efficient, redundant, and resilient intracluster communication, the intracluster zone must contain at least two ports from each node or canister. For IBM SAN Volume Controller nodes with eight ports or more, generally isolate the intracluster traffic by dedicating node ports specifically to internode communication.

The ports to be used for intracluster communication varies according to the machine type-model number and port count. See Figure 2-12 on page 31 for port assignment recommendations.

**NPIV configurations:** On NPIV-enabled configurations, use the physical WWPN for the intracluster zoning.

Only 16 port logins are allowed from one node to any other node in a SAN fabric. Ensure that you apply the correct port masking to restrict the number of port logins. Without port masking, any SAN Volume Controller port and any member of the same zone can be used for intracluster communication, even the port members of IBM SAN Volume Controller to host and IBM SAN Volume Controller to storage zoning.

**Note:** To check whether the login limit is exceeded, count the number of distinct ways by which a port on node X can log in to a port on node Y. This number must not exceed 16. For more information about port masking, see Chapter 8, “Hosts” on page 339.

## 2.4.4 SAN Volume Controller storage zones

The zoning between SAN Volume Controller and other storage is necessary to allow the virtualization of any storage space under the SAN Volume Controller. This storage is referred to as *back-end storage*.

A zone for each back-end storage to each SAN Volume controller node or canister must be created in both fabrics, as shown in Figure 2-14. Doing so reduces the overhead that is associated with many logins. The ports from the storage subsystem must be split evenly across the dual fabrics.

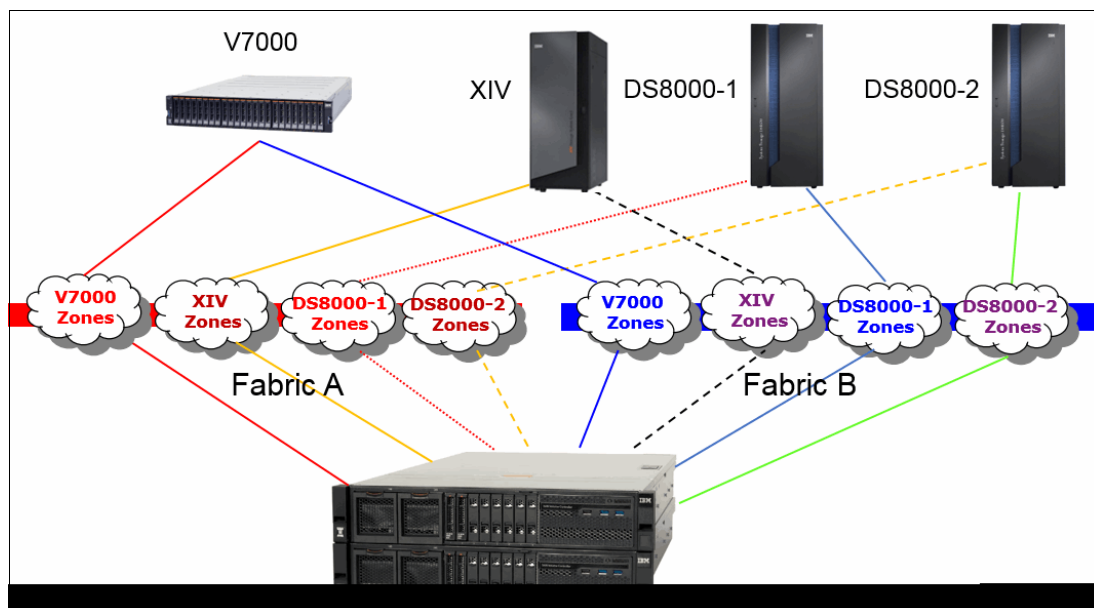


Figure 2-14 Back-end storage zoning

Often, all nodes or canisters in a SAN Volume Controller system should be zoned to the same ports on each back-end storage system, with the following exceptions:

- ▶ When implementing Enhanced Stretched Cluster or HyperSwap configurations where the back-end zoning can be different for the nodes or canisters accordingly to the site definition (see *IBM Spectrum Virtualize and SAN Volume Controller Enhanced Stretched Cluster with VMware*, SG24-8211 and *IBM Storwize V7000, Spectrum Virtualize, HyperSwap, and VMware Implementation*, SG24-8317)
- ▶ When the SAN has a multi-core design that requires special zoning considerations, as described in “Zoning to storage best practice” on page 39

**Important:** Consider the following points:

- ▶ On NPIV enabled systems, use the physical WWPN for the zoning to the back-end controller.
- ▶ For configurations where IBM SAN Volume Controller is virtualizing an IBM Spectrum Virtualize product, enable NPIV on the Spectrum Virtualize product and zone the IBM SAN Volume Controller cluster node ports to the virtual WWPNs on the Spectrum Virtualize storage system.

When two nodes or canisters are zoned to different set of ports for the same storage system, the SAN Volume Controller operation mode is considered degraded. The system then logs errors that request a repair action. This situation can occur if incorrect zoning is applied to the fabric.

Figure 2-15 shows a zoning example (that uses generic aliases) between a two node IBM SAN Volume Controller and a Storwize V5000. Notice that both SAN volume controller nodes can access the same set of Storwize V5000 ports.

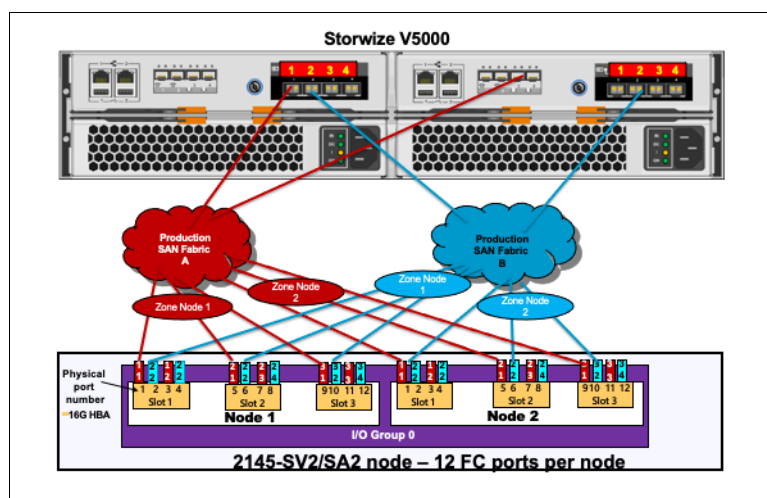


Figure 2-15 V5000 zoning

Each storage controller or model features its own preferred zoning and port placement practices. The generic guideline for all storage is to use the ports that are distributed between the redundant storage components, such as nodes, controllers, canisters, and FA adapters (respecting the port count limit, as described in “Back-end storage port count” on page 41).

The following sections describe the IBM Storage-specific zoning guide lines. Storage vendors other than IBM might have similar preferred practices. For more information, contact your vendor.

## Zoning to storage best practice

In 2.1.2, “ISL considerations” on page 21, we described ISL considerations for ensuring that the SAN Volume Controller is connected to the same physical switches as the back-end storage ports.

In 2.2, “SAN topology-specific guidelines” on page 22, we reviewed some of the options for SAN Design.

This section describes preferred practices for zoning SAN Volume Controller ports to controller ports on each of the different SAN designs.

The high-level best practice is to configure zoning such that the SAN Volume Controller ports are zoned only to the controller ports that are attached to the same switch. For single-core designed fabrics, this practice is not an issue because only one switch is used on each fabric to which the SAN Volume Controller and controller ports are connected. For the mesh and dual-core and other designs in which the SAN Volume Controller is connected to multiple switches in the same fabric, zoning might become an issue.

Figure 2-16 shows the preferred practice zoning on a dual-core fabric. You can see that two zones are used:

- ▶ Zone1 includes only the SAN Volume Controller and back-end ports that are attached to the core switch on the left.
- ▶ Zone2 includes only the SAN Volume Controller and back-end ports that are attached to the core switch on the right.

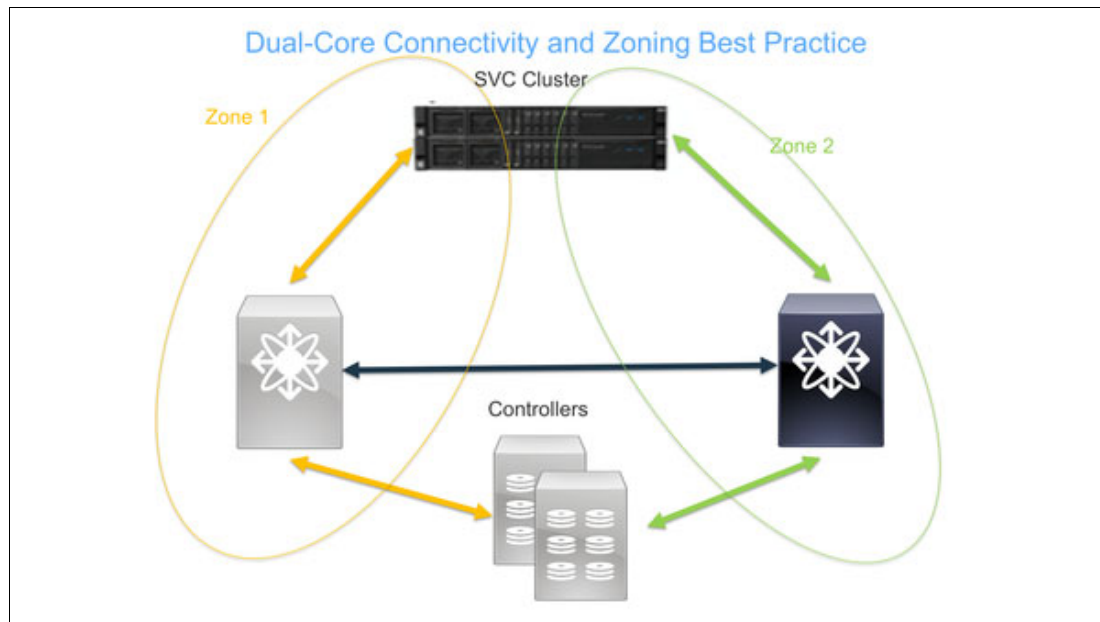


Figure 2-16 Dual core zoning schema

Mesh fabric designs that have the SAN Volume Controller and controller ports connected to multiple switches follow the same general guidelines. Failure to follow this preferred practice recommendation might result in SAN Volume Controller performance impacts to the fabric. Potential effects are described next.

### ***Real-life potential effect of deviation from best practice zoning***

Figure 2-17 shows a design that consists of a dual-core Brocade fabric with the SAN Volume Controller cluster attached to one switch and controllers attached to the other. An IBM GPFS cluster is attached to the same switch as the controllers. This real-world design was used for a client that was experiencing extreme performance problems on its SAN. The client had dual fabrics, each fabric had this same flawed design.

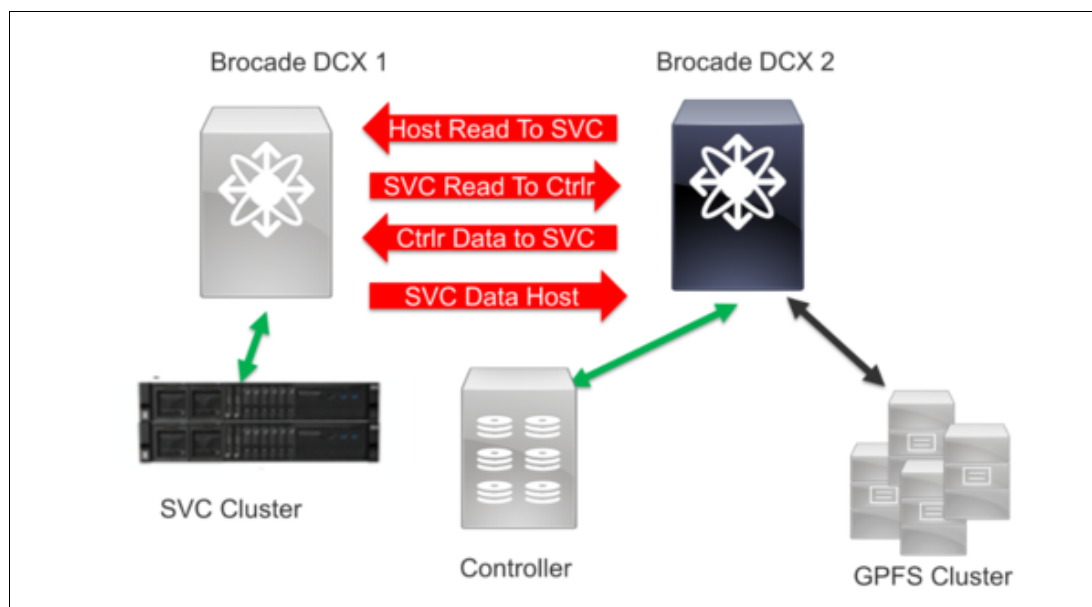


Figure 2-17 ISL traffic overloading

The design violates both the best practices of ensuring SAN Volume Controller and storage ports are connected to the same switches, and zoning the ports as shown in Figure 2-16 on page 39. It also violates the best practice of connecting the host ports (the GPFS cluster) to the same switches as the SAN Volume Controller where possible.

This creates an issue with traffic traversing the ISL unnecessarily, as shown in Figure 2-17. I/O requests from the GPFS cluster must traverse the ISL four times. This design should be corrected such that the SAN Volume Controller, controller, and GPFS cluster ports are all connected to both core switches, and zoning is redone to be in accordance with the example that is shown in Figure 2-16 on page 39.

Again, Figure 2-17 shows an actual customer SAN design. The effect of the extra traffic on the ISL between the core switches from this design caused significant delays in command response time from the GPFS cluster to the SAN Volume Controller and from the SAN Volume Controller to the Controller.

The SAN Volume Controller cluster also logged nearly constant errors against the controller, including disconnecting from controller ports. The SAN switches logged frequent link time-outs and frame drops on the ISL between the switches. Finally, the customer had other devices sharing the ISL that were not zoned to the SAN Volume Controller. These devices also were affected.



## Back-end storage port count

The current firmware available (V8.4 at the time of this writing), sets the limitation of 1024 worldwide node names (WWNNs) per SAN Volume Controller cluster and up to 1024 WWPNS. The rule is that each port represents a WWPNS count on the IBM SAN Volume Controller cluster. However, the WWNN count differs based on the type of storage.

For example, at the time of this writing, EMC DMX/Symmetrix, all HDS storage, and SUN/HP use one WWNN per port. This configuration means that each port appears as a separate controller to the SAN Volume Controller. Therefore, each port connected to the SAN Volume Controller means one WWPNS and a WWNN increment.

IBM storage and EMC Clariion/VNX use one WWNN per storage subsystem, so each appears as a single controller with multiple port WWPNS.

The preferred practice is to assign up to sixteen ports from each back-end storage to the SAN Volume Controller cluster. The reason for this limitation is that with V8.2, the maximum number of ports are recognized by the SAN Volume controller per each WWNN is sixteen. The more ports are assigned, the more throughput is obtained.

In a situation where the back-end storage has hosts direct attached, do not mix the host ports with the SAN Volume Controller ports. The back-end storage ports must be dedicated to the SAN Volume Controller. Therefore, sharing storage ports are functional only during migration and for a limited time. However, if you intend to have some hosts that are permanently directly attached to the back-end storage, you must segregate the SAN Volume Controller ports from the host ports.

## IBM XIV storage subsystem

IBM XIV storage is modular storage and is available as fully or partially populated configurations. XIV® hardware configuration can include between 6 and 15 modules. Each additional module added to the configuration increases the XIV capacity, CPU, memory, and connectivity.

From a connectivity standpoint, four Fibre Channel ports are available in each interface module for a total of 24 Fibre Channel ports in a fully configured XIV system. The XIV modules with FC interfaces are present on modules 4 through module 9. Partial rack configurations do not use all ports, even though they might be physically present.

Table 2-2 lists the XIV port connectivity according to the number of installed modules.

*Table 2-2 XIV connectivity ports as capacity grows*

| XIV modules | Total ports | Port interfaces | Active port modules  |
|-------------|-------------|-----------------|----------------------|
| 6           | 8           | 2               | 4 and 5              |
| 9           | 16          | 4               | 4, 5, 7, and 8       |
| 10          | 16          | 4               | 4, 5, 7, and 8       |
| 11          | 20          | 5               | 4, 5, 7, 8, and 9    |
| 12          | 20          | 5               | 4, 5, 7, 8, and 9    |
| 13          | 24          | 6               | 4, 5, 6, 7, 8, and 9 |
| 14          | 24          | 6               | 4, 5, 6, 7, 8, and 9 |
| 15          | 24          | 6               | 4, 5, 6, 7, 8, and 9 |

**Note:** If the XIV includes the capacity on demand (CoD) feature, all active Fibre Channel interface ports are usable at the time of installation, regardless of how much usable capacity you purchased. For example, if a 9-module system is delivered with six modules active, you can use the interface ports in modules 4, 5, 7, and 8 although, effectively, three of the nine modules are not yet activated through CoD.

To use the combined capabilities of SAN Volume Controller and XIV, you must connect two ports (one per fabric) from each interface module with the SAN Volume Controller ports.

For redundancy and resiliency purposes, select one port from each HBA present on the interface modules. Use port 1 and 3 because both ports are on different HBAs. By default, port 4 is set as a SCSI initiator and is dedicated to XIV replication.

Therefore, if you decide to use port 4 to connect to a SAN Volume controller, you must change its configuration from initiator to target. For more information, see *IBM XIV Storage System Architecture and Implementation*, SG24-7659.

Figure 2-18 shows how to connect an XIV frame to a SAN Volume Controller storage controller.

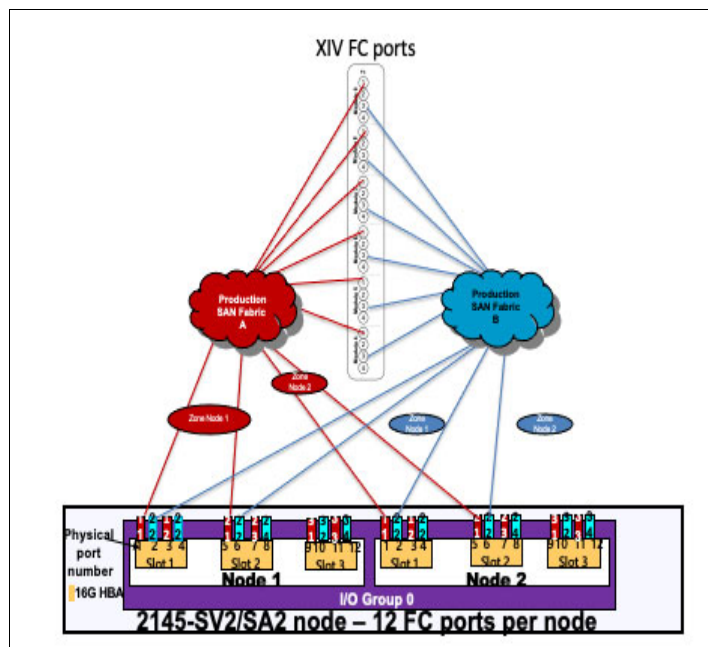


Figure 2-18 XIV Port Cabling

The preferred practice for zoning is to create a single zoning to each SAN Volume Controller node on each SAN fabric. This zone must contain all ports from a single XIV and the IBM SAN Volume Controller node ports that are destined to connect host and back-end storage. All nodes in an IBM SAN Volume Controller cluster must see the same set of XIV host ports.

Notice that Figure 2-18 shows that a single zone is used to each XIV to SAN Volume controller node. For this example, the following zones are used:

- ▶ Fabric A, XIV → SVC Node 1: All XIV fabric A ports to SVC node 1
- ▶ Fabric A, XIV → SVC Node 2: All XIV fabric A ports to SVC node 2
- ▶ Fabric B, XIV → SVC Node 1: All XIV fabric B ports to SVC node 1
- ▶ Fabric B, XIV → SVC Node 2: All XIV fabric B ports to SVC node 2

For more information about other preferred practices and XIV considerations, see Chapter 3, “Storage back-end” on page 67.

**FlashSystem A9000 and A9000R storage systems**

An IBM FlashSystem A9000 system has a fixed configuration with three grid elements, with a total of 12 Fibre Channel (FC) ports. A preferred practice is to restrict ports 2 and 4 of each grid controller for replication and migration use, and use ports 1 and 3 for host access.

However, considering that any replication or migration is done through the IBM Spectrum Virtualize, ports 2 and 4 also can be used for IBM Spectrum Virtualize connectivity. Port 4 must be set to target mode for this to work.

Assuming a dual fabric configuration for redundancy and resiliency purposes, select one port from each HBA present on the grid controller. Therefore, a total of six ports (three per fabric) are used.

Figure 2-19 shows a possible connectivity scheme for IBM SAN Volume Controller 2145-SV2/SA2 nodes and A9000 systems.

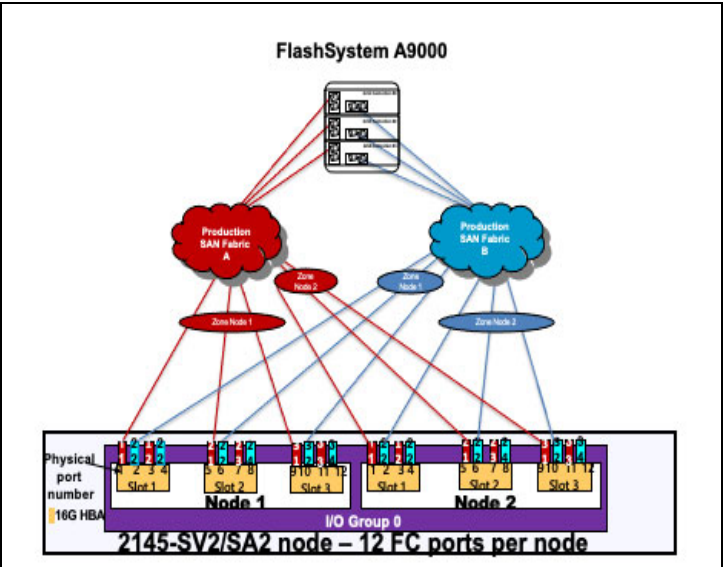


Figure 2-19 A9000 connectivity

The IBM FlashSystem A9000R system has more choices because many configurations are available, as listed in Table 2-3.

Table 2-3 Number of host ports in an IBM FlashSystem A9000R system

| Grid elements | Total host ports available |
|---------------|----------------------------|
| 2             | 8                          |
| 3             | 12                         |
| 4             | 16                         |
| 5             | 20                         |
| 6             | 24                         |

However, IBM Spectrum Virtualize can support only 16 WWPN from any single WWNN. The IBM FlashSystem A9000 or IBM FlashSystem A9000R system has only one WWNN, so you are limited to 16 ports to any IBM FlashSystem A9000R system.

Next is the same table (Table 2-4), but with columns added to show how many and which ports can be used for connectivity. The assumption is a dual fabric, with ports 1 in one fabric, and ports 3 in the other.

For the 4-grid element system, it is possible to attach 16 ports because that is the maximum that Spectrum Virtualize allows. For the 5- and 6-grid element systems, it is possible to use more ports up to the 16 maximum, but that is not recommended because it might create unbalanced work loads to the grid controllers with two ports attached.

*Table 2-4 Host connections to SAN Volume Controller*

| <b>Grid elements</b> | <b>Total host ports available</b> | <b>Total ports that are connected to Spectrum Virtualize</b> | <b>Total ports that are connected to Spectrum Virtualize</b> |
|----------------------|-----------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| 2                    | 8                                 | 8                                                            | All controllers, ports 1 and 3                               |
| 3                    | 12                                | 12                                                           | All controllers, ports 1 and 3                               |
| 4                    | 16                                | 8                                                            | Odd controllers, port 1<br>Even controllers, port 3          |
| 5                    | 20                                | 10                                                           | Odd controllers, port 1<br>Even controllers, port 3          |
| 6                    | 24                                | 12                                                           | Odd controllers, port 1<br>Even controllers, port 3          |

Figure 2-20 shows a possible connectivity scheme for IBM SAN Volume Controller 2145-SV2/SA2 nodes and A9000R systems with up to three grid elements.

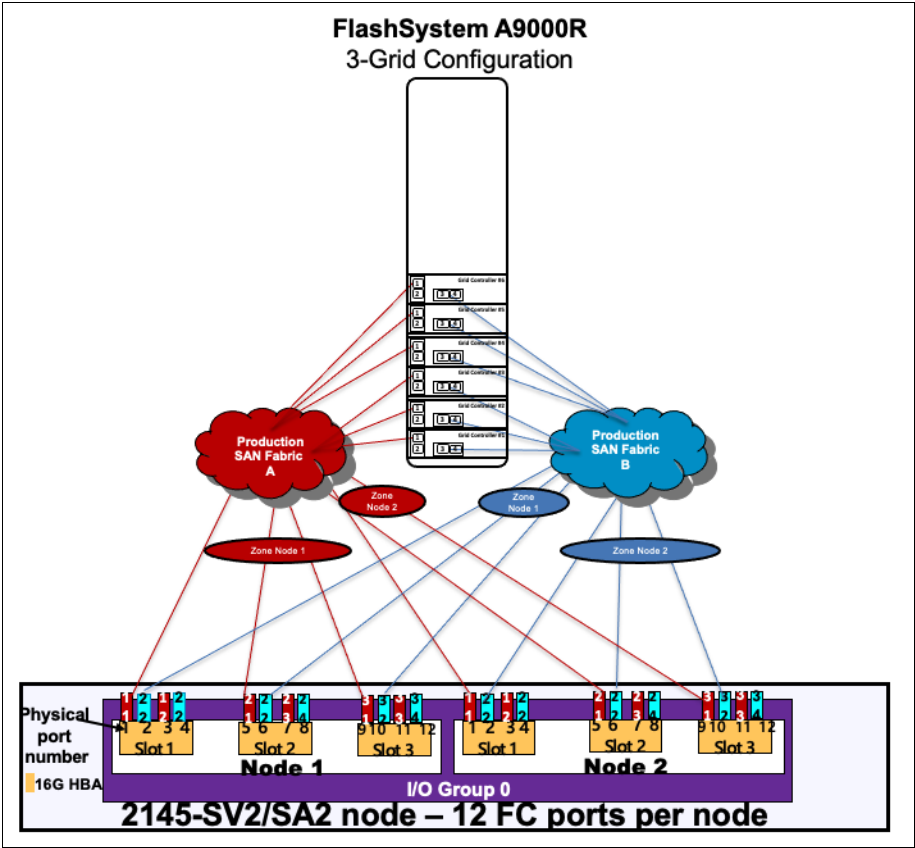


Figure 2-20 A9000 Grid Configuration Cabling

Figure 2-21 shows a possible connectivity schema for IBM SAN Volume Controller 2145-SV2/SA2 nodes and A9000R systems fully configured.

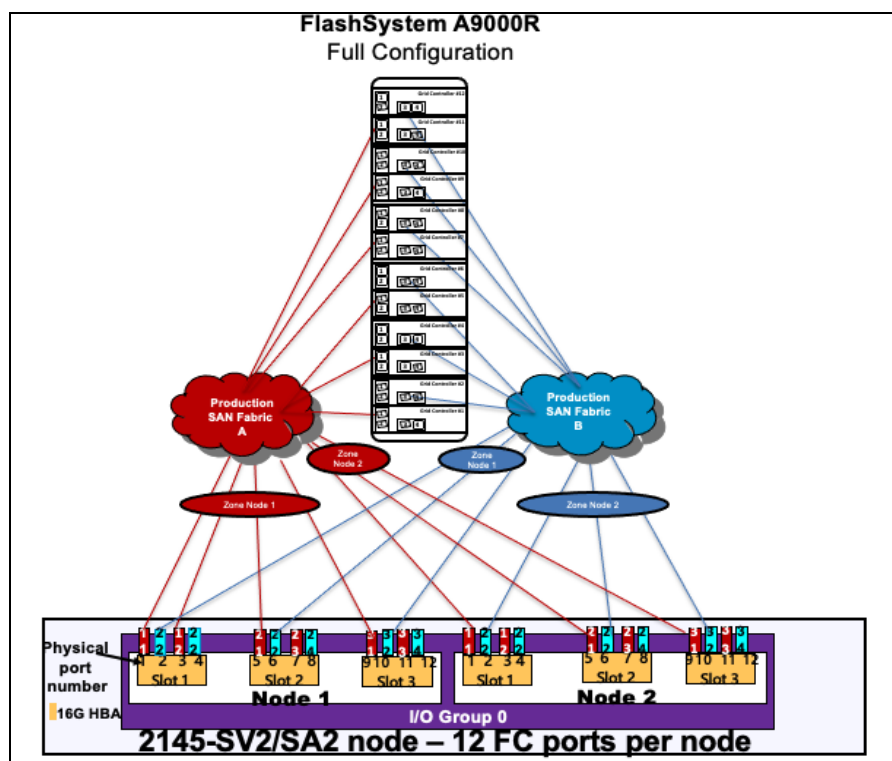


Figure 2-21 Connecting A9000 Fully Configured as a backend controller

For more information about FlashSystem A9000 and A9000R implementation, see *IBM FlashSystem A9000 and IBM FlashSystem A9000R Architecture and Implementation*, SG24-8345.

The SAN Volume Controller layer of replication cannot be changed; therefore, you cannot virtualize SAN Volume Controller behind IBM Storwize. However, IBM Storwize can be changed from storage to replication and from replication to storage layer. If you want to virtualize one IBM Storwize behind another, the IBM Storwize that is used as external storage must have a layer of storage; the IBM Storwize that is performing virtualization must have a layer of replication.

The storage layer and replication layer feature the following differences:

- ▶ In the storage layer, a Storwize family system has the following characteristics and requirements:
  - The system can complete Metro Mirror and Global Mirror replication with other storage layer systems.
  - The system can provide external storage for replication layer systems or SAN Volume Controller.
  - The system cannot use another Storwize family system that is configured with the storage layer as external storage.

- In the replication layer, a Storwize family system has the following characteristics and requirements:
  - The system can complete Metro Mirror and Global Mirror replication with other replication layer systems or SAN Volume Controller.
  - The system cannot provide external storage for a replication layer system or SAN Volume Controller.
  - The system can use another Storwize family system that is configured with storage layer as external storage.

**Note:** To change the layer, you must disable the visibility of every other Storwize or SAN Volume Controller on all fabrics. This process involves deleting partnerships, Remote Copy relationships, and zoning between IBM Storwize and other IBM Storwize or SAN Volume Controller. Then, run the `chsystem -layer` command to set the layer of the system.

For more information about the storage layer, see [IBM Documentation](#).

To zone the Storwize as a back-end storage controller of SAN Volume Controller, every SAN Volume Controller node must access the same Storwize ports as a minimum requirement. Create one zone per SAN Volume Controller node per fabric to the same ports from a Storwize V7000 storage.

Figure 2-22 shows a zone between a 16-port Storwize V7000 and a SAN Volume Controller.

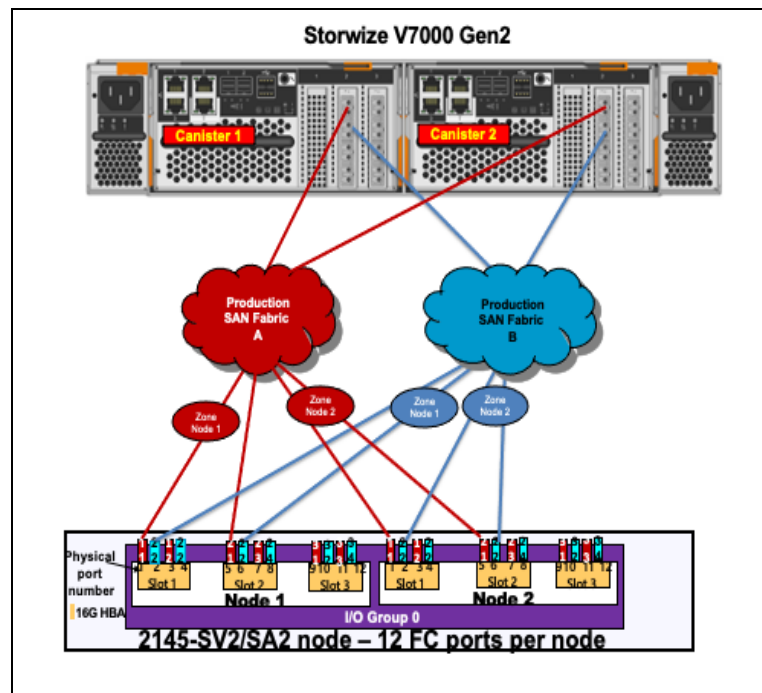


Figure 2-22 V7000 Connected as a Backend Controller

Notice that the ports from Storwize V7000 in Figure 2-22 are split between both fabrics. The odd ports are connected to Fabric A and the even ports are connected to Fabric B. You can also spread the traffic across the Storwize V7000 FC adapters on the same canister. However, it does not significantly increase the availability of the solution, because the mean time between failures (MTBF) of the adapters is not significantly less than that of the non-redundant canister components.

**Note:** If you are using an NPIV-enabled Storwize system as backend storage, only the NPIV ports on the Storwize system must be used for the storage backend zoning.

Connect as many ports as necessary to service your workload to the SAN Volume controller. For information about back-end port limitations and preferred practices, see “Back-end storage port count” on page 41.

## FlashSystem 9100

IBM FlashSystem 9100 also runs IBM Spectrum Virtualize software, and the zoning recommendations are the same as the Storwize V7000. You must cable the odd ports to Fabric A/1 and the even ports to Fabric B/2. If you want to virtualize a FlashSystem 9100 behind an IBM SAN Volume Controller or a Storwize family product, you also must change the layer of the FlashSystem 9100 to replication. If the FlashSystem 9100 is also used to provide storage to hosts, do not mix the ports that are zoned to IBM SAN Volume Controller with hosts.

Figure 2-23 shows a FlashSystem 9100 as an IBM SAN Volume Controller back-end zone example.

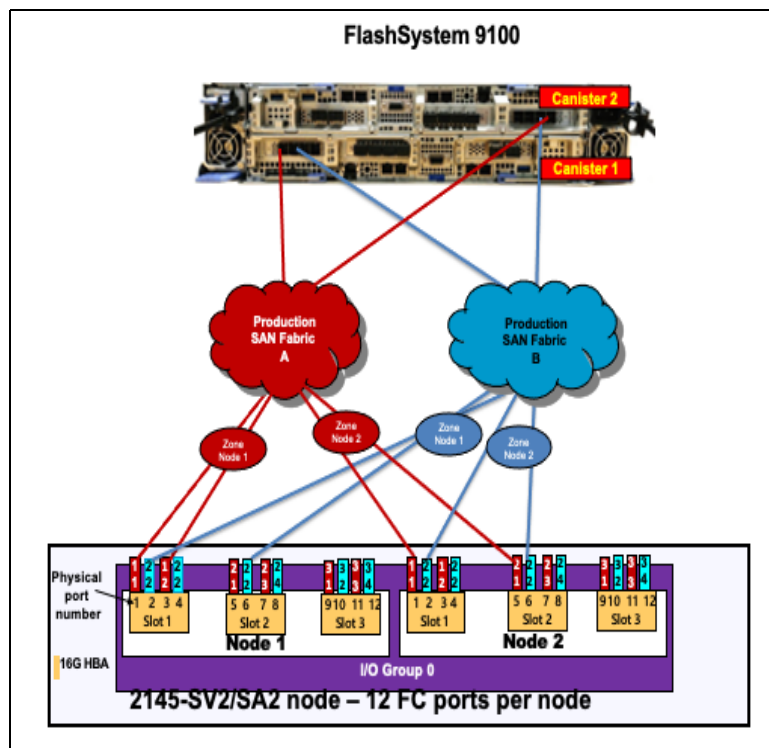


Figure 2-23 FS9100 as a back-end controller

**Note:** If you use an NPIV enabled Storwize system as back-end storage, the NPIV ports on the Storwize system are used for the storage back-end zoning.

Connect as many ports as necessary to service your workload to the SAN Volume controller. For more information about back-end port limitations and preferred practices, see “Back-end storage port count” on page 41.



## FlashSystem 900

IBM FlashSystem 900 is an all-flash storage array that provides extreme performance and can sustain highly demanding throughput and low latency across its FC interfaces. It includes up to 16 ports of 8 Gbps or eight ports of 16 Gbps FC. It also provides enterprise-class reliability, large capacity, and green data center power and cooling requirements.

The main advantage of integrating FlashSystem 900 with SAN Volume Controller is to combine the extreme performance of IBM FlashSystem with the SAN Volume Controller enterprise-class solution such as tiering, mirroring, IBM FlashCopy, thin provisioning, IBM Real-time Compression and Copy Services.

Before starting, work closely with your IBM Sales, pre-sales, and IT architect to properly size the solution by defining the proper number of SAN Volume Controller I/O groups/cluster and FC ports that are necessary according to your servers and application workload demands.

To maximize the performance that you can achieve when deploying the FlashSystem 900 with SAN Volume Controller, carefully consider the assignment and usage of the FC HBA ports on SAN Volume Controller as described in 2.3.2, “Port naming and distribution” on page 30. The FlashSystem 900 ports must be dedicated to the SAN Volume Controller workload, so do not mix direct attached hosts on FlashSystem 900 with SAN Volume Controller ports.

Connect the FlashSystem 900 to the SAN network by completing the following steps:

1. Connect FlashSystem 900 odd ports to odd SAN fabric (or SAN Fabric A) and the even ports from to even SAN fabric (or SAN fabric B).
2. Create one zone for each SAN Volume Controller node with all FlashSystem 900 ports on each fabric.

Figure 2-24 shows a 16-port FlashSystem 900 zoning to a SAN Volume Controller.

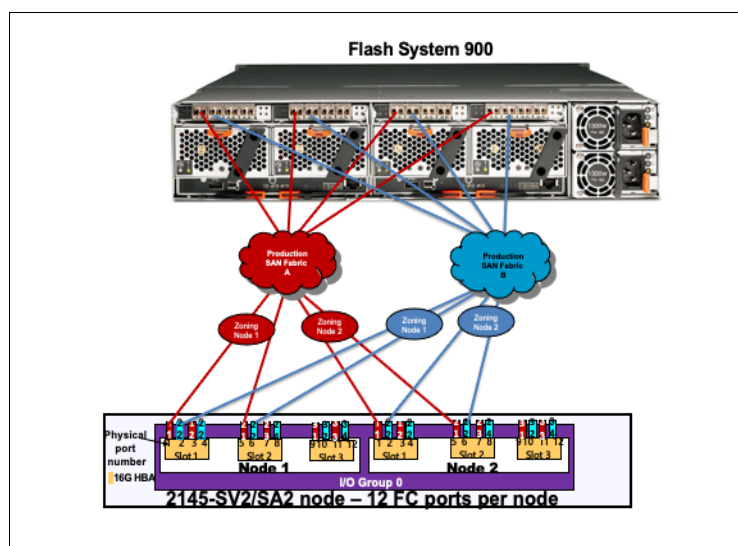


Figure 2-24 FlashSystem 900 connectivity to IBM SAN Volume Controller Cluster

Notice that after the FlashSystem 900 is zoned to two SAN Volume Controller nodes, four zones exist, with one zone per node and two zones per fabric.

You can decide to share or not the SAN Volume Controller ports with other back-end storage. However, it is important to monitor the buffer credit use on SAN Volume Controller switch ports and, if necessary, modify the buffer credit parameters to properly accommodate the traffic to avoid congestion issues.

For more information about FlashSystem 900 best practices, see Chapter 3, “Storage back-end” on page 67.

## IBM DS8900F

The IBM DS8000 family is a high-performance, high-capacity, highly secure, and resilient series of disk storage systems. The DS8900F family is the latest and most advanced of the DS8000 series offerings to date. The high availability, multiplatform support, including IBM z Systems, and simplified management tools help provide a cost-effective path to an on-demand world.

From a connectivity perspective, the DS8900F family is scalable. Two different types of host adapters are available: 16 GFC and 32 GFC. Both can auto-negotiate their data transfer rate down to 8 Gbps full-duplex data transfer. The 16 GFC and 32 GFC host adapters are all 4-port adapters.

Both adapters contain a high-performance application-specific integrated circuit (ASIC). To ensure maximum data integrity, it supports metadata creation and checking. Each FC port supports a maximum of 509 host login IDs and 1,280 paths. This configuration enables the creation of large storage area networks (SANs).

**Tip:** The general best practices guidelines for the use of 16 GFC or 32GFC technology in DS8900F and SAN Volume Controller, consider the use of IBM SAN Volume Controller maximum of 16 ports to DS8900F. Also, ensuring that more ranks can be assigned to the SAN Volume Controller than the number of slots that are available on that host ensures that the ports are not oversubscribed.

On other side, a single 16/32 GFC host adapter does not provide full line rate bandwidth with all ports active:

- ▶ 16 GFC host adapter - 3300 MBps Read/1730 MBps Write
- ▶ 32 GFC host adapter - 6500 MBps Read/3500 MBps Write

The DS8910F model 993 configuration supports a maximum of eight host adapters. The DS8910F model 994 configurations support a maximum of 16 host adapters in the base frame. The DS8950F model 996 configurations support a maximum of 16 host adapters in the base frame and an extra 16 host adapters in the DS8950F model E96.

Host adapters are installed in slots 1, 2, 4, and 5 of the I/O bay. Figure 2-25 shows the locations for the host adapters in the DS8900F I/O bay.

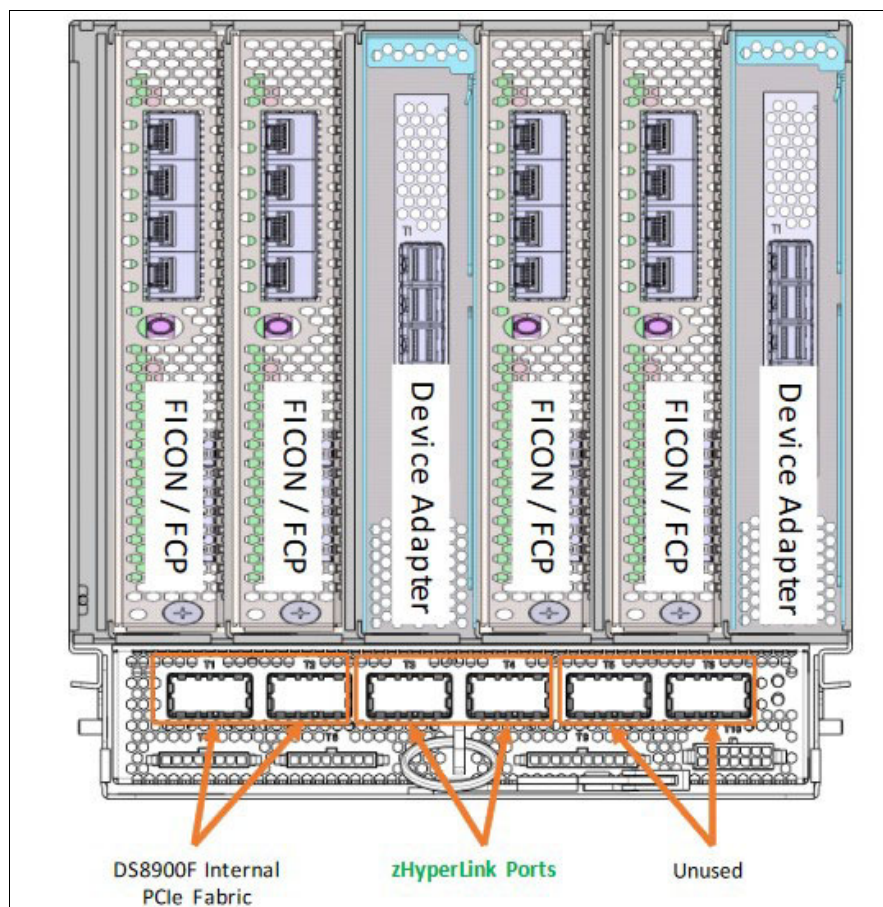


Figure 2-25 DS8900F I/O adapter layout

The system supports an intermix of both adapter types up to the maximum number of ports, as listed in Table 2-5.

Table 2-5 DS8900F port configuration

| Model     | Minimum/maximum host adapters | Minimum/maximum host adapters ports |
|-----------|-------------------------------|-------------------------------------|
| 994       | 2/16                          | 8/64                                |
| 996       | 2/16                          | 8/64                                |
| 996 + E96 | 2/32                          | 8/128                               |

**Important:** Each of the ports on a DS8900F host adapter can be independently configured for FCP or IBM FICON®. The type of port can be changed through the DS8900F Data Storage Graphical User Interface (DS GUI) or by using Data Storage Command-Line Interface (DS CLI) commands. To work with SAN and SAN Volume Controller, use the Small Computer System Interface- Fibre Channel Protocol (SCSI- FCP); Fibre Channel (FC)-switched fabric. FICON is for IBM Z® systems only.

For more information about DS8900F hardware, port, and connectivity, see *IBM DS8900F Architecture and Implementation*, SG24-8456.

Despite the wide DS8900F port availability, to attach a DS8900F series to a SAN Volume Controller, use Disk Magic to know how many host adapters are required according to your workload, and spread the ports across different HBAs for redundancy and resiliency proposes. However, consider the following points as a place to start for a single San Volume Controller cluster configuration:

- Smaller or equal than 16 arrays, use two host adapters - 8 FC ports

**Note:** For redundancy, the recommendation is to use four host adapters as a minimum.

- From 17 - 48 arrays, use four host adapters - 16 FC ports.
- Greater than 48 arrays, use eight host adapters - 16 FC ports. This configuration also matches for the high-performance, most demanding environments.

**Note:** To check the current code MAX limitation, search for the term “configuration limits and restrictions” for your code level and IBM Spectrum Virtualize 8.4, see [this IBM Support web page](#).

Figure 2-26 shows the connectivity between a SAN Volume Controller and a DS8886.

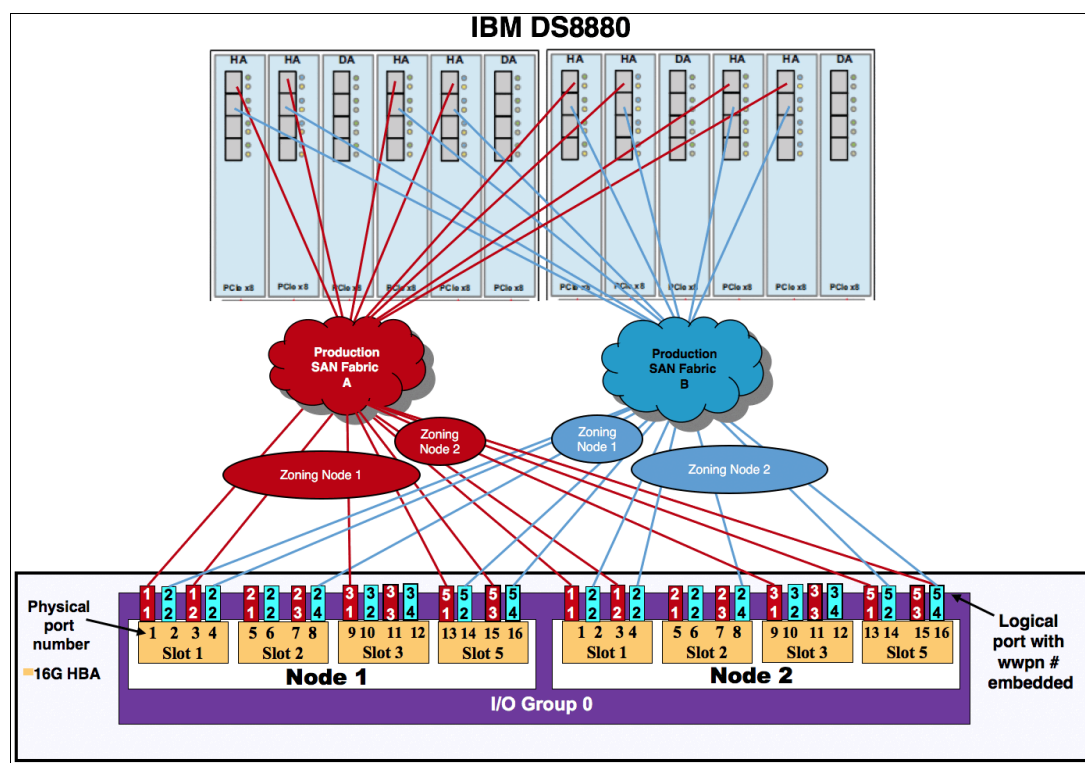


Figure 2-26 DS8886 to SAN volume controller connectivity

**Note:** Figure 2-26 is valid example to be use for DS8900F to SAN Volume Controller connectivity as well.

Notice that in Figure 2-26 on page 52, 16 ports are zoned to the SAN Volume Controller and the ports are spread across the different HBAs that are available on the storage.

To maximize performance, the DS8900F ports must be dedicated to the SAN Volume Controller connections. However, the SAN Volume Controller ports must be shared with hosts so that you can obtain the maximum full duplex performance from these ports. For more information about port usage and assignments, see 2.3.2, “Port naming and distribution” on page 30.

Create one zone per SAN Volume Controller node per fabric. The SAN Volume Controller must access the same storage ports on all nodes. Otherwise, the DS8900F operation status is set to degraded on the SAN Volume Controller.

After the zoning steps, you must configure the *host connections* by using the DS8900F Data Storage Graphical User Interface (DS GUI) or by using Data Storage Command-Line Interface (DS CLI) commands, to all IBM SAN Volume Controller nodes WWPNs to create a single *Volume Group* that adds all IBM SAN Volume Controller cluster ports within this Volume Group.

For more information about Volume Group, Host Connection, and DS8000 administration, see *IBM DS8900F Architecture and Implementation*, SG24-8456.

The specific preferred practices to present DS8880 LUNs as back-end storage to the SAN Volume Controller are described in Chapter 3, “Storage back-end” on page 67.

## 2.4.5 SAN Volume Controller host zones

The preferred practice to connect a host into a SAN volume Controller is creating a single zone to each host port. This zone must contain the host port and *one* port from each SAN Volume Controller node that the host must access. Although two ports from each node per SAN fabric are in a usual dual-fabric configuration, ensure that the host accesses only one of them, as shown in Figure 2-27.

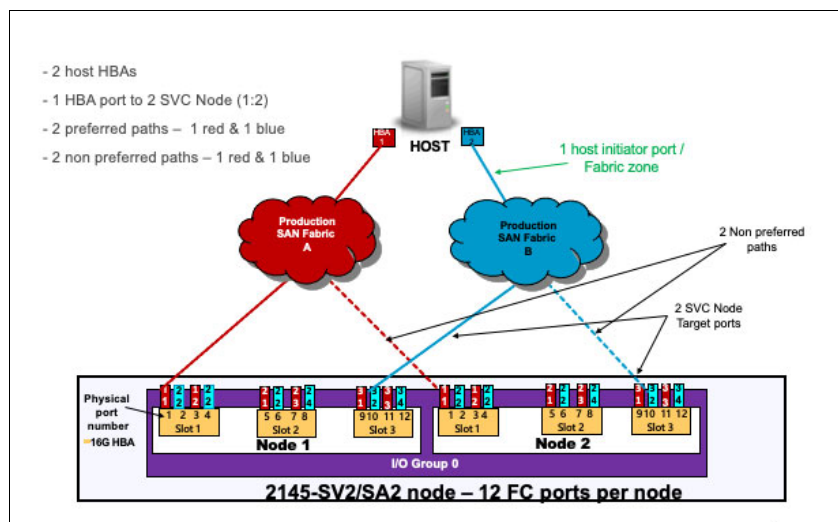


Figure 2-27 Host Zoning to IBM SAN Volume Controller nodes

This configuration provides four paths to each volume, being two preferred paths (one per fabric) and two non-preferred paths. Four paths is the number of paths (per volume) for which multipathing software, such as AIXPCM, SDDPCM, and SDDDSM, and the SAN Volume Controller, are optimized to work.

**NPIV consideration:** All of the recommendations in this section also apply to NPIV-enabled configurations. For more information about the systems that are supported by the NPIV, see the following IBM Support web pages:

- ▶ [IBM SVC 8.2.1 Configuration Limits](#)
- ▶ [IBM V7000 8.2.1 Configuration Limits](#)

When the recommended number of paths to a volume are exceeded, path failures sometimes are not recovered in the required amount of time. In some cases, too many paths to a volume can cause excessive I/O waits, resulting in application failures and, under certain circumstances, it can reduce performance.

**Note:** Eight paths by volume is also supported. However, this design provides no performance benefit and, in some circumstances, can reduce performance. Also, it does not significantly improve reliability nor availability. However, fewer than four paths does not satisfy the minimum redundancy, resiliency, and performance requirements.

To obtain the best overall performance of the system and to prevent overloading, the workload to each SAN Volume Controller port must be equal. Having the same amount of workload typically involves zoning approximately the same number of host FC ports to each SAN Volume Controller FC port.

### **Hosts with four or more host bus adapters**

If you have four HBAs in your host instead of two HBAs, more planning is required. Because eight paths is not an optimum number, configure your SAN Volume Controller host definitions (and zoning) as though the single host is two separate hosts. During volume assignment, you alternate which volume was assigned to one of the “pseudo hosts.”

The reason for not assigning one HBA to each path is because the SAN Volume Controller I/O group works as a cluster. When a volume is created, one node is assigned as preferred and the other node solely serves as a backup node for that specific volume. It means that using one HBA to each path will never balance the workload for that particular volume. Therefore, it is better to balance the load by I/O group instead so that the volume is assigned to nodes automatically.



Figure 2-28 shows an example of a four port host zoning.

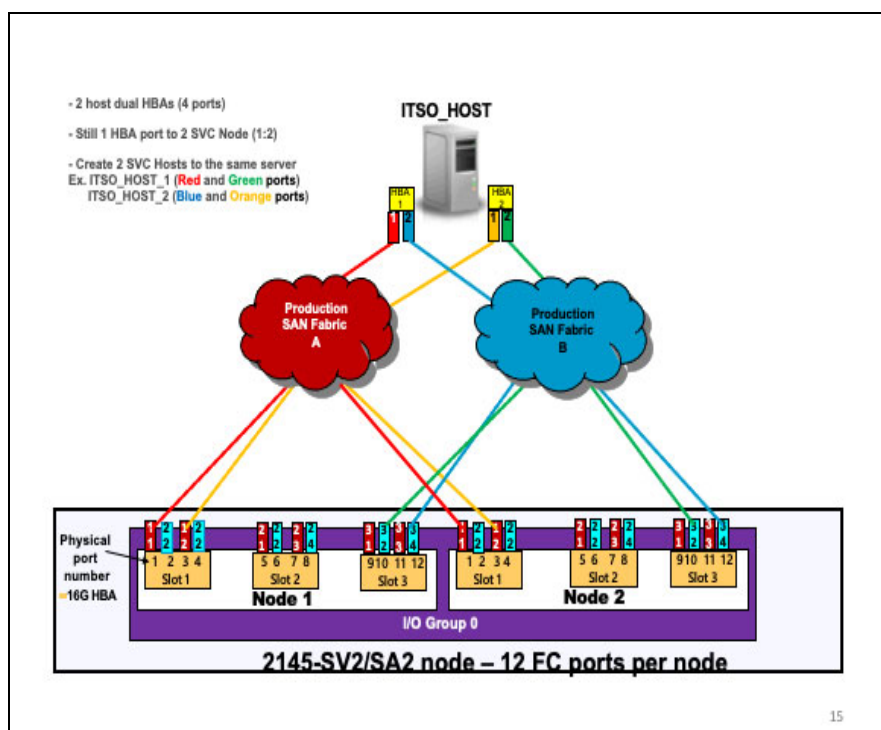


Figure 2-28 4 Port Host Zoning

Because the optimal number of volume paths is four, you must create two or more hosts on SAN Volume Controller. During volume assignment, alternate which volume is assigned to one of the “pseudo-hosts” in a round-robin fashion.

**Note:** Pseudo-hosts is not a defined function or feature of SAN Volume Controller. To create a pseudo-host, you must add another host ID to the SAN Volume Controller host configuration. Instead of creating one host ID with four WWPNs, you define two hosts with two WWPNs; therefore, you must pay extra attention to the SCSI IDs that are assigned to each of the pseudo-hosts to avoid having two different volumes from the same storage subsystem with the same SCSI ID.

## ESX Cluster zoning

For ESX Clusters, you must create separate zones for each host node in the ESX Cluster, as shown in Figure 2-29.

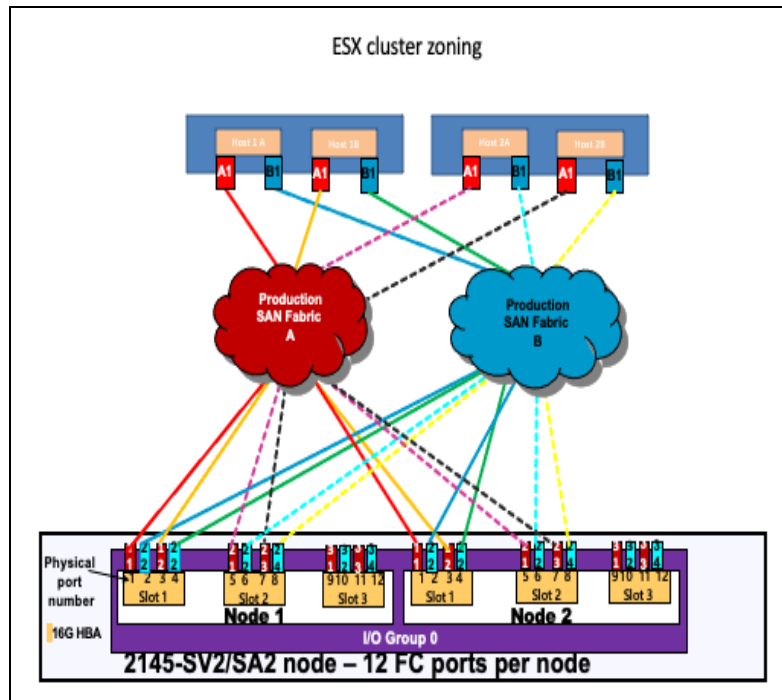


Figure 2-29 ESX Cluster Zoning

Ensure that you apply the following preferred practices to your ESX VMware clustered hosts configuration:

- ▶ Zone a single ESX cluster in a manner that avoids ISL I/O traversing.
- ▶ Spread multiple host clusters evenly across the SAN Volume Controller node ports and I/O Groups.
- ▶ Map LUNs and volume evenly across zoned ports, alternating the preferred node paths evenly for optimal I/O spread and balance.
- ▶ Create separate zones for each host node in SAN Volume Controller and on the ESX cluster.

## AIX VIOs: LPM zoning

When zoning IBM AIX VIOs to IBM Spectrum Virtualize, you must plan carefully. Because of its complexity, it is common to create more than four paths to each Volume and MDisk or not provide for proper redundancy. The following preferred practices can help you to have a non-degraded path error on IBM Spectrum Virtualize with four paths per volume:

- ▶ Create two separate and isolated zones on each fabric for each LPAR.
- ▶ Do not put both the active and inactive LPAR WWPNs in either the same zone or same IBM Spectrum Virtualize host definition.
- ▶ Map LUNs to the virtual host FC HBA port WWPNs, not the physical host FCA adapter WWPN.
- ▶ When using NPIV, generally make no more than a ratio of one physical adapter to eight Virtual ports. This configuration avoids I/O bandwidth oversubscription to the physical adapters.



- Create a pseudo host in IBM Spectrum Virtualize host definitions that contain only two virtual WWPNs (one from each fabric), as shown in Figure 2-30.
- Map the LUNs/volumes to the pseudo LPARs (active and inactive) in a round-robin fashion.

Figure 2-30 shows a correct SAN connection and zoning for LPARs.

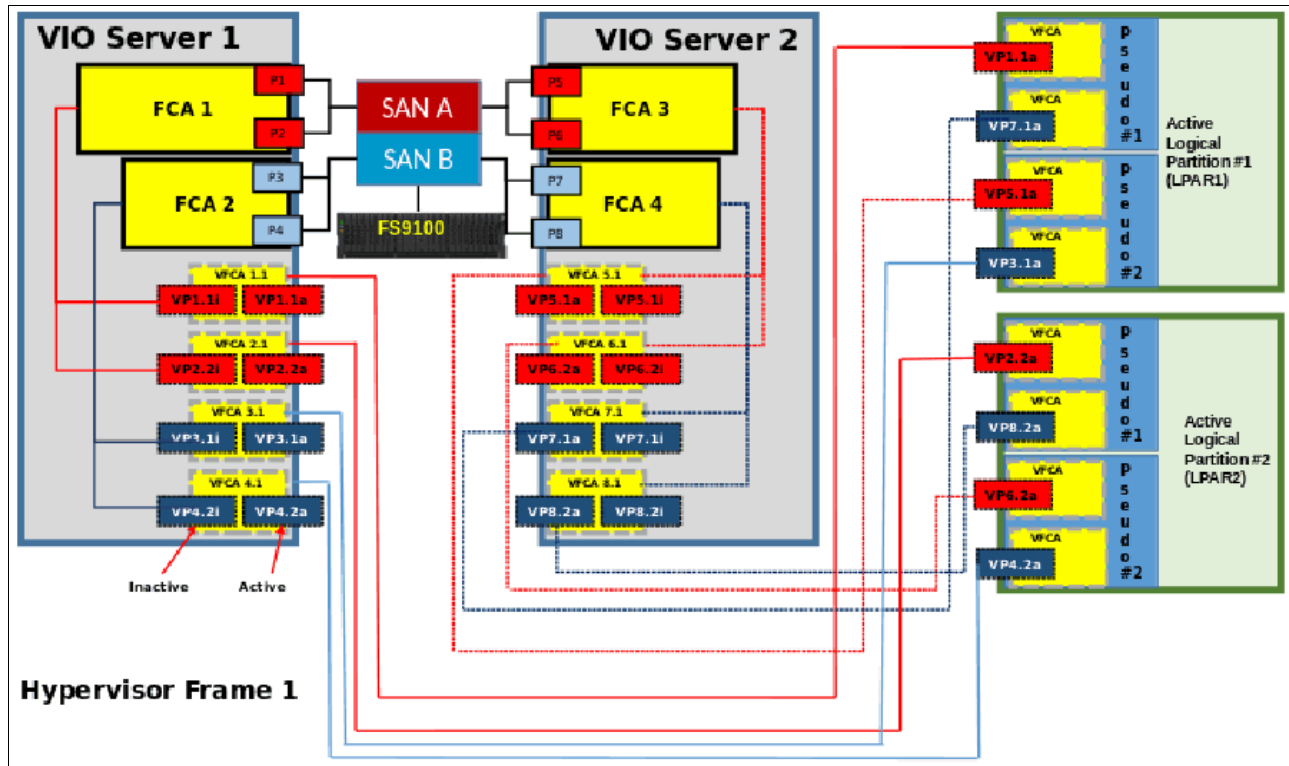


Figure 2-30 LPARs SAN connections

During Live Partition Migration (LPM), both inactive and active ports are active. When LPM is complete, the previously active ports show as inactive and the previously inactive ports show as active.

Figure 2-31 shows a Live partition migration from the hypervisor frame to another frame.

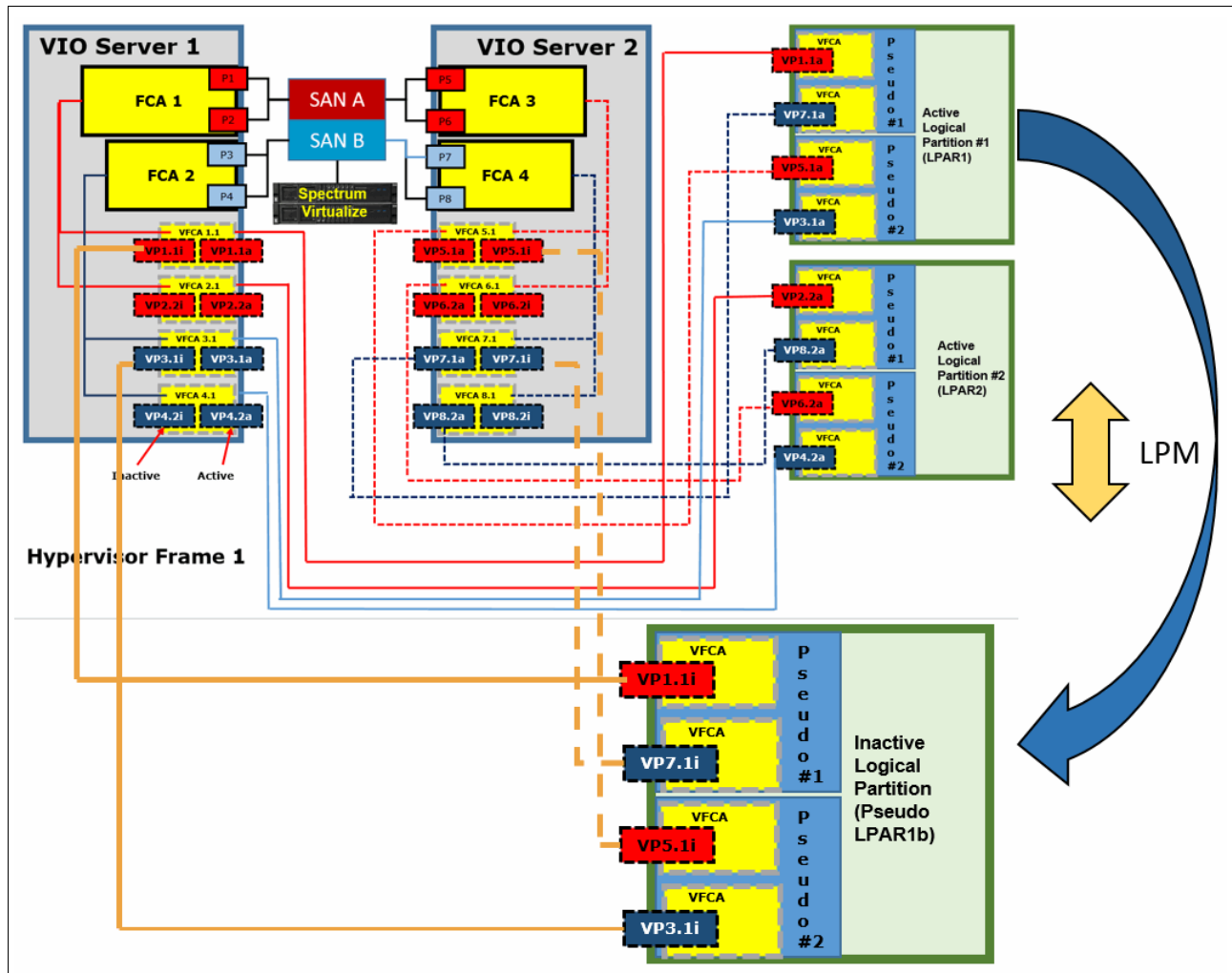


Figure 2-31 Live partition migration

**Note:** During LPM, the number of paths doubles from 4 to 8. Starting with eight paths per LUN/volume results in an unsupported 16 paths during LPM, which can lead to I/O interruption.

## 2.4.6 Hot Spare Node zoning considerations

IBM Spectrum Virtualize V8.1 introduced the Hot Spare Node (HSN) feature that provides a higher availability for SAN Volume Controller clusters by automatically swapping a spare node into the cluster if the cluster detects a failing node. Also the maintenance procedures, like code updates and hardware upgrades, benefit from this feature avoiding prolonged loss of redundancy during the node maintenance.

For more information about hot spare nodes, see *IBM Spectrum Virtualize: Hot Spare Node and NPIV Target Ports*, [REDP-5477](#).

For the Hot Spare Node feature to be fully effective requires the NPIV feature enabled. In an NPIV enabled cluster, each physical port is associated with two WWPNs. When the port initially logs into the SAN it uses the normal WWPN (*primary port*), which does not change from previous releases or from NPIV disabled mode. When the node has completed its startup and is ready to begin processing I/O, the *NPIV target ports* log on to the fabric with the second WWPN.

Special zoning requirements must be considered when implementing the HSN functionality.

### **Host zoning with HSN**

Hosts should be zoned with NPIV target ports only. Spare nodes ports must not be included in the host zoning.

### **Intercluster and intracluster zoning with HSN**

Communications between IBM Spectrum Virtualize nodes, including between different clusters, takes place over primary ports. Spare nodes ports must be included in the intracluster zoning likewise the other nodes.

Similarly, when a spare node comes online, its primary ports are used for Remote Copy relationships and as such must be zoned with the remote cluster.

### **Backend controllers zoning with HSN**

Backend controllers must be zoned to the primary ports on IBM Spectrum Virtualize nodes. Where a spare node is in use, that nodes ports must be included in the backend zoning, likewise the other nodes.

**Note:** Currently the zoning configuration for spare nodes is not policed while the spare is inactive and no errors will be logged if the zoning or backend configuration is incorrect.

### **Backend controller configuration with HSN**

IBM Spectrum Virtualize uses the primary ports to communicate with the backend controller, including the spare. This means that all MDisk must be mapped to all IBM Spectrum Virtualize nodes, including spares. For IBM Spectrum Virtualize based backend controllers, such as Storwize V7000, it is recommended that the host clusters functionality is used, with each node forming one host within this cluster. This will ensure that each volume is mapped identically to each IBM Spectrum Virtualize node.

## **2.4.7 Zoning with multiple SAN Volume Controller clustered systems**

Unless two separate SAN Volume Controller systems participate in a mirroring relationship, configure all zoning so that the two systems do not share a zone. If a single host requires access to two different clustered systems, create two zones with each zone to a separate system.

The back-end storage zones must also be separate, even if the two clustered systems share a storage subsystem. You also must zone separate I/O groups if you want to connect them in one clustered system. Up to four I/O groups can be connected to form one clustered system.

## 2.4.8 Split storage subsystem configurations

In some situations, a storage subsystem might be used for SAN Volume Controller attachment and direct-attach hosts. In this case, pay attention during the LUN masking process on the storage subsystem. Assigning the same storage subsystem LUN to a host and the SAN Volume Controller can result in swift data corruption.

If you perform a migration into or out of the SAN Volume Controller, make sure that the LUN is removed from one place *before* it is added to another place.

## 2.5 Distance extension for Remote Copy services

To implement Remote Copy services over distance, the following choices are available:

- ▶ Optical multiplexors, such as Dense Wavelength Division Multiplexing (DWDM) or Coarse Wavelength Division Multiplexing (CWDM) devices
- ▶ Long-distance SFPs and XFPs
- ▶ FC-to-IP conversion boxes
- ▶ Native IP-based replication with SAN Volume Controller code

Of these options, the optical varieties of distance extension are preferred. IP distance extension introduces more complexity, is less reliable, and has performance limitations. However, optical distance extension is impractical in many cases because of cost or unavailability.

### 2.5.1 Optical multiplexors

Optical multiplexors can extend your SAN up to hundreds of kilometers at high speeds. For this reason, they are the preferred method for long-distance expansion. When you are deploying optical multiplexing, make sure that the optical multiplexor is certified to work with your SAN switch model. The SAN Volume Controller has no allegiance to a particular model of optical multiplexor.

If you use multiplexor-based distance extension, closely monitor your physical link error counts in your switches. Optical communication devices are high-precision units. When they shift out of calibration, you start to see errors in your frames.

### 2.5.2 Long-distance SFPs or XFPs

Long-distance optical transceivers have the advantage of extreme simplicity. Although no expensive equipment is required, a few configuration steps are necessary. Ensure that you use transceivers that are designed for your particular SAN switch *only*. Each switch vendor supports only a specific set of SFP or XFP transceivers, so it is unlikely that Cisco SFPs will work in a Brocade switch.

### 2.5.3 Fibre Channel over IP

Fibre Channel over IP (FCIP) conversion is by far the most common and least expensive form of distance extension. FCIP is a technology that allows FC routing to be implemented over long distances by using the TCP/IP protocol. In most cases, FCIP is implemented in Disaster Recovery scenarios with some kind of data replication between the primary and secondary site.

FCIP is a tunneling technology, which means FC frames are encapsulated in the TCP/IP packets. As such, it is not apparent to devices that are connected through the FCIP link. To use FCIP, you need some kind of tunneling device on both sides of the TCP/IP link that integrates FC and Ethernet connectivity. Most of the SAN vendors offer FCIP capability through stand-alone devices (multiprotocol routers) or by using blades that are integrated in the director class product. SAN Volume Controller systems support FCIP connection.

An important aspect of the FCIP scenario is the IP link quality. With IP-based distance extension, you must dedicate bandwidth to your FC to IP traffic if the link is shared with other IP traffic. Because the link between two sites is low-traffic or used only for email, do not assume that this type of traffic is always the case. The design of FC is sensitive to congestion and you do not want a spyware problem or a DDOS attack on an IP network to disrupt your SAN Volume Controller.

Also, when you are communicating with your organization's networking architects, distinguish between megabytes per second (MBps) and megabits per second (Mbps). In the storage world, bandwidth often is specified in MBps, but network engineers specify bandwidth in Mbps. If you fail to specify MB, you can end up with an impressive-sounding 155 Mbps OC-3 link, which supplies only 15 MBps or so to your SAN Volume Controller. If you include the safety margins, this link is not as fast as you might hope, so ensure that the terminology is correct.

Consider the following points when you are planning for your FCIP TCP/IP links:

- ▶ For redundancy purposes use as many TCP/IP links between sites as you have fabrics in each site that you want to connect. In most cases, there are two SAN FC fabrics in each site, so you need two TCP/IP connections between sites.
- ▶ Try to dedicate TCP/IP links only for storage interconnection. Separate them from other LAN/WAN traffic.
- ▶ Make sure that you have a service level agreement (SLA) with your TCP/IP link vendor that meets your needs and expectations.
- ▶ If you do not use Global Mirror with Change Volumes (GMCV), make sure that you have sized your TCP/IP link to sustain peak workloads.
- ▶ The use of SAN Volume Controller internal Global Mirror (GM) simulation options can help you test your applications before production implementation. You can simulate the GM environment within one SAN Volume Controller system without partnership with another. Run the **chsystem** command with the following parameters to perform GM testing:
  - `gminterdelaysimulation`
  - `gmintradelaysimulation`

For more information about GM planning, see Chapter 6, "Copy services" on page 219.

- ▶ If you are not sure about your TCP/IP link security, enable Internet Protocol Security (IPSec) on the all FCIP devices. IPSec is enabled on the Fabric OS level, so you do not need any external IPSec appliances.

In addition to planning for your TCP/IP link, consider adhering to the following preferred practices:

- ▶ Set the link bandwidth and background copy rate of partnership between your replicating SAN Volume Controller to a value *lower* than your TCP/IP link capacity. Failing to set this rate can cause an unstable TCP/IP tunnel, which can lead to stopping all your Remote Copy relations that use that tunnel.
- ▶ The best case is to use GMCV when replication is done over long distances.
- ▶ Use compression on corresponding FCIP devices.
- ▶ Use at least two ISLs from your local FC switch to local FCIP router.
- ▶ On a Brocade SAN, use the Integrated Routing feature to avoid merging fabrics from both sites.

For more information about FCIP, see the following publications:

- ▶ *IBM System Storage b-type Multiprotocol Routing: An Introduction and Implementation*, SG24-7544
- ▶ *IBM/Cisco Multiprotocol Routing: An Introduction and Implementation*, SG24-7543

## 2.5.4 SAN extension with Business Continuity configurations

Spectrum Virtualize Enhanced Stretched Cluster and HyperSwap technologies provide Business Continuity solutions over metropolitan areas with distances up to 300 km (186.4 miles). These Business Continuity solutions over metropolitan areas are achieved by using a SAN extension over WDM technology.

Furthermore, to avoid single points of failure, multiple WDMs and physical links are implemented. When implementing these solutions, particular attention must be paid in the intercluster connectivity set-up.

**Important:** HyperSwap and Stretched clusters require implementing dedicated private fabrics for the internode communication between the sites. For more information about the requirements, see *IBM Spectrum Virtualize HyperSwap SAN Implementation and Design Best Practices*, REDP-5597.

Consider a typical implementation of an Enhanced Stretched Cluster that uses ISLs, as shown in Figure 2-32.

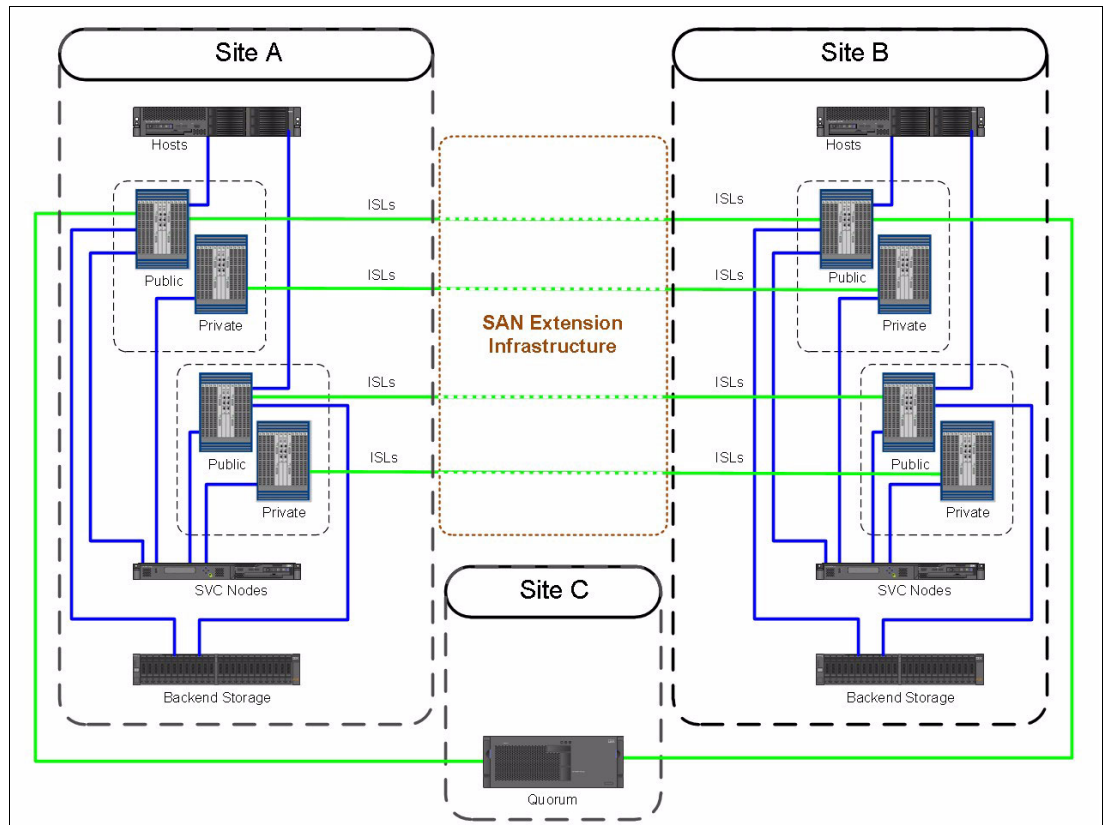


Figure 2-32 Typical Enhanced Stretched Cluster configuration

In this configuration, the intercluster communication is isolated in a Private SAN that interconnects Site A and Site B through a SAN extension infrastructure that consists of two DWDMs. For redundancy reasons, assume that two ISLs are used for each fabric for the Private SAN extension.

Two possible configurations are available to interconnect the Private SANs. In Configuration 1 (see Figure 2-33), one ISL per fabric is attached to each DWDM. In this case, the physical paths Path A and Path B are used to extend both fabrics.

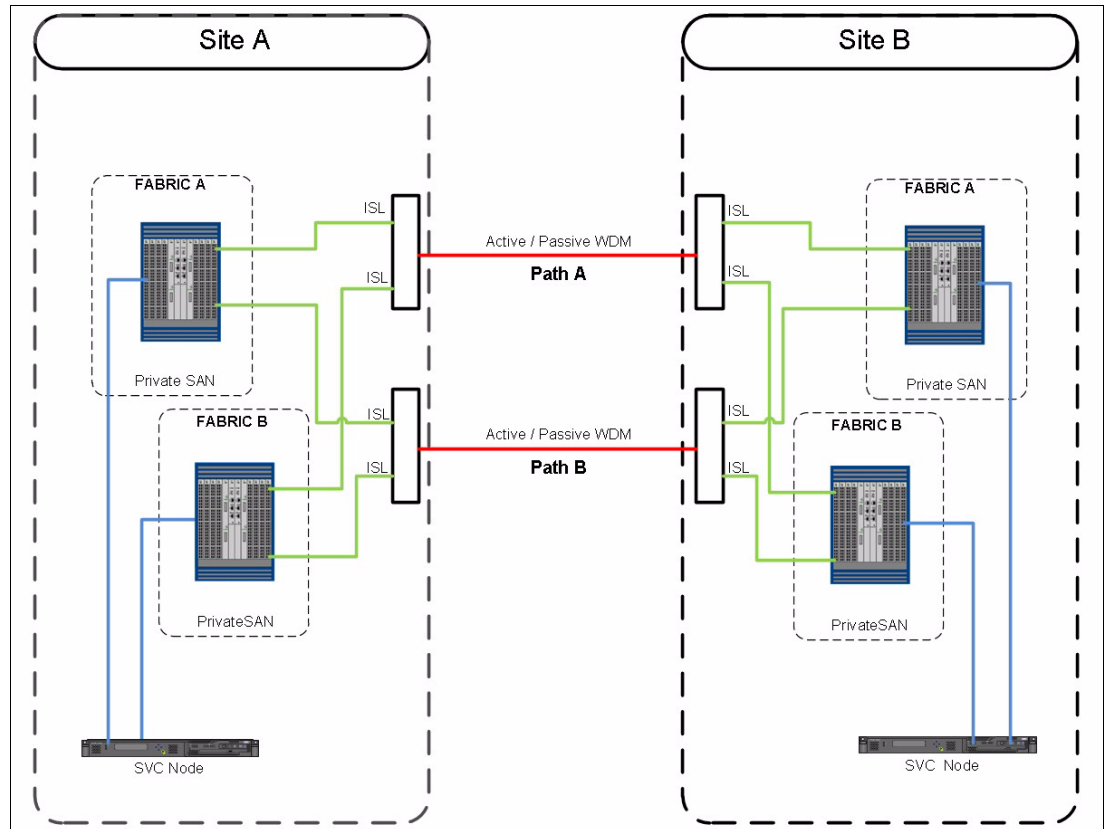


Figure 2-33 Configuration 1: physical paths shared among the fabrics



In Configuration 2 (see Figure 2-34), ISLs of fabric A are attached only to Path A, while ISLs of fabric B are attached only to Path B. In this case, the physical paths are not shared between the fabrics.

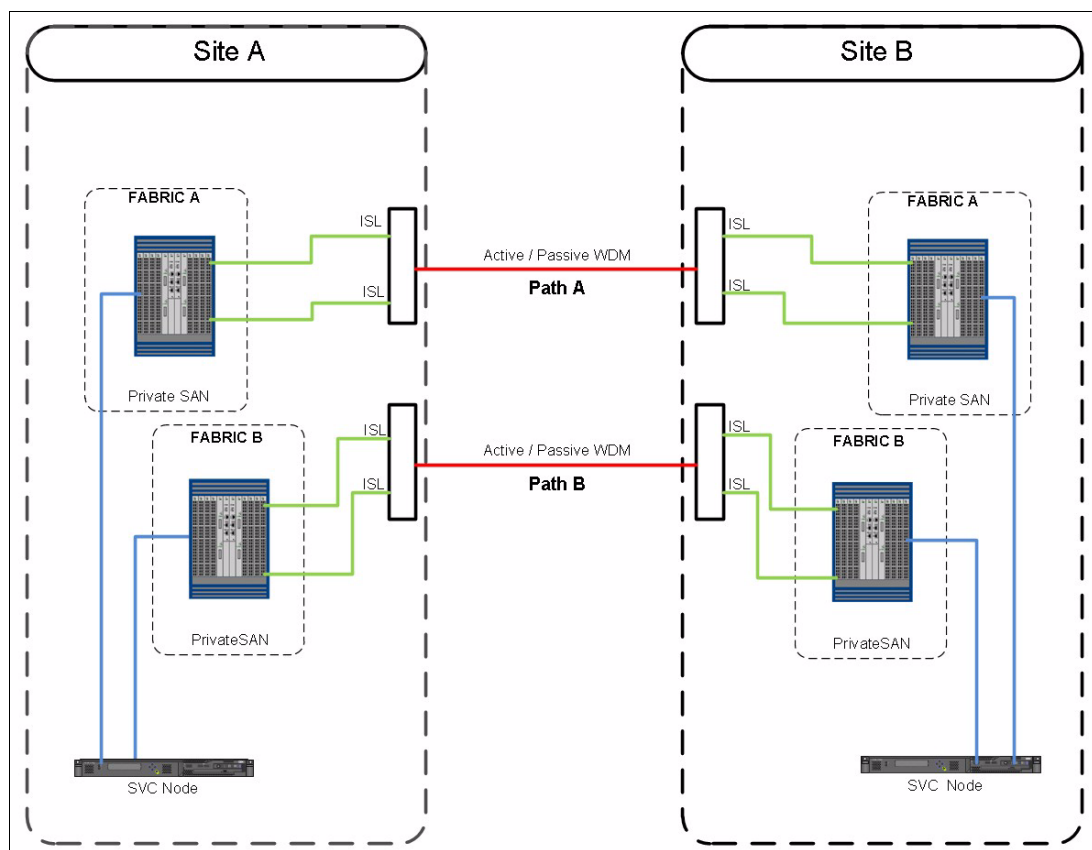


Figure 2-34 Configuration 2: physical paths not shared among the fabrics

With Configuration 1, in case of failure of one of the physical paths, both fabrics are simultaneously affected and a fabric reconfiguration occurs because of an ISL loss. This situation can lead to a temporary disruption of the intracluster communication and, in the worst case, to a split brain condition. To mitigate this situation, link aggregation features such as Brocade ISL trunking can be implemented.

With Configuration 2, a physical path failure leads to a fabric segmentation of one of the two fabrics, leaving the other fabric unaffected. In this case the intracluster communication would be guaranteed through the unaffected fabric.

Summarizing, the recommendation is to fully understand the implication of a physical path or DWDM loss in the SAN extension infrastructure and implement the suitable architecture to avoid a simultaneous impact.

## 2.5.5 Native IP replication

It is possible to implement native IP-based replication. *Native* means that SAN Volume Controller does not need any FCIP routers to create a partnership. This partnership is based on the Internet Protocol network and not on the FC network. For more information about native IP replication, see Chapter 6, “Copy services” on page 219.

To enable native IP replication, SAN Volume Controller implements the Bridgeworks SANSlide network optimization technology. For more information about this solution, see *IBM SAN Volume Controller and Storwize Family Native IP Replication*, REDP-5103.

## 2.6 Tape and disk traffic that share the SAN

If you have free ports on your core switch, you can place tape devices (and their associated backup servers) on the SAN Volume Controller SAN. However, do not put tape and disk traffic on the same FC HBA.

Do not put tape ports and backup servers on different switches. Modern tape devices have high-bandwidth requirements. Placing tape ports and backup servers on different switches can quickly lead to SAN congestion over the ISL between the switches.

## 2.7 Switch interoperability

SAN Volume Controller is flexible as far as switch vendors are concerned. All of the node connections on a particular SAN Volume Controller clustered system must go to the switches of a single vendor. That is, you must not have several nodes or node ports plugged into vendor A and several nodes or node ports plugged into vendor B.

SAN Volume Controller supports some combinations of SANs that are made up of switches from multiple vendors in the same SAN. However, this approach is not preferred in practice. Despite years of effort, interoperability among switch vendors is less than ideal because FC standards are not rigorously enforced. Interoperability problems between switch vendors are notoriously difficult and disruptive to isolate. Also, it can take a long time to obtain a fix. For these reasons, run only multiple switch vendors in the same SAN long enough to migrate from one vendor to another vendor, if this setup is possible with your hardware.

You can run a mixed-vendor SAN if you have agreement from both switch vendors that they fully support attachment with each other. However, Brocade does *not* support interoperability with any other vendors.

Interoperability between Cisco switches and Brocade switches is not recommended, except during fabric migrations, and then only if you have a back-out plan in place. Also, when connecting BladeCenter switches to a core switch, consider the use of the N-Port ID Virtualization (NPIV) technology.

When you have SAN fabrics with multiple vendors, pay special attention to any particular requirements. For example, observe from which switch in the fabric the zoning must be performed.



## Storage back-end

This chapter describes the aspects and practices to consider when system's external back-end storage is planned, configured, and managed.

External storage is acquired by Spectrum Virtualize by virtualizing a separate IBM or third-party storage system, which are attached with FC or iSCSI.

**Note:** IBM SAN Volume Controller that is built on SV1 nodes supports SAS-attached expansions with solid-state drives (SSDs) and hard disk drives (HDDs). However, nodes SV2 and SA2 do not support internal storage.

For more information about configuring internal storage that is attached to SV1 nodes, see *IBM FlashSystem Best Practices and Performance Guidelines*, SG24-8503.

This chapter includes the following topics:

- ▶ 3.1, "General considerations for managing external storage" on page 68
- ▶ 3.2, "Controller-specific considerations" on page 74
- ▶ 3.3, "Quorum disks" on page 89

## 3.1 General considerations for managing external storage

IBM SAN Volume Controller can virtualize external storage that is presented to the system. External back-end storage systems (or *controllers* in Spectrum Virtualize terminology) provide their logical volumes (LVs), which are detected by IBM SAN Volume Controller as MDisks and can be used in storage pools.

This section covers aspects of planning and managing external storage that is virtualized by IBM SAN Volume Controller.

External back-end storage can be connected to IBM SAN Volume Controller with FC (SCSI) or iSCSI. NVMe-FC back-end attachment is not supported because it provides no performance benefits for IBM SAN Volume Controller. The main advantage of NVMe solution is seen as a reduction in the CPU cycles that are needed on the host level, where the Fibre Channel HBA is found, to handle the interrupts.

For external backend, IBM SAN Volume Controller acts as a host. All Spectrum Virtualize Fibre Channel drivers are implemented from day one as polling drivers, not interrupt-driven drivers. Thus, on the storage side, almost no latency savings are gained by switching from SCSI to NVMe as a protocol.

### 3.1.1 Storage controller path selection

When a managed disk (MDisk) logical unit (LU) is accessible through multiple storage system ports, the system ensures that all nodes that access this LU coordinate their activity and access the LU through the same storage system port.

An MDisk path that is presented to the storage system for all system nodes must meet the following criteria:

- ▶ The system node:
  - Is a member of a storage system
  - Includes Fibre Channel or iSCSI connections to the storage system port
  - Successfully discovered the LU
- ▶ The port selection process did not cause the system node to exclude access to the MDisk through the storage system port

When the IBM SAN Volume Controller node selects a set of ports to access the storage system, the two types of path selection that are described in the next sections are supported to access the MDisks. A type of path selection is determined by external system type and cannot be changed.

To determine which algorithm is used for a specific back-end system, see [System Storage Interoperation Center \(SSIC\)](#), as shown in Figure 3-1.



Figure 3-1 SSIC example

### Round-robin path algorithm

With the round-robin path algorithm, each MDisk uses one path per target port per IBM SAN Volume Controller node. Therefore, in cases of storage systems that do not feature a preferred controller (such as XIV or DS8000), each MDisk uses all of the available FC ports of that storage controller.

With a round-robin compatible storage controller, there is no need to create as many volumes as there are storage FC ports. Every volume, and therefore MDisk, uses all available IBM SAN Volume Controller ports.

This configuration results in a significant performance increase because the MDisk is no longer bound to one back-end FC port. Instead, it can issue I/Os to many back-end FC ports in parallel. Particularly, the sequential I/O within a single extent can benefit from this feature.

Additionally, the round-robin path selection improves resilience to specific storage system failures. For example, if one of the back-end storage system FC ports encounters some performance problems, the I/O to MDisk is sent through other ports. Moreover, because I/Os to MDisk are sent through all back-end storage FC ports, the port failure can be detected more quickly.

**Preferred practice:** If your storage system supports the round-robin path algorithm, zone as many FC ports from the back-end storage controller as possible. IBM SAN Volume Controller supports up to 16 FC ports per storage controller. For more information about FC port connections and zoning guidelines, see your storage system documentation.

Example 3-1 shows a storage controller that supports round-robin path selection.

*Example 3-1 Round robin enabled storage controller*

---

```
IBM_2145:SVC-ITS0:superuser>lsmdisk 4
id 4
name mdisk4
...
preferred_WWPN
active_WWPN many ? <<< Round Robin Enabled
```

---

### **MDisk group balanced and controller balanced**

Although round-robin path selection provides optimized and balanced performance with minimum configuration required, some storage systems still require manual intervention to achieve the same goal.

With storage subsystems, such as IBM DS5000 and DS3000 (or other Active-Passive type systems), IBM SAN Volume Controller accesses an MDisk LU through one of the ports on the preferred controller. To best use the back-end storage, it is important to ensure that the number of LUs that is created is a multiple of the connected FC ports and aggregate all LUs to a single MDisk group.

Example 3-2 shows a storage controller that supports MDisk group balanced path selection.

*Example 3-2 MDisk group balanced path selection (no round robin enabled) storage controller*

---

```
IBM_2145:SVC-ITS0:superuser>lsmdisk 5
id 5
name mdisk5
...
preferred_WWPN
active_WWPN 20110002AC00C202 ? <<< indicates MDisk group balancing
```

---

## **3.1.2 Guidelines for creating optimal back-end configuration**

Most of the back-end controllers aggregate spinning or SSDs into RAID arrays, then join arrays into pools. Logical volumes are created on those pools and provided to hosts.

When connected to external back-end storage, IBM SAN Volume Controller acts as a host. It is important to create back-end controller configuration that provides performance and resiliency because IBM SAN Volume Controller relies on back-end storage when serving I/O to host systems that are attached to it.

If your back-end system includes homogeneous storage, create the required number of RAID arrays (usually RAID 6 or RAID 10 are recommended) with an equal number of drives. The type and geometry of array depends on the back-end controller vendor's recommendations. If your back-end controller can spread the load stripe across multiple arrays in a resource pool (for example, by striping), create a single pool and add all arrays there.

On back-end systems with mixed drives, create a separate resource pool for each drive technology (and keep drive technology type in mind because you must assign the correct tier for an MDisk when it is used by IBM SAN Volume Controller).

Create a set of fully allocated logical volumes from the back-end system storage pool (or pools). Each volume is detected as MDisk on IBM SAN Volume Controller. The number of logical volumes to create depends the type of drives that are used by your back-end controller.

### **Back-end controller with spinning drives**

If your backend uses spinning drives, volume number calculation must be based on a queue depth. Queue depth is the number of outstanding I/O requests of a device.

For optimal performance, spinning drives need 8 - 10 concurrent I/O at the device, and this need does not change with drive rotation speed. Therefore, we want to ensure that in a highly loaded system, any IBM SAN Volume Controller MDisk can queue up approximately 8 I/O per back-end system drive.

IBM FlashSystem queue depth per MDisk is approximately 60 (the exact maximum that is seen on a real system can vary, depending on the circumstances; however, for the purpose of this calculation, it does not matter). This queue depth per MDisk number leads to the *HDD Rule of 8*. According to this rule, to achieve 8 I/O per drive and with queue depth 60 per MDisk from IBM FlashSystem, a back-end array with  $60/8 = 7.5$  that is approximately equal to eight physical drives is optimal, or we need one logical volume per every eight drives in an array

#### **Example 1**

Consider the following example:

- ▶ The backend controller to be virtualized is IBM Storwize V5030 with 64 NL-SAS 8 TB drives.
- ▶ The system is homogeneous.
- ▶ According to recommendations that are described in the “Array Considerations” section in *Implementing the IBM FlashSystem with IBM Spectrum Virtualize V8.4*, SG24-8465, a single DRAID6 array is created at Storwize and installed in a storage pool.
- ▶ The HDD rule of 8 tells us that we want  $64/8 = 8$  MDisks. Therefore, eight volumes are created from a pool to present to IBM SAN Volume Controller and then are assigned to the nearline tier.

### **All-flash back-end controllers**

For All-Flash controllers, the considerations are more of I/O distribution across IBM SAN Volume Controller ports and processing threads than of queue depth per drive. Because most all-flash arrays that are put behind the virtualizer include high I/O capabilities, we want to make sure that we are giving IBM SAN Volume Controller the optimal chance to spread the load and evenly make use of its internal resources so that queue depths are less of a concern (because of the lower latency per I/O).

For all-flash back-end arrays, IBM recommends creating 32 logical volumes from the array capacity. By doing so, the queue depths can be kept high enough and the work is spread across the virtualizer resources.

For smaller set ups with a low number of SSDs, this number can be reduced to 16 logical volumes (which results in 16 MDisks) or even eight volumes.

#### **Example 2**

Consider the following example:

- ▶ Backend controllers to be virtualized are IBM FlashSystem 5030 with 24 Tier1 7.6 TB drives IBM FlashSystem 900.

- ▶ Virtualizer needs a pool with two storage tiers.
- ▶ On IBM FlashSystem 5030, a single DRAID6 array is created and then added to a storage pool.
- ▶ Using this all-flash rule, 32 volumes are created to present as MDisks. However, because it is small setup, the number of volumes is reduced to 16.
- ▶ On IBM FlashSystem 900, all micro-latency modules are joined into a RAID5 array and added to a storage pool. FlashSystem 900 is Tier0 solution; therefore, the all-flash rule is used and 32 volumes are created to present as MDisks.
- ▶ On virtualizer, 16 MDisks are added from IBM FlashSystem 5030 as Tier1 flash, and 32 MDisks as Tier0 flash, to a single multitier pool.

### Large setup considerations

For controllers, such as IBM DS8000 and XIV, you can use all-flash rule of 32. However, with installations involving such kinds of back-end controllers, it might be necessary to consider a maximum queue depth per back-end controller port, which is set to 1000 for most supported high-end storage systems.

With high-end controllers, queue depth per MDisk can be calculated by using the following formula:

$$Q = ((P \times C) / N) / M$$

Where:

|   |                                                                                                                                      |
|---|--------------------------------------------------------------------------------------------------------------------------------------|
| Q | Calculated queue depth for each MDisk                                                                                                |
| P | Number of back-end controller host ports (unique WWPNs) that are zoned to IBM SAN Volume Controller (minimum is 2 and maximum is 16) |
| C | Maximum queue depth per WWPN, which is 1000 for controllers, such as XIV or DS8000                                                   |
| N | Number of nodes in the IBM SAN Volume Controller cluster (2, 4, 6, or 8)                                                             |
| M | Number of volumes that are presented by back-end controller and detected as MDisks                                                   |

Because we know that we want to achieve  $Q = 60$ , we can calculate the number of volumes that we need to create as:

$$M = (P \times C) / (N \times Q) \quad ? \quad (16 \times P) / N$$

#### Example 3

A 4-node IBM SAN Volume Controller is used with 12 host ports on the IBM XIV System.

By using the  $Q = ((P \times C) / N) / M$  formula, we must create  $M = (16 \times 12)/4 = 48$  volumes on IBM XIV to obtain balanced, high-performing configuration.

### 3.1.3 Considerations for compressing and deduplicating back-end

IBM SAN Volume Controller supports over-provisioning on selected back-end controllers. Therefore, if back-end storage performs data deduplication or data compression on LUs provisioned from it, they still can be used as external MDisks on IBM SAN Volume Controller.

The implementation steps for thin-provisioned MDisks are the same as for fully allocated storage controllers. Extreme caution should be used when planning capacity for such configurations.



IBM SAN Volume Controller detects if the MDisk is thin-provisioned, its total physical capacity that is used, and remaining physical capacity. It also detects if SCSI UNMAP commands are supported by the back-end.

By sending SCSI UNMAP commands to thin-provisioned MDisks, the system marks data that is no longer used. Then, the garbage collection processes on the backend can free unused capacity and reallocate it to free space.

The use of a suitable compression or data deduplication ratio is key to achieving a stable environment. If you are unsure about the real compression or data deduplication ratio, contact your IBM technical sales representative for more information.

The nominal capacity from a compression and deduplication enabled storage system is not fixed and it varies based on the nature of the data. Always use a conservative data reduction ratio for the initial configuration.

The use of an incorrect ratio for capacity assignment can cause an out of space situation. If the MDisks do not provide enough capacity, IBM SAN Volume Controller disables access to all the volumes in the storage pool, as shown in the following example:

- ▶ Assumption 1: Sizing is performed with an optimistic 5:1 rate
- ▶ Assumption 2: Real rate is 3:1
- ▶ Physical Capacity: 20 TB
- ▶ Calculated capacity:  $20 \text{ TB} \times 5 = 100 \text{ TB}$
- ▶ Volume assigned from compression or deduplication enabled storage subsystem to SAN Volume Controller or Storwize: 100 TB
- ▶ Real usable capacity:  $20 \text{ TB} \times 3 = 60 \text{ TB}$

If the hosts try to write more than 60 TB data to the storage pool, the storage subsystem cannot provide any more capacity. All volumes that are used as IBM Spectrum Virtualize or Storwize Managed Disks and all related pools go offline.

Thin-provisioned back-end storage must be carefully monitored. It is necessary to set up capacity alerts to be aware of the real remaining physical capacity.

Also, the best practice is to have an emergency plan for “Out Of Physical Space” situation on the back-end controller to know what steps must be taken to recover. The plan also must be prepared during the initial implementation phase.

## 3.2 Controller-specific considerations

This section discusses implementation information that is related to supported back-end systems. For more information about general requirements, see [IBM Documentation](#).

### 3.2.1 Considerations for DS8000 series

In this section, we discuss considerations for the DS800 series.

#### Interaction between DS8000 and IBM SAN Volume Controller

It is important to know DS8000 drive virtualization process; that is, the process of preparing physical drives for storing data that belongs to a volume that is used by a host (in this case, the IBM SAN Volume Controller).

In this regard, the basis for virtualization begins with the physical drives of DS8000, which are mounted in storage enclosures. Virtualization builds upon the physical drives as a series of the following layers:

- ▶ Array sites
- ▶ Arrays
- ▶ Ranks
- ▶ Extent pools
- ▶ Logical volumes
- ▶ Logical subsystems

Array sites are the building blocks that are used to define arrays, which are data storage systems for block-based, file-based, or object based storage. Instead of storing data on a server, storage arrays use multiple drives that are managed by a central management and can store a huge amount of data. In general terms, eight identical drives that have the same capacity, speed, and drive class are called the array site. When an array is created, the RAID level, array type, and array configuration are defined. RAID 5, RAID 6, and RAID 10 levels are supported.

**Important:** Normally, the RAID 6 is highly preferred and is the default while the Data Storage Graphical Interface (DS GUI) is used. As with large drives in particular, the RAID rebuild times (after one drive failure) get ever larger. The use of RAID 6 reduces the danger of data loss because of a double-RAID failure. For more information, see [IBM Documentation](#).

A *rank*, which is a logical representation for the physical array, is relevant for an IBM SAN Volume Controller because of the creation of fixed block (FB) pool for each array that you want to virtualize. Ranks in DS8000 are defined in a one-to-one relationship to arrays. It is for this reason that a rank is defined as using only one array.

A fixed block rank that features an extent size of 1 GiB is a large extent; an extent size of 16 MiB is a small extent.

An *extent pool* or storage pool in DS8000 is a logical construct to add the extents from a set of ranks, which forms a domain for extent allocation to a logical volume.

In synthesis, a *logical volume* consists of a set of extents from one extent pool or storage pool. DS8900F supports up to 65,280 logical volumes.

A logical volume that is composed of fix block extents is called LUN. A fix block LUN consist of one or more 1 GiB (large) extents, or one or more 16 MiB (small) extents from one FB extent pool. Although a LUN cannot cross extent pools, it can have extents from multiple ranks within the same extent pool.

**Important:** DS8000 Copy Services do not support FB logical volumes larger than 2 TiB. Do not create a LUN that is larger than 2 TiB if you want to use Copy Services for the LUN unless the LUN is integrated as Managed Disks in an IBM FlashSystem. Use IBM Spectrum Virtualize Copy Services instead. Based on the considerations, the following maximum LUN sizes are used to create at DS8900F and present to IBM FlashSystem:

- ▶ 16 TB LUN with large extents (1 GiB)
- ▶ 16 TB LUN with small extent (16 MiB) for DS8880F with version or edition R8.5 or later, and for DS8900F R9.0 or later

Logical subsystems (or LSS) are another logical construct, and mostly used with fixed block volumes. Thus, 255 LSSs as a maximum can exist on DS8900F. For more information, see [IBM Documentation](#).

The concepts of virtualization of DS8900F for IBM FlashSystem or IBM SAN Volume Controller are shown in Figure 3-2.

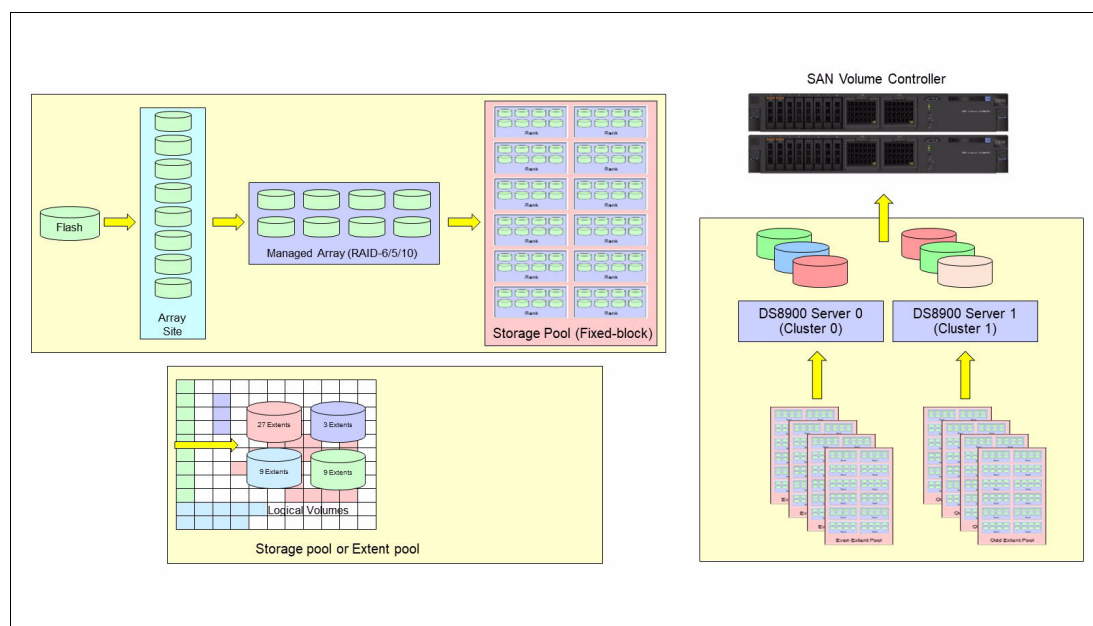


Figure 3-2 DS8900 virtualization concepts focus to IBM SAN Volume Controller

## Connectivity considerations

The number of DS8000 ports to be used is at least eight. With large and workload intensive configurations, consider using up to 16 ports, which is the maximum that is supported by IBM SAN Volume Controller.

Generally, use ports from different host adapters and, if possible, from different I/O enclosures. This configuration is also important because during a DS8000 LIC update, a host adapter port might need to be taken offline. This configuration allows the IBM SAN Volume Controller I/O to survive a hardware failure on any component on the SAN path.

For more information about SAN preferred practices and connectivity, see Chapter 2, “Storage area network” on page 19.

## **Defining storage**

To optimize the DS8000 resource usage, use the following guidelines:

- ▶ Distribute capacity and workload across device adapter pairs.
- ▶ Balance the ranks and extent pools between the two DS8000 internal servers to support the corresponding workloads on them.
- ▶ Spread the logical volume workload across the DS8000 internal servers by allocating the volumes equally on rank groups 0 and 1.
- ▶ Use as many disks as possible. Avoid idle disks, even if all storage capacity is not to be used initially.
- ▶ Consider the use of multi-rank extent pools.
- ▶ Stripe your logical volume across several ranks (the default for multi-rank extent pools).

## **Balancing workload across DS8000 series controllers**

When you configure storage on the DS8000 series disk storage subsystem, ensure that ranks on a device adapter (DA) pair are evenly balanced between odd and even extent pools. If you do not ensure that the ranks are balanced, uneven device adapter loading can lead to a considerable performance degradation.

The DS8000 series controllers assign server (controller) affinity to ranks when they are added to an extent pool. Ranks that belong to an even-numbered extent pool have an affinity to server0, and ranks that belong to an odd-numbered extent pool have an affinity to server1.

Figure 3-3 on page 77 shows an example of a configuration that results in a 50% reduction in available bandwidth. Notice how arrays on each of the DA pairs are accessed only by one of the adapters. In this case, all ranks on DA pair 0 are added to even-numbered extent pools, which means that they all have an affinity to server0. Therefore, the adapter in server1 is sitting idle. Because this condition is true for all four DA pairs, only half of the adapters are actively performing work. This condition can also occur on a subset of the configured DA pair.

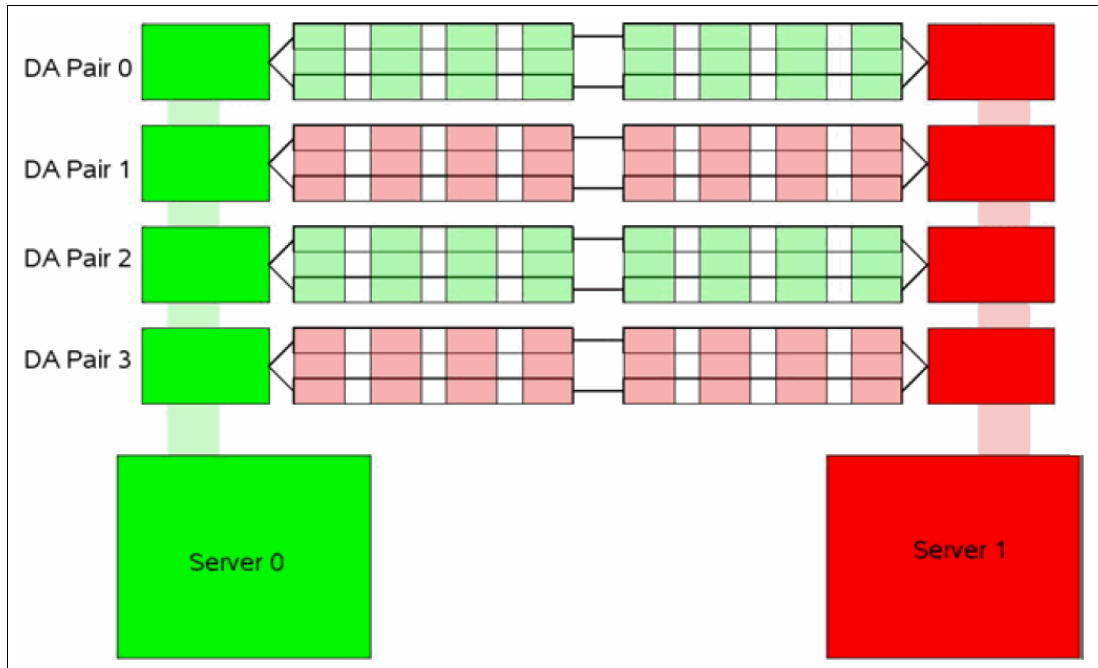


Figure 3-3 DA pair reduced bandwidth configuration

Example 3-3 shows what this invalid configuration resembles from the CLI output of the **lsarray** and **lsrank** commands. The arrays that are on the same DA pair contain the same group number (0 or 1), meaning that they have affinity to the same DS8000 series server. Here, server0 is represented by group0, and server1 is represented by group1.

As an example of this situation, consider arrays A0 and A4, which are attached to DA pair 0. In this example, both arrays are added to an even-numbered extent pool (P0 and P4) so that both ranks have affinity to server0 (represented by group0), which leaves the DA in server1 idle.

Example 3-3 Command output for the **lsarray** and **lsrank** commands

```
dscli> lsarray -l
Date/Time: Oct 20, 2016 12:20:23 AM CEST IBM DSCLI Version: 7.8.1.62 DS: IBM.2107-75L2321
Array  State  Data   RAID type  arsite  Rank  DA  Pair  DDMcap(10^9B)  diskclass
=====
A0    Assign  Normal  5 (6+P+S)  S1      R0    0   146.0  ENT
A1    Assign  Normal  5 (6+P+S)  S9      R1    1   146.0  ENT
A2    Assign  Normal  5 (6+P+S)  S17     R2    2   146.0  ENT
A3    Assign  Normal  5 (6+P+S)  S25     R3    3   146.0  ENT
A4    Assign  Normal  5 (6+P+S)  S2      R4    0   146.0  ENT
A5    Assign  Normal  5 (6+P+S)  S10     R5    1   146.0  ENT
A6    Assign  Normal  5 (6+P+S)  S18     R6    2   146.0  ENT
A7    Assign  Normal  5 (6+P+S)  S26     R7    3   146.0  ENT

dscli> lsrank -l
Date/Time: Oct 20, 2016 12:22:05 AM CEST IBM DSCLI Version: 7.8.1.62 DS: IBM.2107-75L2321
ID  Group State  datastate Array RAIDtype  extpoolID  extpoolnam  stgtype  exts  usedexts
=====
R0   0 Normal  Normal   A0      5    P0      extpool0   fb      779    779
R1   1 Normal  Normal   A1      5    P1      extpool1   fb      779    779
R2   0 Normal  Normal   A2      5    P2      extpool2   fb      779    779
R3   1 Normal  Normal   A3      5    P3      extpool3   fb      779    779
R4   0 Normal  Normal   A4      5    P4      extpool4   fb      779    779
```

|    |   |        |        |    |   |    |          |    |     |     |
|----|---|--------|--------|----|---|----|----------|----|-----|-----|
| R5 | 1 | Normal | Normal | A5 | 5 | P5 | extpool5 | fb | 779 | 779 |
| R6 | 0 | Normal | Normal | A6 | 5 | P6 | extpool6 | fb | 779 | 779 |
| R7 | 1 | Normal | Normal | A7 | 5 | P7 | extpool7 | fb | 779 | 779 |

Figure 3-4 shows a configuration that balances the workload across all four DA pairs.

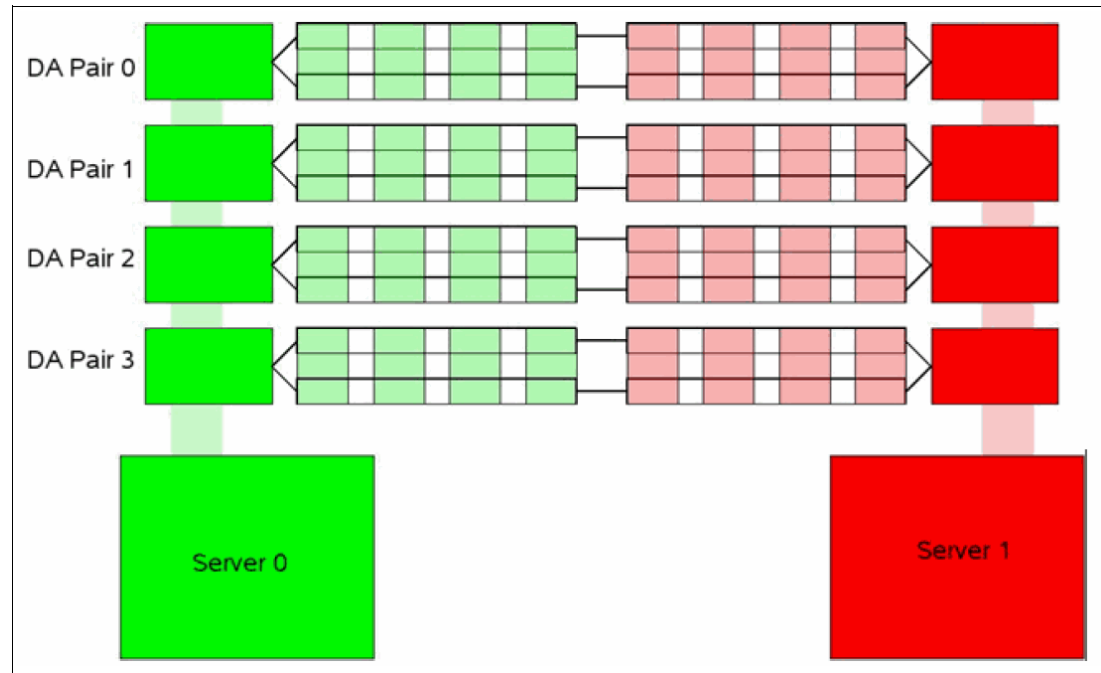


Figure 3-4 DA pair correct configuration

Figure 3-5 shows what a correct configuration resembles the CLI output of the **lsarray** and **lsrank** commands. Notice that the output shows that this configuration balances the workload across all four DA pairs with an even balance between odd and even extent pools. The arrays that are on the same DA pair are split between groups 0 and 1.

```
dscli> lsarray -l
Date/Time: Oct 20, 2016 10:15:43 AM CEST IBM DSCLI Version: 7.8.1.62 DS: IBM.2107-75L2321
Array State Data RAID type arsite Rank DA Pair DDMcap(10^9B) diskclass
=====
```

|    |        |        |           |    |    |   |        |     |
|----|--------|--------|-----------|----|----|---|--------|-----|
| A0 | Assign | Normal | 5 (6+P+S) | S1 | R0 | 0 | 1200.0 | ENT |
| A1 | Assign | Normal | 5 (6+P+S) | S2 | R1 | 1 | 1200.0 | ENT |
| A2 | Assign | Normal | 5 (6+P+S) | S3 | R2 | 2 | 1200.0 | ENT |
| A3 | Assign | Normal | 5 (6+P+S) | S4 | R3 | 3 | 1200.0 | ENT |
| A4 | Assign | Normal | 5 (6+P+S) | S5 | R4 | 0 | 1200.0 | ENT |
| A5 | Assign | Normal | 5 (6+P+S) | S6 | R5 | 1 | 1200.0 | ENT |
| A6 | Assign | Normal | 5 (6+P+S) | S7 | R6 | 2 | 1200.0 | ENT |
| A7 | Assign | Normal | 5 (6+P+S) | S8 | R7 | 3 | 1200.0 | ENT |

```
=====
dscli> lsrank -l
Date/Time: Oct 20, 2016 10:20:23 AM CEST IBM DSCLI Version: 7.8.1.62 DS: IBM.2107-75L2321
ID Group State datastate Array RAIDtype extpoolID extpoolnam stgtype exts usedexts encryptgrp marray
=====
```

|    |   |        |        |    |   |    |          |    |      |      |   |     |
|----|---|--------|--------|----|---|----|----------|----|------|------|---|-----|
| R0 | 0 | Normal | Normal | A0 | 5 | P0 | extpool0 | fb | 6348 | 6348 | - | MA1 |
| R1 | 1 | Normal | Normal | A1 | 5 | P1 | extpool1 | fb | 6348 | 6348 | - | MA2 |
| R2 | 0 | Normal | Normal | A2 | 5 | P2 | extpool2 | fb | 6348 | 6348 | - | MA3 |
| R3 | 1 | Normal | Normal | A3 | 5 | P3 | extpool3 | fb | 6348 | 6348 | - | MA4 |
| R4 | 1 | Normal | Normal | A4 | 5 | P5 | extpool5 | fb | 6348 | 6348 | - | MA5 |
| R5 | 0 | Normal | Normal | A5 | 5 | P4 | extpool4 | fb | 6348 | 6348 | - | MA6 |
| R6 | 1 | Normal | Normal | A6 | 5 | P7 | extpool7 | fb | 6348 | 6348 | - | MA7 |
| R7 | 0 | Normal | Normal | A7 | 5 | P6 | extpool6 | fb | 6348 | 6348 | - | MA8 |

Figure 3-5 The lsarray and lsrank command output

## DS8000 series ranks to extent pools mapping

In the DS8000 architecture, extent pools are used to manage one or more ranks. An extent pool is visible to both processor complexes in the DS8000 storage system, but it is directly managed by only one of them. You must define a minimum of two extent pools with one extent pool that is created for each processor complex to fully use the resources. You can use the following approaches:

- One-to-one approach: One rank per extent pool configuration

With this approach, DS8000 is formatted in 1:1 assignment between ranks and extent pools. This configuration disables any DS8000 storage pool striping or auto-rebalancing activity, if it is enabled. You can create one or two volumes in each extent pool exclusively on one rank only and put all of those volumes into one IBM SAN Volume Controller storage pool. SAN Volume Controller stripes across all of these volumes and balances the load across the RAID ranks by that method. No more than two volumes per rank are needed with this approach. Therefore, the rank size determines the volume size.

Systems often are configured with at least two storage pools: One (or two) containing MDisk of all the 6+P RAID 5 ranks of the DS8000 storage system, and the other one (or more) containing the slightly larger 7+P RAID 5 ranks. This approach maintains equal load balancing across all ranks when the IBM SAN Volume Controller striping occurs because each MDisk in a storage pool is the same size.

The IBM SAN Volume Controller extent size is the stripe size that is used to stripe across all these single-rank MDisk.

This approach delivered good performance and has its justifications. However, it also includes a few minor drawbacks. A natural skew can exist, such as a small file of a few hundred KiB that is heavily accessed. When you have more than two volumes from one rank, but not as many IBM SAN Volume Controller storage pools, the system might start striping across many entities that are effectively in the same rank, depending on the storage pool layout. Such striping should be avoided.

An advantage of this approach is that it delivers more options for fault isolation and control over where a specific volume and extent are located.

- Many-to-one approach: Multi-rank extent pool configuration

A more modern approach is to create a few DS8000 extent pools; for example, two DS8000 extent pools. Use DS8000 storage pool striping or automated Easy Tier rebalancing to help prevent from overloading individual ranks.

Create at least two extent pools for each tier to balance the extent pools by Tier and Controller affinity. Mixing different tiers on the same extent pool is effective only when Easy Tier is activated on the DS8000 pools. However, when virtualized, tier management has more advantages when handled by the IBM SAN Volume Controller. For more information about choosing which level to run Easy Tier, see “External controller tiering considerations” on page 167.

You need only one volume size with this multi-rank approach because enough space is available in each large DS8000 extent pool. The maximum number of back-end storage ports to be presented to the IBM SAN Volume Controller is 16. Each port represents a path to the IBM SAN Volume Controller. Therefore, when sizing the number of LUN/MDisk to be presented to the IBM SAN Volume Controller, the suggestion is to present at least 2 - 4 volumes per path. Therefore, using the maximum of 16 paths, create 32, 48, or 64 DS8000 volumes, which maintains a good queue depth for this configuration.

To maintain the highest flexibility and for easier management, large DS8000 extent pools are beneficial. However, if the DS8000 installation is dedicated to shared-nothing environments, such as Oracle ASM, IBM DB2® warehouses, or General Parallel File System (GPFS), use the single-rank extent pools.

## LUN masking

For a storage controller, all IBM SAN Volume Controller nodes must detect the same set of LUs from all target ports that logged in. If target ports are visible to the nodes or canisters that do not have the same set of LUs assigned, IBM SAN Volume Controller treats this situation as an error condition and generates error code 1625.

You must validate the LUN masking from the storage controller and then, confirm the correct path count from within the IBM SAN Volume Controller.

The DS8000 series controllers perform LUN masking that is based on the volume group. Example 3-4 shows the output of the **showvolgrp** command for volume group (V0), which contains 16 LUNs that are being presented to a two-node IBM SAN Volume Controller cluster.

*Example 3-4 Output of the showvolgrp command*

```
dscli> showvolgrp V0
Date/Time: Oct 20, 2016 10:33:23 AM BRT IBM DCLI Version: 7.8.1.62 DS: IBM.2107-75FPX81
Name ITS0_SVC
ID V0
Type SCSI Mask
Vols 1001 1002 1003 1004 1005 1006 1007 1008 1101 1102 1103 1104 1105 1106 1107 1108
```

Example 3-5 shows output for the **lshostconnect** command from the DS8000 series. In this example, you can see that four ports of the two-node cluster are assigned to the same volume group (V0) and therefore, are assigned to the same four LUNs.

*Example 3-5 Output for the lshostconnect command*

```
dscli> lshostconnect -volgrp v0
Date/Time: Oct 22, 2016 10:45:23 AM BRT IBM DCLI Version: 7.8.1.62 DS: IBM.2107-75FPX81
Name ID WWPN HostType Profile portgrp volgrpID ESSIOport
=====
ITS0_SVC_N1C1P4 0001 500507680C145232 SVC San Volume Controller 1 V0 all
ITS0_SVC_N1C2P3 0002 500507680C235232 SVC San Volume Controller 1 V0 all
ITS0_SVC_N2C1P4 0003 500507680C145231 SVC San Volume Controller 1 V0 all
ITS0_SVC_N2C2P3 0004 500507680C235231 SVC San Volume Controller 1 V0 all
```

From Example 3-5, you can see that only the IBM SAN Volume Controller WWPNs are assigned to V0.

**Attention:** Data corruption can occur if the same LUN is assigned to IBM SAN Volume Controller nodes and other devices, such as hosts attached to DS8000.

Next, you see how the IBM SAN Volume Controller detects these LUNs if the zoning is properly configured. The Managed Disk Link Count (mdisk\_link\_count) represents the total number of MDisks that are presented to the IBM SAN Volume Controller cluster by that specific controller.

Example 3-6 shows the general details of the output storage controller by using the system CLI.

*Example 3-6 Output of the lscontroller command*

```
IBM_2145:SVC-ITS0:superuser>svcinfolcontroller DS8K75FPX81
id 1
controller_name DS8K75FPX81
WWNN 5005076305FFC74C
mdisk_link_count 16
max_mdisk_link_count 16
```



```

degraded no
vendor_id IBM
product_id_low 2107900
...
WWPN 500507630500C74C
path_count 16
max_path_count 16
WWPN 500507630508C74C
path_count 16
max_path_count 16

```

---

## IBM SAN Volume Controller MDisk and storage pool considerations

The recommended practice is to create a single IBM SAN Volume Controller storage pool per DS8900F system. This configuration simplifies management, and increases overall performance.

An example of preferred configuration is shown in Figure 3-6. Four Storage pools or Extent pools (one even and one odd) of DS8900F are joined into one IBM SAN Volume Controller storage pool.

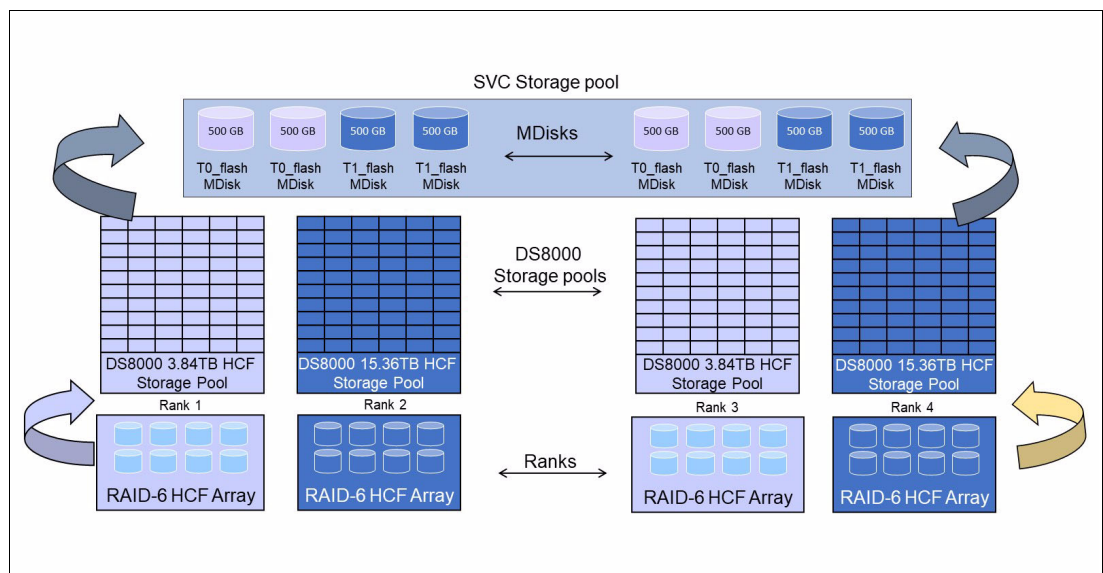


Figure 3-6 Four DS8900F extent pools as one IBM SAN Volume Controller storage pool

To determine how many logical volumes must be created to present to IBM SAN Volume Controller as MDisk, see 3.1.2, “Guidelines for creating optimal back-end configuration” on page 70.

## 3.2.2 IBM XIV Storage System considerations

XIV Gen3 volumes can be provisioned to IBM SAN Volume Controller by way of iSCSI and FC. However, it is preferred to implement FC attachment for performance and stability considerations, unless a dedicated IP infrastructure for storage is available.

### Host options and settings for XIV systems

You must use specific settings to identify IBM SAN Volume Controller systems as hosts to XIV systems. An XIV node within an XIV system is a single WWPN.

An XIV node is considered to be a single SCSI target. Each host object that is created within the XIV System must be associated with the same LUN map.

From an IBM SAN Volume Controller perspective, an XIV Type Number 281x controller can consist of more than one WWPN. However, all are placed under one worldwide node number (WWNN) that identifies the entire XIV system.

### **Creating a host object for IBM SAN Volume Controller for an IBM XIV**

A single host object with all WWPNs of IBM SAN Volume Controller nodes can be created when implementing IBM XIV. This technique makes the host configuration easier to configure. However, the ideal host definition is to consider each node IBM SAN Volume Controller as a host object, and create a cluster object to include all nodes or canisters.

When implemented in this manner, statistical metrics are more effective because performance can be collected and analyzed on IBM SAN Volume Controller node level.

For more information about creating a host on XIV, see *IBM XIV Gen3 with IBM System Storage SAN Volume Controller and Storwize V7000*, REDP-5063.

### **Volume considerations**

As modular storage, XIV storage can be presented 6 - 15 modules in a configuration. Each module that is added to the configuration increases the XIV capacity, CPU, memory, and connectivity. The XIV system currently supports the following configurations:

- ▶ 28 - 81 TB when 1 TB drives are used
- ▶ 55 - 161 TB when 2 TB disks are used
- ▶ 84 - 243 TB when 3 TB disks are used
- ▶ 112 - 325 TB when 4 TB disks are used
- ▶ 169 - 489 TB when 6 TB disks are used

Figure 3-7 shows how XIV configuration varies according to the number of modules present on the system.

| Rack Configuration                                      |                                             |                                              |                                               |                                               |                                               |                                               |                                               |                                               |
|---------------------------------------------------------|---------------------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| Total number of modules<br>(Configuration type)         | 6<br>partial                                | 9<br>partial                                 | 10<br>partial                                 | 11<br>partial                                 | 12<br>partial                                 | 13<br>partial                                 | 14<br>partial                                 | 15<br>full                                    |
| Total number of data modules                            | 3                                           | 3                                            | 4                                             | 5                                             | 6                                             | 7                                             | 8                                             | 9                                             |
| Total number of interface modules                       | 3                                           | 6                                            | 6                                             | 6                                             | 6                                             | 6                                             | 6                                             | 6                                             |
| Number of <i>active</i> interface modules               | 2                                           | 4                                            | 4                                             | 5                                             | 5                                             | 6                                             | 6                                             | 6                                             |
| Interface module 9 state                                |                                             | Disabled                                     | Disabled                                      | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       |
| Interface module 8 state                                |                                             | Enabled                                      | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       |
| Interface module 7 state                                |                                             | Enabled                                      | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       |
| Interface module 6 state                                | Disabled                                    | Disabled                                     | Disabled                                      | Disabled                                      | Disabled                                      | Enabled                                       | Enabled                                       | Enabled                                       |
| Interface module 5 state                                | Enabled                                     | Enabled                                      | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       |
| Interface module 4 state                                | Enabled                                     | Enabled                                      | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       | Enabled                                       |
| FC ports                                                | 8                                           | 16                                           | 16                                            | 20                                            | 20                                            | 24                                            | 24                                            | 24                                            |
| iSCSI ports (1 Gbps – mod 114)                          | 6                                           | 14                                           | 14                                            | 18                                            | 18                                            | 22                                            | 22                                            | 22                                            |
| iSCSI ports (10 Gbps – mod 214)                         | 4                                           | 8                                            | 8                                             | 10                                            | 10                                            | 12                                            | 12                                            | 12                                            |
| Number of disks                                         | 72                                          | 108                                          | 120                                           | 132                                           | 144                                           | 156                                           | 168                                           | 180                                           |
| Usable capacity (1 / 2 / 3 / 4 / 6 TB)                  | 28 TB<br>55 TB<br>84 TB<br>112 TB<br>169 TB | 44 TB<br>88 TB<br>132 TB<br>177 TB<br>267 TB | 51 TB<br>102 TB<br>154 TB<br>207 TB<br>311 TB | 56 TB<br>111 TB<br>168 TB<br>225 TB<br>338 TB | 63 TB<br>125 TB<br>190 TB<br>254 TB<br>382 TB | 67 TB<br>134 TB<br>203 TB<br>272 TB<br>409 TB | 75 TB<br>149 TB<br>225 TB<br>301 TB<br>453 TB | 81 TB<br>161 TB<br>243 TB<br>325 TB<br>489 TB |
| # of CPUs (one per Module)                              | 6                                           | 9                                            | 10                                            | 11                                            | 12                                            | 13                                            | 14                                            | 15                                            |
| Memory (24 GB per module w 1/2/3 TB)                    | 144 GB                                      | 216 GB                                       | 240 GB                                        | 264 GB                                        | 288 GB                                        | 312 GB                                        | 336 GB                                        | 360 GB                                        |
| Memory (48 GB per module w 4/6 TB)                      | 288 GB                                      | 432 GB                                       | 480 GB                                        | 528 GB                                        | 576 GB                                        | 624 GB                                        | 672 GB                                        | 720 GB                                        |
| {Optional for 1, 2, 3, 4, 6 TB XIVs} 400 GB Flash Cache | 2.4 TB                                      | 3.6 TB                                       | 4.0 TB                                        | 4.4 TB                                        | 4.8 TB                                        | 5.2 TB                                        | 5.6 TB                                        | 6.0 TB                                        |
| {Optional for 4, 6 TB XIVs} 800 GB Flash Cache          | 4.8 TB                                      | 7.2 TB                                       | 8.0 TB                                        | 8.8 TB                                        | 9.2 TB                                        | 10.4 TB                                       | 11.2 TB                                       | 12.0 TB                                       |
| Power (kVA) - Model 281x-214 / with SSD                 | 2.5 / 2.6                                   | 3.6 / 3.9                                    | 4.0 / 4.3                                     | 4.3 / 4.6                                     | 4.7 / 5.09                                    | 5.0 / 5.4                                     | 5.5 / 5.8                                     | 5.8 / 6.2                                     |

Figure 3-7 XIV rack configuration: 281x-214

Although XIV has its own queue depth characteristics for direct host attachment, the best practices that are described in 3.1.2, “Guidelines for creating optimal back-end configuration” on page 70 are preferred when you virtualize XIV with IBM Spectrum Virtualize.

Suggested volume sizes and quantities for IBM SAN Volume Controller on the XIV systems with different drive capacities are listed in Table 3-1.

Table 3-1 XIV minimum volume size and quantity recommendations

| Modules | XIV host ports | Volume size (GB)<br>1 TB drives | Volume size (GB)<br>2 TB drives | Volume size (GB)<br>3 TB drives | Volume size (GB)<br>4 TB drives | Volume size (GB)<br>6 TB drives | Volume quantity | Vols to XIV host ports |
|---------|----------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-----------------|------------------------|
| 6       | 4              | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 17              | 4.3                    |
| 9       | 8              | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 27              | 3.4                    |
| 10      | 8              | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 31              | 3.9                    |
| 11      | 10             | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 34              | 3.4                    |
| 12      | 10             | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 39              | 3.9                    |
| 13      | 12             | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 41              | 3.4                    |
| 14      | 12             | 1600                            | 3201                            | 4852                            | 6401                            | 9791                            | 46              | 3.8                    |

## Other considerations

This section highlights the following restrictions for the use of the XIV system as back-end storage for the IBM SAN Volume Controller:

- ▶ Volume mapping

When mapping a volume, you must use the same LUN ID to all IBM SAN Volume Controller nodes. Therefore, map the volumes to the cluster, not to individual nodes.

- ▶ XIV Storage pools

When creating an XIV storage pool, define the Snapshot Size as zero (0). Snapshot space does not need to be reserved because it is not recommended to use XIV snapshots on LUNs mapped as MDisk. The snapshot functions are used on IBM SAN Volume Controller level.

Because all LUNs on a single XIV system share performance and capacity characteristics, use a single IBM SAN Volume Controller storage pool for a single XIV system.

- ▶ Thin provisioning

XIV thin provisioning pools are not supported by IBM SAN Volume Controller. Instead, you must use a regular pool.

- ▶ Copy functions for XIV models

You cannot use advanced copy functions for XIV models, such as taking a snapshot and remote mirroring, with disks that are managed by the IBM SAN Volume Controller.

For more information about configuration of XIV behind IBM SAN Volume Controller, see *IBM XIV Gen3 with IBM System Storage SAN Volume Controller and Storwize V7000*, REDP-5063.

## 3.2.3 IBM FlashSystem A9000/A9000R considerations

IBM FlashSystem A9000 and IBM FlashSystem A9000R use industry-leading data reduction technology that combines inline, real-time pattern matching and removal, data deduplication, and compression. Compression also uses hardware cards inside each grid controller.

Compression can easily provide a 2:1 data reduction saving rate on its own, which effectively doubles the system storage capacity. Combined with pattern removal and data deduplication services, IBM FlashSystem A9000/A9000R can yield an effective data capacity of five times the original usable physical capacity.

Deduplication can be implemented on the IBM SAN Volume Controller by attaching an IBM FlashSystem A9000/A9000R as external storage instead of the use of IBM Spectrum Virtualize Data Reduction Pool (DRP)-level deduplication.

Next, we describe several considerations when you are attaching an IBM FlashSystem A9000/A9000R system as a back-end controller.

### Volume considerations

IBM FlashSystem A9000/A9000R designates resources to data reduction. Because it is always on, it is advised that data reduction be done in the IBM FlashSystem A9000/A9000R only and *not* in the Spectrum Virtualize cluster. Otherwise, needless extra latency occurs as IBM FlashSystem A9000/A9000R tries to reduce the data.

Estimated data reduction is important because that helps determine volume size. Always attempt to use a conservative data reduction ratio when attaching A9000/A9000R because the storage pool goes offline if the back-end storage runs out of capacity.

To determine the controller volume size:

- ▶ Calculate effective capacity by reducing the measured data reduction ratio (for example, if the data reduction estimation tool provides a ratio of 4:1, use 3.5:1 for calculations) and multiply it to physical capacity.
- ▶ Determine the number of connected FC ports by using Table 3-2 and Table 3-3.
- ▶ Consider that the volume size is equal to effective capacity divided by the number of ports taken twice (effective capacity/path\*2)

The remaining usable capacity can be added to the storage pool after the system reaches a stable data reduction ratio.

*Table 3-2 Host connections for A9000*

| Number of controllers | Total FC ports available | Total ports that are connected to SAN Volume Controller | Connected port                 |
|-----------------------|--------------------------|---------------------------------------------------------|--------------------------------|
| 3                     | 12                       | 6                                                       | All controllers, ports 1 and 3 |

*Table 3-3 Host connections for A9000R*

| Grid element | Number of controllers | Total FC ports available | Total ports that are connected to SAN Volume Controller | Connected ports                                                                                                     |
|--------------|-----------------------|--------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 2            | 4                     | 16                       | 8                                                       | All controllers, ports 1 and 3                                                                                      |
| 3            | 6                     | 24                       | 12                                                      | All controllers, ports 1 and 3                                                                                      |
| 4            | 8                     | 32                       | 8                                                       | <ul style="list-style-type: none"> <li>▶ Controllers 1 - 4, port 1</li> <li>▶ Controllers 5 - 8, port 3</li> </ul>  |
| 5            | 10                    | 40                       | 10                                                      | <ul style="list-style-type: none"> <li>▶ Controllers 1 - 5, port 1</li> <li>▶ Controllers 6 - 10, port 3</li> </ul> |
| 6            | 12                    | 48                       | 12                                                      | <ul style="list-style-type: none"> <li>▶ Controllers 1 - 6, port 1</li> <li>▶ Controllers 7 - 12, port 3</li> </ul> |

It is important not to run out of hard capacity on the back-end storage because the storage pool can go offline. Close monitoring of the FlashSystem A9000/A9000R is important. If you start to run out of space, you can use the migration functions of Spectrum Virtualize to move data to another storage system. Consider the following examples:

▶ **Example 1**

In this example, a FlashSystem A9000 with 57 TB of usable capacity, or 300 TB of effective capacity, is used at the standard 5.26:1 data efficiency ratio.

We ran the data reduction tool on a good representative sample of the volumes that we are virtualizing. We know that we have a data reduction ratio of 4.2:1 and as an extra precaution, we use 4:1 for more calculations. The 4 x 57 calculation results in 228 TB. Divide this result by 12 (six paths x 2), and 19 TB is available per volume.

▶ **Example 2**

In this example, a five grid element FlashSystem A9000R that uses 29 TB Flash enclosures is used. It has a total usable capacity of 145 TB.

We use 10 paths and have not run any of the estimation tools on the data. However, we know that the host is not compressing the data. We assume a compression ratio of 2:1; 2 x 145 gives 290, and divided by 20 gives 14.5 TB per volume.

In this case, if we see that we are getting a much better data reduction ratio than we planned for, we can always create volumes and make them available to Spectrum Virtualize.

The biggest concern with the number of volumes is ensuring adequate queue depth is available. Because the maximum volume size on the FlashSystem A9000/A9000R is 1 PB and we are ensuring two volumes per path, we can create a few larger volumes and still have good queue depth and not have numerous volumes to manage.

### Other considerations

Spectrum Virtualize can detect that the IBM FlashSystem A9000 controller uses deduplication technology. It also shows that the Deduplication attribute of the managed disk as Active.

Deduplication status is important because it allows IBM Spectrum Virtualize to enforce the following restrictions:

- ▶ Storage pools with deduplicated MDisks should contain only MDisks from the same IBM FlashSystem A9000 or IBM FlashSystem A9000R storage controller.
- ▶ Deduplicated MDisks cannot be mixed in an Easy Tier enabled storage pool.

## 3.2.4 FlashSystem 5000, 5100, 7200, 9100, and 9200 considerations

**Note:** The recommendations that are described in this section apply to a solution when IBM FlashSystem family system is virtualized by IBM SAN Volume Controller.

### Connectivity considerations

It is expected that NPIV is enabled on both systems: the one that is virtualizing storage, and on the one that works as a back-end zone “host” or “virtual” WWPNs of the back-end system to physical WWPNs of the front-end, or virtualizing system.

For more information about SAN and zoning preferred practices, see Chapter 2, “Storage area network” on page 19.

### System layers

Spectrum Virtualize systems feature the concept of system layers. Two layers exist: storage and replication. Systems that are configured into a storage layer can work as a back-end storage. Systems that are configured into replication layer can virtualize another IBM FlashSystem cluster and use them as a back-end controller.

Systems that are configured with the same layer can be replication partners; systems in the different layers cannot.

IBM SAN Volume Controller is configured to replication layer and it cannot be changed.

### Automatic configuration

IBM FlashSystem family systems that are running code version 8.3x and above can be automatically configured for optimal performance as a back-end storage behind IBM SAN Volume Controller.

Automatic configuration wizard must be used on a system that has no volumes, pools, and host objects configured. An available wizard configures internal storage devices, creates volumes, and maps the to the host object, which represents the IBM SAN Volume Controller.

## Array and disk pool considerations

The back-end IBM FlashSystem family system can have a hybrid configuration that contains FlashCore Modules and SSDs and spinning drives.

Internal storage that is attached to the back-end system must be joined into RAID arrays. You might need one or more DRAID6 arrays, depending on the number and the type of available drives. For more information about RAID recommendations, see the “Array considerations” section in *Implementing the IBM FlashSystem with IBM Spectrum Virtualize V8.4*, SG24-8465.

Consider creating a separate disk pool for each type (tier) of storage and use the Easy Tier function on a front-end system. Front-end FlashSystem family systems cannot monitor Easy Tier activity of the back-end storage. If Easy Tier is enabled on the front-end and back-end systems, they independently rebalance the hot areas according to their own heat map. This process causes a rebalance over a rebalance, and such a situation can eliminate the performance benefits of extent reallocation. For this reason, Easy Tier must be enabled only on one level (the front end is preferred).

For more information about recommendations for Easy Tier with external storage, see Chapter 4, “Storage pools” on page 91.

For most use cases, Standard pools are preferred to DRPs on the back-end storage. The front end performs the reduction, if planned. Data reduction on both levels is *not* recommended because it adds processing overhead and does not result in capacity savings.

If Easy Tier is disabled on the back-end as we advised, the back-end FlashSystem pool extent size is not a performance concern.

## Volume considerations

Volumes in IBM FlashSystem can be created as *striped* or *sequential*. The general rule is to create striped volumes. Volumes on back-end system must be fully allocated.

To determine a number of volumes to create on back-end IBM FlashSystem to provide a virtualizer as MDisks, see the general rules that are described in 3.1.2, “Guidelines for creating optimal back-end configuration” on page 70.

When virtualizing back-end with spinning drives, perform queue depth calculations. For all flash solutions, create 32 volumes from the available pool capacity, which can be reduced to 16 or even 8 for small arrays (for example, if you have 16 or less flash drives in a back-end pool). For FCM arrays, the number of volumes is also governed by load distribution. 32 volumes out of a pool with an FCM array are recommended.

When choosing volume size, consider which system (front-end or back-end) performs compression. If data is compressed and deduplicated on the front-end SAN Volume Controller, FCMs cannot compress it further, which results in a 1:1 compression ratio. Therefore, the back-end volume size is calculated from the pool physical capacity divided by the number of volumes (16 or more).

Consider the following examples:

- ▶ Example 1: FlashSystem 9100 with 24 x 19.2 TB modules

This configuration provides raw disk capacity of 460 TB, with 10+P+Q DRAID6 and one distributed spare. The physical array capacity is 365 TB or 332 TiB. Because it is not recommended to provision more than 85% of a physical flash, we have 282 TiB.

Because we do not expect any compression on FCM (the back-end is getting data that is already compressed by the upper levels), we provision storage to upper level assuming 1:1 compression, which means we create 32 volumes  $282\text{TiB} / 32 = 8.8\text{ TiB}$  each.

If the front-end system is not compressing data, space savings will be achieved with FCM hardware compression. Use compression estimation tools to determine the expected compression ratio and use a smaller ratio for further calculations (for example, if you expect 4.5:1 compression, use 4.3:1). Determine the volume size using the calculated effective pool capacity.

- Example 2: Storwize V7000 Gen3 with 12 x 9.6 TB modules

This configuration provides raw disk capacity of 115 TB, with 9+P+Q DRAID6 and one distributed spare. The physical capacity is 85 TB or 78 TiB.

Because it is not recommended to provision more than 85% of a physical flash, we have 66 TiB. The compression estimator shows that we can achieve a 3.2:1 compression ratio, decreasing in and assuming 3:1, we have  $66\text{ TiB} \times 3 = 198\text{ TiB}$  of effective capacity.

Create 16 volumes,  $198\text{ TiB} / 16 = 12.4\text{ TiB}$  each. If the compression ratio is higher than expected, we can create and provision two more front-end volumes.

### 3.2.5 IBM FlashSystem 900 considerations

The main advantage of integrating FlashSystem 900 with IBM Spectrum Virtualize is to combine the extreme performance of IBM FlashSystem 900 with the Spectrum Virtualize enterprise-class solution, such as tiering, volume mirroring, deduplication, and copy services.

When you configure the IBM FlashSystem 900 as a backend for Spectrum Virtualize family systems, you must remember the considerations that are described next.

#### Defining storage

IBM FlashSystem 900 supports up to 12 IBM MicroLatency® modules. IBM MicroLatency modules are installed in the IBM FlashSystem 900 based on the following configuration guidelines:

- A minimum of four MicroLatency modules must be installed in the system. RAID 5 is the only supported configuration of the IBM FlashSystem 900.
- The system supports configurations of 4, 6, 8, 10, and 12 MicroLatency modules in RAID 5.
- All MicroLatency modules that are installed in the enclosure must be identical in capacity and type.
- For optimal airflow and cooling, if fewer than 12 MicroLatency modules are installed in the enclosure, populate the module bays beginning in the center of the slots and adding on either side until all 12 slots are populated.

The array configuration is performed during system setup. The system automatically creates MDisk/arrays and defines the RAID settings based on the number of flash modules in the system. The default supported RAID level is RAID 5.

#### Volume considerations

To fully use all Spectrum Virtualize system resources, create 32 volumes (or 16 volumes if FlashSystem 900 is not fully populated). This way, all CPU cores, nodes, and FC ports of the virtualizer are fully used.



However, one important factor must be considered when volumes are created from a pure FlashSystem 900 MDisk storage pool. FlashSystem 900 can process I/Os much faster than traditional storage. In fact, sometimes they are even faster than cache operations because with cache, all I/Os to the volume must be mirrored to another node in I/O group.

This operation can take as much as 1 millisecond while I/Os that are issued directly (which means without cache) to the FlashSystem 900 can take 100 - 200 microseconds. Therefore, it might be recommended to disable Spectrum Virtualize cache to optimize for maximum IOPS in some rare use case.

You must keep the cache *enabled* in the following situations:

- ▶ If volumes from FlashSystem 900 pool are:
  - Compressed
  - In a Metro/Global Mirror relationship
  - In a FlashCopy relationship (source or target)
- ▶ If the same pool has MDisk from FlashSystem 900 and contains MDisk from other back-end controllers.

For more information, see *Implementing IBM FlashSystem 900*, SG24-8271.

### 3.2.6 Path considerations for third-party storage with EMC VMAX and Hitachi Data Systems

Although many third-party storage options are available and supported, this section highlights the pathing considerations for EMC VMAX and Hitachi Data Systems (HDS).

When presented to the IBM SAN Volume Controller, most storage controllers are recognized as a single WWNN per controller. However, for some EMC VMAX and HDS storage controller types, the system recognizes each port as a different WWNN. For this reason, each storage port, when zoned to an IBM SAN Volume Controller, appears as a different external storage controller.

IBM Spectrum Virtualize supports a maximum of 16 WWNNs per storage system. Therefore, it is preferred to connect up to 16 storage ports to IBM SAN Volume Controller.

For more information about determining the number of logical volumes or LUNs to be configured on third-party storage, see 3.1.2, “Guidelines for creating optimal back-end configuration” on page 70.

## 3.3 Quorum disks

**Note:** This section does not cover IP-attached quorum. For information about these quorums, see Chapter 7, “Business continuity” on page 329.

A system uses a quorum disk for the following purposes:

- ▶ To break a tie when a SAN fault occurs, when half of the nodes that were a member of the system are present.
- ▶ To hold a copy of important system configuration data.

After internal drives are prepared to be added to an array, or external MDisks become managed, a small portion of its capacity is reserved for quorum data. Its size is less than 0.5 GiB for a drive and not less than one pool extent for an MDisk.

Three devices from all available internal drives and managed MDisks are selected for the *quorum disk* role. They store system metadata, which is used for cluster recovery after a disaster. Despite only three devices that are designated as quorums, capacity for quorum data is reserved on each of them because the designation might change (for example, if the quorum disk fails).

Only one of those disks is selected as the active quorum disk (it is used as a tie-breaker). If as a result of a failure, the cluster is split in half and both parts lose sight of each other (for example, the inter-site link failed in a HyperSwap cluster with two I/O groups), they appeal to the tie-breaker, active quorum device. The half of the cluster nodes that can reach and reserve the quorum disk after the split occurs, lock the disk and continue to operate. The other half stops its operation. This design prevents both sides from becoming inconsistent with each other.

The storage device must match following criteria to be considered a quorum candidate:

- ▶ The internal drive or module should be a member of an array or a “Candidate”; drives in “Unused” state cannot be quorums. The MDisk must be in “Managed” state; “Unmanaged” or “Image” MDisks cannot be quorums.
- ▶ External MDisks cannot be provisioned over iSCSI, only FC.
- ▶ An MDisk must be presented by a disk subsystem, LUNs from which are supported to be quorum disks.

The system uses the following rules when selecting quorum devices:

- ▶ Fully connected candidates are preferred over partially connected candidates.  
In a multiple enclosure environment, MDisks are preferred over drives.
- ▶ Drives are preferred over MDisks.  
If only one control enclosure and no external storage exist in the cluster, drives are considered first.
- ▶ Drives from a different control enclosure are to be preferred over a second drive from the same enclosure.  
If IBM SAN Volume Controller contains more than one IOgroup, at least one of the candidates from each group is selected.

To become an active quorum device (tie-break device), it must be visible to all nodes in a cluster.

In practice, these rules mean that in a standard topology cluster when you attach at least one back-end storage controller that supports quorum and imported MDisks from it as *Managed* type, quorums including active quorum disk are assigned automatically. If all your MDisks are image-mode or unmanaged, your cluster operates without quorum device, unless you deployed IP-based quorum.

For more information about quorum device recommendations in a stretched cluster environment, see Chapter 7, “Business continuity” on page 329.

To list IBM SAN Volume Controller quorum devices, run the **lsquorum** command. To move quorum assignment, run the **chquorum** command.



## Storage pools

This chapter highlights considerations when you are planning storage pools for an IBM SAN Volume Controller implementation. It explains various pool configuration options, including Easy Tier, data reduction pools (DRPs), and provides best practices on the implementation and an overview of some typical operations with MDisks.

This chapter includes the following topics:

- ▶ 4.1, “Introduction to pools” on page 92
- ▶ 4.2, “Storage pool planning considerations” on page 112
- ▶ 4.3, “Data reduction pool best practices” on page 119
- ▶ 4.4, “Operations with storage pools” on page 127
- ▶ 4.5, “Considerations when using encryption” on page 137
- ▶ 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148

## 4.1 Introduction to pools

In general, a storage pool or pool, sometimes still referred to by its familiar name of *managed disk group*, is a grouping of storage capacity that is used to provision volumes and logical units (LUNs) that then can be made visible to hosts.

IBM SAN Volume Controller supports the following types of pools:

- ▶ Standard pools (parent pools and child pools)
- ▶ DRPs (parent pools and Quotaless child pools)

Standard pools were available since the initial release of IBM Spectrum Virtualize in 2003 and can include fully allocated or thin provisioned volumes.

Real-time Compression (RTC) is allowed only with standard pools on some older IBM SAN Volume Controller hardware models and should not be implemented in new configurations.

**Note:** The latest node hardware does not support Real-time Compression.

SA2 and SV2 IBM SAN Volume Controller node hardware do not support the use of RTC volumes. To migrate a system to use these node types, all RTC volumes must be removed (migrated) to uncompressed standard pool volumes, or into a DRP.

IBM FlashSystems using standard pools cannot be configured with Real-time Compression.

DRPs represent a significant enhancement to the storage pool concept because the virtualization layer is primarily a simple layer that runs the task of lookups between virtual and physical extents. With the introduction of data reduction technology, compression, and deduplication, it has become more of a requirement to have an uncomplicated way to stay thin.

DRPs increase infrastructure capacity usage by using new efficiency functions and reducing storage costs. The pools enable you to automatically deallocate (not to be confused with deduplicate) and reclaim capacity of thin-provisioned volumes that contain deleted data. In addition, for the first time, the pools enable this reclaimed capacity to be reused by other volumes.

Either pool type can be made up of different tiers. A tier defines a performance characteristic of that subset of capacity in the pool. Often, no more than three tier types are defined in a pool (fastest, average, and slowest). The tiers and their usage are managed automatically by the Easy Tier function.

### 4.1.1 Standard pool

Standard pools, sometimes also referred to as *traditional* storage pools, are a way of providing storage in IBM SAN Volume Controller. They use a fixed allocation unit of an extent. Standard pools are still a valid method to providing capacity to hosts. For more information about guidelines for implementing standard pools, see 4.2, “Storage pool planning considerations” on page 112.

IBM SAN Volume Controller can define parent and child pools. A *parent* pool has all the capabilities and functions of a normal IBM SAN Volume Controller pool. A *child* pool is a logical subdivision of a storage pool or managed disk group. Like a parent pool, a child pool supports volume creation and migration.

When creating a child pool in a standard parent pool, the user must specify a capacity limit for the child pool. This limit allows for a quota of capacity to be allocated to the child pool. This capacity is reserved for the child pool and detracts from the available capacity in the parent pool. This process is different than the method with which child pools are implemented in a DRP (see “Quotaless data reduction child pool” on page 98).

A child pool inherits its tier setting from the parent pool. Changes to a parent’s tier setting are inherited by child pools.

A child pool supports the Easy Tier function if Easy Tier is enabled on the parent pool. The child pool also inherits Easy Tier status, pool status, capacity information, and back-end storage information. The I/O activity of the parent pool is the sum of the I/O activity on itself and any child pools.

## Parent pools

Parent pools receive their capacity from MDisk. To track the space that is available on an MDisk, the system divides each MDisk into chunks of equal size. These chunks are called *extents* and are indexed internally. The choice of extent size affects the total amount of storage that is managed by the system. The extent size remains constant throughout the lifetime of the parent pool.

All MDisk in a pool are split into extents of the same size. Volumes are created from the extents that are available in the pool. You can add MDisk to a pool at any time to increase the number of extents that are available for new volume copies or to expand volume copies. The system automatically balances volume extents between the MDisk to provide the best performance to the volumes.

You cannot use the volume migration functions to migrate volumes between parent pools that feature different extent sizes. However, you can use volume mirroring to move data to a parent pool that has a different extent size.

Consider choosing extent size wisely according to your future needs. Small extents limit your overall usable capacity, but the use of a larger extent size can waste storage. For example, if you select an extent size of 8 GiB, but then create only a 6 GiB volume, one entire extent is allocated to this volume (8 GiB) and hence 2 GiB goes unused.

When you create or manage a parent pool, consider the following general guidelines:

- ▶ Ensure that all MDisk that are allocated to the same tier of a parent pool are the same RAID type. This configuration ensures that the same resiliency is maintained across that tier. Similarly, for performance reasons, do not mix RAID types within a tier. The performance of all volumes is reduced to the lowest achiever in the tier and a mis-match of tier members can result in I/O convoying effects where everything is waiting on the slowest member.
- ▶ An MDisk can be associated with only one parent pool.
- ▶ You should specify a warning capacity for a pool. A warning event is generated when the amount of space that is used in the pool exceeds the warning capacity. The warning threshold is especially useful with thin-provisioned volumes that are configured to automatically use space from the pool.
- ▶ Volumes are associated with only one pool, except during any migration between parent pools.
- ▶ Volumes that are allocated from a parent pool are by default striped across all the storage that is placed into that parent pool. Wide striping can provide performance benefits.

- ▶ You can add MDiskS to a parent pool that are in unmanaged mode only. When MDiskS are added to a parent pool, their mode changes from unmanaged to managed.
- ▶ You can delete MDiskS from a parent pool under the following conditions:
  - Volumes are not using any of the extents that are on the MDisk.
  - Enough free extents are available elsewhere in the pool to move any extents that are in use from this MDisk.
  - The system ensures that all extents that are used by volumes in the child pool are migrated to other MDiskS in the parent pool to ensure that data is not lost.

**Important:** Before you remove MDiskS from a parent pool, ensure that the parent pool has enough capacity for any child pools that are associated with the parent pool.

- ▶ If the parent pool is deleted, you cannot recover the mapping that existed between extents that are in the pool or the extents that the volumes use. If the parent pool includes associated child pools, you must delete the child pools first and return its extents to the parent pool. After the child pools are deleted, you can delete the parent pool. The MDiskS that were in the parent pool are returned to unmanaged mode and can be added to other parent pools. Because the deletion of a parent pool can cause a loss of data, you must force the deletion if volumes are associated with it.

**Important:** Deleting a child or parent pool is unrecoverable.

If you force delete a pool, all volumes in that pool are deleted, even if they are mapped to a host and still in use. Use extreme caution when force deleting any pool objects because the volume to extent mapping cannot be recovered after the delete is processed.

Force deleting a storage pool is possible only by using the command-line tools. For more information, see the **rmmdiskgrp** command help.

- ▶ When deleting a pool with mirrored volumes, if the volume is mirrored and the synchronized copies of the volume are all in the same pool, the mirrored volume is destroyed when the storage pool is deleted. If the volume is mirrored and a synchronized copy exists in another pool, the volume remains after the pool is deleted.

You cannot delete a pool or child pool if Volume Delete Protection is enabled. In code versions 8.3.1 and later, Volume Delete Protection is enabled by default; however, the granularity of protection was improved. You can specify delete protection to be enabled or disabled on a per pool basis, rather than on a system basis as was previously the case.

## Child pools

Instead of being created directly from MDiskS, child pools are created from existing capacity that is allocated to a parent pool. As with parent pools, volumes can be created that specifically use the capacity that is allocated to the child pool. Child pools are similar to parent pools with similar properties and can be used for volume copy operation.

Child pools are created with fully allocated physical capacity; that is, the physical capacity that is applied to the child pool is reserved from the parent pool, just as though you created a fully allocated volume of the same size in the parent pool.

The capacity of the child pool must be smaller than the free capacity that is available to the parent pool. The allocated capacity of the child pool is no longer reported as the free space of its parent pool. Instead, the parent pool reports the entire child pool as used capacity. You must monitor the used capacity of the child pool instead.

When you create or work with a child pool, consider the following general guidelines:

- ▶ Child pools are created automatically by IBM Spectrum Connect VASA client to implement VMware vVols.
- ▶ As with parent pools, you can specify a warning threshold that alerts you when the capacity of the child pool is reaching its upper limit. Use this threshold to ensure that access is not lost when the capacity of the child pool is close to its allocated capacity.
- ▶ On systems with encryption enabled, child pools can be created to migrate volumes in a non-encrypted pool to encrypted child pools. When you create a child pool after encryption is enabled, an encryption key is created for the child pool, even when the parent pool is not encrypted. You can then use volume mirroring to migrate the volumes from the non-encrypted parent pool to the encrypted child pool.
- ▶ Ensure that any child pools that are associated with a parent pool include enough capacity for the volumes that are in the child pool before removing MDisk from a parent pool. The system automatically migrates all extents that are used by volumes to other MDisk in the parent pool to ensure that data is not lost.
- ▶ You cannot shrink the capacity of a child pool to less than its real capacity. The system uses reserved extents from the parent pool that use multiple extents. The system also resets the warning level when the child pool is shrunk, and issues a warning if the level is reached when the capacity is shrunk.
- ▶ The system supports migrating a copy of volumes between child pools within the same parent pool or migrating a copy of a volume between a child pool and its parent pool. Migrations between a source and target child pool with different parent pools are not supported. However, you can migrate a copy of the volume from the source child pool to its parent pool. The volume copy can then be migrated from the parent pool to the parent pool of the target child pool. Finally, the volume copy can be migrated from the target parent pool to the target child pool.
- ▶ When migrating a volume between parent and child pool (with the same encryption key or no encryption) results in a “nocopy” migration; that is, the data does not move. Instead, the extents are reallocated to the child or parent pool and the accounting of the used space is corrected. That is, the free extents are reallocated to the child or parent to ensure the total capacity that is allocated to the child pool remains unchanged.
- ▶ A special form of *quotaless* data reduction child pool can be created from a data reduction parent pool. For more information, see “Quotaless data reduction child pool” on page 98.

## SCSI unmap in a standard pool

A standard pool can use SCSI unmap space reclamation, but not as efficiently as a DRP.

When a host submits a SCSI **unmap** command to a volume in a standard pool, the system changes the unmap into a write\_same of zeros. This **unmap** command becomes an internal special command and can be handled by different layers in the system.

For example, the cache does not mirror the data; instead, it passes the special reference to zeros. The RTC functions reclaim those areas (assuming 32 KB or larger) and shrink the volume allocation.

The back-end layers also submit the `write_same` of zeros to the internal or external MDisk devices. For a Flash or SSD-based MDisk, this process results in the device freeing the capacity back to its available space. Therefore, it shrinks the used capacity on Flash or SSD, which helps to improve efficiency of garbage collection (on device) and performance.

For Nearline SAS drives, the `write_same` of zeros commands can overload the drives, which can result in performance problems.

**Important:** A standard pool does not shrink its used space as the result of a SCSI `unmap` command. The back-end capacity might shrink its used space, but the pool used capacity does not shrink.

The exception is with RTC volumes where the reused capacity of the volume might shrink; however, the pool allocation to that RTC volume remains unchanged. It means that an RTC volume can reuse that unmapped space first before requesting more capacity from the thin provisioning code.

### Thin provisioned volumes in a standard pool

A thin provisioned volume presents a different capacity to mapped hosts than the capacity that the volume uses in the storage pool. IBM SAN Volume Controller supports thin-provisioned volumes in standard pools.

**Note:** Although DRPs fundamentally support thin provisioned volumes, they are used with compression and deduplication. The use of thin-provisioned volumes without more data reduction should be avoided in DRP.

In standard pools, thin provisioned volumes are created as a specific volume type, which is based on capacity savings criteria. These properties are managed at the volume level.

The virtual capacity of a thin provisioned volume is typically significantly larger than its real capacity. Each system uses the real capacity to store data that is written to the volume, and metadata that describes the thin-provisioned configuration of the volume. As more information is written to the volume, more of the real capacity is used.

The system identifies read operations to unwritten parts of the virtual capacity and returns zeros to the server without using any real capacity. For more information about storage system, pool, and volume capacity metrics, see Chapter 9, “Monitoring” on page 353.

Thin-provisioned volumes can also help simplify server administration. Instead of assigning a volume with some capacity to an application and increasing that capacity as the needs of the application change, you can configure a volume with a large virtual capacity for the application. You can then increase or shrink the real capacity as the application needs change, without disrupting the application or server.

It is important to monitor physical capacity if you want to provide more space to your hosts than you have physically available in your IBM SAN Volume Controller. For more information about monitoring the physical capacity of your storage, and the difference between thin provisioning and over-allocation, see 9.4, “Creating alerts for IBM Spectrum Control and IBM Storage Insights” on page 391.



## Thin provisioning on top of Flash Core Modules

If you use the compression functions that are provided by the IBM Flash Core Modules (FCM) in your FlashSystem as a mechanism to add data reduction to a standard pool while maintaining the maximum performance, take care to understand the capacity reporting, in particular if you want to thin provision on top of the FCMs.

The FCM RAID array reports its theoretical maximum capacity, which can be as large as 4:1. This capacity is the maximum that can be stored on the FCM array. However, it might not reflect the compression savings that you achieve with your data.

It is recommended that you start conservatively, especially if you are allocating this capacity to IBM SAN Volume Controller or another IBM FlashSystem (the virtualizer).

You must first understand your expected compression ratio. In an initial deployment, allocate approximately 50% fewer savings. You can easily add “volumes” to the back-end storage system to present as new external “MDisk” capacity to the virtualizer later if your compression ratio is met or bettered.

For example, you have 100 TiB of physical usable capacity in an FCM RAID array before compression. Your compression results show savings of approximately 2:1, which suggests that you can write 200 TiB of volume data to this RAID array.

Start at 150 TiB of volumes that are mapped to as external MDisks to the virtualizer. Monitor the real compression rates and usage and over time add in the other 50 TiB of MDisk capacity to the same virtualizer pool. Be sure to leave spare space for unexpected growth, and consider the guidelines that are outlined in Chapter 3, “Storage back-end” on page 67.

If you often over-provision your hosts at much higher rates, you *can* use a standard pool and create thin provisioned volumes in that pool. However, be careful that you do not run out of space. You now need to monitor the backend controller pool usage *and* the virtualizer pool usage in terms of volume thin provisioning over-allocation. In essence, you are double accounting with the thin provisioning; that is, expecting 2:1 on the FCM compression, and then whatever level you over-provision at the volumes.

If you know that your hosts rarely grow to use the provisioned capacity, this process can be safely done; however, the risk comes from run-away applications (writing large amounts of capacity) or a trigger happy administrator suddenly enabling application encryption and writing to fill the entire capacity of the thin-provisioned volume.

### 4.1.2 Data reduction pools

IBM SAN Volume Controller uses innovative DRPs that incorporate deduplication and hardware-accelerated compression technology, plus SCSI **unmap** support. It also uses all of the thin provisioning and data efficiency features that you expect from IBM Spectrum Virtualize-based storage to potentially reduce your CAPEX and OPEX. Also, all of these benefits extend to over 500 heterogeneous storage arrays from multiple vendors.

DRPs were designed with space reclamation being a fundamental consideration. DRPs provide the following benefits:

- ▶ Log Structured Array allocation (redirect on all overwrites)
- ▶ Garbage collection to free whole extents
- ▶ Fine-grained (8 KB) chunk allocation/deallocation within an extent.
- ▶ SCSI **unmap** and write same (Host) with automatic space reclamation
- ▶ Support for “back-end” unmap and write same
- ▶ Support compression

- ▶ Support deduplication
- ▶ Support for traditional fully allocated volumes

Data reduction can increase storage efficiency and reduce storage costs, especially for flash storage. Data reduction reduces the amount of data that is stored on external storage systems and internal drives by compressing and deduplicating capacity and providing the ability to reclaim capacity that is no longer used.

The potential capacity savings that compression alone can provide are shown directly in the GUI interfaces by way of the included “comprestimator” functions. Since version 8.4 of the Spectrum Virtualize software, comprestimator is always on and you can see the overall expected savings in the dashboard summary view. The specific savings per volume in the volumes views also are available.

To estimate potential total capacity savings that data reduction technologies (compression and deduplication) can provide on the system, use the Data Reduction Estimation Tool (DRET). This tool is a command line, host-based utility that analyzes user workloads that are to be migrated to a new system. The tool scans target workloads on all attached storage arrays, consolidates these results, and generates an estimate of potential data reduction savings for the entire system.

You download DRET and its readme file to a Windows client and follow the installation instructions in the readme file. This file also describes how to use DRET with various host servers.

See [FixCentral](#) and search under IBM SAN Volume Controller to find the tool and its readme file.

To use data reduction technologies on the system, you must create a DRP, and create compressed or compressed and deduplicated volumes.

For more information, see 4.1.4, “Data reduction estimation tools” on page 105.

### **Quotaless data reduction child pool**

From version 8.4, DRP added support for a special type of child pool, which is known as a *quotaless child pool*.

The concepts and high-level description of parent-child pools are the same as for standard pools, with the following major exceptions:

- ▶ You cannot define a capacity or quota for a DRP child pool.
- ▶ A DRP child pool shares the encryption key of its parent.
- ▶ Capacity warning levels cannot be set on a DRP child pool. Instead, you must rely on the warning levels of the DRP parent pool.
- ▶ A DRP child pool uses space from the DRP parent pool as volumes are written to that belong to the child pool.
- ▶ Child and parent pools share the same data volume; therefore, data is deduplicated between parent and child volumes.
- ▶ A DRP child pool can use 100% of the capacity of the parent pool.
- ▶ The **migratevdisk** commands can now be used between parent and child pools. Because they share the encryption key, this operation becomes a “nocopy” operation.

To create a DRP child pool, use the new pool type of `child_quotaless`.

Because a DRP share capacity between volumes (when deduplication is used), it is virtually impossible to attribute capacity ownership of a specific grain to a specific volume because it might be used by two more volumes, which is the value proposition of deduplication. This process results in the differences between standard and DRP child pools.

Object-based access control (OBAC) or multi-tenancy can now be applied to DRP child pools or volumes because OBAC requires a child pool to function.

VMware vVols support for DRP is not yet supported or certified at the time of this writing; however, it is now technically possible because vVols support requires child pools.

## SCSI unmap

DRPs support end-to-end unmap functions. Space that is freed from the hosts by means of a SCSI **unmap** command results in the used space of the volume and pool reducing.

For example, a user deletes a small file on a host, which the operating system turns into a SCSI **unmap** for the blocks that made up the file. Similarly, a large amount of capacity can be freed if the user deletes (or Vmotions) a volume that is part of a data store on a host. This process might result in many contiguous blocks being freed. Each of these contiguous blocks results in a SCSI **unmap** command being sent to the storage device.

In a DRP, when the IBM SAN Volume Controller receives a SCSI **unmap** command, the result is that the capacity is freed that is allocated within that contiguous chunk. The deletion is asynchronous, and the unmapped capacity is first added to the “reclaimable” capacity, which is later physically freed by the garbage collection code. For more information, see 4.1.5, “Understanding capacity use in a DRP” on page 110.

Similarly, deleting a volume at the DRP level frees all of the capacity back to the pool. The DRP also marks those blocks as “reclaimable” capacity, which the garbage collector later frees back to unused space. After the garbage collection frees an entire extent, a new SCSI **unmap** command is issued to the backend MDisk device.

Unmap can help ensure good MDisk performance; for example, Flash drives can reuse the space for wear leveling and to maintain a healthy capacity of “pre-erased” (ready to be used) blocks.

Virtualization devices, such as IBM SAN Volume Controller with external storage, can also pass on unmap information to other storage systems; for example, when extents are deleted or migrated.

## Enabling, monitoring, throttling, and disabling SCSI unmap

By default, host-based unmap support is disabled on all products other than the FlashSystem 9000 series. Backend unmap is enabled by default on all products.

To enable or disable host-based unmap, run the following command:

```
chsystem -hostunmap on|off
```

To enable or disable backend unmap run the following command:

```
chsystem -backendunmap on|off
```

You can check how much SCSI unmap processing is occurring on a per-volume or per-pool basis by using the performance statistics. This information can be viewed with Spectrum Control or Storage Insights.

**Note:** SCSI **unmap** might add workload to the backend storage.

Performance monitoring helps to notice possible effects and if SCSI **unmap** workload is affecting performance, consider taking the necessary steps and consider the data rates that are observed. It might be expected to see GiBps of unmap if you just deleted many volumes.

You can throttle the amount of “offload” operations (such as SCSI **unmap**) by way of the per node settings for offload throttle; for example:

```
mkthrottle -type offload -bandwidth 500
```

This setting limits each node to 500MiBps of offload commands.

You can also stop the IBM SAN Volume Controller from processing SCSI **unmap** operations for one or more host systems. You might find an over-zealous host, or not have the ability to configure the settings on some of your hosts. To modify a host to disable unmap, change the host type:

```
chhost -type generic_no_unmap <host_id_or_name>
```

If you experience severe performance problems as a result of SCSI **unmap**, you can disable SCSI **unmap** on the entire IBM SAN Volume Controller for the frontend (host), backend, or both.

## Fully allocated volumes in a DRP

It is possible to create fully allocated volumes in a DRP.

A fully allocated volume uses the entire capacity of the volume. That is, when created, that space is reserved (used) from the DRP and is not available for other volumes in the DRP.

Data is not be deduplicated or compressed in a fully allocated volume. Similarly, because it does not use the internal fine-grained allocation functions, the allocation and performance are the same or better than a fully allocated volume in a standard pool.

## Compressed and deduplicated volumes in a DRP

It is possible to create compressed only volumes in a DRP.

A compressed volume is by its nature thin-provisioned. A compressed volume uses only its compressed data size in the pool. The volume grows only as you write data to it.

It is possible (but *not* recommended) to create a deduplicated only volume in a DRP. A deduplicated volume is thin provisioned in nature. The extra processing that is required to also compress the deduplicated block is minimal; therefore, it is recommended to create a compressed and deduplicated volume rather than only a deduplicated volume.

The DRP first looks for deduplication matches; then, it compresses the data before writing to the storage.

## Thin provisioned only volumes in a DRP

It is *not* recommended to create a thin provisioned only volume in a DRP.

Thin provisioned volumes use the fine grained allocation functions of DRP. The main benefit of DRP is in the data reduction functions (compressed and deduplication). Therefore, if you want to create a thin provisioned volume in a DRP, create a compressed volume.

**Note:** In some cases, when the back-end storage is thin-provisioned or data-reduced, the GUI might not even the option to create only a thin provisioned volume in a DRP. This issue occurs because it is highly recommended that this option is not used because it can cause extreme capacity monitoring headaches with a high probability of running out of space.

## DRP internal details

DRPs consist of various internal metadata volumes and it is important to understand how these metadata volumes are used and mapped to user volumes. Every user volume features a corresponding journal, forward lookup, and directory volume.

The internal layout of a DRP is different than a standard pool. A standard pool creates volume objects within the pool. Some fine grained internal metadata is stored within a thin provisioned or Real-time Compressed volume in a standard pool. Overall, the pool contains volume objects.

A DRP reports volumes to the user in the same way as a standard pool; however, it defines a directory volume internally for each user volume that is created within the pool. The directory points to grains of data that are stored in the Customer Data Volume. All volumes in a single DRP use the same Customer Data Volume to store their data. Therefore, deduplication is possible across volumes in a single DRP.

Other internal volumes are created, one per DRP. One Journal Volume is created per I/O group that can be used for recovery purposes, and to replay metadata updates if needed. One Reverse Lookup Volume exists per I/O group that is used by garbage collection.

Figure 4-1 shows the difference between DRP volumes and volumes in standard pools.

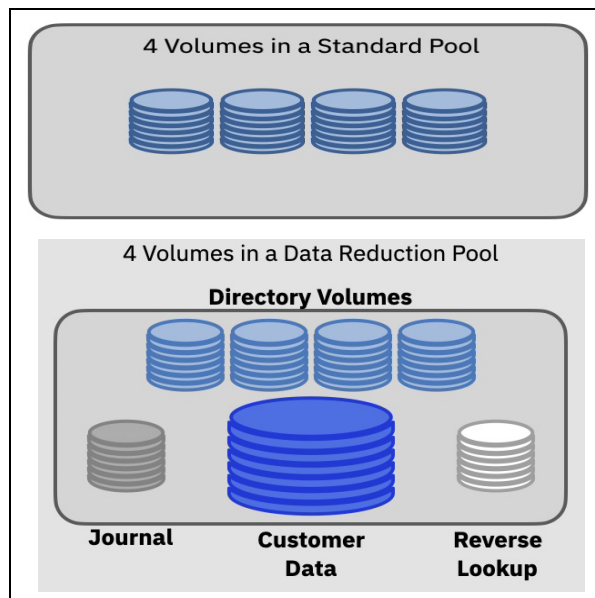


Figure 4-1 Standard and DRP volumes

The Customer Data Volume uses greater than 97% of pool capacity. The I/O pattern is a large sequential write pattern (256 KB) that is coalesced into full stride writes, and you typically see a short random read pattern.

Directory Volumes occupy approximately 1% of pool capacity. They typically have a short 4 KB random read/write I/O. The Journal Volume occupies approximately 1% of pool capacity, and shows large sequential write I/O (256 KB typically).

Journal Volumes are read for recovery scenarios only (for example, T3 recovery). Reverse Lookup Volumes are used by the Garbage Collection process and occupy less than 1% of pool capacity. Reverse Lookup Volumes use a short, semi-random read/write pattern.

The process of reclaiming space is called *garbage collection* (see Figure 4-2). As a result of compression and deduplication, overwriting host writes does not always use the same amount of space that the previous data used. This issue leads to these writes always occupying new space on back-end storage while the old data is still in its original location. The primary task of garbage collection is to track all of the regions that were invalidated, and to make this capacity usable for new writes.

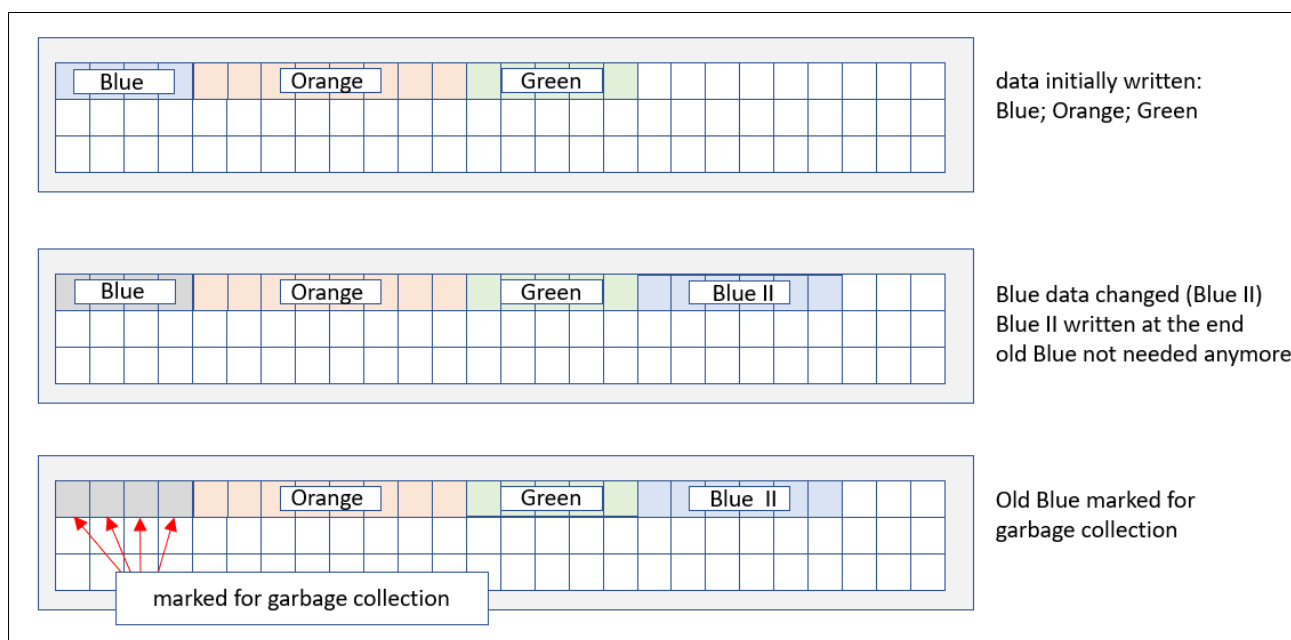


Figure 4-2 Garbage Collection principle

For garbage collection, stored data is divided into regions. As data is overwritten, a record is kept of which areas of those regions were invalidated. Regions that feature many invalidated parts are potential candidates for garbage collection. When most of a region includes invalidated data, it is fairly inexpensive to move the remaining data to another location, which frees the entire region.

DRPs include built-in services to enable garbage collection of unused blocks. Therefore, many smaller unmaps end up enabling a much larger chunk (extent) to be freed back to the pool. Trying to fill small holes is inefficient because too many I/Os are needed to keep reading and rewriting the directory. Therefore, garbage collection waits until an extent has many small holes and moves the remaining data in the extent, compact, and rewrite. When an empty extent is available, it can be freed back to the virtualization layer (and backend with UNMAP) or starts writing into the extent with new data (or rewrites).

The reverse lookup metadata volume tracks the extent usage, or more importantly the holes that are created by overwrites or unmaps. Garbage collection looks for extents with the most unused space. After an entire extent has all valid data moved elsewhere, it can be freed back to the set of unused extents in that pool, or it can be reused for new written data.

Because garbage collection must move data to free regions, it is suggested that you size pools to keep a specific amount of free capacity available. This practice ensures that some free space always is available for garbage collection. For more information, see 4.1.5, “Understanding capacity use in a DRP” on page 110.

### 4.1.3 Standard pools versus DRPs

When it comes to designing pools during the planning of an IBM SAN Volume Controller project, it is important to know all requirements, and to understand the upcoming workload of the environment. Because the IBM SAN Volume Controller is very in creating and using pools, this section describes how to determine which types of pool or setup you can use.

Among the things that you should be aware of is the planned environment, you must know the following information:

- ▶ Is your data compressible?
- ▶ Is your data deduplicable?
- ▶ What are the workload and performance requirements:
  - Read/write ratio
  - Blocksize
  - IOPS, MBps, and response time
- ▶ Flexibility for the future
- ▶ Thin provisioning

#### Determining whether your data is compressible

Compression is one option of DRPs, and the deduplication algorithm is used to reduce the on-disk footprint of data that is written to by thin provisioning. In IBM SAN Volume Controller, this compression is an inline compression or a deduplication approach rather than attempting to compress as a background task. DRP provides unmap support at the pool and volume level, and out-of-space situations can be managed at the DRP pool level.

Compression can be enabled in DRPs on a per-volume basis, and thin provisioning is a prerequisite. The input size changed to a fixed 8 KB. Compression is suited to Flash workloads (IOPS) and a typical 2:1 compression ratio will result in ~4 KB operations and Streaming 256 KB chunks of compressed 8 KB blocks for consistent write performance, by allowing the cache to build full stride writes enabling the most efficient RAID throughput.

Data compression techniques depend on the type of data that must be compressed and on the wanted performance. Effective compression savings generally rely on the accuracy of your planning and the understanding if the specific data is compressible or not. Several methods are available to decide whether your data is compressible, including the following examples:

- ▶ General assumptions
- ▶ Tools

#### ***General assumptions***

IBM SAN Volume Controller compression is lossless. As the name implies, it involves no loss of information. It can be losslessly compressed and the original data can be recovered after the compress or expend cycle. Good compression savings can be achieved in the following environments (and others):

- ▶ Virtualized Infrastructure
- ▶ Database and Data Warehouse

- ▶ Home Directory, Shares, and shared project data
- ▶ CAD/CAM
- ▶ Oil and Gas data
- ▶ Log data
- ▶ Software development
- ▶ Text and some picture files

However, if the data is compressed in some cases, the savings are less, or even negative. Pictures (for example, GIF, JPG, and PNG), audio (MP3 and WMA) and video or audio (AVI and MPG) and even compressed databases data might not be good candidates for compression.

Table 4-1 lists the compression ratio of common data types and applications that provide high compression ratios.

*Table 4-1 Compression ratios of common data types*

| Data types/applications          | Compression ratio |
|----------------------------------|-------------------|
| Databases                        | Up to 80%         |
| Server or Desktop Virtualization | Up to 75%         |
| Engineering Data                 | Up to 70%         |
| Email                            | Up to 80%         |

Also, do not compress encrypted data (for example, compression on host or application). Compressing encrypted data does not show much savings because it contains pseudo random data. The compression algorithm relies on patterns to gain efficient size reduction. Because encryption destroys such patterns, the compression algorithm cannot provide much data reduction.

For more information about compression, see 4.3.1, “Data reduction pools with IBM FlashSystem NVMe attached drives” on page 119.

**Note:** Saving assumptions are based on the type of data are imprecise. We advise that you determine compression savings by using suitable tools.

## Determining whether your data is a deduplication candidate

Deduplication is done by using hash tables to identify previously written copies of data. If duplicate data is found, the algorithm references to the previously found data instead of writing the data to disk.

Consider the following points:

- ▶ Deduplication uses 8 KiB deduplication grains and an SHA-1 hashing algorithm.
- ▶ DRPs build 256 KiB chunks of data that consist of multiple deduplicated and compressed 8 KiB grains.
- ▶ DRPs write contiguous 256 KiB chunks, which allows for efficient write streaming with the capability for cache and RAID to operate on full stride writes.
- ▶ DRPs provide deduplication then, compress capability.
- ▶ The scope of deduplication is within a DRP within an I/O Group.



### ***General assumptions***

Some environments have data with high deduplication savings; therefore, they are candidates for deduplication.

Good deduplication savings can be achieved in several environments, such as virtual desktop and some virtual machine environments. Therefore, they might be good candidates for deduplication.

IBM provides the Data Reduction Estimate Tool (DRET) to help determine the deduplication capacity saving benefits you see.

## **4.1.4 Data reduction estimation tools**

IBM provides the following tools to estimate the savings you see by using data reduction technologies:

- ▶ **Comprestimator**

This tool is built into the IBM SAN Volume Controller. It reports the expected compression savings on a per-volume basis in the GUI and command line.

- ▶ **Data Reduction Estimation Tool**

The DRET tool must be installed on and used to scan the volumes that are mapped to a host and is primarily used to assess the deduplication savings. It is the most accurate way to determine the estimated savings; however, it must scan all of your volumes to provide an accurate summary.

### **Comprestimator**

Comprestimator is provided in the following ways:

- ▶ As a stand-alone, host-based command-line utility. It can be used to estimate the expected compression for block volumes where you do not have an IBM Spectrum Virtualize product providing those volumes.
- ▶ Integrated into the IBM SAN Volume Controller. In software versions before 8.4, triggering volumes (or all volumes) sampling was done manually.
- ▶ Integrated into the IBM SAN Volume Controller and always on, in versions 8.4 and later.

### ***Host-based Comprestimator***

The tool is available at this [IBM Support web page](#).

IBM SAN Volume Controller Comprestimator is a command-line and host-based utility that can be used to estimate an expected compression rate for block devices.

### ***Integrated Comprestimator: Software levels before 8.4.0***

IBM SAN Volume Controller also features an integrated Comprestimator tool that is available through the management GUI and command-line interface. If you are considering applying compression on non-compressed volumes in an IBM SAN Volume Controller, you can use this tool to evaluate if compression will generate capacity savings.

To access the Comprestimator tool in management GUI, select **Volumes** → **Volumes**. If you want to analyze all of the volumes in the system, click **Actions** → **Capacity Savings** → **Estimate Compression Savings**.

If you want to select a list of volumes, click **Actions** → **Capacity Savings** → **Analyze** to evaluate only the capacity savings of the selected volumes, as shown in Figure 4-3.

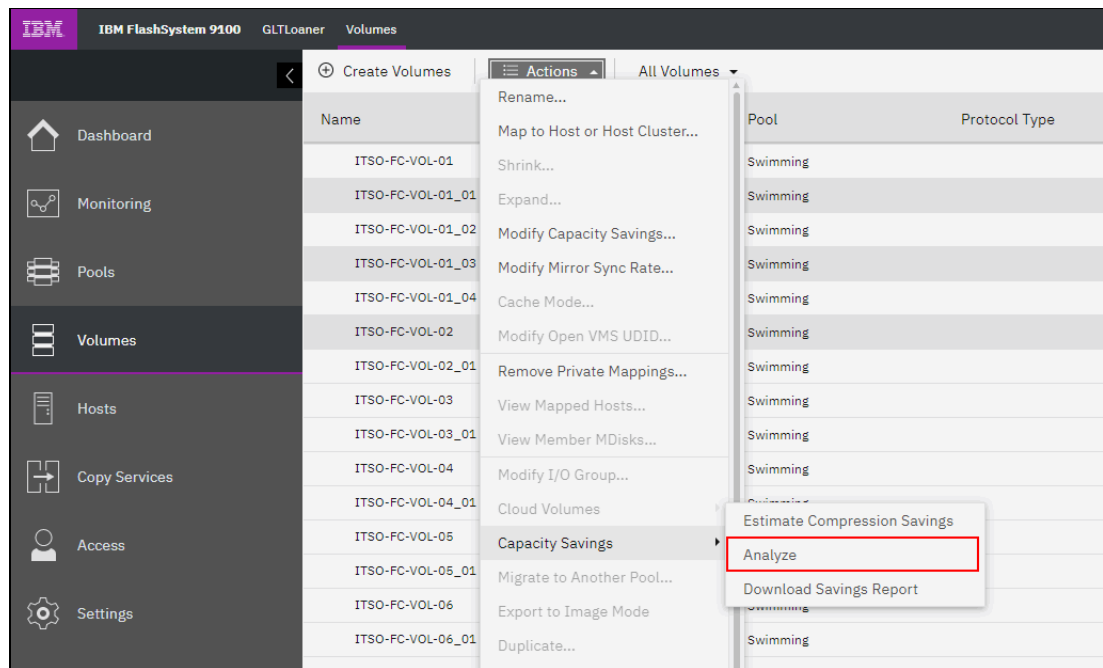


Figure 4-3 Capacity savings analysis

To display the results of the capacity savings analysis, click **Actions** → **Capacity Savings** → **Download Savings Report**, as shown in Figure 4-3, or run the `lsvdiskanalysis` command in the command-line, as shown in Example 4-1.

*Example 4-1 Results of capacity savings analysis*

```
IBM_FlashSystem:superuser>lsvdiskanalysis TESTVOL01
id 64
name TESTVOL01
state estimated
started_time 201127094952
analysis_time 201127094952
capacity 600.00GB
thin_size 47.20GB
thin_savings 552.80GB
thin_savings_ratio 92.13
compressed_size 21.96GB
compression_savings 25.24GB
compression_savings_ratio 53.47
total_savings 578.04GB
total_savings_ratio 96.33
margin_of_error 4.97
IBM_FlashSystem:superuser>
```

The following actions are preferred practices:

- ▶ After you run Comprestimator, consider applying compression on only those volumes that show greater than or equal to 25% capacity savings. For volumes that show less than 25% savings, the trade-off between space saving and hardware resource consumption to compress your data might not make sense. With DRPs, the penalty for the data that cannot be compressed is no longer seen; however, the DRP includes overhead in grain management.
- ▶ After you compress your selected volumes, review which volumes have the most space saving benefits from thin provisioning rather than compression. Consider moving these volumes to thin provisioning only. This configuration requires some effort, but saves hardware resources that are then available to give better performance to those volumes, which achieves more benefit from compression than thin provisioning.

As shown in Figure 4-4, customize the Volume view to get all the metrics you might need to help make your decision.

The screenshot shows the 'All Volumes' view in the IBM Spectrum Storage Manager interface. A table lists various volumes with columns for Name, State, Pool, and Host. A custom metrics overlay is displayed, allowing selection of metrics to show in the Host column. The selected metrics include Protocol Type, UID, Host Mappings, Capacity, Real Capacity, Used Capacity, Compression Savings, Capacity Savings, Estimated Compression Savings, Estimated Compression Savings %, Estimated Thin Savings, and Estimated Thin Savings %.

| Name                  | State  | Pool  | Host              |
|-----------------------|--------|-------|-------------------|
| VOL04                 | Online | Swim  | 78000000000000... |
| VOL03                 | Online | Pool1 | 78000000000000... |
| VOL02                 | Online | Pool1 | 78000000000000... |
| VOL01                 | Online | Swim  | 78000000000000... |
| TestVol_7             | Online | Swim  | 78000000000000... |
| TestVol_6             | Online | Swim  | 78000000000000... |
| TestVol_5             | Online | Swim  | 78000000000000... |
| TestVol_4             | Online | Swim  | 78000000000000... |
| TestVol_3             | Online | Swim  | 78000000000000... |
| TestVol_2             | Online | Swim  | 78000000000000... |
| TestVol_1             | Online | Swim  | 78000000000000... |
| TestVol_0             | Online | Swim  | 78000000000000... |
| ITSO-FC-VOL-10        | Online | Swim  | 78000000000000... |
| ITSO-FC-VOL-08-Target | Online | Swim  | 78000000000000... |
| ITSO-FC-VOL-08-Source | Online | Swim  | 78000000000000... |
| ITSO-FC-VOL-07_01     | Online | Swim  | 78000000000000... |

Figure 4-4 Customized view

### Integrated comprestimator: Software version 8.4 onwards

Because the newer code levels include an always-on comprestimator, you can view the expected capacity savings in the main dashboard view, pool views, and volume views. You do not need to first trigger the “estimate” or “analyze” tasks; these tasks are performed automatically as background tasks.

### Data Reduction Estimation Tool

IBM provides the DRET to support deduplication and compression. The host-based CLI tool scans target workloads on various older storage arrays (from IBM or another company), merges all scan results, and then, provides an integrated system-level data reduction estimate for your IBM SAN Volume Controller planning.

The DRET uses advanced mathematical and statistical algorithms to perform an analysis with a low memory “footprint”. The utility runs on a host that can access the devices to be analyzed. It performs only read operations, so it has no effect on the data that is stored on the device. Depending on the environment configuration, in many cases the DRET is used on more than one host to analyze more data types.

It is important to understand block device behavior when analyzing traditional (fully-allocated) volumes. Traditional volumes that were created without initially zeroing the device might contain traces of old data on the block device level. Such data is not accessible or viewable on the file system level. When the DRET is used to analyze such volumes, the expected reduction results reflect the savings rate to be achieved for all the data on the block device level, including traces of old data.

Regardless of the block device type being scanned, it is also important to understand a few principles of common file system space management. When files are deleted from a file system, the space they occupied before the deletion becomes free and available to the file system. This space freeing occurs even though the data on the disk was not removed but rather the file system index and pointers were updated to reflect this change.

When the DRET is used to analyze a block device that is used by a file system, all underlying data in the device is analyzed, regardless of whether this data belongs to files that were deleted from the file system. For example, you can fill a 100 GB file system and make it 100% used, then delete all the files in the file system, which makes it 0% used. When scanning the block device that is used for storing the file system in this example, the DRET (or any other utility) accesses the data that belongs to the files that are deleted.

To reduce the effect of the block device and file system behavior, it is recommended to use the DRET to analyze volumes that contain as much active data as possible rather than volumes that are mostly empty of data. This use increases the accuracy level and reduces the risk of analyzing old data that is deleted, but might still have traces on the device.

The DRET can be downloaded from this [IBM Fix Central web page](#).

DRET Command line is shown in Example 4-2.

*Example 4-2 DRET command line*

---

```
Data-Reduction-Estimator -d <device> [-x Max MBps] [-o result data filename] [-s
Update interval] [--command scan|merge|load|partialscan] [--mergefiles Files
to merge] [--loglevel Log Level] [--batchfile batch file to process] [-h]
```

---

The DRET can be used on the following client operating systems:

- ▶ Windows 2008 Server and Windows 2012
- ▶ Red Hat Enterprise Linux Version 5.x, 6.x, and 7.x (64-bit)
- ▶ UBUNTU 12.04
- ▶ ESX 5.0, 5.5, 6.0
- ▶ AIX 6.1, 7.1
- ▶ Solaris 10

**Note:** According to the results of the DRET tool, use DRPs to use data deduplication savings that are available, unless performance requirements exceed what DRP can deliver.

Do not enable deduplication if the data set is not expected to provide deduplication savings.

## Determining the workload and performance requirements

An important factor in sizing and planning for an IBM SAN Volume Controller environment is the knowledge of the workload characteristics of that specific environment.

Sizing and performance is affected by the following workloads, among others:

- Read/Write ratio

Read/Write (%) ratio affects performance because higher writes cause more IOPS to the DRP. To effectively size an environment, the Read/Write ratio should be considered. During a write I/O, when data is written to the DRP, it is stored on the data disk, the forward lookup structure is updated, and the I/O is completed.

DRPs use metadata. Even when no volumes are in the pool, some of the space in the pool is used to store the metadata. The space that is allocated to metadata is relatively small. Regardless of the type of volumes that the pool contains, metadata is always stored separately from customer data.

In DRPs, the maintenance of the metadata results in I/O amplification. I/O amplification occurs when a single host-generated read or write I/O results in more than one back-end storage I/O request because of advanced functions. A read request from the host results in two I/O requests, a directory lookup and a data read. A write request from the host results in three I/O requests, a directory lookup, a directory update, and a data write. Therefore, DRPs create *more* IOPS on the FCMs or drives.

- Block size

The concept of a block size is simple and the effect on storage performance might be distinct. Block size effects might also affect overall performance; therefore, consider larger blocks to affect performance more than smaller blocks. Understanding and considering block sizes in the design, optimization, and operation of the storage system sizing leads to a more predictable behavior of the entire environment.

**Note:** Where possible, limit the maximum transfer size that is sent to the IBM SAN Volume Controller to no more than 256 KiB. This limitation is a general best practice and not specific to DRP only.

- IOPS, MBps, and response time

Storage constraints are IOPS, throughput, and latency, and it is crucial to correctly design the solution or plan for a setup for speed and bandwidth. Suitable sizing requires knowledge about the expected requirements.

- Capacity

During the planning of an IBM SAN Volume Controller environment, capacity (physical) must be sized accordingly. Compression and deduplication might save space, but metadata uses little space. For optimal performance, our recommendation is to use the DRP to a maximum of 85%.

Consider monitoring storage infrastructure requirements with monitoring or management software, such as IBM Spectrum Control or IBM Storage Insights, before planning a new environment. At busy times, the peak workload, such as IOPS, MBps, and peak response time, gives you an understanding of the required workload plus expected growth. Also, consider allowing enough room regarding the performance that is required during planned and unplanned events (upgrades and possible defects or failures).

It is important to understand the relevance of application response time rather than internal response time with required IOPS or throughput. Typical OLTP applications require IOPS and low latency as well.

Do not place capacity over performance while designing or planning a storage solution. Even if capacity might be sufficient, the environment can suffer from low performance. Deduplication and compression might satisfy capacity needs, but aim for performance and for robust application performance.

To size an IBM SAN Volume Controller environment, your IBM account team or IBM Business Partner must access IBM Storage Modeller. The tool can be used to determine that DRPs are going to give suitable bandwidth and latency. If the data does not deduplicate (according to the DRET) the volume can also be fully allocated or compressed only.

### Flexibility for the future

During the planning and configuration of storage pools, the decision must be made which pools to create. Because the IBM SAN Volume Controller enables you to create standard pools or DRPs, you must decide which type best fits the requirements.

Verify if performance requirements meet the capabilities of the specific pool type (see “Determining the workload and performance requirements” on page 108. We cover the dependencies with child pools regarding vVols in 4.3.3, “Data reduction pools and VMware vVols” on page 122, and “DRP restrictions” on page 123.

If other important factors do not lead you to choose standard pools, DRPs are the right choice. Use of DRPs can increase storage efficiency and reduce costs because they reduce the amount of data that is stored on hardware and reclaim previously used storage resources that are no longer needed by host systems.

Also, DRPs provide great flexibility for future use because they add the ability of compression and deduplication of data at the volume level in a specific pool, even if these features are initially not used at creation time.

Remember that it is not possible to convert a pool. If you must change the pool type (standard pool to DRP, or vice versa) it is an offline process and you must migrate your data, as described in 4.3.7, “Data migration with DRP” on page 125.

**Note:** We recommend the use of DRPs with fully allocated volumes if the restrictions and capacity do not affect your environment. For more information about the restrictions, see “DRP restrictions” on page 123.

## 4.1.5 Understanding capacity use in a DRP

In this section, we preset some new capacity terms regarding a DRP.

After a reasonable period, the overall free space that is reported by a DRP likely are 15 - 20%. The garbage collection algorithm must balance the need to free space with the overhead of performing garbage collection. Therefore, the incoming write/overwrite rates and any unmap operations dictate how much “reclaimable space” is present at any time.

The capacity in a DRP consists of the components that are listed in Table 4-2.

Table 4-2 *DRP capacity uses*

| Use                   | Description                                                               |
|-----------------------|---------------------------------------------------------------------------|
| Reduced customer data | The data that is written to the DRP, in compressed and deduplicated form. |

| Use                  | Description                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fully allocated data | The amount of capacity that is allocated to fully allocated volumes (assumed to be 100% written).                                                                              |
| Free                 | The amount of free space, not in use by any volume.                                                                                                                            |
| Reclaimable data     | The amount of garbage in the pool. This data is old (overwritten) yet to be freed data or data that is unmapped but not yet freed or associated with recently deleted volumes. |
| Metadata             | Approximately 1 - 3% overhead for DRP metadata volumes.                                                                                                                        |

Balancing how much garbage collection is done versus how much free space is available dictates how much reclaimable space is present at any time. The system dynamically adjusts the target rate of garbage collection to maintain a suitable amount of free space.

An example steady state DRP is shown in Figure 4-5.

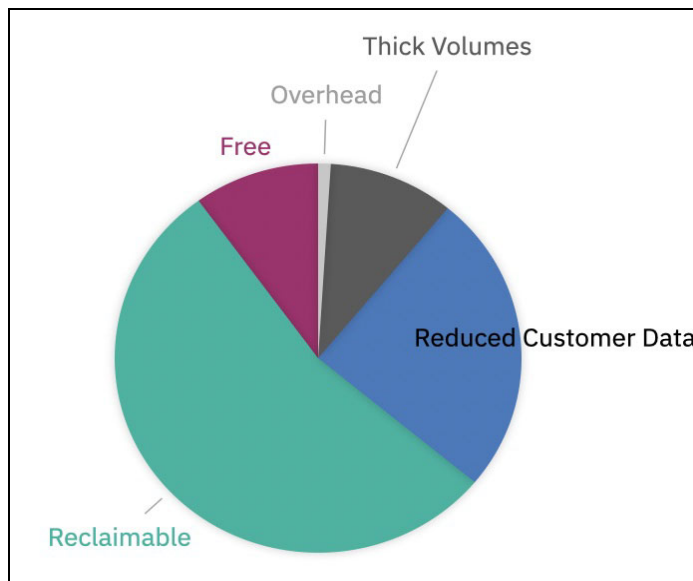


Figure 4-5 DRP capacity use example

Consider the following points:

- ▶ If you create a large capacity of fully allocated volumes in a DRP, you are taking this capacity directly from free space only. This configuration can result in triggering heavy garbage collection if little free space and a large amount of reclaimable space remains.
- ▶ If you do create a large number of fully allocated volumes and experience degraded performance because of garbage collection, reduce the work that is required by temporarily deleting any as yet unused fully allocated volumes.
- ▶ When deleting a fully allocated volume, the capacity is returned directly to free space.
- ▶ When deleting a thin provisioned volume (compressed or deduplicated), the following two-phase approach can be used:
  - a. The grains must be inspected to determine if this volume was the last volume that was referencing this grain (deduplicated) and can be freed. If not, the grain references must be updated and the grain might need to be re-homed to belong to one of the remaining volumes that still requires this grain.

- b. When all grains that are to be deleted are identified, these grains are returned to the “reclaimable” capacity. It is the responsibility of garbage collection to convert them into free space.

The garbage collection process runs in the background, and attempts to maintain a sensible amount of free space. If little free space is available and you delete many volumes, the garbage collection code might trigger many backend data movements and result in performance issues.

- The act of deleting a volume does not immediately return any free space.
- If you are at risk of running out of space, but much reclaimable space exists, you can force garbage collection to work harder by creating a temporary fully allocated volume to reduce the amount of real free space and trigger more garbage collection.

**Important:** Use extreme caution when using up all or most of the free space with fully allocated volumes. Garbage collection requires free space to coalesce data blocks into entire extents and hence free capacity. If little free space is available, the garbage collector must work even harder to free space.

- It can be worth creating some “get out of jail free” fully allocated volumes in a DRP. These types of volumes reserve some space that you can quickly return to the free space resources if you reach a point where you are close to running out of space, or when garbage collection is struggling to free capacity in an efficient manner.

Consider the following points:

- Such volumes are never be mapped to hosts.
- Such volumes are labeled accordingly, as shown in the following example:

RESERVED\_CAPACITY\_DO\_NOT\_USE

## 4.2 Storage pool planning considerations

The implementation of storage pools in an IBM SAN Volume Controller requires a holistic approach that involves application availability and performance considerations. Usually, a trade-off between these two aspects must be considered.

In this section, the main best practices in the storage pool planning activity are described. Most of these practices apply to standard and DRP pools, except where otherwise specified. For more information about best practices for DRPs, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.

### 4.2.1 Planning for availability

By design, IBM Spectrum Virtualize-based storage systems take the entire storage pool offline if a single MDisk in that storage pool goes offline, which means that storage pool’s quantity and size define the failure domain. Reducing the hardware failure domain for back-end storage is only part of your considerations. When you are determining the storage pool layout, you must also consider application boundaries and dependencies to identify any availability benefits that one configuration might have over another.



Sometimes, reducing the hardware failure domain, such as placing the volumes of an application into a single storage pool, is not always an advantage from the application perspective. Alternatively, splitting the volumes of an application across multiple storage pools increases the chances of having an application outage if one of the storage pools that is associated with that application goes offline.

Finally, increasing the number of pools to reduce the failure domain is not always a viable option. For example, in FlashSystems with configurations without expansion enclosures, the number of physical drives is limited (up to 24). Creating arrays reduces the usable space because of spare and protection capacity.

For example, consider a single I/O group FlashSystem configuration with 24 7.68 TB NVMe drives. In a case of a single array DRAID6 creation, the available physical capacity is 146.3 TB, while creating two arrays DRAID6 provides 137.2 TB of available physical capacity with a reduction of 9.1 TB.

When virtualizing external storage, remember that the failure domain is defined by the external storage itself, rather than by the pool definition on the front-end system. For instance, if you provide 20 MDisks from external storage and all of these MDisks are using the same physical arrays, the failure domain becomes the total capacity of these MDisks, no matter how many pools you have distributed them across.

The following actions are the starting preferred practices when planning storage pools for availability:

- ▶ Create separate pools for internal storage and external storage, unless you are creating a hybrid pool managed by Easy Tier (see 4.2.5, “External pools” on page 118).
- ▶ Create a storage pool for each external virtualized storage subsystem, unless you are creating a hybrid pool managed by Easy Tier (see 4.2.5, “External pools” on page 118).
- ▶ Use dedicated pools for image mode volumes.

**Limitation:** Image Mode volumes are *not* supported with DRPs.

When you are selecting storage subsystems, the decision often comes down to the ability of the storage subsystem to be more reliable and resilient, and meet application requirements. While IBM Spectrum Virtualize does not provide any physical level-data redundancy for virtualized external storages, the availability characteristics of the storage subsystems’ controllers have the most impact on the overall availability of the data that is virtualized by IBM Spectrum Virtualize.

## 4.2.2 Planning for performance

When planning storage pools for performance the capability to stripe across disk arrays is one of the most important advantages IBM Spectrum Virtualize provides. To implement performance-oriented pools, create large pools with many arrays rather than more pools with few arrays. This approach usually works better for performance than spreading the application workload across many smaller pools, because typically the workload is not evenly distributed across the volumes, and then across the pools.

Also, adding arrays to a pool (rather than creating one) can be a way to improve the overall performance if the added arrays have the same or better performance characteristics than the existing arrays.

In IBM FlashSystem configurations, arrays that are built from FCM and SAS SSD drives feature different characteristics in terms of performance and data reduction capabilities. For this reason, consider the following recommendations when FCM and SAS SSD arrays are used in the same pool:

- ▶ Enable the Easy Tier function (see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148). The Easy Tier treats the two array technologies as different tier (`tier0_flash` for FCM arrays and `tier1_flash` for SAS-SSD arrays); therefore, the resulting pool is a multitiered pool with inter-tier balancing enabled.
- ▶ Strictly monitor the FCM physical usage. As Easy Tier moves the data between the tiers, the compression ratio can vary frequently and an out-of-space condition can be reached, even without changing the data contents.

The number of arrays that is required in terms of performance must be defined in the presales or solution design phase. When the environment is resized, remember that adding too many arrays to a single storage pool increases the failure domain; therefore, it is important to find the trade-off between the performance, availability, and scalability cost of the solution.

The use of the external virtualization capabilities can boost the performance of the back-end storage systems, first by way of the wide striping across multiple arrays, and second, by adding read/write cache capability. It is typically understood that wide striping can add approximately 10% IOPs performance to the backend system by using these mechanisms.

Another factor is the ability of your virtualized storage subsystems to be scaled up or scaled out. For example, IBM System Storage DS8000 series is a scale-up architecture that delivers the best performance per unit, and the IBM FlashSystem series can be scaled out with enough units to deliver the same performance.

With a virtualized system, debate exists as to whether to scale out back-end system, or add them as individual systems behind IBM SAN Volume Controller. Either case is valid; however, adding individual controllers is likely to allow IBM SAN Volume Controller to generate more I/O (based on queuing and port usage algorithms) and it is recommended to add each controller (I/O Group) of an IBM FlashSystem back-end as its own controller; that is, do not cluster the IBM FlashSystem when as an external storage controller behind another Spectrum Virtualize product, such as IBM SAN Volume Controller. Adding each controller (I/O Group) of an IBM FlashSystem backend as its own controller adds management IP addresses and configurations. However, it also provides the best scalability in terms of IBM SAN Volume Controller performance.

A significant consideration when you compare native performance characteristics between storage subsystem types is the amount of scaling that is required to meet the performance objectives. Although lower-performing subsystems typically can be scaled to meet performance objectives, the extra hardware that is required lowers the availability characteristics of the IBM SAN Volume Controller cluster.

All storage subsystems possess an inherent failure rate. Therefore, the failure rate of a storage pool becomes the failure rate of the storage subsystem times the number of units.

The following actions are the starting preferred practices when planning storage pools for performance:

- ▶ Create a dedicated storage pool with dedicated resources if there is a specific performance application request.
- ▶ When using external storage in an Easy Tier enabled pool, do not intermix MDisk in the same tier with different performance characteristics.

- ▶ In a FlashSystem clustered environment, create storage pools with IOgrp or Control Enclosure affinity. That means you have to use only arrays or MDisks supplied by the internal storage that is directly connected to one IOgrp SAS chain only. This configuration avoids unnecessary IOgrp-to-IOgrp communication traversing the SAN and consuming Fibre Channel bandwidth.
- ▶ Use dedicated pools for image mode volumes.

**Limitation:** Image Mode volumes are *not* supported with DRPs.

- ▶ For those Easy Tier enabled storage pools, always allow some free capacity for Easy Tier to deliver better performance.
- ▶ Consider implementing child pools when you must have a logical division of your volumes for each application set. Cases exist in which you want to subdivide a storage pool, but maintain a larger number of MDisks in that pool. Child pools are logically similar to storage pools, but allow you to specify one or more subdivided child pools. Thresholds and throttles can be set independently per child pool.

### Cache partitioning

The system automatically defines a logical cache partition per storage pool. Child pools do not count towards such partitioning. The cache partition number matches the storage pool ID.

A cache partition is a logical threshold that stops any single partition from using the entire cache resource. This partition is provided as a protection mechanism and has no real bearing on performance in normal operations. Only when a storage pool becomes overloaded does partitioning activate and essentially slows down write operations in that pool to the same speed that the backend can handle. By *overloaded*, we mean that the front-end write throughput is greater than back-end storage in that pool can sustain. This situation generally must be avoided.

In recent versions of IBM Spectrum Control, the cache partition fullness is reported and can be monitored. You should not see partitions reaching 100% full. If you do, this issue suggests that the corresponding storage pool is in an overload situation and workload must be moved from that pool, or storage capability must be added to that pool.

## 4.2.3 Planning for capacity

Capacity planning is never an easy task. Capacity monitoring is more complex with the advent of data reduction. It is important to understand the different terminology that is used to report usable, used and free capacity.

The terminology and its reporting in the GUI changed in recent versions and is listed in Table 4-3.

Table 4-3 Capacity terminology in 8.4.0

| Old term          | New term             | Definition                                                                                                                                                                     |
|-------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical capacity | Usable capacity      | The amount of capacity that is available for storing data on a system, pool, array, or MDisk after formatting and RAID techniques are applied.                                 |
| Volume capacity   | Provisioned capacity | The total capacity of all volumes in the system                                                                                                                                |
| N/A               | Written capacity     | The total capacity that is written to the volumes in the system. This capacity is shown as a percentage of the provisioned capacity and is reported before any data reduction. |

The *usable capacity* describes the amount of capacity that can be written to on the system, which includes any back-end data reduction (that is, the “virtual” capacity that is reported to the system).

**Note:** In DRP, rsize, used, and tier capacities are not reported per volume. These capacities are reported at the parent pool level only because of the complexities of deduplication capacity reporting.

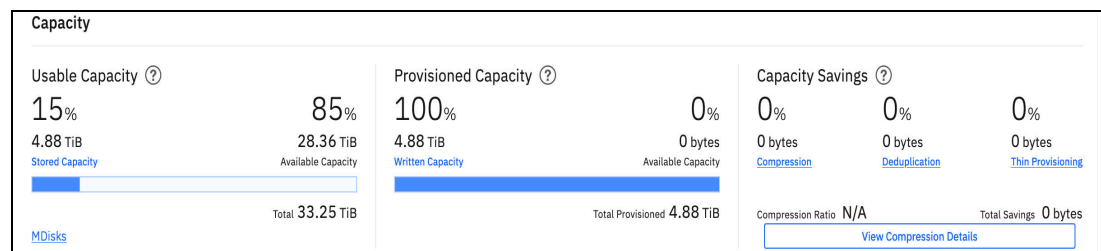


Figure 4-6 Example dashboard capacity view

For FCMs, this capacity is the maximum capacity that can be written to the system. However, for smaller capacity drives (4.8 TB), this capacity reports 20 TiB as usable. The usable capacity might be lower because of the data reduction that is achieved from the FCM compression.

Plan to achieve the default 2:1 compression, which is approximately 10 TiB of usable space on average. Careful monitoring of the data reduction must be considered if you plan to provision to the maximum stated usable capacity when these small capacity FCMs are used.

The larger FCM (9.6 TB and above) report just over 2:1 usable capacity; therefore, 22,44 and 88 for the 9.6, 19.2 and 38.4 TB modules, respectively.

The provisioned capacity shows the total provisioned capacity in terms of the volume allocations. This capacity is the “virtual” capacity that is allocated to fully allocated and thin provisioned volumes. Therefore, it is in theory the capacity that can be written to if all volumes were filled 100% by the using system.

The written capacity is the amount of data that was written into the provisioned capacity. For fully allocated volumes, this capacity always is 100% of the provisioned capacity. For thin provisioned (including data reduced volumes) this written capacity is the amount of data the host wrote to the volumes.

The final set of capacity numbers relate to the data reduction. This information is reported in two ways: as the savings from DRP (compression and deduplication) that is provided at the DRP level, or the FCM compression savings (see Figure 4-7).

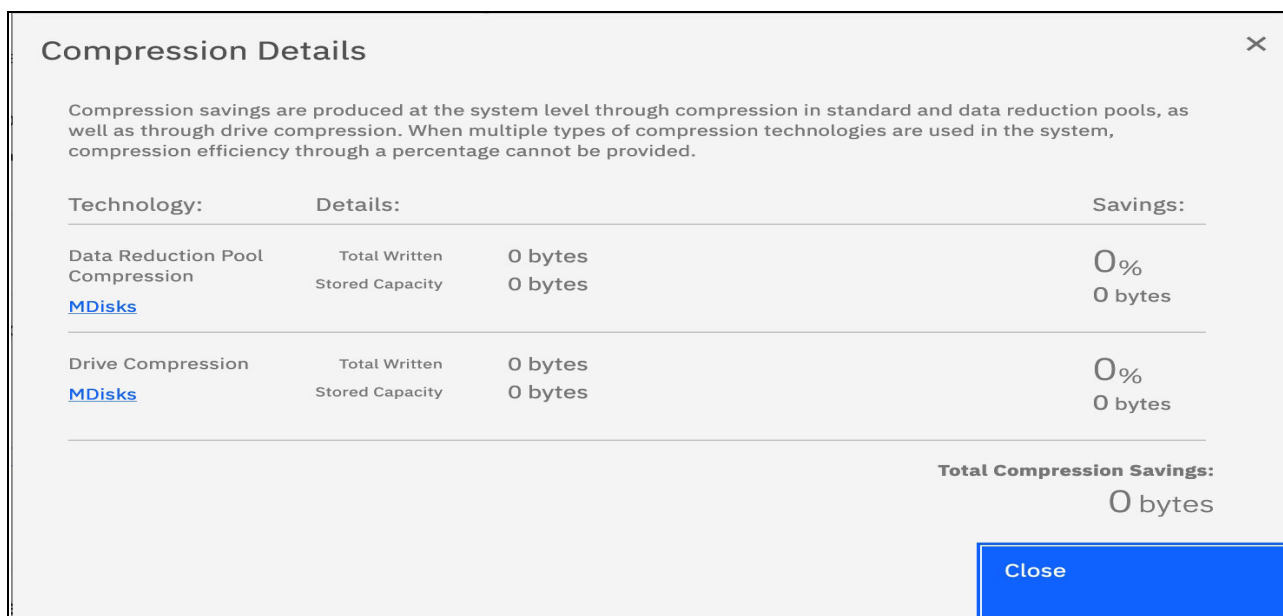


Figure 4-7 Compression Savings dashboard report

## 4.2.4 Extent size considerations

When adding MDisks to a pool they are logically divided into chunks of equal size. These chunks are called *extents* and are indexed internally. Extent sizes can be 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, or 8192 MB. IBM Spectrum Virtualize architecture can manage  $2^{22}$  extents for a system, and therefore the choice of extent size affects the total amount of storage that can be addressed. see this [link](#) for the capacity limits per extent size.

When planning for the extent size of a pool, remember that you cannot change the extent size later, it must remain constant throughout the lifetime of the pool.

For pool extent size planning, consider the following recommendations:

- ▶ For standard pools, 1 GB often is suitable.
- ▶ For DRPs, use 4 GB (for more information about considerations for the extent size on DRP, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148).
- ▶ With Easy Tier enabled hybrid pools, consider smaller extent sizes to better use the higher tier resources and therefore, provide better performance.
- ▶ Keep the same extent size for all pools if possible. The extent-based migration function is not supported between pools with different extent sizes. However, you can use volume mirroring to create copies between storage pools with different extent sizes.

**Limitation:** Extent-based migrations from standard pools to DRPs are not supported unless the volume is fully allocated.

## 4.2.5 External pools

IBM SAN Volume Controller-based storage systems can virtualize external storage systems. In this section, we describe some special considerations when configuring storage pools with external storage.

### Availability considerations

IBM SAN Volume Controller external storage virtualization feature provides many advantages through consolidation of storage, you must understand the availability implications that storage component failures can have on availability domains within the IBM SAN Volume Controller cluster.

IBM Spectrum Virtualize offers significant performance benefits through its ability to stripe across back-end storage volumes. However, consider the effects that various configurations have on availability.

When you select MDisk for a storage pool, performance is often the primary consideration. However, in many cases, the availability of the configuration is traded for little or no performance gain.

IBM SAN Volume Controller must take the entire storage pool offline if a single MDisk in that storage pool goes offline. Consider an example where you have 40 external arrays of 1 TB each for a total capacity of 40 TB with all 40 arrays in the same storage pool.

In this case, you place the entire 40 TB of capacity at risk if one of the 40 arrays fails (which causes the storage pool to go offline). If you then spread the 40 arrays out over some of the storage pools, the effect of an array failure (an offline MDisk) affects less storage capacity, which limits the failure domain.

To ensure optimum availability to well-designed storage pools, consider the following preferred practices:

- ▶ It is recommended that each storage pool must contain only MDisk from a single storage subsystem. An exception exists when you are working Easy Tier hybrid pools. For more information, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.
- ▶ It is suggested that each storage pool contains only MDisk from a single storage tier (SSD or Flash, Enterprise, or NL\_SAS) unless you are working with Easy Tier hybrid pools. For more information, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.

When you are selecting storage subsystems, the decision often comes down to the ability of the storage subsystem to be more reliable and resilient, and meet application requirements.

IBM Spectrum Virtualize does not provide any physical-level data redundancy for virtualized external storages. The availability characteristics of the storage subsystems’ controllers have the most impact on the overall availability of the data that is virtualized by IBM Spectrum Virtualize.

## Performance considerations

Performance is also a determining factor, where adding IBM SAN Volume Controller as a frontend results in considerable gains. Another factor is the ability of your virtualized storage subsystems to be scaled up or scaled out. For example, IBM System Storage DS8000 series is a scale-up architecture that delivers the best performance per unit, and the IBM FlashSystem series can be scaled out with enough units to deliver the same performance.

A significant consideration when you compare native performance characteristics between storage subsystem types is the amount of scaling that is required to meet the performance objectives. Although lower-performing subsystems typically can be scaled to meet performance objectives, the extra hardware that is required lowers the availability characteristics of the IBM SAN Volume Controller cluster.

All storage subsystems possess an inherent failure rate. Therefore, the failure rate of a storage pool becomes the failure rate of the storage subsystem times the number of units.

## Number of MDisks per pool

In the previous sections of this chapter, we discussed the availability and performance considerations when creating external pools; however, the number of MDisks per pool also can affect these measures.

The back-end storage access is controlled through MDisks where the IBM SAN Volume Controller acts as a host to the back-end controller systems. Just as you must consider volume queue depths when accessing storage from a host, so these systems must calculate queue depths to maintain high throughput capability while ensuring the lowest possible latency.

For more information about the queue depth algorithm and the rules about how many MDisks to present for an external pool, see “Volume considerations” on page 82.

This section describes how many “volumes” to create on the backend controller (that are seen as MDisks by the virtualizing controller) based on the type and number of drives (HDD, SSD, and so on).

## 4.3 Data reduction pool best practices

In this section, we describe the DRP planning and implementation best practices.

### 4.3.1 Data reduction pools with IBM FlashSystem NVMe attached drives

The compression ratios that are listed in Table 4-1 on page 104 are the maximum achievable ratios considering the effective capacity that is available. Depending on the pool filling, the actual compression ratio can be higher.

For more information about estimating the compression ratio for a specific workload, see “Determining whether your data is a deduplication candidate” on page 104.

**Important:** If you plan to use DRP with FCM storage, assume zero extra compression from the FCMs. That is, use the reported physical or usable capacity from the RAID array as the usable capacity in the pool and ignore the above maximum effective capacity.

Assuming zero extra compression from the FCMs is the result of the DRP function sending compressed data to the FCM and thus the data reduction (effective) capacity savings are reported at the front-end pool level and the back-end pool capacity is almost 1:1 for the physical capacity.

Some small amount of other compression savings might be seen because of the compression of the DRP metadata on the FCMs.

When providing industry standard NVMe-attached flash drives capacity for the DRP, some considerations must be reviewed.

The main point to consider is whether the data is deduplicable. Tools are available to provide some estimation of deduplication ratio (see “Determining whether your data is a deduplication candidate” on page 104). In this section, we consider DRP configurations with IBM FCM drives:

- ▶ Data is deduplicable. In this case, the recommendation is to use compressed and deduplicated volume type. The double compression (first from DRP and then from FCMs) does not affect the performance and the overall compression ratio.
- ▶ If you are not conducting deduplication (because it is not a good candidate for deduplication), you might use standard pools (instead of DRP with FCM), and allow the FCM hardware complete the compression because the overall achievable throughput is higher.

With standard off-the-shelf NVMe drive (which do not support inline compression), the following similar considerations apply:

- ▶ Data is deduplicable. In this case, the recommendation is to use a compressed and deduplicated volume type. The DRP compression technology has more than enough compression bandwidth for these purposes, so compress makes sense.
- ▶ Data is not deduplicable. In this case, the recommendation is to use a compressed volume type only. Again, the internal compression technology provides enough compression bandwidth for these purposes.

**Note:** In general, avoid creating DRP volumes that are only deduplicated. When using DRP, volumes are fully allocated or deduplicated and compressed.

Various configuration items affect the performance of compression on the system. To attain high compression ratios and performance on your system, ensure that the following guidelines are met:

- ▶ Use FCM compression, unless your data deduplicates well with IBM FlashSystem Family products that support FCMs.
- ▶ With SSD and HDD, use DRP, deduplicate if applicable with IBM FlashSystem 5100, 7000, and 9000 family.
- ▶ The use of a small amount (1- 3%) of SCM capacity in a DRP significantly improves DRP metadata performance. Because the directory data is the most frequently accessed data in the DRP and the design of DRP maintains directory data on the same extents, Easy Tier quickly promotes the metadata extents to the fastest available tier.



- ▶ Never create a DRP with only Nearline SAS capacity. If you want to predominantly use NL SAS drives, ensure that you have a small amount of Flash or SCM capacity for the metadata.
- ▶ In general, DRP is avoided on FlashSystem 5030 unless you have little performance expectations or requirements. The FlashSystem 5030 has no extra offload hardware and uses the internal CPU to provide the compress and decompress engine. Using DRP in FlashSystem 5030 features limited throughput capability and is suitable for low throughput workloads only. Latency also is adversely affected in most cases.
- ▶ Do not compress encrypted data. That is, if the application or operating system provides encryption, do not attempt to use DRP volumes. Data at-rest encryption, which is provided by IBM SAN Volume Controller, is still possible because the encryption is performed after the data is reduced. If host-based encryption is unavoidable, assume that no data reduction is possible. That is, ensure a 1:1 mapping is used of physical to effective capacity.
- ▶ Although DRP and FCM do not have penalties in terms of performance if data cannot be compressed (that is, you can attempt to compress all data), the outlined extra overhead of managing DRP volumes can be avoided by using standard pools or fully allocated volumes if no data reduction benefits are realized.
- ▶ You can use tools that estimate the compressible data, or use commonly known ratios for common applications and data types. Storing these data types on compressed volumes saves disk capacity and improves the benefit of the use of compression on your system. For more information, see “Determining whether your data is compressible” on page 103.
- ▶ Avoid the use of any client, file system, or application based-compression with the system compression. If this avoidance is not possible, use a standard pool for these volumes.
- ▶ Never use DRP on the IBM SAN Volume Controller and virtualized external storage at same time (DRP over DRP). In all cases, use DRP at the virtualizer level rather than the back-end storage as this simplifies capacity management and reporting.

### 4.3.2 DRP and external storage considerations

In general, avoid any configuration that attempts to perform data reduction at two levels.

The recommended configuration is to run DRP only at the IBM SAN Volume Controller that is acting as the virtualizer. For storage that is behind the virtualizer, provision fully allocated volumes to the virtualizer.

By running in this configuration you ensure that:

- ▶ The virtualizer understands the real physical capacity available and can warn and avoid out of space situations (where access is lost because of a lack of space)
- ▶ Capacity monitoring can be wholly performed on the virtualizer level because it sees the true physical and effective capacity usage.
- ▶ The virtualizer performs efficient data reduction on previously unreduced data. Generally, the virtualizer includes offload hardware and more CPU resource than the back-end storage systems because it does not need to deal with RAID, and so on.

If you cannot avoid back-end data reduction (for example, the backend storage controller cannot disable its data reduction features) ensure that:

- ▶ You do not excessively over provision the physical capacity on the backend.  
For example, you have 100 TiB of real capacity. Start by presenting only 100 TiB of volumes to the IBM SAN Volume Controller. Monitor the data reduction on the back-end controller. If your data is reducing well over time, increase the capacity that is provisioned to the IBM SAN Volume Controller.  
  
This configuration ensures that you can monitor and validate your data reduction rates and avoid panic if you do not achieve the expected rates and presented too much capacity to IBM SAN Volume Controller.
- ▶ You do not run DRP on top of the backend device. Because the back-end device attempts to reduce the data, use a standard pool or fully allocated volumes in the IBM SAN Volume Controller DRP.
- ▶ You understand that IBM SAN Volume Controller now does not know the real capacity usage. You must monitor and watch for out of space at the backend storage controller and the IBM SAN Volume Controller.

**Important:** Never run DRP on top of DRP. This configuration is wasteful and causes performance problems with no extra capacity savings.

### 4.3.3 Data reduction pools and VMware vVols

At the time of this writing, DRPs are not supported or certified with VMware vVols; therefore, choose standard pools instead.

### 4.3.4 Data reduction pool configuration limits

The limitations of DRPs (IBM SAN Volume Controller version 8.4.0) at the time of this writing are listed in Table 4-4. Since version 8.2.0, the software does not support 2145-CG8 or earlier node types; only 2145-DH8 or later nodes support versions since 8.2.0.

Table 4-4 DRP properties

| Property                                         | Maximum number                    | Comments                                                      |
|--------------------------------------------------|-----------------------------------|---------------------------------------------------------------|
| DRPs per system                                  | 4                                 |                                                               |
| MDisks per DRP                                   | 128                               |                                                               |
| Volumes per DRP                                  | 10,000 -<br>(Number of DRPs x 12) |                                                               |
| Extents per I/O group per DRP                    | 128K                              |                                                               |
| Compressed volume copies in DRPs per system      | N/A                               | No limit is imposed here beyond the volume copy limit per DRP |
| Compressed volume copies in DRPs per I/O group   | N/A                               | No limit is imposed here beyond the volume copy limit per DRP |
| Deduplicated volume copies in DRPs per system    | N/A                               | No limit is imposed here beyond the volume copy limit per DRP |
| Deduplicated volume copies in DRPs per I/O group | N/A                               | No limit is imposed here beyond the volume copy limit per DRP |

| Property                                                  | Maximum number | Comments                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum volume capacity (fully allocated or data reduced) | 256 TB         | <ul style="list-style-type: none"> <li>▶ Maximum size for an individual fully-allocated volume.</li> <li>▶ Maximum size is dependent on the extent size of the storage pool.</li> <li>▶ Comparison Table: Maximum Volume, MDisk, and System capacity for each. <a href="#">extent size</a>.</li> </ul> |

For more information, see [IBM SAN Volume Controller Support Information](#).

### 4.3.5 DRP provisioning considerations

In this section, we describe several practices that must be considered when the DRP implementation is planned.

#### DRP restrictions

Consider the following important restrictions when planning for a DRP implementation:

- ▶ Maximum number of supported DRPs: 4
- ▶ VVols is not currently supported in DRP.
- ▶ Volume shrinking is not supported in DRP with thin or compressed volumes.
- ▶ Non-Disruptive Volume Move (NDVM) is not supported by DRP volumes.
- ▶ The volume copy split of a Volume Mirror in a different I/O Group is not supported for DRP thin or compressed volumes.
- ▶ Image and sequential mode VDisk are not supported in DRP.
- ▶ No extent level migration is allowed between DRP unless volumes are fully allocated.  
Volume migration for any volume type is permitted is between a quotaless child and its parent DRP pool.
- ▶ A maximum of 128 K extents per Customer Data Volume per I/O Group can be used.  
Therefore, the pool extent size dictates the maximum physical capacity in a pool after data reduction.
- ▶ Use 4 GB extent size or above.
- ▶ Recommended pool size is at least 20 TB.
- ▶ Lower than 1 PB per I/O group is used.
- ▶ Your pool should be no more than 85% occupied.

The following considerations apply to DRPs only:

- ▶ The real, used, free, or tier capacity is not reported per volume for DRP volumes; instead, only information at a pool level is available.
- ▶ Cache mode is always read/write on compressed or deduplicated volumes.
- ▶ Autoexpand is always on.
- ▶ Specific volume capacity cannot be placed on specific MDisk.

## Extent size considerations

With DRPs, the number of extents available per pool is limited by the internal structure of the pool (see 4.1.2, “Data reduction pools” on page 97) and specifically by the size of the data volume. Currently, the maximum number of extents that is supported for a data volume is 128 K. As shown in Figure 4-1 on page 101, one data volume is allotted per pool. Table 4-5 lists the maximum size per pool by extent size and I/O group number.

*Table 4-5 Pool size by extent size and IO group number*

| Extent Size | Max size with one I/O group | Max size with two I/O groups | Max size with three I/O group | Max size with four I/O group |
|-------------|-----------------------------|------------------------------|-------------------------------|------------------------------|
| 1024        | 128 TB                      | 256 TB                       | 384 TB                        | 512 TB                       |
| 2048        | 256 TB                      | 512 TB                       | 768 TB                        | 1024 TB                      |
| 4096        | 512 TB                      | 1024 TB                      | 1536 TB                       | 2048 TB                      |
| 8192        | 1024 TB                     | 2048 TB                      | 3072 TB                       | 4096 TB                      |

Considering that the extent size cannot be changed after the pool is created, it is recommended to carefully plan the extent size according to the environment capacity requirements. For most configurations, we recommend a 4 GB extent for DRP.

## Pool capacity requirements

A minimum capacity also must be provisioned in a DRP to provide capacity for the internal metadata structures. Table 4-6 lists the minimum capacity that is required by extent size and I/O group number.

*Table 4-6 Minimum recommended pool size by extent size and IO group number*

| Extent Size | Min size with one I/O group | Min size with two I/O group | Min size with three I/O group | Min size with four I/O group |
|-------------|-----------------------------|-----------------------------|-------------------------------|------------------------------|
| 1024        | 255 GB                      | 516 GB                      | 780 GB                        | 1052 GB                      |
| 2048        | 510 GB                      | 1032 GB                     | 1560 GB                       | 2104 GB                      |
| 4096        | 1020 GB                     | 2064 GB                     | 3120 GB                       | 4208 GB                      |
| 8192        | 2040 GB                     | 4128 GB                     | 6240 GB                       | 8416 GB                      |

The values that are listed in Table 4-6 represent the minimum required capacity for a DRP to create a single volume.

When sizing a DRP, it is important to remember that the garbage collection process is running all of the time to reclaim the unused space, and optimizing the extents usage. For more information about the garbage collection process, see “DRP internal details” on page 101.

This process then requires a specific amount of free space to work efficiently. For this reason, it is recommended to keep approximately 15% free space in a DRP pool. For more information, see this [IBM Support web page](#).

### 4.3.6 Standard and DRP pools coexistence

Although homogeneous configurations in terms of pool type are preferable, no technical reason exists to avoid the use of standard and DRP pools in the same system. In some circumstances, this coexistence is unavoidable. Consider the following scenarios:

- ▶ IBM SAN Volume Controller installation that requires VMware VVols support and data reduction capabilities for other environments. This scenario requires the definition of standard *and* DRP pools because of the restriction of DRPs regarding the VVols (see “DRP restrictions” on page 123).

In this case, the standard pool is used for VVols environments only, while the DRP is used for the other environments. Some data reduction capability can be achieved for the VVols standard pool by using the in-line data compression that is provided by the IBM FCMs on FlashSystem.

- ▶ IBM SAN Volume Controller installation requires an external pool for image mode volumes and data reduction capabilities for other environments. Also, this scenario requires the definition of standard and DRP pools because of the restriction of DRP regarding the Image mode volumes (see “DRP restrictions” on page 123).

In this case, the standard pool is used for Image mode volumes only (optionally, with the write cache disabled if needed for the back-end native copy services usage). For more information, see Chapter 6, “Copy services” on page 219. DRP is used for all the other environments.

- ▶ IBM SAN Volume Controller installation that uses a FlashSystem system as an external pool that uses DRP capabilities. In this scenario, the external pool must be a standard pool, as recommended in 4.3.2, “DRP and external storage considerations” on page 121. In this case, the internal storage can be defined in a separate DRP enabling the data reduction capabilities, if needed.
- ▶ IBM SAN Volume Controller installation requiring more than four pools.

### 4.3.7 Data migration with DRP

As described in “DRP restrictions” on page 123, extent level migration (such as migrate volume or migrate extent functions) to and from a DRP is not supported. Two options are available for an IBM SAN Volume Controller configuration when you plan to move data to or from a DRP and make use of data reduced volumes: host-based migrations or volume mirroring based migrations.

#### Host-based migration

Host-based migrations use operating system features or software tools running on the hosts to move data concurrently to the normal host operations. VMware vMotion and AIX Logical Volume Mirroring are just two examples of such features. When this approach is used, a specific amount of capacity on the target pool is required to provide the migration target volumes. The process can be summarized by the following steps:

1. Create the target volumes of the migration in the target pool. Depending on the migration technique, the size and the amount of the volumes can be different from the original volumes. For example, we can migrate two 2 TB VMware data store volumes in one single 4 TB data store volume.
2. Map the target volumes to the host.
3. Rescan the HBAs to attach the new volumes to the host.
4. Activate the data move or mirroring feature from the old volumes to the new volumes.

5. Wait until the copy is complete.
6. Detach the old volumes from the host.
7. Unmap and remove the old volumes from the IBM SAN Volume Controller.

When migrating data to a DRP, consider the following options:

- ▶ Migrate directly to compressed or deduplicated volumes. With this option, the migration duration mainly depends on the host migration throughput capabilities. Consider that the target volumes are subject to high write workloads that can use numerous resources because of the compression and deduplication tasks. To avoid any potential performance effects on the workload, limit the migration throughput at the host level or, if this option is not possible, implement the throttling function at the volume level.
- ▶ Migrate first to fully allocated volumes and then, convert them to compressed or deduplicated volumes. Also, with this option, the migration duration mainly depends on the host capabilities, but usually more throughput can be sustained because no compression and deduplication overhead is incurred. The space saving conversion can be done by using the volume mirroring feature.

### Volume mirroring-based migration

The volume mirroring feature can be used to migrate data from a pool to another pool and at the same time, change the space saving characteristics of a volume. Like host-based migration, volume mirroring-based migration requires free capacity on the target pool, but it is not needed to create volumes manually. Volume mirroring migration is a three-step process:

1. Add a volume copy on the DRP and specify the wanted data reduction features.
2. Wait until the copies are synchronized.
3. Remove the original copy.

With volume mirroring, the throughput of the migration activity can be adjusted at a volume level by specifying the Mirror Sync Rate parameter. Therefore, if performance is affected, the migration speed can be lowered or even suspended.

**Note:** Volume \mirroring supports only two copies of a volume. If a configuration uses both copies, one of the copies must be removed first before you start the migration.

The volume copy split of a Volume Mirror in a different I/O Group is not supported for DRP thin or compressed volumes.

## 4.4 Operations with storage pools

In this section, we describe some guidelines for the typical operation with pools, which apply to standard and DRP pool type.

### 4.4.1 Creating data reduction pools

This section describes how to create DRPs.

#### Using the management GUI

To create DRPs, complete the following steps:

1. Create a DRP:
  - a. In the management GUI, select **Pools** → **Pools**.
  - b. In the Pools window, click **Create**.
  - c. In the Create Pool window, enter a name of the pool and select **Data Reduction**.
  - d. Click **Create** (see Figure 4-8).

Create Pool

Pool Name:  
Pool0

Extent Size: 4.00 GiB Maximum Capacity: 512.00 TiB

Additional Options:  
☒ Data Reduction

**Warning**  
If the usable capacity usage of a data reduction pool exceeds more than 85%, I/O performance can be affected. The system needs 15% of usable capacity available in data reduction pools to ensure that capacity reclamation can be performed efficiently.

? Need Help Cancel Create

Figure 4-8 Create Pool window

2. Complete the following steps to create a Data Reduction child pool:
  - a. In the management GUI, select **Pools** → **Pools**.
  - b. Right-click the parent pool in which you want to create the child pool (see Figure 4-9).

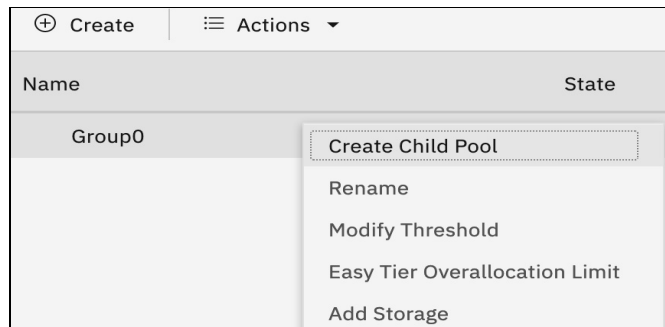


Figure 4-9 Right-click parent pool actions menu

- c. Select **Create Child Pool**.
  - d. Enter a name for the child pool.
  - e. Click **Create** (see Figure 4-10).

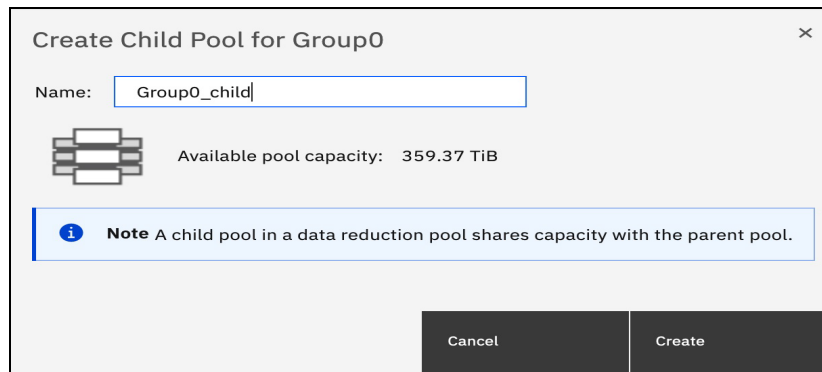


Figure 4-10 Create Child Pool window

3. Add storage to a parent DRP by completing the following steps:
  - a. In the management GUI, select **Pools** → **Pools**.
  - b. Right-click the DRP that you created and select **Add Storage**.
  - c. Select from the available storage and allocate capacity to the pool. Click **Assign**.
4. Create fully allocated, compressed, deduplicated, or a combination of compressed and deduplicated volumes in the DRP and map them to hosts by completing the following steps:
  - a. In the management GUI, select **Volumes** → **Volumes**.
  - b. In the Volumes window, click **Create Volumes**.
  - c. On the Create Volume window, select the type of volume that you want to create.
  - d. Enter the following information for the volume:
    - Pool
 

Select a DRP from the list. Compressed, thin-provisioned, and deduplicated volumes and copies must be in DRPs.



- Volume details

Enter the quantity, capacity, and name for the volumes that you are creating.

- Capacity savings

Select **None** (fully allocated) or **Compressed**. When Compressed is selected, you can also select to use deduplication for the volume that you create.

**Note:** If your system contains self-compressed drives, ensure that the volume is created with Compression enabled. If not, the system cannot calculate accurate available physical capacity.

- e. Click **Create and Map** (see Figure 4-11).

Figure 4-11 Create Volumes window

**Note:** Select **Create** to create the volumes in the DRP without mapping to hosts. If you want to map volumes to hosts later, select **Hosts** → **Hosts** → **Add Hosts**.

- f. In the **Create Mapping** window, select **Host** to display all hosts that are available for mapping. Hosts must support SCSI **unmap** commands. Verify that the selected host type supports SCSI **unmap** commands. Click **Next**.
- g. From version 8.3.1, the system attempts to map the SCSI LUN ID as the same on all Host clusters. If you want to assign specific IDs, select the **Self Assign** option.

- h. Verify the volume, and then, click **Map Volumes** (see Figure 4-12).

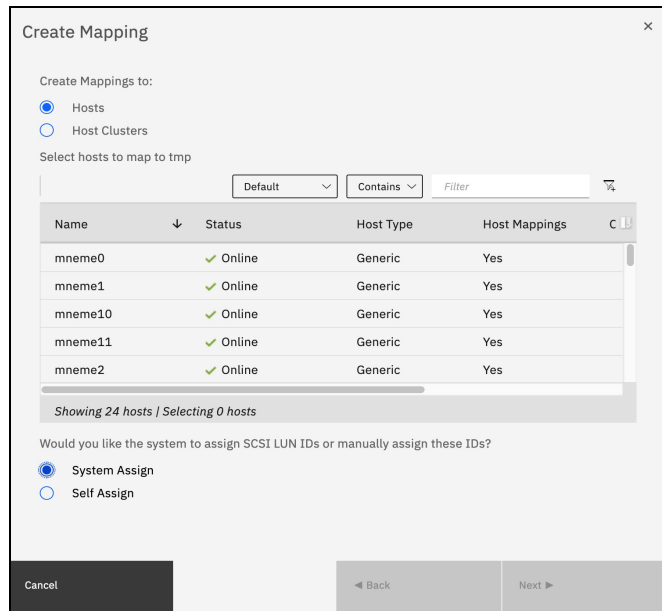


Figure 4-12 Create Mapping window

## Using the command-line interface

Complete the following steps:

1. To create a DRP, enter the following command:

```
mkmdiskgrp -name pool_name -ext extent_size -datareduction yes
```

Where *pool\_name* is the name of the pool and *extent\_size* is the extent size of the pool. DRPs are created as parent pools only, not child pools.

2. To create a compressed volume within a DRP, enter the following command:

```
mkvolume -name name -pool storage_pool_name -size disk_size -compressed
```

Where *name* is the name of the new volume, *storage\_pool\_name* is the name of the DRP, and *disk\_size* is the capacity of the volume.

3. To map the volume to a host, enter the following command:

```
mkvdiskhostmap -host host_name vdisk_name
```

Where *host\_name* is the name of the host and *vdisk\_name* is the name of the volume.

Monitor the physical capacity of DRPs in the management GUI by selecting **Pools** → **Pools**. In the command-line interface, run the `lsmdiskgrp` command to display the physical capacity of a DRP.

### 4.4.2 Adding external MDisks to storage pools

If MDisks are being added to an IBM SAN Volume Controller cluster, it is likely because you want to provide more capacity. In Easy Tier enabled pools, the storage pool balancing feature ensures that the newly added MDisks are automatically populated with extents that come from the other MDisks. Therefore, no manual intervention is required to rebalance the capacity across the available MDisks.

**Important:** When adding external MDisk, the system does not know to which tier the MDisk belongs. You must ensure that you specify or change the tier type to match the tier type of the MDisk.

This specification is vital to ensure that Easy Tier keeps a pool as a single tier pool and balances across all MDisk, or Easy Tier adds the MDisk to the correct tier in a multitier pool.

Failure to set the correct tier type creates a performance problem that might be difficult to diagnose in the future.

The tier\_type can be changed by way of the command line by using:

```
chmdisk -tier <new_tier> <mdisk>
```

For more information, see 4.6.9, “Easy Tier settings” on page 165

Adding MDisk to storage pools is a simple task, but it is suggested that you perform some checks in advance especially when adding external MDisk.

### Checking access to new MDisk

Be careful when you add external MDisk to storage pools to ensure that the availability of the storage pool is not compromised by adding a faulty MDisk. The reason is that loss of access to a single MDisk causes the entire storage pool to go offline.

In IBM Spectrum Virtualize, a feature is available that tests an MDisk automatically for reliable read/write access before it is added to a storage pool so that no user action is required. The test fails under the following conditions:

- ▶ One or more nodes cannot access the MDisk through the chosen controller port.
- ▶ I/O to the disk does not complete within a reasonable time.
- ▶ The SCSI inquiry data that is provided for the disk is incorrect or incomplete.
- ▶ The IBM Spectrum Virtualize cluster suffers a software error during the MDisk test.

Image-mode MDisk are not tested before they are added to a storage pool because an offline image-mode MDisk does not take the storage pool offline. Therefore, the suggestion here is to use a dedicated storage pool for each image mode MDisk. This preferred practice makes it easier to discover what the MDisk is going to be virtualized as, and reduce the chance of human error.

### Persistent reserve

A common condition where external MDisk can be configured by IBM SAN Volume Controller, but cannot perform read/write, is when a persistent reserve is left on a LUN from a previously attached host.

In this condition, rezone the back-end storage and map them back to the host that is holding the reserve. Alternatively, map them to another host that can remove the reserve by using a utility, such as the Microsoft Windows SDD Persistent Reserve Tool.

## 4.4.3 Renaming MDisk

After you discover MDisk, rename them from their IBM SAN Volume Controller default name. This process can help during problem isolation and avoid confusion that can lead to an administrative error by using a naming convention for MDisk that associates the MDisk with the controller and array.

When multiple tiers of storage are on the same IBM SAN Volume Controller cluster, you might also want to indicate the storage tier in the name. For example, you can use R5 and R10 to differentiate RAID levels, or you can use T1, T2, and so on, to indicate the defined tiers.

**Preferred practice:** Use a naming convention for MDisks that associates the MDisk with its corresponding controller and array within the controller, such as DS8K\_<extent pool name/id>\_<volume id>.

#### 4.4.4 Removing MDisks from storage pools

You might want to remove MDisks from a storage pool (for example, when you decommission a storage controller). When you remove MDisks from a storage pool, consider whether to manually migrate extents from the MDisks. It is also necessary to ensure that you remove the correct MDisks.

**Sufficient space:** The removal occurs only if sufficient space is available to migrate the volume data to other extents on other MDisks that remain in the storage pool. After you remove the MDisk from the storage pool, it takes time to change the mode from managed to unmanaged, depending on the size of the MDisk that you are removing.

When you remove the MDisk that consists of internal disk drives from the storage pool on an IBM FlashSystem, the MDisk is deleted. This process also deletes the array on which this MDisk was built, and converts all drives that were included in this array to a candidate state. You can now use those disk drives to create another array of a different size and RAID type, or you can use them as hot spares.

##### Migrating extents from the MDisk to be deleted

If an MDisk contains volume extents, you must move these extents to the remaining MDisks in the storage pool. Example 4-3 shows how to list the volumes that have extents on an MDisk by using the CLI.

*Example 4-3 Listing of volumes that have extents on an MDisk to be deleted*

```
IBM_2145:itsosvcc11:admin>lsmdiskextent mdisk14
id          number_of_extents  copy_id
5           16          0
3           16          0
6           16          0
8           13          1
9           23          0
8           25          0
```

**DRP restriction:** The `lsmdiskextent` command does not provide accurate extent usage for thin-provisioned or compressed volumes on DRPs.

Specify the **-force** flag in the `rmmdisk` command, or select the corresponding option in the GUI. Both actions cause IBM SAN Volume Controller to automatically move all used extents on the MDisk to the remaining MDisks in the storage pool.

Alternatively, you might want to manually perform the extent migrations. Otherwise, the automatic migration randomly allocates extents to MDisks (and areas of MDisks). After all of the extents are manually migrated, the MDisk removal can proceed without the **-force** flag.

External MDisks must appear to the IBM SAN Volume Controller cluster as unmanaged before their controller LUN mapping is removed. Unmapping LUNs from IBM SAN Volume Controller that are still part of a storage pool results in the storage pool going offline and affects all hosts with mappings to volumes in that storage pool.

The UID is unique across all MDisks on all controllers. However, the controller LUN is unique only within a specified controller and for a specific host. Therefore, when you use the controller LUN, check that you are managing the correct storage controller and that you are looking at the mappings for the correct IBM SAN Volume Controller host object.

How to correlate back-end volumes (LUNs) to MDisks is described next.

The correct correlation between the back-end volume (LUN) with the external MDisk is crucial to avoid mistakes and possible outages. You can correlate the back-end volume with MDisk for DS8000 series, XIV, and FlashSystem V7000 storage controllers.

The LUN ID only uniquely identifies LUNs within the same storage controller. If multiple storage devices are attached to the same IBM SAN Volume Controller cluster, the LUN ID must be combined with the worldwide node name (WWNN) attribute to uniquely identify LUNs within the IBM SAN Volume Controller cluster.

To get the WWNN of the DS8000 controller, take the first 16 digits of the MDisk UID and change the first digit from 6 to 5, such as 6005076305ffc74c to 5005076305ffc74c. When detected as IBM SAN Volume Controller ctrl\_LUN\_#, the DS8000 LUN is decoded as 40XX40YY00000000, where XX is the logical subsystem (LSS) and YY is the LUN within the LSS. As detected by the DS8000, the LUN ID is the four digits starting from the 29th digit, as in the Example 4-4.

[illegible]

In Example 4-4, you can identify the MDisk supplied by the DS8000, which is LUN ID 1007.

Identify the XIV volumes by using the volume serial number and the LUN that is associated with the host mapping. The example in this section uses the following values:

- Complete the following steps:

- ### Example 4-5 The `lscontroller` command

[illegible]

- ### FlashSystem volumes

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Complete the following steps to correlate the IBM FlashSystem volumes with the external MDisks seen by the virtualizer:

1. From the back-end IBM FlashSystem side first, check the Volume UID field for the volume that was presented to the virtualizer, as shown in Figure 4-13.

The screenshot shows a window titled "Properties for Volume" with a close button (X) in the top right corner. Below the title bar, there are two tabs: "Volume Overview" (selected) and "Copy 0". The main area displays various volume properties in a two-column layout.

|                            |          |                        |                                  |
|----------------------------|----------|------------------------|----------------------------------|
| <b>Name:</b>               | VD_SVC_4 | <b>Cache mode:</b>     | Enabled                          |
| <b>Volume ID:</b>          | 5        | <b>Cache state:</b>    | Not empty                        |
| <b>State:</b>              | ✓ Online | <b>UDID (OpenVMS):</b> | N/A                              |
| <b>Capacity:</b>           | 9.70 TiB | <b>Volume UID:</b>     | 6005076810810026D800000000000008 |
| <b>IOPS limit:</b>         | Disabled | <b>I/O group:</b>      | Caching: io_grp0                 |
| <b>Bandwidth limit:</b>    | Disabled |                        | Accessible: io_grp0              |
| <b>Encrypted:</b>          | No       | <b>Preferred node:</b> | node1                            |
| <b>FlashCopy mappings:</b> | 0        | <b>Protocol type:</b>  | SCSI                             |
| <b>Mirror sync rate:</b>   | 50       |                        |                                  |

Figure 4-13 FlashSystem volume details

2. On the Host Maps tab (see Figure 4-14), check the SCSI ID number for the specific volume. This value is used to match the virtualizer ctrl\_LUN\_# (in hexadecimal format).

The screenshot shows a window titled "Host Details: SVC" with a close button (X) in the top right corner. Below the title bar, there are three tabs: "Overview", "Mapped Volumes" (selected), and "Port Definitions". The main area is titled "Volumes Mapped to the Host" and contains a table with columns: SCSI ID, Name, UID, and Caching I/O... There are also filters for "Default", "Contains", and a "Filter" input field. The table lists six mappings, with the first one (SCSI ID 5) selected.

| SCSI ID | Name      | UID                              | Caching I/O... |
|---------|-----------|----------------------------------|----------------|
| 5       | VD_SVC_4  | 6005076810810026D800000000000008 | 0              |
| 2       | VD_SVC... | 6005076810810026D800000000000005 | 0              |
| 3       | VD_SVC... | 6005076810810026D800000000000006 | 0              |
| 4       | VD_SVC... | 6005076810810026D800000000000007 | 0              |
| 0       | VD_SVC... | 6005076810810026D800000000000003 | 0              |
| 1       | VD_SVC... | 6005076810810026D800000000000004 | 0              |

Showing 6 mappings | Selecting 1 mapping

Figure 4-14 FlashSystem volume details for host maps

3. At the virtualizer, review the MDisk details (see Figure 4-15) and compare the MDisk UID field with the FlashSystem Volume UID. The first 32 bits should be the same.

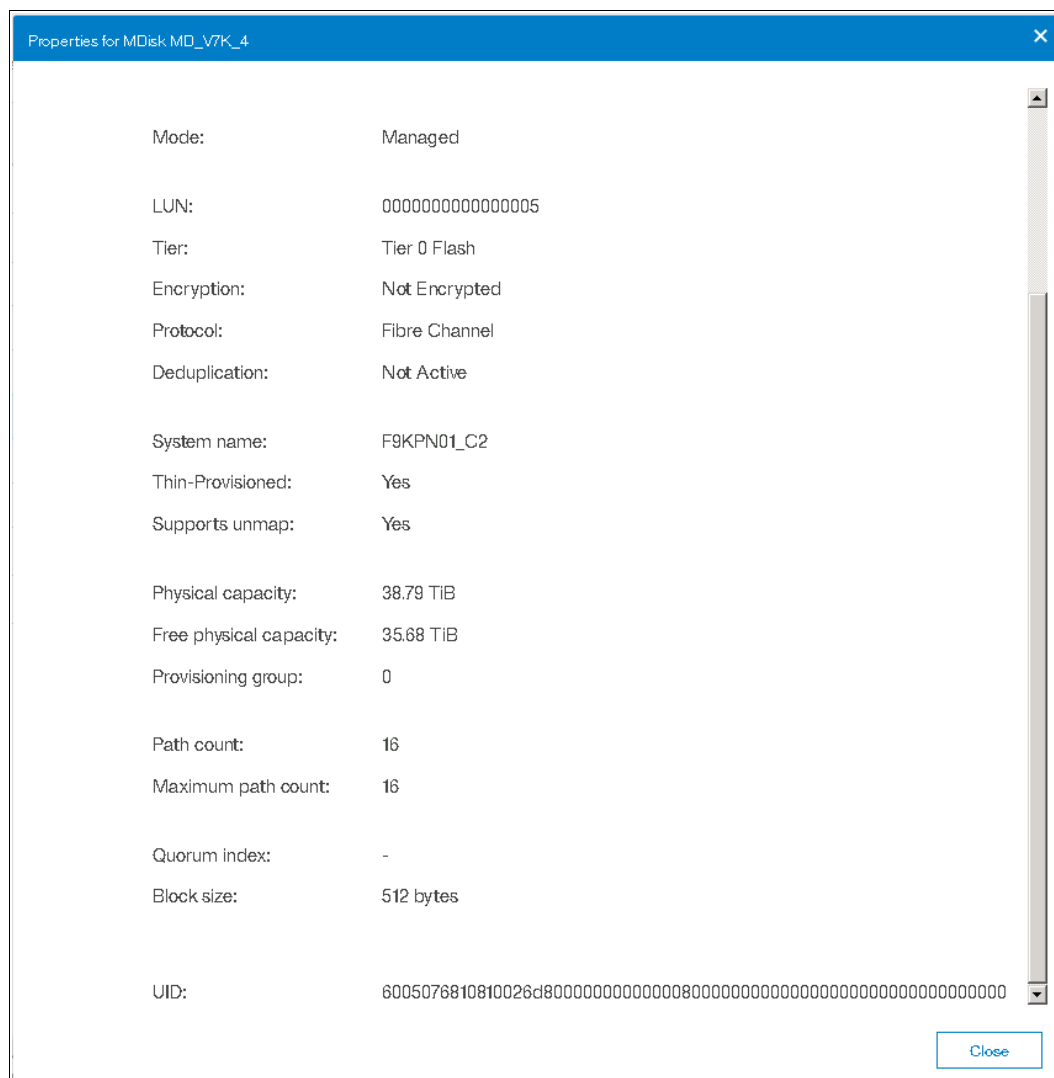


Figure 4-15 IBM SAN Volume Controller MDisk details for IBM FlashSystem volumes

4. Double-check that the virtualizer ctrl\_LUN\_# is the IBM FlashSystem SCSI ID number in hexadecimal format. In this example, the number is 0000000000000005.

#### 4.4.5 Remapping-managed MDisk

Generally, you do not unmap managed external MDisk from IBM SAN Volume Controller because this process causes the storage pool to go offline. However, if managed MDisk were unmapped from IBM SAN Volume Controller for a specific reason, the LUN must present the same attributes to IBM SAN Volume Controller before it is mapped back. Such attributes include UID, subsystem identifier (SSID), and LUN\_ID.

If the LUN is mapped back with different attributes, IBM SAN Volume Controller recognizes this MDisk as a new MDisk. In this case, the associated storage pool does *not* come back online. Consider this situation for storage controllers that support LUN selection because selecting a different LUN ID changes the UID. If the LUN was mapped back with a different LUN ID, it must be mapped again by using the previous LUN ID.



#### 4.4.6 Controlling extent allocation order for volume creation

When creating a volume on a standard pool, the allocation of extents is performed by using a round robin algorithm, which takes one extent from each MDisk in the pool in turn.

The first MDisk to allocate an extent from is chosen in a pseudo-random way rather than always starting from the same MDisk. The pseudo-random algorithm avoids the situation where the “striping effect” that is inherent in a round-robin algorithm places the first extent for many volumes on the same MDisk.

Placing the first extent of several volumes on the same MDisk might lead to poor performance for workloads that place a large I/O load on the first extent of each volume or that create multiple sequential streams.

However, this allocation pattern is unlikely to remain for long because Easy Tier balancing moves the extents to balance the load evenly across all MDisk in the tier. The hot and cold extents also are moved between tiers.

In a multitier pool, the middle tier is used by default for volume creation. If free space is not available in the middle tier, the cold tier is used if it exists. If it does not exist, the hot tier is used. For more information about Easy Tier, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.

**DRP restriction:** With compressed or deduplicated volumes on DRP, it is not possible to check the extent distribution across the MDisks and initially a minimal number of extents are allocated to the volume based on the rsize parameter.

### 4.5 Considerations when using encryption

IBM SAN Volume Controller since 2145-DH8 and all IBM FlashSystem support optional encryption of data at rest. This support protects against the potential exposure of sensitive user data and user metadata that is stored on discarded, lost, or stolen storage devices. To use encryption on the system, an encryption license is required for each IBM SAN Volume Controller I/O Group that support encryption.

**Note:** Check whether you have the required IBM Security™ Key Lifecycle Manager licenses available. Consider redundancy/high availability regarding Key Lifecycle Manager servers.

#### 4.5.1 General considerations

USB encryption, key server encryption, or both can be enabled on the system. The system supports IBM Security Key Lifecycle Manager version 2.6.0 or later for enabling encryption with a key server. To encrypt data that is stored on drives, the IBM SAN Volume Controller I/O Groups that can encrypt must be licensed and configured to use encryption.

When encryption is activated and enabled on the system, valid encryption keys must be present on the system when the system unlocks the drives or the user generates a new key. If USB encryption is enabled on the system, the encryption key must be stored on USB flash drives that contain a copy of the key that was generated when encryption was enabled. If key server encryption is enabled on the system, the key is retrieved from the key server.

It is not possible to convert the data to an encrypted copy. You can use the volume migration function to migrate the data to an encrypted storage pool or encrypted child pool. Alternatively, you can also use the volume mirroring function to add a copy to an encrypted storage pool or encrypted child pool and delete the decrypted copy after the migration.

**Note:** Hot Spare Nodes also need encryption licenses if they are to be used to replace the failed nodes that support encryption.

Before you activate and enable encryption, you must determine the method of accessing key information during times when the system requires an encryption key to be present. The system requires an encryption key to be present during the following operations:

- ▶ System start
- ▶ System restart
- ▶ User-initiated rekey operations
- ▶ System recovery

The following factors must be considered when planning for encryption:

- ▶ Physical security of the system
- ▶ Need and benefit of manually accessing encryption keys when the system requires
- ▶ Availability of key data
- ▶ Encryption license is purchased, activated, and enabled on the system
- ▶ Using Security Key Lifecycle Manager clones

**Note:** It is suggested that IBM Security Key Lifecycle Manager version 2.7.0 or later is used for any new clone end-points that are created on the system.

For more information about configuring IBM Spectrum Virtualize encryption, see the following publications:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8467
- ▶ *Implementing the IBM FlashSystem with IBM Spectrum Virtualize V8.4*, SG24-8465

## 4.5.2 Hardware and software encryption

Encryption can be performed on IBM SAN Volume Controller devices by using one of two methods: hardware encryption or software encryption. Both methods of encryption protect against the potential exposure of sensitive user data that is stored on discarded, lost, or stolen media. Both methods also can facilitate the warranty return or disposal of hardware.

The method that is used for encryption is chosen automatically by the system based on the placement of the data (see Figure 4-16).

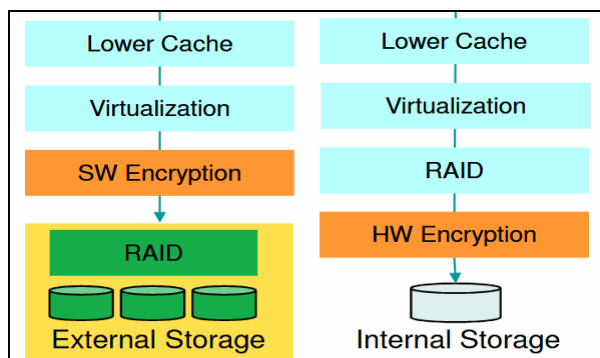


Figure 4-16 Encryption placement in lower layers of the IBM SAN Volume Controller software stack

### Hardware encryption only storage pool

Hardware encryption features the following characteristics:

- ▶ Algorithm is built in SAS chip for all SAS attached drives, or built into the drive itself for NVMe attached drives (FCM, IS NVMe, and SCM)
- ▶ No system overhead
- ▶ Only available to direct attached SAS disks
- ▶ Can be enabled only when you create internal arrays
- ▶ Child pools cannot be encrypted if the parent storage pool is not encrypted
- ▶ Child pools are automatically encrypted if the parent storage pool is encrypted but can have different encryption keys
- ▶ DRP child pools can use only the same encryption key as their parent

### Software encryption only storage pool

Software encryption features the following characteristics:

- ▶ The algorithm is running at the interface device driver
- ▶ Uses special CPU instruction set and engines (AES\_NI)
- ▶ Allows encryption for virtualized external storage controllers, which cannot self-encrypt
- ▶ Less than 1% system overhead
- ▶ Only available to virtualized external storage
- ▶ Can be enabled only when you create storage pools and child pools that consist of virtualized external storage
- ▶ Child pools can be encrypted even if the parent storage pool is not encrypted

## Mixed encryption in a storage pool

It is possible to mix hardware and software encryption in a storage pool, as shown in Figure 4-17.

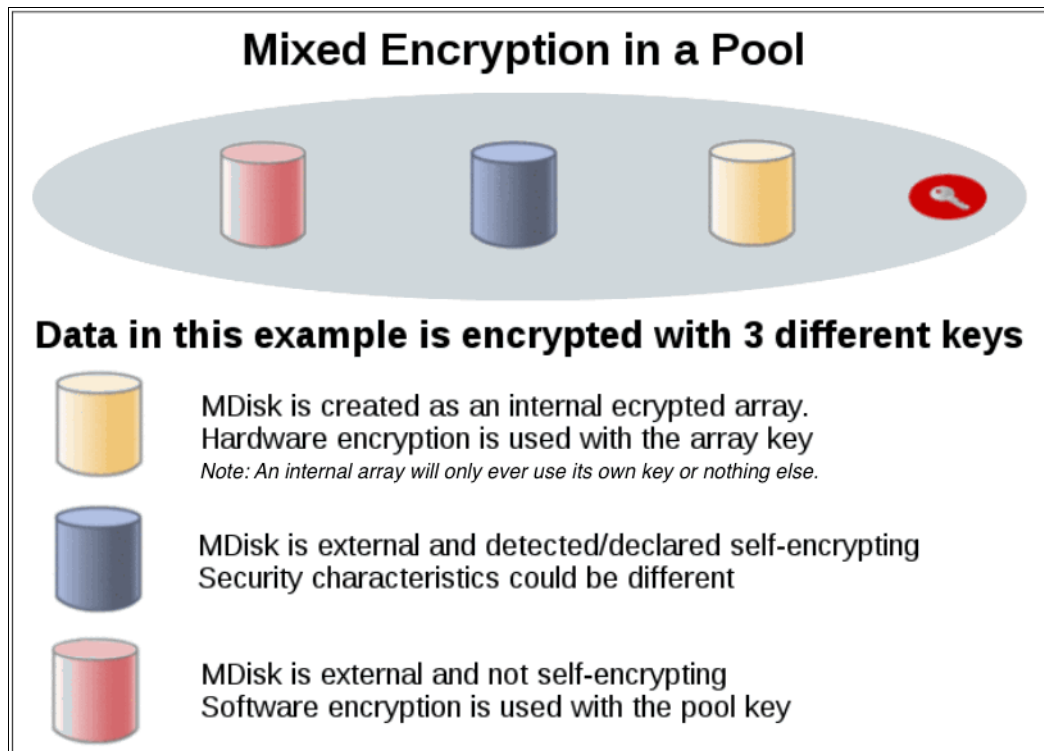


Figure 4-17 Mixed encryption in a storage pool

However, if you want to create encrypted child pools from a decrypted storage pool that contain a mix of internal arrays and external MDisks, the following restrictions apply:

- ▶ The parent pool must not contain any decrypted internal arrays.
- ▶ All IBM SAN Volume Controller nodes in the system must support software encryption and have encryption license that is activated.

**Note:** An encrypted child pool that is created from a decrypted parent storage pool reports as decrypted if the parent pool contains any decrypted internal arrays. Remove these arrays to ensure that the child pool is fully encrypted.

The general rule is not to mix different types of MDisks in a storage pool, unless it is intended to use the Easy Tier tiering function. In this scenario, the internal arrays must be encrypted if you want to create encrypted child pools from a decrypted parent storage pool. All the methods of encryption use the same encryption algorithm, the same key management infrastructure, and the same license.

**Note:** Always implement encryption on the self-encryption capable back-end storage, such as IBM FlashSystem, IBM Storwize, IBM XIV, IBM FlashSystem A9000, and IBM DS8000, to avoid potential system overhead.

Declare or identify the self-encrypted virtualized external MDisk as encrypted on IBM SAN Volume Controller by specifying the **-encrypt** option to **yes** with the **chmdisk** command, as shown in Example 4-6. This configuration is important to avoid IBM SAN Volume Controller trying to encrypt them again.

*Example 4-6 Command to declare/identify a self-encrypted MDisk from a virtualized external storage*

---

```
IBM_2145:ITS0_DH8_A:superuser>chmdisk -encrypt yes mdisk0
```

---

**Note:** It is important to declare and identify the self-encrypted MDisk from a virtualized external storage before creating an encrypted storage pool or child pool on IBM SAN Volume Controller.

### 4.5.3 Encryption at rest with USB keys

The following section describes the characteristics of using USB flash drives for encryption and the available options to access the key information.

USB flash drives have the following characteristics:

- ▶ Physical access to the system is required to process a rekeying operation
- ▶ No mechanical components to maintain with almost no read operations or write operations to the USB flash drive
- ▶ Inexpensive to maintain and use
- ▶ Convenient and easy to have multiple identical USB flash drives available as backups

Two options are available for accessing key information that is on USB flash drives:

- ▶ USB flash drives are left inserted in the system always

If you want the system to restart automatically, a USB flash drive must be left inserted in all the nodes on the system. When you power on, all nodes then have access to the encryption key. This method requires that the physical environment where the system is located is secure. If the location is secure, it prevents an unauthorized person from making copies of the encryption keys, stealing the system, or accessing data that is stored on the system.

- ▶ USB flash drives are not left inserted into the system except as required

For the most secure operation, do not keep the USB flash drives inserted into the nodes on the system. However, this method requires that you manually insert the USB flash drives that contain copies of the encryption key in the nodes during operations that the system requires an encryption key to be present. USB flash drives that contain the keys must be stored securely to prevent theft or loss.

During operations that the system requires an encryption key to be present, the USB flash drives must be inserted manually into each node so data can be accessed. After the system completes unlocking the drives, the USB flash drives must be removed and stored securely to prevent theft or loss.

### 4.5.4 Encryption at rest with key servers

The following section describes the characteristics of using key servers for encryption and essential recommendations for key server configuration with IBM SAN Volume Controller.

## Key servers

Key servers have the following characteristics:

- ▶ Physical access to the system is not required to process a rekeying operation
- ▶ Support for businesses that have security requirements not to use USB ports
- ▶ Strong key generation
- ▶ Key self-replication and automatic backups
- ▶ Implementations follow an open standard that aids in interoperability
- ▶ Audit detail
- ▶ Ability to administer access to data separately from storage devices

Encryption key servers create and manage encryption keys that are used by the system. In environments with many systems, key servers distribute keys remotely without requiring physical access to the systems. A key server is a centralized system that generates, stores, and sends encryption keys to the system. If the key server provider supports replication of keys among multiple key servers, you can specify up to 4 key servers (one master and three clones) that connect to the system over both a public network or a separate private network.

The system supports enabling encryption using an IBM Security Key Lifecycle Manager key server. All key servers must be configured on the IBM Security Key Lifecycle Manager before defining the key servers in the management GUI. IBM Security Key Lifecycle Manager supports Key Management Interoperability Protocol (KMIP), which is a standard for encryption of stored data and management of cryptographic keys.

IBM Security Key Lifecycle Manager can be used to create managed keys for the system and provide access to these keys through a certificate. If you are configuring multiple key servers, use IBM Security Key Lifecycle Manager 2.6.0.2 or later. The additional key servers (clones) support more paths when delivering keys to the system; however, during rekeying only the path to the primary key server is used. When the system is rekeyed, secondary key servers are unavailable until the primary has replicated the new keys to these secondary key servers.

Replication must complete before keys can be used on the system. You can either schedule automatic replication or complete it manually with IBM Security Key Lifecycle Manager. During replication, key servers are not available to distribute keys or accept new keys. The time a replication completes on the IBM Security Key Lifecycle Manager depends on the number of key servers that are configured as clones, and the amount of key and certificate information that is being replicated.

The IBM Security Key Lifecycle Manager issues a completion message when the replication completes. Verify that all key servers contain replicated key and certificate information before keys are used on the system.

## Recommendations for key server configuration

The following section provides some essential recommendations for key server configuration with IBM SAN Volume Controller.

### ***Transport Layer Security***

Define the IBM Security Key Lifecycle Manager to use Transport Layer Security version 2 (TLSv2).

The default setting on IBM Security Key Lifecycle Manager since version 3.0.1 is TLSv1.2, but the IBM SAN Volume Controller only supports version 2. On the IBM Security Key Lifecycle Manager, set the value to SSL\_TLSv2, which is a set of protocols that includes TLSv1.2.

For more information about definitions, see [IBM Documentation](#).

Example 4-7 shows the example of a SKLMConfig.properties configuration file. The default path on a Linux based server is:

/opt/IBM/WebSphere/AppServer/products/sklm/config/SKLMConfig.properties.

*Example 4-7 Example of a SKLMConfig.properties configuration file*

---

```
#Mon Nov 20 18:37:01 EST 2017
KMIPLListener.ssl.port=5696
Audit.isSyslog=false
Audit.syslog.server.host=
TransportListener.ssl.timeout=10
Audit.handler.file.size=10000
user.gui.init.config=true
config.keystore.name=defaultKeyStore
tklm.encryption.password=D1181E14054B1E1526491F152A4A1F3B16491E3B160520151206
Audit.event.types=runtime,authorization,authentication,authorization_terminate,resource_management,key_management
tklm.lockout.enable=true
enableKeyRelease=false
TransportListener.tcp.port=3801
Audit.handler.file.name=logs/audit/sklm_audit.log
config.keystore.batchUpdateTimer=60000
Audit.eventQueue.max=0
enableClientCertPush=true
debug=none
tklm.encryption.keysize=256
TransportListener.tcp.timeout=10
backup.keycert.before.serving=false
TransportListener.ssl.protocols=SSL_TLSv2
Audit.syslog.isSSL=false
cert.validate=false
config.keystore.batchUpdateSize=10000
useSKIDefaultLabels=false
maximum.keycert.expiration.period.in.years=50
config.keystore.ssl.certalias=sklm
TransportListener.ssl.port=441
TransportListener.ssl.vulnerableciphers.patterns=_RC4_,RSA_EXPORT,_DES_
Audit.syslog.server.port=
tklm.lockout.attempts=3
fips=off
Audit.event.outcome=failure
```

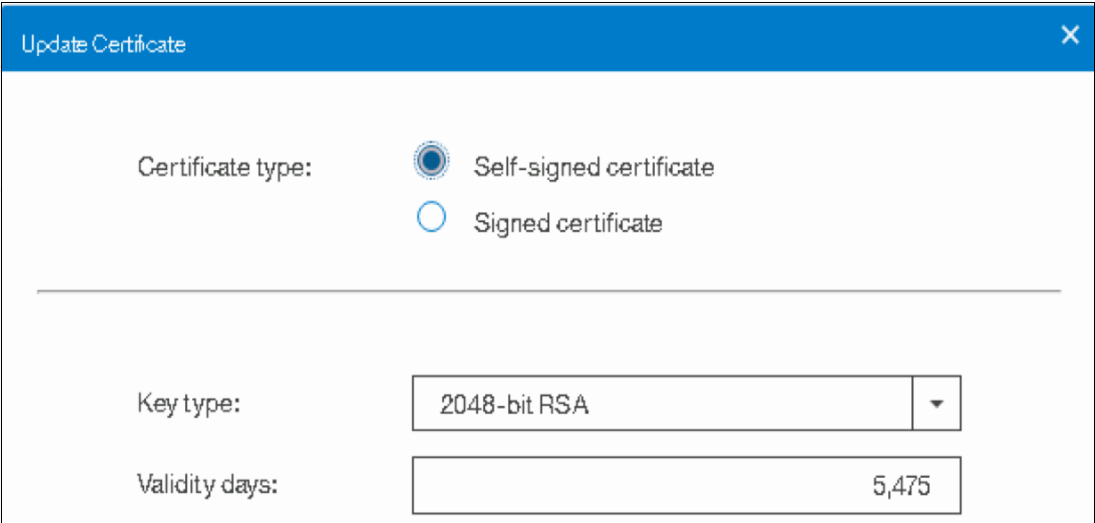
---

***Self-signed certificate type and validity period***

The default certificate type on IBM Security Key Lifecycle Manager server and IBM SAN Volume Controller is RSA. If it is intended to use different certificate type, ensure that you match the certificate type on both ends. The default certificate validity period is 1095 days on IBM Security Key Lifecycle Manager server and 5475 days on IBM SAN Volume Controller.

You can adjust the validity period to comply with specific security policies and always match the certificate validity period on IBM SAN Volume Controller and IBM Security Key Lifecycle Manager server. A mismatch causes a certificate authorization error and leads to unnecessary certificate exchange.

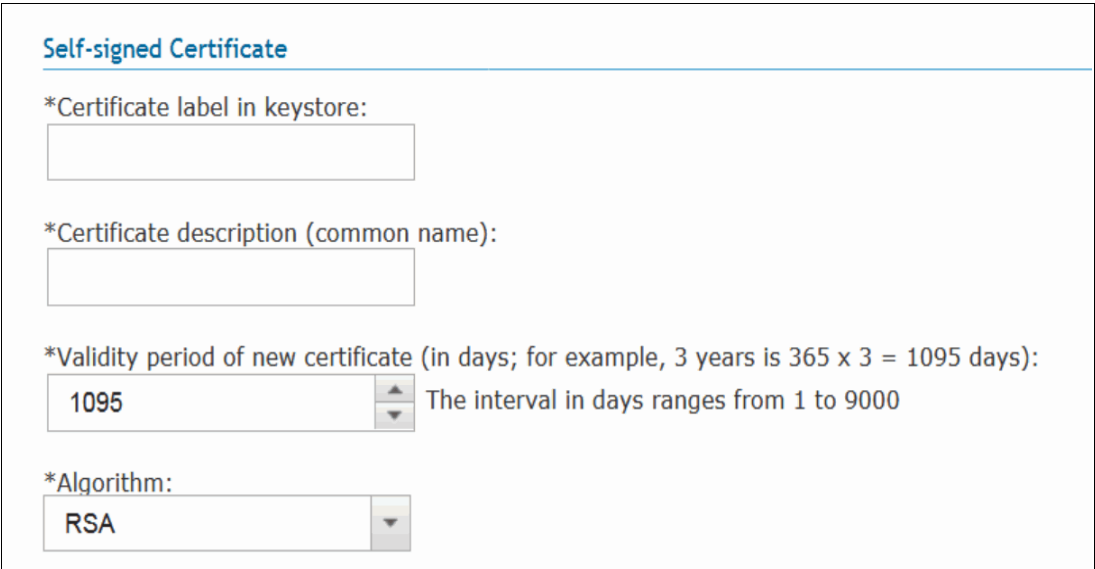
Figure 4-18 shows the default certificate type and validity period on IBM SAN Volume Controller.



The screenshot shows a dialog box titled "Update Certificate" with a close button (X) in the top right corner. The dialog contains two sections. The first section, "Certificate type:", has two radio buttons: "Self-signed certificate" (which is selected) and "Signed certificate". The second section, "Key type:", has a dropdown menu showing "2048-bit RSA". The third section, "Validity days:", has a text input field containing "5,475".

Figure 4-18 Update certificate on IBM SAN Volume Controller

Figure 4-19 shows the default certificate type and validity period on IBM Security Key Lifecycle Manager server.



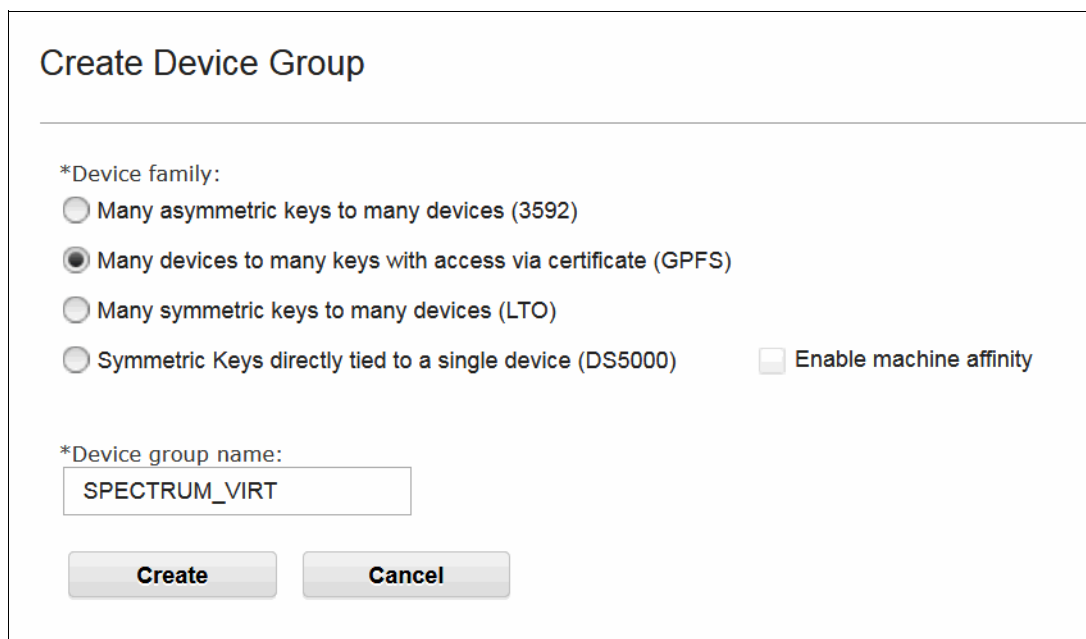
The screenshot shows a form titled "Self-signed Certificate". It contains four fields with asterisks indicating they are required: "\*Certificate label in keystore:" with an empty text box; "\*Certificate description (common name):" with an empty text box; "\*Validity period of new certificate (in days; for example, 3 years is 365 x 3 = 1095 days):" with a spinner box showing "1095" and a note "The interval in days ranges from 1 to 9000"; and "\*Algorithm:" with a dropdown menu showing "RSA".

Figure 4-19 Create self-signed certificate on IBM Security Key Lifecycle Manager server



### Device group configuration

The SPECTRUM\_VIRT device group is not predefined on IBM Security Key Lifecycle Manager; it must be created based on a GPFS device family, as shown in Figure 4-20.



Create Device Group

\*Device family:

☐ Many asymmetric keys to many devices (3592)

☒ Many devices to many keys with access via certificate (GPFS)

☐ Many symmetric keys to many devices (LTO)

☐ Symmetric Keys directly tied to a single device (DS5000) ☐ Enable machine affinity

\*Device group name:

SPECTRUM\_VIRT

Create Cancel

Figure 4-20 Create device group for IBM SAN Volume Controller

By default, IBM SAN Volume Controller has the *SPECTRUM\_VIRT* predefined in the encryption configuration wizard, and *SPECTRUM\_VIRT* contains all of the keys for the managed IBM SAN Volume Controller. However, It is possible to use different device groups if they are GPFS device family based; for example, one device group for each environment (Production or DR). Each device group maintains its own key database, and this approach allows more granular key management.

### Clone servers configuration management

The minimum replication interval on IBM Security Key Lifecycle Manager is one hour, as shown in Figure 4-21 on page 146. It is more practical to perform backup and restore or manual replication for the initial configuration to speed up the configuration synchronization.

Figure 4-21 shows the replication interval.

The screenshot displays the 'Advance Properties' tab of a configuration window. It includes a 'Replication backup destination directory' field with a 'Browse...' button, a 'Maximum number of replication files to keep before rollover' spinner set to 2, a 'Replication Scheduler Section' with radio buttons for 'Replication frequency (in hours)' (selected, set to 24) and 'Daily replication time (in HH:MM format)' (set to 00:00), and a 'Replication Log Section' with fields for 'Replication log file name' (replicationMaster.log), 'Maximum log file size (in KB)' (1000), and 'Maximum number of log files to keep' (30).

Figure 4-21 SKLM Replication Schedule

Also, the rekey process creates a new configuration on the IBM Security Key Lifecycle Manager server, and it is important not to wait for the next replication window but to manually synchronize the configuration to the additional key servers (clones); otherwise, an error message is generated by the IBM SAN Volume Controller system that indicates that the key is missing on the clones.

Example 4-8 shows an example of manually triggered replication.

*Example 4-8 Manually triggered replication*

---

```
/opt/IBM/WebSphere/AppServer/bin/wsadmin.sh -username SKLMAdmin -password
<password> -lang jython -c "print AdminTask.tklmReplicationNow()"
```

---

### **Encryption key management**

Only one active key is available for each encryption enabled IBM SAN Volume Controller system. The previously used key is deactivated after the rekey process. It is possible to delete the deactivated keys to keep the key database tidy and updated.

Figure 4-22 on page 147 shows the keys that are associated with a device group. In this example, the SG247933\_REDB00K device group contains one encryption-enabled IBM FlashSystem, and it has three associated keys. Only one of the keys is activated, and the other two were deactivated after the rekey process.



SG247933\_REDBOOK

The screen below allows you to add or delete certificate and their associated node name. As well as, modify the node name associated with a certificate. New keys can be added and a name associated with that key(s).




Add ▼

Modify

Delete

No filter applied

| Certificate UUID                                         | Name             | Endpoint Count |
|----------------------------------------------------------|------------------|----------------|
| CERTIFICATE-8a89d57-70cfd4f7-adda-4b29-9b1c-89c200fd1745 | sg247933_redbook | 2              |

Total: 1 Selected: 0

No filter applied

| Key UUID                                         | Name                  |
|--------------------------------------------------|-----------------------|
| KEY-8a89d57-15bf8f41-cea6-4df3-8f4e-be0c36318615 | mmm008a89d57000000870 |
| KEY-8a89d57-74edaef9-b6d9-4766-9b39-7e21d9911011 | mmm008a89d5700000086e |
| KEY-8a89d57-ebe5d5a1-8987-4aff-ab58-5f808a078269 | mmm008a89d5700000086f |

Total: 3 Selected: 0

Figure 4-22 Keys associated to a device group

Example 4-9 shows an example to check the state of the keys.

#### Example 4-9 Verify key state

```
/opt/IBM/WebSphere/AppServer/bin/wsadmin.sh -username SKLMAdmin -password
<password> -lang jython
wsadmin>print AdminTask.tklmKeyList('[-uuid
KEY-8a89d57-15bf8f41-cea6-4df3-8f4e-be0c36318615]')
CTGKM0001I Command succeeded.
```

```
uuid = KEY-8a89d57-15bf8f41-cea6-4df3-8f4e-be0c36318615
alias = mmm008a89d57000000870
key algorithm = AES
key store name = defaultKeyStore
key state = ACTIVE
creation date = 18/11/2017, 01:43:27 Greenwich Mean Time
expiration date = null
```

```
wsadmin>print AdminTask.tklmKeyList('[-uuid
KEY-8a89d57-74edaef9-b6d9-4766-9b39-7e21d9911011]')
CTGKM0001I Command succeeded.
```

```
uuid = KEY-8a89d57-74edaef9-b6d9-4766-9b39-7e21d9911011
alias = mmm008a89d5700000086e
key algorithm = AES
key store name = defaultKeyStore
key state = DEACTIVATED
creation date = 17/11/2017, 20:07:19 Greenwich Mean Time
expiration date = 17/11/2017, 23:18:37 Greenwich Mean Time
```

```
wsadmin>print AdminTask.tklmKeyList('[-uuid
KEY-8a89d57-ebe5d5a1-8987-4aff-ab58-5f808a078269]')
```

CTGKM0001I Command succeeded.

```
uuid = KEY-8a89d57-ebe5d5a1-8987-4aff-ab58-5f808a078269
alias = mmm008a89d5700000086f
key algorithm = AES
key store name = defaultKeyStore
key state = DEACTIVATED
creation date = 17/11/2017, 23:18:34 Greenwich Mean Time
expiration date = 18/11/2017, 01:43:32 Greenwich Mean Time
```

---

**Note:** The initial configuration, such as certificate exchange and Transport Layer Security configuration, is only required on the master IBM Security Key Lifecycle Manager server. The restore or replication process duplicates all of the required configurations to the clone servers.

If encryption was enabled on a pre-V7.8.0 code level system and the system is updated to V7.8.x or above, you must run a USB rekey operation to enable key server encryption. Run the **chencryption** command before you enable key server encryption. To perform a rekey operation, run the commands that are shown in Example 4-10.

*Example 4-10 Commands to enable key server encryption option on a system upgraded from pre-7.8.0*

---

```
chencryption -usb newkey -key prepare
chencryption -usb newkey -key commit
```

---

For more information about Encryption with Key Server, see [IBM Documentation](#).

## 4.6 Easy Tier, tiered, and balanced storage pools

Easy Tier was originally developed to provide the maximum performance benefit from a few SSDs or flash drives. Because of their low response times, high throughput, and IOPS-energy-efficient characteristics, SSDs and flash arrays were a welcome addition to the storage system, but initially their acquisition cost per Gigabyte (GB) was more than for HDDs.

By implementing an evolving almost AI-like algorithm, Easy Tier moved the most frequently accessed blocks of data to the lowest latency device. Therefore, it provides an exponential improvement in performance when compared to a small investment in SSD and flash capacity.

The industry moved on in the more than 10 years since Easy Tier was first introduced. The cost of SSD and flash-based technology meant that more users can deploy all-flash environments.

HDD-based large capacity NL-SAS drives are still the most cost-effective online storage devices. Although SSD and flash ended the 15 K RPM and 10 K RPM drive market, it has yet to reach a price point that competes with NL-SAS for lower performing workloads.

The use cases for Easy Tier changed, and most deployments now use “flash and trash” approaches, with 50% or more flash capacity and the remainder using NL-SAS.

Easy Tier also provides balancing within a tier. This configuration ensures that no one single component within a tier of the same capabilities is more heavily loaded than another. It does so to maintain an even latency across the tier and help to provide consistent and predictable performance.

As the industry strives to develop technologies that can enable higher throughput and lower latency than even flash, Easy Tier continues to provide user benefits. For example, Storage Class Memory (SCM) technologies, which were introduced to FlashSystem in 2020, now provide lower latency than even flash, but as with flash when first introduced, at a considerably higher cost of acquisition per GB.

Choosing the correct mix of drives and the data placement is critical to achieve optimal performance at the lowest cost. Maximum value can be derived by placing “hot” data with high I/O density and low response time requirements on the highest tier, while targeting lower tiers for “cooler” data, which is accessed more sequentially and at lower rates.

Easy Tier dynamically automates the ongoing placement of data among different storage tiers. It also can be enabled for internal and external storage to achieve optimal performance.

Also, the Easy Tier feature that is called *storage pool balancing* (introduced in V7.3) automatically moves extents within the same storage tier from overloaded to less loaded managed disks (MDisks). Storage pool balancing ensures that your data is optimally placed among all disks within storage pools.

### 4.6.1 Easy Tier concepts

IBM SAN Volume Controller products implement Easy Tier enterprise storage functions, which were originally designed with the development of Easy Tier on IBM DS8000 enterprise class storage systems. It enables automated subvolume data placement throughout different or within the same storage tiers. This feature intelligently aligns the system with current workload requirements and optimizes the use of high-performance storage, such as SSD, flash, and SCM.

Easy Tier reduces the I/O latency for hot spots, but it does not replace storage cache. Both Easy Tier and storage cache solve a similar access latency workload problem. However, these two methods weigh differently in the algorithmic construction that is based on *locality of reference*, recency, and frequency. Because Easy Tier monitors I/O performance from the device end (after cache), it can pick up the performance issues that cache cannot solve, and complement the overall storage system performance.

The primary benefit of Easy Tier is to reduce latency for hot spots; however, this benefit also includes an added benefit where the remaining “medium” (that is, not cold) data has less contention for its resources and performs better as a result (that is, lower latency).

In addition, Easy Tier can be used in a single tier pool to balance the workload across storage MDisks and ensures an even load on all MDisks in a tier or pool. Therefore, bottlenecks and convoying effects are removed when striped volumes are used. In a multitier pool, each tier is balanced.

In general, the storage environment’s I/O is monitored at a volume level, and the entire volume is always placed inside one suitable storage tier. Determining the amount of I/O, moving part of the underlying volume to an appropriate storage tier, and reacting to workload changes is too complex for manual operation. It is in this situation that the Easy Tier feature can be used.

Easy Tier is a performance optimization function that automatically migrates extents that belong to a volume between different storage tiers (see Figure 4-23) or the same storage tier (see Figure 4-25 on page 157). Because this migration works at the extent level, it is often referred to as *sublogical unit number (LUN) migration*. Movement of the extents is dynamic, nondisruptive, and is not visible from the host perspective. As a result of extent movement, the volume no longer has all its data in one tier; rather, it is in two or three tiers, or is balanced between MDisks in the same tier.

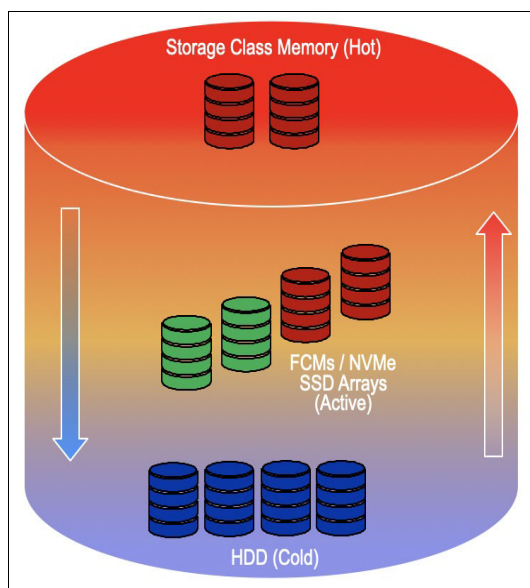


Figure 4-23 Easy Tier single volume, multiple tiers

You can enable Easy Tier on a per volume basis, except for non-fully allocated volumes in a DRP where Easy Tier is always enabled. It monitors the I/O activity and latency of the extents on all Easy Tier enabled volumes.

Based on the performance characteristics, Easy Tier creates an extent migration plan and dynamically moves (promotes) high activity or hot extents to a higher disk tier within the same storage pool. Generally, a new migration plan is generated on a stable system once every 24 hours. Instances might occur when Easy Tier reacts within 5 minutes; for example, when detecting an overload situation. For more information, see “Warm Promote” on page 152.

It also moves (demotes) extents whose activity dropped off, or cooled, from higher disk tier MDisk back to a lower tier MDisk. When Easy Tier runs in a storage pool rebalance mode, it moves extents from busy MDisk to less busy MDisk of the same type.

**Note:** Image mode and sequential volumes are not candidates for Easy Tier automatic data placement because all extents for those types of volumes must be on one specific MDisk and cannot be moved.

## 4.6.2 Easy Tier definitions

Easy Tier measures and classifies each extent into one of its three tiers. It performs this classification process by looking for the outliers in any system:

1. It looks for the *hottest* extents in the pool. These extents contain the most frequently accessed data of a suitable workload type (less than 64 KiB I/O). Easy Tier plans to migrate these extents into whatever set of extents that come from MDisks that are designated as the *hot tier*.
2. It looks for *coldest* extents in the pool, which are classed as having done < 1 I/O in the measurement period. These extents are planned to be migrated onto extents that come from the MDisks that are designated as the *cold tier*.

It is not necessary for Easy Tier to look for extents to place in the *middle tier*. By definition, if something is not designated as “hot” or “cold”, it stays or is moved to extents that come from MDisks in the middle tier.

With these three tier classifications, an Easy Tier pool can be optimized.

### Internal processing

The Easy Tier function includes the following main processes:

- I/O Monitoring

This process operates continuously and monitors volumes for host I/O activity. It collects performance statistics for each extent, and derives averages for a rolling 24-hour period of I/O activity.

Easy Tier makes allowances for large block I/Os; therefore, it considers only I/Os of up to 64 kilobytes (KiB) as migration candidates.

This process is efficient and adds a negligible processing resource use function to the IBM SAN Volume Controller nodes.

- Data Placement Advisor (DPA)

The DPA uses workload statistics to make a cost-benefit decision as to which extents are to be candidates for migration to a higher performance tier.

This process also identifies extents that can be migrated back to a lower tier.

- Data Migration Planner (DMP)

By using the extents that were identified, the DMP builds the extent migration plans for the storage pool. The DMP builds two plans:

- The Automatic Data Relocation (ADR mode) plan to migrate extents across adjacent tiers
- The Rebalance (RB mode) plan to migrate extents within the same tier

- Data migrator

This process involves the movement or migration of the volume’s extents up to, or down from, the higher disk tier. The extent migration rate is capped so that a maximum of up to 12 GiB every five minutes is migrated, which equates to approximately 3.4 TiB per day that is migrated between disk tiers.

**Note:** You can increase the target migration rate to 48 GiB every 5 minutes by temporarily enabling accelerated mode. For more information, see “Easy Tier acceleration” on page 165.

When active, Easy Tier performs the following actions across the tiers:

- ▶ **Promote**  
Moves the hotter extents to a higher performance tier with available capacity. Promote occurs within adjacent tiers.
- ▶ **Demote**  
Demotes colder extents from a higher tier to a lower tier. Demote occurs within adjacent tiers.
- ▶ **Swap**  
Exchanges cold extent in an upper tier with hot extent in a lower tier.
- ▶ **Warm demote**  
Prevents performance overload of a tier by demoting a warm extent to a lower tier. This process is triggered when bandwidth or IOPS exceeds predefined threshold. If you see these operations, it is a trigger to suggest you should add capacity to the higher tier.
- ▶ **Warm promote**  
Introduced with version 7.8, this feature addresses the situation where a lower tier suddenly becomes active. Instead of waiting for the next migration plan, Easy Tier can react immediately. Warm promote acts in a similar way to warm demote. If the 5-minute average performance shows that a layer is overloaded, Easy Tier immediately starts to promote extents until the condition is relieved. This is often referred to as “overload protection”
- ▶ **Cold demote**  
Demotes inactive (or cold) extents that are on a higher performance tier to its adjacent lower-cost tier. In that way Easy Tier automatically frees extents on the higher storage tier before the extents on the lower tier become hot. Only supported between HDD tiers.
- ▶ **Expanded cold demote**  
Demotes appropriate sequential workloads to the lowest tier to better use nearline disk bandwidth.
- ▶ **Auto rebalance**  
Redistributes extents within a tier to balance use across MDisk for maximum performance. This process moves hot extents from highly used MDisk to lesser used MDisk, and exchanges extents between high used MDisk and low used MDisk.

Easy Tier attempts to migrate the most active volume extents up to SSD first.

If a new migration plan is generated before the completion of the previous plan, the previous migration plan and any queued extents that are not yet relocated are abandoned. However, any migrations that are still applicable are included in the new plan.

**Note:** Extent migration occurs between adjacent tiers only. For example, in a three-tiered storage pool, Easy Tier does not move extents from the flash tier directly to the nearline tier and vice versa without moving them first to the enterprise tier.



Easy Tier extent migration types are shown in Figure 4-24.

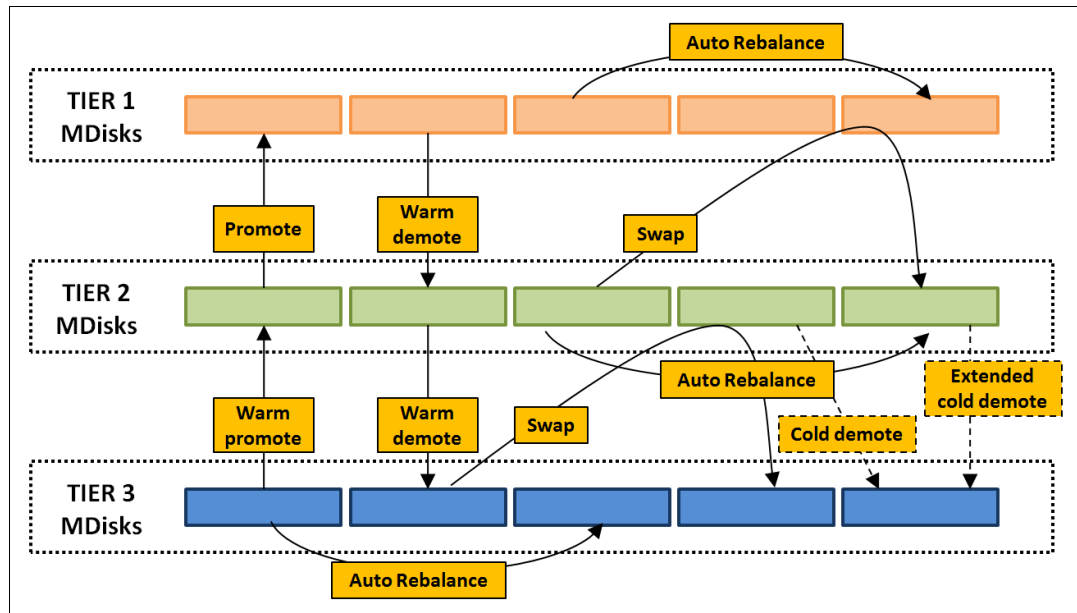


Figure 4-24 Easy Tier extent migration types

### 4.6.3 Easy Tier operating modes

Easy Tier includes the following main operating modes:

- ▶ Off
- ▶ On
- ▶ Automatic
- ▶ Measure

On some FlashSystem 50x0 systems, Easy Tier is a licensed feature. If the license is not present and Easy Tier is set to Auto or On, the system runs in Measure mode.

**Options:** The Easy Tier function can be turned on or off at the storage pool level *and* at the volume level, except for non-fully allocated volumes in a DRP where Easy Tier is always enabled.

#### Easy Tier off mode

With Easy Tier turned off, no statistics are recorded, and no cross-tier extent migration occurs.

#### Measure mode

Easy Tier can be run in an evaluation or measurement-only mode and collects usage statistics for each extent in a storage pool where the Easy Tier value is set to measure.

This collection is typically done for a single-tier pool so that the benefits of adding performance tiers to the pool can be evaluated before any major hardware acquisition.

The heat and activity of each extent can be viewed in the GUI under the **Monitoring** → **Easy Tier Reports**. For more information, see 4.6.10, “Monitoring Easy Tier using the GUI” on page 170.

## Automatic mode

In Automatic mode, the storage pool parameter **-easytier auto** must be set, and the volumes in the pool must include **-easytier on**.

The behavior of Easy Tier depends on the pool configuration. Consider the following points:

- ▶ If the pool contains only MDisks with a single tier type, the pool is in balancing mode.
- ▶ If the pool contains MDisks with more than one tier type, the pool runs automatic data placement and migration in addition to balancing within each tier.

Dynamic data movement is transparent to the host server and application users of the data, other than providing improved performance. Extents are automatically migrated (see “Implementation rules” on page 163).

Some situations might exist in which the Easy Tier setting is “auto”, but the system is running in monitoring mode only; for example, with unsupported tier types or if the Easy Tier license is not yet enabled (see Table 4-9 on page 159).

The GUI provides the same reports as available in measuring mode and the data movement report that shows the breakdown of the migration events that are triggered by Easy Tier. These migrations are reported in terms of the migration types, as described in “Internal processing” on page 151.

## Easy Tier On mode

This mode forces Easy Tier to perform the tasks as in Automatic mode.

For example, when Easy Tier detects an unsupported set of tier types in a pool (see Table 4-9 on page 159), the use of On mode forces Easy Tier to the active state and it performs to the best of its ability. The system raises an alert and an associated Directed Maintenance Procedure guides you to fix the unsupported tier types.

**Important:** Avoid creating a pool with more than three tiers. Although the system attempts to create “buckets”, you might end up with Easy Tier running only in measure mode.

These configurations are unsupported because they can cause a performance problem in the longer term; for example, disparate performance within a single tier.

The ability to override the automatic mode is provided to enable temporary migration from an older set of tiers to new tiers and must be rectified as soon as possible.

## Storage pool balancing

This feature assesses the extents that are written in a pool, and balances them automatically across all MDisks within the pool. This process works along with Easy Tier when multiple classes of disks exist in a single pool. In such a case, Easy Tier moves extents between the different tiers, and storage pool balancing moves extents within the same tier, to enable a balance in terms of workload across all MDisks that belong to a tier.

Balancing is looking to maintain equivalent latency across all MDisks in a tier, which can result in different capacity use across the MDisks. However, performance balancing is preferable to capacity balancing in almost all cases.

The process automatically balances data when new MDisks are added to a pool, even if the pool contains only a single type of drive.

Balancing is automatically active on all storage pools, no matter the Easy Tier setting. For a single tier pool, the Easy Tier state reports as balancing.

**Note:** Storage pool balancing can be used to balance extents when mixing different size disks of the same performance tier. For example, when adding larger capacity drives to a pool with smaller capacity drives of the same class, storage pool balancing redistributes the extents to take advantage of the extra performance of the new MDisks.

## Easy Tier mode settings

The Easy Tier setting can be changed on a storage pool and volume level. Depending on the Easy Tier setting and the number of tiers in the storage pool, Easy Tier services might function in a different way. Table 4-7 lists the possible combinations of Easy Tier setting.

Table 4-7 Easy Tier settings

| Storage pool Easy Tier setting | Number of tiers in the storage pool | Volume copy Easy Tier setting | Volume copy Easy Tier status |
|--------------------------------|-------------------------------------|-------------------------------|------------------------------|
| Off                            | One or more                         | off                           | inactive (see note 2)        |
|                                |                                     | on                            | inactive (see note 2)        |
| Measure                        | One or More                         | off                           | measured (see note 3)        |
|                                |                                     | on                            | measured (see note 3)        |
| Auto                           | One                                 | off                           | measured (see note 3)        |
|                                |                                     | on                            | balanced (see note 4)        |
|                                | Two - four                          | off                           | measured (see note 3)        |
|                                |                                     | on                            | active (see note 5 & 6)      |
|                                | Five                                | any                           | measured (see note 3)        |
|                                |                                     |                               |                              |
| On                             | One                                 | off                           | measured (see note 3)        |
|                                |                                     | on                            | balanced (see note 4)        |
|                                | Two - four                          | off                           | measured (see note 3)        |
|                                |                                     | on                            | active (see note 5)          |
|                                | Five                                | off                           | measured (see note 3)        |
|                                |                                     | on                            | active (see note 6)          |

**Table notes:** The following points correspond to the notations in the last column of Table 4-7 on page 155:

1. If the volume copy is in image or sequential mode, or is being migrated, the volume copy Easy Tier status is measured rather than active.
2. When the volume copy status is inactive, no Easy Tier functions are enabled for that volume copy.
3. When the volume copy status is measured, the Easy Tier function collects usage statistics for the volume, but automatic data placement is not active.
4. When the volume copy status is balanced, the Easy Tier function enables performance-based pool balancing for that volume copy.
5. When the volume copy status is active, the Easy Tier function operates in automatic data placement mode for that volume.
6. When five tiers, or some four-tier configurations are used and Easy Tier is in the on state, this configuration forces Easy Tier to operate but might not behave as expected (see Table 4-9 on page 159).

The default Easy Tier setting for a storage pool is Auto, and the default Easy Tier setting for a volume copy is On. Therefore, Easy Tier functions (except pool performance balancing) are disabled for storage pools with a single tier. Automatic data placement mode is enabled by default for all striped volume copies in a storage pool with two or more tiers.

#### 4.6.4 MDisk tier types

Until now, we discussed the three Easy Tier tier types (“hot”, “medium”, and “cold”). Because these types are generic “buckets” that Easy Tier uses to build a set of extents that belong to each tier, we must tell Easy Tier which MDisks belong to which bucket.

The type of disk and RAID geometry that is used by internal or external MDisks defines their expected performance characteristics. We use these characteristics to help define a tier type for each MDisk in the system.

Five tier types can be assigned. The tables in this section use the numbers from the following list as a shorthand for the tier name:s

1. tier\_scm that represents Storage Class Memory MDisks
2. tier0\_flash that represents enterprise flash technology, including FCM
3. tier1\_flash that represents lower performing tier1 flash technology (lower DWPD)
4. tier\_enterprise that represents enterprise HDD technology (both 10K and 15K RPM)
5. tier\_nearline that represents nearline HDD technology (7.2K RPM)

Consider the following points:

- ▶ Easy Tier is designed to operate with up to three tiers of storage: “hot”, “medium”, “cold”
- ▶ An MDisk can belong to one tier type only.
- ▶ As of this writing, five different MDisk tier types are available.
- ▶ Internal MDisks have their tier type set automatically.
- ▶ External MDisks default to the “enterprise” tier and might need to be changed by the user.
- ▶ The number of MDisk tier types found in a pool determines if the pool is a single-tier pool or a multitier pool.

**Attention:** As described in 4.6.5, “Changing the tier type of an MDisk” on page 160, IBM SAN Volume Controller do not automatically detect the type of external MDisks. Instead, all external MDisks are initially put into the enterprise tier by default. The administrator must then manually change the MDisks tier and add them to storage pools.

## Single-tier storage pools

Figure 4-25 shows a scenario in which a single storage pool is populated with MDisks that are presented by an external storage controller. In this solution, the striped volumes can be measured by Easy Tier, and can benefit from *storage pool balancing* mode, which moves extents between MDisks of the same type.

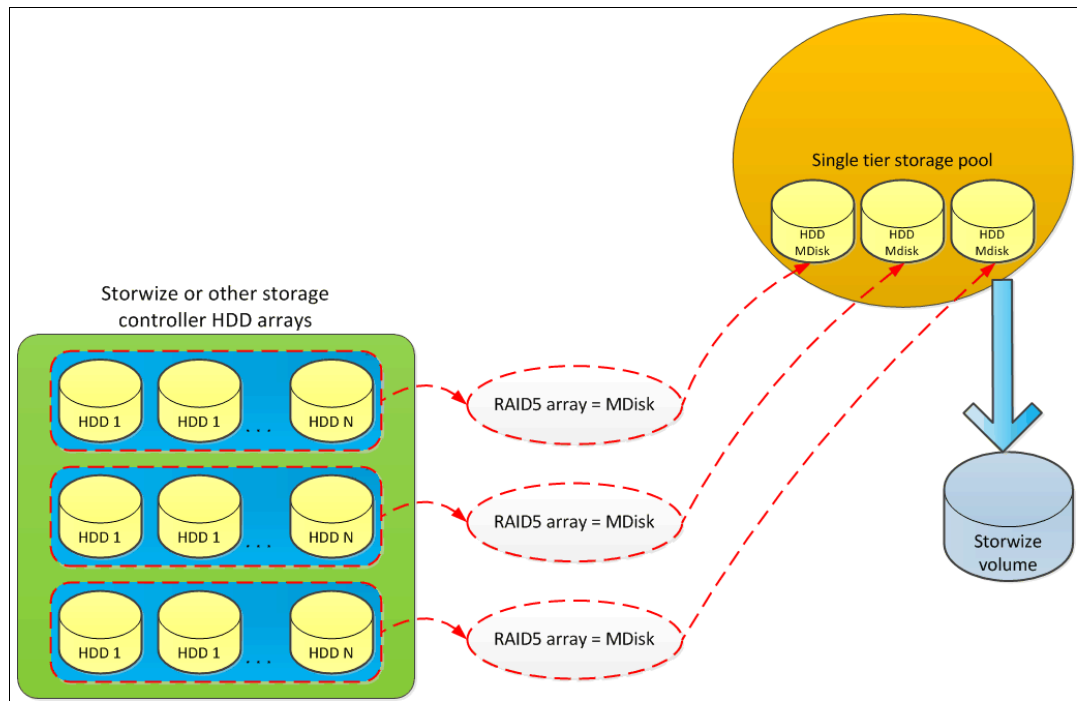


Figure 4-25 Single tier storage pool with striped volume

MDisks that are used in a single-tier storage pool should have the same hardware characteristics. These characteristics include the same RAID type, RAID array size, disk type, disk RPM, and controller performance characteristics.

For external MDisks, attempt to create all MDisks with the same RAID geometry (number of disks). If this issue cannot be avoided, you can modify the Easy Tier load setting to manually balance the workload; however, care must be taken. For more information, see “MDisk Easy Tier load” on page 166

For internal MDisks, the system can tolerate with different geometries because the number of drives is reported to Easy Tier, which then uses the Overload Protection information to balance the workload appropriately. For more information, see 4.6.6, “Easy Tier overload protection” on page 162.

## Multitier storage pools

A multitier storage pool features a mix of MDisks with more than one type of MDisk tier attribute. For example, this pool can be a storage pool that contains a mix of enterprise and SSD MDisks or enterprise and NL-SAS MDisks.

Figure 4-26 shows a scenario in which a storage pool is populated with three different MDisk types (one belonging to an SSD array, one belonging to an SAS HDD array, and one belonging to an NL-SAS HDD array). Although this example shows RAID 5 arrays, other RAID types also can be used.

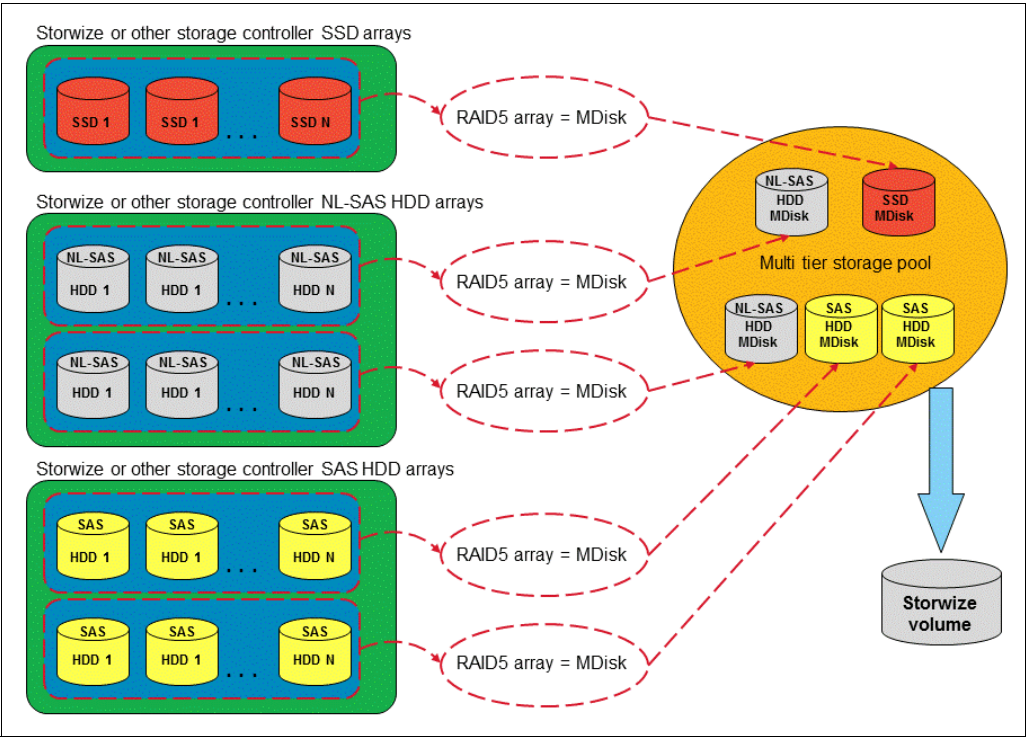


Figure 4-26 Multitier storage pool with striped volume

**Note:** If you add MDisks to a pool and they include or you assign more than three tier types, Easy Tier attempts to group two or more of the tier types into a single “bucket” and use them both as the “middle” or “cold” tier. The groupings are listed in table Table 4-9 on page 159.

However, overload protection and pool balancing might result in a bias on the load being placed on those MDisks despite them being in the same “bucket”.

### Easy Tier mapping to MDisk tier types

The five MDisk tier types are mapped to the three Easy Tier tiers, depending on the pool configuration, as listed in Table 4-8.

Table 4-8 Recommended 3-tier Easy Tier mapping policy

| Tier mix    | 1+2,<br>1+3,<br>1+4,<br>1+5 | 2+3,<br>2+4,<br>2+5 | 3+4,<br>3+5 | 4+5 | 1+2+3,<br>1+2+4,<br>1+2+5 | 1+3+4,<br>1+3+5 | 1+4+5,<br>2+4+5,<br>3+4+5 | 2+3+4,<br>2+3+5 |
|-------------|-----------------------------|---------------------|-------------|-----|---------------------------|-----------------|---------------------------|-----------------|
| Hot Tier    | 1                           | 2                   |             |     | 1                         | 1               | 1 or<br>2 or<br>3         | 2               |
| Middle Tier | 2 or<br>3 or<br>4 or        | 3 or<br>4 or        | 3           | 4   | 2                         | 3               | 4                         | 3               |

| Tier mix  | 1+2,<br>1+3,<br>1+4,<br>1+5 | 2+3,<br>2+4,<br>2+5 | 3+4,<br>3+5 | 4+5 | 1+2+3,<br>1+2+4,<br>1+2+5 | 1+3+4,<br>1+3+5 | 1+4+5,<br>2+4+5,<br>3+4+5 | 2+3+4,<br>2+3+5 |
|-----------|-----------------------------|---------------------|-------------|-----|---------------------------|-----------------|---------------------------|-----------------|
| Cold Tier | 5                           | 5                   | 4 or 5      | 5   | 3 or<br>4 or<br>5         | 4 or<br>5       | 5                         | 4 or<br>5       |

### ***Four- and five-tier pools***

In general, Easy Tier attempts to place tier\_enterprise (4) and tier1\_flash (3) based tiers into the one bucket to reduce the number of tiers defined in a pool to 3 (see Table 4-9).

*Table 4-9 4 and 5 Tier mapping policy4 and 5 Tier mapping policy*

| Tier Mix    | 1+2+3+4,<br>1+2+3+5,<br>1+2+4+5 | 1+3+4+5,<br>2+3+4+5 | 1+2+3+4+5                      |
|-------------|---------------------------------|---------------------|--------------------------------|
| Hot Tier    | Not supported;<br>measure only  | 1 or 2              | Not supported;<br>measure only |
| Middle Tier |                                 | 3 and 4             |                                |
| Cold Tier   |                                 | 5                   |                                |

If you create a pool with all five tiers or one of the unsupported four-tier pools and Easy Tier is set to “auto” mode, Easy Tier enters “measure” mode and measures the statistics but does not move any extents. Manually remove one or more MDiskS to return to a supported tier configuration.

**Important:** Avoid creating a pool with more than three tiers. Although the system attempts to create “buckets”, the result might be that Easy Tier runs in measure mode only.

### ***Temporary unsupported four- or five-tier mapping***

If you need to temporarily have four or five tiers defined in a pool, and you end up with an unsupported configuration, you can force Easy Tier to migrate data by setting the Easy Tier mode to “on”.

**Attention:** Extreme caution should be used and a full understanding of the implications should be made before forcing Easy Tier to run in this mode.

This setting is provided to allow temporary migrations where it is unavoidable to create one of these unsupported configurations. The implications are that long term use in this mode can cause performance issues due to the grouping of unlike MDiskS within a single Easy Tier tier.

For these configurations, the mapping that is listed in Table 4-10 are used by Easy Tier.

*Table 4-10 Unsupported temporary 4 and 5 Tier mapping policy*

| Tier mix    | 1+2+3+4,<br>1+2+3+5 | 1+2+4+5 | 1+2+3+4+5 |
|-------------|---------------------|---------|-----------|
| Hot tier    | 1                   | 1       | 1         |
| Middle tier | 2 & 3               | 2       | 2 & 3     |
| Cold tier   | 4 or 5              | 4 and 5 | 4 and 5   |





It is also possible to change the MDisk tier from the graphical user interface (GUI), but this change applies to external MDisks only. To change the tier, complete the following steps:

1. Click **Pools** → **External Storage** and click the **Plus** sign (+) that is next to the controller that owns the MDisks for which you want to change the tier.
2. Right-click the wanted MDisk and select **Modify Tier** (see Figure 4-27).

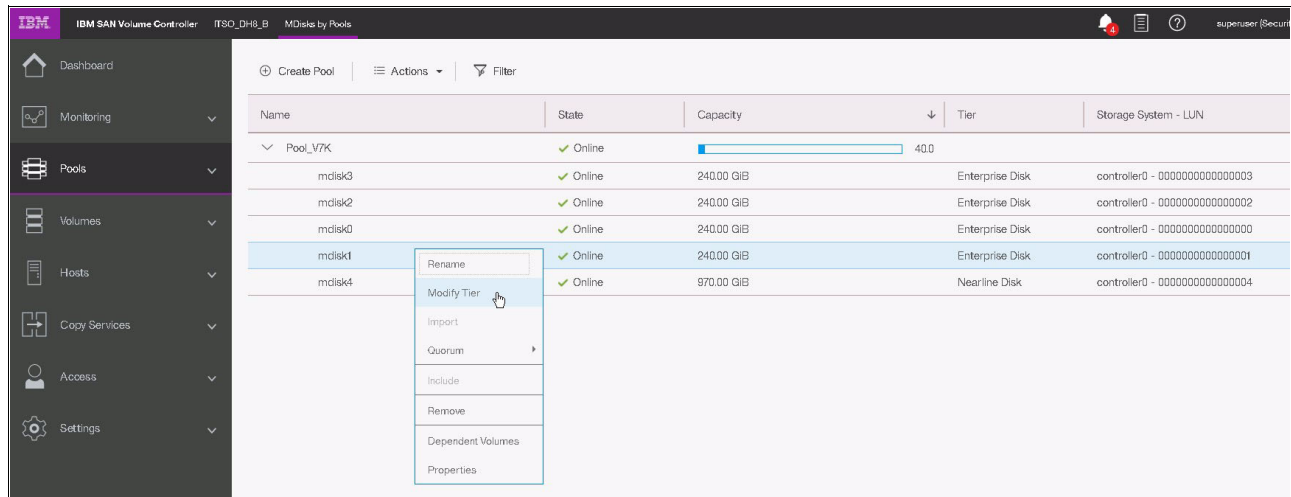


Figure 4-27 Change the MDisk tier

The new window opens with options to change the tier (see Figure 4-28).

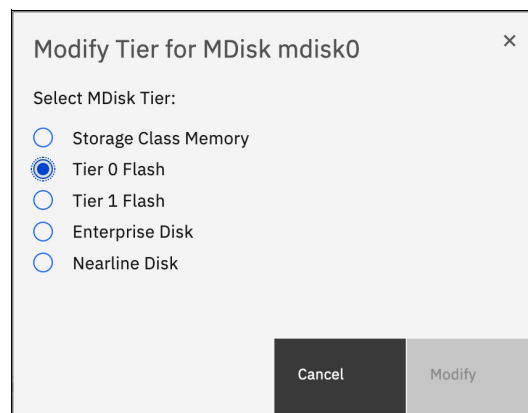


Figure 4-28 Select wanted MDisk tier

This change occurs online and has no effect on hosts or the availability of the volumes.

3. If you do not see the Tier column, right-click the blue title row and select the **Tier** check box, as shown in Figure 4-29.

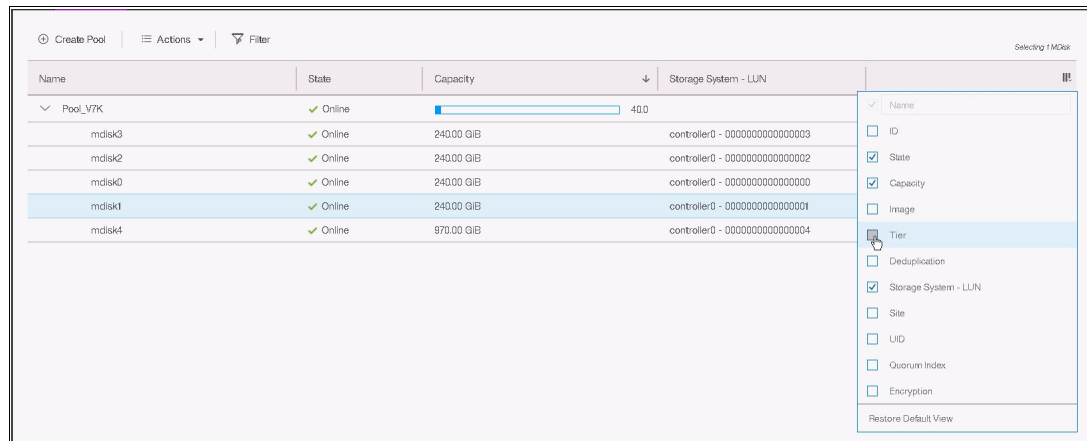


Figure 4-29 Customizing the title row to show the tier column

## 4.6.6 Easy Tier overload protection

Easy Tier is defined as a “greedy” algorithm. That is, it always attempts to improve the performance of the system by moving as much data as possible to the hot tier.

If overload protection is not used, Easy Tier attempts to use every extent on the hot tier. In some cases, this issue leads to overloading the hot tier MDisk and creates a performance problem.

Therefore, Easy Tier implements overload protection to ensure that it does not move too much workload onto the hot tier. If this protection is triggered, no other extents are moved onto that tier while the overload is detected. Extents can still be swapped; therefore, if one extent becomes colder and another hotter, they can be swapped.

To implement overload protection, Easy Tier must understand the capabilities of an MDisk. For internal MDisk, this understanding is handled automatically because the system can instruct Easy Tier as to the type of drive and RAID geometry (for example, 8+P+Q); therefore, the system can calculate the expected performance ceiling for any internal MDisk.

With external MDisk, the only measure or details we have is the storage controller type. Therefore, we know if the controller is an Enterprise, Midrange, or Entry level system and can make some assumptions about the load it can handle.

However, external MDisk cannot automatically have their MDisk tier type or “Easy Tier Load” defined. You must set the tier type manually and (if wanted), modify the load setting. For more information about Easy Tier load, see “MDisk Easy Tier load” on page 166.

Overload Protection is also used by the “warm promote” functionality. If Easy Tier detects a sudden change on a cold tier in which a workload is causing overloading of the cold tier MDisk, it can quickly react and recommend migration of the extents to the middle tier. This feature is useful when provisioning new volumes that overrun the capacity of the middle tier, or when no middle tier is present; for example, with Flash and Nearline only configurations.

## 4.6.7 Removing an MDisk from an Easy Tier pool

When you remove an MDisk from a pool that still includes defined volumes, and that pool is an Easy Tier pool, the extents that are still used on the MDisk you are removing are migrated to other free extents in the pool.

Easy Tier attempts to migrate the extents to another extent within the same tier. However, if not enough space is available in the same tier, Easy Tier selects the highest priority tier with free capacity. The priority is defined as listed in Table 4-11.

Table 4-11 Migration target tier priorities

| Tier of MDisk being removed | Target tier priority (select highest with free capacity) |                 |               |                 |               |
|-----------------------------|----------------------------------------------------------|-----------------|---------------|-----------------|---------------|
|                             | 1                                                        | 2               | 3             | 4               | 5             |
| tier_scm                    | tier_scm                                                 | tier0_flash     | tier1_flash   | tier_enterprise | tier_nearline |
| tier0_flash                 | tier0_flash                                              | tier_scm        | tier1_flash   | tier_enterprise | tier_nearline |
| tier1_flash                 | tier1_flash                                              | tier0_flash     | tier_scm      | tier_enterprise | tier_nearline |
| tier_enterprise             | tier_enterprise                                          | tier1_flash     | tier_nearline | tier0_flash     | tier_scm      |
| tier_nearline               | tier_nearline                                            | tier_enterprise | tier1_flash   | tier0_flash     | tier_scm      |

These tiers are chosen to optimize for the typical migration cases; for example, replacing the Enterprise HDD tier with Tier1 Flash arrays or replacing Nearline HDD with Tier1 Flash arrays.

## 4.6.8 Easy Tier implementation considerations

Easy Tier is included as part of the IBM Spectrum Virtualize code. For Easy Tier to migrate extents between different tier disks, you must have storage available that offers different tiers (for example, a mix of Flash and HDD). With single tier (homogeneous) pools, Easy Tier uses storage pool balancing only.

**Important:** Easy Tier uses the extent migration capabilities of IBM Spectrum Virtualize. These migrations require some free capacity, as an extent first is cloned to a new extent before the old extent is returned to the free capacity in the relevant tier.

It is recommended that a minimum of 16 extents are needed for Easy Tier to operate. However, if only 16 extents are available, Easy Tier can move at most 16 extents at a time.

Do not allocate 100% of the storage pool to volumes or Easy Tier and storage pool balancing does not function.

### Implementation rules

Remember the following implementation and operational rules when you use Easy Tier:

- ▶ Easy Tier automatic data placement is not supported on image mode or sequential volumes. I/O monitoring for such volumes is supported, but you cannot migrate extents on these volumes unless you convert image or sequential volume copies to striped volumes.
- ▶ Automatic data placement and extent I/O activity monitors are supported on each copy of a mirrored volume. Easy Tier works with each copy independently of the other copy.

**Volume mirroring consideration:** Volume mirroring can have different workload characteristics on each copy of the data because reads are normally directed to the primary copy and writes occur to both copies. Therefore, the number of extents that Easy Tier migrates between the tiers might be different for each copy.

- ▶ If possible, the IBM SAN Volume Controller system creates volumes or expands volumes by using extents from MDisk from the HDD tier. However, if necessary, it uses extents from MDisk from the SSD tier.
- ▶ Do not provision the 100% of an Easy Tier enabled pool capacity. Reserve at least 16 extents for each tier for the Easy Tier movement operations.

When a volume is migrated out of a storage pool that is managed with Easy Tier, Easy Tier automatic data placement mode is no longer active on that volume. Automatic data placement is also turned off while a volume is being migrated, even when it is between pools that both have Easy Tier automatic data placement enabled. Automatic data placement for the volume is re-enabled when the migration is complete.

## Limitations

When you use Easy Tier on the IBM SAN Volume Controller system, consider the following limitations:

- ▶ Removing an MDisk by using the **-force** parameter  
When an MDisk is deleted from a storage pool with the **-force** parameter, extents in use are migrated to MDisk in the same tier as the MDisk that is being removed, if possible. If insufficient extents exist in that tier, extents from the other tier are used.
- ▶ Migrating extents  
When Easy Tier automatic data placement is enabled for a volume, you cannot use the **migrateexts** CLI command on that volume.
- ▶ Migrating a volume to another storage pool  
When IBM SAN Volume Controller system migrates a volume to a new storage pool, Easy Tier automatic data placement between the two tiers is temporarily suspended. After the volume is migrated to its new storage pool, Easy Tier automatic data placement between resumes for the moved volume, if appropriate.  
When the system migrates a volume from one storage pool to another, it attempts to migrate each extent to an extent in the new storage pool from the same tier as the original extent. In several cases, such as where a target tier is unavailable, another tier is used based on the same priority rules, as described in 4.6.7, “Removing an MDisk from an Easy Tier pool” on page 163.
- ▶ Migrating a volume to an image mode copy  
Easy Tier automatic data placement does not support image mode. When a volume with active Easy Tier automatic data placement mode is migrated to an image mode volume, Easy Tier automatic data placement mode is no longer active on that volume.
- ▶ Image mode and sequential volumes cannot be candidates for automatic data placement. However, Easy Tier supports evaluation mode for image mode volumes.

## 4.6.9 Easy Tier settings

The Easy Tier setting for storage pools and volumes can be changed from the command-line interface only. All of the changes are done online without effecting hosts or data availability.

### Turning on and off Easy Tier

Run the **chvdisk** command to turn on and off on Easy Tier on selected volumes. Run the **chmdiskgrp** command to change status of Easy Tier on selected storage pools, as shown in Example 4-12.

*Example 4-12 Changing Easy Tier setting*

---

```
IBM_FlashSystem:V7000 Gen 2:superuser>chvdisk -easytier on test_vol_2
IBM_FlashSystem:V7000 Gen 2:superuser>chmdiskgrp -easytier auto test_pool_1
```

---

### Tuning Easy Tier

It is also possible to change more advanced parameters of Easy Tier. These parameters should be used with caution because changing the default values can affect system performance.

#### **Easy Tier acceleration**

The first setting is called *Easy Tier acceleration*. This system-wide setting is disabled by default. Turning on this setting makes Easy Tier move extents up to four times faster than when in default setting. In accelerate mode, Easy Tier can move up to 48 GiB every 5 minutes; in normal mode, it moves up to 12 GiB. Enabling Easy Tier acceleration is advised only during periods of low system activity. The following use cases for acceleration are the most likely:

- ▶ When installing a new system, accelerating Easy Tier quickly reaches a steady state and reduces the time that is needed to reach an optimal configuration. Easy Tier acceleration applies to single-tier and multitier pools. In a single-tier pool, Easy Tier acceleration allows balancing to spread the workload quickly. In a multitier pool, it allows inter-tier movement and balancing within each tier.
- ▶ When adding capacity to the pool, accelerating Easy Tier can quickly spread existing volumes onto the new MDisk by way of pool balancing. It can also help if you added capacity to stop warm demote operations. In this case, Easy Tier knows that specific extents are hot and were demoted only because of a lack of space, or because Overload Protection was triggered.
- ▶ When migrating the volumes between the storage pools in cases where the target storage pool has more tiers than the source storage pool, accelerating Easy Tier can quickly promote or demote extents in the target pool.

This setting can be changed online, without effecting host or data availability. To turn on or off Easy Tier acceleration mode, run the following command:

```
chsystem -easytieracceleration <on/off>
```

**Important:** Do not leave accelerated mode on indefinitely. It is recommended to run in accelerated mode only for a few days to weeks to enable Easy Tier to reach a steady state quickly. After the system is performing fewer migration operations, disable accelerated mode to ensure Easy Tier does not affect system performance.

### ***MDisk Easy Tier load***

The second setting is called *MDisk Easy Tier load*. This setting is set on an individual MDisk basis, and indicates how much load Easy Tier can put on that particular MDisk. This setting was introduced to handle situations where Easy Tier is underutilizing or overutilizing an external MDisk.

This setting cannot be changed for internal MDisk (arrays) because the system can determine the exact load that an internal MDisk can handle based on the drive technology type, the number of drives, and type of RAID in use per MDisk.

For an external MDisk, Easy Tier uses specific performance profiles that are based on the characteristics of the external controller and the tier that is assigned to the MDisk. These performance profiles are generic, which means that they do not consider the back-end configuration. For example, the same performance profile is used for a DS8000 with 300 GB 15K RPM and 1.8 TB 10K RPM.

This feature is provided for advanced users to change the Easy Tier load setting to better align it with a specific external controller configuration.

**Note:** The load setting is used with the MDisk tier type setting to calculate the number concurrent I/O and expected latency from the MDisk. Setting this value incorrectly, or using the wrong MDisk tier type, can have a detrimental effect on overall pool performance.

The following values can be set to each MDisk for the Easy Tier load:

- ▶ Default
- ▶ Low
- ▶ Medium
- ▶ High
- ▶ Very high

The system uses a default setting that is based on a controller performance profile and the MDisk tier setting of the presented MDisk.

Change the default setting to any other value only when you are certain that a specific MDisk is underutilized and can handle more load, or that the MDisk is overutilized and the load should be lowered. Change this setting to very high only for SDD and Flash MDisk.

This setting can be changed online, without effecting the hosts or data availability.

To change this setting, run the following command:

```
chmdisk -easytierload high mdisk0
```

**Important:** Consider the following points:

- ▶ When IBM SAN Volume Controller is used with FlashSystem back-end storage, it is recommended to set the Easy Tier load to high for FlashSystem MDisk other than FlashSystem 50x0 where the default is recommended.

The same is recommended for modern high performance all-flash storage controllers from other vendors.

- ▶ After changing the load setting, make a note of the old and new settings and record the date and time of the change. Use Storage Insights to review the performance of the pool in the coming days to ensure that you did not inadvertently degrade the performance of the pool.

You can also gradually increase the load setting and validate at each change that you are seeing an increase in throughput without a corresponding detrimental increase in latency (and vice versa if you are decreasing the load setting).

## Extent size considerations

The extent size determines the granularity level at which Easy Tier operates, which is the size of the chunk of data that Easy Tier moves across the tiers. By definition, a hot extent refers to an extent that has more I/O workload compared to other extents in the same pool and in the same tier.

It is unlikely that all the data that is contained in an extent has the same I/O workload, and therefore the same temperature. So, moving a hot extent likely also moves data that is not hot. The overall Easy Tier efficiency to put hot data in the proper tier is then inversely proportional to the extent size.

Consider the following practical aspects:

- ▶ Easy Tier efficiency is affecting the storage solution cost-benefit ratio. It is more effective for Easy Tier to place hot data in the top tier. In this case, less capacity can be provided for the relatively more expensive Easy Tier top tier.
- ▶ The extent size determines the bandwidth requirements for Easy Tier background process. The smaller the extent size, the lower the bandwidth consumption.

However, Easy Tier efficiency should not be the only factor considered when choosing the extent size. Manageability and capacity requirement considerations must also be considered.

As a general rule, use the default 1GB (standard pool) or 4GB (DRP) extent size for Easy Tier enabled configurations.

## External controller tiering considerations

IBM Easy Tier is an algorithm that was developed by IBM Almaden Research and made available to many members of the IBM storage family, such as the DS8000, IBM SAN Volume Controller, and FlashSystem products. The DS8000 is the most advanced in Easy Tier implementation and currently provides features that are not yet available for IBM SAN Volume Controller technology, such as Easy Tier Application, Easy Tier Heat Map Transfer, and Easy Tier Control.

In general, the use of Easy Tier at the highest level is recommended; that is, the virtualizer and any backend controller tiering functions should be disabled.

**Important:** Never run tiering at two levels. Doing so causes thrashing and unexpected heat or cold jumps to be observed at both levels.

Consider the following options:

- Easy Tier is done at the virtualizer level.

In this case, complete these steps at the backend level:

- a. Set up homogeneous pools according to the tier technology available.
- b. Create volumes to present to the virtualizer from the homogeneous pool.
- c. Disable tiering functions.

At a virtualizer level, complete the following steps:

- a. Discover the MDisk provided by the backend storage and set the tier properly.
- b. Create hybrid pools that aggregate the MDisk.
- c. Enable the Easy Tier function.

- Easy Tier is done at the backend level.

In this case, complete the following steps at the back-end level:

- a. Set up hybrid pools according to the available tier technology.
- b. Create volumes to present to the virtualizer from the hybrid pools.
- c. Enable the tiering functions.

At virtualizer level, complete the following steps:

- a. Discover the MDisk that are provided by the back-end storage and set the same tier for all.
- b. Create standard pools that aggregate the MDisk.
- c. Disable the Easy Tier function.

Although both of these options provide benefits in term of performance, they have different characteristics.

Option 1 provides some advantages when compared to option 2. One advantage is that Easy Tier can be enabled or disabled at volume level. This feature allows users to decide which volumes benefit from Easy Tier and which do not. With option 2, this goal cannot be achieved.

Another advantage of option 1 is that the volume heat map matches directly to the host workload profile by using the volumes. This option also allows you to use Easy Tier across different storage controllers by using lower performance and cost systems to implement the middle or cold tiers.

With option 2, the volume heat map on the backend storage is based on the IBM SAN Volume Controller workload. Therefore, it does not precisely represent the hosts workload profile because of the effects of the IBM SAN Volume Controller caching.

Finally, with option 1, you can change the extent size to improve the overall Easy Tier efficiency (as described in “Extent size considerations” on page 167).

However, with option 2, especially with DS8000 as the backend, offers some advantages when compared to option 1. For example, when external storage is used, the virtualizer uses generic performance profiles to evaluate the workload that can be placed on a specific MDisk, as described in “MDisk Easy Tier load” on page 166. These profiles might not match the backend capabilities, which can lead to a resource usage that is not optimized. With option 2, this problem rarely occurs because the performance profiles are based on the real back-end configuration.



## Easy Tier and thin-provisioned backend considerations

When a data reduction-capable backend is used in Easy Tier enabled pools, it is important to note that the data reduction ratio on the physical backend can vary over time because of Easy Tier data moving. Easy Tier continuously moves extents across the tiers, and within the same tier, as it attempts to optimize performance. As result, the amount of data that is written to the backend, and therefore the compression ratio, can unpredictably fluctuate over time, even though the data is not modified by the user.

It is not recommended to intermix data reduction-capable and non-data reduction-capable storage in the same tier of a pool with Easy Tier enabled.

## Easy Tier and Remote Copy considerations

When Easy Tier is enabled, the workloads that are monitored on the primary and the secondary system can differ. Easy Tier at the primary system sees a normal workload, and it sees only the write workloads at the secondary system.

This situation means that the optimized extent distribution on the primary system can differ considerably from the one on the secondary system. The optimized extent reallocation that is based on the workload learning on the primary system is not sent to the secondary system now to allow the same extent optimization on both systems based on the primary workload pattern.

In a Disaster Recovery (DR) situation with a failover from the primary site to a secondary site, the extent distribution of the volumes on the secondary system is not optimized to match the primary workload. Easy Tier relearns the production I/O profile and builds a new extent migration plan on the secondary system to adapt to the new production workload.

It eventually achieves the same optimization and level of performance as on the primary system. This task takes some time, so the production workload on the secondary system might not run at its optimum performance during that period. The Easy Tier acceleration feature (see “Easy Tier acceleration” on page 165) can be used to mitigate this situation.

IBM SAN Volume Controller Remote Copy configurations that use NearLine tier at the secondary system must be carefully planned, especially when practicing DR by using FlashCopy. In these scenarios, FlashCopy often is started just before the beginning of the DR test. It is likely that the FlashCopy target volumes are in the NearLine tier because of prolonged inactivity.

When the FlashCopy is started, an intensive workload often is added to the FlashCopy target volumes because of the background and foreground I/Os. This situation can easily lead to overloading, and then possibly performance degradation of the NearLine storage tier if it is not correctly sized in terms of resources.

## Easy Tier on DRP and interaction with garbage collection

DRPs makes use of Log Structured Array (LSA) structures that need garbage collection activity to be done regularly. An LSA always appends new writes to the end of the allocated space (see “DRP internal details” on page 101).

Even if data exists and the write is an overwrite, the new data is not written in that place. Instead, the new write is appended at the end and the old data is marked as needing garbage collected. This process provides the following advantages:

- Writes to a DRP volume are always treated as sequential; therefore, we can build all the 8 KB chunks into a larger 256 KB chunk and destage the writes from cache as full stripe writes or as large as a 256 KB sequential stream of smaller writes.

- Easy Tier with DRP gives the best performance in terms of RAID on back-end systems and on Flash, where it becomes easier for the Flash device to perform its internal garbage collection on a larger boundary.

To improve the Easy Tier efficiency with this write workload profile, we can start to record metadata about how frequently certain areas of a volume are overwritten. The Easy Tier algorithm was modified so that we can then bin sort the chunks into a heat map in terms of rewrite activity, and then group commonly rewritten data onto a single extent. This configuration ensures that Easy Tier operates correctly for not only read data, but write data when data reduction is in use.

Before DRP, write operations to compressed volumes held lower value to the Easy Tier algorithms because writes were always to a new extent; therefore, the previous heat was lost. Now, we can maintain the heat over time and ensure that frequently rewritten data is grouped. This process also aids the garbage collection process where it is likely that large contiguous areas are garbage collected together.

### **Tier sizing considerations**

Tier sizing is a complex task that always requires an environment workload analysis to match the performance and costs expectations.

Consider the following sample configurations that address some or most common customer requirements. The same benefits can be achieved by adding Storage Class Memory to the configuration. In these examples, the top Flash tier can be replaced with an SCM tier, or SCM can be added as the hot tier and the corresponding medium and cold tiers be shifted down to drop the coldest tier:

- 10 - 20% Flash, 80 - 90% Enterprise  
This configuration provides Flash-like performance with reduced costs.
- 5% Tier 0 Flash, 15% Tier 1 Flash, 80% Nearline  
This configuration provides Flash like performance with reduced costs.
- 3 - 5% Flash, 95 - 97% Enterprise  
This configuration provides improved performance compared to a single tier solution, and all data is ensured to have at least enterprise performance. It also removes the requirement for over provisioning for high access density environments.
- 3 - 5% Flash, 25 - 50% Enterprise, 40 - 70% Nearline  
This configuration provides improved performance and density compared to a single tier solution. It also provides significant reduction in environmental costs.
- 20 - 50% Enterprise, 50 - 80% Nearline  
This configuration provides reduced costs and comparable performance to a single tier Enterprise solution.

#### **4.6.10 Monitoring Easy Tier using the GUI**

Since software version 8.3.1, the GUI includes various reports and statistical analysis that can be used to understand what Easy Tier movement, activity, and skew is present in a storage pool. These windows replace the old IBM Storage Tier Advisor Tool (STAT) and STAT Charting Tool.

Unlike previous versions in which you were required to download the necessary log files from the system and upload to the STAT tool, from 8.3.1 onwards, the system continually reports the Easy Tier information and hence the GUI always displays the most up-to-date information.

## Accessing Easy Tier reports

To show the Easy Tier Reports window, in the GUI select **Monitoring** → **Easy Tier Reports**.

**Note:** If the system or Easy Tier was running for less than 24 hours, no data might be available to display.

The Reports window features the following views that can be accessed by using the tabs at the top of the window, which are described next:

- ▶ Data Movement
- ▶ Tier Composition
- ▶ Workload Skew

### Data movement report

The data movement report shows the amount of data that was moved in a specific period. You can change the period by using the drop-down selection on the right side (see Figure 4-30).



Figure 4-30 Easy Tier Data Movement window

The report breaks down the type of movement, which is described in terms of the internal Easy Tier extent movement types (see 4.6.2, “Easy Tier definitions” on page 151).

To aid your understanding and remind you of the definitions, click the **Movement Description** button to view the information window (see Figure 4-30 on page 171).

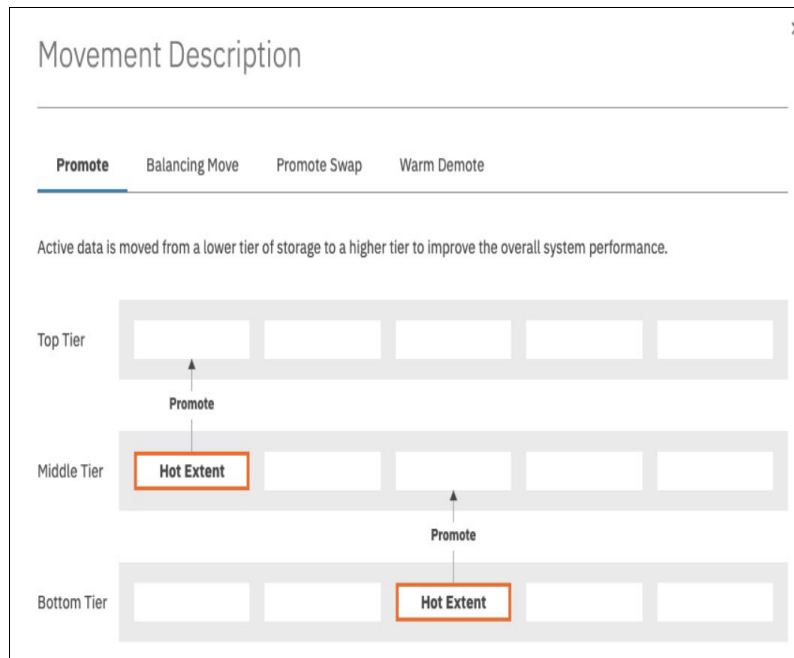


Figure 4-31 Easy Tier Movement description window

**Important:** If you are regularly seeing “warm demote” in the movement data, consider increasing the amount of hot tier that is available. A warm demote suggests that an extent is hot, but not enough capacity or Overload Protection was triggered in the hot tier.

## Tier composition report

The tier composition window (see Figure 4-32) shows how much data in each tier is active versus inactive. In an ideal case, most of your active data is in the hot tier alone. In most cases, the active data set cannot fit in only the hot tier; therefore, expect to also see active data in the middle tier.

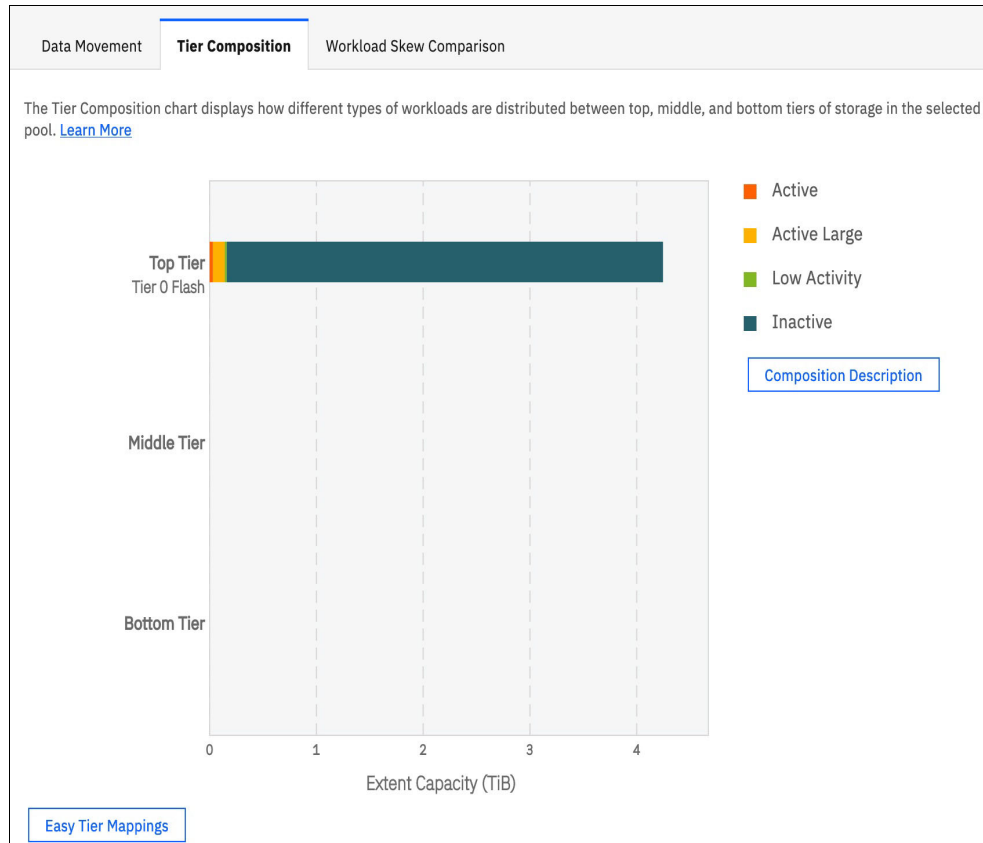


Figure 4-32 Easy Tier (single tier pool): Composition report window

If all active data can fit in the hot tier alone, you see the best possible performance from the system. “Active large” is data that is active but is being accessed at block sizes larger than the 64 KiB for which Easy Tier is optimized. This data is still monitored and can contribute to “expanded cold demote” operations.

The presence of any active data in the cold tier (regularly) suggests that you must increase the capacity or performance in the hot or middle tiers.

In the same way as with the Movement window, you can click **Composition Description** to view the information for each composition type (see Figure 4-33).

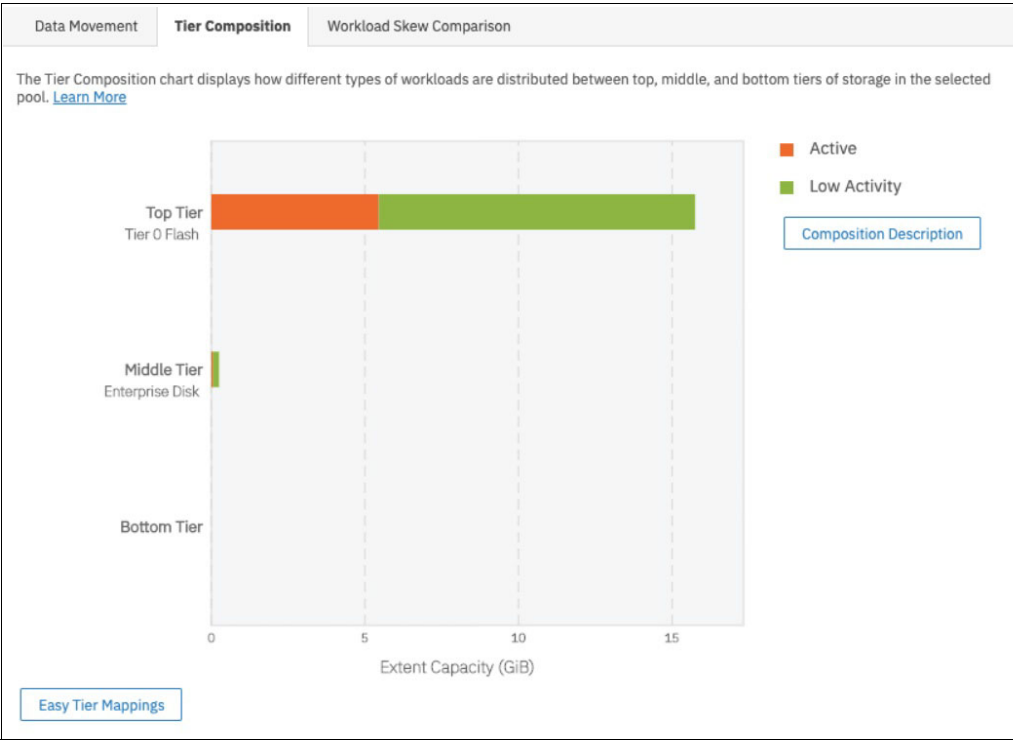


Figure 4-33 Easy tier (multitier pool): Composition window

## Workload skew comparison report

The workload skew report plots the percentage of the workload against the percentage of capacity. The skew shows a good estimate for how much capacity is required in the top tier to realize the most optimal configuration that is based on your workload.

**Tip:** The skew can be viewed when the system is in measuring mode with a single tier pool to help guide the recommended capacity to purchase that can be added to the pool in a hot tier.

A highly skewed workload (the line on the graph rises sharply within the first percentage of capacity) means that a smaller proportional capacity of hot tier is required. A low skewed workload (the line on the graph rises slowly and covers a large percentage of the capacity) requires more hot tier capacity, and consideration to a good performing middle tier when you cannot configure enough hot tier capacity (see Figure 4-34).

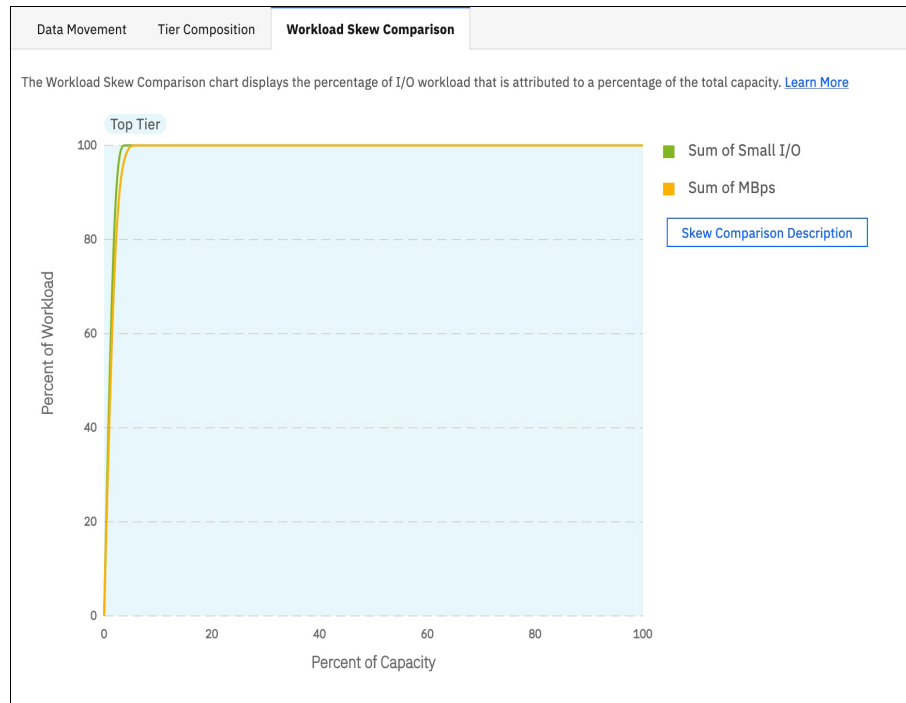


Figure 4-34 Workload skew: Single tier pool

In the first example that is shown in Figure 4-34, you can clearly see that this workload is highly skewed. This single-tier pool uses less than 5% of the capacity but is performing 99% of the workload in terms of IOPs and MBps.

This result is a prime example to add a small amount of faster storage to create a “hot” tier and improve overall pool performance (see Figure 4-35).

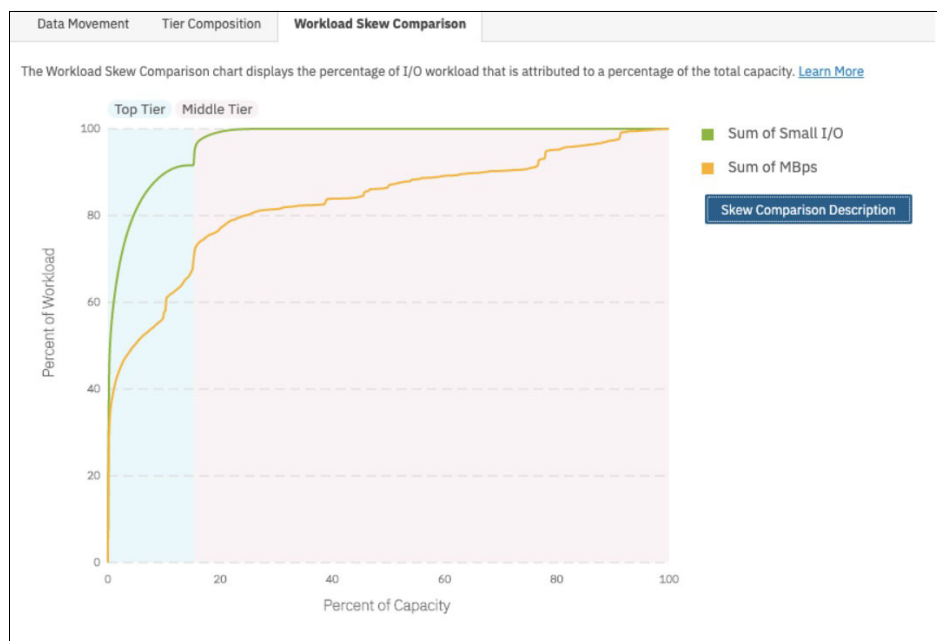


Figure 4-35 Workload skew: Multitier configuration

In this second example that is shown in Figure 4-35, the system is configured as a multitier pool and Easy Tier optimized the data placement for some time. This workload is less skewed than in the first example, with almost 20% of the capacity performing up to 99% of the workload.

Here again it might be worth considering increasing the amount of capacity in the top tier because approximately 10% of the IOPs workload is coming from the middle tier and can be optimized to reduce latency.

The graph that is shown in Figure 4-35 also shows the split between IOPs and MBps. Although the middle tier is not handling much of the IOPs workload, it is providing a reasonably large proportion of the MBps workload.

In these cases, ensure that the middle tier can manage good large block throughput. A case might be made for further improving performance by adding some higher throughput devices as a new middle tier, and demoting the current middle tier to the cold tier; however, this change depends on the types of storage that is used to provide the tiers.

Any new configuration with three tiers must comply with the configuration rules regarding the different types of storage that are supported in three tier configurations (see “Easy Tier mapping to MDisk tier types” on page 158).

If you implemented a new system and you see most of the workload is coming from a middle or cold tier, it might take only a day or two for Easy Tier to complete the migrations after it has initially analyzed the system.

If after a few days a distinct bias still exists to the lower tiers, you might want to consider enabling “Accelerated Mode” for a week or so; however, remember to turn disable this mode after the system reaches a steady state. For more information, see “Easy Tier acceleration” on page 165.





# Volumes

In IBM SAN Volume Controller, a volume is a logical disk that the system presents to attached hosts. This chapter describes the various types of volumes and it provides guidance for managing the properties.

This chapter includes the following topics:

- ▶ 5.1, “Overview of volumes” on page 178
- ▶ 5.2, “Guidance for creating volumes” on page 178
- ▶ 5.3, “Thin-provisioned volumes” on page 182
- ▶ 5.4, “Mirrored volumes” on page 191
- ▶ 5.5, “HyperSwap volumes” on page 195
- ▶ 5.6, “VMware virtual volumes” on page 196
- ▶ 5.7, “Cloud volumes” on page 200
- ▶ 5.8, “Volume migration” on page 202
- ▶ 5.9, “Preferred paths to a volume” on page 206
- ▶ 5.10, “Moving a volume between I/O groups and nodes” on page 207
- ▶ 5.11, “Volume throttling” on page 209
- ▶ 5.12, “Volume cache mode” on page 211
- ▶ 5.13, “Other considerations” on page 214

## 5.1 Overview of volumes

Each volume in IBM SAN Volume Controller can be one of the following types:

- ▶ **Striped**  
A volume that is striped at the extent level. The extents are allocated from each MDisk that is in the storage pool. This volume type is the most frequently used because each I/O to the volume is spread across many external storage MDisks or internal disk drives compared to a sequential volume.
- ▶ **Sequential**  
A volume on which extents are allocated sequentially from one MDisk. This type of volume is rarely used because striped volume is better suited to most of the cases.
- ▶ **Image**  
Type of volume that has a direct relationship with one MDisk. The extents on the volume are directly mapped to the extents on the MDisk. This image is commonly used for data migration from a storage subsystem to an IBM SAN Volume Controller.

Volumes can be created with the following attributes:

- ▶ **Standard provisioned volumes**  
Volumes with no special attributes.
- ▶ **Thin-provisioned volumes**  
Volumes that present a larger capacity to the host than their real capacity.
- ▶ **Compressed volumes**  
Volumes whose data is compressed and optionally deduplicated.
- ▶ **Mirrored volumes**  
A volume can contain a duplicate copy of the data in another volume. Two copies are called a *mirrored volume*.
- ▶ **HyperSwap volumes**  
Volumes that participate in a HyperSwap relationship.
- ▶ **VMware Virtual Volumes (VVols)**  
Volumes that are managed remotely by VMware vCenter.
- ▶ **Cloud volumes**  
Volumes that are enabled for transparent cloud tiering.

## 5.2 Guidance for creating volumes

When creating volumes, consider the following guidelines:

- ▶ Consider the naming rules before you create volumes. It is easier to assign the correct name when the volume is created than to modify it later.
- ▶ Choose which type of volume that you want to create. First, decide whether fully allocated (standard volumes) or thin-provisioned. If you decide to create a thin-provisioned volume, analyze whether you need compression and deduplication enabled.

A fully allocated volume is automatically formatted, which can be a time-consuming process. However, this background process does not impede the immediate use of the volume.

Actions, such as moving, expanding, shrinking, or adding a volume copy are disabled when the specified volume is formatting. Although it is unlikely that you must perform one of these actions after the volume is created, you can disable the format option in the Custom tab of the volume creation window by clearing the **Format volumes** option, as shown in Figure 5-1.

The screenshot shows the 'Create Volumes' window with the 'Custom' tab selected. The 'Quantity' is 1, 'Capacity' is 1 GiB, and 'Name' is VOL01. Under 'Capacity savings', 'None' is selected and 'Deduplicated' is unchecked. A link 'Define another volume' is present. In the 'General' section, the 'Format volume' checkbox is unchecked, and the 'Enabled' label is highlighted with a red box. Below it, 'Cache mode' is set to 'Enabled' and 'OpenVMS UDID' is empty. At the bottom are buttons for 'Need Help', 'Cancel', 'Create and Map', and 'Create'.

Figure 5-1 Volumes format option

You can also create volumes by using the CLI. Example 5-1 shows the command to disable auto formatting option with the **-nofmtdisk** parameter.

*Example 5-1 Volume creation without auto formatting option*

```
superuser>mkvdisk -name VOL01 -mdiskgrp 0 -size 1 -unit gb -vtype striped -iogrp
io_grp0 -nofmtdisk
Virtual Disk, id [52], successfully created
superuser>lsvdisk VOL01
id 52
name VOL01
IO_group_id 0
IO_group_name io_grp0
status online
mdisk_grp_id 0
mdisk_grp_name Swimming
capacity 1.00GB
type striped
formatted no
```

formatting no  
.  
lines removed for brevity

Remember that when you create a volume, it takes some time to completely format it (depending on the volume size). The *syncrate* parameter of the volume specifies the volume copy synchronization rate and can be modified to accelerate the completion of the format process.

For example, the initialization of a 1 TB volume can take more than 120 hours to complete with the default syncrate value 50, or approximately 4 hours if you manually set the syncrate to 100. If you increase the syncrate to accelerate the volume initialization, remember to reduce it again to avoid issues the next time you use volume mirroring to perform a data migration of that volume.

For more information about creating a thin-provisioned volume, see 5.3, “Thin-provisioned volumes” on page 182.

- Each volume includes an I/O group and an associated preferred node. When creating a volume, consider balancing volumes across the I/O groups to balance the load across the cluster.

In configurations where it is not possible to zone a host to multiple I/O groups so that the host can access only one I/O group, the volume must be created in the I/O group to which the host can access.

Also, it is possible to define a list of I/O groups in which a volume can be accessible to hosts. It is recommended that a volume is accessible to hosts by the caching only I/O group. You can have more than one I/O group in the access list of a volume in some scenarios with specific requirements, such as when a volume is migrated to another I/O group.

**Tip:** Migrating volumes across I/O groups can be a disruptive action. Therefore, specify the correct I/O group at the time the volume is created.

- By default, the *preferred node*, which owns a volume within an I/O group, is selected in a load balancing basis. Although it is not easy to estimate the workload when the volume is created, distribute the workload evenly on each node within an I/O group.
- Except for a few cases, the cache mode of a volume is set to read/write. For more information, see 5.12, “Volume cache mode” on page 211.
- The maximum number of volumes per I/O group and system is listed in Table 5-1.

Table 5-1 Maximum number of volumes in IBM SAN Volume Controller

| Volume type                          | Maximum number | Comments                                                                                                                          |
|--------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Standard volumes (VDisks) per system | 10,000         | Each basic volume uses one VDisk, each with one copy.                                                                             |
| Stretched volumes per system         | 5,000          | Each stretched volume uses one VDisk, each with two copies.                                                                       |
| HyperSwap volumes per system         | 2,000          | Each HyperSwap volume uses 4 VDisks, each with one copy, one active-active Remote Copy relationship, and four FlashCopy mappings. |

| Volume type                                                                                                   | Maximum number | Comments                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volumes per I/O group (volumes per caching I/O group)                                                         | 10,000         |                                                                                                                                                                                                                                                                                 |
| Thin-provisioned (space-efficient) volume copies in regular pools per system                                  | -              | No limit is imposed here beyond the volume copies per system limit.                                                                                                                                                                                                             |
| Compressed volume copies in regular pools per system                                                          | 2,048          | Maximum requires an 8-node cluster                                                                                                                                                                                                                                              |
| Compressed volume copies in regular pools per I/O group                                                       | 512            | With 64 GB Memory Option Only                                                                                                                                                                                                                                                   |
| Compressed volume copies in data reduction pools per system                                                   | -              | No limit is imposed here beyond the volume copy limit per data reduction pool.                                                                                                                                                                                                  |
| Compressed volume copies in data reduction pools per I/O group                                                | -              | No limit is imposed here beyond the volume copy limit per data reduction pool.                                                                                                                                                                                                  |
| Deduplicated volume copies in data reduction pools per system                                                 | -              | No limit is imposed here beyond the volume copy limit per data reduction pool.                                                                                                                                                                                                  |
| Deduplicated volume copies in data reduction pools per I/O group                                              | -              | No limit is imposed here beyond the volume copy limit per data reduction pool.                                                                                                                                                                                                  |
| Volumes per storage pool                                                                                      | -              | No limit is imposed beyond the volumes per system limit.                                                                                                                                                                                                                        |
| Fully allocated volume capacity                                                                               | 256 TB         | <ul style="list-style-type: none"> <li>▶ Maximum size for an individual fully allocated volume.</li> <li>▶ Maximum size depends on the extent size of the storage pool.</li> <li>▶ See 5.3.4, "Limits on virtual capacity of thin-provisioned volumes" on page 190.</li> </ul>  |
| Thin-provisioned (space-efficient) per-volume capacity for volumes copies in regular and data reduction pools | 256 TB         | <ul style="list-style-type: none"> <li>▶ Maximum size for an individual thin-provisioned volume.</li> <li>▶ Maximum size depends on the extent size of the storage pool.</li> <li>▶ See 5.3.4, "Limits on virtual capacity of thin-provisioned volumes" on page 190.</li> </ul> |
| Compressed volume capacity in regular pools containing non-Flash storage                                      | 16 TB          | <ul style="list-style-type: none"> <li>▶ Maximum size for an individual compressed volume.</li> <li>▶ Maximum size depends on the extent size of the storage pool.</li> <li>▶ See 5.3.4, "Limits on virtual capacity of thin-provisioned volumes" on page 190.</li> </ul>       |
| Compressed volume capacity in regular pools containing all-Flash storage                                      | 32 TB          | <ul style="list-style-type: none"> <li>▶ Maximum size for an individual compressed volume.</li> <li>▶ Maximum size depends on the extent size of the storage pool.</li> <li>▶ See 5.3.4, "Limits on virtual capacity of thin-provisioned volumes" on page 190.</li> </ul>       |

- The pool extent size does not affect the overall storage performance. A volume occupies an integer number of extents, but its length does not need to be an integer multiple of the extent size. Also, the length does need to be an integer multiple of the block size. Any space that is left over between the last logical block in the volume and the end of the last extent in the volume is unused.

A small extent size is used to minimize this unused space, and to have a finer granularity of the volume space that is occupied on the underlying storage controller. However, determine the best extent size for your storage pools that consider the back-end storage.

**Important:** Volume migration (by using the `migratevdisk` command) between storage pools requires that the source and destination pools feature the same extent size.

## 5.3 Thin-provisioned volumes

A thin-provisioned volume presents a different capacity to mapped hosts than the capacity that the volume uses in the storage pool. The system supports thin-provisioned volumes in both standard pools and data reduction pools (DRPs).

Figure 5-2 shows the basic concept of a thin-provisioned volume.

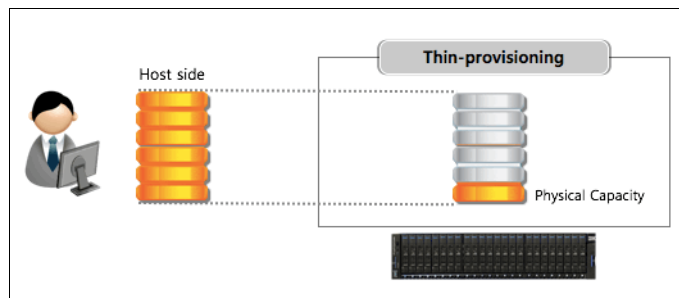


Figure 5-2 Thin-provisioned volume

The different types of volumes in a DRP are shown in Figure 5-3.

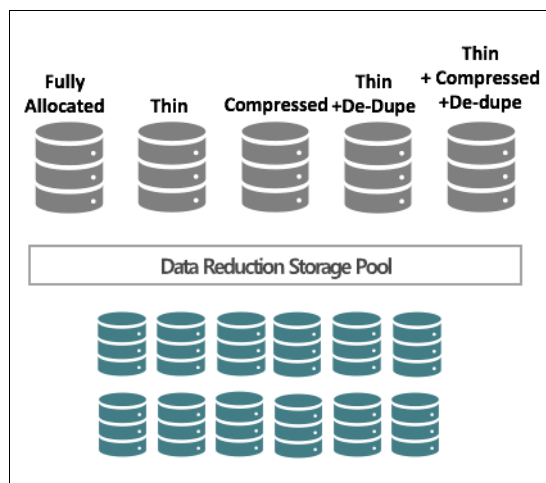


Figure 5-3 Different kinds of volumes in DRP

In standard pools, thin-provisioned volumes are created based on capacity savings criteria. These properties are managed at the volume level. However, in DRPs, all the benefits of thin-provisioning are available to all the volumes that are assigned to the pool. For the thin-provisioned volumes in DRPs, you can configure compression and data deduplication on these volumes, which increases the capacity savings for the entire pool.

You can enhance capacity efficiency for thin-provisioned volumes by monitoring the hosts' use of capacity. When the host indicates that the capacity is no longer needed, the space is released and can be reclaimed by the DRP and it is redistributed automatically. Standard pools do not have these functions.

Figure 5-4 shows the concepts of thin-provisioned volumes. These concepts are described next.

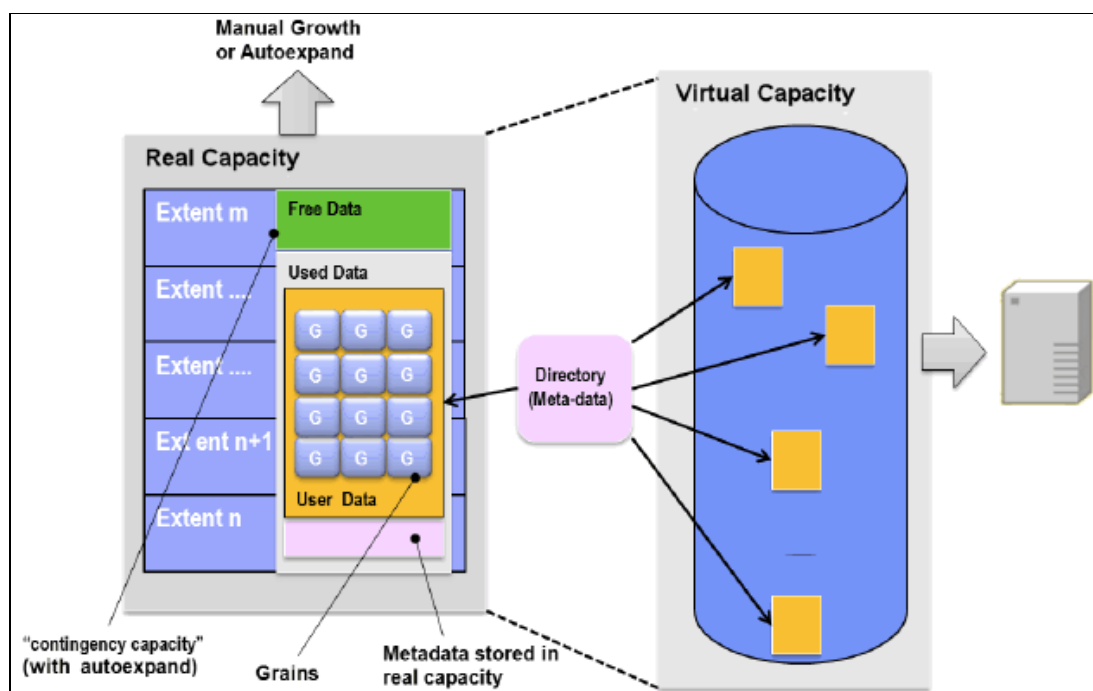


Figure 5-4 Thin-provisioned volume concepts

*Real capacity* defines how much disk space from a pool is allocated to a volume. *Virtual capacity* is the capacity of the volume that is reported to the hosts. A volume's virtual capacity is larger than its real capacity.

Each system uses the real capacity to store data that is written to the volume and metadata that describes the thin-provisioned configuration of the volume. As more information is written to the volume, more of the real capacity is used. The system identifies read operations to unwritten parts of the virtual capacity and returns zeros to the server without the use of any real capacity.

Thin-provisioned volumes are available in two operating modes: autoexpand and noautoexpand. You can switch the mode at any time.

If you select the autoexpand feature, IBM SAN Volume Controller automatically adds a fixed amount of extra real capacity to the thin volume as required. Therefore, the autoexpand feature attempts to maintain a fixed amount of unused real capacity for the volume. We recommend using autoexpand by default to avoid volume offline issues.

This amount is known as the *contingency capacity*. The contingency capacity is initially set to the real capacity that is assigned when the volume is created. If the user modifies the real capacity, the contingency capacity is reset to be the difference between the used capacity and real capacity.

A volume that is created *without* the **autoexpand** feature (and therefore has a zero contingency capacity) goes offline when the real capacity is used. In this case, it must be expanded.

When creating a thin-provisioned volume with compression and deduplication enabled, you must be careful for out of space issues in the volume and pool where the volume is created. Set the warning threshold in the pools that contain thin-provisioned volumes, and in the volume.

**Warning threshold:** When you are working with thin-provisioned volumes, enable the warning threshold (by using email or an SNMP trap) in the storage pool. If you are not using the **autoexpand** feature, you also must enable the warning threshold on the volume level. If the pool or volume runs out of space, the volume goes offline, which results in a loss of access situation.

If you do not want to be concerned with monitoring volume capacity, it is highly recommended that the **autoexpand** option is enabled. Also, when you create a thin-provisioned volume, you must specify the space that is initially allocated to it (**-rsize** option in the CLI) and the grain size.

By default, **rsize** (or real capacity) is set to 2% of the volume virtual capacity, and grain size is 256 KiB. These default values, with **autoexpand** enabled and **warning** disabled options, work in most scenarios. Some instances exist in which you might consider the use of different values to suit your environment.

Example 5-2 shows the command to create a volume with the suitable parameters.

*Example 5-2 Creating thin-provisioned volume*

---

```
superuser>mkvdisk -name VOL02 -mdiskgrp Pool1 -size 100 -unit gb -vtype striped
-iogrp io_grp0 -rsize 2% -autoexpand -warning 0 -grainsize 256
Virtual Disk, id [53], successfully created
superuser>lsvdisk VOL02
id 53
name VOL02
.
lines removed for brevity
.
capacity 100.00GB
.
lines removed for brevity
.
used_capacity 0.75MB
real_capacity 2.02GB
free_capacity 2.01GB
overallocation 4961
autoexpand on
warning 0
grainsize 256
se_copy yes
.
```



lines removed for brevity

A thin-provisioned volume can be converted nondisruptively to a fully allocated volume, or vice versa. Figure 5-5 shows how to modify capacity savings of a volume. You can right-click the volume and select **Modify Capacity Savings**.

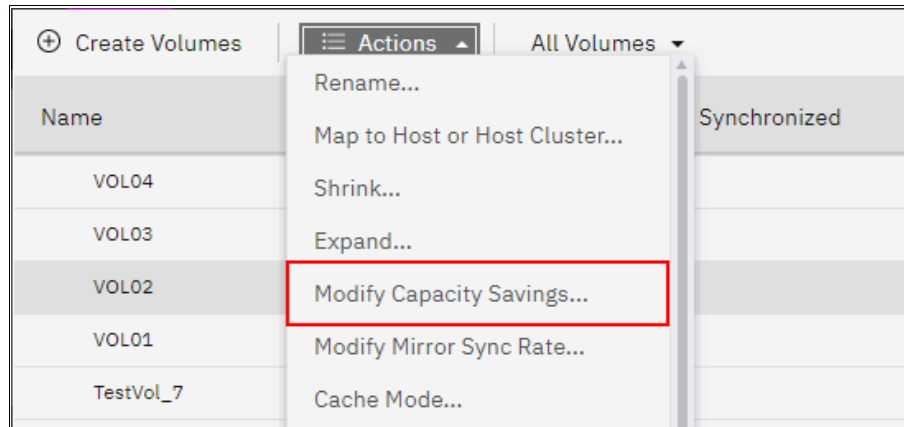


Figure 5-5 Modifying capacity savings of a volume nondisruptively

The fully allocated to thin-provisioned migration procedure uses a zero-detection algorithm so that grains that contain all zeros do not cause any real capacity to be used.

### 5.3.1 Compressed volumes

When you create volumes, you can specify compression as a method to save capacity for the volume. With compressed volumes, data is compressed as it is written to disk, which saves more space. When data is read to hosts, the data is decompressed.

Compression is provided by using DRPs in IBM SAN Volume Controller nodes model SV2 and SA2 only. SV1 nodes support compressed volumes in standard pools and DRPs.

DRPs also reclaim capacity that is not used by hosts that support SCSI **unmap** commands. When these hosts issue SCSI **unmap** commands, a DRP reclaims the released capacity for redistribution.

For compressed volumes in DRPs, the used capacity before compression indicates the total amount of data that is written to volume copies in the storage pool before data reduction occurs. This compression solution provides nondisruptive conversion between compressed and uncompressed volumes. It also eliminates the need for special procedures to deal with compressed volumes.

If you are planning to virtualize volumes that are connected to your hosts directly from any storage subsystems, and you want to know what the space saving you achieve, run the IBM Data Reduction Estimator Tool (DRET). For more information about the tool and to download it, see [this IBM Support web page](#).

DRET tool is a command-line and host-based utility that can be used to estimate an expected compression rate for block devices. This tool can also evaluate capacity savings by using deduplication. For more information, see [this IBM Support web page](#).

IBM SAN Volume Controller also includes an integrated Comprestimator tool, which is available through the management GUI and command-line interface. If you are considering applying compression on non-compressed volumes in an IBM SAN Volume Controller, you can use this tool to evaluate if compression generates enough capacity savings.

For more information, see 4.1.4, “Data reduction estimation tools” on page 105.

As shown in Figure 5-6, customize the Volume view to see the compression savings for a compressed volume, and estimated compression savings for a non-compressed volume that you are planning to migrate.

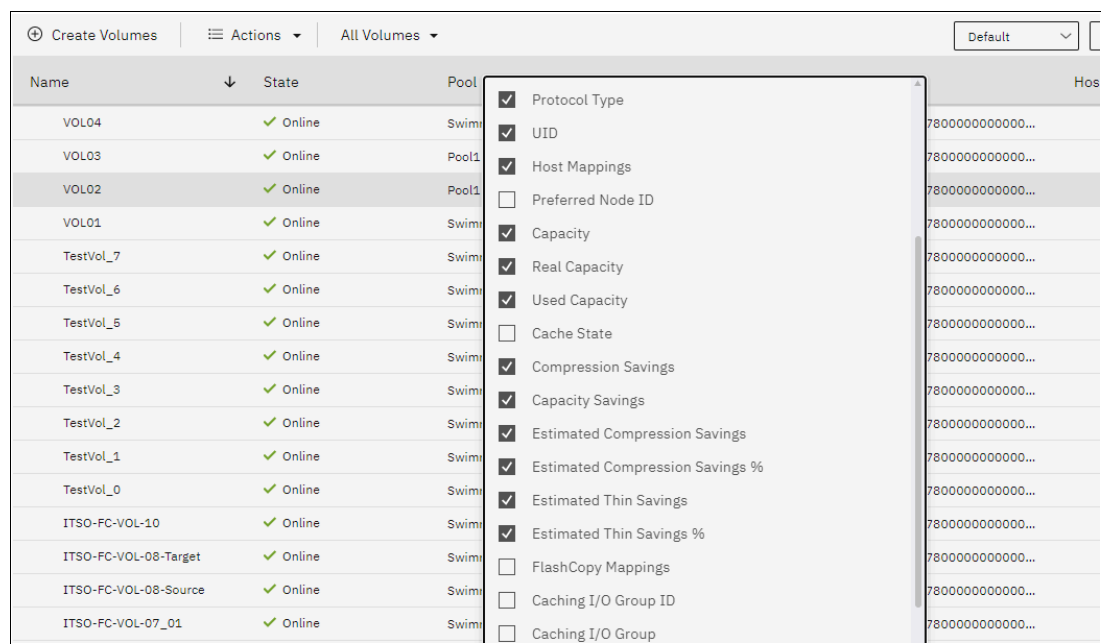


Figure 5-6 Customized view

## 5.3.2 Deduplicated volumes

Deduplication is a data reduction technique for eliminating duplicate copies of data. It can be configured with thin-provisioned and compressed volumes in a DRP for saving capacity.

The deduplication process identifies unique chunks of data, or byte patterns, and stores a signature of the chunk for reference when writing new data chunks. If the new chunk’s signature matches an existing signature, the new chunk is replaced with a small reference that points to the stored chunk. The same byte pattern can occur many times, which results in the amount of data that must be stored being greatly reduced.

If a volume is configured with deduplication and compression, data is deduplicated first and then, compressed. Therefore, deduplication references are created on the compressed data that is stored on the physical domain.

Figure 5-7 shows the settings to create a compressed and deduplicated volume.

Figure 5-7 Creating deduplicated volumes

To create a thin-provisioned volume that uses deduplication, enter the command in the CLI that is shown in Example 5-3.

*Example 5-3 Creating thin-provisioned volume with deduplication option*

```
superuser>mkvolume -name dedup_test_01 -size 10 -unit gb -pool 0 -thin
-deduplicated
Volume, id [55], successfully created
```

To create a compressed volume that uses deduplication, enter the command that is shown in Example 5-4.

*Example 5-4 Creating compressed volume with deduplication option*

```
superuser>mkvolume -name dedup_test_02 -size 10 -unit gb -pool 0 -compressed
-deduplicated
Volume, id [56], successfully created
```

To maximize the space that is available for the deduplication database, the system distributes it between all nodes in the I/O groups that contain deduplicated volumes. Each node holds a distinct portion of the records that are stored in the database.

Depending on the data type stored on the volume, the capacity savings can be significant. Examples of use cases that typically benefit from deduplication are virtual environments with multiple virtual machines running the same operating system and backup servers.

In both cases, it is expected that multiple copies of identical files exist, such as components of the standard operating system or applications that are used in the organization. Conversely, data that is encrypted or compressed at the file system level does not benefit from deduplication because these operations were removed for redundancy.

If you want to evaluate if savings are realized by migrating a set of volumes to deduplicated volumes, you can use DRET, which is a command-line host-based utility for estimating the data reduction saving on block devices. For more information about DRET, see 4.1.4, “Data reduction estimation tools” on page 105.

### 5.3.3 Thin provisioning considerations

Thin provisioning works only if the host limits writes to areas of the volume that store data. For example, if the host performs a low-level format of the entire volume, the host writes to the entire volume and no advantage is gained by using a thin provisioning volume over a fully allocated volume.

Consider the following properties of thin-provisioned volumes that are useful to understand for the rest of the section:

- ▶ When the used capacity first exceeds the volume *warning threshold*, an event is raised, which indicates that real capacity is required. The default warning threshold value is 80% of the volume capacity. To disable warnings, specify 0%.
- ▶ Compressed volumes include an attribute called *uncompressed used capacity* (for standard pools) and *used capacity before reduction* (for a DRP). These volumes are the used capacities before compression or data reduction. They are used to calculate the compression ratio.

#### Thin provisioning and over-allocation

Because thin-provisioned volumes do not store the zero blocks, a storage pool is over-allocated only after the sum of all volume capacities exceeds the size of the storage pool.

Storage administrators likely think about the “out of space” problem. If enough capacity exists on disk to store fully allocated volumes, and you convert them to thin-provisioned volumes, enough space exists to store data (even if the servers writes to every byte of virtual capacity). Therefore, this issue is not going to be a problem for the short term, and you have time to monitor your system and understand how your capacity grows.

#### Monitoring capacity with thin-provisioned volumes

**Note:** It is critical that capacity be monitored when thin-provisioned or compressed volumes are used. Be sure to add capacity *before* running out of space.

If you run out of space on a volume or storage pool, the host that uses the affected volumes cannot perform new write operations to these volumes. Therefore, an application or database that is running on this host becomes unavailable.

In a storage pool with only fully allocated volumes, the storage administrator can easily manage the used and available capacity in the storage pool as its used capacity grows when volumes are created or expanded.

However, in a pool with thin-provisioned volumes, the used capacity can increase at any time if the host file system grows. For this reason, the storage administrator must consider capacity planning carefully. It is critical to put in place volume and pool capacity monitoring.

Tools, such as IBM Spectrum Control and Storage Insights, can display the capacity of a storage pool in real time and graph how it is growing over time. These tools are important because they are used to predict when the pool will run out of space.

IBM SAN Volume Controller also alerts you by including an event in the event log when the storage pool reaches the configured threshold, which is called the *warning level*. The GUI sets this threshold to 80% of the capacity of the storage pool by default.

By using enhanced Call Home and Storage Insights, IBM now has the ability to monitor systems and flag systems that can be short on capacity. Extreme situations result in a support ticket being generated and the customer being contacted.

## What to do if you run out of space in a storage pool

You can use one or a combination of the following options that are available if a storage pool runs out of space:

- Contingency capacity on thin-provisioned volumes

If the storage pool runs out of space, each volume has its own contingency capacity, which is an amount of storage that is reserved by the volume and is sizable. Contingency capacity is defined by the *real capacity* parameter that is specified when the volume is created, which has a default value of 2%.

The contingency capacity protects the volume from going offline when its storage pool runs out of space by having the storage pool use this reserved space first. Therefore, you have some time to repair things before everything starts going offline.

If you want more safety, you might implement a policy of creating volumes with 10% of *real capacity*. Also, remember that you do not need to have the same contingency capacity for every volume.

**Note:** This protection likely solves most immediate problems. However, after you are informed that you ran out of space, a limited amount of time exists to react. You need a plan in place and well-understood about what to do next.

- Have unallocated storage on standby

You can always have spare drives or managed disks ready to be added to whichever storage pool runs out of space within only a few minutes. This capacity gives you some breathing room while you take other actions. The more drives or MDisks you have, the more time you have to solve the problem.

- Sacrificial emergency space volume

Consider the use of a fully-allocated sacrificial emergency space volume in each pool. If the storage pool is running out of space, you can delete or shrink this volume to quickly provide more available space in the pool.

- Move volumes

You can migrate volumes to other pools to free up space. However, data migration on IBM SAN Volume Controller is designed to move slowly to avoid performance problems. Therefore, it might be impossible to complete this migration before your applications go offline.

► Policy-based solutions

No policy is going to solve the problem if you run out of space; however, you can use policies to reduce the likelihood of that ever happening to the point where you feel comfortable doing less of the other options.

You can use these types of policies for thin provisioning:

**Note:** The following policies use arbitrary numbers. These numbers are designed to make the suggested policies more readable. We do not provide any recommended numbers to insert into these policies because they are determined by business risk, and this consideration is different for every client.

- Manage free space such that enough free capacity always is available for your 10 largest volumes to reach 100% full without running out of free space.
- Never over-allocate more than 200%. That is, if you have 100 TB of capacity in the storage pool, the sum of the volume capacities in the same pool must not exceed 200 TB.
- Always start the process of adding capacity when the storage pool reaches 70% full.

### Grain size

The grain size is defined when the thin-provisioned volume is created and can be set to 32 KB, 64 KB, 128 KB, or 256 KB (default). The grain size cannot be changed after the thin-provisioned volume is created.

Smaller granularities can save more space, but they have larger directories. If you select 32 KB for the grain size, the volume size cannot exceed 260,000 GB. Therefore, if you are not going to use the thin-provisioned volume as a FlashCopy source or target volume, use 256 KB by default to maximize performance.

Thin-provisioned volume copies in DRPs feature a grain size of 8 KB. This predefined value cannot be set or changed.

If you are planning to use thin provisioning with FlashCopy, remember that grain size for FlashCopy volumes can be only 64 KB or 256 KB. In addition, to achieve best performance, the grain size for the thin-provisioned volume and FlashCopy mapping must be same. For this reason, it is not recommended to use thin-provisioned volume in DRPS as a FlashCopy source or target volume.

**Note:** The use of thin-provisioned volumes in a DRP for FlashCopy is not recommended.

## 5.3.4 Limits on virtual capacity of thin-provisioned volumes

The extent and grain size factors limit the virtual capacity of thin-provisioned volumes beyond the factors that limit the capacity of regular volumes. For more information about the maximum volume capacity for each extent size on IBM SAN Volume Controller, see this [this IBM Support web page](#).

## 5.4 Mirrored volumes

By using volume mirroring, a volume can have two copies. Each volume copy can belong to a different pool, and each copy features the same virtual capacity as the volume. In the management GUI, an asterisk (\*) indicates the primary copy of the mirrored volume. The primary copy indicates the preferred volume for read requests.

When a server writes to a mirrored volume, the system writes the data to both copies. When a server reads a mirrored volume, the system picks one of the copies to read. If one of the mirrored volume copies is temporarily unavailable (for example, because the storage system that provides the pool is unavailable), the volume remains accessible to servers. The system remembers which areas of the volume are written and resynchronizes these areas when both copies are available.

You can create a volume with one or two copies, and you can convert a non-mirrored volume into a mirrored volume by adding a copy. When a copy is added in this way, the system synchronizes the new copy so that it is the same as the existing volume. Servers can access the volume during this synchronization process.

You can convert a mirrored volume into a non-mirrored volume by deleting one copy or by splitting one copy to create a non-mirrored volume.

The volume copy can be any type: image, striped, or sequential. The volume copy can use thin-provisioning or compression to save capacity. If the copies are in DRPs, you can also use deduplication to the volume copies to increase the capacity savings.

If you are creating a volume, the two copies can be of different types. However, to use deduplication, both copies must be in a DRP. You can add a deduplicated volume copy in a DRP to an existing volume with a copy in a standard pool. You can use this method to migrate existing volume copies to data migration pools.

You can use mirrored volumes for the following reasons:

- ▶ Improving availability of volumes by protecting them from a single storage system failure.
- ▶ Providing concurrent maintenance of a storage system that does not natively support concurrent maintenance.
- ▶ Providing an alternative method of data migration with better availability characteristics. While a volume is migrated by using the data migration feature, it is vulnerable to failures on both the source and target pool. Volume mirroring provides an alternative because you can start with a non-mirrored volume in the source pool, and then, add a copy to that volume in the destination pool.

When the volume is synchronized, you can delete the original copy that is in the source pool. During the synchronization process, the volume remains available, even if a problem occurs with the destination pool.

- ▶ Converting fully allocated volumes to use data reduction technologies, such as thin-provisioning, compression, or deduplication.
- ▶ Converting compressed or thin-provisioned volumes in standard pools to DRPs to improve capacity savings.

When you use volume mirroring, consider how quorum candidate disks are allocated. Volume mirroring maintains some state data on the quorum disks. If a quorum disk is not accessible and volume mirroring cannot update the state information, a mirrored volume might need to be taken offline to maintain data integrity.

To ensure the high availability of the system, ensure that multiple quorum candidate disks are allocated and configured on different storage systems.

When a volume mirror is synchronized, a mirrored copy can become unsynchronized if it goes offline and write I/O requests need to be processed, or if a mirror fast failover occurs. The fast failover isolates the host systems from temporarily slow-performing mirrored copies, which affect the system with a short interruption to redundancy.

**Note:** In standard-provisioned volumes, the primary volume formats before synchronizing to the volume copies. The **-syncrate** parameter on the **mkvdisk** command controls the format and synchronization speed.

You can create a mirrored volume by using the Mirrored option in the Create Volume window, as showing in Figure 5-9 on page 193.

The screenshot shows the 'Create Volumes' window with the 'Mirrored' tab selected. The window title is 'Create Volumes' with a close button (X) in the top right corner. Below the title bar are three tabs: 'Basic', 'Mirrored' (selected), and 'Custom'. A descriptive text states: 'Create preset volumes with copies in multiple pools but at a single site.' The 'Mirrored copies' section contains two rows: 'Copy 0' with a dropdown menu set to 'DBPool' and 'Copy 1' with a dropdown menu set to 'Swimming'. To the right, 'Capacity Details' shows two progress bars; the top one is labeled 'Total 71.07 TiB' and the bottom one 'Total 99.08 TiB'. The 'Volume Details' section includes fields for 'Quantity' (set to 1), 'Capacity' (set to 10 GiB), and 'Name' (set to TESTVOL01). Below these, 'Capacity savings' is set to 'None' and there is an unchecked checkbox for 'Deduplicated'. At the bottom of the window are three buttons: 'Cancel', 'Create and Map', and 'Create'. A help icon (?) is located in the bottom left corner.

Figure 5-8 Mirrored volume creation

You can convert a non-mirrored volume into a mirrored volume by adding a copy, as shown in Figure 5-9 on page 193.



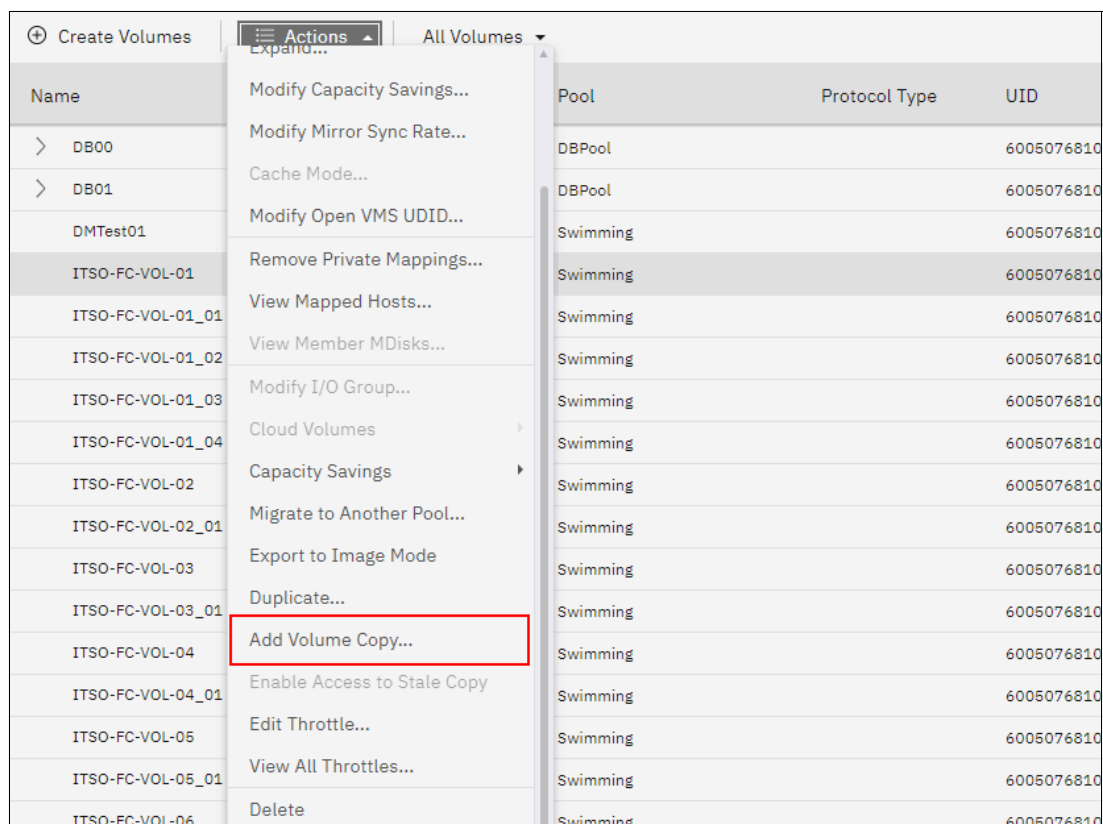


Figure 5-9 Adding a volume copy

## 5.4.1 Write fast failovers

With write fast failovers, the system submits writes to both copies during processing of host write I/O. If one write succeeds and the other write takes longer than 10 seconds, the slower request times-out and ends. The duration of the ending sequence for the slow copy I/O depends on the back-end from which the mirror copy is configured. For example, if the I/O occurs over the Fibre Channel network, the I/O ending sequence typically completes in 10 - 20 seconds.

However, in rare cases, the sequence can take more than 20 seconds to complete. When the I/O ending sequence completes, the volume mirror configuration is updated to record that the slow copy is now no longer synchronized. When the configuration updates finish, the write I/O can be completed on the host system.

The volume mirror stops by using the slow copy for 4 - 6 minutes; subsequent I/O requests are satisfied by the remaining synchronized copy. During this time, synchronization is suspended. Also, the volume's synchronization progress shows less than 100% and decreases if the volume receives more host writes. After the copy suspension completes, volume mirroring synchronization resumes and the slow copy starts synchronizing.

If another I/O request times out on the unsynchronized copy during the synchronization, volume mirroring again stops by using that copy for 4 - 6 minutes. If a copy is always slow, volume mirroring attempts to synchronize the copy again every 4 - 6 minutes and another I/O timeout occurs. The copy is not used for another 4 - 6 minutes and becomes progressively unsynchronized. Synchronization progress gradually decreases as more regions of the volume are written.

If write fast failovers occur regularly, an underlying performance problem can exist within the storage system that is processing I/O data for the mirrored copy that became unsynchronized. If one copy is slow because of storage system performance, multiple copies on different volumes are affected. The copies might be configured from the storage pool that is associated with one or more storage systems. This situation indicates possible overloading or other back-end performance problems.

When you run the `mkvdisk` command to create a volume, the `mirror_write_priority` parameter is set to `latency` by default. Fast failover is enabled. However, fast failover can be controlled by changing the value of the `mirror_write_priority` parameter on the `chvdisk` command. If the `mirror_write_priority` is set to `redundancy`, fast failover is disabled.

The system applies a full SCSI initiator-layer error recovery procedure (ERP) for all mirrored write I/O. If one copy is slow, the ERP can take up to 5 minutes. If the write operation is still unsuccessful, the copy is taken offline. Carefully consider whether maintaining redundancy or fast failover and host response time (at the expense of a temporary loss of redundancy) is more important.

**Note:** Mirrored volumes can be taken offline if no quorum disk is available. This behavior occurs because synchronization status for mirrored volumes is recorded on the quorum disk. To protect against mirrored volumes being taken offline, follow the guidelines for setting up quorum disks.

## 5.4.2 Read fast failovers

Read fast failovers affect how the system processes read I/O requests. A read fast failover determines which copy of a volume the system tries first for a read operation. The primary-for-read copy is the copy that the system tries first for read I/O.

The system submits a host read I/O request to one copy of a volume at a time. If that request succeeds, the system returns the data. If it is not successful, the system retries the request to the other copy volume.

With read fast failovers, when the primary-for-read copy goes slow for read I/O, the system fails over to the other copy. Therefore, the system tries the other copy first for read I/O during the following 4 - 6 minutes. After that attempt, the system reverts to read the original primary-for-read copy.

During this period, if read I/O to the other copy also is slow, the system reverts immediately. Also, if the primary-for-read copy changes, the system reverts to try the new primary-for-read copy. This issue can occur when the system topology changes or when the primary or local copy changes. For example, in a standard topology, the system normally tries to read the primary copy first. If you change the volume's primary copy during a read fast failover period, the system reverts to read the newly set primary copy immediately.

The read fast failover function is always enabled on the system. During this process, the system does not suspend the volumes or make the copies out of sync.

## 5.4.3 Maintaining data integrity of mirrored volumes

Volume mirroring improves data availability by allowing hosts to continue I/O to a volume, even if one of the back-end storage systems fails. However, this mirroring does not affect data integrity. If either of the back-end storage systems corrupts the data, the host is at risk of reading that corrupted data in the same way as for any other volume.

Therefore, before you perform maintenance on a storage system that might affect the data integrity of one copy, it is important to check that both volume copies are synchronized. Then, remove that volume copy before you begin the maintenance.

## 5.5 HyperSwap volumes

HyperSwap volumes create copies on two separate sites for systems that are configured with HyperSwap topology. Data that is written to a HyperSwap volume is automatically sent to both copies so that either site can provide access to the volume if the other site becomes unavailable.

HyperSwap is a system topology that enables Disaster Recovery (DR) and high availability between I/O groups at different locations. Before you configure HyperSwap volumes, the system topology must be configured for HyperSwap and sites must be defined. Figure 5-10 shows an overall view of IBM SAN Volume Controller HyperSwap that is configured with two sites.

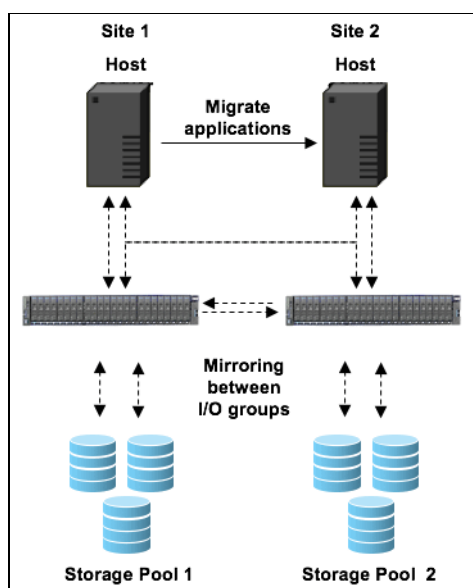


Figure 5-10 Overall HyperSwap diagram

In the management GUI, HyperSwap volumes are configured by specifying volume details, such as quantity, capacity, name, and the method for saving capacity. As with basic volumes, you can choose compression or thin-provisioning to save capacity on volumes. For thin-provisioning or compression, you can also select to use deduplication for the volume that you create. For example, you can create a compressed volume that also uses deduplication to remove duplicated data.

The method for capacity savings applies to all HyperSwap volumes and copies that are created. The volume location displays the site where copies are located, based on the configured sites for the HyperSwap system topology. For each site, specify a pool and I/O group that are used by the volume copies that are created on each site. If you select to deduplicate volume data, the volume copies must be in DRPs on both sites.

The management GUI creates an HyperSwap relationship and change volumes automatically. HyperSwap relationships manage the synchronous replication of data between HyperSwap volume copies at the two sites.

If your HyperSwap system supports self-encrypting drives and the base volume is fully allocated in a DRP, the corresponding change volume is created with compression enabled. If the base volume is in a standard pool, the change volume is created as a thin-provisioned volume.

You can specify a consistency group that contains multiple active-active relationships to simplify management of replication and provide consistency across multiple volumes. A consistency group is commonly used when an application spans multiple volumes. Change volumes maintain a consistent copy of data during resynchronization. Change volumes allow an older copy to be used for DR if a failure occurred on the up-to-date copy before resynchronization completes.

You can also use the `mkvolume` command line to create a HyperSwap volume. The command also defines pools and sites for HyperSwap volume copies and creates the active-active relationship and change volumes automatically. If your HyperSwap system supports self-encrypting drives and the base volume is fully allocated in a DRP, the corresponding change volume is created with compression enabled. If the base volume is in a standard pool, the change volume is created as a thin-provisioned volume.

You can see the relationship between the Master and Auxiliary volume in a 2-site HyperSwap topology in Figure 5-11.

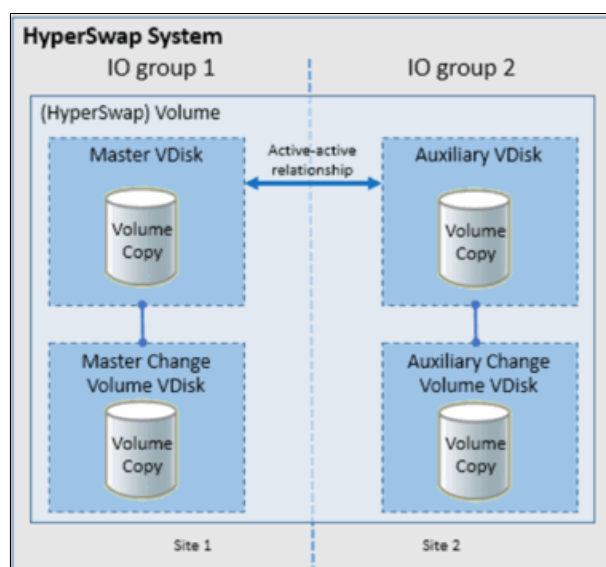


Figure 5-11 Master and Auxiliary volumes

For more information about HyperSwap volumes, see 7.3, “HyperSwap Volumes” on page 334.

## 5.6 VMware virtual volumes

The IBM SAN Volume Controller supports VMware vSphere Virtual Volumes, sometimes referred to as *VVols*, which allows VMware vCenter to automate the management of system objects, such as volumes and pools.

You can assign ownership of Virtual Volumes to IBM Spectrum Connect by creating a user with the VASA Provider security role. IBM Spectrum Connect provides communication between the VMware vSphere infrastructure and the system.

Although you can complete specific actions on volumes and pools that are owned by the VASA Provider security role, IBM Spectrum Connect retains management responsibility for Virtual Volumes.

When virtual volumes are enabled on the system, a utility volume is created to store metadata for the VMware vCenter applications. You can select a pool to provide capacity for the utility volume. With each new volume that is created by the VASA provider, VMware vCenter defines a few kilobytes of metadata that are stored on the utility volume.

The utility volume can be mirrored to a second storage pool to ensure that the failure of a storage pool does not result in loss of access to the metadata. Utility volumes are exclusively used by the VASA provider and cannot be deleted or mapped to other host objects.

**Note:** The utility volume cannot be created in a DRP.

Figure 5-12 provides a high-level overview of the key components that enable the VVols management framework.

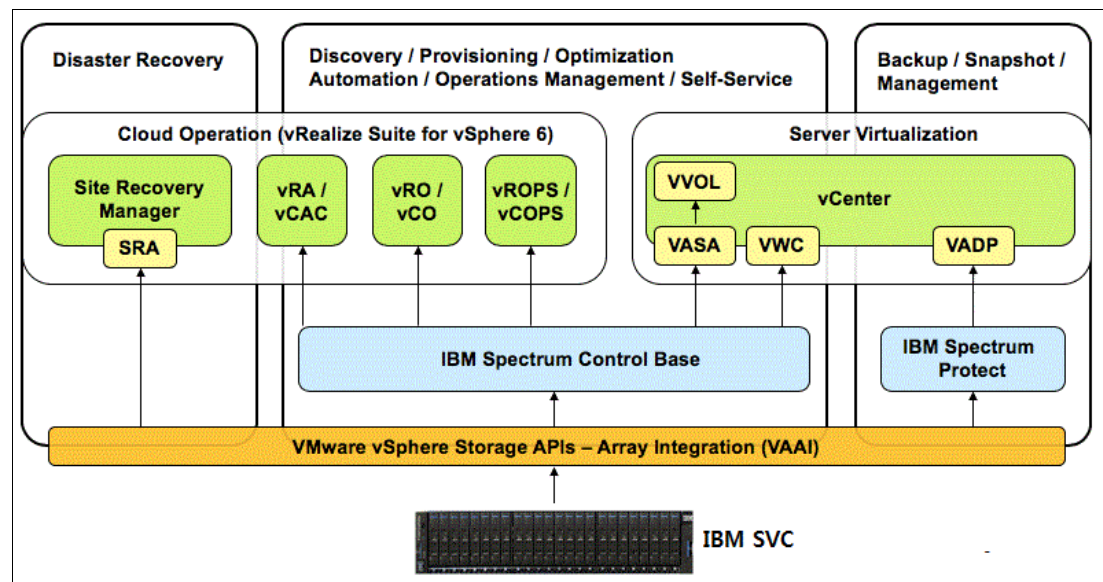


Figure 5-12 Overview of the key components of VMware environment

You can also use data copy through VMware vSphere Storage APIs Array Integration (VAAI), as shown in Figure 5-13.

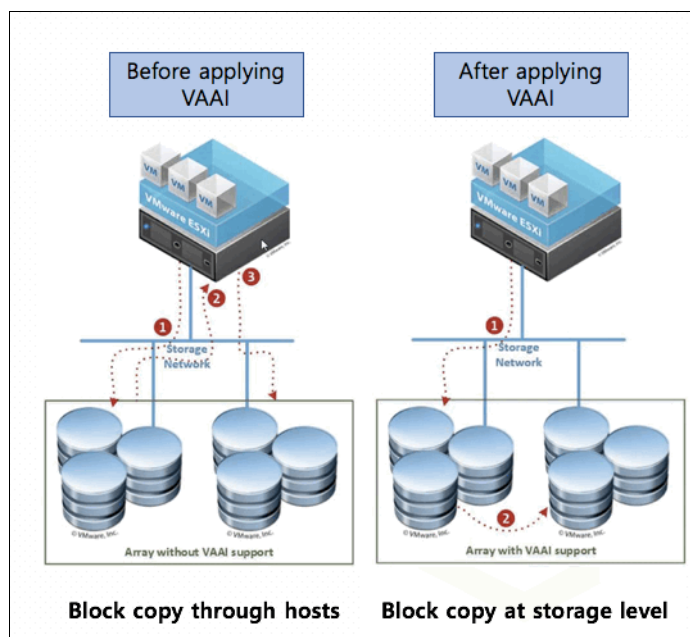


Figure 5-13 VMware vSphere Storage APIs Array Integration (VAAI)

The following are prerequisites must be met before configuring Virtual Volumes:

- ▶ An IBM Spectrum Connect must be set up.
- ▶ VMware vSphere ESXi hosts and vCenter running version 6.0 or later.
- ▶ The Network Time Protocol (NTP) server is configured on IBM SAN Volume Controller and IBM Spectrum Connect.

To start using Virtual Volumes, complete the following steps on the IBM SAN Volume Controller before you configure any settings within the IBM Spectrum Connect server:

1. Enable Virtual Volumes on the IBM SAN Volume Controller:
  - a. In the management GUI, click **Settings** → **System** → **VVOL** and select **On**.
  - b. Select the pool to where the utility volume is stored. If possible, store a mirrored copy of the utility volume in a second storage pool that is in a separate failure domain. The utility volume cannot be created in a DRP.
  - c. Create a user for IBM Spectrum Connect to communicate with the IBM SAN Volume Controller, as shown in Figure 5-14 on page 199.

**Enable VVOL** [X]

Store utility volume here: Pool1 ▼

Store mirrored copy here (optional): Select Pool ▼

Create a user for IBM Spectrum Control Base to connect to this system.

User name: SpectrumUser

Password: ••••••••

Verify password: ••••••••

**Password requirements**

- ✓ Minimum 6 characters long
- ✓ Must not include problematic characters (ex: control characters), or start or end with a space

Cancel Enable

Figure 5-14 Enable VVOL window

2. Create the user account for the IBM Spectrum Connect and the user group with VMware vSphere API for Storage Awareness (VASA) provider role, if they were not set in the previous step:
  - a. Create a user group by clicking **Access** → **Users by Group** → **Create User Group**. Enter the user group name, select **VASA Provider** for the role and click **Create**.
  - b. Create the user account by clicking **Access** → **Users by Group**, select the user group that was created in the previous step, and click **Create User**. Enter the name of the user account, select the user group with VASA Provider role, enter a valid password for the user and click **Create**.
3. For each ESXi host server to use Virtual Volumes, create a host object:
  - a. In the management GUI, select **Hosts** → **Hosts** → **Add Host**.
  - b. Enter the name of the ESXi host server, enter connection information, select **VVOL** for the host type and click **Add Host**.
  - c. If the ESXi host was previously configured, the host type can be changed by modifying the ESXi host type.

**Note:** The user account with VASA Provider role is used by only the IBM Spectrum Connect server to access the IBM SAN Volume Controller and to run the automated tasks that are required for Virtual Volumes. Users must not directly log in to the management GUI or CLI with this type of account and complete system tasks, unless they are directed to by IBM Support.

## 5.7 Cloud volumes

A cloud volume is any volume that is enabled for transparent cloud tiering. After transparent cloud tiering is enabled on a volume, point-in-time copies or snapshots can be created and copied to cloud storage that is provided by a cloud service provider. These snapshots can be restored to the system for DR purposes. Before you create cloud volumes, a valid connection to a supported cloud service provider must be configured.

With transparent cloud tiering, the system supports connections to cloud service providers and the creation of cloud snapshots of any volume or volume group on the system. Cloud snapshots are point-in-time copies of volumes that are created and transferred to cloud storage that is managed by a cloud service provider.

A cloud account defines the connection between the system and a supported cloud service provider. It also must be configured before data can be transferred to or restored from the cloud storage. After a cloud account is configured with the cloud service provider, you determine which volumes you want to create cloud snapshots of and enable transparent cloud tiering on those volumes.

Figure 5-15 shows an example of IBM SAN Volume Controller Transparent Cloud Tiering.

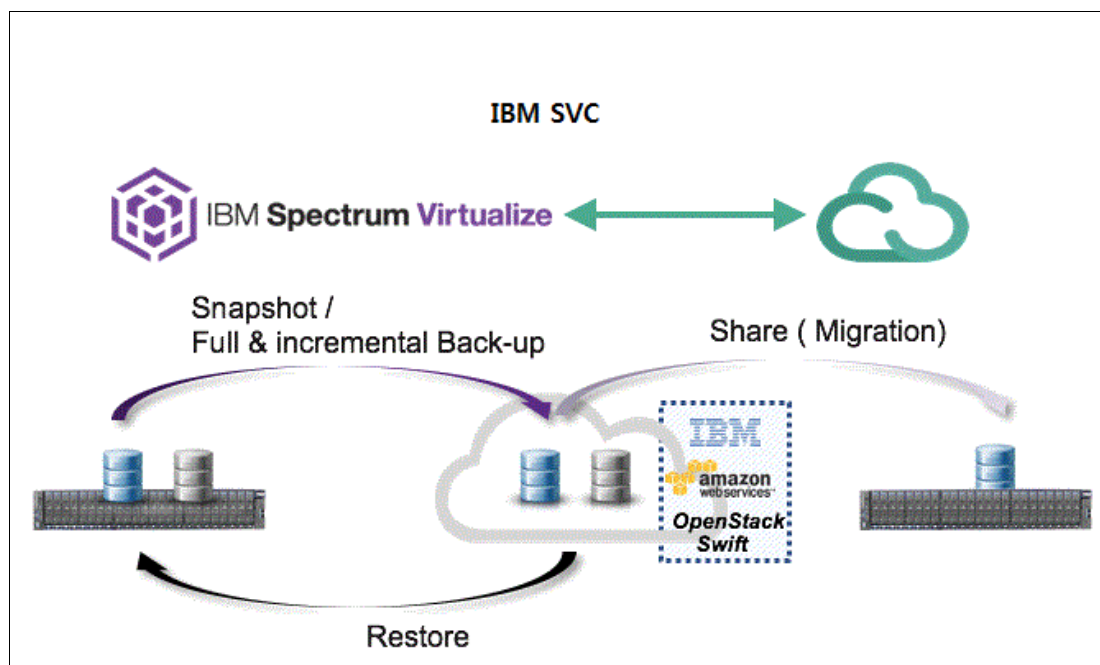


Figure 5-15 Cloud volumes - Transparent Cloud Tiering

A cloud account is an object on the system that represents a connection to a cloud service provider by using a particular set of credentials. These credentials differ depending on the type of cloud service provider that is being specified. Most cloud service providers require the host name of the cloud service provider and an associated password. Some cloud service providers also require certificates to authenticate users of the cloud storage. Public clouds use certificates that are signed by well-known certificate authorities.

Private cloud service providers can use a self-signed certificate or a certificate that is signed by a trusted certificate authority. These credentials are defined on the cloud service provider and passed to the system through the administrators of the cloud service provider.



A cloud account defines whether the system can successfully communicate and authenticate with the cloud service provider by using the account credentials.

If the system is authenticated, it can access cloud storage to copy data to the cloud storage or restore data that is copied to cloud storage back to the system. The system supports one cloud account to a single cloud service provider. Migration between providers is not supported.

The system supports IBM Cloud®, OpenStack Swift and Amazon S3 cloud service providers.

### 5.7.1 Transparent cloud tiering configuration limitations and rules

Consider the following limitations and rules regarding transparent cloud tiering:

- ▶ One cloud account per system.
- ▶ A maximum of 1024 volumes can have cloud-snapshot enabled volumes.
- ▶ The maximum number of active snapshots per volume is 256.
- ▶ The maximum number of volume groups is 512.
- ▶ Cloud volumes cannot be expanded or shrunk.
- ▶ A volume cannot be configured for a cloud snapshot if any of the following conditions exist:
  - The volume is part of a Remote Copy relationship (Metro Mirror, Global Mirror, active-active) master, auxiliary, or change volume. This configuration prevents the cloud snapshot from being used with HyperSwap volumes.
  - The volume is a VMware vSphere Virtual Volumes volume, including IBM FlashCopy owned volumes that are used internally for Virtual Volumes restoration functions.
  - The volume is:
    - A file system volume
    - Associated with any user-owned FlashCopy maps
    - A mirrored volume with copies in different storage pools
    - Being migrated between storage pools
- ▶ A volume cannot be enabled for cloud snapshots if the cloud storage is set to import mode.
- ▶ A volume cannot be enabled for cloud snapshots if the maximum number of cloud volumes exists. The maximum number of cloud volumes on the system is 1024. If the system exceeds this limit, you can disable cloud snapshots on a cloud volume and delete its associated snapshots from the cloud storage to accommodate snapshots on new cloud volumes.
- ▶ A volume cannot be used for a restore operation if it meets any of the following criteria:
  - A Virtual Volume, including FlashCopy volumes that are used internally for Virtual Volumes restoration functions
  - A file system volume
  - Part of a Remote Copy relationship (Metro Mirror, Global Mirror, or active-active) master, auxiliary, or change volume
- ▶ A volume that is configured for backup or is being used for restoration cannot be moved between I/O groups.
- ▶ Only one operation (cloud snapshot, restore, or snapshot deletion) is allowed at a time on a cloud volume.
- ▶ Cloud volume traffic is allowed only through management interfaces (1 G or 10 G).

## 5.7.2 Restoring to the production volume

This is a process where snapshot version is restored to the production volume, which is the original volume from which the snapshots were created. After the restore operation completes, the snapshot version completely replaces the current data that exists on production volume. During the restore operation, the production volume goes offline until it completes. Data is not fully restored to the production volume until the changes are committed.

## 5.7.3 Restore to a new volume

If you do not want to have the production volume offline for the restore, you can restore a cloud snapshot to a new volume. The production volume remains online and host operations are not disrupted.

When the snapshot version is restored to a new volume, you can use the restored data independently of the original volume from which the snapshot was created. If the new volume exists on the system, the restore operation uses the unique identifier (UID) of the new volume. If the new volume does not exist on the system, you must choose whether to use the UID from the original volume or create a UID. If you plan to use the new volume on the same system, use the UID that is associated with the snapshot version that is being restored.

## 5.8 Volume migration

Migrating an image mode volume to managed mode volume or vice versa is done by migrating a volume from one storage pool to another. A volume can also be migrated to a different type of storage pool.

When migrating from image to managed or vice versa, the command varies, as listed in Table 5-2.

*Table 5-2 Migration types and associated commands*

| Storage pool-to-storage pool type         | Command               |
|-------------------------------------------|-----------------------|
| Managed-to-managed or<br>Image-to-managed | <b>migratevdisk</b>   |
| Managed-to-image or<br>Image-to-image     | <b>migratetoimage</b> |

Migrating a volume from one storage pool to another is nondisruptive to the host application that uses the volume. Depending on the workload of IBM SAN Volume Controller, performance might be slightly affected.

The migration of a volume from one storage pool to another storage pool by using **migratevdisk** command is allowed only if both storage pools feature the same extent size. Volume mirroring can be used if a volume must be migrated from one storage pool to another storage pool with different extent sizes.

In the next sections, we provide guidance for migrating volumes by using the methods that were described in this section.

## 5.8.1 Image-type to striped-type volume migration

When you are migrating storage into the IBM SAN Volume Controller, the storage is brought in as *image-type volumes*, which means that the volume is based on a single MDisk. The CLI command that can be used is **migratevdisk**.

Example 5-5 shows the **migratevdisk** command that can be used to migrate an image-type volume to a striped-type volume. The command also can be used to migrate a striped-type volume to a striped-type volume.

*Example 5-5 The migratevdisk command*

---

```
superuser> migratevdisk -mdiskgrp MDG1DS4K -threads 4 -vdisk Migrate_sample
```

---

This command migrates the volume `Migrate_sample` to the storage pool `MDG1DS4K`, and uses four threads when migrating. Instead of using the volume name, you can use its ID number.

You can monitor the migration process by using the **lsmigrate** command, as shown in Example 5-6.

*Example 5-6 Monitoring the migration process*

---

```
superuser> lsmigrate
migrate_type MDisk_Group_Migration
progress 0
migrate_source_vdisk_index 3
migrate_target_mdisk_grp 2
max_thread_count 4
migrate_source_vdisk_copy_id 0
```

---

## 5.8.2 Migrating to image-type volume

An *image-type volume* is a direct, “straight-through” mapping to one image mode MDisk. If a volume is migrated to another MDisk, the volume is represented as being in managed mode during the migration (because it is striped on two MDisks).

It is represented only as an image-type volume after it reaches the state where it is a straight-through mapping. An image-type volume cannot be expanded.

Image-type disks are used to migrate data to an IBM SAN Volume Controller and migrate data out of virtualization. In general, the reason for migrating a volume to an image type volume is to move the data on the disk to a non-virtualized environment.

If the migration is interrupted by a cluster recovery, the migration resumes after the recovery completes.

The **migratetoimage** command migrates the data of a user-specified volume by consolidating its extents (which might be on one or more MDisks) onto the extents of the target MDisk that you specify. After migration is complete, the volume is classified as an image type volume, and the corresponding MDisk is classified as an image mode MDisk.

The managed disk that is specified as the target must be in an *unmanaged* state at the time that the command is run. Running this command results in the inclusion of the MDisk into the user-specified storage pool.

**Remember:** This command cannot be used if the source volume copy is in a child pool or if the target MDisk group that is specified is a child pool. This command does not work if the volume is fast formatting.

The **migratetoimage** command fails if the target or source volume is offline. Correct the offline condition before attempting to migrate the volume.

If the volume (or volume copy) is a target of a FlashCopy mapping with a source volume in an active-active relationship, the new managed disk group must be in the same site as the source volume. If the volume is in an active-active relationship, the new managed disk group must be in the same site as the source volume. Also, the site information for the MDisk being added must be well defined and match the site information for other MDisk in the storage pool.

**Note:** You cannot migrate a volume or volume image between storage pools if cloud snapshot is enabled on the volume.

An encryption key cannot be used when migrating an image mode MDisk. To use encryption (when the MDisk has an encryption key), the MDisk must be self-encrypting before configuring storage pool.

The **migratetoimage** command is useful when you want to use your system as a data mover. For more information about the requirements and specifications for the **migratetoimage** command, see [IBM Documentation](#).

### 5.8.3 Migrating with volume mirroring

Volume mirroring also offers the ability to migrate volumes between storage pools with different extent sizes.

Complete the following steps to migrate volumes between storage pools:

1. Add a copy to the target storage pool.
2. Wait until the synchronization is complete.
3. Remove the copy in the source storage pool.

To migrate from a thin-provisioned volume to a fully allocated volume, the process is similar:

1. Add a target fully allocated copy.
2. Wait for synchronization to complete.
3. Remove the source thin-provisioned copy.

In both cases, if you set the **autodelete** option to yes when creating the volume copy, the source copy is automatically deleted, and you can skip the third step in both processes. The preferred practice on this type of migration is to try not to overload the systems with a high syncrate or with too many migrations at the same time.

**Note:** You cannot use the data migration function to move a volume between storage pools that have different extent sizes. Migration commands fail if the target or source volume is offline, no quorum disk is defined, or the defined quorum disks are unavailable. Correct the offline or quorum disk condition and rerun the command.

The **syncrate** parameter specifies the copy synchronization rate. A value of zero (0) prevents synchronization. The default value is 50. The supported **-syncrate** values and their corresponding rates are listed in Table 5-3.

Table 5-3 Sample syncrate values

| User-specified syncrate attribute value | Data copied per second |
|-----------------------------------------|------------------------|
| 1 - 10                                  | 128 KB                 |
| 11 - 20                                 | 256 KB                 |
| 21 - 30                                 | 512 KB                 |
| 31 - 40                                 | 1 MB                   |
| 41 - 50                                 | 2 MB                   |
| 51 - 60                                 | 4 MB                   |
| 61 - 70                                 | 8 MB                   |
| 71 - 80                                 | 16 MB                  |
| 81 - 90                                 | 32 MB                  |
| 91 - 100                                | 64 MB                  |

We recommend modifying syncrate after monitoring overall bandwidth and latency. Then, if the performance is not affected on migration, increase the syncrate to complete within the allotted time.

You can also use volume mirroring when you migrate a volume from a non-virtualized storage device to IBM SAN Volume Controller. As you can see in Figure 5-16, you first must attach the storage to IBM SAN Volume Controller by using the virtualization solution, which requires some downtime because hosts start to access the volumes through IBM SAN Volume Controller.

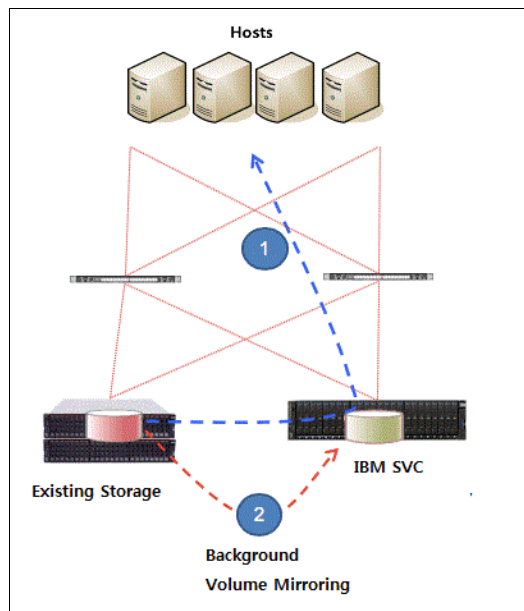


Figure 5-16 Migration with Volume Mirroring

After the storage is correctly attached to IBM SAN Volume Controller, map the image-type volumes to the hosts so that recognizes volumes as though they were accessed through the non-virtualized storage device. Then, you can restart applications. After that process completed, you can use volume mirroring to migrate the volumes to a storage pool with managed MDisks, which creates striped-type copies of each volume in this target pool. Data synchronization in the volume copies then starts in the background.

For more information, see [IBM Documentation](#).

## 5.8.4 Migration from standard pools to Data Reduction Pools

If you want to migrate volumes to DRP, you can move them with volume mirroring between a standard pool and DRP. Hosts I/O operations are not disrupted during migration. Figure 5-17 shows two examples of how you can use volume mirroring to convert volumes to a different type or migrate volumes to a different type of pool.

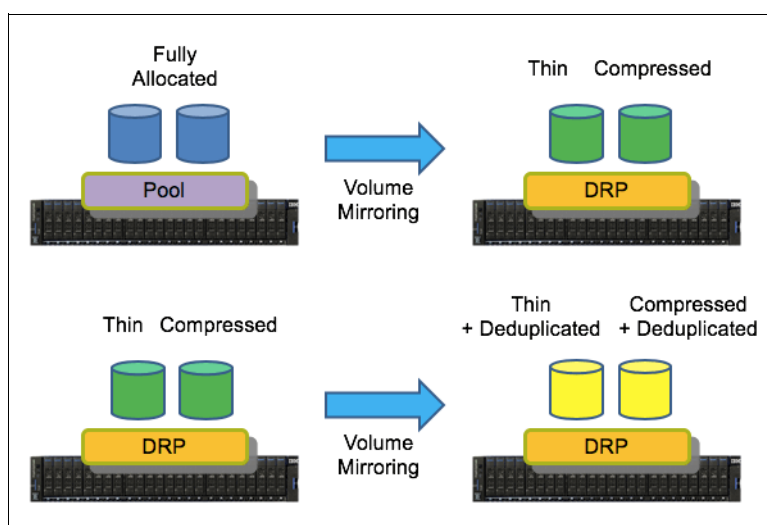


Figure 5-17 Converting volumes with Volume Mirroring

You can also move compressed or thin-provisioned volumes in standard pools to DRPs to simplify management of reclaimed capacity. The DRP tracks the unmap operations of the hosts and reallocates capacity automatically. The system supports volume mirroring to create a copy of the volume in a new DRP. This method creates a copy of the volume in a new DRP and does not disrupt host operations.

## 5.9 Preferred paths to a volume

When a volume is created, it is assigned to an I/O group and assigned a preferred node. The preferred node is the node that normally processes I/Os for the volume. The primary purposes of a preferred node are load balancing and to determine which node destages writes to the backend storage.

Preferred node assignment is normally automatic. The system selects the node in the I/O group that includes the fewest volumes. However, the preferred node can be specified or changed, if needed.

All modern multipathing drivers support Asymmetric Logical Unit Access (ALUA). This access allows the storage to mark certain paths as preferred (paths to the preferred node). ALUA multipathing drivers honor preferred pathing and send I/O to only the other node if the preferred node is not accessible.

Figure 5-18 shows write operations from a host to two volumes with different preferred nodes.

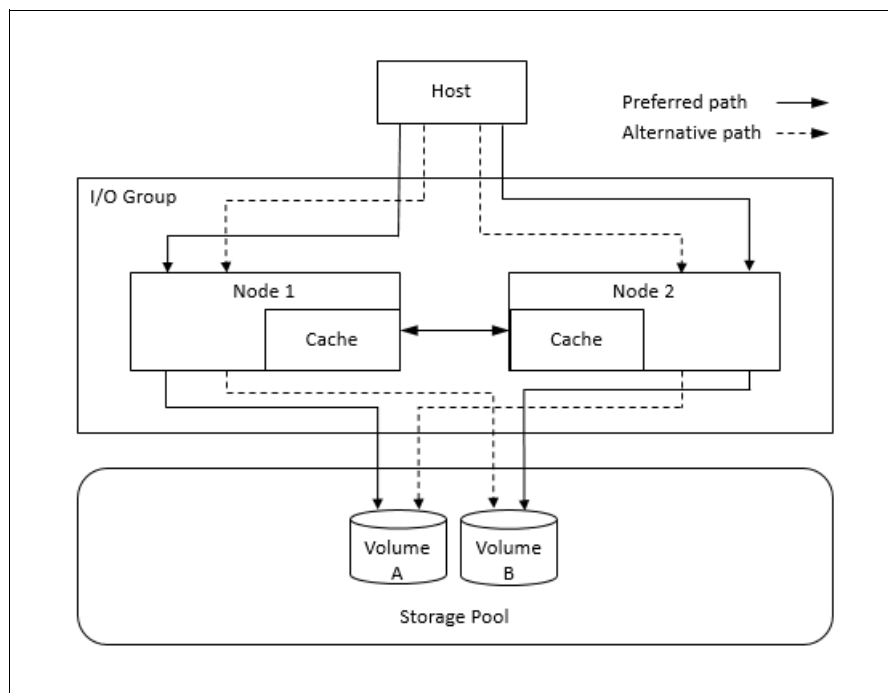


Figure 5-18 Write operations from a host

When debugging performance problems, it can be useful to review the Non-Preferred Node Usage Percentage metric in IBM Spectrum Control or IBM Storage Insights. I/O to the non-preferred node might cause performance problems for the I/O group and can be identified on these tools.

For more information about this performance metric and more in IBM Spectrum Control, see [IBM Spectrum Control 5.4.2 Documentation - Performance metrics for resources that run IBM Spectrum Virtualize](#).

## 5.10 Moving a volume between I/O groups and nodes

To balance the workload across I/O groups and nodes, you can move volumes between I/O groups and nodes.

The change of preferred node of a volume within an I/O group or to another I/O group is a nondisruptive process.

### 5.10.1 Changing the preferred node of a volume within an I/O group

Changing the preferred node within an I/O group can be done with concurrent I/O. However, it can lead to some delay in performance and in the case of some specific operating systems or applications, they might detect some timeouts.

This operation can be done by using the CLI and GUI; however, if you have only one I/O group, this operation is not possible by using the GUI. To change the preferred node within an I/O group by using CLI, use the `movevdisk -node <node_id or node_name> <vdisk_id or vdisk_name>` command.

### 5.10.2 Moving a volume between I/O groups

When moving a volume between I/O groups, it is recommended that the system chooses the volume preferred node in the new I/O group. However, it is possible to manually set the preferred node during this operation by using the GUI and CLI.

Some limitations exist in moving a volume across I/O groups, which is named Non-Disruptive Volume Move (NDVM). These limitations are mostly in Host Cluster environments, and you can check the compatibility at the [IBM SSIC website](#).

**Note:** These migration tasks can be nondisruptive if performed correctly and the hosts that are mapped to the volume support NDVM. The cached data that is held within the system must first be written to disk before the allocation of the volume can be changed.

Modifying the I/O group that services the volume can be done concurrently with I/O operations if the host supports non-disruptive volume move. It also requires a rescan at the host level to ensure that the multipathing driver is notified that the allocation of the preferred node changed and the ports by which the volume is accessed changed. This rescan can be done in the situation where one pair of nodes becomes over-used.

If any host mappings are available for the volume, the hosts must be members of the target I/O group or the migration fails.

Verify that you created paths to I/O groups on the host system. After the system successfully adds the new I/O group to the volume's access set and you moved the selected volumes to another I/O group, detect the new paths to the volumes on the host.

The commands and actions on the host vary depending on the type of host and the connection method used. These steps must be completed on all hosts to which the selected volumes are currently mapped.

**Note:** If the selected volume is performing quick initialization, this wizard is unavailable until quick initialization is complete.



## 5.11 Volume throttling

Volume throttling effectively throttles the number of I/O operations per second (IOPS) or bandwidth (MBps) that can be achieved to and from a specific volume. You might want to use I/O throttling if you have a volume that has an access pattern that adversely affects the performance of other volumes.

For example, volumes that are used for backup or archive operations can have I/O intensive workloads, potentially taking bandwidth from production volumes. Volume throttle can be used to limit I/Os for these types of volumes so that I/O operations for production volumes are not affected.

Figure 5-19 shows the example of volume throttling.



Figure 5-19 Volume throttling for each LUNs

When deciding between using IOPS or bandwidth as the I/O governing throttle, consider the disk access pattern of the application. Database applications often issue large amounts of I/O, but they transfer only a relatively small amount of data. In this case, setting an I/O governing throttle that is based on MBps does not achieve the expected result. Therefore, it is better to set an IOPS limit.

On the other hand, a streaming video application often issues a small amount of I/O, but it transfers large amounts of data. In contrast to the database example, defining an I/O throttle based in IOPS does not achieve a good result. For a streaming video application, it is better to set an MBps limit.

You can edit the throttling value in the menu, as shown in Figure 5-20.

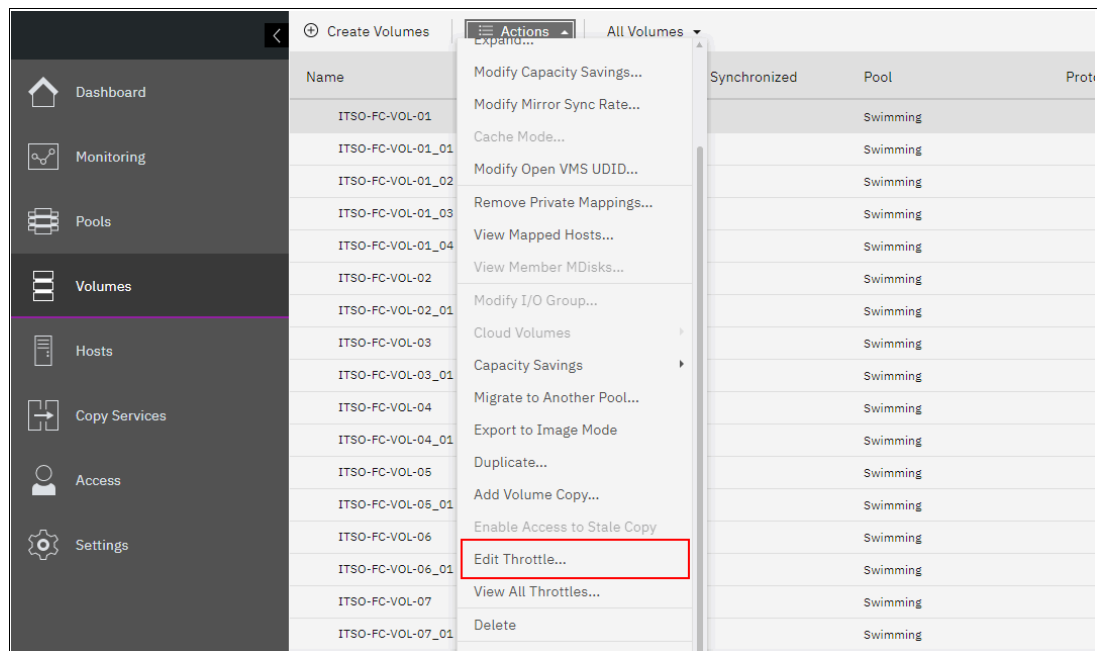


Figure 5-20 Volume Throttling

Figure 5-21 shows both bandwidth and IOPS parameter that can be set.

Figure 5-21 Edit bandwidth and IOPS limit

Throttling at a volume level can be set by using the following commands:

► **mkthrottle**

This command is used to set I/O throttles for volumes that use this command. It must be used with **-type vdisk** parameter, followed by **-bandwidth bandwidth\_limit\_in\_mbdisk** or **-iops iops\_limit** to define MBps and IOPS limits.

► **chvdisk**

When used with **-rate throttle\_rate** parameter, this command specifies the IOPS and MBps limits. The default **throttle\_rate** units are I/Os. To change the **throttle\_rate** units to megabits per second (MBps), specify the **-unitmb** parameter. If **throttle\_rate** value is zero, the throttle rate is disabled. By default, the **throttle\_rate** parameter is disabled.

**Note:** The command **mkthrottle** is used not to create throttles for volumes only, but also for hosts, host clusters, pools, and system offload.

When the IOPS limit is configured on a volume and it is smaller than 100 IOPS, the throttling logic rounds it to 100 IOPS. Even if throttle is set to a value smaller than 100 IOPS, the throttling occurs at 100 IOPS.

After any of the commands that were described thus far are used to set volume throttling, a throttle object is created. Then, you can list your created throttle objects by using the **lsthrottle** command and change their parameters with the **chthrottle** command. Example 5-7 shows some command examples.

*Example 5-7 Throttle commands example*

```
superuser>mkthrottle -type vdisk -bandwidth 100 -vdisk Vol01
Throttle, id [0], successfully created.
superuser>lsthrottle
throttle_id throttle_name object_id object_name throttle_type IOPs_limit
bandwidth_limit_MB
0          throttle0      52          Vol01          vdisk          100
```

```

superuser>chthrottle -iops 1000 throttle0
superuser>lsthrottle
throttle_id throttle_name object_id object_name throttle_type IOPs_limit
bandwidth_limit_MB
0          throttle0    52          Vol01      vdisk        1000        100

superuser>lsthrottle throttle0
id 0
throttle_name throttle0
object_id 52
object_name Vol01
throttle_type vdisk
IOPs_limit 1000
bandwidth_limit_MB 100

```

---

For more information and the procedure to set volume throttling, see [IBM Documentation](#).

## 5.12 Volume cache mode

Cache mode in IBM SAN Volume Controller determines whether read and write operations are stored in cache. For each volume, one of the following cache modes can be used:

- readwrite (enabled)

All read and write I/O operations that are performed by the volume are stored in cache. This default cache mode is used for all volumes. A volume or volume copy that is created from a DRP must have a cache mode of readwrite.

When you create a thin provisioned volume, set the cache mode to readwrite to maximize performance. If you set the mode to none, the system cannot cache the thin-provisioned metadata and performance is decreased. In a DRPs, it is not possible to create a thin-provisioned or compressed volume copy setting for the cache mode that is different than readwrite.

- readonly

All read I/O operations that are performed by the volume are stored in cache.

- none (disabled)

All read and write I/O operations that are performed by the volume are not stored in cache.

By default, when a volume is created, the cache mode is set to readwrite. Disabling cache can affect performance and increase read and write response time.

Figure 5-22 shows write operation behavior when volume cache is activated (readwrite).

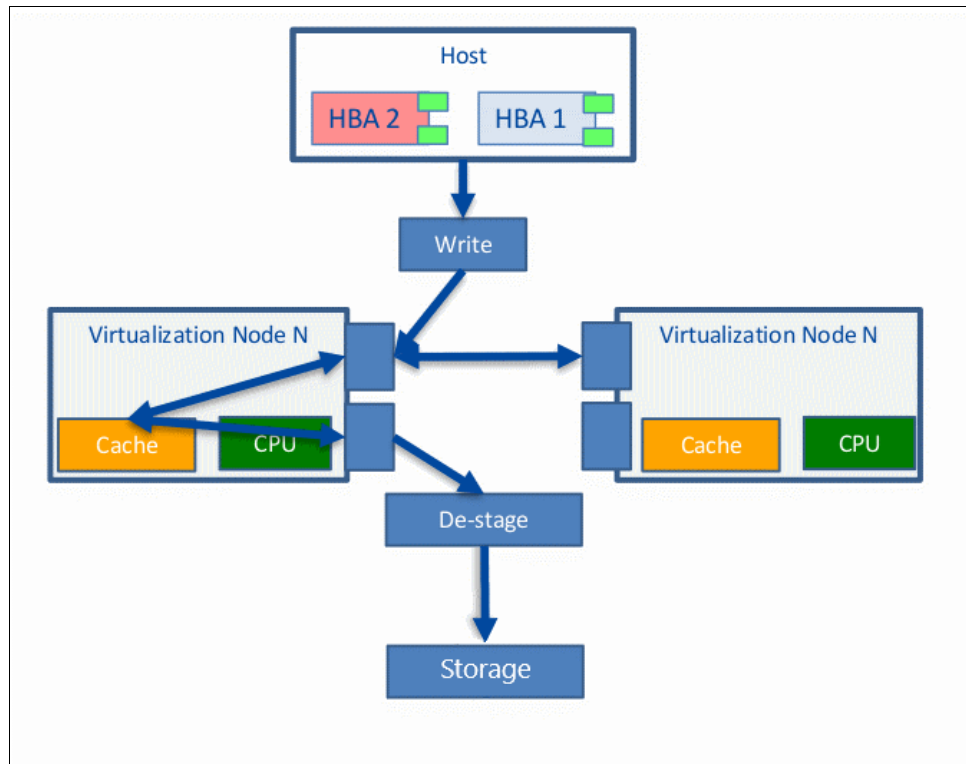


Figure 5-22 Cache activated

Figure 5-23 shows a write operation behavior when volume cache is deactivated (none).

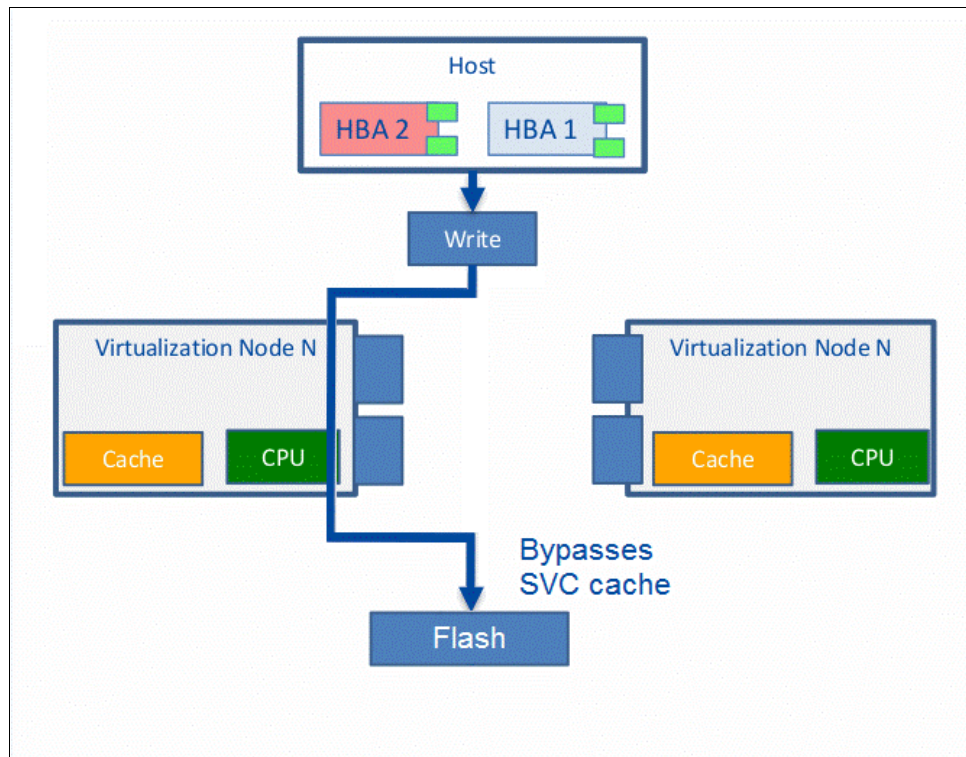


Figure 5-23 Cache deactivated

In most cases, the volume with readwrite cache mode is recommended because disabling cache for a volume can result in performance issues to the host. However, some specific scenarios exist in which it is recommended to disable the readwrite cache.

You might use cache-disabled (none) volumes when you have Remote Copy or FlashCopy in a backend storage controller, and these volumes are virtualized in IBM SAN Volume Controller devices as image VDisks. Another possible use of a cache-disabled volume is when intellectual capital is in copy services automation scripts. Keep the use of cache-disabled volumes to minimum for normal workloads.

You can also use cache-disabled volumes to control the allocation of cache resources. By disabling the cache for specific volumes, more cache resources are available to cache I/Os to other volumes in the same I/O group. For example, a non-critical application that uses volumes in MDisks from all-flash storage.

Even if this application generates many IOPS and requires low response time, you can disable the cache of the volumes in IBM SAN Volume Controller. This action does not cause a greatly affect the application, It also uses fewer resources in the storage system.

**Note:** Volumes with readwrite cache enabled is recommended.

By default, volumes are created with cache mode enabled (read/write), but you can specify the cache mode when the volume is created by using the **-cache** option.

The cache mode of a volume can be concurrently changed (with I/O) by using the **chvdisk** command or GUI, selecting **Volumes** → **Volumes** → **Actions** → **Cache Mode**. Figure 5-24 shows editing cache mode for a volume.

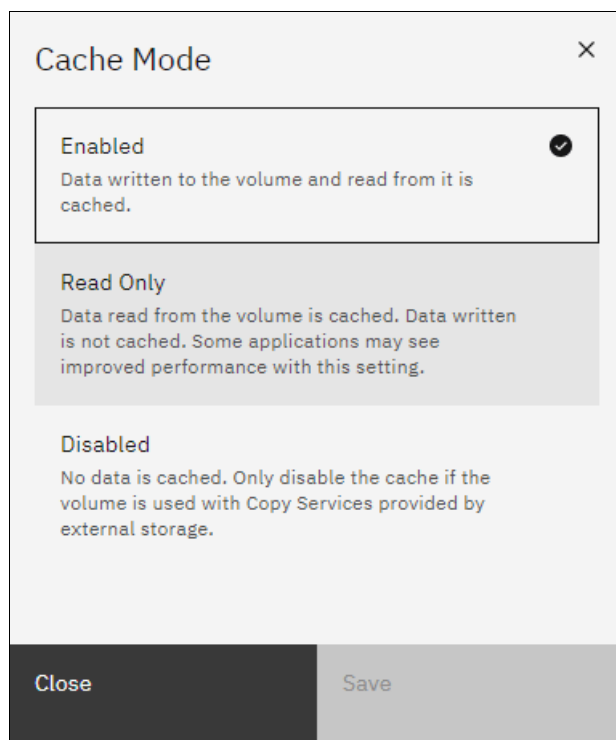


Figure 5-24 Edit cache mode

The command line does not fail I/O to the user, and the command must be allowed to run on any volume. If used correctly without the **-force** flag, the command does not result in a corrupted volume. Therefore, the cache must be flushed and you must discard cache data if the user disables cache on a volume.

Example 5-8 shows an image volume VDISK\_IMAGE\_1 that changed the cache parameter after it was created.

*Example 5-8 Changing the cache mode of a volume*

---

```
superuser>mkvdisk -name VDISK_IMAGE_1 -iogrp 0 -mdiskgrp IMAGE_Test -vtype image
-mdisk D8K_L3331_1108
Virtual Disk, id [9], successfully created
superuser>lsvdisk VDISK_IMAGE_1
id 9
.
lines removed for brevity
.
fast_write_state empty
cache readwrite
.
lines removed for brevity

superuser>chvdisk -cache none VDISK_IMAGE_1
superuser>lsvdisk VDISK_IMAGE_1
id 9
.
lines removed for brevity
.
cache none
.
lines removed for brevity
```

---

In an environment with Copy Services (FlashCopy, Metro Mirror, Global Mirror, and Volume Mirroring) and typical workloads, disabling IBM SAN Volume Controller cache is detrimental to overall performance.

**Attention:** Carefully evaluate the effect to the entire system with quantitative analysis before and after making this change.

## 5.13 Other considerations

This section describes other considerations regarding volumes.

### 5.13.1 Volume protection

You can protect volumes to prevent active volumes or host mappings from being deleted. IBM SAN Volume Controller features a global setting enabled by default that prevents these objects from being deleted if the system detects recent I/O activity. You can set this value to apply to all volumes that are configured on your system, or control whether the system-level volume protection is enabled or disabled on specific pools.

To prevent an active volume from being deleted unintentionally, administrators must enable volume protection. They can also specify a period that the volume must be idle before it can be deleted. If volume protection is enabled and the period is not expired, the volume deletion fails, even if the **-force** parameter is used.

When you delete a volume, the system verifies whether it is a part of a host mapping, FlashCopy mapping, or remote-copy relationship. In these cases, the system fails to delete the volume, unless the **-force** parameter is specified. However, if volume protection is enabled, the **-force** parameter does not delete a volume if it has I/O activity in the last minutes defined in the protection duration time in volume protection.

**Note:** The **-force** parameter overrides the volume dependencies, not the volume protection setting. Volume protection must be disabled to permit a volume or host-mapping deletion if the volume had recent I/O activity.

Consider enabling volume protection by using **chsystem vdiskprotectionenabled yes -vdiskprotectiontime <value\_in\_minutes>**.  
If you want to have volume protection enabled in your system, but disabled in a specific storage pool, you can use the **chmdiskgrp -vdiskprotectionenabled no <pool\_name\_or\_ID>** command.

You can also manage volume protection in GUI by selecting **Settings → System → Volume Protection**, as shown in Figure 5-25.

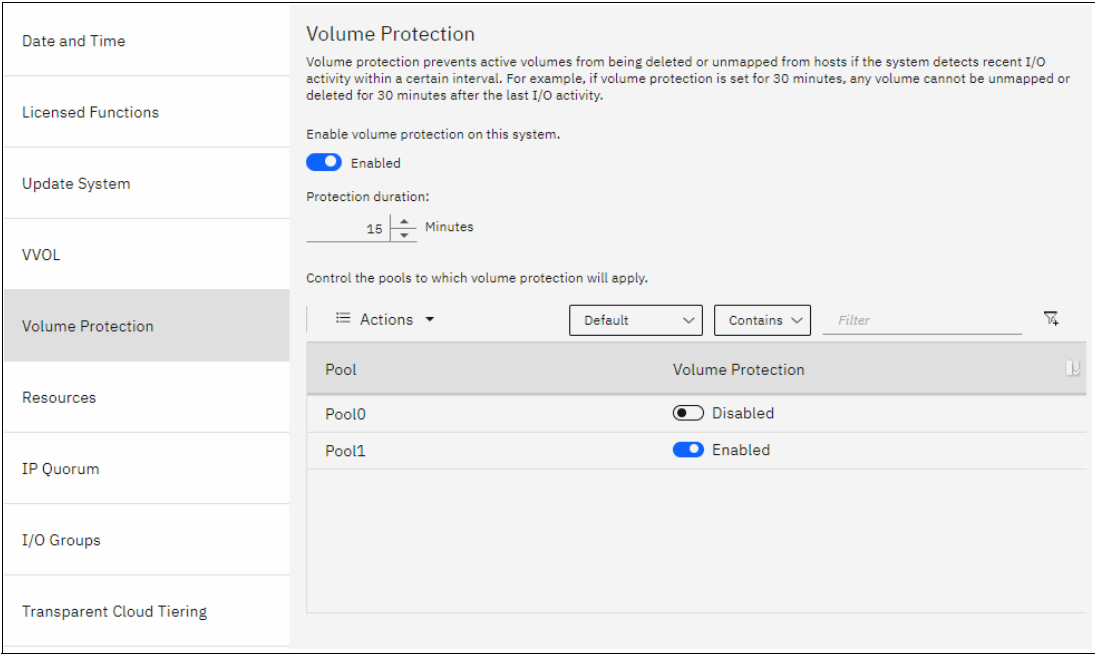


Figure 5-25 Volume Protection

## 5.13.2 Volume resizing

Fully allocated and thin-provisioned volumes can have their sizes increased or decreased. A volume can be expanded with concurrent I/Os for some operating systems. However, *never* attempt to shrink a volume that is in use that contains data because volume capacity is removed from the end of the disk, whether that capacity is in use by a server. A volume cannot be expanded or shrunk during its quick initialization process.

### Expanding a volume

You can expand volumes for the following reasons:

- ▶ To increase the available capacity on a specific volume that is mapped to a host.
- ▶ To increase the size of a volume to make it match the size of the source or master volume so that it can be used in a FlashCopy mapping or Metro Mirror relationship.

Figure 5-26 shows the Expand Volume window.

Expand Volume - ITS0-FC-VOL-01

**i** You selected to expand the provisioned capacity of volume ITS0-FC-VOL-01. This action increases the available capacity in the volume.

Current size: 10 GiB

Expand by: 5 GiB

Final size: 15 GiB

Format: ☒ Enabled

Maximum final size: 106,578.00 GiB

Cancel Expand

Figure 5-26 Expand Volumes

### Shrinking a volume

Volumes can be reduced in size if necessary. If a volume does not contain any data, it is unlikely that you encounter any issues when shrinking its size. However, if a volume is in use and contains data, do not shrink its size because IBM Spectrum Virtualize is unaware if it is removing used or non-used capacity.

**Attention:** When you shrink a volume, capacity is removed from the end of the disk, whether that capacity is in use. Even if a volume includes free capacity, do not assume that only unused capacity is removed when you shrink a volume.



Figure 5-27 shows shrinking volumes.

Shrink Volume - ITSO-FC-VOL-02

⚠ If this volume contains data that is being used, do not attempt to shrink a volume under any circumstances without first backing up your data.  
You selected to shrink the capacity of volume **ITSO-FC-VOL-02**. The system arbitrarily reduces the capacity of the volume by removing one or more extents that are assigned to the volume. You cannot control which extents are removed and cannot ensure that unused capacity is removed.

Current size:  GiB

Shrink by:  GiB ▾

Final size:  GiB

Cancel Shrink

Figure 5-27 Shrink Volumes

### 5.13.3 Migrating from Fibre Channel connections to RDMA over Ethernet connections between nodes

IBM SAN Volume Controller support node-to-node connections that use Ethernet protocols that support remote direct memory access (RDMA) technology, such as RDMA over Converged Ethernet (RoCE) or iWARP. To use these protocols, the system requires that an RDMA-capable adapter is installed on each node and dedicated RDMA-capable Ethernet ports are configured only for node-to-node communication. If your system uses Fibre Channel ports, you can migrate to RDMA-capable Ethernet ports for node-to-node communications.

RDMA technologies, such as RoCE and iWARP, enable the RDMA-capable adapter to transfer data directly between nodes, and bypass the CPU and cache, which makes transfers faster. RDMA technologies provide faster connection and processing time than traditional iSCSI connections.

The following prerequisites must be met for all RDMA-capable Ethernet ports that are used between nodes:

- ▶ All node hardware is installed.
- ▶ The 25-Gbps Ethernet adapter that supports RDMA technology is installed on each node. If you use RDMA-technology for node-to-node communications, ensure that the RDMA-capable adapters use the same technology, such as RoCE or iWARP. These RDMA-capable adapters must be installed in the same slots across all of the nodes of the system. These installation requirements ensure that port identifiers are the same across all nodes in the system.
- ▶ Ethernet cables between each node are connected correctly.
- ▶ The protocol technology on the source and destination adapters is the same.
- ▶ The local and remote IP addresses can be reached.
- ▶ Each IP address for RDMA-capable Ethernet ports and their associated subnet masks are unique on each node.

- ▶ The router is not be placed between nodes that use RDMA-capable Ethernet ports for node-to-node communication.
- ▶ The negotiated speeds on the local and remote adapters are the same.
- ▶ The local and remote port virtual LAN identifiers are the same. Use virtual LAN to create physical separation of networks for unrelated systems, wherever possible. All the ports that are used for node-to node communication must be assigned with one VLAN ID and ports that are used for host attachment must have a different VLAN ID.

If you plan to use VLAN to create this separation, you must configure VLAN support on the all the Ethernet switches in your network before you define the RDMA-capable Ethernet ports on nodes in the system. On each switch in your network, set VLAN to Trunk mode and specify the VLAN ID for the RDMA-ports that are in the same VLAN.

In addition, if VLAN settings for a RDMA-capable Ethernet port must be updated, these settings cannot be updated independently of other configuration settings. Before you update VLAN settings on specific RDMA-capable Ethernet ports, you must unconfigure the port, make any necessary changes to the switch configuration and then, reconfigure RDMA-capable Ethernet ports on each of the nodes in the system.

- ▶ A minimum of two dedicated RDMA-capable Ethernet ports are available for node-to-node communications to ensure best performance and reliability. These ports must be configured for inter-node traffic only and must not be used for host attachment, virtualization of Ethernet-attached external storage, or IP replication traffic.
- ▶ A maximum of four RDMA-capable Ethernet ports per node are available for node-to-node communications.



## Copy services

Copy services are a collection of functions that provide capabilities for disaster recovery (DR), data migration, and data duplication solutions. This chapter provides an overview and the preferred practices of IBM SAN Volume Controller copy services capabilities, including FlashCopy, Metro Mirror and Global Mirror, and Volume Mirroring.

This chapter includes the following topics:

- ▶ 6.1, “Introduction to copy services” on page 220
- ▶ 6.2, “FlashCopy” on page 221
- ▶ 6.3, “Remote Copy services” on page 245
- ▶ 6.4, “Native IP replication” on page 303
- ▶ 6.5, “Volume Mirroring” on page 319

## 6.1 Introduction to copy services

IBM Spectrum Virtualize based systems, including IBM SAN Volume Controller, offer a complete set of copy services functions that provide capabilities for DR, business continuity, data movement, and data duplication solutions.

### 6.1.1 FlashCopy

FlashCopy is a function that allows you to create a point-in-time copy of one of your volumes. This function might be helpful when performing backups or application testing. These copies can be cascaded on one another, read from, written to, and even reversed. These copies are able to conserve storage, if needed, by being space-efficient copies that only record items that have changed from the originals instead of full copies.

### 6.1.2 Metro Mirror and Global Mirror

Metro Mirror and Global Mirror are technologies that enable you to keep a real-time copy of a volume at a remote site that contains another IBM Spectrum Virtualize system. Consider the following points:

- ▶ Metro Mirror creates *synchronous* copies, which means that the original writes are not considered complete until the write to the destination volume is confirmed. The distance between your two sites often is determined by how much latency your applications can manage.
- ▶ Global Mirror creates *asynchronous* copies of your volume, which means that the write is considered complete after it is complete at the local volume. It does not wait for the write to be confirmed at the remote system as Metro Mirror does.

This requirement greatly reduces the latency that is experienced by your applications if the other system is far away. However, it also means that during a failure, the data on the Remote Copy might not include the most recent changes that were committed to the local volume. IBM Spectrum Virtualize provides two type of asynchronous mirroring technology: the standard Global Mirror (referred as *Global Mirror*) and the Global Mirror with Change Volume (GMCV).

### 6.1.3 Volume Mirroring

Volume Mirroring increases the high availability of the storage infrastructure. It also provides the ability to create up to two local copies of a volume. Volume Mirroring can use space from two storage pools, and preferably from two separate back-end disk subsystems.

Primarily, you use this function to insulate hosts from the failure of a storage pool and from the failure of a back-end disk subsystem. During a storage pool failure, the system continues to provide service for the volume from the other copy on the other storage pool, with no disruption to the host.

You can also use Volume Mirroring to migrate from a thin-provisioned volume to a non-thin-provisioned volume, and to migrate data between storage pools of different extent sizes.

## 6.2 FlashCopy

By using the IBM FlashCopy function of the IBM SAN Volume Controller, you can perform a *point-in-time copy* of one or more volumes. This section describes the inner workings of FlashCopy, and provides some preferred practices for its use.

You can use FlashCopy to help you solve critical and challenging business needs that require duplication of data of your source volume. Volumes can remain online and active while you create consistent copies of the data sets. Because the copy is performed at the block level, it operates below the host operating system and its cache. Therefore, the copy is not apparent to the host.

**Important:** Because FlashCopy operates at the block level below the host operating system and cache, those levels do need to be flushed for consistent FlashCopies.

While the FlashCopy operation is performed, the source volume is stopped briefly to initialize the FlashCopy bitmap, and then input/output (I/O) can resume. Although several FlashCopy options require the data to be copied from the source to the target in the background, which can take time to complete, the resulting data on the target volume is presented so that the copy appears to complete immediately.

This process is performed by using a bitmap (or bit array) that tracks changes to the data after the FlashCopy is started, and an indirection layer that enables data to be read from the source volume transparently.

### 6.2.1 FlashCopy use cases

When you are deciding whether FlashCopy addresses your needs, you must adopt a combined business and technical view of the problems that you want to solve. First, determine the needs from a business perspective. Then, determine whether FlashCopy can address the technical needs of those business requirements.

The business applications for FlashCopy are wide-ranging. In the following sections, a short description of the most common use cases is provided.

#### Back up improvements with FlashCopy

FlashCopy does not reduce the time that it takes to perform a backup to traditional backup infrastructure. However, it can be used to minimize and under certain conditions, eliminate application downtime that is associated with performing backups. FlashCopy can also transfer the resource usage of performing intensive backups from production systems.

After the FlashCopy is performed, the resulting image of the data can be backed up to tape as though it were the source system. After the copy to tape is complete, the image data is redundant and the target volumes can be discarded. For time-limited applications, such as these examples, “no copy” or incremental FlashCopy is used most often. The use of these methods puts less load on your infrastructure.

When FlashCopy is used for backup purposes, the target data usually is managed as read-only at the operating system level. This approach provides extra security by ensuring that your target data was not modified and remains true to the source.

## Restore with FlashCopy

FlashCopy can perform a restore from any existing FlashCopy mapping. Therefore, you can restore (or copy) from the target to the source of your regular FlashCopy relationships. It might be easier to think of this method as reversing the direction of the FlashCopy mappings. This capability has the following benefits:

- ▶ There is no need to worry about pairing mistakes because you trigger a restore.
- ▶ The process appears instantaneous.
- ▶ You can maintain a pristine image of your data while you are restoring what was the primary data.

This approach can be used for various applications, such as recovering your production database application after an errant batch process that caused extensive damage.

**Preferred practices:** Although restoring from a FlashCopy is quicker than a traditional tape media restore, do not use restoring from a FlashCopy as a substitute for good archiving practices. Instead, keep one to several iterations of your FlashCopies so that you can near-instantly recover your data from the most recent history. Keep your long-term archive as needed for your business.

In addition to the restore option, which copies the original blocks from the target volume to modified blocks on the source volume, the target can be used to perform a restore of individual files. To do that, you must make the target available on a host. Do not make the target available to the source host because seeing duplicates of disks causes problems for most host operating systems. Copy the files to the source by using the normal host data copy methods for your environment.

## Moving and migrating data with FlashCopy

FlashCopy can be used to facilitate the movement or migration of data between hosts while minimizing downtime for applications. By using FlashCopy, application data can be copied from source volumes to new target volumes while applications remain online. After the volumes are fully copied and synchronized, the application can be stopped and then immediately started on the new server that is accessing the new FlashCopy target volumes.

**Use Case:** FlashCopy can be used to migrate volumes from and to Data Reduction Pools (DRPs), which do not support extent-based migrations.

This method differs from the other migration methods, which are described later in this chapter. Common uses for this capability are host and back-end storage hardware refreshes.

## Application testing with FlashCopy

It is often important to test a new version of an application or operating system that is using actual production data. This testing ensures the highest quality possible for your environment. FlashCopy makes this type of testing easy to accomplish without putting the production data at risk or requiring downtime to create a constant copy.

Create a FlashCopy of your source and use that for your testing. This copy is a duplicate of your production data down to the block level so that even physical disk identifiers are copied. Therefore, it is impossible for your applications to tell the difference.

## 6.2.2 FlashCopy capabilities overview

FlashCopy occurs between a source volume and a target volume in the same storage system. The minimum granularity that IBM SAN Volume Controller supports for FlashCopy is an entire volume. It is not possible to use FlashCopy to copy only part of a volume.

To start a FlashCopy operation, a relationship between the source and the target volume must be defined. This relationship is called *FlashCopy Mapping*.

FlashCopy mappings can be stand-alone or a member of a Consistency Group. You can perform the actions of preparing, starting, or stopping FlashCopy on either a stand-alone mapping or a Consistency Group.

Figure 6-1 shows the concept of FlashCopy mapping.

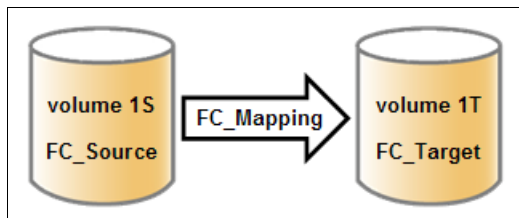


Figure 6-1 FlashCopy mapping

A FlashCopy mapping features a set of attributes and settings that define the characteristics and the capabilities of the FlashCopy. These characteristics are explained next.

### Background copy

The *background copy rate* is a property of a FlashCopy mapping that allows to specify whether a background physical copy of the source volume to the corresponding target volume occurs. A value of 0 disables the background copy. If the FlashCopy background copy is disabled, only data that has changed on the source volume is copied to the target volume. A FlashCopy with background copy disabled is also known as *No-Copy* FlashCopy.

The benefit of using a FlashCopy mapping with background copy enabled is that the target volume becomes a real clone (independent from the source volume) of the FlashCopy mapping source volume after the copy is complete. When the background copy function is not performed, the target volume remains a valid copy of the source data while the FlashCopy mapping remains in place.

Valid values for the background copy rate are 0 - 150. The background copy rate can be defined and changed dynamically for individual FlashCopy mappings.

Table 6-1 shows the relationship of the background copy rate value to the attempted amount of data to be copied per second.

*Table 6-1 Relationship between the rate and data rate per second*

| Value    | Data copied per second |
|----------|------------------------|
| 1 - 10   | 128 KB                 |
| 11 - 20  | 256 KB                 |
| 21 - 30  | 512 KB                 |
| 31 - 40  | 1 MB                   |
| 41 - 50  | 2 MB                   |
| 51 - 60  | 4 MB                   |
| 61 - 70  | 8 MB                   |
| 71 - 80  | 16 MB                  |
| 81 - 90  | 32 MB                  |
| 91 - 100 | 64 MB                  |
| 101-110  | 128 MB                 |
| 111-120  | 256 MB                 |
| 121-130  | 512 MB                 |
| 131-140  | 1024 MB                |
| 141-150  | 2048 MB                |

**Note:** To ensure optimal performance of all IBM Spectrum Virtualize features, it is advised not to exceed a copy rate value of 130.

## FlashCopy Consistency Groups

*Consistency Groups* can be used to help create a consistent point-in-time copy across multiple volumes. They are used to manage the consistency of dependent writes that are run in the application following the correct sequence.

When Consistency Groups are used, the FlashCopy commands are issued to the Consistency Groups. The groups perform the operation on all FlashCopy mappings contained within the Consistency Groups at the same time.



Figure 6-2 shows a Consistency Group that consists of two volume mappings.

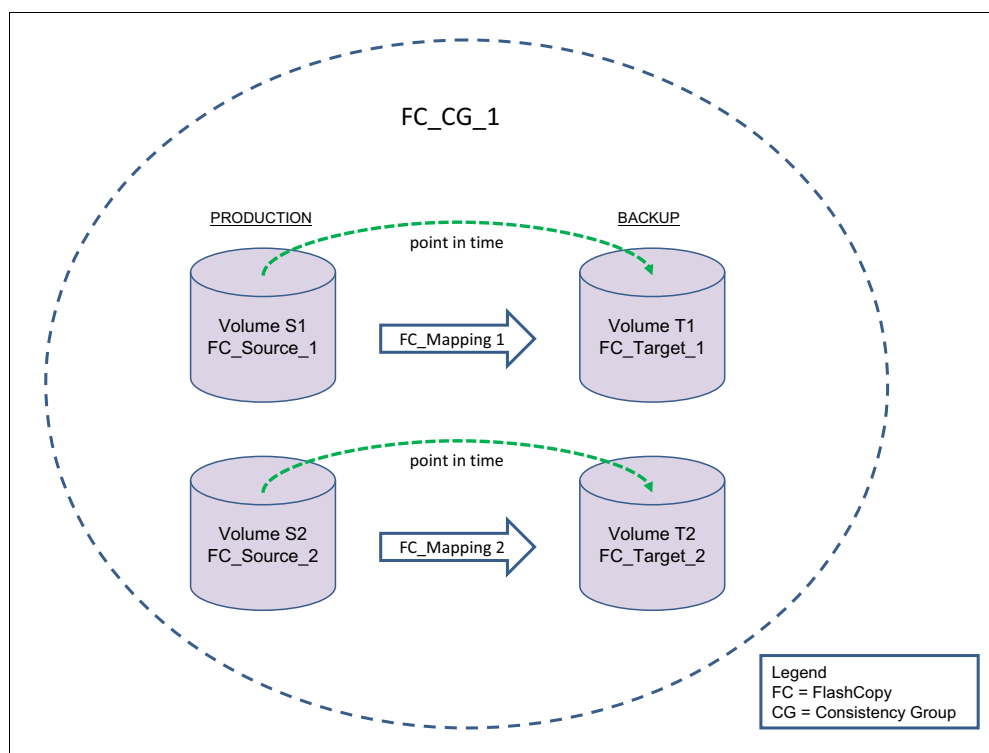


Figure 6-2 Multiple volumes mapping in a Consistency Group

**FlashCopy mapping considerations:** If the FlashCopy mapping has been added to a Consistency Group, it can only be managed as part of the group. This limitation means that FlashCopy operations are no longer allowed on the individual FlashCopy mappings.

### Incremental FlashCopy

Using Incremental FlashCopy, you can reduce the required time of copy. Also, because less data must be copied, the workload put on the system and the back-end storage is reduced.

Incremental FlashCopy does not require that you copy an entire disk source volume every time the FlashCopy mapping is started. It means that only the changed regions on source volumes are copied to target volumes, as shown in Figure 6-3.

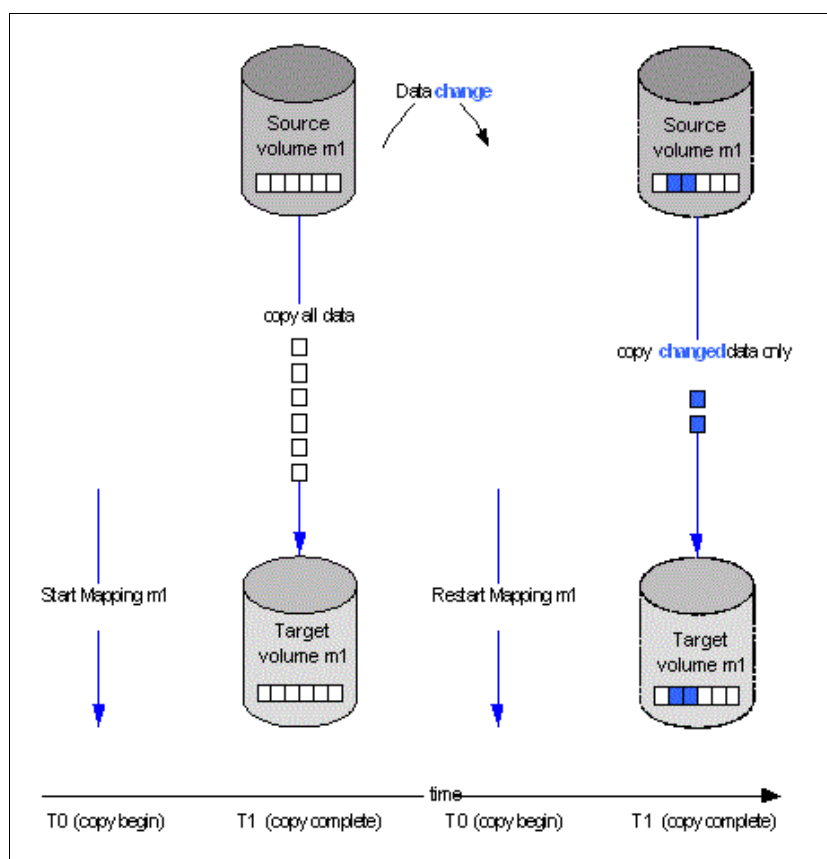


Figure 6-3 Incremental FlashCopy

If the FlashCopy mapping was stopped before the background copy completed, then when the mapping is restarted, the data that was copied before the mapping was stopped will not be copied again. For example, if an incremental mapping reaches 10 percent progress when it is stopped and then it is restarted, that 10 percent of data will not be recopied when the mapping is restarted, assuming that it was not changed.

**Stopping an incremental FlashCopy mapping:** If you are planning to stop an incremental FlashCopy mapping, make sure that the copied data on the source volume is not changed, if possible. Otherwise, you might have an inconsistent point-in-time copy.

A "difference" value is provided in the query of a mapping, which makes it possible to know how much data has changed. This data must be copied when the Incremental FlashCopy mapping is restarted. The difference value is the percentage (0-100 percent) of data that has been changed. This data must be copied to the target volume to get a fully independent copy of the source volume.

An incremental FlashCopy can be defined setting the *incremental* attribute in the FlashCopy mapping.

## Multiple Target FlashCopy

In Multiple Target FlashCopy, a source volume can be used in multiple FlashCopy mappings, while the target is a different volume, as shown in Figure 6-4.

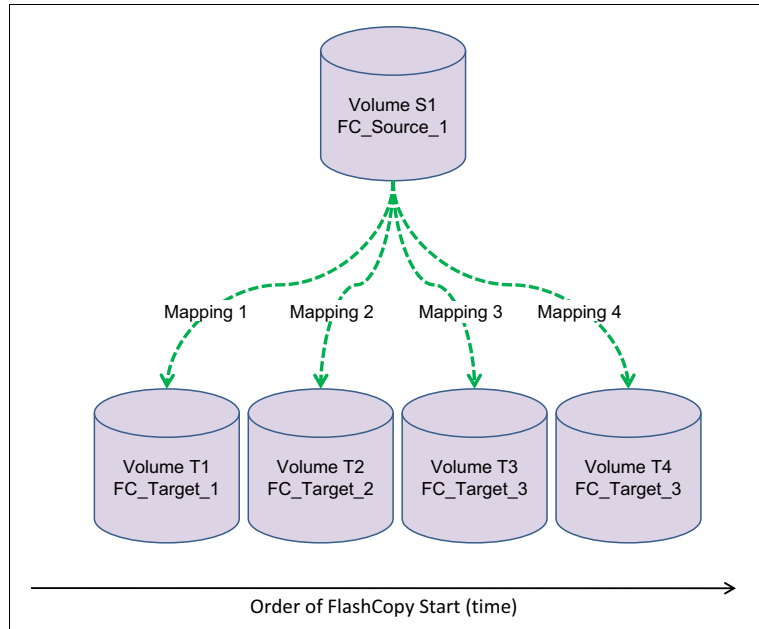


Figure 6-4 Multiple Target FlashCopy

Up to 256 different mappings are possible for each source volume. These mappings are independently controllable from each other. Multiple Target FlashCopy mappings can be members of the same or different Consistency Groups. In cases where all the mappings are in the same Consistency Group, the result of starting the Consistency Group will be to FlashCopy to multiple identical target volumes.

## Cascaded FlashCopy

With Cascaded FlashCopy, you can have a source volume for one FlashCopy mapping and as the target for another FlashCopy mapping; this is referred to as a *Cascaded FlashCopy*. This function is illustrated in Figure 6-5.

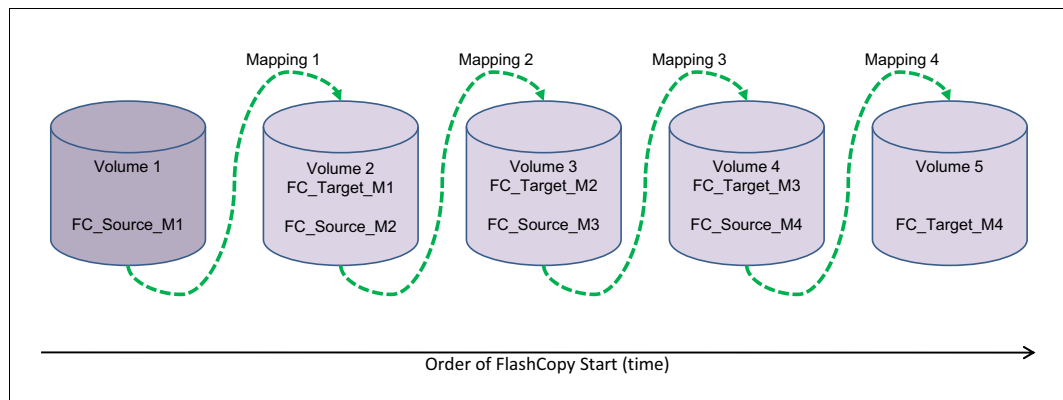


Figure 6-5 Cascaded FlashCopy

A total of 255 mappings are possible for each cascade.

## Reverse FlashCopy

Reverse FlashCopy enables FlashCopy targets to become restore points for the source without breaking the FlashCopy relationship, and without having to wait for the original copy operation to complete. It can be used in combination with the Multiple Target FlashCopy to create multiple rollback points.

A key advantage of the Multiple Target Reverse FlashCopy function is that the reverse FlashCopy does not destroy the original target. This feature enables processes that use the target, such as a tape backup, to continue uninterrupted. IBM Spectrum Virtualize systems also allow you to create an optional copy of the source volume to be made before the reverse copy operation starts. This ability to restore back to the original source data can be useful for diagnostic purposes.

## Thin-provisioned FlashCopy

When a new volume is created, you can designate it as a *thin-provisioned volume*, and it has a virtual capacity and a real capacity.

*Virtual capacity* is the volume storage capacity that is available to a host. *Real capacity* is the storage capacity that is allocated to a volume copy from a storage pool. In a fully allocated volume, the virtual capacity and real capacity are the same. However, in a thin-provisioned volume, the virtual capacity can be much larger than the real capacity.

The virtual capacity of a thin-provisioned volume is typically larger than its real capacity. On IBM Spectrum Virtualize based systems, the real capacity is used to store data that is written to the volume, and metadata that describes the thin-provisioned configuration of the volume. As more information is written to the volume, more of the real capacity is used.

Thin-provisioned volumes can also help to simplify server administration. Instead of assigning a volume with some capacity to an application and increasing that capacity following the needs of the application if those needs change, you can configure a volume with a large virtual capacity for the application. You can then increase or shrink the real capacity as the application needs change, without disrupting the application or server.

When you configure a thin-provisioned volume, you can use the warning level attribute to generate a warning event when the used real capacity exceeds a specified amount or percentage of the total real capacity. For example, if you have a volume with 10 GB of total capacity and you set the warning to 80 percent, an event is registered in the event log when you use 80 percent of the total capacity. This technique is useful when you need to control how much of the volume is used.

If a thin-provisioned volume does not have enough real capacity for a write operation, the volume is taken offline and an error is logged (error code 1865, event ID 060001). Access to the thin-provisioned volume is restored by either increasing the real capacity of the volume or increasing the size of the storage pool on which it is allocated.

You can use thin volumes for cascaded FlashCopy and multiple target FlashCopy. It is also possible to mix thin-provisioned with normal volumes. It can be used for incremental FlashCopy too, but using thin-provisioned volumes for incremental FlashCopy only makes sense if the source *and* target are thin-provisioned.

When thin provisioned volumes are used on DRPs, consider also implementing compression because it provides several benefits:

- ▶ Reduced amount of IO operation to the backend. The amount particularly relevant with poorly performing back-ends.
- ▶ Space efficiency. The compressed data provides more capacity savings.

- Better backend capacity monitoring. DRP pools with thin provisioned, uncompressed volumes do not provide physical allocation information.

Therefore, the recommendation is to always enable compression on DRP thin provisioned volumes.

### ***Thin-provisioned incremental FlashCopy***

The implementation of thin-provisioned volumes does not preclude the use of incremental FlashCopy on the same volumes. It does not make sense to have a fully allocated source volume and then use incremental FlashCopy, which is always a full copy at first, to copy this fully allocated source volume to a thin-provisioned target volume. However, this action is not prohibited.

Consider the following optional configuration:

- A thin-provisioned source volume can be copied incrementally by using FlashCopy to a thin-provisioned target volume. Whenever the FlashCopy is performed, only data that has been modified is recopied to the target. Note that if space is allocated on the target because of I/O to the target volume, this space will not be reclaimed with subsequent FlashCopy operations.
- A fully allocated source volume can be copied incrementally using FlashCopy to another fully allocated volume at the same time as it is being copied to multiple thin-provisioned targets (taken at separate points in time). This combination allows a single full backup to be kept for recovery purposes, and separates the backup workload from the production workload. At the same time, it allows older thin-provisioned backups to be retained.

## **6.2.3 FlashCopy functional overview**

Understanding how FlashCopy works internally helps you to configure it in a way that you want and enables you to obtain more benefits from it.

### **FlashCopy mapping states**

A FlashCopy mapping defines the relationship that copies data between a source volume and a target volume. FlashCopy mappings can be stand-alone or a member of a Consistency Group. You can perform the actions of preparing, starting, or stopping FlashCopy on a stand-alone mapping or a Consistency Group.

A FlashCopy mapping includes an attribute that represents the state of the mapping. The following FlashCopy states are available:

- **Idle\_or\_copied**

Read and write caching is enabled for the source and target. A FlashCopy mapping exists between the source and target, but the source and target behave as independent volumes in this state.

- **Copying**

The FlashCopy indirection layer (see “Indirection layer” on page 232) governs all I/O to the source and target volumes while the background copy is running. The background copy process is copying grains from the source to the target. Reads and writes are run on the target as though the contents of the source were instantaneously copied to the target during the **startfcmaporstartfcconsistgrp** command. The source and target can be independently updated. Internally, the target depends on the source for specific tracks. Read and write caching is enabled on the source and the target.

► Stopped

The FlashCopy was stopped by a user command or by an I/O error. When a FlashCopy mapping is stopped, the integrity of the data on the target volume is lost. Therefore, while the FlashCopy mapping is in this state, the target volume is in the Offline state. To regain access to the target, the mapping must be restarted (the previous point-in-time is lost) or the FlashCopy mapping must be deleted. The source volume is accessible, and read and write caching is enabled for the source. In the Stopped state, a mapping can be prepared again or deleted.

► Stopping

The mapping is in the process of transferring data to a dependent mapping. The behavior of the target volume depends on whether the background copy process completed while the mapping was in the Copying state. If the copy process completed, the target volume remains online while the stopping copy process completes.

If the copy process did not complete, data in the cache is discarded for the target volume. The target volume is taken offline, and the stopping copy process runs.

After the data is copied, a stop complete asynchronous event notification is issued. The mapping moves to the Idle/Copied state if the background copy completed or to the Stopped state if the background copy did not complete. The source volume remains accessible for I/O.

► Suspended

The FlashCopy was in the Copying or Stopping state when access to the metadata was lost. As a result, the source and target volumes are offline and the background copy process was halted.

When the metadata becomes available again, the FlashCopy mapping returns to the Copying or Stopping state. Access to the source and target volumes is restored, and the background copy or stopping process resumes. Unflushed data that was written to the source or target before the FlashCopy was suspended is pinned in cache until the FlashCopy mapping leaves the Suspended state.

► Preparing

The FlashCopy is in the process of preparing the mapping. While in this state, data that is from cache is destaged to disk and a consistent copy of the source exists on disk. At this time, cache is operating in write-through mode and writes to the source volume experience more latency.

The target volume is reported as online, but it does not perform reads or writes. These reads and writes are failed by the SCSI front end.

Before starting the FlashCopy mapping, it is important that any cache at the host level, for example, buffers on the host operating system or application, are also instructed to flush any outstanding writes to the source volume. Performing the cache flush that is required as part of the **startfcmap** or **startfcconsistgrp** command causes I/Os to be delayed waiting for the cache flush to complete. To overcome this problem, FlashCopy supports the **prestartfcmap** or **prestartfcconsistgrp** commands, which prepare for a FlashCopy start while still allowing I/Os to continue to the source volume.

In the Preparing state, the FlashCopy mapping is prepared by completing the following steps:

- a. Flushing any modified write data that is associated with the source volume from the cache. Read data for the source are left in the cache.
- b. Placing the cache for the source volume into write-through mode so that subsequent writes wait until data was written to disk before completing the write command that is received from the host.

- c. Discarding any read or write data that is associated with the target volume from the cache.

► Prepared

While in the Prepared state, the FlashCopy mapping is ready to perform a start. While the FlashCopy mapping is in this state, the target volume is in the Offline state. In the Prepared state, writes to the source volume experience more latency because the cache is operating in write-through mode.

Figure 6-6 shows the FlashCopy mapping state diagram. It shows the states in which a mapping can exist, and which events are responsible for a state change.

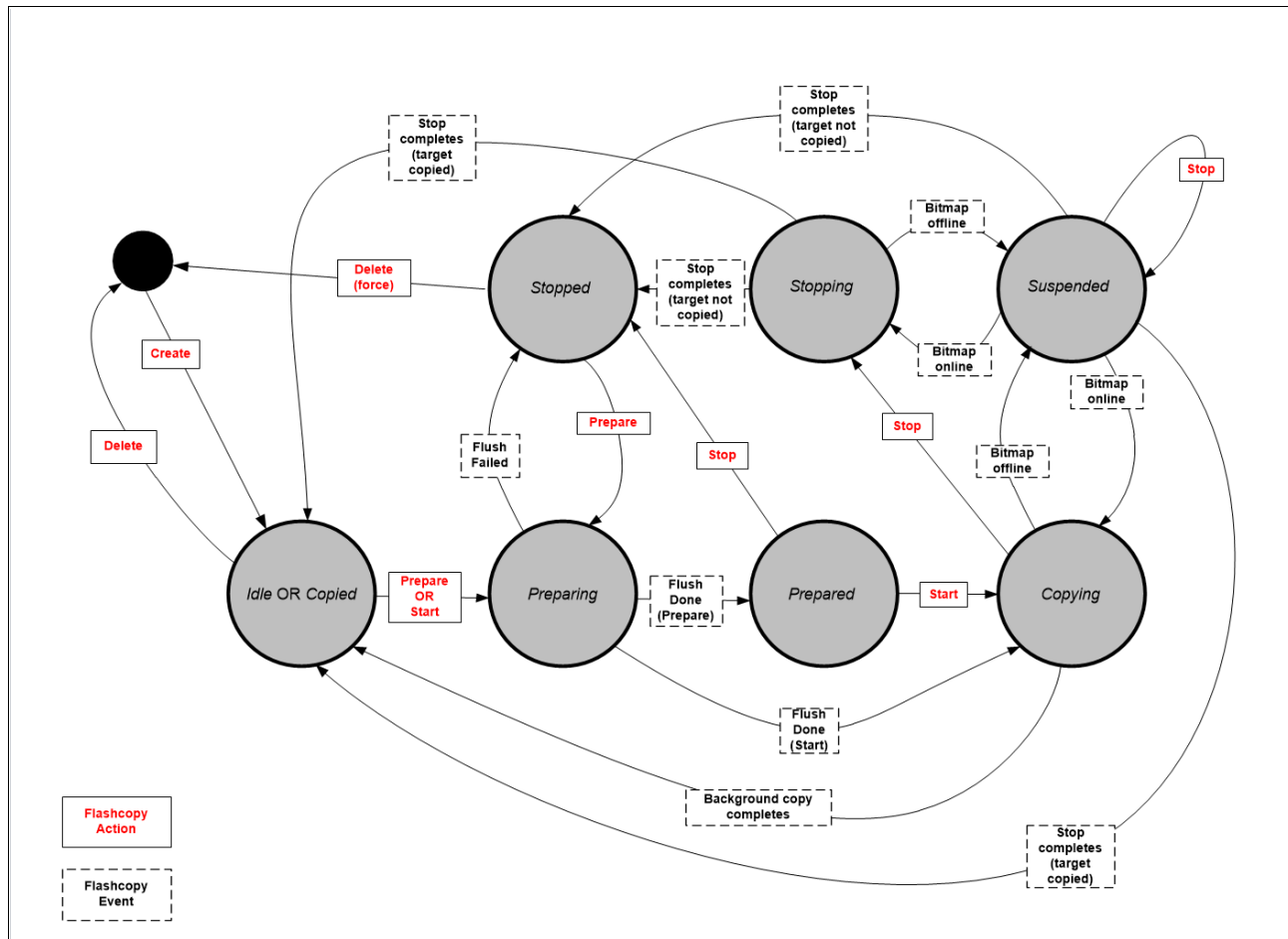


Figure 6-6 FlashCopy mapping state

## FlashCopy bitmaps and grains

A *bitmap* is an internal data structure that is stored in a specific I/O Group that is used to track which data in FlashCopy mappings was copied from the source volume to the target volume. *Grains* are units of data that is grouped together to optimize the use of the bitmap. One bit in each bitmap represents the state of one grain. FlashCopy grain can be 64 KB or 256 KB.

A FlashCopy bitmap takes up the bitmap space in the memory of the I/O group that must be shared with other features' bitmaps (such as Remote Copy bitmaps, Volume Mirroring bitmaps, and RAID bitmaps).

## Indirection layer

The *FlashCopy indirection layer* governs the I/O to the source and target volumes when a FlashCopy mapping is started. This process is done by using a FlashCopy bitmap. The purpose of the FlashCopy indirection layer is to enable both the source and target volumes for read and write I/O immediately after FlashCopy starts.

The following description illustrates how the FlashCopy indirection layer works when a FlashCopy mapping is prepared and then started.

When a FlashCopy mapping is prepared and started, the following sequence is applied:

1. Flush the write cache to the source volume or volumes that are part of a Consistency Group.
2. Put the cache into write-through mode on the source volumes.
3. Discard the cache for the target volumes.
4. Establish a sync point on all of the source volumes in the Consistency Group (creating the FlashCopy bitmap).
5. Ensure that the indirection layer governs all of the I/O to the source volumes and target.
6. Enable the cache on source volumes and target volumes.

FlashCopy provides the semantics of a point-in-time copy that uses the indirection layer, which intercepts I/O that is directed at either the source or target volumes. The act of starting a FlashCopy mapping causes this indirection layer to become active in the I/O path, which occurs automatically across all FlashCopy mappings in the Consistency Group. The indirection layer then determines how each of the I/O is to be routed based on the following factors:

- ▶ The volume and the logical block address (LBA) to which the I/O is addressed
- ▶ Its direction (read or write)
- ▶ The state of an internal data structure, the FlashCopy bitmap

The indirection layer allows the I/O to go through the underlying volume and preserves the point-in-time copy. To do that, the Spectrum Virtualize code uses two different mechanisms:

- ▶ **Copy-on-Write (CoW):** With this mechanism, when a write operation occurs in the source volume, a portion of data (grain) that contains the data to be modified is copied to the target volume before the operation completion.
- ▶ **Redirect-on-Write (RoW):** With this mechanism, when a write operation occurs in the source volume, the data to be modified is written in another area, which leaves the original data unmodified to be used by the target volume.

Spectrum Virtualize implements CoW and RoW logics transparently to the user with the aim to optimize the performance and capacity. By using the RoW mechanism, the performance can improve by reducing the number of physical I/Os for the write operations. A significant capacity saving can be achieved by improving the overall deduplication ratio.

The RoW was introduced with Spectrum Virtualize version 8.4 and it is used in the following conditions:

- ▶ Source and target volumes:
  - Are in the same pool
  - Are in the same IO group
  - Do not participate in a Volume Mirroring relationship
  - Are not fully allocated
- ▶ The pool containing the source and target volumes must be a Data Reduction Pool (DRP)



The CoW is used all the instances in which the RoW is not applicable.

The indirection layer algorithm in the CoW example is listed in Table 6-2.

Table 6-2 FlashCopy indirection layer algorithm

| Volume being accessed | Was the grain been copied? | Host I/O operation                                                                                                                                 |                                                                                                                                                                                                                                     |
|-----------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                            | Read                                                                                                                                               | Write                                                                                                                                                                                                                               |
| Source                | No                         | Read from the source volume.                                                                                                                       | Copy grain to the most recently started target for this source; then, write to the source.                                                                                                                                          |
|                       | Yes                        | Read from the source volume.                                                                                                                       | Write to the source volume.                                                                                                                                                                                                         |
| Target                | No                         | If any newer targets exist for this source in which this grain was copied, read from the oldest of these targets. Otherwise, read from the source. | Hold the write. Check the dependency target volumes to see whether the grain was copied. If the grain is not copied to the next oldest target for this source, copy the grain to the next oldest target. Then, write to the target. |
|                       | Yes                        | Read from the target volume.                                                                                                                       | Write to the target volume.                                                                                                                                                                                                         |

### Interaction with cache

The Spectrum Virtualize technology provides a two layer cache. The cache is divided into *upper cache* and *lower cache*. Upper cache serves mostly as write cache and hides the write latency from the hosts and application. Lower cache is a read/write cache and optimizes I/O to and from disks.

Figure 6-7 shows the IBM Spectrum Virtualize cache architecture.

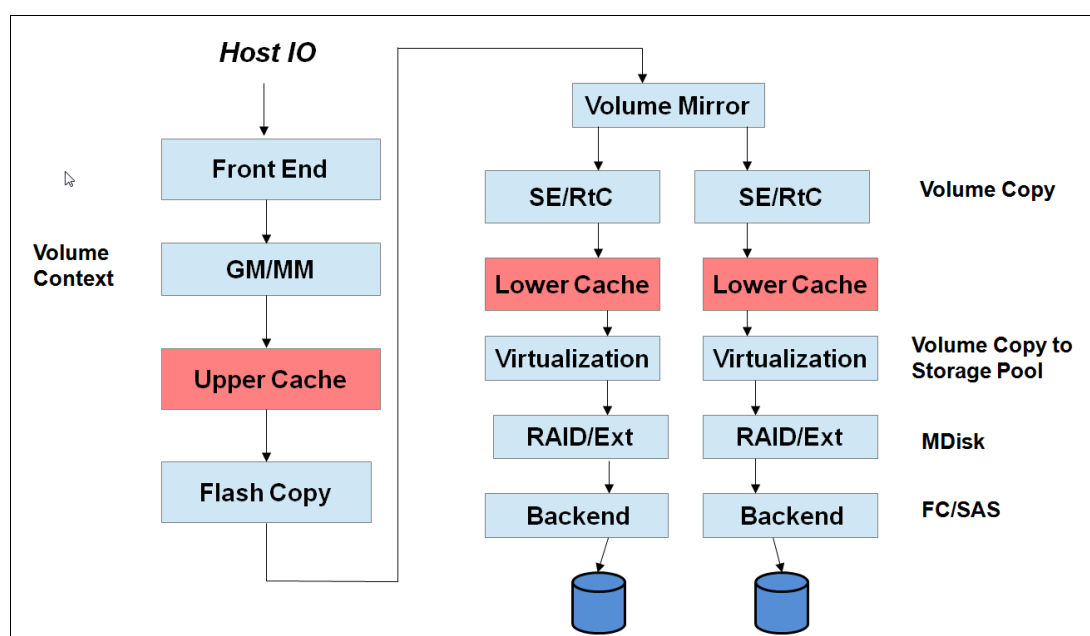


Figure 6-7 New cache architecture

The CoW process might introduce significant latency into write operations. To isolate the active application from this extra latency, the FlashCopy indirection layer is placed logically between the upper and lower cache. Therefore, the extra latency that is introduced by the CoW process is encountered by the internal cache operations only, and not by the application.

The logical placement of the FlashCopy indirection layer is shown in Figure 6-8.

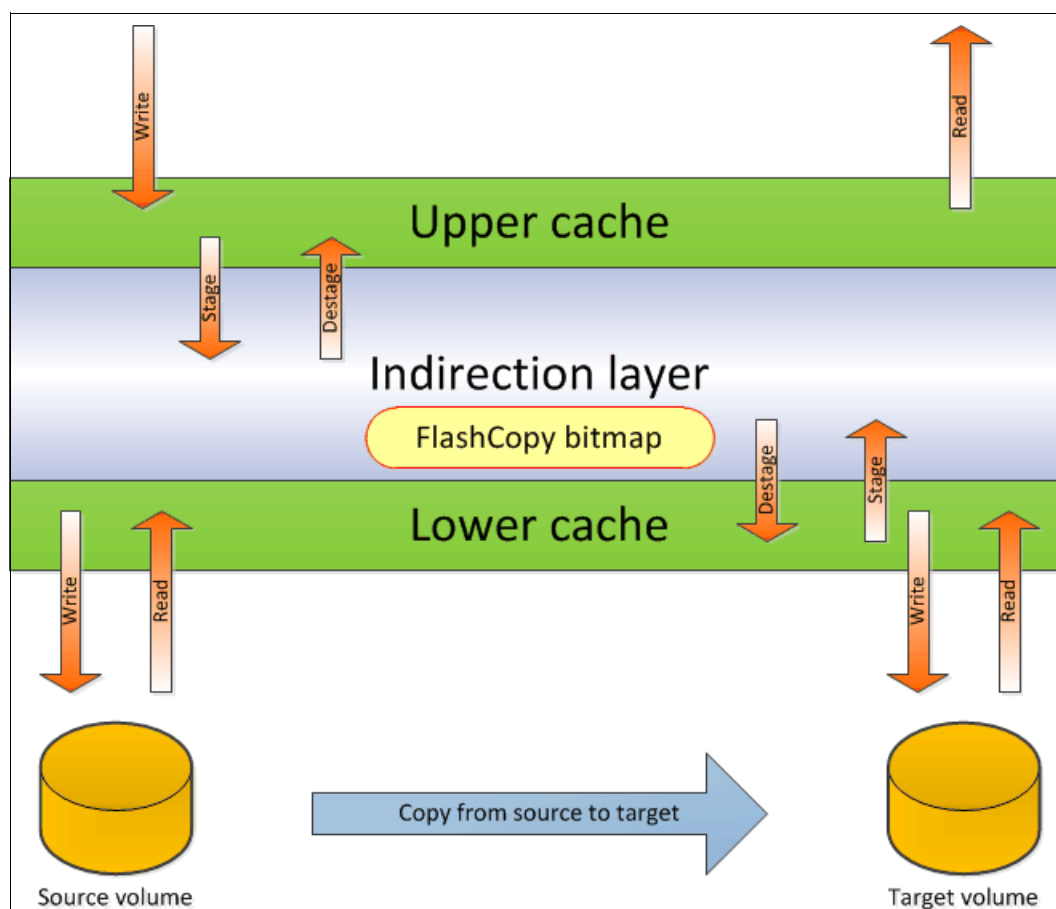


Figure 6-8 Logical placement of the FlashCopy indirection layer

The introduction of the two-level cache provides more performance improvements to the FlashCopy mechanism. Because the FlashCopy layer is now above the lower cache in the IBM Spectrum Virtualize software stack, it can benefit from read prefetching and coalescing writes to back-end storage. Also, preparing FlashCopy is much faster because upper cache write data does not have to go directly to back-end storage, but to the lower cache layer only.

Also, in multi-target FlashCopy, the target volumes of the same image share cache data. This design is opposite to previous IBM Spectrum Virtualize code versions, where each volume had its own copy of cached data.

## Interaction and dependency between Multiple Target FlashCopy mappings

Figure 6-9 shows a set of three FlashCopy mappings that share a source. The FlashCopy mappings target target volumes Target 1, Target 2, and Target 3.

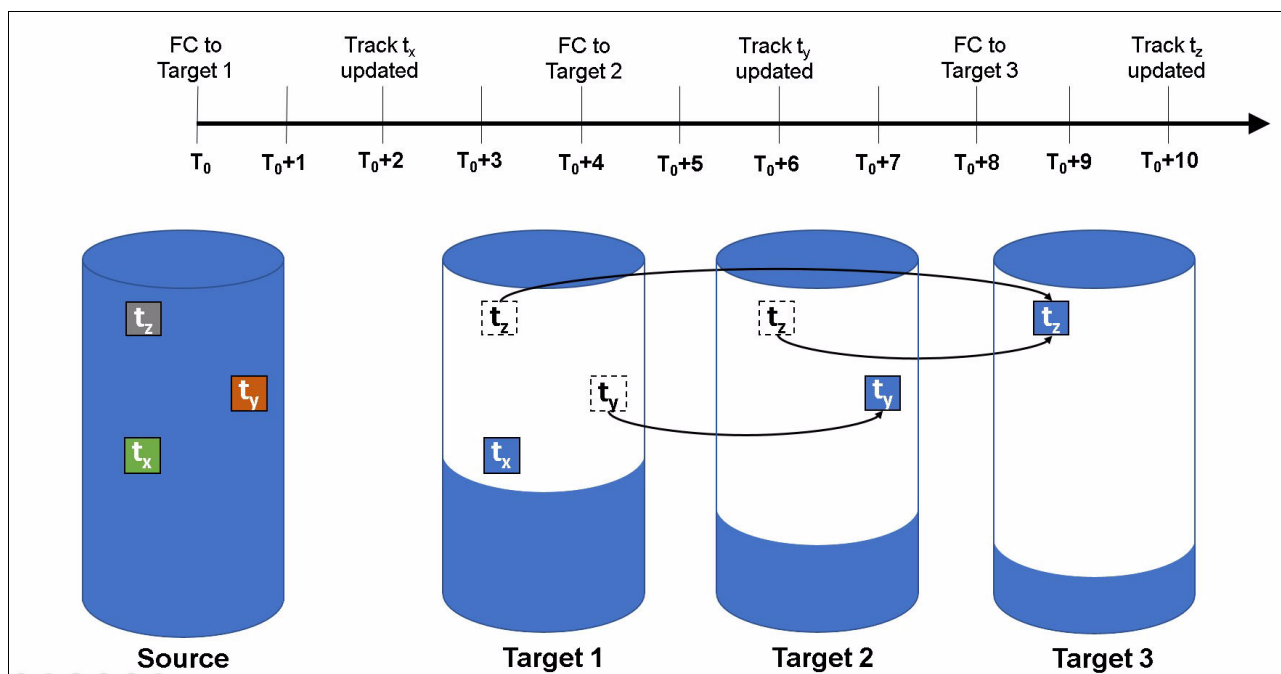


Figure 6-9 Interaction between Multiple Target FlashCopy mappings

Consider the following events timeline:

- ▶ At time  $T_0$ , a FlashCopy mapping is started between the source and the Target 1.
- ▶ At time  $T_0+2$ , the track  $t_x$  is updated in the source. Because this track is not yet copied in background on Target 1, the copy-on-write process copies this track to the Target 1 before being updated on the source.
- ▶ At time  $T_0+4$ , a FlashCopy mapping is started between the source and the Target 2.
- ▶ At time  $T_0+6$ , the track  $t_y$  is updated in the source. Because this track is not yet copied in background on Target 2, the copy-on-write process copies this track to the Target 2 only before being updated on the source.
- ▶ At time  $T_0+8$ , a FlashCopy mapping is started between the source and the Target 3.
- ▶ At time  $T_0+10$ , the track  $t_z$  is updated in the source. Because this track is not yet copied in background on Target 3, the copy-on-write process copies this track to the Target 3 only before being updated on the source.

As a result of this sequence of events, the configuration in Figure 6-9 features the following characteristics:

- ▶ Target 1 is dependent upon Target 2 and Target 3. It remains dependent until all of Target 1 is copied. Because no target depends on Target 1, the mapping can be stopped without need to copy any data to maintain the consistency in the other targets.
- ▶ Target 2 depends on Target 3, and remains dependent until all of Target 2 is copied. Because Target 1 depends on Target 2, if this mapping is stopped, the cleanup process is started to copy all data that is uniquely held on this mapping (that is,  $t_y$ ) to Target 1.

- Target 3 is not dependent on any target, but it has Target 1 and Target 2 depending on it. Therefore, if this mapping is stopped, the cleanup process is started to copy all data that is uniquely held on this mapping (that is,  $t_2$ ) to Target 2.

### Target writes with Multiple Target FlashCopy

A write to an intermediate or newest target volume must consider the state of the grain within its own mapping, and the state of the grain of the next oldest mapping:

- If the grain of the next oldest mapping has not been copied yet, it must be copied before the write is allowed to proceed to preserve the contents of the next oldest mapping. The data that is written to the next oldest mapping comes from a target or source.
- If the grain in the target being written has not yet been copied, the grain is copied from the oldest already copied grain in the mappings that are newer than the target, or the source if none are already copied. After this copy is done, the write can be applied to the target.

### Target reads with Multiple Target FlashCopy

If the grain being read has already been copied from the source to the target, the read simply returns data from the target being read. If the grain has not been copied, each of the newer mappings is examined in turn and the read is performed from the first copy found. If none are found, the read is performed from the source.

## 6.2.4 FlashCopy planning considerations

The FlashCopy function, as with all of the advanced IBM SAN Volume Controller features, offers useful capabilities. However, some basic planning considerations are to be followed for a successful implementation.

### FlashCopy configurations limits

To plan for and implement FlashCopy, you must check the configuration limits and adhere to them. Table 6-3 shows the system limits that apply to the latest version at the time of writing this book.

Table 6-3 *FlashCopy properties and maximum configurations*

| FlashCopy property                       | Maximum | Comment                                                                                                 |
|------------------------------------------|---------|---------------------------------------------------------------------------------------------------------|
| FlashCopy targets per source             | 256     | This maximum is the maximum number of FlashCopy mappings that can exist with the same source volume.    |
| FlashCopy mappings per system            | 10000   | This maximum is the maximum number of FlashCopy mappings per system.                                    |
| FlashCopy Consistency Groups per system  | 500     | This maximum is an arbitrary limit that is policed by the software.                                     |
| FlashCopy volume space per I/O Group     | 4096 TB | This maximum is a limit on the quantity of FlashCopy mappings by using bitmap space from one I/O Group. |
| FlashCopy mappings per Consistency Group | 512     | This limit is due to the time that is taken to prepare a Consistency Group with many mappings.          |

**Configuration limits:** The configuration limits always change with the introduction of new hardware and software capabilities. Check the IBM SAN Volume Controller online documentation for the latest configuration limits.

The total amount of cache memory that is reserved for the FlashCopy bitmaps limits the amount of capacity that can be used as a FlashCopy target. Table 6-4 illustrates the relationship of bitmap space to FlashCopy address space, depending on the size of the grain and the kind of FlashCopy service that is used.

*Table 6-4 Relationship of bitmap space to FlashCopy address space for the specified I/O Group*

| Copy service          | Grain size in KB | 1 MB of memory provides the following volume capacity for the specified I/O Group |
|-----------------------|------------------|-----------------------------------------------------------------------------------|
| FlashCopy             | 256              | 2 TB of target volume capacity                                                    |
| FlashCopy             | 64               | 512 GB of target volume capacity                                                  |
| Incremental FlashCopy | 256              | 1 TB of target volume capacity                                                    |
| Incremental FlashCopy | 64               | 256 GB of target volume capacity                                                  |

**Mapping consideration:** For multiple FlashCopy targets, you must consider the number of mappings. For example, for a mapping with a 256 KB grain size, 8 KB of memory allows one mapping between a 16 GB source volume and a 16 GB target volume. Alternatively, for a mapping with a 256 KB grain size, 8 KB of memory allows two mappings between one 8 GB source volume and two 8 GB target volumes.

When you create a FlashCopy mapping, if you specify an I/O Group other than the I/O Group of the source volume, the memory accounting goes towards the specified I/O Group, not towards the I/O Group of the source volume.

The default amount of memory for FlashCopy is 20 MB. This value can be increased or decreased by using the **chlogrp** command or through the GUI. The maximum amount of memory that can be specified for FlashCopy is 2048 MB (512 MB for 32-bit systems). The maximum combined amount of memory across all copy services features is 2600 MB (552 MB for 32-bit systems).

**Bitmap allocation:** When creating a FlashCopy mapping, you can optionally specify the I/O group where the bitmap is allocated. If you specify an I/O Group other than the I/O Group of the source volume, the memory accounting goes towards the specified I/O Group, not towards the I/O Group of the source volume. This option can be useful when an I/O group is exhausting the memory that is allocated to the FlashCopy bitmaps and no more free memory is available in the I/O group.

## FlashCopy general restrictions

The following implementation restrictions apply to FlashCopy:

- ▶ The size of source and target volumes in a FlashCopy mapping must be the same.
- ▶ Multiple FlashCopy mappings that use the same target volume can be defined, but only one of these mappings can be started at a time. This limitation means that no multiple FlashCopy can be active to the same target volume.
- ▶ Expansion or shrinking of volumes defined in a FlashCopy mapping is not allowed. To modify the size of a source or target volume, first remove the FlashCopy mapping.
- ▶ In a cascading FlashCopy, the grain size of all the FlashCopy mappings that participate must be the same.

- ▶ In a multi-target FlashCopy, the grain size of all the FlashCopy mappings that participate must be the same.
- ▶ In a reverse FlashCopy, the grain size of all the FlashCopy mappings that participate must be the same.
- ▶ No FlashCopy mapping can be added to a consistency group while the FlashCopy mapping status is *Copying*.
- ▶ No FlashCopy mapping can be added to a consistency group while the consistency group status is *Copying*.
- ▶ The use of Consistency Groups is restricted when using Cascading FlashCopy. A Consistency Group serves the purpose of starting FlashCopy mappings at the same point in time. Within the *same* Consistency Group, it is not possible to have mappings with these conditions:
  - The source volume of one mapping is the target of another mapping.
  - The target volume of one mapping is the source volume for another mapping.

These combinations are not useful because within a Consistency Group, mappings cannot be established in a certain order. This limitation renders the content of the target volume undefined. For instance, it is not possible to determine whether the first mapping was established before the target volume of the first mapping that acts as a source volume for the second mapping.

Even if it were possible to ensure the order in which the mappings are established within a Consistency Group, the result is equal to Multi Target FlashCopy (two volumes holding the same target data for one source volume). In other words, a cascade is useful for copying volumes in a certain order (and copying the changed content targets of FlashCopies), rather than at the same time in an undefined order (from within one single Consistency Group).

- ▶ Both source and target volumes can be used as primary in a Remote Copy relationship. For more details about the FlashCopy and the Remote Copy possible interactions see “Interaction between Remote Copy and FlashCopy” on page 277.

## FlashCopy presets

The IBM SAN Volume Controller GUI interface provides three FlashCopy presets (Snapshot, Clone, and Backup) to simplify the more common FlashCopy operations. Figure 6-10 on page 239 shows the preset selection panel in the GUI.

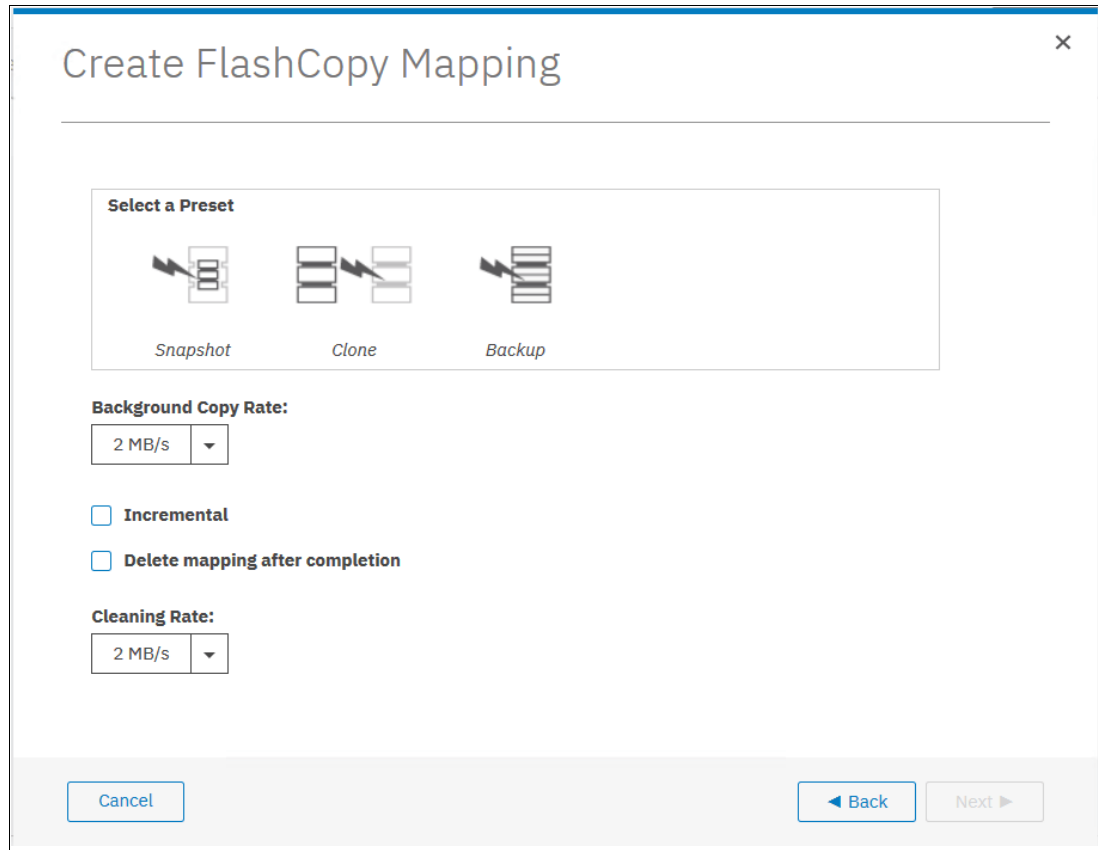


Figure 6-10 GUI FlashCopy Presets

Although these presets meet most FlashCopy requirements, they do not provide support for all possible FlashCopy options. If more specialized options are required that are not supported by the presets, the options must be performed by using CLI commands.

This section describes the three preset options and their use cases.

### **Snapshot**

This preset creates a copy-on-write point-in-time copy. The snapshot is not intended to be an independent copy. Instead, the copy is used to maintain a view of the production data at the time that the snapshot is created. Therefore, the snapshot holds only the data from regions of the production volume that have changed since the snapshot was created. Because the snapshot preset uses thin provisioning, only the capacity that is required for the changes is used.

Snapshot uses the following preset parameters:

- ▶ Background copy: None
- ▶ Incremental: No
- ▶ Delete after completion: No
- ▶ Cleaning rate: No
- ▶ Primary copy source pool: Target pool

A typical use case for the Snapshot is when the user wants to produce a copy of a volume without affecting the availability of the volume. The user does not anticipate many changes to be made to the source or target volume. A significant proportion of the volumes remains unchanged.

By ensuring that only changes require a copy of data to be made, the total amount of disk space that is required for the copy is reduced. Therefore, many Snapshot copies can be used in the environment.

Snapshots are useful for providing protection against corruption or similar issues with the validity of the data. However, they do not provide protection from physical controller failures. Snapshots can also provide a vehicle for performing repeatable testing (including “what-if” modeling that is based on production data) without requiring a full copy of the data to be provisioned.

### ***Clone***

The clone preset creates a replica of the volume, which can then be changed without affecting the original volume. After the copy completes, the mapping that was created by the preset is automatically deleted.

Clone uses the following preset parameters:

- ▶ Background copy rate: 50
- ▶ Incremental: No
- ▶ Delete after completion: Yes
- ▶ Cleaning rate: 50
- ▶ Primary copy source pool: Target pool

A typical use case for the Snapshot is when users want a copy of the volume that they can modify without affecting the original volume. After the clone is established, there is no expectation that it is refreshed or that there is any further need to reference the original production data again. If the source is thin-provisioned, the target is thin-provisioned for the auto-create target.

### ***Backup***

The backup preset creates a point-in-time replica of the production data. After the copy completes, the backup view can be refreshed from the production data, with minimal copying of data from the production volume to the backup volume.

Backup uses the following preset parameters:

- ▶ Background Copy rate: 50
- ▶ Incremental: Yes
- ▶ Delete after completion: No
- ▶ Cleaning rate: 50
- ▶ Primary copy source pool: Target pool

The Backup preset can be used when the user wants to create a copy of the volume that can be used as a backup if the source becomes unavailable. This unavailability can happen during loss of the underlying physical controller. The user plans to periodically update the secondary copy, and does not want to suffer from the resource demands of creating a new copy each time.

Incremental FlashCopy times are faster than full copy, which helps to reduce the window where the new backup is not yet fully effective. If the source is thin-provisioned, the target is also thin-provisioned in this option for the auto-create target.

Another use case, which is not supported by the name, is to create and maintain (periodically refresh) an independent image. This image can be subjected to intensive I/O (for example, data mining) without affecting the source volume’s performance.



## Thin provisioning considerations

When creating FlashCopy with thin provisioned target volumes, the no-copy option is used often. The real size of a thin provisioned volume is an attribute that defines how much physical capacity is reserved for the volume. The real size can vary 0 - 100% of the virtual capacity.

In the case of thin provisioned volumes that are used as FlashCopy targets, it is important to provide a non-zero real size. This issue exists because when the FlashCopy is started, the copy-on-write process requires to allocate capacity on the target volumes. If some capacity is not yet allocated, the write IO can be delayed until the capacity is made available (as with thin provisioned volumes with zero real size). Usually, the write caching hides this effect, but in the case of heavy write workloads, the performance can be affected.

**Important:** Consider the following points:

- ▶ The recommendation with thin provisioned target volumes is to assign at least 2 GB of real capacity.
- ▶ Thin provisioned FlashCopy can greatly benefit from the Redirect-on-Write capability that was introduced with Spectrum Virtualize version 8.4. For more information, see “Indirection layer” on page 232.

## Grain size considerations

When creating a mapping a grain size of 64 KB can be specified as compared to the default 256 KB. This smaller grain size has been introduced specifically for the incremental FlashCopy, even though its use is not restricted to the incremental mappings.

In an incremental FlashCopy, the modified data is identified by using the bitmaps. The amount of data to be copied when refreshing the mapping depends on the grain size. If the grain size is 64 KB, as compared to 256 KB, there might be less data to copy to get a fully independent copy of the source again.

**Incremental FlashCopy:** For incremental FlashCopy, the 64 KB grain size is preferred.

Similar to FlashCopy, the thin provisioned volumes also feature a grain size attribute that represents the size of chunk of storage to be added to used capacity.

The following preferred settings are recommended for thin-provisioned FlashCopy:

- ▶ Thin-provisioned volume grain size is equal to the FlashCopy grain size. If the 256 KB thin-provisioned volume grain size is selected, it is still beneficial to limit the FlashCopy grain size to 64 KB. It is possible to minimize the performance impact to the source volume, although this size increases the I/O workload on the target volume.
- ▶ Thin-provisioned volume grain size must be 64 KB for the best performance and the best space efficiency.

The exception is where the thin target volume is going to become a production volume (and is likely to be subjected to ongoing heavy I/O). In this case, the 256 KB thin-provisioned grain size is preferable because it provides better long-term I/O performance at the expense of a slower initial copy.

**FlashCopy limitation:** Configurations with large numbers of FlashCopy and Remote Copy relationships might be forced to choose a 256 KB grain size for FlashCopy to avoid constraints on the amount of bitmap memory.

Cascading FlashCopy and Multi Target FlashCopy require all the mappings that participate with the FlashCopy chain to include the same grain size (for more information, see “FlashCopy general restrictions” on page 237).

### Volume placement considerations

The source and target volumes placement among the pools and the I/O groups must be planned to minimize the effect of the underlying FlashCopy processes. In normal condition (that is with all the nodes/canisters fully operative), the FlashCopy background copy workload distribution follows this schema:

- ▶ The preferred node of the source volume is responsible for the background copy read operations.
- ▶ The preferred node of the target volume is responsible for the background copy write operations.

Table 6-5 shows how the backend I/O operations are distributed across the nodes.

*Table 6-5 Workload distribution for backend I/O operations*

|                                                                    | <b>Read from source</b>                     | <b>Read from target</b>                     | <b>Write to source</b>                                                                                                           | <b>Write to target</b>                                                                                                |
|--------------------------------------------------------------------|---------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Node that performs the back-end I/O if the grain is copied         | Preferred node in source volume's I/O group | Preferred node in target volume's I/O group | Preferred node in source volume's I/O group                                                                                      | Preferred node in target volume's I/O group                                                                           |
| Node that performs the back-end I/O if the grain is not yet copied | Preferred node in source volume's I/O group | Preferred node in source volume's I/O group | The preferred node in the source volume's I/O group reads and writes, and the preferred node in target volume's I/O group writes | The preferred node in the source volume's I/O group reads, and the preferred node in target volume's I/O group writes |

The data transfer among the source and the target volume's preferred nodes occurs through the node-to-node connectivity. Consider the following volume placement alternatives:

- ▶ Source and target volumes use the same preferred node.  
In this scenario, the node that is acting as preferred for source and target volume manages all the read and write FlashCopy operations. Only resources from this node are used for the FlashCopy operations, and no node-to-node bandwidth is used.
- ▶ Source and target volumes use the different preferred node.  
In this scenario, both nodes that are acting as preferred nodes manage read and write FlashCopy operations according to the schemes described in this section. The data that is transferred between the two preferred nodes goes through the node-to-node network.

In most cases, option 1 (source and target volumes use the same preferred node) is preferred. With the IBM SAN Volume Controller system with multiple I/O groups in Enhanced Stretched Cluster configuration, an alternative option is valid.

In this example, the preferred node placement can follow the location of the source and target volumes on the backend storage. For example, if the source volume is on site A and the target volume is on site B, the target volumes preferred node must be in site B. Placing the target volumes preferred node in site A causes the redirection of the FlashCopy write operation through the node-to-node network.

Placement on the back-end storage is mainly driven by the availability requirements. Generally, use different back-end storage controllers or arrays for the source and target volumes.

**DRP Optimized Snapshots:** To use the Redirect-on-Write capability that was introduced with Spectrum Virtualize version 8.4, check the volume placement restrictions that are described in “Indirection layer” on page 232.

## Background copy considerations

The background copy process uses internal resources such as CPU, memory, and bandwidth. This copy process tries to reach the target copy data rate for every volume according to the background copy rate parameter setting (as reported in Table 6-1 on page 224).

If the copy process is unable to achieve these goals, it starts contending resources to the foreground I/O (that is the I/O coming from the hosts). As result, both background copy and foreground I/O will tend to see an increase in latency and therefore reduction in throughput compared to the situation when the bandwidth not been limited. Degradation is graceful. Both background copy and foreground I/O continue to make progress, and will not stop, hang, or cause the node to fail.

To avoid any impact on the foreground I/O, that is in the hosts response time, carefully plan the background copy activity, taking in account the overall workload running in the systems. The background copy basically reads and writes data to managed disks. Usually, the most affected component is the back-end storage. CPU and memory are not normally significantly affected by the copy activity.

The theoretical added workload due to the background copy is easily estimable. For instance, starting 20 FlashCopy with a background copy rate of 70 each adds a maximum throughput of 160 MBps for the reads and 160 MBps for the writes.

The source and target volumes distribution on the back-end storage determines where this workload is going to be added. The duration of the background copy depends on the amount of data to be copied. This amount is the total size of volumes for full background copy or the amount of data that is modified for incremental copy refresh.

Performance monitoring tools like IBM Spectrum Control can be used to evaluate the existing workload on the back-end storage in a specific time window. By adding this workload to the foreseen background copy workload, you can estimate the overall workload running toward the back-end storage. Disk performance simulation tools, such as Disk Magic and StorM, can be used to estimate the effect, if any, of the added back-end workload to the host service time during the background copy window. The outcomes of this analysis can provide useful hints for the background copy rate settings.

When performance monitoring and simulation tools are not available, use a conservative and progressive approach. Consider that the background copy setting can be modified at any time, even when the FlashCopy is already started. The background copy process can even be completely stopped by setting the background copy rate to 0.

Initially set the background copy rate value to add a limited workload to the backend (for example less than 100 MBps). If no effects on hosts are noticed, the background copy rate value can be increased. Do this process until you see negative effects. Note that the background copy rate setting follows an exponential scale, so changing for instance from 50 to 60 doubles the data rate goal from 2 MBps to 4 MBps.

## **Cleaning process and Cleaning Rate**

The Cleaning Rate is the rate at which the data is copied among dependent FlashCopies such as Cascaded and Multi Target FlashCopy. The Cleaning process aims to release the dependency of a mapping in such a way that it can be stopped immediately (without going to the stopping state). The typical use case for setting the Cleaning Rate is when it is required to stop a Cascaded or Multi Target FlashCopy that is not the oldest in the FlashCopy chain. In this case to avoid the stopping state lasting for a long time, the cleaning rate can be adjusted accordingly.

There is an interaction between the background copy rate and the Cleaning Rate settings:

- ▶ Background copy = 0 and Cleaning Rate = 0  
No background copy or cleaning take place. When the mapping is stopped, it goes into stopping state and a cleaning process starts with the default cleaning rate, which is 50 or 2 MBps.
- ▶ Background copy > 0 and Cleaning Rate = 0  
The background copy takes place at the background copy rate but no cleaning process is started. When the mapping is stopped, it goes into stopping state and a cleaning process starts with the default cleaning rate (50 or 2 MBps).
- ▶ Background copy = 0 and Cleaning Rate > 0  
No background copy takes place, but the cleaning process runs at the cleaning rate. When the mapping is stopped, the cleaning completes (if not yet completed) at the cleaning rate.
- ▶ Background copy > 0 and Cleaning Rate > 0  
The background copy takes place at the background copy rate but no cleaning process is started. When the mapping is stopped, it goes into stopping state and a cleaning process starts with the specified cleaning rate.

Regarding the workload considerations for the cleaning process, the same guidelines as for background copy apply.

## **Host and application considerations to ensure FlashCopy integrity**

Because FlashCopy is at the block level, it is necessary to understand the interaction between your application and the host operating system. From a logical standpoint, it is easiest to think of these objects as “layers” that sit on top of one another. The application is the topmost layer, and beneath it is the operating system layer.

Both of these layers feature various levels and methods of caching data to provide better speed. Because IBM SAN Volume Controller, and therefore FlashCopy, sit below these layers, they are unaware of the cache at the application or operating system layers.

To ensure the integrity of the copy that is made, it is necessary to flush the host operating system and application cache for any outstanding reads or writes before the FlashCopy operation is performed. Failing to flush the host operating system and application cache produces what is referred to as a *crash consistent* copy.

The resulting copy requires the same type of recovery procedure, such as log replay and file system checks, that is required following a host crash. FlashCopies that are crash consistent often can be used following file system and application recovery procedures.

**Note:** Although the best way to perform FlashCopy is to flush host cache first, some companies, such as Oracle, support the use of snapshots without it, as stated in Metalink note 604683.1.

Various operating systems and applications provide facilities to stop I/O operations and ensure that all data is flushed from host cache. If these facilities are available, they can be used to prepare for a FlashCopy operation. When this type of facility is not available, the host cache must be flushed manually by quiescing the application and unmounting the file system or drives.

**Preferred practice:** From a practical standpoint, when you have an application that is backed by a database and you want to make a FlashCopy of that application's data, it is sufficient in most cases to use the write-suspend method that is available in most modern databases. You can use this method because the database maintains strict control over I/O.

This method is as opposed to flushing data from both the application and the backing database, which is always the suggested method because it is safer. However, this method can be used when facilities do not exist or your environment includes time sensitivity.

## 6.3 Remote Copy services

IBM SAN Volume Controller technology offers various Remote Copy services functions that address DR and Business Continuity needs.

*Metro Mirror* is designed for metropolitan distances with a zero recovery point objective (RPO), which is zero data loss. This objective is achieved with a synchronous copy of volumes. Writes are not acknowledged until they are committed to both storage systems. By definition, any vendors' synchronous replication makes the host wait for write I/Os to complete at both the local and remote storage systems, and includes round-trip network latencies. *Metro Mirror* has the following characteristics:

- ▶ Zero RPO
- ▶ Synchronous
- ▶ Production application performance that is affected by round-trip latency

*Global Mirror* technologies are designed to minimize the network latency effects by replicating asynchronously. Spectrum Virtualize provides two type of asynchronous mirroring technology: the standard *Global Mirror* (simply referred as *Global Mirror*) and the *Global Mirror with Change Volume (GMCV)*.

With the *Global Mirror*, writes are acknowledged as soon as they can be committed to the local storage system, sequence-tagged, and passed on to the replication network. This technique allows *Global Mirror* to be used over longer distances. By definition, any vendors' asynchronous replication results in an RPO greater than zero. However, for *Global Mirror*, the RPO is quite small, typically anywhere from several milliseconds to some number of seconds.

Although Global Mirror is asynchronous, the RPO is still small, and thus the network and the remote storage system must both still be able to cope with peaks in traffic. Global Mirror has the following characteristics:

- ▶ Near-zero RPO
- ▶ Asynchronous
- ▶ Production application performance that is affected by I/O sequencing preparation time

GMCV provides an option to replicate point-in-time copies of volumes. This option generally requires lower bandwidth because it is the average rather than the peak throughput that must be accommodated. The RPO for Global Mirror with Change Volumes is higher than traditional Global Mirror. Global Mirror with Change Volumes includes the following characteristics:

- ▶ Larger RPO
- ▶ Point-in-time copies
- ▶ Asynchronous
- ▶ Possible system performance effect because point-in-time copies are created locally

Successful implementation of Remote Copy depends on taking a holistic approach in which you consider all components and their associated properties. The components and properties include host application sensitivity, local and remote SAN configurations, local and remote system and storage configuration, and the intersystem network.

### 6.3.1 Remote copy use cases

Data replication techniques are the foundations of DR and Business Continuity solutions. In addition to these common use cases, Remote Copy technologies can be used in other data movement scenarios, as described next.

#### Storage systems renewal

Remote copy functions can be used to facilitate the migration of data between storage systems while minimizing downtime for applications. By using remote copy, application data can be copied from a Spectrum Virtualize based system to another, while applications remain online. After the volumes are fully copied and synchronized, the application can be stopped and then immediately started on the new storage system.

#### Data center moving

Remote copy functions can be used to move data between IBM Spectrum Virtualize-based systems to facilitate data centers moving operations. By using remote copy, application data can be copied from volumes in a source data centers to volumes in another data center while applications remain online. After the volumes are fully copied and synchronized, the applications can be stopped and then immediately started in the target data center.

### 6.3.2 Remote copy functional overview

In this section, the terminology and the basic functional aspects of the Remote Copy services are presented.

#### Common terminology and definitions

When such a breadth of technology areas is covered, the same technology component can have multiple terms and definitions. This document uses the following definitions:

- ▶ *Local system or master system*

The system on which the foreground applications run.

► *Local hosts*

Hosts that run on the foreground applications.

► *Master volume or source volume*

The local volume that is being mirrored. The volume has nonrestricted access. Mapped hosts can read and write to the volume.

► *Intersystem link or intersystem network*

The network that provides connectivity between the local and the remote site. It can be a Fibre Channel network (SAN), an IP network, or a combination of the two.

► *Remote system or auxiliary system*

The system that holds the remote mirrored copy.

► *Auxiliary volume or target volume*

The remote volume that holds the mirrored copy. It is read-access only.

► *Remote copy*

A generic term that is used to describe a Metro Mirror or Global Mirror relationship in which data on the source volume is mirrored to an identical copy on a target volume. Often the two copies are separated by some distance, which is why the term *remote* is used to describe the copies. However, having remote copies is not a prerequisite. A Remote Copy relationship includes the following states:

– Consistent relationship

A Remote Copy relationship where the data set on the target volume represents a data set on the source volumes at a certain point.

– Synchronized relationship

A relationship is *synchronized* if it is consistent *and* the point that the target volume represents is the current point. The target volume contains identical data as the source volume.

► *Synchronous Remote Copy*

Writes to the source and target volumes that are committed in the foreground before confirmation is sent about completion to the local host application. Metro Mirror is a synchronous Remote Copy type.

► *Asynchronous remote copy*

A foreground write I/O is acknowledged as complete to the local host application before the mirrored foreground write I/O is cached at the remote system. Mirrored foreground writes are processed asynchronously at the remote system, but in way that in the remote system a consistent copy is always present. Global Mirror and GMCV are asynchronous Remote Copy types.

► The *background copy* process manages the initial synchronization or resynchronization processes between source volumes to target mirrored volumes on a remote system.

► *Foreground I/O* reads and writes I/O on a local SAN, which generates a mirrored foreground write I/O that is across the intersystem network and remote SAN.

Figure 6-11 shows some of the concepts of remote copy.

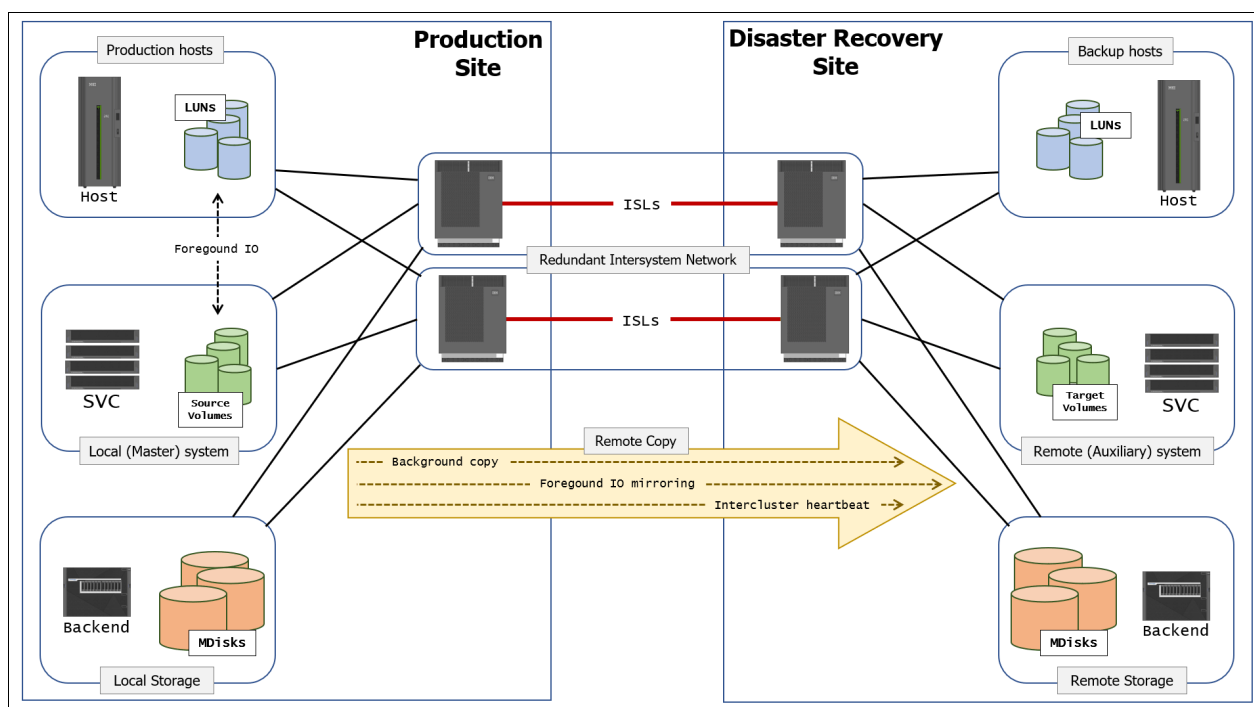


Figure 6-11 Remote copy components and applications

A successful implementation of intersystem Remote Copy services significantly depends on the quality and configuration of the intersystem network.

## Remote Copy partnerships and relationships

A Remote Copy *partnership* is a partnership that is established between a master (local) system and an auxiliary (remote) system, as shown in Figure 6-12 on page 249.

Partnerships are established between two systems by running the **mkfcpartnership** or **mkippartnership** command once from each end of the partnership. The parameters that must be specified are the remote system name (or ID), the available bandwidth (in Mbps), and the maximum background copy rate as a percentage of the available bandwidth. The background copy parameter determines the maximum speed of the initial synchronization and resynchronization of the relationships.

**Tip:** To establish a fully functional Metro Mirror or Global Mirror partnership, run the **mkfcpartnership** or **mkippartnership** command from both systems.



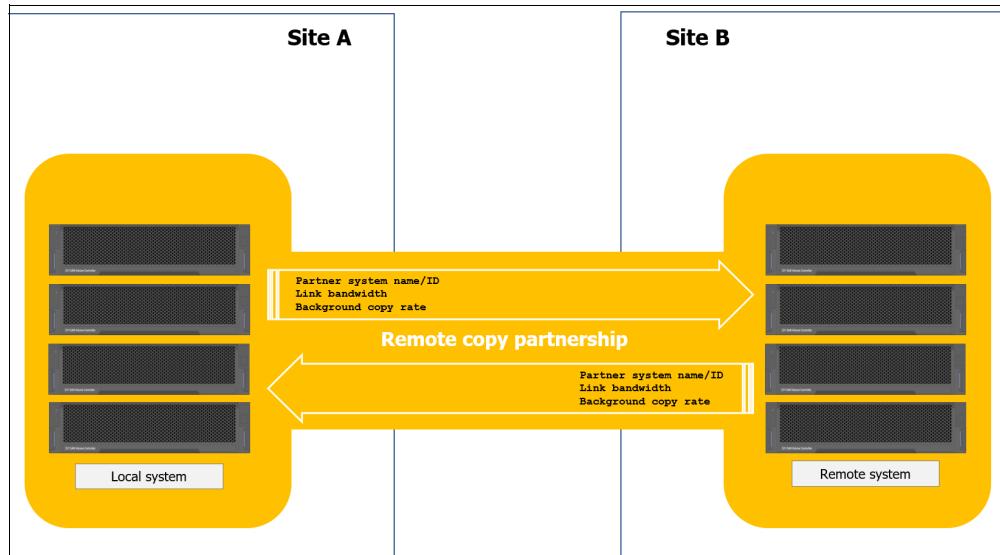


Figure 6-12 Remote copy partnership

In addition to the background copy rate setting, the initial synchronization can be adjusted at relationship level with the **relationship\_bandwidth\_limit** parameter. The **relationship\_bandwidth\_limit** is a system-wide parameter that sets the maximum bandwidth that can be used to initially synchronize a single relationship.

After background synchronization or resynchronization is complete, a Remote Copy relationship provides and maintains a consistent mirrored copy of a source volume to a target volume.

### ***Copy directions and default roles***

When a Remote Copy relationship is created, the source volume is assigned the role of the *master*, and the target volume is assigned the role of the *auxiliary*. This design implies that the initial copy direction of mirrored foreground writes and background resynchronization writes (if applicable) is from master to auxiliary. When a Remote Copy relationship is initially started, the master volume assumes the role of *primary* volume, while the auxiliary volume became *secondary* volumes.

After the initial synchronization is complete, you can change the copy direction (see Figure 6-13) by switching the roles of primary and secondary. The ability to change roles is used to facilitate DR.

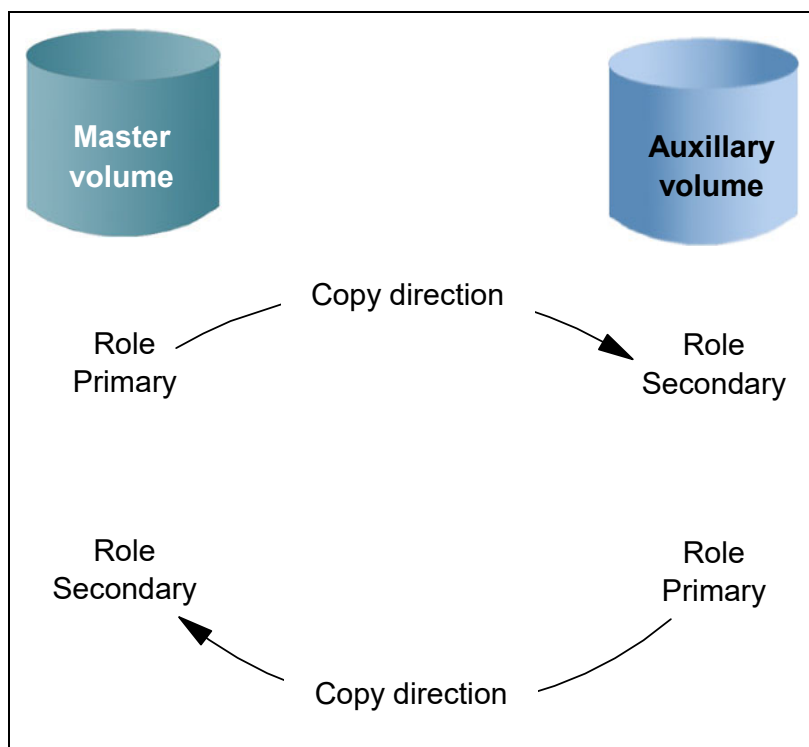


Figure 6-13 Role and direction changes

**Attention:** When the direction of the relationship is changed, the primary and secondary roles of the volumes are altered. A consequence is that the read/write properties are also changed; that is, the master volume takes on a secondary role and becomes read-only.

### Consistency Groups

A Consistency Group is a collection of relationships that can be treated as one entity. This technique is used to preserve write order consistency across a group of volumes that pertain to one application, for example, a database volume and a database log file volume.

After a Remote Copy relationship is added into a Consistency Group, you cannot manage the relationship in isolation from the Consistency Group. For example, issuing a **stoprcrelationship** command on the stand-alone volume fails because the system knows that the relationship is part of a Consistency Group.

Similarly to the Remote Copy relationships, also a Consistency Group, when created, assigns the role of *master* to the source storage system and *auxiliary* to the target storage system.

Note the following points regarding Consistency Groups:

- ▶ Each volume relationship can belong to only one Consistency Group.
- ▶ Volume relationships can also be stand-alone, that is, not in any Consistency Group.
- ▶ Consistency Groups can also be created and left empty, or can contain one or many relationships.
- ▶ You can create up to 256 Consistency Groups on a system.

- ▶ All relationships roles in a Consistency Group must have matching master and auxiliary systems as defined in the Consistency Group.
- ▶ All relationships in a Consistency Group have the same copy direction and state.
- ▶ Each Consistency Group is either for Metro Mirror or for Global Mirror relationships, but not both. This choice is determined by the first volume relationship that is added to the Consistency Group.

**Consistency Group consideration:** A Consistency Group relationship does not have to be in a directly matching I/O group number at each site. A Consistency Group owned by I/O group 1 at the local site does not have to be owned by I/O group 1 at the remote site. If you have more than one I/O group at either site, you can create the relationship between any two I/O groups. This technique spreads the workload, for example, from local I/O group 1 to remote I/O group 2.

### ***Streams***

Consistency Groups can also be used as a way to spread replication workload across multiple streams within a partnership.

The Metro or Global Mirror partnership architecture allocates traffic from each Consistency Group in a round-robin fashion across 16 streams. That is, cg0 traffic goes into stream0, and cg1 traffic goes into stream1, and so on.

Any volume that is *not* in a Consistency Group also goes into stream0. You might want to consider creating an empty Consistency Group 0 so that stand-alone volumes do not share a stream with active Consistency Group volumes.

It can also pay to optimize your streams by creating more Consistency Groups. Within each stream, each batch of writes must be processed in tag sequence order and any delays in processing any particular write also delays the writes behind it in the stream. Having more streams (up to 16) reduces this kind of potential congestion.

Each stream is sequence-tag-processed by one node; therefore, generally you want to create at least as many Consistency Groups as you have IBM SAN Volume Controller nodes, and, ideally, perfect multiples of the node count.

### **Layer concept**

The *layer* is an attribute of Spectrum Virtualize based systems, which allows you to create partnerships among different Spectrum Virtualize products. Consider the following points regarding layers:

- ▶ IBM SAN Volume Controller is always in the *Replication* layer.
- ▶ By default, Storwize/FlashSystem products are in the *Storage* layer.
- ▶ A system can form partnerships only with systems in the same layer.
- ▶ An IBM SAN Volume Controller can virtualize a Storwize/FlashSystem system only if the Storwize is in Storage layer.
- ▶ A Storwize/FlashSystem system in the Replication layer can virtualize a Storwize/FlashSystem system in the Storage layer.

Figure 6-14 shows the concept of layers.

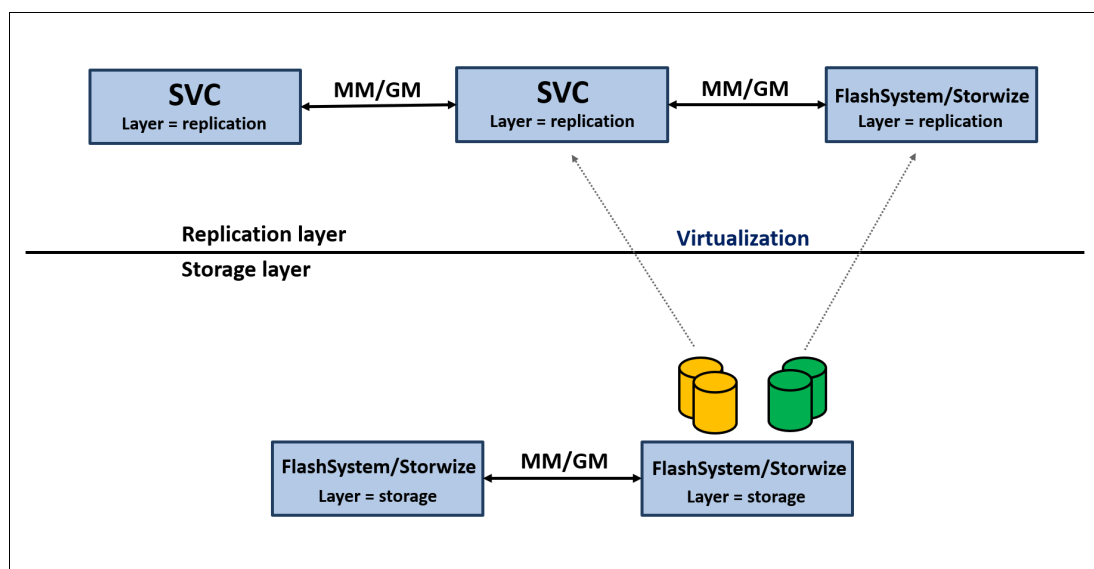


Figure 6-14 Conceptualization of layers

Generally, changing the layer is performed only at initial setup time or as part of a major reconfiguration. To change the layer of a Storwize/FlashSystem system, the system must meet the following preconditions:

- ▶ The Storwize/FlashSystem system must not have any Storwize/FlashSystem host objects defined, and must not be virtualizing any other Storwize/FlashSystem controllers.
- ▶ The Storwize/FlashSystem system must not be visible to any other IBM SAN Volume Controller or Storwize/FlashSystem system in the SAN fabric, which might require SAN zoning changes.
- ▶ The Storwize/FlashSystem system must not have any system partnerships defined. If it uses Metro Mirror or Global Mirror, the partnerships and relationships must be removed first.

Changing a Storwize/FlashSystem system from Storage layer to Replication layer can be performed only by using the CLI. After you are certain that all of the preconditions are met, issue the following command:

```
chsystem -layer replication
```

## Partnership topologies

Spectrum Virtualize allows various partnership topologies, as shown in Figure 6-15. Each box represents an IBM Spectrum Virtualize based system.

The set of systems directly or indirectly connected form the *connected set*. A system can be partnered with up to three remote systems. No more than four systems can be in the same connected set is allowed.

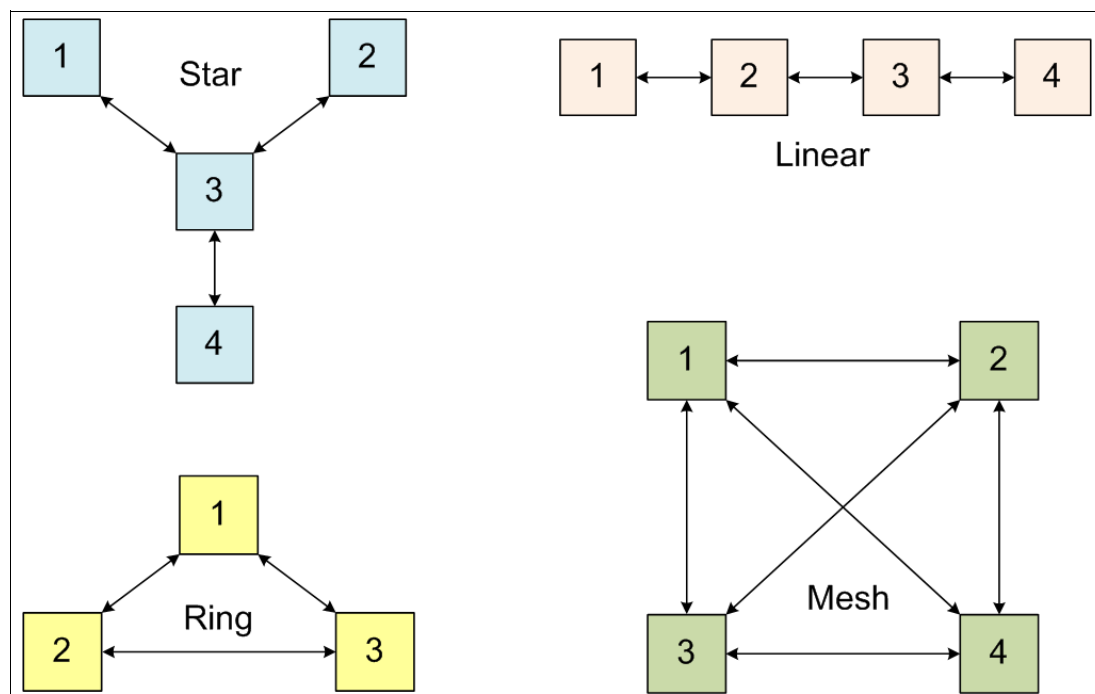


Figure 6-15 Supported topologies for Remote Copy partnerships

### Star topology

A star topology can be used, for example, to share a centralized DR system (3, in this example) with up to three other systems; for example replicating  $1 \rightarrow 3$ ,  $2 \rightarrow 3$ , and  $4 \rightarrow 3$ .

### Ring topology

A ring topology (3 or more systems) can be used to establish a one-in, one-out implementation. For example, the implementation can be  $1 \rightarrow 2$ ,  $2 \rightarrow 3$ ,  $3 \rightarrow 1$  to spread replication loads evenly among three systems.

### Linear topology

A linear topology of two or more sites is also possible. However, it is often simpler to create partnerships between system 1 and system 2, and separately between system 3 and system 4.

### Mesh topology

A fully connected mesh topology is where every system has a partnership to each of the three other systems. This topology allows flexibility in that volumes can be replicated between any two systems.

**Topology considerations:** Although systems can have up to three partnerships, any one volume can be part of only a single relationship. That is, you cannot establish a multi-target Remote Copy relationship for a volume. However, three-site replication is possible with the introduction of the IBM Spectrum Virtualize 3-Site Replication. For more information, see *IBM Spectrum Virtualize 3-Site Replication*, SG24-8474. Although various topologies are supported, it is advisable to keep your partnerships as simple as possible, which in most cases means system pairs or a star.

### ***Intrasystem versus intersystem***

Although Remote Copy services are available for intrasystem, it has no functional value for production use. Intrasystem Metro Mirror provides the same capability with less overhead. However, leaving this function in place simplifies testing and allows for experimentation and testing. For example, you can validate server failover on a single test system.

**Intrasystem remote copy:** Intrasystem Remote Copy is not supported on IBM Spectrum Virtualize systems that run version 6 or later.

### **Metro Mirror functional overview**

Metro Mirror provides synchronous replication. It is designed to ensure that updates are committed to both the primary and secondary volumes before sending an acknowledgment (Ack) of the completion to the server.

If the primary volume fails completely for any reason, Metro Mirror is designed to ensure that the secondary volume holds the same data as the primary did immediately before the failure.

Metro Mirror provides the simplest way to maintain an identical copy on both the primary and secondary volumes. However, as with any synchronous copy over long distance, there can be a performance impact to host applications due to network latency.

Metro Mirror supports relationships between volumes that are up to 300 km apart. Latency is an important consideration for any Metro Mirror network. With typical fiber optic round-trip latencies of 1 ms per 100 km, you can expect a minimum of 3 ms extra latency, due to the network alone, on each I/O if you are running across the 300 km separation.

Figure 6-16 shows the order of Metro Mirror write operations.

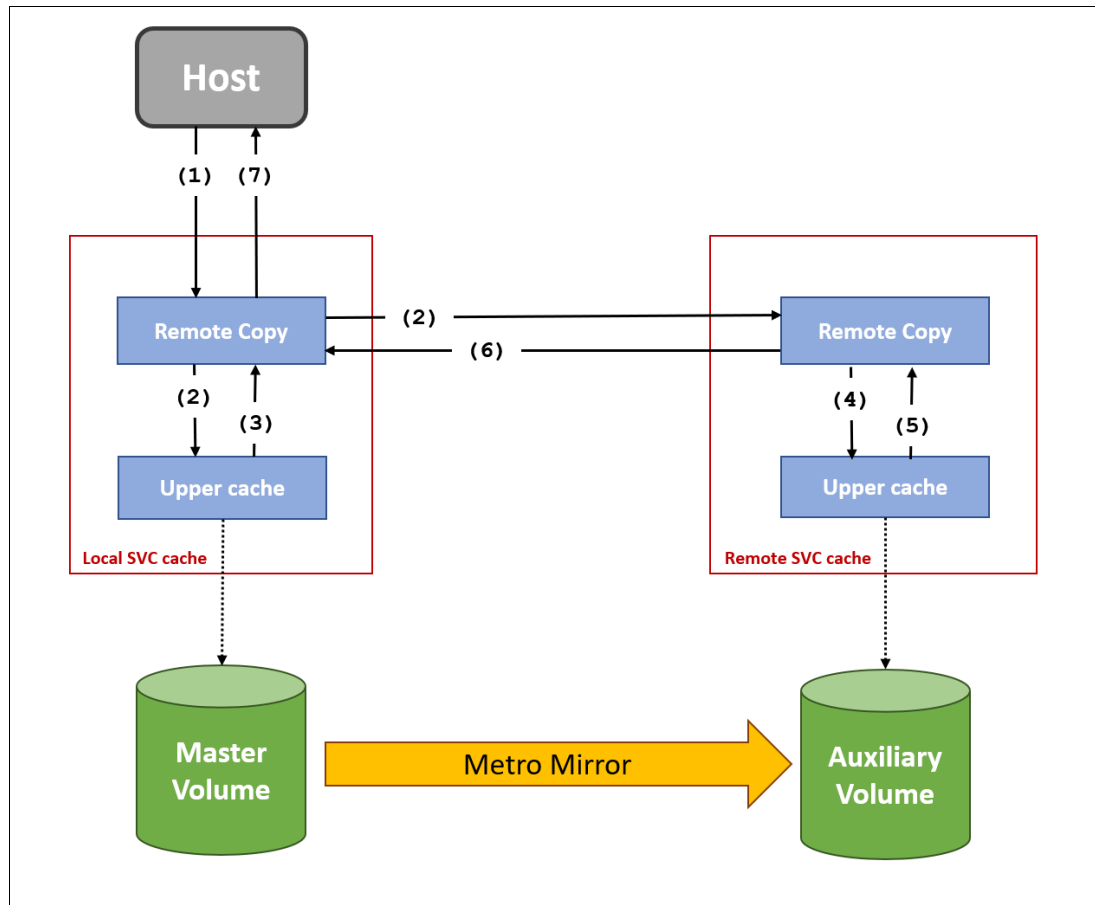


Figure 6-16 Metro Mirror relationship write sequence

The write operation sequence includes the following steps:

1. The write operation is started by the host and intercepted by the Remote Copy component of the local system cache.
2. The write operation is simultaneously written in the upper cache component and sent to the remote system.
3. The write operation on local system upper cache is acknowledged back to Remote Copy component on local system.
4. The write operation is written in the upper cache component of the remote system. This operation is started when the data arrives from the local system and do not depend on operation ongoing in the local system.
5. The write operation on remote system upper cache is acknowledged back to Remote Copy component on remote system.
6. The remote write operation is acknowledged back to Remote Copy component on local system.
7. The write operation is acknowledged back to the host.

For a write to be considered as committed, the data must be written in local and remote systems cache. Although de-staging to disk is a natural part of I/O management, it is not generally in the critical path for a Metro Mirror write acknowledgment.

## **Global Mirror functional overview**

Global Mirror provides asynchronous replication. It is designed to reduce the dependency on round-trip network latency by acknowledging the primary write in parallel with sending the write to the secondary volume.

If the primary volume fails completely for any reason, Global Mirror is designed to ensure that the secondary volume holds the same data as the primary did at a point a short time before the failure. That short period of data loss is typically between 10 ms and 10 seconds, but varies according to individual circumstances.

Global Mirror provides a way to maintain a write-order-consistent copy of data at a secondary site only slightly behind the primary. Global Mirror has minimal impact on the performance of the primary volume.

Although Global Mirror is an asynchronous Remote Copy technique, foreground writes at the local system and mirrored foreground writes at the remote system are not wholly independent of one another. IBM Spectrum Virtualize implementation of Global Mirror uses algorithms to maintain a consistent image at the target volume always.

They achieve this image by identifying sets of I/Os that are active concurrently at the source, assigning an order to those sets, and applying these sets of I/Os in the assigned order at the target. The multiple I/Os within a single set are applied concurrently.

The process that marshals the sequential sets of I/Os operates at the remote system, and therefore is not subject to the latency of the long-distance link.

Figure 6-17 on page 257 shows that a write operation to the master volume is acknowledged back to the host that issues the write before the write operation is mirrored to the cache for the auxiliary volume.



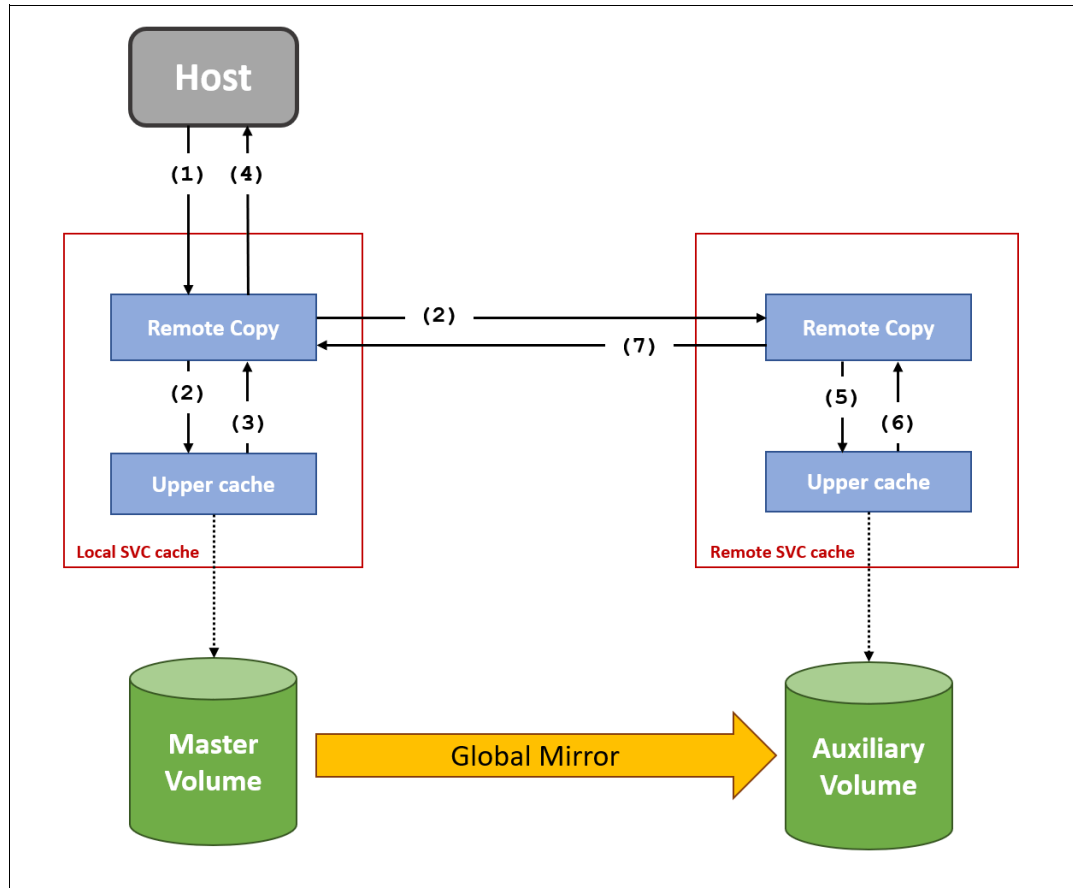


Figure 6-17 Global Mirror relationship write operation

The write operation sequence includes the following steps:

1. The write operation is started by the host and intercepted by the Remote Copy component of the local system cache.
2. The Remote Copy component on local system completes the sequence tagging and the write operation is simultaneously written in the upper cache component and sent to the remote system (along with the sequence number).
3. The write operation on local system upper cache is acknowledged back to Remote Copy component on local system.
4. The write operation is acknowledged back to the host.
5. The Remote Copy component on remote system started the write operation to the upper cache component according with the sequence number. This operation is started as soon as the data arrives from the local system and do not depend on operation ongoing in the local system.
6. The write operation on remote system upper cache is acknowledged back to Remote Copy component on remote system.
7. The remote write operation is acknowledged back to Remote Copy component on local system.

With Global Mirror, a confirmation is sent to the host server before the host receives a confirmation of the completion at the auxiliary volume. The GM function identifies sets of write I/Os that are active concurrently at the primary volume. It then assigns an order to those sets, and applies these sets of I/Os in the assigned order at the auxiliary volume.

Further writes might be received from a host when the secondary write is still active for the same block. In this case, although the primary write might complete, the new host write on the auxiliary volume is delayed until the previous write is completed. Finally, any delay in step 2 on page 257 is reflected in write delay on primary volume.

### ***Write ordering***

Many applications that use block storage are required to survive failures, such as a loss of power or a software crash. They are also required to not lose data that existed before the failure. Because many applications must perform many update operations in parallel to that storage block, maintaining write ordering is key to ensuring the correct operation of applications after a disruption.

An application that performs a high volume of database updates is often designed with the concept of dependent writes. Dependent writes ensure that an earlier write completes before a later write starts. Reversing the order of dependent writes can undermine the algorithms of the application and can lead to problems, such as detected or undetected data corruption.

### ***Colliding writes***

Colliding writes are defined as new write I/Os that overlap existing active write I/Os.

The original Global Mirror algorithm required only a single write to be active on any 512-byte LBA of a volume. If another write was received from a host while the auxiliary write was still active, the new host write was delayed until the auxiliary write was complete (although the master write might complete). This restriction was needed if a series of writes to the auxiliary must be retried (which is known as *reconstruction*). Conceptually, the data for reconstruction is from the master volume.

If multiple writes were allowed to be applied to the master for a sector, only the most recent write included the correct data during reconstruction. If reconstruction was interrupted for any reason, the intermediate state of the auxiliary was inconsistent.

Applications that deliver such write activity do not achieve the performance that Global Mirror is intended to support. A volume statistic is maintained about the frequency of these collisions. The original Global Mirror implementation was modified to allow multiple writes to a single location to be outstanding in the Global Mirror algorithm.

A need still exists for master writes to be serialized. The intermediate states of the master data must be kept in a non-volatile journal while the writes are outstanding to maintain the correct write ordering during reconstruction. Reconstruction must never overwrite data on the auxiliary with an earlier version. The colliding writes of volume statistic monitoring are now limited to those writes that are not affected by this change.

Figure 6-18 shows a colliding write sequence.

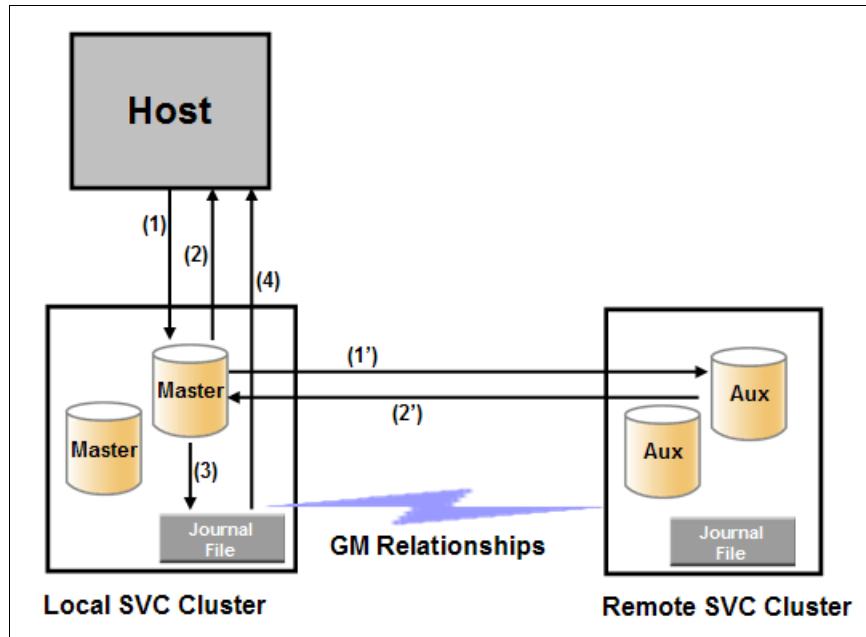


Figure 6-18 Colliding writes

The following numbers correspond to the numbers that are shown in Figure 6-18:

1. A first write is performed from the host to LBA X.
2. A host is provided acknowledgment that the write is complete, even though the mirrored write to the auxiliary volume is not yet completed.  
The first two actions (1 and 2) occur asynchronously with the first write.
3. A second write is performed from the host to LBA X. If this write occurs before the host receives acknowledgment (2), the write is written to the journal file.
4. A host is provided acknowledgment that the second write is complete.

### Global Mirror Change Volumes functional overview

Global Mirror with Change Volumes (GMCV) provides asynchronous replication that is based on point-in-time copies of data. It is designed to allow for effective replication over lower bandwidth networks and to reduce any effect on production hosts.

Metro Mirror and Global Mirror both require the bandwidth to be sized to meet the peak workload. Global Mirror with Change Volumes must only be sized to meet the average workload across a cycle period.

Figure 6-19 shows a high-level conceptual view of Global Mirror with Change Volumes. GMCV uses FlashCopy to maintain image consistency and to isolate host volumes from the replication process.

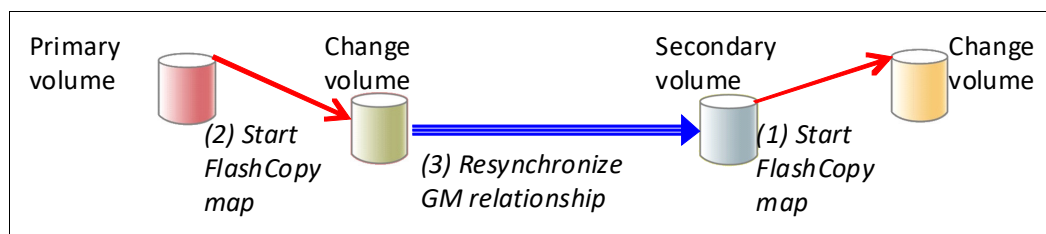


Figure 6-19 Global Mirror with Change Volumes

Global Mirror with Change Volumes also only sends one copy of a changed grain that might have been rewritten many times within the cycle period.

If the primary volume fails completely, GMCV is designed to ensure that the secondary volume holds the same data as the primary did at a specific time. That period of data loss is typically between 5 minutes and 24 hours, but varies according to your design choices.

Change Volumes hold point-in-time copies of 256 KB grains. If any of the disk blocks in a grain change, that grain is copied to the change volume to preserve its contents. Change Volumes are also maintained at the secondary site so that a consistent copy of the volume is always available even when the secondary volume is being updated.

Primary and Change Volumes are always in the same I/O group and the Change Volumes are always thin-provisioned. Change Volumes cannot be mapped to hosts and used for host I/O, and they cannot be used as a source for any other FlashCopy or Global Mirror operations.

Figure 6-20 shows how a Change Volume is used to preserve a point-in-time data set, which is then replicated to a secondary site. The data at the secondary site is in turn preserved by a Change Volume until the next replication cycle has completed.

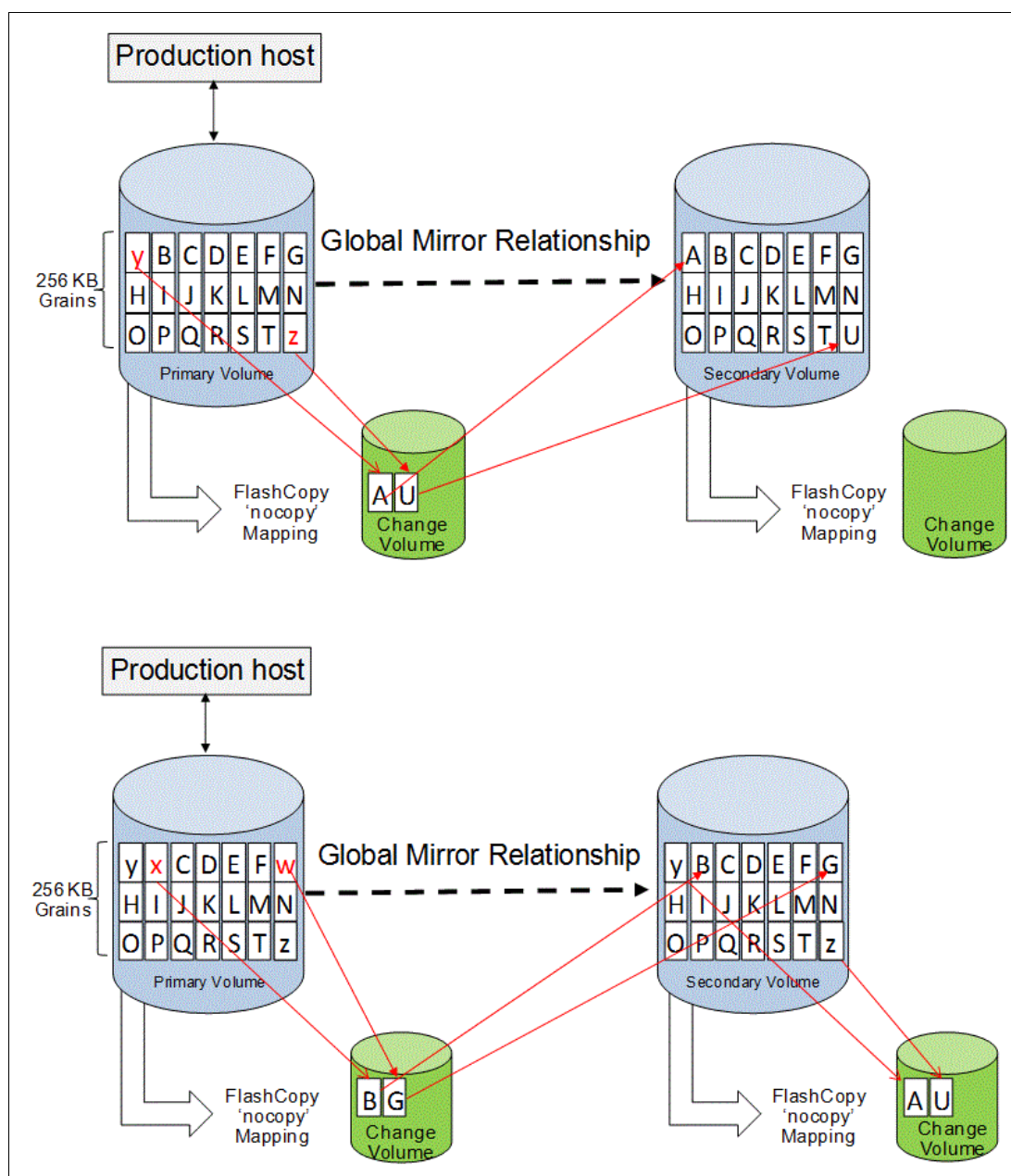


Figure 6-20 Global Mirror with Change Volumes uses FlashCopy point-in-time copy technology

**FlashCopy mapping note:** These FlashCopy mappings are not standard FlashCopy volumes and are not accessible for general use. They are internal structures that are dedicated to supporting Global Mirror with Change Volumes.

The options for `-cyclingmode` are `none` and `multi`.

Specifying or taking the default `none` means that Global Mirror acts in its traditional mode without Change Volumes.

Specifying **multi** means that Global Mirror starts cycling based on the cycle period, which defaults to 300 seconds. The valid range is from 60 seconds to 24\*60\*60 seconds (86,400 seconds = one day).

If all of the changed grains cannot be copied to the secondary site within the specified time, then the replication is designed to take as long as it needs and to start the next replication as soon as the earlier one completes. You can choose to implement this approach by deliberately setting the cycle period to a short amount of time, which is a perfectly valid approach. However, remember that the shorter the cycle period, the less opportunity there is for peak write I/O smoothing, and the more bandwidth you need.

The **-cyclingmode** setting can only be changed when the Global Mirror relationship is in a stopped state.

### ***Recovery point objective using Change Volumes***

RPO is the maximum tolerable period in which data might be lost if you switch over to your secondary volume.

If a cycle completes within the specified cycle period, then the RPO is not more than 2x cycle long. However, if it does not complete within the cycle period, then the RPO is not more than the sum of the last two cycle times.

The current RPO can be determined by looking at the **1srcrelationship** freeze time attribute. The freeze time is the time stamp of the last primary Change Volume that has completed copying to the secondary site. Note the following example:

1. The cycle period is the default of 5 minutes and a cycle is triggered at 6:00 AM. At 6:03 AM, the cycle completes. The freeze time also is 6:00 AM, and the RPO is 3 minutes.
2. The cycle starts again at 6:05 AM. The RPO now is 5 minutes. The cycle is still running at 6:12 AM, and the RPO is now up to 12 minutes because 6:00 AM is still the freeze time of the last complete cycle.
3. At 6:13 AM, the cycle completes and the RPO now is 8 minutes because 6:05 AM is the freeze time of the last complete cycle.
4. Because the cycle period has been exceeded, the cycle immediately starts again.

## **6.3.3 Remote copy network planning**

Remote copy partnerships and relationships do not work reliably if the connectivity on which they are running is configured incorrectly. This section focuses on the intersystem network, giving an overview of the remote system connectivity options.

### **Terminology**

The intersystem network is specified in terms of *latency* and *bandwidth*. These parameters define the capabilities of the link regarding the traffic that is on it. They must be chosen so that they support all forms of traffic, including mirrored foreground writes, background copy writes, and intersystem heartbeat messaging (node-to-node communication).

*Link latency* is the time that is taken by data to move across a network from one location to another and is measured in milliseconds. The latency measures the time spent to send the data and to receive the acknowledgment back (Round Trip Time - RTT).

*Link bandwidth* is the network capacity to move data as measured in millions of bits per second (Mbps) or billions of bits per second (Gbps).

The term *bandwidth* is also used in the following context:

- ▶ Storage bandwidth: The ability of the back-end storage to process I/O. Measures the amount of data (in bytes) that can be sent in a specified amount of time.
- ▶ Remote copy partnership bandwidth (parameter): The rate at which background write synchronization is attempted (unit of MBps).

*Intersystem connectivity* supports mirrored foreground and background I/O. A portion of the link is also used to carry traffic that is associated with the exchange of low-level messaging between the nodes of the local and remote systems. A *dedicated amount* of the link bandwidth is required for the exchange of heartbeat messages and the initial configuration of intersystem partnerships.

*Fibre Channel* connectivity is the standard connectivity that is used for the Remote Copy intersystem networks. It uses the Fibre Channel protocol and SAN infrastructures to interconnect the systems.

*Native IP* connectivity is connectivity option that is based on standard TCP/IP infrastructures that are provided by IBM Spectrum Virtualize technology.

### **SCSI write latency**

Typical SCSI behavior for a write is shown in Figure 6-21.

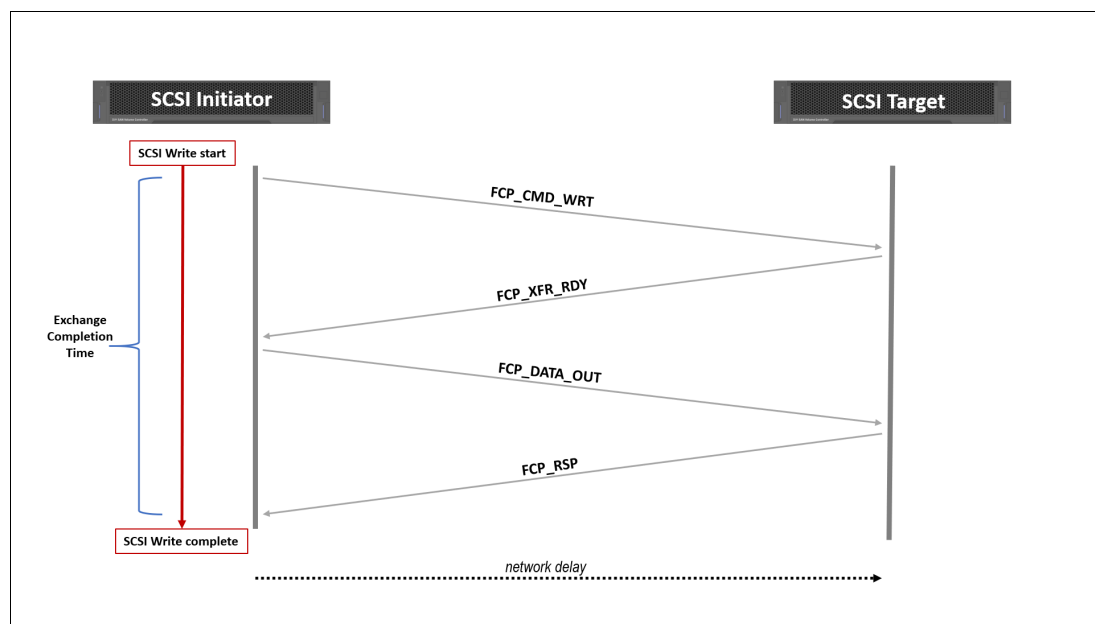


Figure 6-21 SCSI write operation

A standards-based SCSI write is a two-step process. First, the write command is sent across the network to the target. The first round trip is essentially asking transfer permission from the target. The target responds with an acceptance (FCP\_XFR\_RDY). The initiator waits until it receives a response from the target before starting the second step; that is, sending the data (FCP\_DATA\_OUT).

Overall, two round trips are required to complete the write; therefore, the write takes at least  $2 \times \text{RTT}$ , plus the time for the data out.

Within the confines of a data center, where the latencies are measured in microseconds ( $\mu$ sec), no issues exist. However, across a geographical network where the latencies are measured in milliseconds (ms), the overall service time can be significantly affected.

Considering that the network delay over fiber optics per kilometer (km) is approximately 5  $\mu$ sec (10  $\mu$ sec RTT), the resulting minimum service time for a SCSI write is 20  $\mu$ sec per every km of distance; for example, a SCSI write over 50 km has a minimum service time of 1000  $\mu$ sec (that is, 1 ms).

## Network latency considerations

The maximum supported round-trip latency between sites depends on the type of partnership between systems. Table 6-6 lists the maximum round-trip latency. This restriction applies to all variants of remote mirroring.

Table 6-6 Maximum round trip

| Partnership |           |            |
|-------------|-----------|------------|
| FC          | 1 Gbps IP | 10 Gbps IP |
| 250 ms      | 80 ms     | 10 ms      |

More configuration requirements and guidelines apply to systems that perform remote mirroring over extended distances, where the round-trip time is greater than 80 ms. If you use remote mirroring between systems with 80 - 250 ms round-trip latency, you must meet the following additional requirements:

- The RC buffer size setting must be 512 MB on each system in the partnership. This setting can be accomplished by running the `chsystem -rcbuffersize 512` command on each system.

**Important:** Changing this setting is disruptive to Metro Mirror and Global Mirror operations. Use this command only before partnerships are created between systems, or when all partnerships with the system are stopped.

- Two Fibre Channel ports on each node that will be used for replication must be dedicated for replication traffic. This configuration can be achieved by using SAN zoning and port masking.
- SAN zoning should be applied to provide separate intrasystem zones for each local-remote I/O group pair that is used for replication. See “Remote system ports and zoning considerations” on page 271 for further zoning guidelines.

## Link bandwidth that is used by internode communication

IBM Spectrum Virtualize uses part of the bandwidth for its internal intersystem heartbeat. The amount of traffic depends on how many nodes are in each of the local and remote systems. Table 6-7 shows the amount of traffic (in megabits per second) that is generated by different sizes of systems.

Table 6-7 IBM Spectrum Virtualize intersystem heartbeat traffic (megabits per second)

| Local or remote system | Two nodes | Four nodes | Six nodes | Eight nodes |
|------------------------|-----------|------------|-----------|-------------|
| Two nodes              | 5         | 6          | 6         | 6           |
| Four nodes             | 6         | 10         | 11        | 12          |
| Six nodes              | 6         | 11         | 16        | 17          |



| Local or remote system | Two nodes | Four nodes | Six nodes | Eight nodes |
|------------------------|-----------|------------|-----------|-------------|
| Eight nodes            | 6         | 12         | 17        | 21          |

These numbers represent the total traffic between the two systems when *no* I/O is occurring to a mirrored volume on the remote system. Half of the data is sent by one system, and half of the data is sent by the other system. The traffic is divided evenly over all available connections. Therefore, if you have two redundant links, half of this traffic is sent over each link during fault-free operation.

If the link between the sites is configured with redundancy to tolerate single failures, size the link so that the bandwidth and latency statements continue to be accurate even during single failure conditions.

### Network sizing considerations

Proper network sizing is essential for the Remote Copy services operations. Failing to estimate the network sizing requirements can lead to poor performance in Remote Copy services and the production workload.

Consider that intersystem bandwidth should be capable of supporting the combined traffic of the following items:

- ▶ Mirrored foreground writes, as generated by your server applications at peak times
- ▶ Background write synchronization, as defined by the Global Mirror bandwidth parameter
- ▶ Intersystem communication (*heartbeat messaging*)

Calculating the required bandwidth is essentially a question of mathematics based on your current workloads; therefore, it is advisable to start by assessing your current workloads.

### ***Metro Mirror and Global Mirror network sizing***

With the Metro Mirror, because of its synchronous nature, the amount of replication bandwidth that is required to mirror a specific foreground write data throughput is not less than the foreground write data throughput itself.

Not having write buffering resources, the Global Mirror tends to mirror the foreground write when they are committed in cache (see “Global Mirror functional overview” on page 256); therefore, the bandwidth requirements are similar to Metro Mirror.

For a suitable bandwidth sizing with Metro Mirror or Global Mirror, you must know your peak write workload to at least a 5-minute interval. This information can be easily gained from tools, such as IBM Spectrum Control. Finally, you must allow for the background copy, intercluster communication traffic, and a safe margin for unexpected peaks and workload growth.

**Recommendation:** Do not compromise on bandwidth or network quality when planning a Metro Mirror or Global Mirror deployment. If bandwidth is likely to be an issue in your environment, consider Global Mirror with Change Volumes.

As an example, consider a business with the following I/O profile:

- ▶ Average write size 8 KB (= 8 x 8 bits/1024 = 0.0625 Mb).
- ▶ For most of the day from 8 AM - 8 PM, the write activity is approximately 1500 writes per second.
- ▶ Twice a day (once in the morning and once in the afternoon), the system bursts up to 4500 writes per second for up to 10 minutes.

This example is intended to represent a general traffic pattern that might be common in many medium-sized sites. Furthermore, 20% of bandwidth must be left available for the background synchronization.

Metro Mirror or Global Mirror require bandwidth on the instantaneous peak of 4500 writes per second, as shown in the following example:

$4500 \times 0.0625 = 282 \text{ Mbps} + 20\% \text{ resync allowance} + 5 \text{ Mbps heartbeat} = 343 \text{ Mbps}$   
dedicated plus any safety margin plus growth

### ***GMCV network sizing***

The GMCV is typically less demanding in terms of bandwidth requirements for several reasons.

First, by using its journaling capabilities, the GMCV provides a way to maintain point-in-time copies of data at a secondary site where insufficient bandwidth is available to replicate the peak workloads in real time.

Another factor that can reduce the bandwidth that is required for GMCV is that it only sends one copy of a changed grain, which might be rewritten many times within the cycle period.

The GMCV network sizing is a trade off between RPO, journal capacity, and network bandwidth. A direct relation between the RPO and the physical occupancy of the change volumes exists: the lower the RPO, the less capacity is used by change volumes. However, higher RPO often requires less network bandwidth.

For a suitable bandwidth sizing with GMCV, you must know your average write workload during the cycle time. This information can be easily gained from tools, such as IBM Spectrum Control.

Finally, you must allow for the background resync workload, intercluster communication traffic, and a safe margin for unexpected peaks and workload growth.

As an example, consider a business with the following I/O profile:

- ▶ Average write size 8 KB ( $= 8 \times 8 \text{ bits}/1024 = 0.0625 \text{ Mb}$ ).
- ▶ For most of the day from 8 AM - 8 PM, the write activity is approximately 1500 writes per second.
- ▶ Twice a day (once in the morning and once in the afternoon), the system bursts up to 4500 writes per second for up to 10 minutes.
- ▶ Outside of the 8 AM - 8 PM window, there is little or no I/O write activity.

This example is intended to represent a general traffic pattern that might be common in many medium-sized sites. Furthermore, 20% of bandwidth must be left available for the background synchronization. Consider the following sizing exercises:

- ▶ GMCV peak 30-minute cycle time

If we look at this time broken into 10-minute periods, the peak 30-minute period is made up of one 10-minute period of 4500 writes per second, and two 10-minute periods of 1500 writes per second. The average write rate for the 30-minute cycle period can then be expressed mathematically as follows:

$(4500 + 1500 + 1500) / 3 = 2500 \text{ writes/sec}$  for a 30-minute cycle period

The minimum bandwidth that is required for the cycle period of 30 minutes is as follows:

$2500 \times 0.0625 = 157 \text{ Mbps} + 20\% \text{ resync allowance} + 5 \text{ Mbps heartbeat} = 195 \text{ Mbps}$   
dedicated plus any safety margin plus growth

► **GMCV peak 60-minute cycle time**

For a cycle period of 60 minutes, the peak 60-minute period is made up of one 10-minute period of 4500 writes per second, and five 10-minute periods of 1500 writes per second. The average write for the 60-minute cycle period can be expressed as follows:

$$(4500 + 5 \times 1500) / 6 = 2000 \text{ writes/sec for a 60-minute cycle period}$$

The minimum bandwidth that is required for a cycle period of 60 minutes is as follows:

$$2000 \times 0.0625 = 125 \text{ Mbps} + 20\% \text{ resync allowance} + 5 \text{ Mbps heartbeat} = 155 \text{ Mbps}$$

dedicated plus any safety margin plus growth

► **GMCV with daily cycle time**

Now consider whether the business does not have aggressive RPO requirements and does not want to provide dedicated bandwidth for Global Mirror. However, the network is available and unused at night, so Global Mirror can use that network. An element of risk exists here; that is, if the network is unavailable for any reason, GMCV cannot keep running during the day until it catches up. Therefore, you must allow a much higher resync allowance in your replication window; for example, 100 percent.

A GMCV replication that is based on daily point-in-time copies at 8 PM each night, and replicating until 8 AM at the latest likely requires at least the following bandwidth:

$$(9000 + 70 \times 1500) / 72 = 1584 \times 0.0625 = 99 \text{ Mbps} + 100\% + 5 \text{ Mbps heartbeat} = 203 \text{ Mbps}$$

at night plus any safety margin plus growth, non-dedicated, time-shared with daytime traffic

The central principle of sizing is that you must know your write workload. For Metro Mirror and Global Mirror, you must know the peak write workload. For GMCV, you must know the average write workload.

**GMCV bandwidth:** In the examples that were described thus far, the bandwidth estimation for the GMCV is based on the assumption that the write operations occurs in such a way that a change volume grain (that has a size of 256 KB) is changed before it is transferred to the remote site. In a real-world scenario, this situation is unlikely to occur.

Often, only a portion of a grain is changed during a GMCV cycle, but the transfer process always copies the entire grain to the remote site. This behavior can lead to an unforeseen processor burden in the transfer bandwidth that, in the edge case, can be even higher than the one required for a standard Global Mirror.

### ***Global Mirror and GMCV coexistence considerations***

Global Mirror and GMCV relationships can be defined in the same system. With these configurations, particular attention must be paid to bandwidth sizing and the partnership settings.

The two Global Mirror technologies use the available bandwidth in different ways. Regular Global Mirror uses the amount of bandwidth that is needed to sustain the write workload of the replication set. The GMCV uses the fixed amount of bandwidth as defined in the partnership as background copy.

For this reason, during GMCV cycle creation, a fixed part of the bandwidth is allocated for the background copy and only the remaining part of the bandwidth is available for Global Mirror. To avoid bandwidth contention, which can lead to a 1920 error (see 6.3.6, “1920 error” on page 291) or delayed GMCV cycle creation, the bandwidth must be sized to consider both requirements.

Ideally, in these cases the bandwidth is enough to accommodate the peak write workload for the Global Mirror replication set plus the estimated bandwidth that is needed to fulfill the RPO of GMCV. If these requirements cannot be met because of bandwidth restrictions, the least affecting option is to increase the GMCV cycle period and then reduce the background copy rate to minimize the chance of a 1920 error.

**Note:** These considerations also apply to configurations in which multiple IBM Spectrum Virtualize-based systems are sharing bandwidth resources.

## **Fibre Channel connectivity considerations**

Remember the following considerations when you use Fibre Channel (FC) technology for the intersystem network:

- ▶ Redundancy
- ▶ Basic topology and problems
- ▶ Distance extensions options
- ▶ Hops
- ▶ Buffer credits
- ▶ Remote system ports and zoning considerations

### ***Redundancy***

The intersystem network must adopt the same policy toward redundancy as for the local and remote systems to which it is connecting. The ISLs must have redundancy, and the individual ISLs must provide the necessary bandwidth in isolation.

### ***Basic topology and problems***

Because of the nature of Fibre Channel, you must avoid ISL congestion whether within individual SANs or across the intersystem network. Although FC (and IBM SAN Volume Controller) can handle an overloaded host or storage array, the mechanisms in FC are ineffective for dealing with congestion in the fabric in most circumstances. The problems that are caused by fabric congestion can range from dramatically slow response time to storage access loss. These issues are common with all high-bandwidth SAN devices and are inherent to FC. They are not unique to the IBM Spectrum Virtualize products.

When an FC network becomes congested, the FC switches stop accepting more frames until the congestion clears. They can also drop frames. Congestion can quickly move upstream in the fabric and clog the end devices from communicating anywhere.

This behavior is referred to as *head-of-line blocking*. Although modern SAN switches internally have a nonblocking architecture, head-of-line-blocking still exists as a SAN fabric problem. Head-of-line blocking can result in IBM SAN Volume Controller nodes that cannot mirror their write caches because you have a single congested link that leads to an edge switch.

### ***Distance extensions options***

The following choices are available to implement remote mirroring over a distance by using the FC:

- ▶ *Optical multiplexors*, such as dense wavelength division multiplexing (DWDM) or coarse wavelength division multiplexing (CWDM) devices.

Optical multiplexors can extend a SAN up to hundreds of kilometers (or miles) at high speeds. For this reason, they are the preferred method for long-distance expansion. If you use multiplexor-based distance extension, closely monitor your physical link error counts in your switches. Optical communication devices are high-precision units. When they shift out of calibration, you begin to see errors in your frames.

- Long-distance Small Form-factor Pluggable (SFP) transceivers and XFPs

Long-distance optical transceivers have the advantage of extreme simplicity. You do not need any expensive equipment, and you have only a few configuration steps to perform. However, ensure that you use only transceivers that are designed for your specific SAN switch.

- Fibre Channel-to-IP conversion boxes

Fibre Channel over IP (FCIP) is by far the most common and least expensive form of distance extension. It is also complicated to configure. Relatively subtle errors can result in severe performance implications.

With IP-based distance extension, you must dedicate bandwidth to your FCIP traffic if the link is shared with other IP traffic. Do not assume that because the link between two sites has low traffic or is used only for email, this type of traffic is always the case. FC is far more sensitive to congestion than most IP applications.

Also, when you are communicating with the networking architects for your organization, make sure to distinguish between *megabytes per second* as opposed to *megabits per second*. In the storage world, bandwidth often is specified in megabytes per second (MBps), and network engineers specify bandwidth in megabits per second (Mbps).

Of these options, the optical distance extension is the preferred method. IP distance extension introduces more complexity, is less reliable, and includes performance limitations. However, optical distance extension can be impractical in many cases because of cost or unavailability.

For more information about supported SAN routers and FC extenders, see [IBM Documentation](#).

### **Hops**

The hop count is not increased by the intersite connection architecture. For example, if you have a SAN extension that is based on DWDM, the DWDM components are not apparent to the number of hops. The hop count limit within a fabric is set by the fabric devices (switch or director) operating system. It is used to derive a frame hold time value for each fabric device.

This hold time value is the maximum amount of time that a frame can be held in a switch before it is dropped or the fabric is busy condition is returned. For example, a frame might be held if its destination port is unavailable. The hold time is derived from a formula that uses the error detect timeout value and the resource allocation timeout value. It is considered that every extra hop adds about 1.2 microseconds of latency to the transmission.

As of this writing, IBM SAN Volume Controller copy services support three hops when protocol conversion exists. Therefore, if you use DWDM extended between primary and secondary sites, three SAN directors or switches can exist between the primary and secondary systems.

### **Buffer credits**

SAN device ports need memory to temporarily store frames as they arrive, assemble them in sequence, and deliver them to the upper layer protocol. The number of frames that a port can hold is called its *buffer credit*. Fibre Channel architecture is based on a flow control that ensures a constant stream of data to fill the available pipe.

When two FC ports begin a conversation, they exchange information about their buffer capacities. An FC port sends only the number of buffer frames for which the receiving port gives credit. This method avoids overruns and provides a way to maintain performance over distance by filling the pipe with in-flight frames or buffers.

The following types of transmission credits are available:

- **Buffer\_to\_Buffer Credit**

During login, N\_Ports and F\_Ports at both ends of a link establish its Buffer to Buffer Credit (BB\_Credit).

- **End\_to\_End Credit**

In the same way during login, all N\_Ports establish End-to-End Credit (EE\_Credit) with each other. During data transmission, a port must not send more frames than the buffer of the receiving port can handle before you receive an indication from the receiving port that it processed a previously sent frame. Two counters are used: BB\_Credit\_CNT and EE\_Credit\_CNT. Both counters are initialized to zero during login.

**FC Flow Control:** Each time that a port sends a frame, it increments BB\_Credit\_CNT and EE\_Credit\_CNT by one. When it receives R\_RDY from the adjacent port, it decrements BB\_Credit\_CNT by one. When it receives ACK from the destination port, it decrements EE\_Credit\_CNT by one.

At any time, if BB\_Credit\_CNT becomes equal to the BB\_Credit, or EE\_Credit\_CNT becomes equal to the EE\_Credit of the receiving port, the transmitting port stops sending frames until the respective count is decremented.

The previous statements are true for Class 2 service. Class 1 is a dedicated connection. Therefore, BB\_Credit is not important, and only EE\_Credit is used (EE Flow Control). However, Class 3 is an unacknowledged service. Therefore, it uses only BB\_Credit (BB Flow Control), but the mechanism is the same in all cases.

Here, you see the importance that the number of buffers has in overall performance. You need enough buffers to ensure that the transmitting port can continue to send frames without stopping to use the full bandwidth, which is true with distance. The total amount of buffer credit needed to optimize the throughput depends on the link speed and the average frame size.

For example, consider an 8 Gbps link connecting two switches that are 100 km apart. At 8 Gbps, a full frame (2148 bytes) occupies about 0.51 km of fiber. In a 100 km link, you can send 198 frames before the first one reaches its destination. You need an ACK to go back to the start to fill EE\_Credit again. You can send another 198 frames before you receive the first ACK.

You need at least 396 buffers to allow for nonstop transmission at 100 km distance. The maximum distance that can be achieved at full performance depends on the capabilities of the FC node that is attached at either end of the link extenders, which are vendor-specific. A match should occur between the buffer credit capability of the nodes at either end of the extenders.

### ***Remote system ports and zoning considerations***

Ports and zoning requirements for the remote system partnership have changed over time. The current preferred configuration is based on the [Preferred configuration Flash Alert](#).

The preferred practice for the IBM SAN Volume Controller is to provision dedicated node ports for local node-to-node traffic (by using port masking) and isolate Remote Copy traffic between the local nodes from other local SAN traffic.

**Remote port masking:** To isolate the node-to-node traffic from the Remote Copy traffic, the local and remote port masking implementation is preferable.

This configuration of local node port masking is less of a requirement on non-clustered FlashSystem systems, where traffic between node canisters in an I/O group is serviced by the dedicated PCI inter-canister link in the enclosure.

The following guidelines apply to the remote system connectivity:

- ▶ The minimum requirement to establish a Remote Copy partnership is to connect at least one node per system. When remote connectivity among all the nodes of both systems is not available, the nodes of the local system not participating to the remote partnership will use the node/nodes defined in the partnership as a bridge to transfer the replication data to the remote system.

This replication data transfer occurs through the node-to-node connectivity. Note that this configuration, even though supported, allows the replication traffic to go through the node-to-node connectivity and this is not recommended.

- ▶ Partnered systems should use the same number of nodes in each system for replication.
- ▶ For maximum throughput, all nodes in each system should be used for replication, both in terms of balancing the preferred node assignment for volumes and for providing intersystem Fibre Channel connectivity.
- ▶ Where possible, use the minimum number of partnerships between systems. For example, assume site A contains systems A1 and A2, and site B contains systems B1 and B2. In this scenario, creating separate partnerships between pairs of systems (such as A1-B1 and A2-B2) offers greater performance for Global Mirror replication between sites than a configuration with partnerships defined between all four systems.

For zoning, the following rules apply for the remote system partnership:

- ▶ For Remote Copy configurations where the round-trip latency between systems is less than 80 milliseconds, zone two Fibre Channel ports on each node in the local system to two Fibre Channel ports on each node in the remote system.
- ▶ For Remote Copy configurations where the round-trip latency between systems is more than 80 milliseconds, apply SAN zoning to provide separate intrasystem zones for each local-remote I/O group pair that is used for replication, as shown in Figure 6-22.

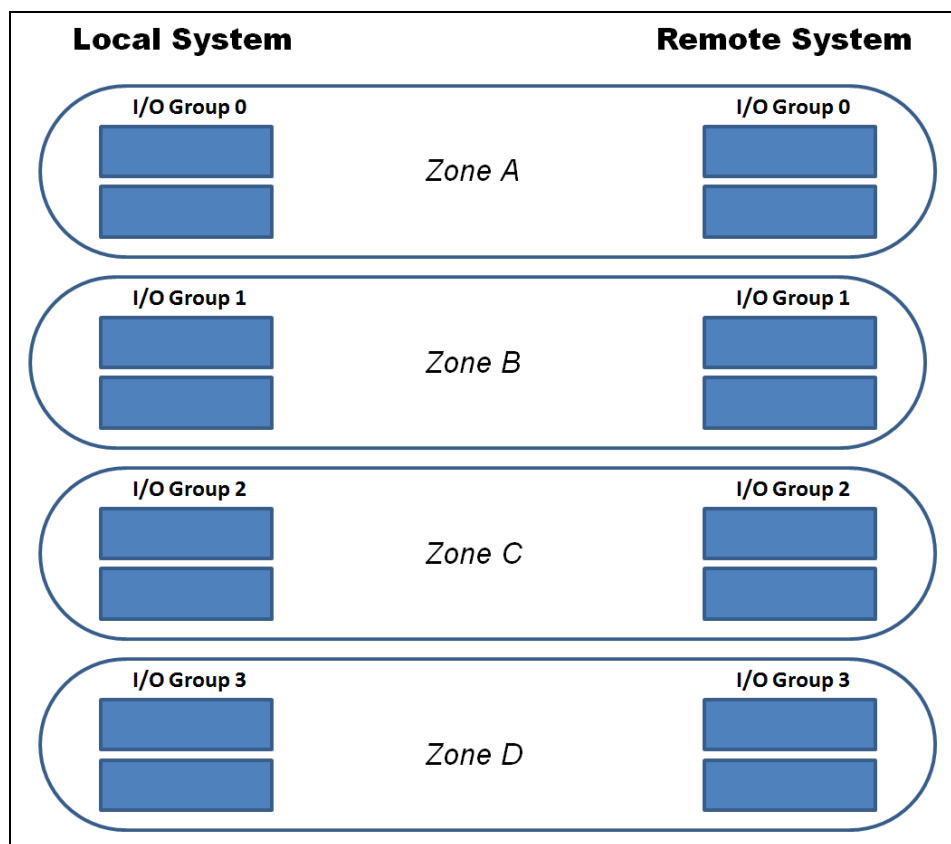


Figure 6-22 Zoning scheme for >80 ms Remote Copy partnerships

**NPIV:** IBM SAN Volume Controller with the NPIV feature enabled provides virtual WWPN for the host zoning. Those WWPNs are intended for host zoning only and cannot be used for the Remote Copy partnership.

### ***SAN Extension design considerations***

DR solutions that are based on Remote Copy technologies require reliable SAN extensions over geographical links. To avoid single points of failure, multiple physical links often are implemented. When implementing these solutions, specific attention must be paid in the Remote Copy network connectivity set up.



Consider a typical implementation of a Remote Copy connectivity that uses ISLs that is shown in Figure 6-23.

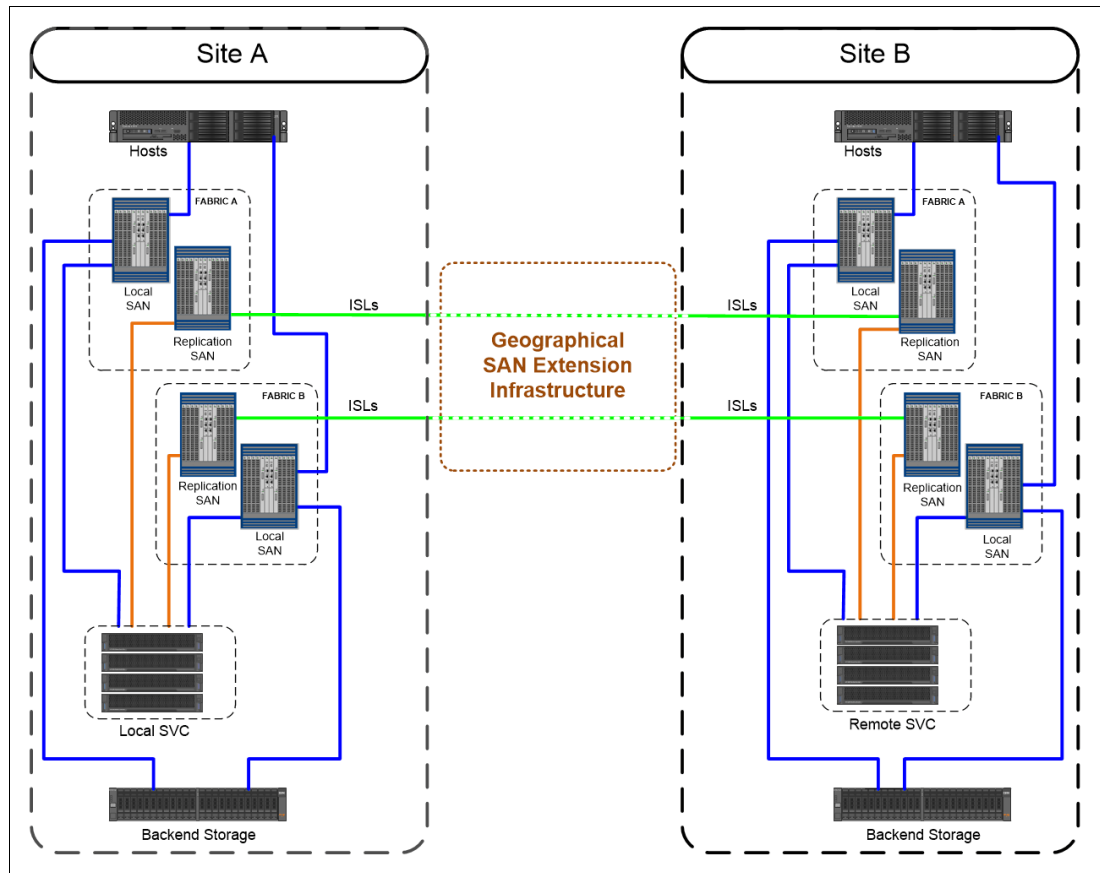


Figure 6-23 Typical Remote Copy network configuration

In this configuration, the Remote Copy network is isolated in a Replication SAN that interconnects Site A and Site B through a SAN extension infrastructure that uses two physical links. For redundancy reasons, assume that two ISLs are used for each fabric for the Replication SAN extension.

Two possible configurations are available to interconnect the Replication SANs. In Configuration 1 (see Figure 6-24), one ISL per fabric is attached to each physical link through xWDM or FCIP routers. In this example, the physical paths Path A and Path B are used to extend both fabrics.

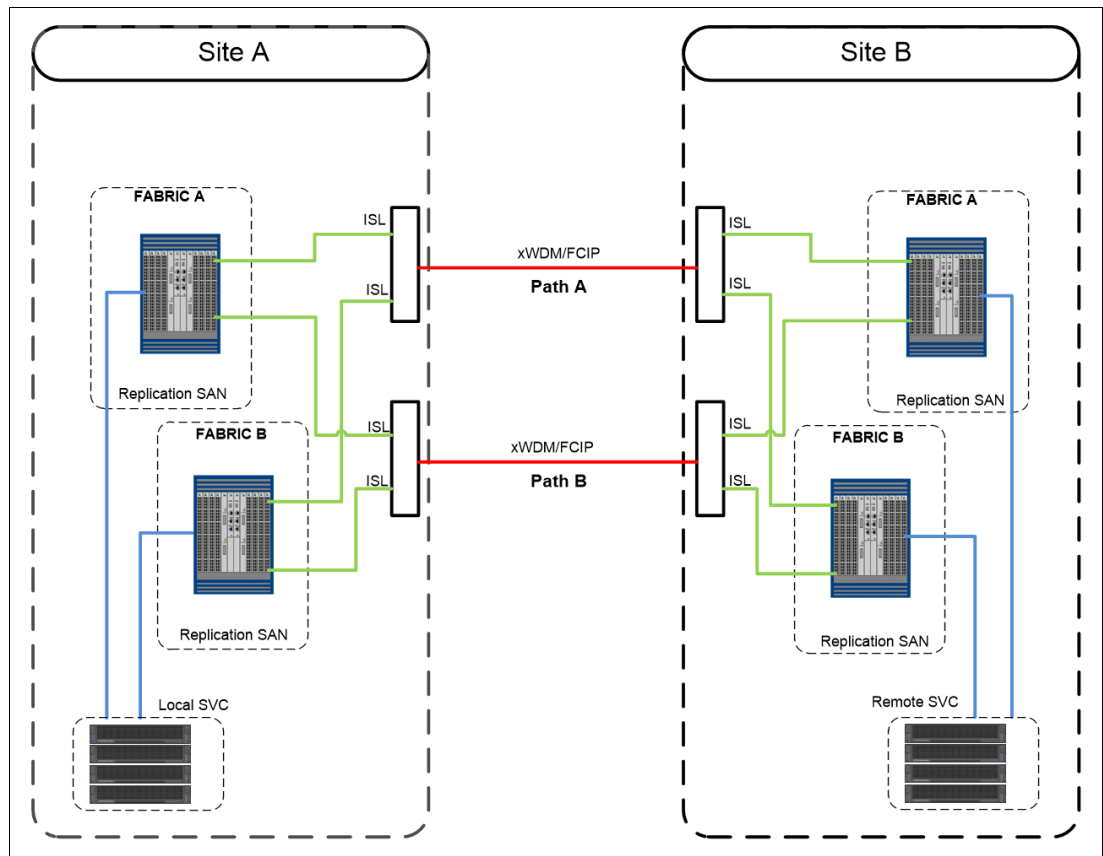


Figure 6-24 Configuration 1: physical paths shared among the fabrics



## 6.3.4 Remote copy services planning

When you plan for Remote Copy services, you must keep in mind the considerations that are outlined in the following sections.

### Remote copy configurations limits

To plan for and implement Remote Copy services, you must check the configuration limits and adhere to them. Table 6-8 lists the limits for a system that apply to IBM SAN Volume Controller version 8.4 for the supported system. Check the online documentation because these limits can change over time.

Table 6-8 Remote copy maximum limits

| Remote copy property                                                       | Maximum | Apply to                                 | Comment                                                                                                               |
|----------------------------------------------------------------------------|---------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Remote copy (Metro Mirror and Global Mirror) relationships per system      | 10000   | All models                               | This configuration can be any mix of Metro Mirror and Global Mirror relationships.                                    |
| Active-Active Relationships                                                | 2000    | All models                               | This limit is for the number of HyperSwap volumes in a system.                                                        |
| Remote copy relationships per consistency group                            | None    | All models                               | No limit is imposed beyond the Remote copy relationships per system limit. Applies to Global Mirror and Metro Mirror. |
| GMCV relationships per consistency group                                   | 200     | All models                               |                                                                                                                       |
| Remote copy consistency groups per system                                  | 256     | All models                               |                                                                                                                       |
| Total Metro Mirror and Global Mirror volume capacity per I/O group         | 1024 TB | All models                               | This limit is the total capacity for all master and auxiliary volumes in the I/O group.                               |
| Total number of Global Mirror with Change Volumes relationships per system | 256     | SAN Volume Controller models DH8         | 60 s cycle time                                                                                                       |
|                                                                            | 1500    | SAN Volume Controller models DH8         | 300 s cycle time                                                                                                      |
|                                                                            | 256     | SAN Volume Controller models SV1 and SV2 | 60 s cycle time                                                                                                       |
|                                                                            | 2500    | SAN Volume Controller models SV1 and SV2 | 300 s cycle time                                                                                                      |

Similar to FlashCopy, the Remote Copy services require memory to allocate the bitmap structures that are used to track the updates while volume are suspended or synchronizing. The default amount of memory for Remote Copy services is 20 MB. This value can be increased or decreased by using the **chlogrp** command or the GUI. The maximum amount of memory that can be specified for Remote Copy services is 512 MB. The grain size for the Remote Copy services is 256 KB.

## Remote copy general restrictions

To use Metro Mirror and Global Mirror, you must adhere to the following rules:

- ▶ You must have the same size for source and target volume when defining a Remote Copy relationship. However, the target volume can be a different type (image, striped, or sequential mode) or have different cache settings (cache-enabled or cache-disabled).
- ▶ You cannot move Remote Copy source or target volumes to different I/O groups.
- ▶ Remote copy volumes can be resized with the following restrictions:
  - Apply to Metro Mirror and Global Mirror only; GMCV is not supported.
  - The Remote copy Consistency Protection feature is not allowed and must be removed before resizing the volumes.
  - No active FlashCopy is allowed.
  - The Remote Copy relationship must be in synchronized status.
  - The resize order must ensure that the target volume always is larger than the source volume.

**Note:** The volume expansion for Metro Mirror and Global Mirror volumes was introduced with Spectrum Virtualize version 7.8.1 with some restrictions. In the first implementation (up to version 8.2.1), only thin provisioned or compressed volumes were supported. With version 8.2.1, non-mirrored fully allocated volumes were supported. With version 8.4, all restrictions on the volume type were removed.

- ▶ You can mirror intrasystem Metro Mirror or Global Mirror only between volumes in the same I/O group.

**Intrasystem remote copy:** The intrasystem Global Mirror is not supported on IBM Spectrum Virtualize based systems that are running version 6 or later.

- ▶ Global Mirror is not recommended for cache-disabled volumes that are participating in a Global Mirror relationship.

## Changing the Remote Copy type

Changing the Remote Copy type for a relationship is an easy task. It is enough to stop the relationship (if it is active) and change the properties to set the new Remote Copy type. Do not forget to create the change volumes if a change from Metro Mirror or Global Mirror to Global Mirror Change Volumes occurs.

## Interaction between Remote Copy and FlashCopy

Remote Copy functions can be used with the FlashCopy function so that you can have both operating concurrently on the same volume. The following combinations between Remote Copy and FlashCopy are possible:

- ▶ Remote copy source:
  - Can be a FlashCopy source.
  - Can be a FlashCopy target with the following restrictions:
    - A FlashCopy target volume cannot be updated while it is the source volume of a Metro Mirror or Global Mirror relationship that is actively mirroring. A FlashCopy mapping cannot be started while the target volume is in an active Remote Copy relationship.

- The I/O group for the FlashCopy mappings must be the same as the I/O group for the FlashCopy target volume (that is, the I/O group of the Remote copy source).
- Remote copy target:
- A Remote Copy target can be a FlashCopy source.
  - A Remote Copy target can be a FlashCopy target with the restriction that a FlashCopy mapping must be in the `idle_copied` state when its target volume is the target volume of an active Metro Mirror or Global Mirror relationship.

When implementing FlashCopy functions for volumes in GMCV relationships, remember that FlashCopy multi-target mappings are created. As described in “Interaction and dependency between Multiple Target FlashCopy mappings” on page 235, this configuration results in dependent mappings that can affect the cycle formation because of the cleaning process (see “**Cleaning process and Cleaning Rate**” on page 244). With such configurations, it is advised to set the Cleaning Rate accordingly. This consideration also applies to Consistency Protection volumes and HyperSwap configurations.

## Native backend controller copy functions considerations

The IBM Spectrum Virtualize technology provides a widespread set of copy services functions that cover most of the clients requirements.

However, some storage controllers can provide specific copy services capabilities that are not available with the current version of IBM Spectrum Virtualize software. The IBM SAN Volume Controller technology addresses these situations by using cache-disabled image mode volumes that virtualize LUNs that are participating with the native backend controller’s copy services relationships.

Keeping the cache disabled ensures data consistency throughout the I/O stack, from the host to the backend controller. Otherwise, by leaving the cache enabled on a volume, the underlying controller does not receive any write I/Os as the host writes them. Instead, IBM SAN Volume Controller caches them and processes them later. This process can have more ramifications if a target host depends on the write I/Os from the source host as they are written.

**Note:** Native copy services are not supported on all storage controllers. For more information about the known limitations, see [This IBM Support web page](#).

As part of its copy services function, the storage controller might take a LUN offline or suspend reads or writes. Because IBM SAN Volume Controller does not recognize why this event occurs, it might log errors. For this reason, if the IBM SAN Volume Controller must detect the LUN, ensure that you keep that LUN in the unmanaged state until full access is granted.

Native backend controller copy services also can be used for LUNs that are not managed by the IBM SAN Volume Controller. Accidental incorrect configurations of the backend controller copy services that involve IBM SAN Volume Controller-attached LUNs can produce unpredictable results.

For example, if you accidentally use a LUN with IBM SAN Volume Controller data on it as a point-in-time target LUN, you can corrupt that data. Moreover, if that LUN was a managed disk in a managed disk group with striped or sequential volumes on it, the managed disk group might be brought offline. This situation, in turn, makes all of the volumes that belong to that group go offline, which leads to a widespread host access disruption.

## Remote Copy and code upgrade considerations

When you upgrade system software in which the system participates in one or more intersystem relationships, upgrade only one cluster at a time. That is, do not upgrade the systems concurrently.

**Attention:** Upgrading both systems concurrently is not monitored by the software upgrade process.

Allow the software upgrade to complete one system before it is started on the other system. Upgrading both systems concurrently can lead to a loss of synchronization. In stress situations, it can further lead to a loss of availability.

Usually, preexisting Remote Copy relationships are unaffected by a software upgrade that is performed correctly. However, always check in the target code release notes for special considerations on the copy services.

Even if it is not a best practice, a Remote Copy partnership can be established, with some restriction, among systems with different IBM Spectrum Virtualize versions. For more information about a compatibility table for intersystem Metro Mirror and Global Mirror relationships between IBM Spectrum Virtualize code levels, see this [IBM Support web page](#).

## Volume placement considerations

You can optimize the distribution of volumes within I/O groups at the local and remote systems to maximize performance.

Although defined at a system level, the partnership bandwidth, and consequently the background copy rate, is evenly divided among the cluster's I/O groups. The available bandwidth for the background copy can be used by either node, or shared by both nodes within the I/O Group.

This bandwidth allocation is independent from the number of volumes for which a node is responsible. Each node, in turn, divides its bandwidth evenly between the (multiple) Remote Copy relationships with which it associates volumes that are performing a background copy.

### ***Volume preferred node***

Conceptually, a connection (path) goes between each node on the primary system to each node on the remote system. Write I/O, which is associated with remote copying, travels along this path. Each node-to-node connection is assigned a finite amount of Remote Copy resource and can sustain only in-flight write I/O to this limit.

The node-to-node in-flight write limit is determined by the number of nodes in the remote system. The more nodes that exist at the remote system, the lower the limit is for the in-flight write I/Os from a local node to a remote node. That is, less data can be outstanding from any one local node to any other remote node. Therefore, to optimize performance, Global Mirror volumes must have their preferred nodes distributed evenly between the nodes of the systems.

The preferred node property of a volume helps to balance the I/O load between nodes in that I/O group. This property is also used by Remote Copy to route I/O between systems.

The IBM SAN Volume Controller node that receives a write for a volume is normally the preferred node of the volume. For volumes in a Remote Copy relationship, that node is also responsible for sending that write to the preferred node of the target volume. The primary preferred node is also responsible for sending any writes that relate to the background copy. Again, these writes are sent to the preferred node of the target volume.

Each node of the remote system has a fixed pool of Remote Copy system resources for *each* node of the primary system. That is, each remote node has a separate queue for I/O from each of the primary nodes. This queue is a fixed size and is the same size for every node. If preferred nodes for the volumes of the remote system are set so that every combination of primary node and secondary node is used, Remote Copy performance is maximized.

Figure 6-26 shows an example of Remote Copy resources that are not optimized. Volumes from the local system are replicated to the remote system. All volumes with a preferred node of node 1 are replicated to the remote system, where the target volumes also have a preferred node of node 1.

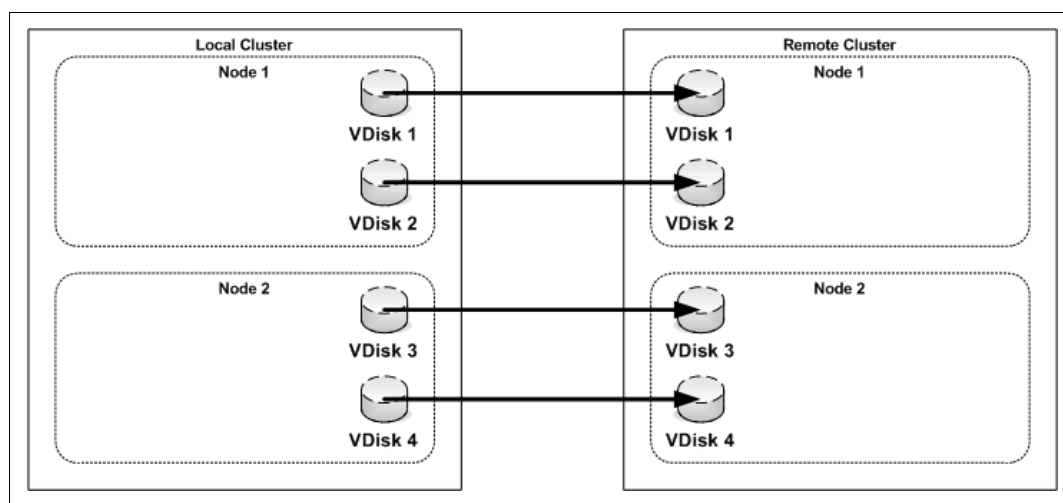


Figure 6-26 Remote copy resources that are not optimized

With this configuration, the resources for remote system node 1 that are reserved for local system node 2 are not used. The resources for local system node 1 that are used for remote system node 2 also are not used.



If the configuration changes to the configuration that is shown in Figure 6-27, all Remote Copy resources for each node are used, and Remote Copy operates with better performance.

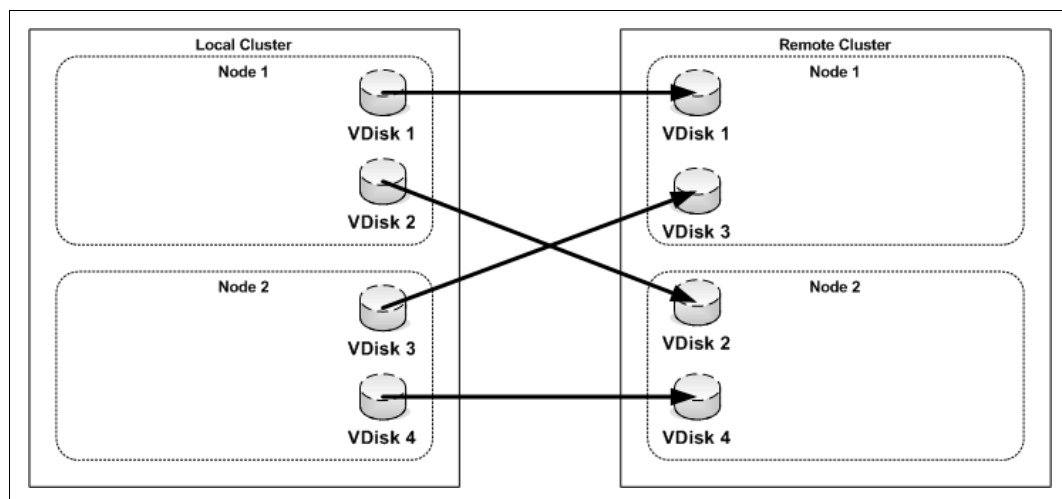


Figure 6-27 Optimized Global Mirror resources

### **GMCV change volumes placement considerations**

The change volumes in a GMCV configuration are essentially thin provisioned volumes that are used as FlashCopy targets. For this reason, the same considerations apply that are described in “Volume placement considerations” on page 242.

The change volumes can be compressed to reduce the amount of space used; however, the change volumes might be subject to heavy write workload in the primary and secondary system.

Therefore, the placement on the back-end is critical to provide adequate performance. Consider the use of DRPs for the change volumes only if doing so is beneficial in terms of space savings.

**Trick:** The internal FlashCopy that is used by the GMCV is 256 KB grain. However, it is possible to force a 64 KB grain by creating a FlashCopy with 64 KB grain from the GMCV volume and a dummy target volume before to assigning the change volume to the relationship. This process can be done to the source and target volumes. After the CV assignment is done, the dummy FlashCopy can be deleted.

### **Background copy considerations**

The Remote Copy partnership bandwidth parameter *explicitly* defines the rate at which the background copy is attempted, but also *implicitly* affects foreground I/O. Background copy bandwidth can affect foreground I/O latency in one of the following ways:

- Increasing latency of foreground I/O

If the Remote Copy partnership bandwidth parameter is set too high for the actual intersystem network capability, the background copy resynchronization writes use too much of the intersystem network. It starves the link of the ability to service synchronous or asynchronous mirrored foreground writes. Delays in processing the mirrored foreground writes increase the latency of the foreground I/O as perceived by the applications.

- Read I/O overload of primary storage

If the Remote Copy partnership background copy rate is set too high, the added read I/Os that are associated with background copy writes can overload the storage at the primary site and delay foreground (read and write) I/Os.

- Write I/O overload of auxiliary storage

If the Remote Copy partnership background copy rate is set too high for the storage at the secondary site, the background copy writes overload the auxiliary storage. Again, they delay the synchronous and asynchronous mirrored foreground write I/Os.

**Important:** An increase in the peak foreground workload can have a detrimental effect on foreground I/O. It does so by pushing more mirrored foreground write traffic along the intersystem network, which might not have the bandwidth to sustain it. It can also overload the primary storage.

To set the background copy bandwidth optimally, consider all aspects of your environments, starting with the following most significant contributing resources:

- Primary storage
- Intersystem network bandwidth
- Auxiliary storage

Provision the most restrictive of these three resources between the background copy bandwidth and the peak foreground I/O workload. Perform this provisioning by calculation or by determining experimentally how much background copy can be allowed before the foreground I/O latency becomes unacceptable.

Then, reduce the background copy to accommodate peaks in workload. In cases where the available network bandwidth cannot sustain an acceptable background copy rate, consider alternatives to the initial copy, as described in “Initial synchronization options and Offline Synchronization” on page 283.

Changes in the environment, or loading of it, can affect the foreground I/O. IBM SAN Volume Controller technology provides a means to monitor, and a parameter to control, how foreground I/O is affected by running Remote Copy processes. IBM Spectrum Virtualize software monitors the delivery of the mirrored foreground writes. If latency or performance of these writes extends beyond a (predefined or client-defined) limit for a period, the Remote Copy relationship is suspended (see 6.3.6, “1920 error” on page 291).

Finally, with Global Mirror Change Volume, the cycling process that transfers the data from the local to the remote system is a background copy task (see also “Global Mirror and GMCV coexistence considerations” on page 267). For this reason, the background copy rate, and the `relationship_bandwidth_limit` setting affects the available bandwidth not only during the initial synchronization, but also during the normal cycling process.

**Background copy bandwidth allocation:** As described in “Volume placement considerations” on page 279, the available bandwidth of a Remote Copy partnership is evenly divided among the cluster’s I/O Groups. In a case of unbalanced distribution of the remote copies among the I/O groups, the partnership bandwidth is adjusted to reach the wanted background copy rate.

For example, consider a 4-I/O groups cluster that features a partnership bandwidth of 4,000 Mbps and a background copy percentage of 50. The expected maximum background copy rate for this partnership is then 250 MBps. Having the available bandwidth evenly divided among the I/O groups, every I/O group in this cluster can theoretically synchronize data at a maximum rate of about 62 MBps (50% of 1,000 Mbps). Now, in an edge case where only volumes from one I/O group are replicated, the partnership bandwidth is adjusted to 16000 Mbps to reach the full background copy rate (250 MBps).

### ***Initial synchronization options and Offline Synchronization***

When creating a Remote Copy relationship, the following options regarding the initial synchronization process are available:

- ▶ The `not_synchronized` option is the default. With this option, when a Remote Copy relationship is started, a full data synchronization at the background copy rate occurs between the source and target volumes. It is the simplest approach in that it requires no other administrative activity, apart from issuing the necessary IBM SAN Volume Controller commands. However, in some environments, the available bandwidth makes this option unsuitable.
- ▶ The `already_synchronized` option does not force any data synchronization when the relationship is started. The administrator must ensure that the source and target volumes contain identical data before a relationship is created. The administrator can perform this check in one of the following ways:
  - Create both volumes with the security delete feature to change all data to zero.
  - Copy a complete tape image (or other method of moving data) from one disk to the other.

In either technique, no write I/O must occur to the source and target volume before the relationship is established. The administrator must then complete the following actions:

- Create the relationship with the `already_synchronized` settings (`-sync` option).
- Start the relationship.

**Attention:** If you do not perform these steps correctly, the Remote Copy reports the relationship as being *consistent*, when it is not. This setting is likely to make any auxiliary volume useless.

By understanding the methods to start a Metro Mirror and Global Mirror relationship, you can use one of them as a means to implement the Remote Copy relationship that saves bandwidth.

Consider a situation in which you have a large source volume (or many source volumes) that contain active data and that you want to replicate to a remote site. Your planning shows that the mirror initial sync time takes too long (or is too costly if you pay for the traffic that you use). In this case, you can set up the sync by using another medium that is less expensive. This synchronization method is called *Offline Synchronization*.

This example uses tape media as the source for the initial sync for the Metro Mirror relationship or the Global Mirror relationship target before it uses Remote Copy services to maintain the Metro Mirror or Global Mirror. This example does not require downtime for the hosts that use the source volumes.

Before you set up Global Mirror relationships and save bandwidth, complete the following steps:

1. Ensure that the hosts are up and running and are using their volumes normally. No Metro Mirror relationship nor Global Mirror relationship is defined yet.

Identify all of the volumes that become the source volumes in a Metro Mirror relationship or in a Global Mirror relationship.

2. Establish the Remote Copy partnership with the target IBM Spectrum Virtualize based system.

To set up Global Mirror relationships and save bandwidth, complete the following steps:

1. Define a Metro Mirror relationship or a Global Mirror relationship for each source disk. When you define the relationship, ensure that you use the **-sync** option, which stops the system from performing an initial sync.

**Attention:** If you do not use the **-sync** option, all of these steps are redundant because the IBM Spectrum Virtualize system performs a full initial synchronization anyway.

2. Stop each mirror relationship by using the **-access** option, which enables write access to the target volumes. You need this write access later.
3. Copy the source volume to the alternative media by using the **dd** command to copy the contents of the volume to tape. Another option is to use your backup tool (for example, IBM Spectrum Protect) to make an image backup of the volume.

**Change tracking:** Although the source is being modified while you are copying the image, the IBM SAN Volume Controller is tracking those changes. The image that you create might have some of the changes and is likely to also miss some of the changes.

When the relationship is restarted, the IBM SAN Volume Controller applies all of the changes that occurred since the relationship stopped in step 2. After all the changes are applied, you have a consistent target image.

4. Ship your media to the remote site and apply the contents to the targets of the Metro Mirror or Global Mirror relationship. You can mount the Metro Mirror and Global Mirror target volumes to a UNIX server and use the **dd** command to copy the contents of the tape to the target volume.

If you used your backup tool to make an image of the volume, follow the instructions for your tool to restore the image to the target volume. Remember to remove the mount if the host is temporary.

**Tip:** It does not matter how long it takes to get your media to the remote site and perform this step. However, the faster you can get the media to the remote site and load it, the quicker IBM Spectrum Virtualize system starts running and maintaining the Metro Mirror and Global Mirror.

5. Unmount the target volumes from your host. When you start the Metro Mirror and Global Mirror relationship later, the IBM SAN Volume Controller stops write access to the volume while the mirror relationship is running.
6. Start your Metro Mirror and Global Mirror relationships. The relationships must be started with the **-clean** parameter. In this way, any changes that are made on the secondary volume are ignored, and only changes made on the clean primary volume are considered when synchronizing the primary and secondary volumes.
7. While the mirror relationship catches up, the target volume is not usable at all. When it reaches `ConsistentSynchronized` status, your remote volume is ready for use in a disaster.

## Back-end storage considerations

To reduce the overall solution costs, it is a common practice to provide the remote systems with lower performance characteristics compared to the local system, especially when using asynchronous Remote Copy technologies. This attitude can be risky especially when using the Global Mirror technology where the application performances at the primary system can indeed be limited by the performance of the remote system.

The preferred practice is to perform an accurate back-end resource sizing for the remote system to fulfill the following capabilities:

- ▶ The peak application workload to the Global Mirror or Metro Mirror volumes
- ▶ The defined level of background copy
- ▶ Any other I/O that is performed at the remote site

## Remote Copy tunable parameters

Several commands and parameters help to control Remote Copy and its default settings. You can display the properties and features of the systems by using the **lssystem** command. Also, you can change the features of systems by using the **chssystem** command.

### *relationshipbandwidthlimit*

The **relationshipbandwidthlimit** is an optional parameter that specifies the new background copy bandwidth in the range 1 - 1000 MBps. The default is 25 MBps. This parameter operates system-wide, and defines the maximum background copy bandwidth that any relationship can adopt. The existing background copy bandwidth settings that are defined on a partnership continue to operate, with the lower of the partnership and volume rates attempted.

**Important:** Do not set this value higher than the default without establishing that the higher bandwidth can be sustained.

The **relationshipbandwidthlimit** also applies to Metro Mirror relationships.

### *gmlinktolerance and gmmxhostdelay*

The **gmlinktolerance** and **gmmxhostdelay** parameters are critical in the system for deciding internally whether to terminate a relationship due to a performance problem. In most cases, these two parameters need to be considered in tandem. The defaults are not normally changed unless a specific reason existed to do so.

The **gmlinktolerance** parameter can be thought of as how long you allow the host delay to go on being significant before you decide to terminate a Global Mirror volume relationship. This parameter accepts values of 20 - 86,400 seconds in increments of 10 seconds. The default is 300 seconds. You can disable the link tolerance by entering a value of zero for this parameter.

The **gmmaxhostdelay** parameter can be thought of as the maximum host I/O impact that is due to Global Mirror. That is, how long does that local I/O take with Global Mirror turned off, and how long does it take with Global Mirror turned on. The difference is the host delay due to Global Mirror tag and forward processing.

Although the default settings are adequate for most situations, increasing one parameter while reducing another might deliver a tuned performance environment for a particular circumstance.

Example 6-1 shows how to change **gmlinktolerance** and the **gmmaxhostdelay** parameters using the **chsystem** command.

*Example 6-1 Changing gmlinktolerance to 30 and gmmaxhostdelay to 100*

---

```
chsystem -gmlinktolerance 30
chsystem -gmmaxhostdelay 100
```

---

**Test and monitor:** Thoroughly test and carefully monitor the host impact of any changes such as these before putting them into a live production environment.

For more information and settings considerations about the **gmlinktolerance** and **gmmaxhostdelay** parameters, see 6.3.6, “1920 error” on page 291.

### ***rcbuffersize***

The **rcbuffersize** parameter was introduced with the version 6.2 code level so that systems with intense and bursty write I/O do not fill the internal buffer while Global Mirror writes were undergoing sequence tagging.

**Important:** Do not change the **rcbuffersize** parameter except under the direction of IBM Support.

Example 6-2 shows how to change **rcbuffersize** to 64 MB by using the **chsystem** command. The default value for **rcbuffersize** is 48 MB and the maximum is 512 MB.

*Example 6-2 Changing rcbuffersize to 64 MB*

---

```
chsystem -rcbuffersize 64
```

---

Remember that any additional buffers you allocate are taken away from the general cache.

### ***maxreplicationdelay and partnershipexclusionthreshold***

The **maxreplicationdelay** system-wide parameter defines a maximum latency (in seconds) for any individual write that passes through the Global Mirror logic. If a write is hung for that time (for example, because of a rebuilding array on the secondary system), Global Mirror stops the relationship (and any containing consistency group), triggering a 1920 error.

The **partnershipexclusionthreshold** parameter was introduced to allow users to set the timeout for an I/O that triggers a temporarily dropping of the link to the remote cluster. The value must be a number from 30 - 315.

**Important:** Do not change the **partnershipexclusionthreshold** parameter except under the direction of IBM Support.

For more information and settings considerations about the `maxreplicationdelay` parameter, see 6.3.6, “1920 error” on page 291.

### ***Link delay simulation parameters***

Even though Global Mirror is an asynchronous replication method, there can be an impact to server applications due to Global Mirror managing transactions and maintaining write order consistency over a network. To mitigate this impact, as a testing and planning feature, Global Mirror allows you to simulate the effect of the round-trip delay between sites by using the following parameters:

- The `gminterclusterdelaysimulation` parameter

This optional parameter specifies the intersystem delay simulation, which simulates the Global Mirror round-trip delay between two systems in milliseconds. The default is 0. The valid range is 0 - 100 milliseconds.

- The `gmintraclusterdelaysimulation` parameter

This optional parameter specifies the intrasystem delay simulation, which simulates the Global Mirror round-trip delay in milliseconds. The default is 0. The valid range is 0 - 100 milliseconds.

## **6.3.5 Multiple site remote copy**

The most common use cases for the Remote Copy functions are DR solutions. Code level 8.3.1 introduced more DR capabilities, such as the Spectrum Virtualize 3-site replication that provides a solution for co-ordinated DR across three sites in various topologies. A complete discussion about the DR solutions that are based on IBM Spectrum Virtualize technology is beyond the intended scope of this book.

For an overview of the DR solutions with the IBM Spectrum Virtualize copy services see, *IBM System Storage SAN Volume Controller and Storwize V7000 Replication Family Services*, SG24-7574. For more information about 3-site replication, see *IBM Spectrum Virtualize 3-Site Replication*, SG24-8474.

Another typical Remote Copy use case is the data movement among distant locations as required, for example, for data center relocation and consolidation projects. In these scenarios, the IBM Spectrum Virtualize Remote Copy technology is especially effective when combined with the image copy feature that allows data movement among storage systems of different technology or vendor.

Mirroring scenarios involving multiple sites can be implemented by using a combination of Spectrum Virtualize capabilities, as described next.

## IBM SAN Volume Controller Enhanced Stretched Cluster three site mirroring

With SAN Volume Controller Enhanced Stretched Cluster, Remote Copy services can be combined with Volume Mirroring to implement three site solutions, as shown in Figure 6-28.

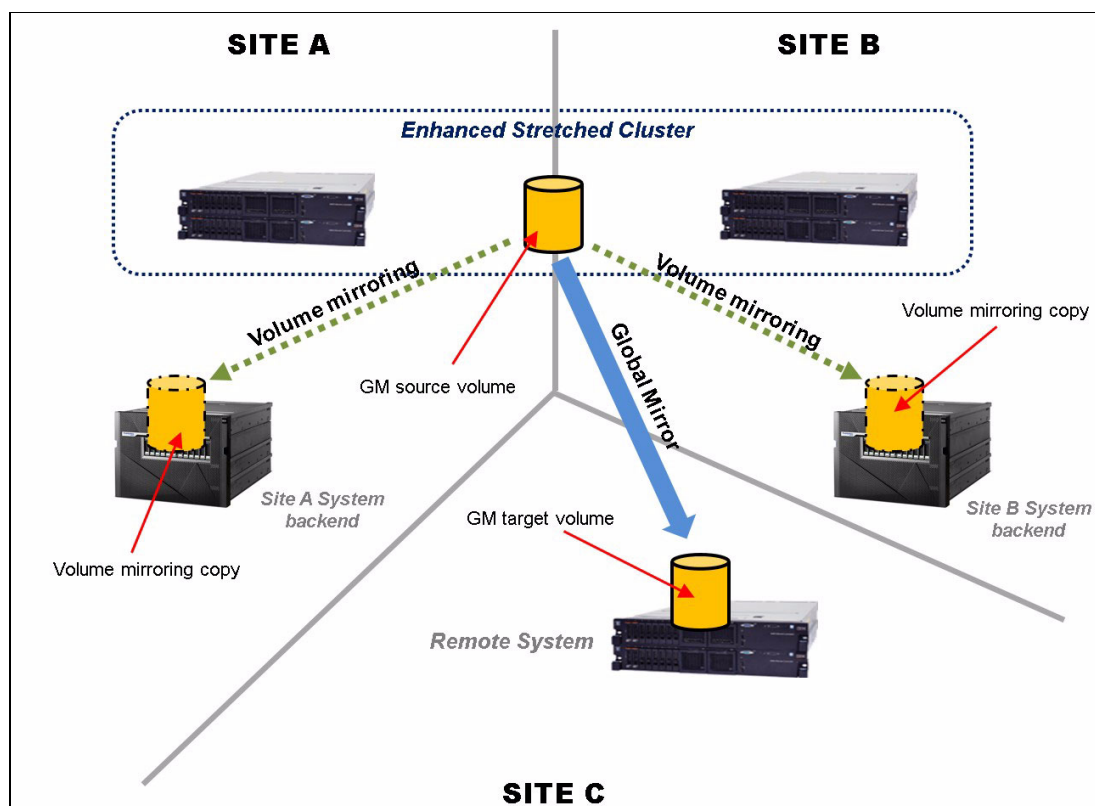


Figure 6-28 Three-site configuration with Enhanced Stretched Cluster

Three-site configurations also can be implemented by using special cascading configurations, as described next.

### Performing cascading copy service functions

Cascading copy service functions that use IBM SAN Volume Controller are not directly supported. However, you might require a three-way (or more) replication by using copy service functions (synchronous or asynchronous mirroring). You can address this requirement by using IBM SAN Volume Controller copy services and by combining IBM SAN Volume Controller copy services (with image mode cache-disabled volumes) and native storage controller copy services.

**DRP limitation:** As of this writing, the image mode VDisk is not supported with DRPs.



### ***Cascading with native storage controller copy services***

Figure 6-29 shows the configuration for three-site cascading by using the native storage controller copy services with IBM SAN Volume Controller Remote Copy functions.

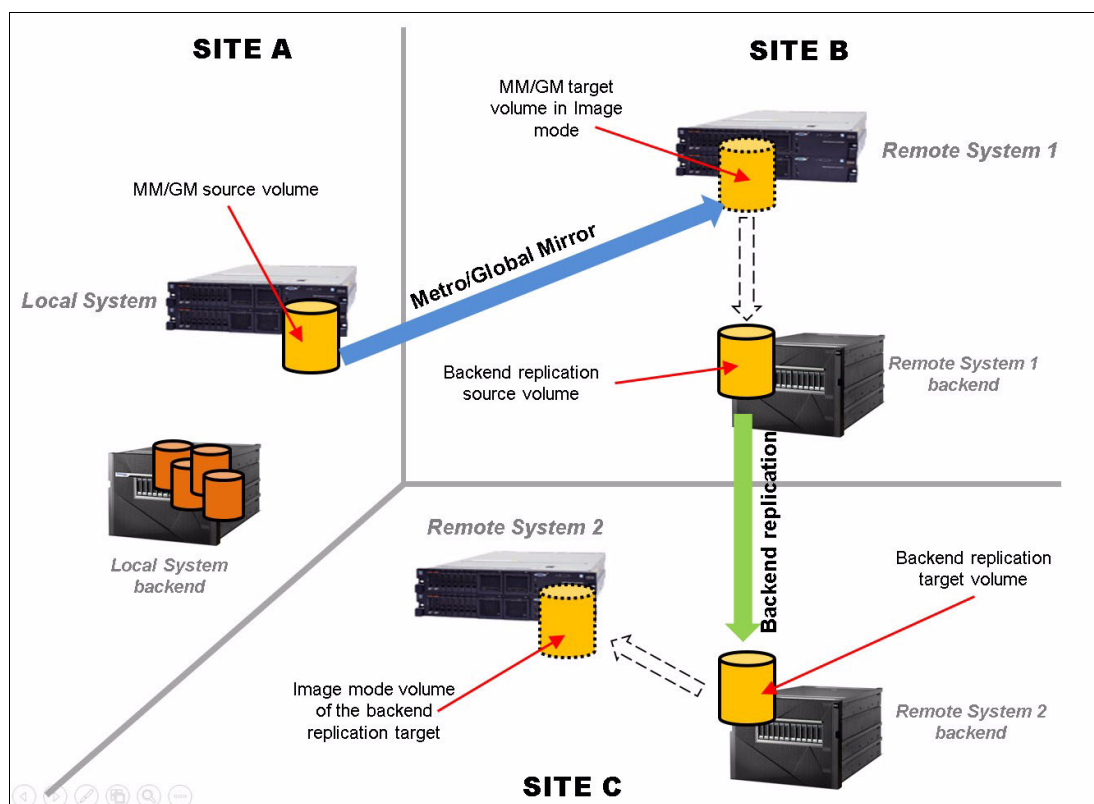


Figure 6-29 Using three-way copy services

As shown in Figure 6-29, the primary site uses IBM SAN Volume Controller Remote Copy functions (Global Mirror or Metro Mirror) at the secondary site. Therefore, if a disaster occurs at the primary site, the storage administrator enables access to the target volume (from the secondary site) and the business application continues processing.

While the business continues processing at the secondary site, the storage controller copy services replicate to the third site. This configuration is allowed under the following conditions:

- ▶ The backend controller native copy services must be supported by SAN Volume Controller (see “Native backend controller copy functions considerations” on page 278).
- ▶ The source and target volumes that are used by the back-end controller native copy services must be imported to the IBM SAN Volume Controller as image-mode volumes with the cache disabled.

### ***Cascading with IBM SAN Volume Controller systems copy services***

Remote copy services cascading is allowed with the Spectrum Virtualize 3-site replication capability (see *IBM Spectrum Virtualize 3-Site Replication*, SG24-8474). However, a cascading-like solution is also possible by combining the IBM SAN Volume Controller copy services. These Remote Copy services implementations are useful in three (or more) site DR solutions and data center moving scenarios.

In the configuration that is shown in Figure 6-30, a Global Mirror (Metro Mirror can also be used) solution is implemented between the Local System in Site A (the production site) and the Remote System 1 that is in Site B (the primary disaster recover site). A third system, Remote System 2, is in Site C, which is the secondary disaster recover site. Connectivity is provided between Site A and Site B, between Site B and Site C, and optionally between Site A and Site C.

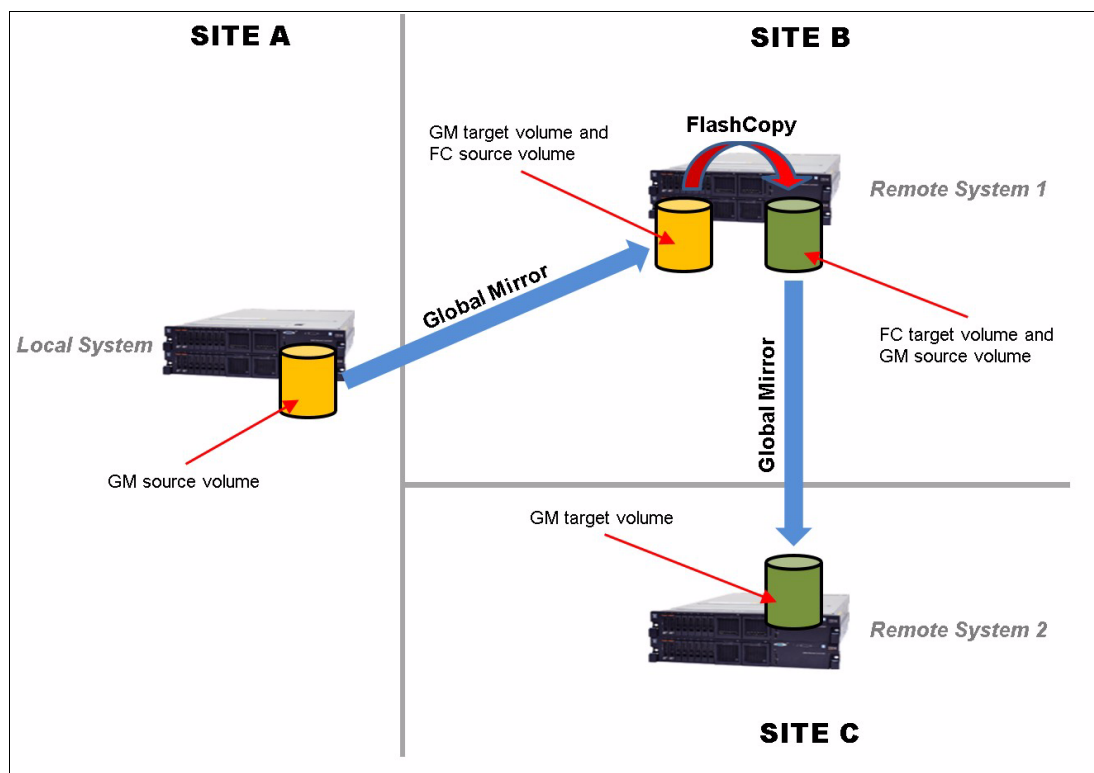


Figure 6-30 Cascading-like infrastructure

To implement a cascading-like solution, complete the following steps:

1. Set up phase. Complete the following steps to initially set up the environment:
  - a. Create the Global Mirror relationships between the Local System and Remote System 1.
  - b. Create the FlashCopy mappings in the Remote System 1 by using the target Global Mirror volumes as FlashCopy source volumes. The FlashCopy must be incremental.
  - c. Create the Global Mirror relationships between Remote System 1 and Remote System 2 by using the FlashCopy target volumes as Global Mirror source volumes.
  - d. Start the Global Mirror from Local System to Remote System 1.

After the Global Mirror is in ConsistentSynchronized state, you are ready to create the cascading.
2. Consistency point creation phase. Complete the following steps whenever a consistency point creation in Site C is required:
  - a. Check whether the Global Mirror between Remote System 1 and Remote System 2 is in stopped or idle status; if it is not, stop the Global Mirror.
  - b. Stop the Global Mirror between the Local System to Remote System 1.
  - c. Start the FlashCopy in Remote Site 1.

- d. Resume the Global Mirror between the Local System and Remote System 1.
- e. Start or resume the Global Mirror between Remote System 1 and Remote System 2.

The first time that these operations are performed, a full copy between Remote System 1 and Remote System 2 occurs. Later executions of these operations perform incremental resynchronizations instead. After the Global Mirror between Remote System 1 and Remote System 2 is in Consistent Synchronized state, the consistency point in Site C is created. The Global Mirror between Remote System 1 and Remote System 2 can now be stopped to be ready for the next consistency point creation.

### 6.3.6 1920 error

An IBM SAN Volume Controller generates a 1920 error message whenever a Metro Mirror or Global Mirror relationship stops because of adverse conditions. The adverse conditions, if left unresolved, might affect performance of foreground I/O.

A 1920 error can result for many reasons. The condition might be the result of a temporary failure, such as maintenance on the intersystem connectivity, unexpectedly higher foreground host I/O workload, or a permanent error because of a hardware failure. It is also possible that not all relationships are affected and that multiple 1920 errors can be posted.

The 1920 error might be triggered for Metro Mirror and Global Mirror relationships. However, in Metro Mirror configurations the 1920 error is associated only with a permanent I/O error condition. For this reason, the main focus of this section is 1920 errors in a Global Mirror configuration.

#### Internal Global Mirror control policy and raising 1920 errors

Although Global Mirror is an asynchronous Remote Copy service, the local and remote sites have some interplay. When data comes into a local volume, work must be done to ensure that the remote copies are consistent. This work can add a delay to the local write. Normally, this delay is low. The IBM SAN Volume Controller code implements many control mechanisms that mitigate the impacts of the Global Mirror to the foreground I/Os.

#### ***gmmaxhostdelay and gmlinktolerance***

The **gmlinktolerance** parameter helps to ensure that hosts do not perceive the latency of the long-distance link, regardless of the bandwidth of the hardware that maintains the link or the storage at the secondary site. In terms of nodes and backend characteristics, the system configuration must be provisioned so that when combined, they can support the maximum throughput that is delivered by the applications at the primary that uses Global Mirror.

If the capabilities of the system configuration are exceeded, the system becomes backlogged and the hosts receive higher latencies on their write I/O. Remote copy in Global Mirror implements a protection mechanism to detect this condition and halts mirrored foreground write and background copy I/O. Suspension of this type of I/O traffic ensures that misconfiguration or hardware problems (or both) do not affect host application availability.

Global Mirror attempts to detect and differentiate between backlogs that occur because of the operation of the Global Mirror protocol. It does not examine the general delays in the system when it is heavily loaded, where a host might see high latency even if Global Mirror were disabled.

Global Mirror uses the **gmmaxhostdelay** and **gmlinktolerance** parameters to monitor Global Mirror protocol backlogs in the following ways:

- ▶ Users set the **gmmaxhostdelay** and **gmlinktolerance** parameters to control how software responds to these delays. The **gmmaxhostdelay** parameter is a value in milliseconds that can go up to 100.
- ▶ Every 10 seconds, Global Mirror samples all of the Global Mirror writes and determines how much of a delay it added. If the delay added by at least a third of these writes is greater than the **gmmaxhostdelay** setting, that sample period is marked as *bad*.
- ▶ Software keeps a running count of bad periods. Whenever a bad period occurs, this count goes up by one. Whenever a good period occurs, this count goes down by 1, to a minimum value of 0. For example, 10 bad periods, followed by five good periods, followed by 10 bad periods, results in a bad period count of 15.
- ▶ The **gmlinktolerance** parameter is defined in seconds. Because bad periods are assessed at intervals of 10 seconds, the maximum bad period count is the **gmlinktolerance** parameter value that is divided by 10. For example, with a **gmlinktolerance** value of 300, the maximum bad period count is 30. When maximum bad period count is reached, a 1920 error is reported.

Under a light I/O load, a single bad write can become significant. For example, if only one write I/O is performed for every 10 and this write is considered slow, the bad period count increments.

An edge case is achieved by setting the **gmmaxhostdelay** and **gmlinktolerance** parameters to their minimum settings (1 ms and 20 s). With these settings, you need only two consecutive bad sample periods before a 1920 error condition is reported. Consider a foreground write I/O that has a light I/O load. For example, a single I/O happens in the 20 s. With unlucky timing, a single bad I/O results (that is, a write I/O that took over 1 ms in remote copy), and it spans the boundary of two, 10-second sample periods. This single bad I/O theoretically can be counted as 2 x the bad periods and trigger a 1920 error.

A higher **gmlinktolerance** value, **gmmaxhostdelay** setting, or I/O load might reduce the risk of encountering this edge case.

### ***maxreplicationdelay and partnershipexclusionthreshold***

The **maxreplicationdelay** and **partnershipexclusionthreshold** parameters provide more performance protection mechanisms when Remote Copy services (Metro Mirror and Global Mirror) are used.

The **maxreplicationdelay** system-wide attribute configures how long a single write can be outstanding from the host before the relationship is stopped, which triggers a 1920 error. It can protect the hosts from seeing timeouts because of secondary hung I/Os.

This parameter is mainly intended to protect from secondary system issues. It does not help with ongoing performance issues, but can be used to limit the exposure of hosts to long write response times that can cause application errors. For example, setting **maxreplicationdelay** to 30 means that if a write operation for a volume in a Remote Copy relationship does not complete within 30 seconds, the relationship is stopped, triggering a 1920 error.

This happens even if the cause of the write delay is not related to the remote copy. For this reason the **maxreplicationdelay** settings can lead to false positive 1920 error triggering.

In addition to the 1920 error, the specific event ID 985004 is generated with the text “Maximum replication delay exceeded”.

The **maxreplicationdelay** values can be 0 - 360 seconds. Setting **maxreplicationdelay** to 0 disables the feature.

The **partnershipexclusionthreshold** is a system-wide parameter that sets the timeout for an I/O that triggers a temporarily dropping of the link to the remote system. Similar to **maxreplicationdelay**, the **partnershipexclusionthreshold** attribute provides some flexibility in a part of replication that tries to shield a production system from hung I/Os on a secondary system.

In an IBM SAN Volume Controller, a *node assert* (restart with a 2030 error) occurs if any individual I/O takes longer than 6 minutes. To avoid this situation, some actions are attempted to clean up anything that might be hanging I/O before the I/O gets to 6 minutes.

One of these actions is temporarily dropping (for 15 minutes) the link between systems if any I/O takes longer than 5 minutes 15 seconds (315 seconds). This action often removes hang conditions caused by replication problems. The **partnershipexclusionthreshold** parameter introduced the ability to set this value to a time lower than 315 seconds to respond to hung I/O more swiftly. The **partnershipexclusionthreshold** value must be a number in the range 30 - 315.

If an I/O takes longer the **partnershipexclusionthreshold** value, a 1720 error is triggered (with an event ID 987301) and any regular Global Mirror or Metro Mirror relationships stop on the next write to the primary volume.

**Important:** Do not change the **partnershipexclusionthreshold** parameter except under the direction of IBM Support.

To set the **maxreplicationdelay** and **partnershipexclusionthreshold** parameters, the **chsystem** command must be used, as shown in Example 6-3.

*Example 6-3 maxreplicationdelay and partnershipexclusionthreshold setting*

---

```
IBM_2145:SVC_ESC:superuser>chsystem -maxreplicationdelay 30
IBM_2145:SVC_ESC:superuser>chsystem -partnershipexclusionthreshold 180
```

---

The **maxreplicationdelay** and **partnershipexclusionthreshold** parameters do not interact with the **gmlinktolerance** and **gmmaxhostdelay** parameters.

## Troubleshooting 1920 errors

When you are troubleshooting 1920 errors that are posted across multiple relationships, you must diagnose the cause of the earliest error first. You must also consider whether other higher priority system errors exist and fix these errors because they might be the underlying cause of the 1920 error.

The diagnosis of a 1920 error is assisted by SAN performance statistics. To gather this information, you can use IBM Spectrum Control with a statistics monitoring interval of 1 or 5 minutes. Also, turn on the internal statistics gathering function, **I0stats**, in IBM SAN Volume Controller. Although not as powerful as IBM Spectrum Control, **I0stats** can provide valuable debug information if the **snap** command gathers system configuration data close to the time of failure.

The following are the main performance statistics to investigate for the 1920 error:

- **Write I/O Rate and Write Data Rate**

For volumes that are primary volumes in relationships, these statistics are the total amount of write operations submitted per second by hosts on average over the sample period, and the bandwidth of those writes. For secondary volumes in relationships, this is the average number of replicated writes that are received per second, and the bandwidth that these writes consume. Summing the rate over the volumes you intend to replicate gives a coarse estimate of the replication link bandwidth required.

- **Write Response Time and Peak Write Response Time**

On primary volumes, these are the average time (in milliseconds) and peak time between a write request being received from a host, and the completion message being returned. The write response time is the best way to show what kind of write performance that the host is seeing.

If a user complains that an application is slow, and the stats show the write response time leap from 1 ms to 20 ms, the two are most likely linked. However, some applications with high queue depths and low to moderate workloads will not be affected by increased response times. Note that this being high is an effect of some other problem. The peak is less useful, as it is sensitive to individual glitches in performance, but it can show more detail of the distribution of write response times.

On secondary volumes, these statistics describe the time for the write to be submitted from the replication feature into the system cache, and should normally be of a similar magnitude to those on the primary volume. Generally, the write response time should be below 1 ms for a fast-performing system.

- **Global Mirror Write I/O Rate**

This statistic shows the number of writes per second, the (regular) replication feature is processing for this volume. It applies to both types of Global Mirror and to Metro Mirror, but in each case only for the secondary volume. Because writes are always separated into 32 KB or smaller tracks before replication, this setting might be different from the Write I/O Rate on the primary volume (magnified further because the samples on the two systems will not be aligned, so they will capture a different set of writes).

- **Global Mirror Overlapping Write I/O Rate**

This statistic monitors the amount of overlapping I/O that the Global Mirror feature is handling for regular Global Mirror relationships. That is where an LBA is written again after the primary volume has been updated, but before the secondary volume has been updated for an earlier write to that LBA. To mitigate the effects of the overlapping I/Os, a journaling feature has been implemented, as discussed in “Colliding writes” on page 258.

- **Global Mirror secondary write lag**

This statistic is valid for regular Global Mirror primary and secondary volumes. For primary volumes, it tracks the length of time in milliseconds that replication writes are outstanding from the primary system. This amount includes the time to send the data to the remote system, consistently apply it to the secondary non-volatile cache, and send an acknowledgment back to the primary system.

For secondary volumes, this statistic records only the time that is taken to consistently apply it to the system cache, which is normally up to 20 ms. Most of that time is spent coordinating consistency across many nodes and volumes. Primary and secondary volumes for a relationship tend to record times that differ by the round-trip time between systems. If this statistic is high on the secondary system, look for congestion on the secondary system’s fabrics, saturated auxiliary storage, or high CPU utilization on the secondary system.

- **Write-cache Delay I/O Rate**

These statistics show how many writes were not instantly accepted into the system cache because cache was full. It is a good indication that the write rate is faster than the storage can cope with. If this amount starts to increase on auxiliary storage while primary volumes suffer from increased Write Response Time, it is possible that the auxiliary storage is not fast enough for the replicated workload.

- **Port to Local Node Send Response Time**

The time in milliseconds that it takes this node to send a message to other nodes in the same system (which will mainly be the other node in the same I/O group) and get an acknowledgment back. This amount should be well below 1 ms, with values below 0.3 ms being essential for regular Global Mirror to provide a Write Response Time below 1 ms.

This requirement is necessary because up to three round-trip messages within the local system will happen before a write completes to the host. If this number is higher than you want, look at fabric congestion (Zero Buffer Credit Percentage) and CPU Utilization of all nodes in the system.

- **Port to Remote Node Send Response Time**

This value is the time in milliseconds that it takes to send a message to nodes in other systems and get an acknowledgment back. This amount is not separated out by remote system, but for environments that have replication to only one remote system. This amount should be close to the low-level ping time between your sites. If this starts going significantly higher, it is likely that the link between your systems is saturated, which usually causes high Zero Buffer Credit Percentage as well.

- **Sum of Port-to-local node send response time and Port-to-local node send queue time**

The time must be less than 1 ms for the primary system. A number in excess of 1 ms might indicate that an I/O group is reaching its I/O throughput limit, which can limit performance.

- **System CPU Utilization**

These values show how heavily loaded the nodes in the system are. If any core has high utilization (say, over 90%) and there is an increase in write response time, it is possible that the workload is being CPU limited. You can resolve this by upgrading to faster hardware, or spreading out some of the workload to other nodes and systems.

- **Zero Buffer Credit Percentage or Port Send Delay IO Percentage**

This is the fraction of messages that this node attempted to send through Fibre Channel ports that had to be delayed because the port ran out of buffer credits. If you have a long link from the node to the switch it is attached to, there might be benefit in getting the switch to grant more buffer credits on its port.

It is more likely to be the result of congestion on the fabric, because running out of buffer credits is how Fibre Channel performs flow control. Normally, this value is well under 1%. From 1 - 10% is a concerning level of congestion, but you might find the performance acceptable. Over 10% indicates severe congestion. This amount is also called out on a port-by-port basis in the port-level statistics, which gives finer granularity about where any congestion might be.

When looking at the port-level statistics, high values on ports used for messages to nodes in the same system are much more concerning than those on ports that are used for messages to nodes in other systems.

► **Back-end Write Response Time**

This value is the average response time in milliseconds for write operations to the back-end storage. This time might include several physical I/O operations, depending on the type of RAID architecture.

Poor back-end performances on secondary system are a frequent cause of 1920 errors, while it is not so common for primary systems. Exact values to watch out for depend on the storage technology, but usually the response time should be less than 50 ms. A longer response time can indicate that the storage controller is overloaded. If the response time for a specific storage controller is outside of its specified operating range, investigate for the same reason.

**Focus areas for 1920 errors**

The causes of 1920 errors might be numerous. To fully understand the underlying reasons for posting this error, consider the following components that are related to the Remote Copy relationship:

- The intersystem connectivity network
- Primary storage and remote storage
- IBM SAN Volume controller nodes
- Storage area network

***Data collection for diagnostic purposes***

A successful diagnosis depends on the collection of the following data at both systems:

- The **snap** command with **livedump** (triggered at the point of failure)
- I/O Stats running (if possible)
- IBM Spectrum Control performance statistics data (if possible)
- The following information and logs from other components:
  - Intersystem network and switch details:
    - Technology
    - Bandwidth
    - Typical measured latency on the Intersystem network
    - Distance on all links (which can take multiple paths for redundancy)
    - Whether trunking is enabled
    - How the link interfaces with the two SANs
    - Whether compression is enabled on the link
    - Whether the link dedicated or shared; if so, the resource and amount of those resources they use
    - Switch Write Acceleration to check with IBM for compatibility or known limitations
    - Switch Compression, which should be transparent but complicates the ability to predict bandwidth
  - Storage and application:
    - Specific workloads at the time of 1920 errors, which might not be relevant, depending upon the occurrence of the 1920 errors and the volumes that are involved
    - RAID rebuilds
    - Whether 1920 errors are associated with Workload Peaks or Scheduled Backup



## ***Intersystem network***

For diagnostic purposes, ask the following questions about the intersystem network:

- ▶ Was network maintenance being performed?

Consider the hardware or software maintenance that is associated with intersystem network, such as updating firmware or adding more capacity.

- ▶ Is the intersystem network overloaded?

You can find indications of this situation by using statistical analysis with the help of I/O stats, IBM Spectrum Control, or both. Examine the internode communications, storage controller performance, or both. By using IBM Spectrum Control, you can check the storage metrics for the Global Mirror relationships were stopped, which can be tens of minutes depending on the **gmLinktolerance** and **maxreplicationdelay** parameters.

Diagnose the overloaded link by using the following methods:

- Look at the statistics generated by the routers or switches near your most bandwidth-constrained link between the systems

Exactly what is provided, and how to analyze it varies depending on the equipment used.

- Look at the port statistics for high response time in the internode communication

An overloaded long-distance link causes high response times in the internode messages (the *Port to remote node send response time* statistic) that are sent by IBM Spectrum Virtualize. If delays persist, the messaging protocols exhaust their tolerance elasticity and the Global Mirror protocol is forced to delay handling new foreground writes while waiting for resources to free up.

- Look at the port statistics for buffer credit starvation

The Zero Buffer Credit Percentage and Port Send Delay IO Percentage statistics can be useful here too, because you normally have a high value here as the link saturates. Only look at ports that are replicating to the remote system.

- Look at the volume statistics (before the 1920 error is posted):

- Target volume write throughput approaches the link bandwidth.

If the write throughput on the target volume is equal to your link bandwidth, your link is likely overloaded. Check what is driving this situation. For example, does peak foreground write activity exceed the bandwidth, or does a combination of this peak I/O and the background copy exceed the link capacity?

- Source volume write throughput approaches the link bandwidth.

This write throughput represents only the I/O that is performed by the application hosts. If this number approaches the link bandwidth, you might need to upgrade the link's bandwidth. Alternatively, reduce the foreground write I/O that the application is attempting to perform, or reduce the number of Remote Copy relationships.

- Target volume write throughput is greater than the source volume write throughput.

If this condition exists, the situation suggests a high level of background copy and mirrored foreground write I/O. In these circumstances, decrease the background copy rate parameter of the Global Mirror partnership to bring the combined mirrored foreground I/O and background copy I/O rates back within the remote links bandwidth.

- Look at the volume statistics (after the 1920 error is posted):
  - Source volume write throughput after the Global Mirror relationships were stopped.

If write throughput increases greatly (by 30% or more) after the Global Mirror relationships are stopped, the application host was attempting to perform more I/O than the remote link can sustain.

When the Global Mirror relationships are active, the overloaded remote link causes higher response times to the application host. This overload, in turn, decreases the throughput of application host I/O at the source volume. After the Global Mirror relationships stop, the application host I/O sees a lower response time, and the true write throughput returns.

To resolve this issue, increase the remote link bandwidth, reduce the application host I/O, or reduce the number of Global Mirror relationships.

### ***Storage controllers***

Investigate the primary and remote storage controllers, starting at the remote site. If the back-end storage at the secondary system is overloaded, or another problem is affecting the cache there, the Global Mirror protocol fails to keep up. Similarly, the problem exhausts the (**gm1inktolerance**) elasticity and has a similar effect at the primary system.

In this situation, ask the following questions:

- Are the storage controllers at the remote system overloaded (performing slowly)?

Use IBM Spectrum Control to obtain the back-end write response time for each MDisk at the remote system. A response time for any individual MDisk that exhibits a sudden increase of 50 ms or more, or that is higher than 100 ms, generally indicates a problem with the back-end. When 1920 error is triggered by the max replication delay exceeded condition, check the peak back-end write response time to see if it exceeded the **maxreplicationdelay** value is around the 1920 occurrence.

However, if you followed the specified back-end storage controller requirements and were running without problems until recently, the error is most likely caused by a decrease in controller performance because of maintenance actions or a hardware failure of the controller. Check whether an error condition is on the storage controller, for example, media errors, a failed physical disk, or a recovery activity, such as RAID array rebuilding that uses more bandwidth.

If an error occurs, fix the problem and then restart the Global Mirror relationships.

If no error occurs, consider whether the secondary controller can process the required level of application host I/O. You might improve the performance of the controller in the following ways:

- Adding more or faster physical disks to a RAID array.
- Changing the RAID level of the array.
- Changing the cache settings of the controller and checking that the cache batteries are healthy, if applicable.
- Changing other controller-specific configuration parameter.

- Are the storage controllers at the primary site overloaded?

Analyze the performance of the primary back-end storage by using the same steps that you use for the remote back-end storage. The main effect of bad performance is to limit the amount of I/O that can be performed by application hosts. Therefore, you must monitor back-end storage at the primary site regardless of Global Mirror.

In case of 1920 error triggered by the max replication delay exceeded condition, check the peak back-end write response time to see if it has exceeded the **maxreplicationdelay** value around the 1920 occurrence.

However, if bad performance continues for a prolonged period, a false 1920 error might be flagged.

### **Node**

For SAN Volume controller node hardware, the possible cause of the 1920 errors might be from a heavily loaded secondary or primary system. If this condition persists, a 1920 error might be posted.

Global Mirror needs to synchronize its I/O processing across all nodes in the system to ensure data consistency. If any node is running out of CPU, it can affect all relationships. So check the CPU cores usage statistic. If it looks higher when there is a performance problem, then running out of CPU bandwidth might be causing the problem. Of course, CPU usage goes up when the IOPS going through a node goes up, so if the workload increases, you expect to see CPU usage increase.

If there is an increase in CPU usage on the secondary system but no increase in IOPS, and volume write latency increases too, it is likely that the increase in CPU usage has caused the increased volume write latency. In that case, try to work out what might have caused the increase in CPU usage (for example, starting many FlashCopy mappings). Consider moving that activity to a time with less workload. If there is an increase in both CPU usage and IOPS, and the CPU usage is close to 100%, then that node might be overloaded. A *Port-to-local node send queue time* value higher than 0.2 ms often denotes CPU cores overloading.

In a primary system, if it is sufficiently busy, the write ordering detection in Global Mirror can delay writes enough to reach a latency of **gmmmaxhostdelay** and cause a 1920 error. Stopping replication potentially lowers CPU usage, and also lowers the opportunities for each I/O to be delayed by slow scheduling on a busy system.

Solve overloaded nodes by upgrading them to newer, faster hardware if possible, or by adding more I/O groups/control enclosures (or systems) to spread the workload over more resources.

### **Storage area network**

Issues and congestions both in local and remote SANs can lead to 1920 errors. The *Port to local node send response time* is the key statistic to investigate on. It captures the round-trip time between nodes in the same system. Anything over 1.0 ms is surprisingly high, and will cause high secondary volume write response time. Values greater than 1 ms on primary system will cause an impact on write latency to Global Mirror primary volumes of 3 ms or more.

If you checked CPU cores usage on all nodes, and it is near 100%, a high Port to local node send response time means that fabric congestion or a slow-draining Fibre Channel device exists.

Good indicators of SAN congestion are the Zero Buffer Credit Percentage and Port Send Delay IO Percentage on the port statistics (for more information about, for more information about Buffer Credit, see “Buffer credits” on page 269). If any port is seeing over 10% zero buffer credits or delay I/Os, that issues causes a problem for all I/O, not just Global Mirror writes. Values from 1 - 10% are moderately high and might contribute to performance issues.

For both primary and secondary systems, congestion on the fabric from other slow-draining devices becomes much less of an issue when only dedicated ports are used for node-to-node traffic within the system. However, this only really becomes an option on systems with more than four ports per node. Use port masking to segment your ports.

### ***FlashCopy considerations***

Check that FlashCopy mappings are in the prepared state. Check whether the Global Mirror target volumes are the sources of a FlashCopy mapping and whether that mapping was in the prepared state for an extended time.

Volumes in the prepared state are cache disabled, so their performance is impacted. To resolve this problem, start the FlashCopy mapping, which reenables the cache and improves the performance of the volume and of the Global Mirror relationship.

Consider also that FlashCopy can add significant workload to the back-end storage, especially when the background copy is active (see “Background copy considerations” on page 243). In cases where the remote system is used to create golden or practice copies for DR testing, the workload that is added by the FlashCopy background processes can overload the system. This overload can lead to poor Remote Copy performances and then to a 1920 error.

Careful planning of the back-end resources is particularly important with these kinds of scenarios. Reducing the FlashCopy background copy rate can also help to mitigate this situation. Furthermore, note that the FlashCopy copy-on-write process adds some latency by delaying the write operations on the primary volumes until the data is written to the FlashCopy target.

This process does not directly affect the Remote Copy operations because it is logically placed below the Remote Copy processing in the I/O stack, as shown in Figure 6-7 on page 233. Nevertheless, in some circumstances (especially write-intensive environments), the copy-on-write process tends to stress some of the system’s internal resources, such as CPU and memory. This condition can also affect the remote copy, which competes for the same resources and eventually leads to 1920 errors.

### ***FCIP considerations***

When you get a 1920 error, always check the latency first. The FCIP routing layer can introduce latency if it is not properly configured. If your network provider reports a much lower latency, you might have a problem at your FCIP routing layer. Most FCIP routing devices have built-in tools to enable you to check the RTT. When you are checking latency, remember that TCP/IP routing devices (including FCIP routers) report RTT by using standard 64-byte ping packets.

In Figure 6-31 on page 301, you can see why the effective transit time must be measured only by using packets that are large enough to hold an FC frame, or 2148 bytes (2112 bytes of payload and 36 bytes of header). Allow estimated resource requirements to be a safe amount because various switch vendors have optional features that might increase this size. After you verify your latency by using the proper packet size, proceed with normal hardware troubleshooting.

Look at the second largest component of your RTT, which is *serialization delay*. Serialization delay is the amount of time that is required to move a packet of data of a specific size across a network link of a certain bandwidth. The required time to move a specific amount of data decreases as the data transmission rate increases.

Figure 6-31 shows the orders of magnitude of difference between the link bandwidths. It is easy to see how 1920 errors can arise when your bandwidth is insufficient. Never use a TCP/IP ping to measure RTT for FCIP traffic.

| Packet Size | Link Size | Serialization Delay (Time Required to Send Data) | Unit         |
|-------------|-----------|--------------------------------------------------|--------------|
| 64          | 256 Kbps  | 2.0E+03                                          | microseconds |
| 64          | 1.5 Mbps  | 3.4E+02                                          | microseconds |
| 64          | 100 Mbps  | 5.1E+00                                          | microseconds |
| 64          | 155 Mbps  | 3.3E+00                                          | microseconds |
| 64          | 622 Mbps  | 8.2E-01                                          | microseconds |
| 64          | 1 Gbps    | 5.1E-04                                          | microseconds |
| 64          | 10 Gbps   | 5.1E-05                                          | microseconds |
| 1500        | 256 Kbps  | 4.7E+04                                          | microseconds |
| 1500        | 1.5 Mbps  | 8.0E+03                                          | microseconds |
| 1500        | 100 Mbps  | 1.2E+02                                          | microseconds |
| 1500        | 155 Mbps  | 7.7E+01                                          | microseconds |
| 1500        | 622 Mbps  | 1.9E+01                                          | microseconds |
| 1500        | 1 Gbps    | 1.2E+01                                          | microseconds |
| 1500        | 10 Gbps   | 1.2E+00                                          | microseconds |
| 2148        | 256 Kbps  | 6.7E+04                                          | microseconds |
| 2148        | 1.5 Mbps  | 1.1E+04                                          | microseconds |
| 2148        | 100 Mbps  | 1.7E+02                                          | microseconds |
| 2148        | 155 Mbps  | 1.1E+02                                          | microseconds |
| 2148        | 622 Mbps  | 2.8E+01                                          | microseconds |
| 2148        | 1 Gbps    | 1.7E+01                                          | microseconds |
| 2148        | 10 Gbps   | 1.7E-03                                          | microseconds |

Figure 6-31 Effect of packet size (in bytes) versus the link size

As shown in Figure 6-31, the amount of time in microseconds that is required to transmit a packet across network links of varying bandwidth capacity is compared. The following packet sizes are used:

- ▶ 64 bytes: The size of the common ping packet
- ▶ 1500 bytes: The size of the standard TCP/IP packet
- ▶ 2148 bytes: The size of an FC frame

Finally, your path maximum transmission unit (MTU) affects the delay that is incurred to get a packet from one location to another location. An MTU might cause fragmentation, or be too large and cause too many retransmits when a packet is lost.

## Recovery after 1920 errors

After a 1920 error occurs, the Global Mirror auxiliary volumes are no longer in a Consistent Synchronized state. You must establish the cause of the problem and fix it before you restart the relationship.

When the relationship is restarted, you must resynchronize it. During this period, the data on the Metro Mirror or Global Mirror auxiliary volumes on the secondary system is inconsistent, and your applications cannot use the volumes as backup disks. To address this data consistency exposure on the secondary system, a FlashCopy of the auxiliary volumes can be created to maintain a consistent image until the Global Mirror (or the Metro Mirror) relationships are synchronized again and back in a consistent state.

IBM Spectrum Virtualize provides the Remote Copy *Consistency Protection* feature that automates this process. When Consistency Protection is configured, the relationship between the primary and secondary volumes does not go in to the Inconsistent copying status once restarted. Instead, the system uses a secondary *change volume* to automatically copy the previous consistent state of the secondary volume.

The relationship automatically moves to the Consistent copying status as the system resynchronizes and protects the consistency of the data. The relationship status changes to Consistent synchronized when the resynchronization process completes. For more information about the Consistency Protection feature, see *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8467.

To ensure that the system can handle the background copy load, delay restarting the Metro Mirror or Global Mirror relationship until a quiet period occurs. If the required link capacity is unavailable, you might experience another 1920 error, and the Metro Mirror or Global Mirror relationship might stop in an inconsistent state.

Copy services tools, like IBM Copy Services Manager (CSM), or manual scripts can be used to automatize the relationships to restart after a 1920 error. CSM implements a logic to avoid recurring restart operations in case of a persistent problem. CSM attempts an automatic restart for every occurrence of 1720/1920 error a certain number of times (determined by the **gmlinktolerance** value) within a 30 minute time period.

If the number of allowable automatic restarts is exceeded within the time period, CSM will not automatically restart GM on the next 1720/1920 error. Furthermore, with CSM it is possible to specify the amount of time, in seconds, in which the tool will wait after an 1720/1920 error before automatically restarting the GM. Further details about IBM Copy Services Manager can be found on the [IBM Copy Services Manager Home Page](#).

**Tip:** When implementing automatic restart functions, it is advised to preserve the data consistency on GM target volumes during the resynchronization using features like FlashCopy or Consistency Protection.

## Adjusting the Global Mirror settings

Although the default values are valid in most configurations, the settings of the **gmlinktolerance** and **gmmaxhostdelay** can be adjusted to accommodate particular environment or workload conditions.

For example, Global Mirror is designed to look at average delays. However, some hosts such as VMware ESX might not tolerate a single I/O getting old, for example, 45 seconds, before it decides to reboot. Given that it is better to terminate a Global Mirror relationship than it is to reboot a host, you might want to set **gmlinktolerance** to something like 30 seconds and then compensate so that you do not get too many relationship terminations by setting **gmmaxhostdelay** to something larger, such as 100 ms.

If you compare the two approaches, the default (**gmlinktolerance 300**, **gmmaxhostdelay 5**) is a rule that “If more than one third of the I/Os are slow and that happens repeatedly for 5 minutes, then terminate the busiest relationship in that stream.” In contrast, the example of **gmlinktolerance 30**, **gmmaxhostdelay 100** is a rule that “If more than one third of the I/Os are extremely slow and that happens repeatedly for 30 seconds, then terminate the busiest relationship in the stream.”

Therefore, one approach is designed to pick up general slowness, and the other approach is designed to pick up shorter bursts of extreme slowness that might disrupt your server environment. The general recommendation is to change the **gmlinktolerance** and **gmmaxhostdelay** values progressively and evaluate the overall impact to find an acceptable compromise between performances and Global Mirror stability.

You can even disable the **gmLinktolerance** feature by setting the **gmLinktolerance** value to 0. However, the **gmLinktolerance** parameter cannot protect applications from extended response times if it is disabled. You might consider disabling the **gmLinktolerance** feature in the following circumstances:

- ▶ During SAN maintenance windows, where degraded performance is expected from SAN components and application hosts can withstand extended response times from Global Mirror volumes.
- ▶ During periods when application hosts can tolerate extended response times and it is expected that the **gmLinktolerance** feature might stop the Global Mirror relationships. For example, you are testing usage of an I/O generator that is configured to stress the back-end storage. Then, the **gmLinktolerance** feature might detect high latency and stop the Global Mirror relationships. Disabling the **gmLinktolerance** parameter stops the Global Mirror relationships at the risk of exposing the test host to extended response times.

Another tunable parameter that interacts with the GM is the **maxreplicationdelay**. Note that the **maxreplicationdelay** settings do not mitigate the 1920 error occurrence because it actually adds a trigger to the 1920 error itself. However, the **maxreplicationdelay** provides users with a fine granularity mechanism to manage the hung I/Os condition and it can be used in combination with **gmLinktolerance** and **gmmaxhostdelay** settings to better address particular environment conditions.

In the above VMware example, an alternative option is to set the **maxreplicationdelay** to 30 seconds and leave the **gmLinktolerance** and **gmmaxhostdelay** settings to their default. With these settings, the **maxreplicationdelay** timeout effectively handles the hung I/Os conditions, while the **gmLinktolerance** and **gmmaxhostdelay** settings still provide an adequate mechanism to protect from ongoing performance issues.

## 6.4 Native IP replication

The native IP replication feature enables replication between any IBM Spectrum Virtualize products by using the built-in networking ports or optional 1/10 Gb adapter.

Native IP replication uses SANslide technology that was developed by Bridgeworks Limited of Christchurch, UK. They specialize in products that can bridge storage protocols and accelerate data transfer over long distances. Adding this technology at each end of a wide area network (WAN) TCP/IP link significantly improves the use of the link.

This improvement is realized by applying patented artificial intelligence (AI) to hide latency that is normally associated with WANs. Doing so can greatly improve the performance of mirroring services, in particular Global Mirror with Change Volumes (GMCV) over long distances.

### 6.4.1 Native IP replication technology

Remote Mirroring over IP communication is supported on the IBM Spectrum Virtualize systems by using Ethernet communication links. The IBM Spectrum Virtualize Software IP replication uses the innovative Bridgeworks SANslide technology to optimize network bandwidth and utilization. This new function enables the use of a lower-speed and lower-cost networking infrastructure for data replication.

Bridgeworks' SANSlide technology, which is integrated into the IBM Spectrum Virtualize Software, uses artificial intelligence to help optimize network bandwidth use and adapt to changing workload and network conditions. This technology can improve remote mirroring network bandwidth usage up to three times. It can enable clients to deploy a less costly network infrastructure, or speed up remote replication cycles to enhance DR effectiveness.

With an Ethernet network data flow, the data transfer can slow down over time. This condition occurs because of the latency that is caused by waiting for the acknowledgment of each set of packets that are sent. The next packet set cannot be sent until the previous packet is acknowledged, as shown in Figure 6-32.

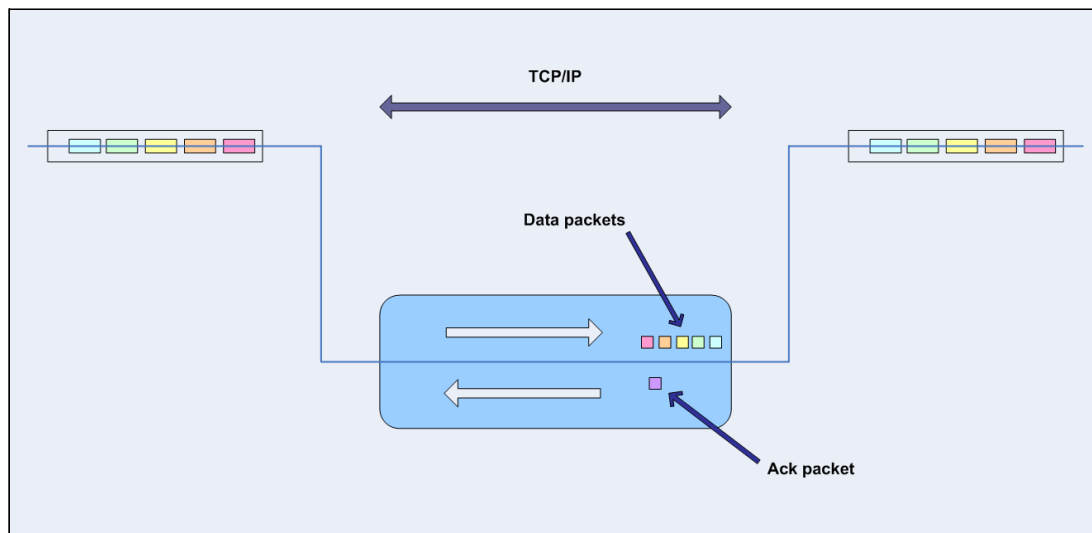


Figure 6-32 Typical Ethernet network data flow

However, by using the embedded IP replication, this behavior can be eliminated with the enhanced parallelism of the data flow. This parallelism uses multiple virtual connections (VCs) that share IP links and addresses.

The artificial intelligence engine can dynamically adjust the number of VCs, receive window size, and packet size as appropriate to maintain optimum performance. While the engine is waiting for one VC's ACK, it sends more packets across other VCs. If packets are lost from any VC, data is automatically retransmitted, as shown in Figure 6-33.

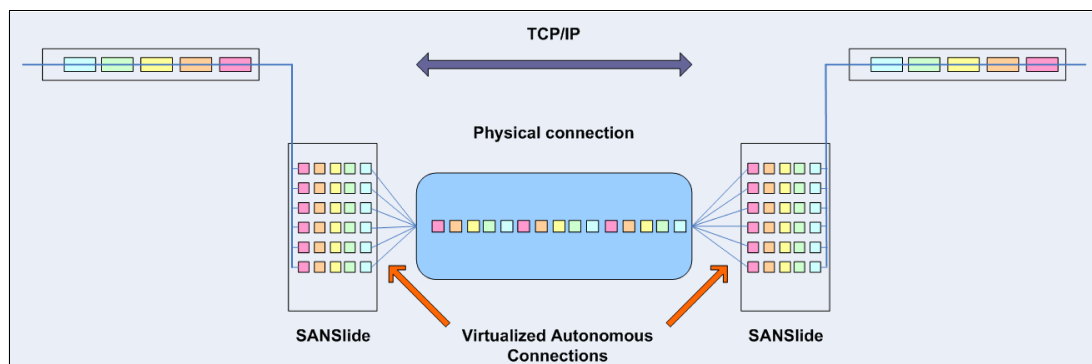


Figure 6-33 Optimized network data flow by using Bridgeworks SANSlide technology



For more information about this technology, see *IBM SAN Volume Controller and Storwize Family Native IP Replication*, REDP-5103.

Metro Mirror, Global Mirror, and Global Mirror Change Volume are supported with native IP partnership.

## 6.4.2 IP partnership limitations

The following prerequisites and assumptions must be considered before IP partnership between two IBM Spectrum Virtualize systems can be established:

- ▶ The systems have 7.2 or later code levels.
- ▶ The systems feature the necessary licenses that enable Remote Copy partnerships to be configured between two systems. No separate license is required to enable IP partnership.
- ▶ The storage SANs are configured correctly and the correct infrastructure to support the systems in Remote Copy partnerships over IP links is in place.
- ▶ The two systems must be able to ping each other and perform the discovery.
- ▶ The maximum number of partnerships between the local and remote systems, including both IP and Fibre Channel (FC) partnerships, is limited to the current maximum that is supported, which is three partnerships (four systems total).
- ▶ Only a single partnership over IP is supported.
- ▶ A system can have simultaneous partnerships over FC and IP, but with separate systems. The FC zones between two systems must be removed before an IP partnership is configured.
- ▶ The use of WAN optimization devices, such as Riverbed, is not supported in IP partnership configurations that contain a SAN Volume Controller.
- ▶ IP partnerships are supported with 25, 10 and 1 Gbits links. However, the intermix on a single link is not supported.
- ▶ The maximum supported round-trip time is 80 milliseconds (ms) for 1 Gbps links.
- ▶ The maximum supported round-trip time is 10 ms for 25 and 10 Gbps links.
- ▶ The minimum supported link bandwidth is 10 Mbps.
- ▶ The inter-cluster heartbeat traffic uses 1 Mbps per link.
- ▶ Only nodes from two I/O Groups can have ports that are configured for an IP partnership.
- ▶ Migrations of Remote Copy relationships directly from FC-based partnerships to IP partnerships are not supported.
- ▶ IP partnerships between the two systems can be over IPv4 or IPv6 only, but not both.
- ▶ Virtual LAN (VLAN) tagging of the IP addresses that are configured for Remote Copy is supported.
- ▶ Management IP and Internet SCSI (iSCSI) IP on the same port can be in a different network.
- ▶ An added layer of security is provided by using Challenge Handshake Authentication Protocol (CHAP) authentication.
- ▶ Direct attached systems configurations are supported with the following restrictions:
  - Only two direct attach link are allowed.
  - The direct attach links must be on the same I/O group.

- Use two port groups, where a port group contains only the two ports that are directly linked.
- ▶ Transmission Control Protocol (TCP) ports 3260 and 3265 are used for IP partnership communications. Therefore, these ports must be open in firewalls between the systems.
- ▶ Network address translation (NAT) between systems that are being configured in an IP Partnership group is not supported.
- ▶ Only a single Remote Copy data session per Remote Copy port group can be established. It is intended that only one connection (for sending or receiving Remote Copy data) is made for each independent physical link between the systems.

**Note:** A physical link is the physical IP link between the two sites, A (local) and B (remote). Multiple IP addresses on local system A can be connected (by Ethernet switches) to this physical link. Similarly, multiple IP addresses on remote system B can be connected (by Ethernet switches) to the same physical link. At any point, only a single IP address on cluster A can form an RC data session with an IP address on cluster B.

- ▶ The maximum throughput is restricted based on the use of 1 Gbps or 10 Gbps Ethernet ports. The output varies based on distance (for example, round-trip latency) and quality of communication link (for example, packet loss). The maximum achievable throughput is the following rate for each port:
  - One 1 Gbps port can transfer up to 110 MBs
  - One 10 Gbps port can transfer up to 500 MBs

Table 6-9 lists the IP replication limits.

Table 6-9 IP replication limits

| Remote copy property                      | Maximum  | Apply to   | Comment                                                                                                              | Remote copy property                      |
|-------------------------------------------|----------|------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Inter-system IP partnerships per system   | 1        | All models | A system can be partnered with up to three remote systems. A maximum of one of those can be IP and the other two FC. | Inter-system IP partnerships per system   |
| I/O groups per system in IP partnerships  | 2        | All models | The nodes from a maximum of two I/O groups per system can be used for IP partnership.                                | I/O groups per system in IP partnerships  |
| Inter-site links per IP partnership       | 2        | All models | A maximum of two inter site links can be used between two IP partnership sites.                                      | Inter-site links per IP partnership       |
| Ports per node                            | 1        | All models | A maximum of one port per node can be used for IP partnership.                                                       | Ports per node                            |
| IP partnership Software Compression Limit | 140 MBps | All models |                                                                                                                      | IP partnership Software Compression Limit |

### 6.4.3 VLAN support

VLAN tagging is supported for iSCSI host attachment and IP replication. Hosts and remote-copy operations can connect to the system through Ethernet ports. Each traffic type has different bandwidth requirements, which can interfere with each other if they share IP connections. VLAN tagging creates two separate connections on the same IP network for different types of traffic. The system supports VLAN configuration on both IPv4 and IPv6 connections.

When the VLAN ID is configured for the IP addresses that are used for iSCSI host attach or IP replication, the suitable VLAN settings on the Ethernet network and servers must be configured correctly to avoid connectivity issues. After the VLANs are configured, changes to the VLAN settings disrupt iSCSI and IP replication traffic to and from the partnerships.

During the VLAN configuration for each IP address, the VLAN settings for the local and failover ports on two nodes of an I/O Group can differ. To avoid any service disruption, switches must be configured so that the failover VLANs are configured on the local switch ports and the failover of IP addresses from a failing node to a surviving node succeeds. If failover VLANs are not configured on the local switch ports, no paths are available to the Storwize V7000 system during a node failure and the replication fails.

Consider the following requirements and procedures when implementing VLAN tagging:

- ▶ VLAN tagging is supported for IP partnership traffic between two systems.
- ▶ VLAN provides network traffic separation at the layer 2 level for Ethernet transport.
- ▶ VLAN tagging by default is disabled for any IP address of a node port. You can use the CLI or GUI to set the VLAN ID for port IPs on both systems in the IP partnership.
- ▶ When a VLAN ID is configured for the port IP addresses that are used in Remote Copy port groups, appropriate VLAN settings on the Ethernet network must also be properly configured to prevent connectivity issues.

Setting VLAN tags for a port is disruptive. Therefore, VLAN tagging requires that you stop the partnership first before you configure VLAN tags. Then, restart again when the configuration is complete.

### 6.4.4 IP compression

IBM SAN Volume Controller can use the IP compression capability to speed up replication cycles, or to reduce bandwidth usage.

This feature reduces the volume of data that must be transmitted during Remote Copy operations by using compression capabilities that are similar to those capabilities that are experienced with Real-time Compression implementations.

**No License:** IP compression feature does not require an RtC software license.

The data compression is made within the IP replication component of the IBM Spectrum Virtualize code. It can be used with all the Remote Copy technology (Metro Mirror, Global Mirror, and Global Mirror Change Volume). The IP compression feature provides two kinds of compression mechanisms: the hardware compression and software compression.

The IP compression can be enabled on hardware configurations that support hardware-assisted compression acceleration engines. The hardware compression is active when compression accelerator engines are available; otherwise, software compression is used.

Hardware compression makes use of currently underused compression resources. The internal resources are shared between data and IP compression. Software compression uses the system CPU and might affect heavily used systems.

To evaluate the benefits of the IP compression, the Comprestimator tool can be used to estimate the compression ratio of the data to be replicated. The IP compression can be enabled and disabled without stopping the Remote Copy relationship by using the **mkippartnership** and **chpartnership** commands with the **-compress** parameter. Also, in systems with replication enabled in both directions, the IP compression can be enabled in only one direction. IP compression is supported for IPv4 and IPv6 partnerships.

## 6.4.5 Remote copy groups

This section describes Remote Copy groups (or Remote Copy port groups) and different ways to configure the links between the two remote systems. The two systems can be connected to each other over one link or, at most, two links. To address the requirement to enable the systems to know about the physical links between the two sites, the concept of Remote Copy port groups was introduced.

Remote copy port group ID is a numerical tag that is associated with an IP port of system to indicate to which physical IP link it is connected. Multiple IBM Spectrum Virtualize nodes can be connected to the same physical long-distance link; therefore, they must share a Remote Copy port group ID.

In scenarios with two physical links between the local and remote clusters, two Remote Copy port group IDs must be used to designate which IP addresses are connected to which physical link. This configuration must be done by the system administrator by using the GUI or the **cfgportip** CLI command. The relationship between the physical links and the Remote Copy group IDs is not monitored by the IBM Spectrum Virtualize code. Therefore, two different Remote Copy groups can be used with a single physical link and vice versa.

**Remember:** IP ports on both partners must be configured with identical Remote Copy port group IDs for the partnership to be established correctly.

The system IP addresses that are connected to the same physical link should be designated with identical Remote Copy port groups. The IBM Spectrum Virtualize systems supports three Remote Copy groups: 0, 1, and 2.

By default, the IP addresses are in Remote Copy port group 0. Ports in port group 0 are not considered for creating Remote Copy data paths between two systems. For partnerships to be established over IP links directly, IP ports must be configured in Remote Copy group 1 if a single inter-site link exists, or in Remote Copy groups 1 and 2 if two inter-site links exist.

You can assign one IPv4 address and one IPv6 address to each Ethernet port on the IBM Spectrum Virtualize and Storwize family systems. Each of these IP addresses can be shared between iSCSI host attach and the IP partnership. The user must configure the required IP address (IPv4 or IPv6) on an Ethernet port with a Remote Copy port group.

The administrator might want to use IPv6 addresses for Remote Copy operations and use IPv4 addresses on that same port for iSCSI host attach. This configuration also implies that for two systems to establish an IP partnership, both systems must have IPv6 addresses that are configured.

Administrators can choose to dedicate an Ethernet port for IP partnership only. In that case, host access must be specifically disabled for that IP address and any other IP address that is configured on that Ethernet port.

### **Failover operations within and between port groups**

Within one remote-copy port group, only one port from each system is selected for sending and receiving Remote Copy data at any one time. Therefore, on each system, at most one port for each remote-copy port group is reported as used.

If the IP partnership becomes unable to continue over an IP port, the system fails over to another port within that remote-copy port group. Some reasons this might occur are the switch to which it is connected fails, the node goes offline, or the cable that is connected to the port is unplugged.

For the IP partnership to continue during a failover, multiple ports must be configured within the remote-copy port group. If only one link is configured between the two systems, configure two ports (one per node) within the remote-copy port group. You can configure these two ports on two nodes within the same I/O group or within separate I/O groups.

While failover is in progress, no connections in that remote-copy port group exist between the two systems in the IP partnership for a short time. Typically, failover completes within 30 seconds to 1 minute. If the systems are configured with two remote-copy port groups, the failover process within each port group continues independently of each other.

The disadvantage of configuring only one link between two systems is that, during a failover, a discovery is initiated. When the discovery succeeds, the IP partnership is reestablished. As a result, the relationships might stop, in which case a manual restart is required. To configure two intersystem links, you must configure two remote-copy port groups.

When a node fails in this scenario, the IP partnership can continue over the other link until the node failure is rectified. Failback then happens when both links are again active and available to the IP partnership. The discovery is triggered so that the active IP partnership data path is made available from the new IP address.

In a two-node system, or if there is more than one I/O Group and the node in the other I/O group has IP ports pre-configured within the remote-copy port group, the discovery is triggered. The discovery makes the active IP partnership data path available from the new IP address.

## **6.4.6 Supported configuration examples**

Multiple IP partnership configurations are available depending on the number of physical links and the number of nodes. In the following sections, some example configurations are described.

### **Single inter-site link configurations**

Consider two 2-node systems in IP partnership over a single inter-site link (with failover ports configured), as shown in Figure 6-34 on page 310.

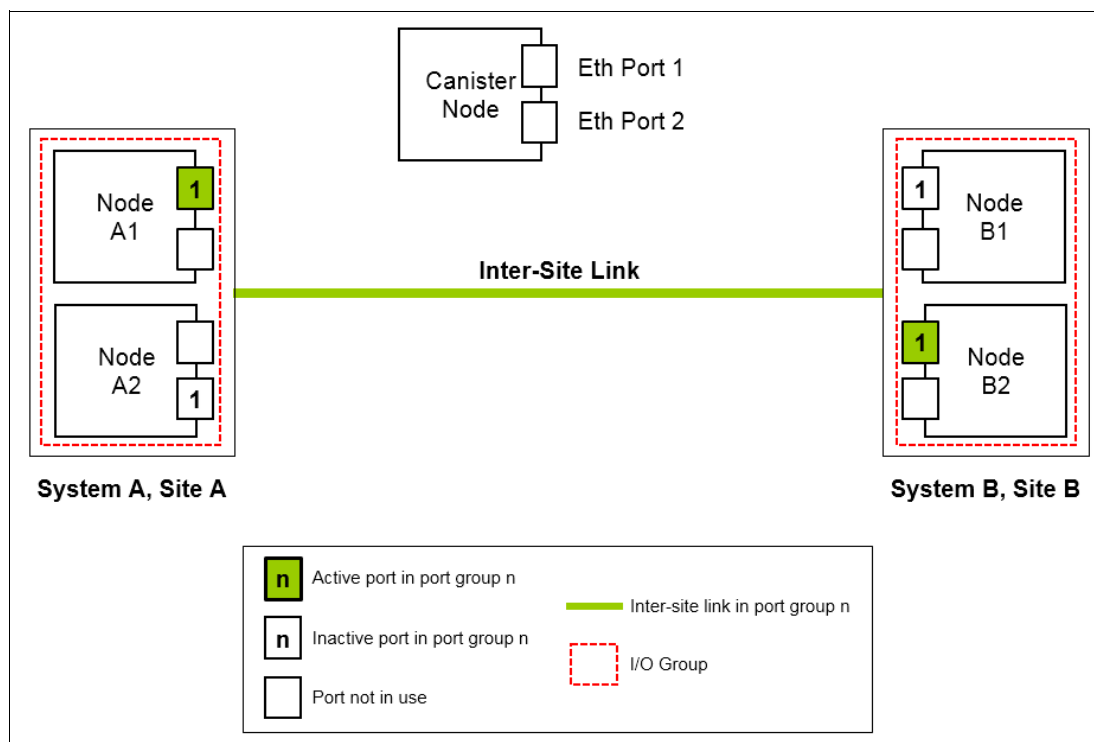


Figure 6-34 Only one Remote Copy group on each system and nodes with failover ports configured

Figure 6-34 shows two systems: System A and System B. A single Remote Copy port group 1 is configured on two Ethernet ports, one each on Node A1 and Node A2 on System A. Similarly, a single Remote Copy port group is configured on two Ethernet ports on Node B1 and Node B2 on System B.

Although two ports on each system are configured for Remote Copy port group 1, only one Ethernet port in each system actively participates in the IP partnership process. This selection is determined by a path configuration algorithm that is designed to choose data paths between the two systems to optimize performance.

The other port on the partner node in the I/O Group behaves as a standby port that is used during a node failure. If Node A1 fails in System A, IP partnership continues servicing replication I/O from Ethernet Port 2 because a failover port is configured on Node A2 on Ethernet Port 2.

However, it might take some time for discovery and path configuration logic to reestablish paths post failover. This delay can cause partnerships to change to Not\_Present for that time. The details of the particular IP port that is actively participating in IP partnership is provided in the **1sportip** output (reported as used).

This configuration has the following characteristics:

- ▶ Each node in the I/O group has the same Remote Copy port group that is configured. However, only one port in that Remote Copy port group is active at any time at each system.
- ▶ If Node A1 in System A or Node B2 in System B fails in the respective systems, IP partnerships rediscovery is triggered and continues servicing the I/O from the failover port.
- ▶ The discovery mechanism that is triggered because of failover might introduce a delay where the partnerships momentarily change to the Not\_Present state and recover.

Figure 6-35 shows a configuration with two 4-node systems in IP partnership over a single inter-site link (with failover ports configured).

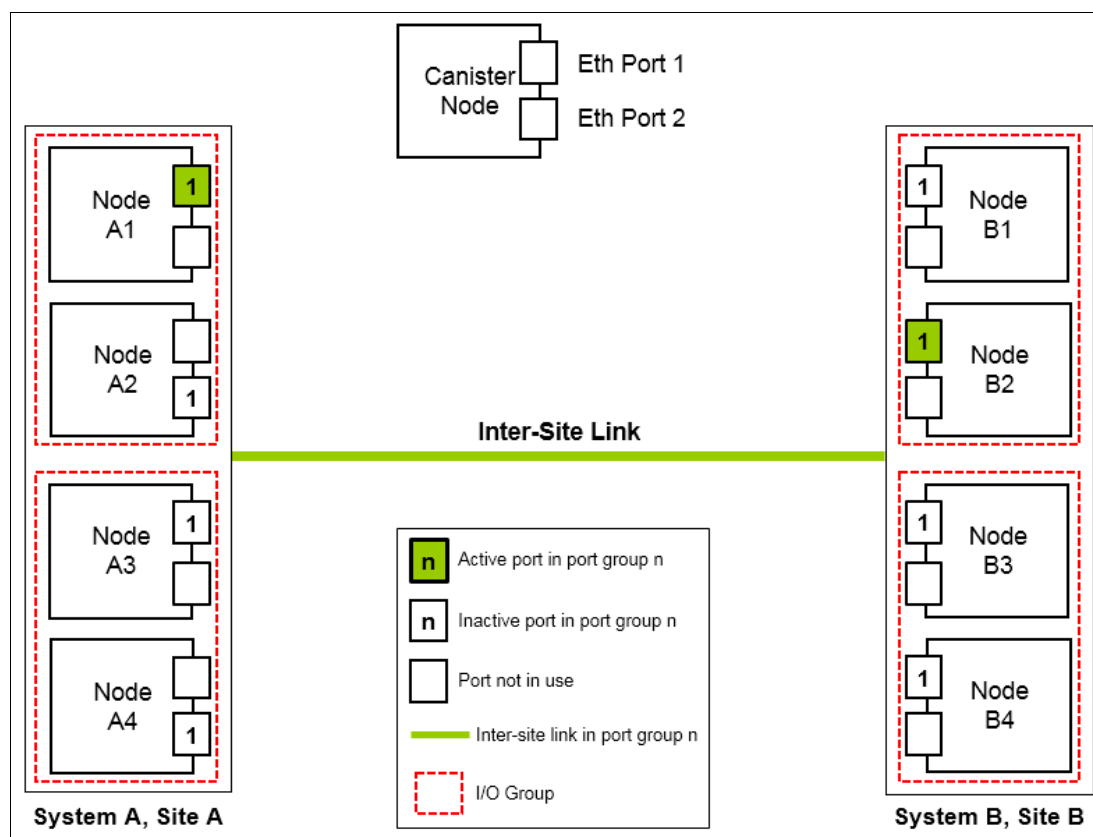


Figure 6-35 Multinode systems single inter-site link with only one Remote Copy port group

Figure 6-35 shows two 4-node systems: System A and System B. A single Remote Copy port group 1 is configured on nodes A1, A2, A3, and A4 on System A, Site A, and on nodes B1, B2, B3, and B4 on System B, Site B.

Although four ports are configured for Remote Copy group 1, only one Ethernet port in each Remote Copy port group on each system actively participates in the IP partnership process. Port selection is determined by a path configuration algorithm. The other ports play the role of standby ports.

If Node A1 fails in System A, the IP partnership selects one of the remaining ports that is configured with Remote Copy port group 1 from any of the nodes from either of the two I/O groups in System A. However, it might take some time (generally seconds) for discovery and path configuration logic to reestablish paths post failover. This process can cause partnerships to change to the Not\_Present state.

This result causes Remote Copy relationships to stop. The administrator might need to manually verify the issues in the event log and start the relationships or Remote Copy consistency groups, if they do not automatically recover. The details of the specific IP port actively participating in the IP partnership process is provided in the **Isportip** view (reported as used). This configuration has the following characteristics:

- Each node has the Remote Copy port group that is configured in both I/O groups. However, only one port in that Remote Copy port group remains active and participates in IP partnership on each system.

- If Node A1 in System A or Node B2 in System B encounter some failure in the system, IP partnerships discovery is triggered and continues servicing the I/O from the failover port.
- The discovery mechanism that is triggered because of failover might introduce a delay where the partnerships momentarily change to the `Not_Present` state and then recover.
- The bandwidth of the single link is used completely.

An eight-node system in IP partnership with four-node system over single inter-site link is shown in Figure 6-36.

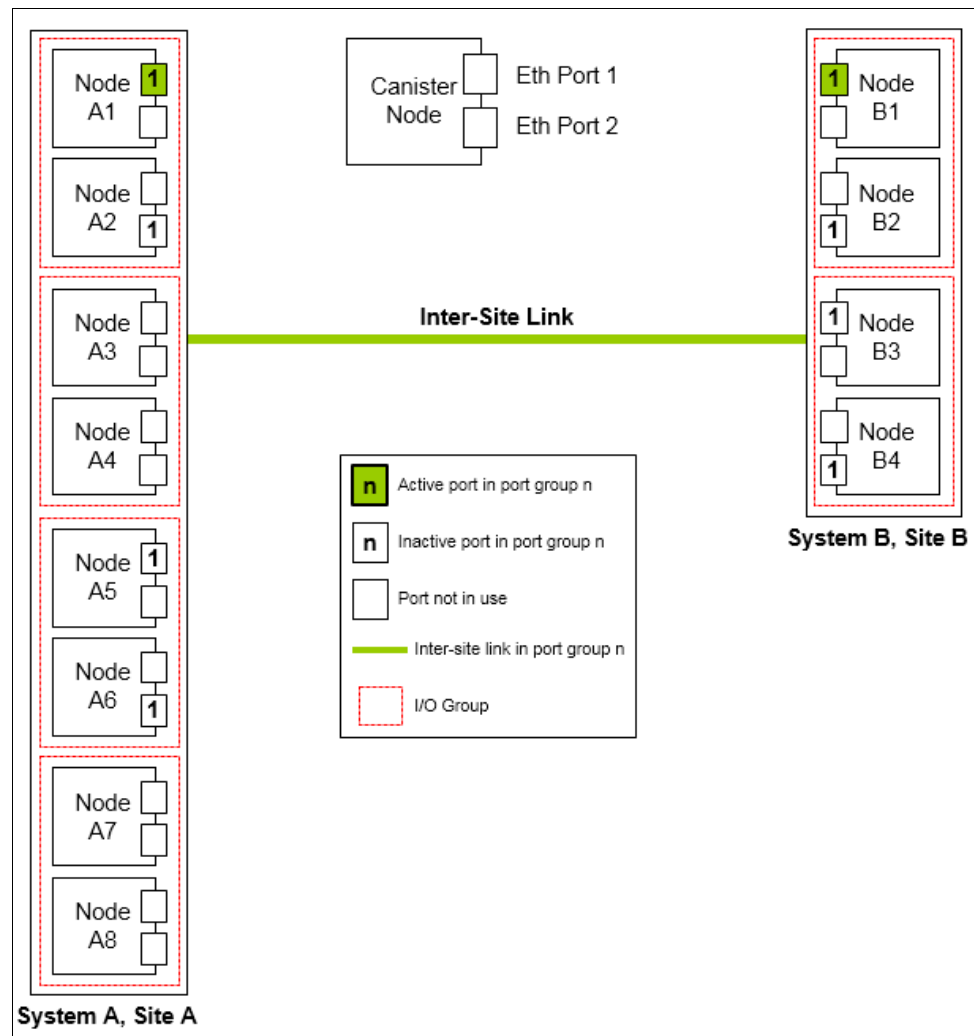


Figure 6-36 Multinode systems single inter-site link with only one Remote Copy port group

Figure 6-36 shows an eight-node system (System A in Site A) and a four-node system (System B in Site B). A single Remote Copy port group 1 is configured on nodes A1, A2, A5, and A6 on System A at Site A. Similarly, a single Remote Copy port group 1 is configured on nodes B1, B2, B3, and B4 on System B.

Although four I/O groups (eight nodes) are in System A, any two I/O groups at maximum are supported to be configured for IP partnerships. If Node A1 fails in System A, IP partnership continues using one of the ports that is configured in Remote Copy port group from any of the nodes from either of the two I/O groups in System A.



However, it might take some time for discovery and path configuration logic to reestablish paths post-failover. This delay might cause partnerships to change to the Not\_Present state.

This process can lead to Remote Copy relationships stopping. The administrator must manually start them if the relationships do not auto-recover. The details of which particular IP port is actively participating in IP partnership process is provided in `lspportip` output (reported as used).

This configuration includes the following characteristics:

- ▶ Each node has the Remote Copy port group that is configured in both the I/O groups that are identified for participating in IP Replication. However, only one port in that Remote Copy port group remains active on each system and participates in IP Replication.
- ▶ If the Node A1 in System A or the Node B2 in System B fails in the system, the IP partnerships trigger discovery and continue servicing the I/O from the failover ports.
- ▶ The discovery mechanism that is triggered because of failover might introduce a delay where the partnerships momentarily change to the Not\_Present state and then recover.
- ▶ The bandwidth of the single link is used completely.

## Two inter-site link configurations

A two 2-node system with two inter-site links configuration is depicted in Figure 6-37.

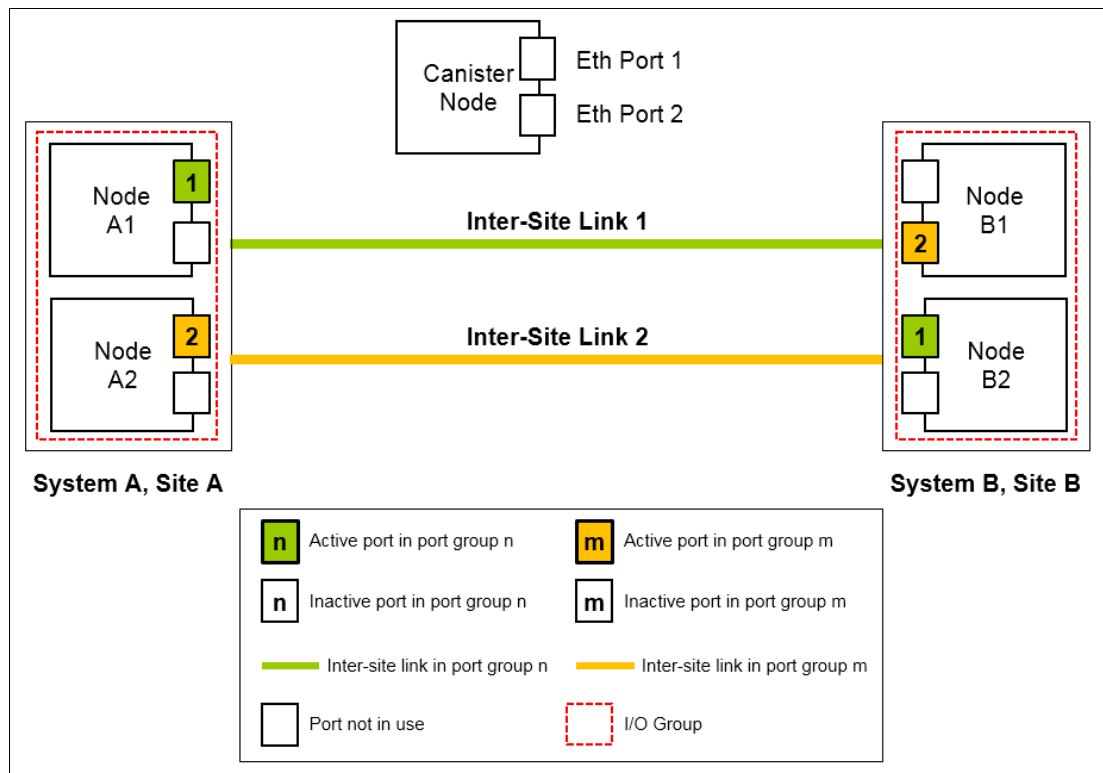


Figure 6-37 Dual links with two Remote Copy groups on each system configured

As shown in Figure 6-37 on page 313, Remote Copy port groups 1 and 2 are configured on the nodes in System A and System B because two inter-site links are available. In this configuration, the failover ports are not configured on partner nodes in the I/O group. Rather, the ports are maintained in different Remote Copy port groups on both of the nodes. They can remain active and participate in IP partnership by using both of the links.

However, if either of the nodes in the I/O group fail (that is, if Node A1 on System A fails), the IP partnership continues from only the available IP port that is configured in Remote Copy port group 2. Therefore, the effective bandwidth of the two links is reduced to 50% because only the bandwidth of a single link is available until the failure is resolved.

This configuration includes the following characteristics:

- ▶ Two inter-site links exist, and two Remote Copy port groups are configured.
- ▶ Each node has only one IP port in Remote Copy port group 1 or 2.
- ▶ Both IP ports in the two Remote Copy port groups participate simultaneously in IP partnerships. Therefore, both of the links are used.
- ▶ During node failure or link failure, the IP partnership traffic continues from the other available link and the port group. Therefore, if two links of 10 Mbps each are available and you have 20 Mbps of effective link bandwidth, bandwidth is reduced to 10 Mbps only during a failure.
- ▶ After the node failure or link failure is resolved and failback happens, the entire bandwidth of both of the links is available as before.

A configuration with two 4-node systems in IP partnership with dual inter-site links is shown in Figure 6-38.

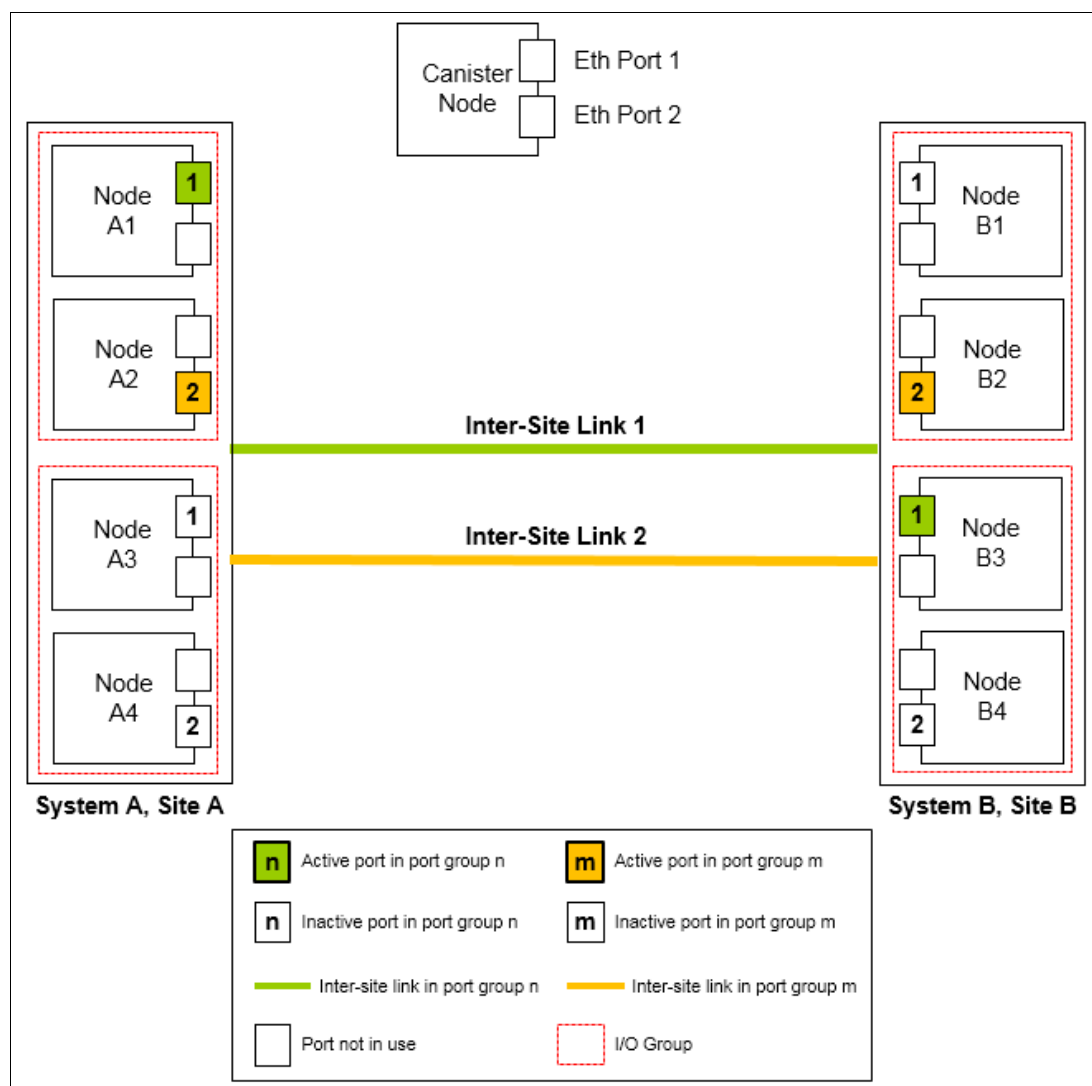


Figure 6-38 Multinode systems with dual inter-site links between the two systems

Figure 6-38 shows two 4-node systems: System A and System B. This configuration is an extension of the configuration that is shown in Figure 6-37 on page 313 to a multinode, multi-I/O group environment.

As seen in this configuration, two I/O groups are used. Each node in the I/O group features a single port that is configured in Remote Copy port groups 1 or 2.

Although two ports are configured in Remote Copy port groups 1 and 2 on each system, only one IP port in each Remote Copy port group on each system actively participates in IP partnership. The other ports that are configured in the same Remote Copy port group act as standby ports during a failure. Which port in a configured Remote Copy port group participates in IP partnership at any moment is determined by a path configuration algorithm.

In this configuration, if Node A1 fails in System A, IP partnership traffic continues from Node A2 (that is, Remote Copy port group 2). At the same time, the failover also causes discovery in Remote Copy port group 1. Therefore, the IP partnership traffic continues from Node A3 on which Remote Copy port group 1 is configured. The details of the particular IP port that is actively participating in IP partnership process is provided in the **lspportip** output (reported as used).

This configuration includes the following characteristics:

- ▶ Each node has the Remote Copy port group that is configured in the I/O groups 1 or 2. However, only one port per system in both Remote Copy port groups remains active and participates in IP partnership.
- ▶ Only a single port per system from each configured Remote Copy port group participates simultaneously in IP partnership. Therefore, both of the links are used.
- ▶ During node failure or port failure of a node that is actively participating in IP partnership, IP partnership continues from the alternative port because another port is in the system in the same Remote Copy port group, but in a different I/O Group.
- ▶ The pathing algorithm can start discovery of available port in the affected Remote Copy port group in the second I/O group and pathing is reestablished. This process restores the total bandwidth; therefore, both of the links are available to support IP partnership.

Finally, an eight-node system in IP partnership with a four-node system over dual inter-site links is shown in Figure 6-39.

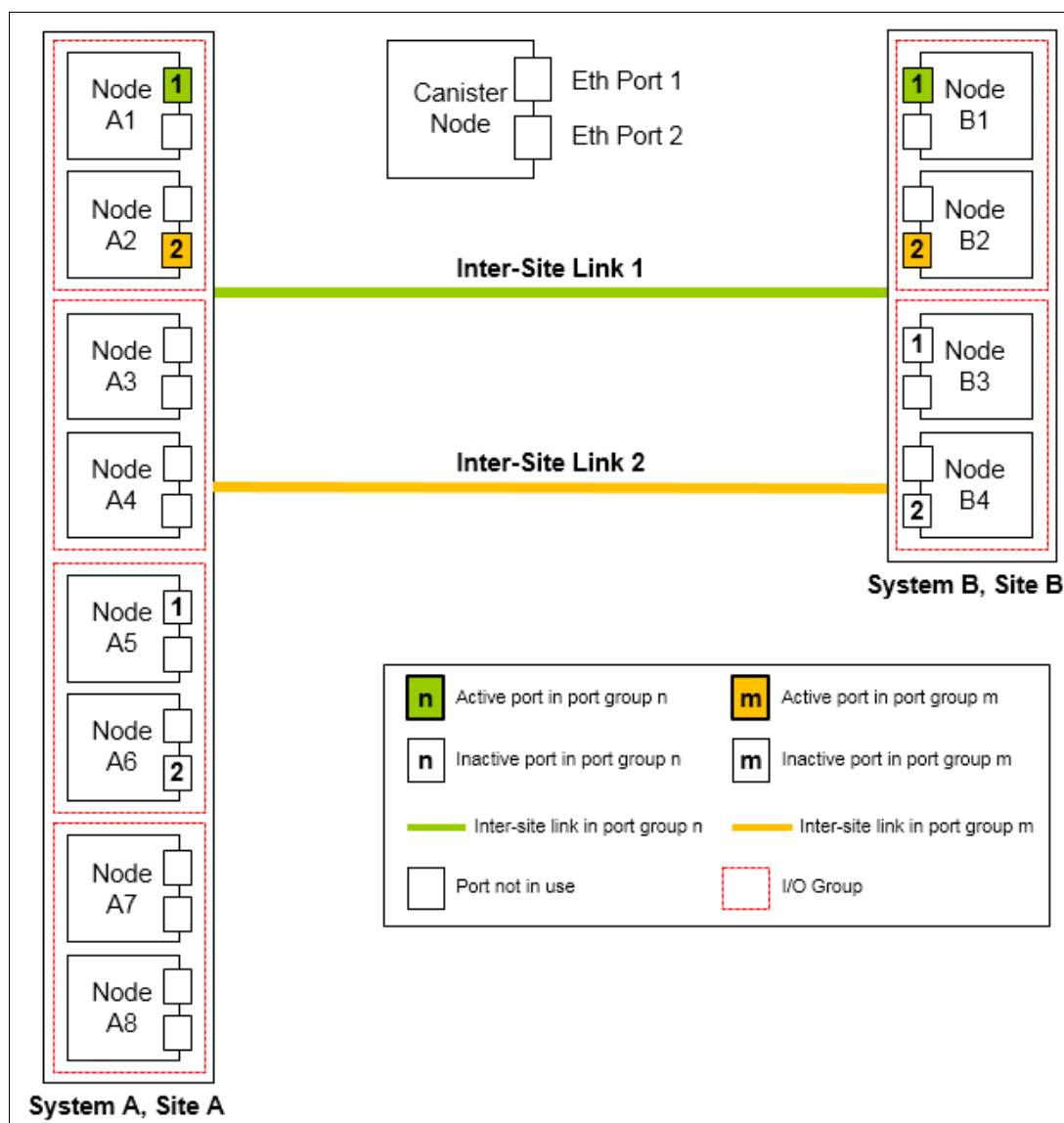


Figure 6-39 Multinode systems with dual inter-site links between the two systems

Figure 6-39 shows an eight-node System A in Site A and a four-node System B in Site B. Because a maximum of two I/O groups in IP partnership is supported in a system (although four I/O groups (eight nodes) are used), nodes from only two I/O groups are configured with Remote Copy port groups in System A. The remaining or all of the I/O groups can be configured to be Remote Copy partnerships over FC.

In this configuration, two links and two I/O groups are configured with Remote Copy port groups 1 and 2. However, path selection logic is managed by an internal algorithm. Therefore, this configuration depends on the pathing algorithm to decide which of the nodes actively participate in IP partnership. Even if Node A5 and Node A6 are configured with Remote Copy port groups properly, active IP partnership traffic on both of the links can be driven from Node A1 and Node A2 only.

If Node A1 fails in System A, IP partnership traffic continues from Node A2 (that is, Remote Copy port group 2). The failover also causes IP partnership traffic to continue from Node A5 on which Remote Copy port group 1 is configured. The details of the particular IP port actively participating in IP partnership process is provided in the **lspportip** output (reported as used).

This configuration includes the following characteristics:

- ▶ Two I/O Groups with nodes are in those I/O groups that are configured in two Remote Copy port groups because two inter-site links for participating in IP partnership are used. However, only one port per system in a particular Remote Copy port group remains active and participates in IP partnership.
- ▶ One port per system from each Remote Copy port group participates in IP partnership simultaneously. Therefore, both of the links are used.
- ▶ If a node or port on the node that is actively participating in IP partnership fails, the Remote Copy (RC) data path is established from that port because another port is available on an alternative node in the system with the same Remote Copy port group.
- ▶ The path selection algorithm starts discovery of available ports in the affected Remote Copy port group in the alternative I/O groups and paths are reestablished. This process restores the total bandwidth across both links.
- ▶ The remaining or all of the I/O groups can be in Remote Copy partnerships with other systems.

**Remote port copy groups:** As described in this section, configuring two Remote Copy port group provides more bandwidth and resilient configurations in case of a link failure. Two Remote Copy port groups can be configured also with a single physical link. This configuration make sense only if the total link bandwidth exceeds the aggregate bandwidth of two Remote Copy port groups together. Creating two Remote Copy port groups when the link bandwidth does not provide the aggregate throughput can lead to network resources contention and bad link performance.

### 6.4.7 Native IP replication performance consideration

Several factors affect the performance of an IP partnership. Some of these factors are latency, link speed, number of intersite links, host I/O, MDisk latency, and hardware. Since the introduction, many improvements were made to make IP replication better performing and more reliable.

Nevertheless, in the presence of poor quality networks that experience significant packet loss and high latency, the usable bandwidth might decrease considerably.

Figure 6-40 shows the throughput trend for a 1 Gbps port in respect of the packet loss ratio and the latency.

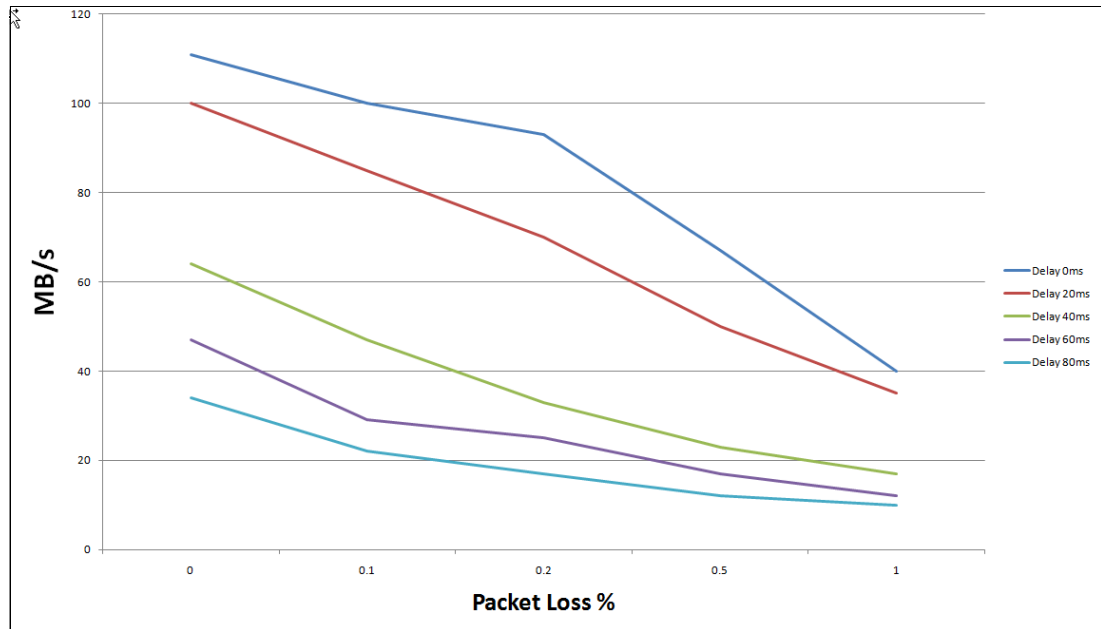


Figure 6-40 1 Gbps port throughput trend

The chart shows how the combined effect of the packet loss and the latency can lead to a throughput reduction of more than 85%. For these reasons, the IP replication option should only be considered for the replication configuration not affected by poor quality and poor performing networks. Because of its characteristic of low-bandwidth requirement, the Global Mirror Change Volume is the preferred solution with the IP replication.

The following recommendations might help improve this performance when compression and IP partnership are used in the same system:

- ▶ If you require more than 100 MBps throughput per intersite link with IP partnership on a node that uses compression, consider IBM SAN Volume Controller nodes SV1 or newer.
- ▶ Use a different port for iSCSI host I/O and IP partnership traffic. Also, use a different VLAN ID for iSCSI host I/O and IP partnership traffic.

## 6.5 Volume Mirroring

By using Volume Mirroring, you can have two physical copies of a volume that provide a basic RAID-1 function. These copies can be in the same storage pool or in different storage pools, with different extent sizes of the storage pool. Typically the two copies are allocated in different storage pools.

The first storage pool contains the original (primary volume copy). If one storage controller or storage pool fails, a volume copy is not affected if it has been placed on a different storage controller or in a different storage pool.

If a volume is created with two copies, both copies use the same virtualization policy. However, you can have two copies of a volume with different virtualization policies. In combination with *thin-provisioning*, each mirror of a volume can be thin-provisioned, compressed or fully allocated, and in striped, sequential, or image mode.

A mirrored (secondary) volume has all of the capabilities of the primary volume copy. It also has the same restrictions (for example, a mirrored volume is owned by an I/O Group, just as any other volume). This feature also provides a *point-in-time copy* function that is achieved by “splitting” a copy from the volume. However, the mirrored volume does not address other forms of mirroring based on Remote Copy (Global or Metro Mirror functions), which mirrors volumes across I/O Groups or clustered systems.

One copy is the primary copy, and the other copy is the secondary copy. Initially, the first volume copy is the primary copy. You can change the primary copy to the secondary copy if required.

Figure 6-41 provides an overview of Volume Mirroring.

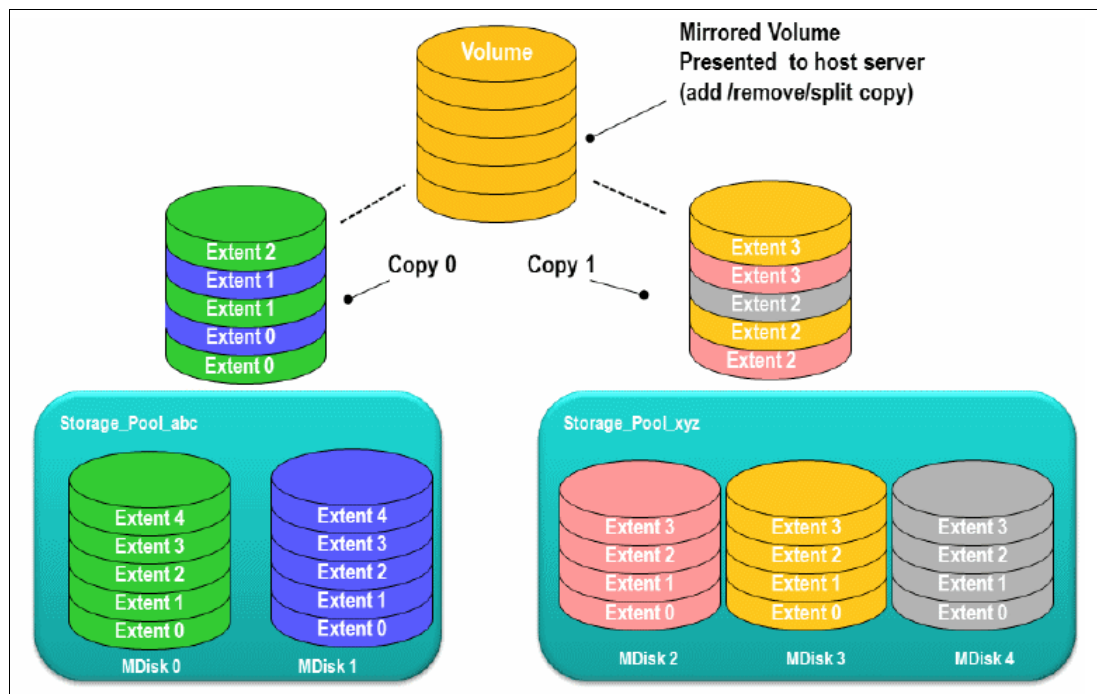


Figure 6-41 Volume Mirroring overview

### 6.5.1 Read and write operations

Read and write operations behavior depends on the status of the copies and on other environment settings. During the initial synchronization or a resynchronization, only one of the copies is in synchronized status, and all the reads are directed to this copy. The write operations are directed to both copies.

When both copies are synchronized, the write operations are again directed to both copies. The read operations usually are directed to the primary copy, unless the system is configured in Enhanced Stretched Cluster topology. With this system topology and the enablement of site awareness capability, the concept of primary copy still exists, but is not more relevant. The read operation follows the site affinity.

For example, consider an Enhanced Stretched Cluster (ESC) configuration with mirrored volumes with one copy in Site A and the other in Site B. If a host I/O read is attempted to a mirrored disk through a Spectrum Virtualize Node in Site A, then the I/O read is directed to the copy in Site A, if available. Similarly, a host I/O read attempted through a node in Site B goes to the Site B copy.



**Important:** With IBM SAN Volume Controller ESC, keep consistency between Hosts, Nodes, and Storage Controller site affinity as long as possible to ensure the best performance.

During back-end storage failure, consider the following points:

- ▶ If one of the mirrored volume copies is temporarily unavailable, the volume remains accessible to servers.
- ▶ The system remembers which areas of the volume are written and resynchronizes these areas when both copies are available.
- ▶ The remaining copy can service read I/O when the failing one is offline, without user intervention.

## 6.5.2 Volume mirroring use cases

Volume Mirroring offers the capability to provide extra copies of the data that can be used for High Availability solutions and data migration scenarios. You can convert a non-mirrored volume into a mirrored volume by adding a copy. When a copy is added using this method, the cluster system synchronizes the new copy so that it is the same as the existing volume. You can convert a mirrored volume into a non-mirrored volume by deleting one copy or by splitting one copy to create a new non-mirrored volume.

**Access:** Servers can access the volume during the synchronization processes described.

You can use mirrored volumes to provide extra protection for your environment or to perform a migration. This solution offers several options:

- ▶ Stretched Cluster configurations

Standard and Enhanced Stretched Cluster IBM SAN Volume Controller configuration uses the Volume Mirroring feature to implement the data availability across the sites.

- ▶ Export to Image mode

This option allows you to move storage from *managed mode* to *image mode*. This option is useful if you use IBM SAN Volume Controller as a migration device. For example, suppose vendor A's product cannot communicate with vendor B's product; however, you must migrate data from vendor A to vendor B.

The use of Export to image mode allows you to migrate data by using the Copy Services functions and then return control to the native array, while maintaining access to the hosts.

- ▶ Import to Image mode

This option allows you to import an existing storage MDisk or logical unit number (LUN) with its existing data from an external storage system, without putting metadata on it. The existing data remains intact. After you import it, the volume mirroring function can be used to migrate the storage to the other locations, while the data remains accessible to your hosts.

- ▶ Volume cloning using Volume Mirroring and then using the Split into New Volume option

This option allows any volume to be cloned without any interruption to the host access. You have to create two mirrored copies of data and then break the mirroring with the split option to make two independent copies of data. This option does not apply to already mirrored volumes.

- Volume pool migration using the volume mirroring option

This option allows any volume to be moved between storage pools without any interruption to the host access. You might use this option to move volumes as an alternative to the Migrate to Another Pool function. Compared to the Migrate to Another Pool function, volume mirroring provides more manageability because it can be suspended and resumed anytime, and also it allows you to move volumes among pools with different extent sizes. This option does not apply to mirrored volumes.

**Use Case:** Volume Mirroring can be used to migrate volumes from and to DRP, which do not support extent based migrations. See 4.3.7, “Data migration with DRP” on page 125.

- Volume capacity saving change

This option allows you to modify the capacity saving characteristics of any volume from standard to thin provisioned or compressed and vice versa, without any interruption to host access. This option works the same as the volume pool migration but specifying a different capacity saving for the newly created copy. This option does not apply to mirrored volumes.

When you use Volume Mirroring, consider how quorum candidate disks are allocated. Volume Mirroring maintains some state data on the quorum disks. If a quorum disk is not accessible and Volume Mirroring is unable to update the state information, a mirrored volume might need to be taken offline to maintain data integrity. To ensure the high availability of the system, ensure that multiple quorum candidate disks, which are allocated on different storage systems, are configured.

**Quorum disk consideration:** Mirrored volumes can be taken offline if there is no quorum disk available. This behavior occurs because synchronization status for mirrored volumes is recorded on the quorum disk. To protect against mirrored volumes being taken offline, follow the guidelines for setting up quorum disks.

The following are other Volume Mirroring usage cases and characteristics:

- Creating a mirrored volume:
  - The maximum number of copies is two.
  - Both copies are created with the same virtualization policy by default.

To have a volume mirrored using different policies, you need to add a volume copy with a different policy to a volume that has only one copy.
  - Both copies can be located in different storage pools. The first storage pool that is specified contains the primary copy.
  - It is not possible to create a volume with two copies when specifying a set of MDisk.
- Add a volume copy to a volume:
  - The volume copy to be added can have a different space allocation policy.
  - Two existing volumes with one copy each cannot be merged into a single mirrored volume with two copies.
- Remove a volume copy from a mirrored volume:
  - The volume remains with only one copy.
  - It is not possible to remove the last copy from a volume.

- ▶ Split a volume copy from a mirrored volume and create a new volume with the split copy:
    - This function is only allowed when the volume copies are synchronized. Otherwise, use the **-force** command.
    - It is not possible to recombine the two volumes after they have been split.
    - Adding and splitting in one workflow enables migrations that are not currently allowed.
    - The split volume copy can be used as a means for creating a point-in-time copy (clone).
  - ▶ Repair/validate in three ways. This compares volume copies and performs these functions:
    - Reports the first difference found. It can iterate by starting at a specific LBA by using the **-startlba** parameter.
    - Creates virtual medium errors where there are differences. This is a useful in case of back-end data corruption.
    - Corrects the differences that are found (reads from primary copy and writes to secondary copy).
  - ▶ View to list volumes affected by a back-end disk subsystem being offline:
    - Assumes that a standard use is for mirror between disk subsystems.
    - Verifies that mirrored volumes remain accessible if a disk system is being shut down.
    - Reports an error in case a quorum disk is on the back-end disk subsystem.
  - ▶ Expand or shrink a volume:
    - This function works on both of the volume copies at once.
    - All volume copies always have the same size.
    - All copies must be synchronized before expanding or shrinking them.
- DRP limitation:** DRP do not support shrinking thin or compressed volumes.
- ▶ Delete a volume. When a volume gets deleted, all copies get deleted.
  - ▶ Migration commands apply to a specific volume copy.
  - ▶ Out-of-sync bitmaps share the bitmap space with FlashCopy and Metro Mirror/Global Mirror. Creating, expanding, and changing I/O groups might fail if there is insufficient memory.
  - ▶ GUI views contain volume copy identifiers.

### 6.5.3 Mirrored volume components

Consider the following points regarding mirrored volume components:

- ▶ A mirrored volume is always composed of two copies (copy 0 and copy1).
- ▶ A volume that is not mirrored consists of a single copy (which for reference might be copy 0 or copy 1).

A mirrored volume looks the same to upper-layer clients as a non-mirrored volume. That is, upper layers within the cluster software, such as FlashCopy and Metro Mirror/Global Mirror, and storage clients, do not know whether a volume is mirrored. They all continue to handle the volume as they did before without being aware of whether the volume is mirrored.

Figure 6-42 shows the attributes of a volume and Volume Mirroring.

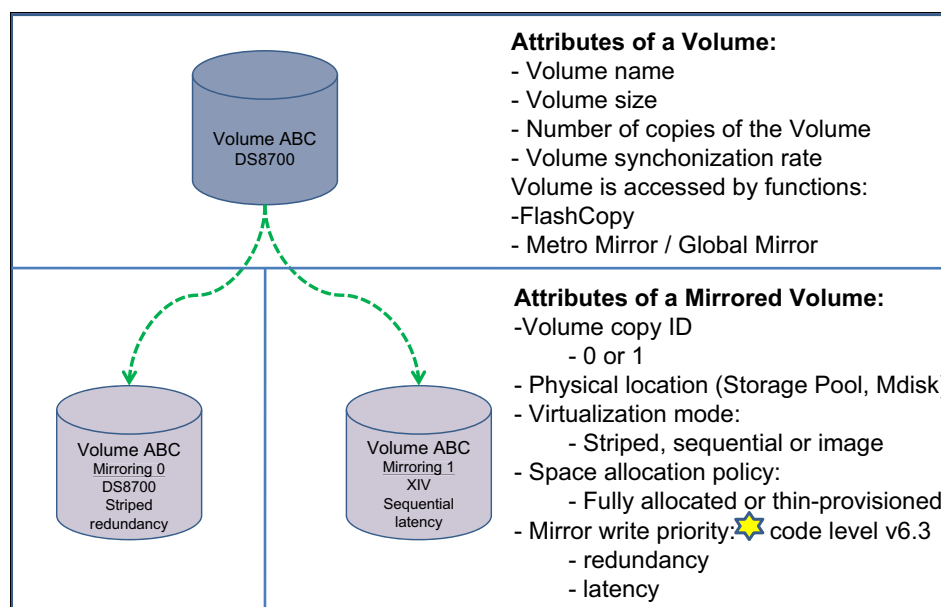


Figure 6-42 Attributes of a volume and Volume Mirroring

In Figure 6-42, XIV and IBM DS8700 illustrate that a mirrored volume can use different storage devices.

## 6.5.4 Volume Mirroring synchronization options

As soon as a volume is created with two copies, copies are in the *out-of-sync* state. The primary volume copy (located in the first specified storage pool) is defined as in sync and the secondary volume copy as out of sync. The secondary copy is synchronized through the synchronization process.

This process runs at the default synchronization rate of 50 (see Table 6-10 on page 325), or at the defined rate while creating or modifying the volume (see 6.5.5, “Volume Mirroring performance considerations” on page 325 for the effect of the copy rate setting). After the synchronization process is completed, the volume mirroring copies are *in-sync* state.

By default, a format process is started when a mirrored volume is created. This process ensures that the volume data is zeroed, which avoids in this way to access to data still present on reused extents.

This format process runs in background at the defined synchronization rate (see Table 6-10 on page 325). Before Spectrum Virtualize version 8.4, the format processing overwrite with zeros only the Copy 0 and then synchronize the Copy 1. With version 8.4 or later, the format process is started concurrently to both volume mirroring copies, which avoids the second synchronization step.

You can specify that a volume is synchronized (**-createsync** parameter), even if it is not. The use of this parameter can cause data corruption if the primary copy fails and leaves an unsynchronized secondary copy to provide data.

The use of this parameter also can cause loss of read stability in unwritten areas if the primary copy fails, data is read from the primary copy, and then different data is read from the secondary copy.

To avoid data loss or read stability loss, use this parameter only for a primary copy that is formatted and not written to. When the **-createsync** setting is used, the initial formatting is also skipped.

Another example use case for **-createsync** is for a newly created mirrored volume where both copies are thin provisioned or compressed because no data has been written to disk and unwritten areas return zeros (0). If the synchronization between the volume copies has been lost, the resynchronization process is incremental. This term means that only grains that have been written to need to be copied, and then get synchronized volume copies again.

The progress of the volume mirror synchronization can be obtained from the GUI or by using the **lsvdisksyncprogress** command.

## 6.5.5 Volume Mirroring performance considerations

Because the writes of mirrored volumes always occur to both copies, mirrored volumes put more workload on the cluster, the back-end disk subsystems, and the connectivity infrastructure. The mirroring is symmetrical, and writes are only acknowledged when the write to the last copy completes. The result is that if the volumes copies are on storage pools with different performance characteristics, the slowest storage pool determines the performance of writes to the volume. This performance applies when writes must be destaged to backend.

**Tip:** Locate volume copies of one volume on storage pools of the same or similar characteristics. Usually, if only good read performance is required, you can place the primary copy of a volume in a storage pool with better performance. Because the data is always only read from one volume copy, reads are not faster than without Volume Mirroring.

However, be aware that this is only true when both copies are synchronized. If the primary is out of sync, reads are submitted to the other copy.

Synchronization between volume copies has a similar impact on the cluster and the back-end disk subsystems as FlashCopy or data migration. The synchronization rate is a property of a volume that is expressed as a value of 0 - 100. A value of 0 disables synchronization. Table 6-10 shows the relationship between the rate value and the data copied per second.

*Table 6-10 Relationship between the rate value and the data copied per second*

| User-specified rate attribute value per volume | Data copied per second           |
|------------------------------------------------|----------------------------------|
| 0                                              | Synchronization is disabled      |
| 1 - 10                                         | 128 KB                           |
| 11 - 20                                        | 256 KB                           |
| 21 - 30                                        | 512 KB                           |
| 31 - 40                                        | 1 MB                             |
| 41 - 50                                        | 2 MB ** 50% is the default value |
| 51 - 60                                        | 4 MB                             |
| 61 - 70                                        | 8 MB                             |
| 71 - 80                                        | 16 MB                            |
| 81 - 90                                        | 32 MB                            |

| User-specified rate attribute value per volume | Data copied per second |
|------------------------------------------------|------------------------|
| 91 - 100                                       | 64 MB                  |

**Rate attribute value:** The rate attribute is configured on each volume that you want to mirror. The default value of a new volume mirror is 50 percent.

In large, IBM SAN Volume Controller configurations, the settings of the copy rate can considerably affect the performance in scenarios where a back-end storage failure occurs. For example, consider a scenario in which a failure of a back-end storage controller is affecting one copy of 300 mirrored volumes. The host continues the operations by using the remaining copy.

When the failed controller comes back online, the resynchronization process for all the 300 mirrored volumes starts at the same time. With a copy rate of 100 for each volume, this process adds a theoretical workload of 18.75 GBps, which drastically overloads the system.

The general suggestion for the copy rate settings is then to evaluate the effect of massive resynchronization and set the parameter accordingly. Consider setting the copy rate to high values for initial synchronization only, and with a limited number of volumes at a time. Alternatively, consider defining a volume provisioning process that allows the safe creation of already synchronized mirrored volumes, as described in 6.5.4, “Volume Mirroring synchronization options” on page 324.

## Volume mirroring I/O Time-out configuration

A mirrored volume has pointers to the two copies of data, usually in different storage pools, and each write completes on both copies before the host receives I/O completion status. For a synchronized mirrored volume, if a write I/O to a copy has failed or a long timeout has expired, then system has completed all available controller level Error Recovery Procedures (ERPs). In this case, that copy is taken offline and goes out of sync. The volume remains online and continues to service I/O requests from the remaining copy.

The *Fast Failover* feature isolates hosts from temporarily poorly-performing back-end storage of one Copy at the expense of a short interruption to redundancy. The fast failover feature behavior is that during normal processing of host write I/O, the system submits writes to both copies with a timeout of 10 seconds (20 seconds for stretched volumes). If one write succeeds and the other write takes longer than 5 seconds, then the slow write is stopped. The Fibre Channel abort sequence can take around 25 seconds.

When the stop is completed, one copy is marked as out of sync and the host write I/O completed. The overall fast failover ERP aims to complete the host I/O in around 30 seconds (40 seconds for stretched volumes).

The fast failover can be set for *each* mirrored volume by using the **chvdisk** command and the **mirror\_write\_priority** attribute settings:

- ▶ *Latency* (default value): A short timeout prioritizing low host latency. This option enables the fast failover feature.
- ▶ *Redundancy*: A long timeout prioritizing redundancy. This option indicates a copy that is slow to respond to a write I/O can use the full ERP time. The response to the I/O is delayed until it completes to keep the copy in sync if possible. This option disables the fast failover feature.

Volume Mirroring ceases to use the slow copy for 4 - 6 minutes, and subsequent I/O data is not affected by a slow copy. Synchronization is suspended during this period. After the copy suspension completes, Volume Mirroring resumes, which allows I/O data and synchronization operations to the slow copy that will, typically, quickly complete the synchronization.

If another I/O times out during the synchronization, then the system stops using that copy again for 4 - 6 minutes. If one copy is always slow, then the system tries it every 4 - 6 minutes and the copy gets progressively more out of sync as more grains are written. If fast failovers are occurring regularly, there is probably an underlying performance problem with the copy's back-end storage.

The preferred **mirror\_write\_priority** setting for the Enhanced Stretched Cluster configurations is *latency*.

## 6.5.6 Bitmap space for out-of-sync volume copies

The grain size for the synchronization of volume copies is 256 KB. One grain takes up one bit of bitmap space. 20 MB of bitmap space supports 40 TB of mirrored volumes. This relationship is the same as the relationship for copy services (Global and Metro Mirror) and standard FlashCopy with a grain size of 256 KB (Table 6-11).

Table 6-11 Relationship of bitmap space to Volume Mirroring address space

| Function         | Grain size in KB | 1 byte of bitmap space gives a total of | 4 KB of bitmap space gives a total of | 1 MB of bitmap space gives a total of | 20 MB of bitmap space gives a total of | 512 MB of bitmap space gives a total of |
|------------------|------------------|-----------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------|-----------------------------------------|
| Volume Mirroring | 256              | 2 MB of volume capacity                 | 8 GB of volume capacity               | 2 TB of volume capacity               | 40 TB of volume capacity               | 1024 TB of volume capacity              |

**Shared bitmap space:** This bitmap space on one I/O group is shared between Metro Mirror, Global Mirror, FlashCopy, and Volume Mirroring.

The command to create Mirrored Volumes can fail if there is not enough space to allocate bitmaps in the target I/O Group. To verify and change the space allocated and available on each I/O Group with the CLI, see the Example 6-4.

Example 6-4 A *lsiogrp* and *chiogrp* command example

```
IBM_2145:SVC_ESC:superuser>lsiogrp io_grp0
id 0
.
lines removed for brevity
.
remote_copy_free_memory 19.9MB
mirroring_total_memory 20.0MB
mirroring_free_memory 20.0MB
raid_total_memory 40.0MB
raid_free_memory 40.0MB
.
lines removed for brevity
.
```

```

IBM_2145:SVC_ESC:superuser>chiogrp -feature mirror -size 64 io_grp0
IBM_2145:SVC_ESC:superuser>lsiogrp io_grp0
id 0
.
lines removed for brevity
.
remote_copy_free_memory 19.9MB
mirroring_total_memory 64.0MB
mirroring_free_memory 64.0MB
.
lines removed for brevity
.

```

---

To verify and change the space allocated and available on each I/O Group with the GUI, see Figure 6-43.

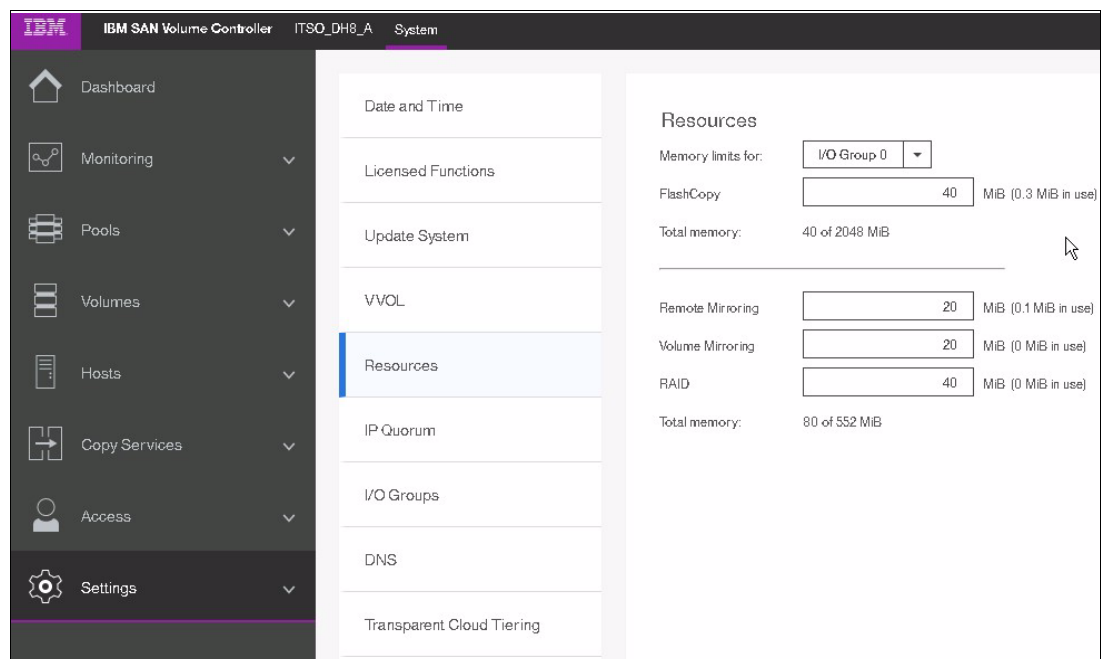


Figure 6-43 IOgrp feature example





## Business continuity

Business continuity (BC) and continuous application availability are among the most important requirements for many organizations. Advances in virtualization, storage, and networking made enhanced business continuity possible. Information technology solutions can now manage both planned and unplanned outages, and provide the flexibility and cost efficiencies that are available from cloud-computing models.

This chapter briefly describes the Stretched Cluster, Enhanced Stretched Cluster, and HyperSwap solutions for IBM Spectrum Virtualize systems. Technical details or implementation guidelines are not presented in this chapter because they are described in separate publications.

**Important:** This book was written specifically for IBM SAN Volume Controller systems. For IBM FlashSystems products, Stretched Cluster and Enhanced Stretched Cluster topologies do not apply. For more information about IBM FlashSystems, see *IBM FlashSystem Best Practices and Performance Guidelines*, SG24-8503.

This book does not cover the three-site replication solutions that are available with the IBM Spectrum Virtualize code version 8.3.1 or later. For more information, see *Spectrum Virtualize 3-Site Replication*, SG24-8474.

This chapter includes the following topics:

- ▶ 7.1, “Business continuity topologies” on page 330
- ▶ 7.2, “Third site and IP quorum” on page 332
- ▶ 7.3, “HyperSwap Volumes” on page 334
- ▶ 7.4, “Comparison of business continuity solutions” on page 335

## 7.1 Business continuity topologies

IBM SAN Volume Controller systems support three different cluster topologies: Standard, Stretched, and HyperSwap. In this chapter, we describe the Stretched and HyperSwap topologies that provide high availability (HA) functions.

### 7.1.1 Business continuity with Stretched Cluster

Within standard implementations of IBM Spectrum Virtualize, all controller nodes are physically installed in the same location. To fulfill the different HA needs of customers, the Stretched Cluster configuration was introduced, in which each node from the same I/O Group is physically installed at a different site.

When implemented, this configuration can be used to maintain access to data on the system, even if failures occur at different levels, such as the SAN, back-end storage, IBM Spectrum Virtualize controller, or data center power.

Stretched Cluster is considered a HA solution because both sites work as instances of the production environment (no standby location is used). Combined with application and infrastructure layers of redundancy, Stretched Clusters can provide enough protection for data that requires availability and resiliency.

When IBM Spectrum Virtualize was first introduced, the maximum supported distance between nodes within an I/O Group was 100 meters (328 feet). With the evolution of code and the introduction of new features, stretched cluster configurations were enhanced to support distances up to 300 km (186.4 miles). These geographically dispersed solutions use specific configurations that use Fibre Channel (FC) or Fibre Channel over IP (FC/IP) switch, or Multiprotocol Router (MPR) inter-switch links (ISLs) between different locations.

### 7.1.2 Business continuity with Enhanced Stretched Cluster

IBM Spectrum Virtualize V7.2 introduced the Enhanced Stretched Cluster feature that further improved the Stretched Cluster configurations. The Enhanced Stretched Cluster introduced the *site awareness* concept for nodes and external storage, and the Disaster Recovery (DR) feature that enables you to effectively manage rolling disaster scenarios.

With IBM Spectrum Virtualize V7.5, the site awareness concept was extended to hosts. This extension enables more efficiency for host I/O traffic through the SAN, and easier host path management.

Stretched Cluster and Enhanced Stretched Cluster solutions can also be combined with other replication techniques, such as Metro Mirror or Global Mirror, which allows a three-site configuration for HA and DR purposes.

In a stretched system configuration, each site is defined as an independent failure domain. If one site experiences a failure, the other site can continue to operate without disruption.

You also must configure a third site to host a quorum device that provides an automatic tie-break if a link failure occurs between the two main sites (for more information, see 7.2, “Third site and IP quorum” on page 332). The main site can be in the same room or across rooms in the data center, buildings on the same campus, or buildings in different cities. Different kinds of sites protect against different types of failures.

### Two sites within a single location

If each site is a different power phase within a single location or data center, the system can survive the failure of any single power domain. For example, one node can be placed in one rack installation and the other node can be in another rack. Each rack is considered a separate site with its own power phase. In this case, if power was lost to one of the racks, the partner node in the other rack can be configured to process requests and effectively provide availability to data, even when the other node is offline because of a power disruption.

### Two sites at separate locations

If each site is a different physical location, the system can survive the failure of any single location. These sites can span shorter distances (for example, two sites in the same city), or they can be spread farther geographically, such as two sites in separate cities. If one site experiences a site-wide disaster, the other site can remain available to process requests.

If configured correctly, the system continues to operate after the loss of one site. The key prerequisite is that each site contains only one node from each I/O group. However, placing one node from each I/O group in different sites for a stretched system configuration does *not* provide HA. You must also configure the suitable mirroring technology and ensure that all configuration requirements for those technologies are correctly configured.

**Note:** For best results, configure an enhanced stretched system to include at least two I/O groups (four nodes). A system with only one I/O group cannot maintain mirroring of data or uninterrupted host access in the presence of node failures or system updates.

## 7.1.3 Business continuity with HyperSwap

The HyperSwap HA feature that is available in the IBM Spectrum Virtualize and FlashSystems products enables business continuity during a hardware failure, power outage, connectivity problem, or other disasters, such as fire or flooding.

It provides highly available volumes that are accessible through two sites that are up to 300 km (186.4 miles) apart. A fully independent copy of the data is maintained at each site.

When data is written by hosts at either site, both copies are synchronously updated before the write operation is completed. HyperSwap automatically optimizes itself to minimize data that is transmitted between sites, and to minimize host read and write latency. For more information about the optimization algorithm, see 7.3, “HyperSwap Volumes” on page 334.

HyperSwap includes the following key features:

- ▶ Works with all IBM Spectrum Virtualize products, except for IBM FlashSystem 5010.
- ▶ Uses intra-cluster synchronous Remote Copy (named Active-Active Metro Mirror) capability along with change volumes and access I/O group technologies.
- ▶ Makes a host's volumes accessible across two IBM Spectrum Virtualize I/O groups in a clustered system by using the Active-Active Metro Mirror relationship. The volumes are presented as a single volume to the host.
- ▶ Works with the standard multipathing drivers that are available on various host types, with no other host support required to access the highly available volumes.

The IBM Spectrum Virtualize HyperSwap configuration requires that at least one control enclosure is implemented in each location. Therefore, a minimum of two control enclosures for each cluster are needed to implement HyperSwap. Configuration with three or four control enclosures is also supported for the HyperSwap.

## 7.2 Third site and IP quorum

In stretched cluster or HyperSwap configurations, you can use a third, independent site to house a quorum device to act as the tie-breaker in case of split-brain scenarios. The quorum device can also hold a backup copy of the cluster metadata to be used in specific situations that might require a full cluster recovery.

To use a quorum disk as the quorum device, this third site must have FC or iSCSI connectivity between an external storage system and the IBM Spectrum Virtualize cluster. Sometimes, this third site quorum disk requirement can be expensive in terms of infrastructure and network costs. For this reason, a less demanding solution that is based on a Java application was introduced with the release V7.6, and is known as *IP quorum application*.

Initially, IP quorum was used only as a tie-breaker solution. However, it was expanded to store cluster configuration metadata with the release V8.2.1, fully serving as an alternative for quorum disk devices. To use an IP quorum application as the quorum device for the third site, no FC connectivity is used. It can be run on any host at the third site, as shown in the Figure 7-1.

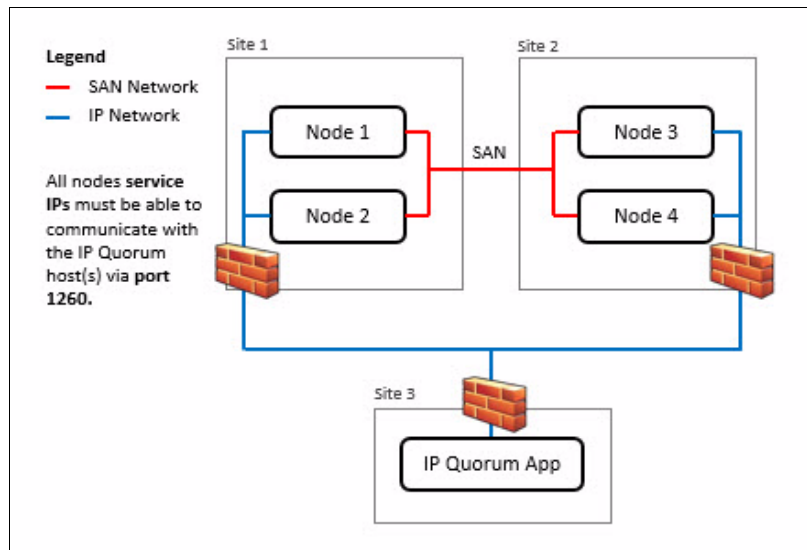


Figure 7-1 IP Quorum network layout

However, the following strict requirements on the IP network must be met when using it:

- Connectivity must exist from the servers that are running an IP quorum application to the service IP addresses of all nodes or node canisters. The network must also deal with possible security implications of exposing the service IP addresses because this connectivity also can be used to access the service assistant interface if the IP network security is configured incorrectly.
- On each server that runs an IP quorum application, ensure that only authorized users can access the directory that contains the IP quorum application. Because metadata is stored in the directory in a readable format, ensure that access to the IP quorum application and the metadata is restricted to authorized users only.
- Port 1260 is used by the IP quorum application to communicate from the hosts to all nodes or enclosures.
- The maximum round-trip delay must not exceed 80 milliseconds (ms), which means 40 ms each direction.

- ▶ If you are configuring the IP quorum application without a quorum disk for metadata, a minimum bandwidth of 2 MBps is ensured for traffic between the system and the quorum application. If your system uses an IP quorum application with quorum disk for metadata, a minimum bandwidth of 64 MBps is ensured for traffic between the system and the quorum application.
- ▶ Ensure that the directory that stores an IP quorum application with metadata contains at least 250 MB of available capacity.

Quorum devices are also required at the sites 1 and 2, and can be disk-based quorum devices or IP quorum applications. The maximum number of IP quorum applications that can be deployed is five.

**Important:** Do *not* host the quorum disk devices or IP quorum applications on storage that is provided by the system it is protecting because this storage is paused for I/O in a tie-break situation.

For more information about IP Quorum requirements and installation, including supported Operating Systems and Java runtime environments (JREs), see [IBM Documentation](#).

**Note:** The IP Quorum configuration process was integrated into the IBM Spectrum Virtualize GUI and can be found at **Settings** → **Systems** → **IP Quorum**.

For more information about quorum disk devices, see 3.3, “Quorum disks” on page 89.

## 7.2.1 Quorum modes

With the release of IBM Spectrum Virtualize V8.3, a new configuration option was added to the IP Quorum functions, called *quorum mode*. By default, the IP quorum mode is set to Standard. In Stretched or HyperSwap clusters, this mode can be changed to Preferred or Winner.

This configuration allows you to specify which site resumes I/O after a disruption, based on the applications that run on each site or other factors. For example, you can specify whether a selected site is the preferred for resuming I/O, or if the site automatically “wins” in tie-break scenarios.

### Preferred mode

If only one site runs critical applications, you can configure this site as *Preferred*. During a split-brain situation, the system delays processing tie-break operations on other sites that are not specified as “preferred”. That is, the designated preferred site has a timed advantage when a split-brain situation is detected, and starts racing for the quorum device a few seconds before the non-preferred sites. Therefore, the likelihood of reaching the quorum device first is higher. If the preferred site is damaged or cannot reach the quorum device, the other sites can attempt to win the tie-break and continue I/O.

### Winner mode

This configuration is recommended for use when no third site is available for a quorum device to be installed. In this case, when a split-brain situation is detected, the site that is configured as the winner always is the site to continue processing I/O, regardless of the failure and its condition. The nodes at the non-winner site always lose the tie-break and stop processing I/O requests until the fault is fixed.

## 7.3 HyperSwap Volumes

*HyperSwap Volumes* consist of a Master Volume and a Master Change Volume (CV) in one site, and an Auxiliary Volume and an Auxiliary CV in the other system site. An active-active synchronous mirroring relationship exists between the two sites. As with a regular Metro Mirror relationship, the active-active relationship keeps the Master Volume and Auxiliary Volume synchronized.

The relationship uses the CVs as journaling volumes during any resynchronization process. The Master CV must be in the same I/O Group as the Master Volume. It also is recommended that it is in the same pool as the Master Volume. A similar practice applies to the Auxiliary CV and the Auxiliary Volume. For more information about the Change Volume, see “Global Mirror functional overview” on page 256.

The HyperSwap Volume always uses the unique identifier (UID) of the Master Volume. The HyperSwap Volume is assigned to the host by mapping only the Master Volume, even though access to the Auxiliary Volume is ensured by the HyperSwap function.

Figure 7-2 shows how the HyperSwap Volume is implemented.

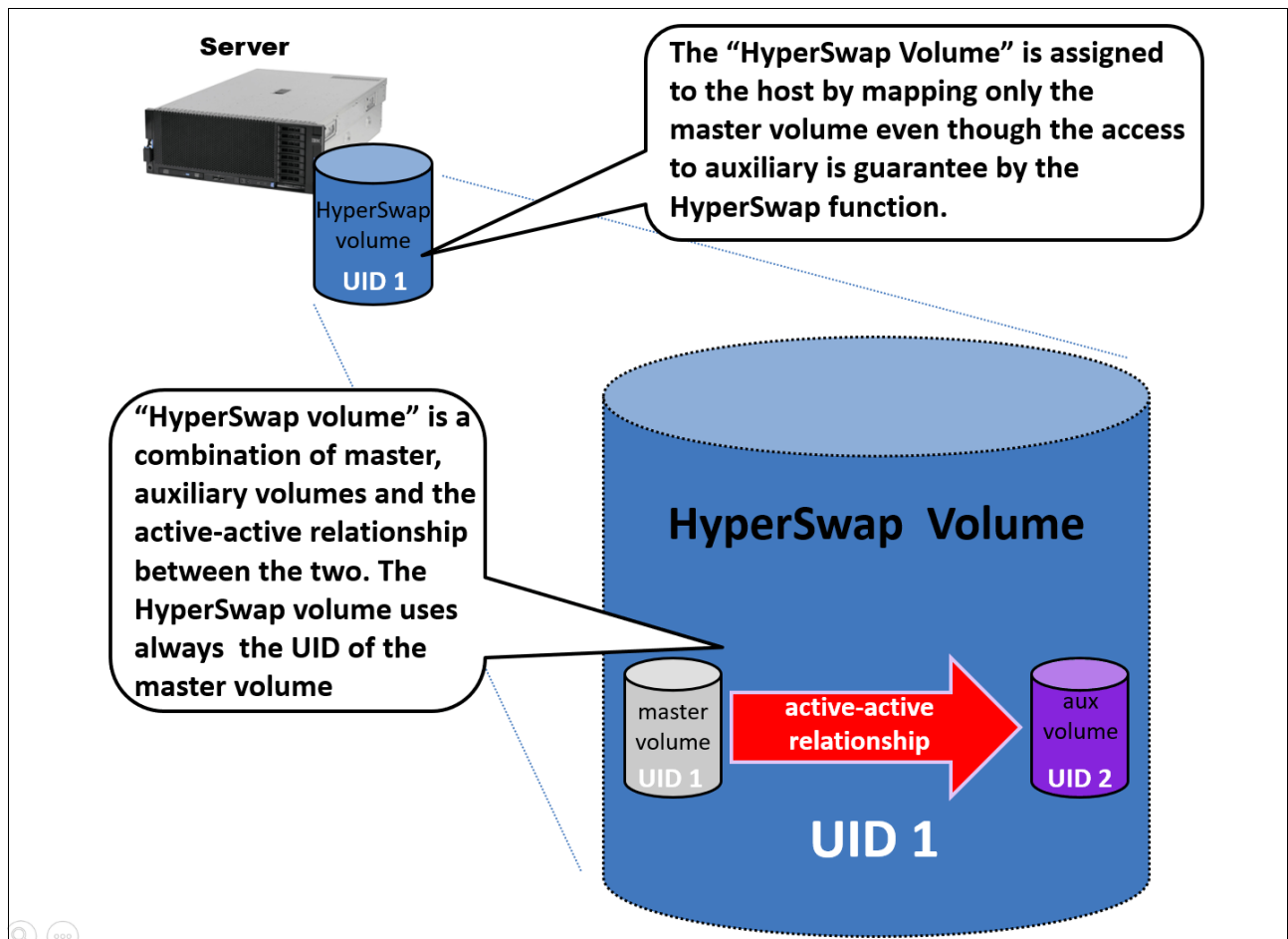


Figure 7-2 HyperSwap Volume

The active-active synchronous replication workload traverses the SAN by using the node-to-node communication. Master and Auxiliary Volumes also have a specific role of Primary or Secondary. Master or Auxiliary Volumes are Primary or Secondary, based on the Metro Mirror active-active relationship direction.

Starting with the IBM Spectrum Virtualize 8.3.1 code level, reads are always done in the local copy of the volume. Write operations are always routed to the Primary copy. Therefore, hosts that access the Secondary copy for writes might experience an increased latency in the I/O operations. As a mitigation of this behavior, if sustained workload (that is, more than 75% of I/O operations for at least 20 minutes) is running over Secondary volumes, the HyperSwap function switches the direction of the active-active relationships, which swaps the Secondary volume to Primary and vice versa.

**Note:** Frequent or continuous primary to secondary volume swap can lead to performance degradation. Avoid constantly switching the workload between sites at the host level.

## 7.4 Comparison of business continuity solutions

The business continuity solutions that are described in this section feature different characteristics in terms of implementation and features. Table 7-1 provides a comparison of these business continuity solutions that can help to identify the most fitting solution to a specific environment and needs.

Table 7-1 Business continuity solutions comparison

|                                                                           | Standard Stretched Cluster                                                                                | Enhanced Stretched Cluster                            | HyperSwap                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The function is available on these products                               | IBM Spectrum Virtualize only                                                                              | IBM Spectrum Virtualize only                          | <ul style="list-style-type: none"> <li>► All IBM Spectrum Virtualize based products, except for the IBM FlashSystem 5010</li> <li>► Requires IBM Spectrum Virtualize cluster with two or more I/O Groups</li> </ul> |
| Complexity of configuration                                               | Command-line interface (CLI) or graphical user interface (GUI) on a single system; simple object creation | CLI or GUI on a single system; simple object creation | CLI or GUI on a single system; simple object creation                                                                                                                                                               |
| The number of sites on which data is stored                               | Two                                                                                                       | Two                                                   | Two                                                                                                                                                                                                                 |
| Distance between sites                                                    | Up to 300 km (186.4 miles)                                                                                | Up to 300 km (186.4 miles)                            | Up to 300 km (186.4 miles)                                                                                                                                                                                          |
| Maintained independent copies of data                                     | Two                                                                                                       | Two                                                   | Two (four if you use more Volume Mirroring to two pools in each site)                                                                                                                                               |
| Technology for host to access multiple copies and automatically fail over | Standard host multipathing driver                                                                         | Standard host multipathing driver                     | Standard host multipathing driver                                                                                                                                                                                   |

|                                                                                                | <b>Standard Stretched Cluster</b>                                   | <b>Enhanced Stretched Cluster</b>                                                                                                                    | <b>HyperSwap</b>                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cache that is retained if only one site is online?                                             | Yes, if spare node is used, no otherwise                            | Yes, if spare node is used, no otherwise                                                                                                             | Yes                                                                                                                                                                                             |
| Host-to-storage-system path optimization                                                       | Manual configuration of preferred node                              | Manual configuration of preferred node for each volume before version 7.5; automatic configuration that is based on host site as HyperSwap from V7.5 | Automatic configuration based on host site (requires Asymmetric Logical Unit Access (ALUA)/Target Port Group Support (TPGS) support from the multipathing driver)                               |
| Synchronization and resynchronization of copies                                                | Automatic                                                           | Automatic                                                                                                                                            | Automatic                                                                                                                                                                                       |
| Stale consistent data is retained during resynchronization for DR?                             | No                                                                  | No                                                                                                                                                   | Yes                                                                                                                                                                                             |
| Scope of failure and resynchronization                                                         | Single volume                                                       | Single volume                                                                                                                                        | One or more volumes; the scope is user-configurable                                                                                                                                             |
| Ability to use FlashCopy with an HA solution                                                   | Yes (although no awareness of the site locality of the data)        | Yes (although no awareness of the site locality of the data)                                                                                         | Limited: You can use FlashCopy maps with a HyperSwap Volume as a source; avoids sending data across link between sites                                                                          |
| Ability to use Metro Mirror, Global Mirror, or Global Mirror Change Volume with an HA solution | One Remote Copy; it can maintain current copies on up to four sites | One Remote Copy; it can maintain current copies on up to four sites                                                                                  | Support for 3-site solutions fully available with IBM Spectrum Virtualize release V8.4 or later.                                                                                                |
| Maximum number of highly available volumes                                                     | 5,000                                                               | 5,000                                                                                                                                                | 1,250 in the IBM FlashSystems 5000/5100 products, or any other product running the code level 8.3.1 or lower.<br>2,000 in other IBM Spectrum Virtualize products with code level 8.4 or higher. |
| Minimum required paths for each logical unit (LUN) for each host port                          | Two                                                                 | Two                                                                                                                                                  | Four                                                                                                                                                                                            |
| Minimum number of I/O Groups                                                                   | One                                                                 | One I/O Group is supported, but it is recommended to have two or more I/O Groups.                                                                    | Two                                                                                                                                                                                             |
| Rolling disaster support                                                                       | No                                                                  | Yes                                                                                                                                                  | Yes                                                                                                                                                                                             |
| Licensing                                                                                      | Included in base product                                            | Included in base product                                                                                                                             | Requires Remote Mirroring license for volumes. Exact license requirements might vary by product.                                                                                                |



## 7.4.1 Other considerations and general recommendations

A business continuity solution implementation requires special considerations in the infrastructure and network setup. In Stretched and HyperSwap topologies, the communication between the IBM Spectrum Virtualize controllers must be optimal and free of errors for best performance, as the internode messaging and cache mirroring is done across the sites. Have a dedicated private SAN for internode communication so that it is not affected by regular SAN activities.

One other important recommendation is to review the site attribute of all the components to ensure they are accurate. With the site awareness algorithm present in the IBM Spectrum Virtualize code, optimizations are done to reduce the cross-site workload. If this attribute is missing or not accurate, increased unnecessary cross-site traffic might occur, which can lead to higher response time to the applications.

To further increase the system's resiliency, plan to have at least one hot-spare node per site. This configuration helps reduce the time that an I/O group remains with a single controller node when a node fails, or during planned actions, such as system code upgrades or hardware maintenance.

For more information, see the following resources:

- ▶ Hot-spare nodes, see *IBM Spectrum Virtualize: Hot-Spare Node and NPIV Target Ports*, REDP-5477
- ▶ Implementation guidelines for the Enhanced Stretched Cluster, see *IBM Spectrum Virtualize and SAN Volume Controller Enhanced Stretched Cluster with VMware*, SG24-8211
- ▶ HyperSwap configurations, see *IBM Spectrum Virtualize HyperSwap SAN Implementation and Design Best Practices*, REDP-5597

For detailed step-by-step configurations, see the following IBM Documentation web pages:

- ▶ [Stretched system configuration details](#)
- ▶ [HyperSwap system configuration details](#)





# Hosts

This chapter provides general guidelines and best practices around configuring host systems. The primary reference for host configuration is available at [IBM Documentation](#).

For more information about host attachment, see [IBM Documentation](#) and click **Configuring → Host attachment**.

For more information about hosts that are connected by using Fibre Channel, see Chapter 2, SAN Design. Host connectivity is a key consideration in overall SAN design.

Before attaching a new host, confirm that the host is supported by the IBM Spectrum Virtualize storage. For more information, see [IBM System Storage Interoperation Center \(SSIC\)](#).

The host configuration guidelines apply equally to all IBM Spectrum Virtualize systems. As such, the product name will often be referred to as an IBM Spectrum Virtualize system.

This chapter includes the following topics:

- ▶ 8.1, “General configuration guidelines” on page 340
- ▶ 8.2, “Host pathing” on page 343
- ▶ 8.3, “I/O queues” on page 344
- ▶ 8.4, “Host clusters” on page 345
- ▶ 8.5, “AIX hosts” on page 347
- ▶ 8.6, “Virtual I/O server hosts” on page 348
- ▶ 8.7, “Windows hosts” on page 348
- ▶ 8.8, “Linux hosts” on page 349
- ▶ 8.9, “Oracle Solaris hosts” on page 349
- ▶ 8.10, “VMware server hosts” on page 351

## 8.1 General configuration guidelines

The following subsections complement the content in Chapter 2, “Storage area network” on page 19.

### 8.1.1 Number of paths

It is generally recommended that the total number of Fibre Channel paths per volume be limited to four paths. Adding paths does not significantly increase redundancy and it tends to bog down the host with path management. Too many paths can increase failover time.

### 8.1.2 Host ports

Each host uses two ports from two different host bus adapters (HBAs). These ports are in separate SAN fabrics and zoned to one target port of each node or node canister. When the volumes are created, they are assigned to an I/O group and the resulting path count between the volume and the host should be four.

**Preferred practice:** Keep Fibre Channel tape (including Virtual Tape Libraries) and Fibre Channel disks on separate HBAs. These devices have two different data patterns when operating in their optimum mode. Switching between them can cause unwanted processor usage and performance slowdown for the applications.

### 8.1.3 Port masking

In general, Fibre Channel ports should be dedicated to certain functions. Hosts should only be zoned to ports designated for host I/O.

For more information about port masking, see Chapter 2, “Storage area network” on page 19.

### 8.1.4 N-port ID virtualization

IBM Spectrum Virtualize now uses N-port ID virtualization (NPIV) by default. This use reduces failover time and allows for features like hot spare nodes.

For more information about configuring NPIV, see Chapter 2, “Storage area network” on page 19.

### 8.1.5 Host to I/O group mapping

An *I/O group* consists of two nodes or node canisters that share management of volumes within the cluster. Use a single I/O group (iogrp) for all volumes that are allocated to a specific host. This guideline results in the following benefits:

- ▶ Minimizes port fan-outs within the SAN fabric
- ▶ Maximizes the potential host attachments to IBM Spectrum Virtualize because maximums are based on I/O groups
- ▶ Reduces the number of target ports that must be managed within the host

## 8.1.6 Volume size as opposed to quantity

In general, host resources, such as memory and processing time, are used up by each storage LUN that is mapped to the host. For each extra path, more memory can be used, and a portion of more processing time is also required. The user can control this effect by using fewer larger LUNs rather than many small LUNs. However, you might need to tune queue depths and I/O buffers to support controlling the memory and processing time efficiently.

If a host does not have tunable parameters, such as on the Windows operating system, the host does not benefit from large volume sizes. AIX greatly benefits from larger volumes with a smaller number of volumes and paths that are presented to it.

## 8.1.7 Host volume mapping

Host mapping is the process of controlling which hosts can access specific volumes within the system. IBM Spectrum Virtualize always presents a specific volume with the same SCSI ID on all host ports. When a volume is mapped, IBM Spectrum Virtualize software automatically assigns the next available SCSI ID if none is specified. In addition, a unique identifier, called the *UID*, is on each volume.

You can allocate the operating system volume of the SAN boot as the lowest SCSI ID (zero for most hosts), and then allocate the various data disks. If you share a volume among multiple hosts, consider controlling the SCSI ID so that the IDs are identical across the hosts. This consistency ensures ease of management at the host level and prevents potential issues during IBM Spectrum Virtualize updates and even node restarts, mostly for ESX operating systems.

If you are using image mode to migrate a host to IBM Spectrum Virtualize, allocate the volumes in the same order that they were originally assigned on the host from the back-end storage.

The **lshostvdiskmap** command displays a list of VDisk (volumes) that are mapped to a host. These volumes are recognized by the specified host. Example 8-1 shows the syntax of the **lshostvdiskmap** command that is used to determine the SCSI ID and the UID of volumes.

*Example 8-1 The lshostvdiskmap command*

---

```
svcinfo lshostvdiskmap -delim
```

---

Example 8-2 shows the results of using the **lshostvdiskmap** command.

*Example 8-2 Output of using the lshostvdiskmap command*

---

```
svcinfo lsvdiskhostmap -delim : EEXCLS_HBin01
id:name:SCSI_id:host_id:host_name:wwpn:vdisk_UID
950:EEXCLS_HBin01:14:109:HDMCENTEX1N1:10000000C938CFDF:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:109:HDMCENTEX1N1:10000000C938D01F:600507680191011D4800000000000466
950:EEXCLS_HBin01:13:110:HDMCENTEX1N2:10000000C938D65B:600507680191011D4800000000000466
950:EEXCLS_HBin01:13:110:HDMCENTEX1N2:10000000C938D3D3:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:111:HDMCENTEX1N3:10000000C938D615:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:111:HDMCENTEX1N3:10000000C938D612:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:112:HDMCENTEX1N4:10000000C938CFBD:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:112:HDMCENTEX1N4:10000000C938CE29:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:113:HDMCENTEX1N5:10000000C92EE1D8:600507680191011D4800000000000466
950:EEXCLS_HBin01:14:113:HDMCENTEX1N5:10000000C92EDFFE:600507680191011D4800000000000466
```

---

**Note:** Example 8-2 shows the same volume that is mapped to five different hosts, but host 110 features a different SCSI ID than the other four hosts. This example is a non-recommended practice that can lead to loss of access in some situations because of SCSI ID mismatch.

### 8.1.8 Server adapter layout

If your host system includes multiple internal I/O buses, place the two adapters that are used for IBM Spectrum Virtualize cluster access on two different I/O buses to maximize the availability and performance. When purchasing a server, always have two cards instead of one. For example, two dual port HBA cards are preferred over one quad port HBA card because you can spread the I/O and add redundancy.

### 8.1.9 Host status improvements

As of 8.3, IBM Spectrum Virtualize provides an alternative for reporting host status.

Previously, a host was marked as *degraded* if one of the host ports logged off the fabric. However, examples exist in which this marking might be normal and can cause confusion.

At the host level, a new **status\_policy** setting is available that can be set to **complete** or **redundant**. The **complete** setting uses the original host status definitions. With the **redundant** setting, a host is not reported as *degraded* unless not enough ports are available for redundancy.

### 8.1.10 Considerations for NVMe over Fibre Channel host attachments

As of 8.3, IBM Spectrum Virtualize now supports a single host initiator port by using both SCSI and NVMe connections to the storage.

Asymmetric Namespace Access was added to the FC-NVMe protocol standard. This addition gives it functions that are similar to ALUA. As a result, FC-NVMe can now be used in stretched clusters.

IBM Spectrum Virtualize code 8.4.0 allows a maximum of 32 NVMe hosts, if no other types of hosts are attached. IBM Spectrum Virtualize code does not monitor or enforce these limits. If you are planning to use NVMe hosts with IBM SAN Volume Controller, see this [IBM Support](#) web page.

### 8.1.11 Considerations for iSER host attachments

On IBM SAN Volume Controller, iSER (iSCSI Extensions for RDMA) hosts with different operating systems can be attached to the system. iSER is a network protocol that extends the Internet Small Computer System Interface (iSCSI) to use Remote Direct Memory Access (RDMA).

If you are planning to use iSER hosts in your IBM SVC, see the following web pages as you plan your environment:

- ▶ [IBM Documentation](#)
- ▶ [IBM Support](#)

## 8.2 Host pathing

Each host mapping associates a volume with a host object and allows all HBA ports in the host object to access the volume. You can map a volume to multiple host objects.

When a mapping is created, multiple paths normally exist across the SAN fabric from the hosts to the IBM Spectrum Virtualize system. Most operating systems present each path as a separate storage device. Therefore, multipathing software is required on the host. The multipathing software manages the paths that are available to the volume, presents a single storage device to the operating system, and provides failover in the case of a lost path.

If your IBM Spectrum Virtualize system uses NPIV, the multipathing driver is *not* need to perform the failover.

### 8.2.1 Path selection

I/O for a specific volume is handled exclusively by the nodes in a single I/O group. Although both nodes in the I/O group can service the I/O for the volume, the system prefers to use a consistent node, which is called the *preferred node*. The primary purposes of the use of a preferred node are load balancing and to determine which node de-stages writes to the backend storage.

When a volume is created, an I/O group and preferred node are defined and can optionally be set by the administrator. The owner node for a volume is the preferred node when both nodes are available.

IBM Spectrum Virtualize uses Asymmetric Logical Unit Access (ALUA) as do most multipathing drivers. Therefore, the multipathing driver gives preference to paths to the preferred node. Most modern storage systems use ALUA.

**Note:** Some competitors claim that ALUA means that IBM Spectrum Virtualize is effectively an active-passive cluster. This claim is not true. Both nodes in IBM Spectrum Virtualize can and do service I/O concurrently.

In the odd chance that an I/O goes to the non-preferred node, that node services the I/O without issue.

## 8.3 I/O queues

Host operating system and HBA software must have a way to fairly prioritize I/O to the storage. The host bus might run faster than the I/O bus or external storage. Therefore, you must have a way to queue I/O to the devices. Each operating system and host adapter use unique methods to control the I/O queue. The unique method to control I/O queue can be host adapter-based, memory and thread resources-based, or based on the number of commands that are outstanding for a device.

### 8.3.1 Queue depths

*Queue depth* is used to control the number of concurrent operations that occur on different storage resources. Queue depth is the number of I/O operations that can be run in parallel on a device.

Queue depths apply at various levels of the system: at the disk or flash level, the storage controller level, and the per volume and HBA level on the host. For example, each IBM Spectrum Virtualize node has a queue depth of 10,000. A typical disk drive operates efficiently at a queue depth of 8. Most host volume queue depth defaults around 32.

Guidance for limiting queue depths in large SANs that was described in previous documentation is replaced with calculations for overall I/O group-based queue depth considerations.

No hard and fast rule exists for what value is set per host HBA or per volume. The requirements for your environment are driven by the intensity of each workload.

You want to ensure that no one application or host can run away and use the entire controller queue. However, if you have a specific host application that requires the lowest latency and highest throughput, consider giving it a proportionally larger share than others.

Consider the following points:

- ▶ A single IBM Spectrum Virtualize Fibre Channel port accepts a maximum concurrent queue depth of 2048.
- ▶ A single IBM Spectrum Virtualize node accepts a maximum concurrent queue depth of 10,000. After this depth is reached, it reports a full status for the queue.
- ▶ Host HBA queue depths should be set to the maximum (typically, 1024).
- ▶ Host queue depth should be controlled through the per volume value:
  - A typical random workload volume should use approximately 32
  - To limit the workload of a volume use 4 or less
  - To maximize throughput and give a higher share to a volume, use 64

The total workload capability can be calculated by multiplying the number of volumes by their respective queue depths and summing. With low latency storage, a workload of over 1 million IOPs can be achieved with a concurrency on a single IO Group of 1000.

For more information about queue depths, see the following IBM Documentation web pages:

- ▶ [FC hosts](#)
- ▶ [iSCSI hosts](#)
- ▶ [iSER hosts](#)



## 8.4 Host clusters

IBM Spectrum Virtualize supports host clusters. This feature allows multiple hosts to access the same set of volumes.

Volumes that are mapped to that host cluster are assigned to all members of the host cluster with the same SCSI ID. A typical use-case is to define a host cluster that contains all the WWPNs that belong to the hosts that are participating in a host operating system-based cluster, such as IBM PowerHA®, Microsoft Cluster Server (MSCS), or VMware ESXi clusters.

The following commands can be used to manage host clusters:

- ▶ `lshostcluster`
- ▶ `lshostclustermember`
- ▶ `lshostclustervolumemap`
- ▶ `addhostclustermember`
- ▶ `chhostcluster`
- ▶ `mkhost` (with parameter `-hostcluster` to create the host in one existing cluster)
- ▶ `rmhostclustermember`
- ▶ `rmhostcluster`
- ▶ `rmvolumehostclustermap`

Starting with IBM Spectrum Virtualize 8.1, host clusters can be added by using the GUI. It allows you to let the system assign the SCSI IDs for the volumes or you can manually assign them. For ease of management purposes, it is suggested to use separate ranges of SCSI IDs for hosts and host clusters.

For example, you can use SCSI IDs 0 - 99 to non-cluster host volumes, and above 100 for the cluster host volumes. When you choose the option **System Assign**, the system automatically assigns the SCSI IDs starting from the first available in the sequence. If you choose **Self Assign**, the system enables you to select the SCSI IDs manually for each volume. On the right side of the window, it shows the SCSI IDs that are used by the selected host or host cluster (see Figure 8-1).

| Name    | SCSI ID | Caching I/O G |
|---------|---------|---------------|
| test_01 | 2       | 0             |
| test_02 | 3       | 0             |
| test_03 | 4       | 0             |
| test_04 | 5       | 0             |
| test_05 | 6       | 0             |

| Type of Mapping | SCSI ID |
|-----------------|---------|
| Shared          | 0       |
| Shared          | 1       |

Figure 8-1 SCSI ID assignment on volume mappings

**Note:** Although extra care is always recommended when dealing with hosts, IBM Spectrum Virtualize does not allow you to join a host into a host cluster if it includes a volume mapping with a SCSI ID that also exists in the host cluster:

```
IBM_2145:ITS0-SVCLab:superuser>addhostclustermember -host ITS0_HOST3  
ITS0_CLUSTER1
```

CMMVC9068E Hosts in the host cluster have conflicting SCSI ID's for their private mappings.

```
IBM_2145:ITS0-SVCLab:superuser>
```

### 8.4.1 Persistent reservations

To prevent hosts from sharing storage inadvertently, establish a storage reservation mechanism. The mechanisms for restricting access to IBM Spectrum Virtualize volumes use the SCSI-3 persistent reserve commands or the SCSI-2 reserve and release commands.

The host software uses several methods to implement host clusters. These methods require sharing the volumes on IBM Spectrum Virtualize between hosts. To share storage between hosts, maintain control over accessing the volumes. Some clustering software uses software locking methods.

You can choose other methods of control by the clustering software or by the device drivers to use the SCSI architecture reserve or release mechanisms. The multipathing software can change the type of reserve that is used from an earlier reserve to persistent reserve, or remove the reserve.

*Persistent reserve* refers to a set of SCSI-3 standard commands and command options that provide SCSI initiators with the ability to establish, preempt, query, and reset a reservation policy with a specified target device. The functions that are provided by the persistent reserve commands are a superset of the original reserve or release commands.

The persistent reserve commands are incompatible with the earlier reserve or release mechanism. Also, target devices can support only reservations from the earlier mechanism or the new mechanism. Attempting to mix persistent reserve commands with earlier reserve or release commands results in the target device returning a reservation conflict error.

Earlier reserve and release mechanisms (SCSI-2) reserved the entire LUN (volume) for exclusive use down a single path. This approach prevents access from any other host or even access from the same host that uses a different host adapter. The persistent reserve design establishes a method and interface through a reserve policy attribute for SCSI disks. This design specifies the type of reservation (if any) that the operating system device driver establishes before it accesses data on the disk.

The following possible values are supported for the reserve policy:

- ▶ No\_reserve: No reservations are used on the disk.
- ▶ Single\_path: Earlier reserve or release commands are used on the disk.
- ▶ PR\_exclusive: Persistent reservation is used to establish *exclusive host access* to the disk.
- ▶ PR\_shared: Persistent reservation is used to establish *shared host access* to the disk.

When a device is opened (for example, when the AIX **varyonvg** command opens the underlying hdisks), the device driver checks the object data manager (ODM) for a `reserve_policy` and a `PR_key_value`. The driver then opens the device. For persistent reserve, each host that is attached to the shared disk must use a unique registration key value.

## 8.4.2 Clearing reserves

It is possible to accidentally leave a reserve on the IBM Spectrum Virtualize volume or on the IBM Spectrum Virtualize MDisk during migration into IBM Spectrum Virtualize, or when disks are reused for another purpose. Several tools are available from the hosts to clear these reserves. You can also clear the IBM Spectrum Virtualize volume reserves by removing all the IBM Spectrum Virtualize MDisk reserves

There are instances in which a host image mode migration appears to succeed, but problems occur when the volume is opened for read or write I/O. The problems can result from not removing the reserve on the MDisk before image mode migration is used in IBM Spectrum Virtualize. You cannot clear a leftover reserve on an IBM Spectrum Virtualize MDisk from IBM Spectrum Virtualize. You must clear the reserve by mapping the MDisk back to the owning host and clearing it through host commands, or through back-end storage commands as advised by IBM technical support.

## 8.5 AIX hosts

This section describes various topics that are specific to AIX hosts.

### 8.5.1 Multipathing support

SDD PCM is no longer supported. Use the default AIX PCM. For more information, see this [IBM Support web page](#).

### 8.5.2 Configuration recommendations for AIX

These device settings can be changed by using the **chdev** AIX command:

```
reserve_policy=no_reserve
```

The default reserve policy is **single\_path** (SCSI-2 reserve). Unless a specific need exists for reservations, use **no\_reserve**:

```
algorithm=shortest_queue
```

If coming from SDD PCM, AIX defaults to **fail\_over**. You cannot set the algorithm to **shortest\_queue** unless the reservation policy is **no\_reserve**:

```
queue_depth=32
```

The default queue depth is 20. IBM recommends 32:

```
rw_timeout=30
```

IBM recommends the following values:

- ▶ For `queue_depth`: 32
- ▶ For `rw_timeout`: 30

Note that 60 was the default for the SDD PCM and 30 is the default for AIX PCM.

## 8.6 Virtual I/O server hosts

In this section, we discuss the considerations for virtual I/O server hosts.

### 8.6.1 Multipathing support

Subsystem Device Driver Path Control Module (SDDPCM) is no longer supported. Use the default AIX PCM. For more details see this [IBM Support web page](#).

### 8.6.2 Physical and logical volumes

Virtual SCSI (VSCSI) is based on a client/server relationship. The Virtual I/O Server (VIOS) owns the physical resources and acts as the server or target device. Physical adapters with attached disks (in this case, volumes on IBM Spectrum Virtualize) on the VIOS partition can be shared by one or more partitions. These partitions contain a virtual SCSI client adapter that detects these virtual devices as standard SCSI-compliant devices and LUNs.

You can create the following types of volumes on a VIOS:

- ▶ Physical volume (PV) VSCSI hdisks
- ▶ Logical volume (LV) VSCSI hdisks

PV VSCSI hdisks are entire LUNs from the VIOS perspective. If you are concerned about failure of a VIOS and have configured redundant VIOSs for that reason, you must use PV VSCSI hdisks. Therefore, PV VSCSI hdisks are entire LUNs that are volumes from the virtual I/O client perspective. An LV VSCSI hdisk cannot be served up from multiple VIOSs.

LV VSCSI hdisks are in LVM volume groups on the VIOS and cannot span PVs in that volume group or be striped LVs. Because of these restrictions, use PV VSCSI hdisks.

### 8.6.3 Identifying a disk for use as a virtual SCSI disk

The VIOS uses the following methods to uniquely identify a disk for use as a virtual SCSI disk:

- ▶ Unique device identifier (UDID)
- ▶ IEEE volume identifier
- ▶ Physical volume identifier (PVID)

Each of these methods can result in different data formats on the disk. The preferred disk identification method for volumes is the use of UDIDs.

## 8.7 Windows hosts

This section describes various topics that are specific to Microsoft Windows hosts.

## 8.7.1 Multipathing support

Use Microsoft MPIO with MS DSM (Microsoft Device Specific Module), which is included in the Windows Server operating system. The older Subsystem Device Driver Device Specific Module (SDDDSM) is no longer supported.

## 8.7.2 Windows configuration

For more information about configuring Windows hosts, see [IBM Documentation](#).

## 8.8 Linux hosts

IBM Spectrum Virtualize supports Linux hosts that use native DM-MP multipathing. Veritas DMP is also available for specific kernels.

For more information about configuring Linux hosts, see [IBM Documentation](#).

**Note:** Occasionally we see that storage administrators modify parameters in `multipath.conf` to address some perceived shortcoming in the DM-MP configuration. This issue can create unintended and unexpected behaviors. The recommendations that are provided in IBM Documentation are optimal for most configurations.

## 8.9 Oracle Solaris hosts

Two options are available for multipathing support on Solaris hosts: Symantec Veritas Volume Manager and Oracle Solaris MPxIO (Storage Multipathing). The option that you choose depends on your file system requirements and the operating system levels in the latest interoperability matrix.

For more information, see [IBM System Storage Interoperation Center \(SSIC\)](#).

IBM SDD is no longer supported because its features are now available natively in the multipathing driver for Solaris MPxIO.

For more information about host attachment for Solaris hosts, see [IBM Documentation](#).

### 8.9.1 Solaris MPxIO

SAN boot and clustering support is available for V5.9, V5.10, and 5.11, depending on the multipathing driver and HBA choices. Support for load balancing of the MPxIO software is included in IBM SAN Volume Controller that is running IBM Spectrum Virtualize 8.2.

If you want to run MPxIO on your Sun SPARC host, configure your IBM Spectrum Virtualize host object with the type attribute set to `tpgs`, as shown in the following example:

```
svctask mkhost -name new_name_arg -hbawwpn wwpn_list -type tpgs
```

In this command, `-type` specifies the type of host. Valid entries are `hpux`, `tpgs`, `generic`, `openvms`, `adminlun`, and `hide_secondary`. The `tpgs` option enables an extra target port unit. The default is `generic`.

## 8.9.2 Symantec Veritas Volume Manager

When you are managing IBM Spectrum Virtualize storage in Symantec volume manager products, you must install an Array Support Library (ASL) on the host so that the volume manager is aware of the storage subsystem properties (active/active or active/passive). If the suitable ASL is not installed, the volume manager did not claim the LUNs. Usage of the ASL is required to enable the special failover or failback multipathing that IBM Spectrum Virtualize requires for error recovery.

Use the commands that are shown in Example 8-3 to determine the basic configuration of a Symantec Veritas server.

---

### *Example 8-3 Determining the Symantec Veritas server configuration*

---

```
pkginfo -l (lists all installed packages)
showrev -p |grep vxvm (to obtain version of volume manager)
vxddladm listsupport (to see which ASLs are configured)
vxdisk list
vxdmpadm listctrl all (shows all attached subsystems, and provides a type where possible)
vxdmpadm getsubpaths ctrl=cX (lists paths by controller)
vxdmpadm getsubpaths dmpnodename=cxtxdxs2' (lists paths by LUN)
```

---

The commands that are shown in Example 8-4 and Example 8-5 determine whether the IBM Spectrum Virtualize is correctly connected. They show at a glance which ASL is used (native DMP (Dynamic Multi-Pathing) ASL or SDD ASL). Example 8-4 shows what you see when Symantec Volume Manager correctly accesses IBM Spectrum Virtualize by using the SDD pass-through mode ASL.

---

### *Example 8-4 Symantec Volume Manager that uses SDD pass-through mode ASL*

---

```
# vxdmpadm list enclosure all
ENCLR_NAME ENCLR_TYPE ENCLR_SNO STATUS
=====
OTHER_DISKS OTHER_DISKS OTHER_DISKS CONNECTED
VPATH_SANVC0 VPATH_SANVC 0200628002faXX00 CONNECTED
```

---

Example 8-5 shows what you see when IBM Spectrum Virtualize is configured by using native DMP ASL.

---

### *Example 8-5 IBM Spectrum Virtualize that is configured by using native ASL*

---

```
# vxdmpadm listenclosure all
ENCLR_NAME ENCLR_TYPE ENCLR_SNO STATUS
=====
OTHER_DISKS OTHER_DSKSI OTHER_DISKS CONNECTED
SAN_VC0 SAN_VC 0200628002faXX00 CONNECTED
```

---

## 8.9.3 DMP multipathing

For the latest ASL levels to use native DMP, see the array-specific module table that is available at [this web page](#).

To check the installed Symantec Veritas version, enter the following command:

```
showrev -p |grep vxvm
```

To check which IBM ASLs are configured into the Volume Manager, enter the following command:

```
vxddladm listsupport |grep -i ibm
```

After you install a new ASL by using the **pkgadd** command, restart your system or run the **vxdtl enable** command. To list the ASLs that are active, enter the following command:

```
vxddladm listsupport
```

## 8.10 VMware server hosts

To determine the various VMware ESXi levels that are supported, see the [IBM System Storage Interoperation Center \(SSIC\)](#).

### 8.10.1 Configuring VMware

For more information about specific configuration best practices for VMware, see [IBM Documentation](#).

VMware features a built-in multipathing driver that supports IBM Spectrum Virtualize ALUA-preferred path algorithms.

Consider the following points:

- ▶ The storage array type should be ALUA (VMW\_SATP\_ALUA).
- ▶ Path selection policy should be RoundRobin.
- ▶ The Round Robin IOPS should be changed from 1000 to 1 so that I/Os are evenly distributed across as many ports on the system as possible. For more information about how to change this setting, see [this web page](#).

### 8.10.2 Multipathing configuration maximums

The VMware multipathing software supports the following maximum configuration:

- ▶ A total of 256 SCSI devices
- ▶ Up to 32 paths to each volume
- ▶ Up to 4096 paths per server

**Tip:** Each path to a volume equates to a single SCSI device.

For more information about a complete list of maximums, see [this web page](#).

For more information about IBM i-related considerations, see Appendix A, “IBM i considerations” on page 529.







# Monitoring

Monitoring in a storage environment is crucial and is part of what often is called *storage governance*.

With a robust and reliable storage monitoring system, you can realize significant financial savings and minimize pain in your operation by monitoring and predicting usage bottlenecks in your virtualized storage environment.

It is also possible to use the data that is collected from monitoring to create strategies and apply configurations to improve performance, tuning connections, and tools usability.

This chapter provides suggestions and the basic concepts of how to implement a storage monitoring system for IBM Spectrum Virtualize by using their specific functions or external IBM Tools.

This chapter includes the following topics:

- ▶ 9.1, “Generic monitoring” on page 354
- ▶ 9.2, “Performance monitoring” on page 357
- ▶ 9.3, “Capacity metrics for block storage systems” on page 377
- ▶ 9.4, “Creating alerts for IBM Spectrum Control and IBM Storage Insights” on page 391
- ▶ 9.5, “Important metrics” on page 399
- ▶ 9.6, “Performance support package” on page 401
- ▶ 9.7, “Metro and Global Mirror monitoring with IBM Copy Services Manager and scripts” on page 402
- ▶ 9.8, “Monitoring Tier1 SSD” on page 404

## 9.1 Generic monitoring

With IBM Spectrum Virtualize, you can implement generic monitoring using specific functions that are integrated with the product itself without adding any external tools or cost.

**Note:** At the time of this writing, a hot spare node that is in `online_spare` status shows as failed or offline in most scripts. It also shows on IBM Spectrum Control 5.2.15, which was released before the Hot Spare Node feature. If you use manual scripts, add the value `online_spare` as a healthy condition as well.

### 9.1.1 Monitoring with the GUI

The management GUI is the primary tool that is used to service your system. Regularly monitor the status of the system by using the management GUI. If you suspect a problem, use the management GUI first to diagnose and resolve the problem.

Use the views that are available in the management GUI to verify the status of the system, the hardware devices, the physical storage and the available volumes. The **Monitoring** → **Events** window provides access to all problems that exist on the system. Use the **Recommended Actions** filter to display the most important events that need to be resolved.

If a service error code exists for the alert, you can run a fix procedure that assists you in resolving the problem. These fix procedures analyze the system and provide more information about the problem. These actions also ensure that the required changes do not cause volumes to be inaccessible to the hosts and automatically perform configuration changes that are required to return the system to its optimum state. If any interaction is required, they suggest actions to take and guide you through those actions that automatically manage the system where necessary. If the problem is fixed, the alert is excluded.

#### Email notification

It is possible to add multiple email addresses (you can also use your email distribution list) to receive notifications from the storage. You also can set notification options for each email box that you added with different sets of information (see Figure 9-1).

| Email Users              |                                     |                                     |                                     |                                     |      |   |   |
|--------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------|---|---|
| Email Address            | Notifications                       |                                     |                                     |                                     |      |   |   |
|                          | Error                               | Warning                             | Info                                | Inventory                           |      |   |   |
| joedoe@yourcompany.com   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | Test | + | - |
| jondoe@yourcompany.com   | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Test | + | - |
| maillist@yourcompany.com | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | Test | + | - |

Figure 9-1 E-mail notification options

From a monitoring perspective, email notification is one of the most common and important tools that you can use and set up. From the notification events, you can validate if your system is running under normal status or needs attention.

## Call Home

The Call Home feature transmits operational and event-related data to you and IBM through a Simple Mail Transfer Protocol (SMTP) server or cloud services connection. When configured, this function alerts IBM service personnel about hardware failures and potentially serious configuration or environmental issues.

You can configure the Cloud Call Home option that alerts IBM Support for any problem reported by the system. For more information, see *Implementing the IBM System Storage SAN Volume Controller with IBM Spectrum Virtualize Version 8.4*, [SG24-8491](#).

## SNMP notification

Simple Network Management Protocol (SNMP) is a standard protocol for managing networks and exchanging messages. The system can send SNMP messages that notify personnel about an event. You can use an SNMP manager to view the SNMP messages that are sent by the IBM SAN Volume Controller.

The MIB file describes the format of the SNMP messages that are sent by the system. Use this MIB file to configure a network management program to receive SNMP event notifications that are sent from an IBM Spectrum Virtualize system. This MIB file is suitable for use with SNMP messages from all versions of IBM Spectrum Virtualize.

IBM Spectrum Virtualize MIB file for IBM SAN Volume Controller 8.4.0 is available at [Management Information Base \(MIB\) file for SNMP](#).

## Syslog notification

The syslog protocol is a standard protocol for forwarding log messages from a sender to a receiver on an IP network. The IP network can be IPv4 or IPv6. The system can send Syslog messages that notify personnel about an event. You can configure a syslog server to receive log messages from various systems and store them in a central repository.

Figure 9-2 on page 356 shows the new syslog grid layout from the IBM Spectrum Virtualize GUI. It is possible to configure multiple syslog servers and monitor the communication between IBM Spectrum Virtualize to the syslog server from the syslog panel.

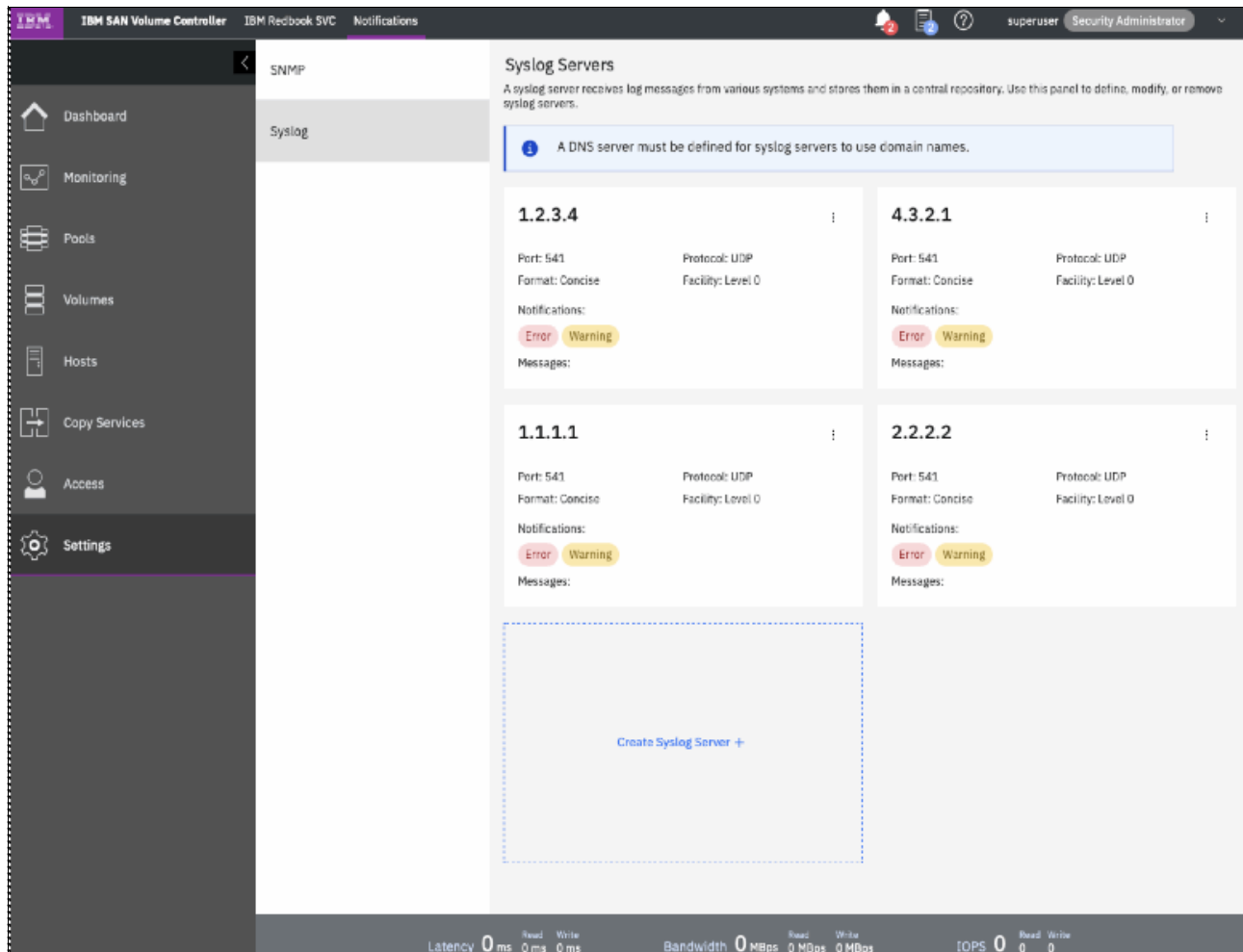


Figure 9-2 Syslog Configuration

**Note:** Starting on version 8.4, FQDN can be used for services, such as Syslog, LDAP and NTP.

## IBM Storage Mobile Dashboard

IBM Storage Mobile Dashboard is a no-charge application that provides basic monitoring capabilities for IBM storage systems. You can securely check the health and performance status of your IBM Spectrum Virtualize system by viewing events and performance metrics.

To install IBM Storage Mobile Dashboard on an iOS device, open the App Store and search for “IBM Storage Mobile Dashboard”.

## 9.1.2 Monitoring using quotas and alert

In an IBM Spectrum Virtualize system, the space usage of storage pools and thin provisioned or compressed volumes can be monitored by setting some specific quota alerts.

### Storage pool

You can set a warning such that when the pool capacity reaches this quota setting, an alert is issued. This setting generates a warning when the used capacity in the storage pool first exceeds the specified threshold. You can specify a `disk_size` integer, which defaults to megabytes (MB) unless the `-unit` parameter is specified. Or you can specify a `disk_size%`, which is a percentage of the storage pool size. To disable warnings, specify 0 or 0%. The default value is 0.

### Volumes

Thin provisioned and compressed volumes near their size limits are monitored at specified thresholds to preserve data integrity. If the volume trespasses this threshold, one option is create a non-compressed mirror of the data (if one does not exist), set it as the default volume, and delete the primary copy.

## 9.2 Performance monitoring

The ability to collect historical performance metrics is essential to properly monitor and manage storage subsystems and it is for IBM Spectrum Virtualize systems as well. During troubleshooting and performance tuning, the historical data can be used as a parameter.

Performance statistics are useful not only to debug or prevent some potential bottlenecks, but to make capacity planning for future growth easier, as shown in Figure 8-1 on page 345.

The next sections describe the performance analysis tools that are integrated with IBM Spectrum Virtualize systems and what IBM external tools are available to collect performance statistics to allow historical retention as well.

### 9.2.1 Performance monitoring with the GUI

In IBM Spectrum Virtualize, real-time performance statistics provide short-term status information for your systems. The statistics are shown as graphs in the management GUI.

You can use system statistics to monitor the bandwidth of all the volumes, interfaces, and MDisks that are used on your system. You can also monitor the overall CPU usage for the system. These statistics also summarize the overall performance health of the system.

You can monitor changes to stable values or differences between related statistics, such as the latency between volumes and MDisks. These differences can then be further evaluated by performance diagnostic tools.

With system-level statistics, you also can quickly view bandwidth of volumes, interfaces, and MDisks. Each of these graphs displays the current bandwidth in megabytes per second and a view of bandwidth over time.

Each data point can be accessed to determine its individual bandwidth use and to evaluate whether a specific data point might represent performance impacts. For example, you can monitor the interfaces, such as for Fibre Channel or SAS interfaces, to determine whether the host data-transfer rate is different from the expected rate.

You can also select node-level statistics, which can help you determine the performance impact of a specific node. As with system statistics, node statistics help you to evaluate whether the node is operating within normal performance metrics.

The CPU utilization graph shows the current percentage of CPU usage and specific data points on the graph that show peaks in utilization. If compression is being used, you can monitor the amount of CPU resources that are being used for compression and the amount that is available to the rest of the system.

The Interfaces graph displays data points for Fibre Channel (FC), iSCSI, serial-attached SCSI (SAS), and IP Remote Copy interfaces. You can use this information to help determine connectivity issues that might affect performance.

The Volumes and MDisks graphs on the Performance window show four metrics: Read, Write, Read latency, and Write latency. You can use these metrics to help determine the overall performance health of the volumes and MDisks on your system. Consistent unexpected results can indicate errors in configuration, system faults, or connectivity issues.

Each graph represents 5 minutes of collected statistics, which are updated every 5 seconds. They also provide a means of assessing the overall performance of your system, as shown in Figure 9-3.

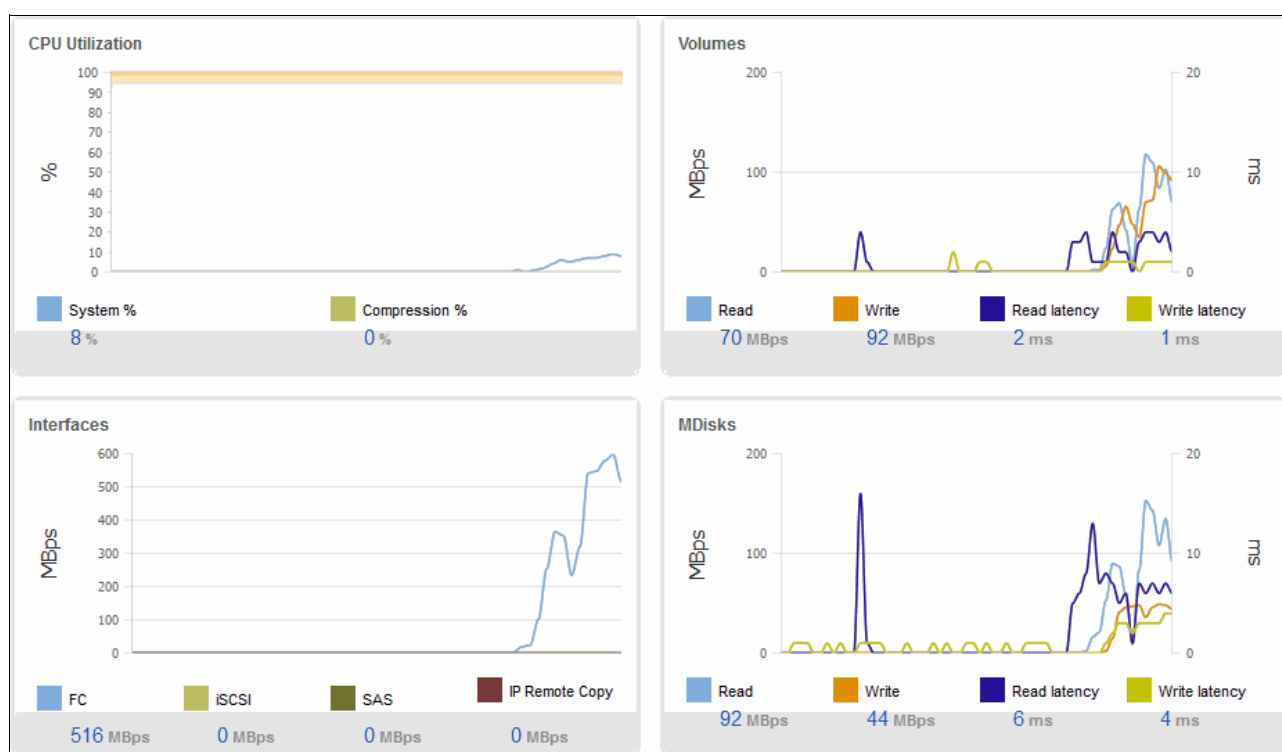


Figure 9-3 Monitoring GUI example

You can then choose the metrics that you want to be displayed, as shown in Figure 9-4.

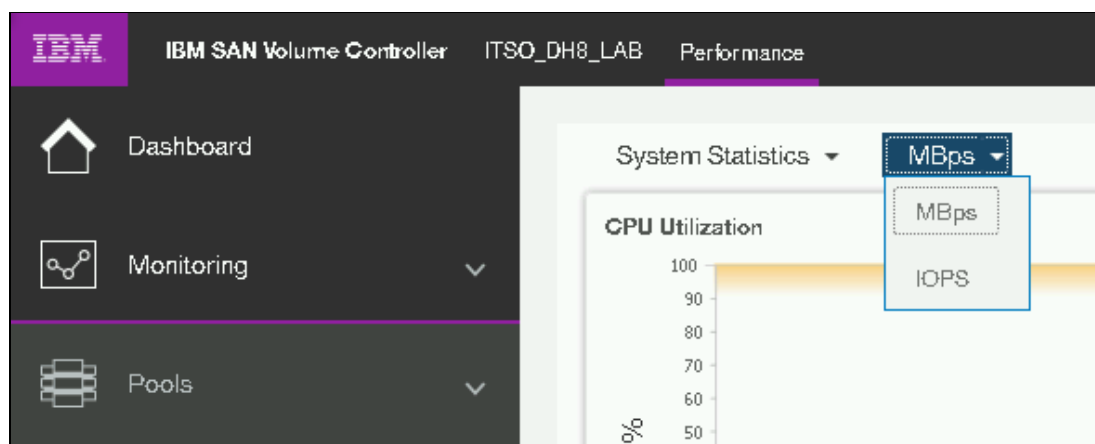


Figure 9-4 Selecting metrics

You can also obtain a quick overview by using the GUI option **System** → **Dashboard**, as shown in Figure 9-5.

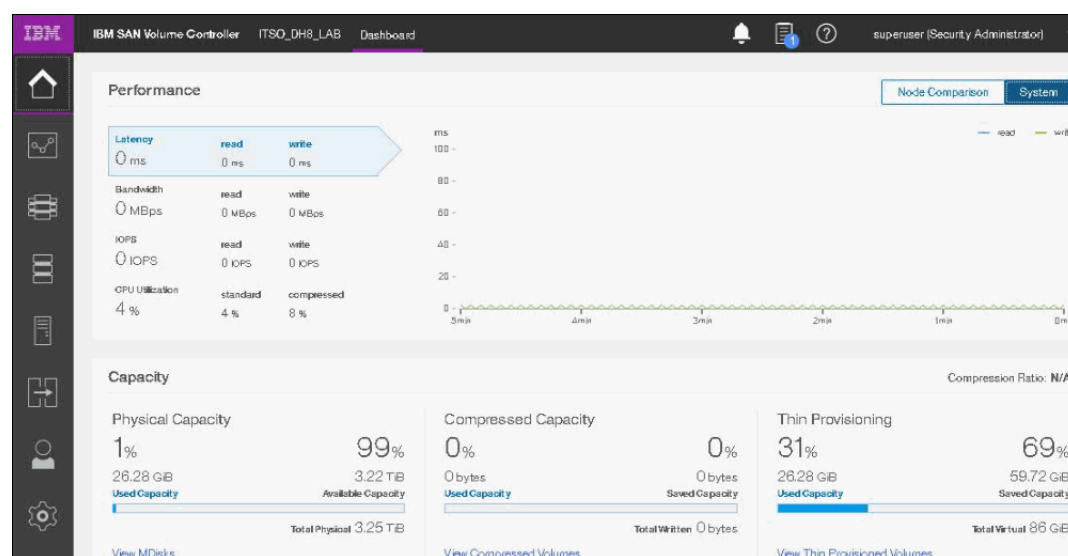


Figure 9-5 Dashboard option

## 9.2.2 Performance monitoring with IBM Spectrum Control

IBM Spectrum Control is an on-premises storage management, monitoring, and reporting solution. It uses the metadata that it collects about vendors' storage devices to provide services, such as custom alerting, analytics, and replication management. Both IBM Spectrum Control and IBM Storage Insights monitor storage systems, but IBM Spectrum Control also monitors hypervisors, fabrics, and switches to provide you with unique analytics and insights into the topology of your storage network.

It also provides more granular collection of performance data, with 1-minute intervals rather than the 5-minute intervals in IBM Storage Insights or IBM Storage Insights Pro. For more information about IBM Storage Insights, see 9.2.3, "Performance monitoring with IBM Storage Insights" on page 363.

Because IBM Spectrum Control is an on-premises tool, it does not send the metadata about monitored devices off-site, which is ideal for dark shops and sites that do not want to open ports to the cloud.

For more information about the capabilities of IBM Spectrum Control, see the [IBM Spectrum Control documentation](#).

For more information about pricing and other purchasing information, see [this web page](#).

**Note:** If you use IBM Spectrum Control or manage IBM block storage systems, you can access the no-charge version of IBM Storage Insights. For more information, see [Getting Started with IBM Storage Insights](#).

IBM Spectrum Control offers several reports that you can use to monitor IBM SAN Volume Controller systems to identify performance problems. IBM Spectrum Control provides improvements to the web-based user interface that is designed to offer easy access to your storage environment.

IBM Spectrum Control provides a large amount of detailed information about IBM SAN Volume Controller. The next sections provide some basic suggestions about what metrics need to be monitored and analyzed to debug potential bottleneck problems. In addition, which alerts need to be set to be notified when some specific metrics exceed limits that are considered important for this specific environment.

For more information about the installation, configuration, and administration of IBM Spectrum Control (including how to add a storage system), see the following web pages:

- ▶ [IBM Support](#)
- ▶ [Installing IBM Spectrum Control 5.4.2](#)

**Note** IBM Spectrum Control 5.3.0 or higher is recommended for monitoring IBM SAN Volume Controller Version 8.4.

## IBM Spectrum Control Dashboard

The performance dashboard provides Key Performance Indicators (in earlier releases, Best Practice Performance Guidelines) for the critical monitoring metrics. These guidelines do not represent the maximum operating limits of the related components. Instead, they suggest limits that are selected with an emphasis on maintaining a stable and predictable performance profile.

The dashboard displays the last 24 hours from the active viewing time and date. Selecting an individual element from the chart overlays the corresponding 24 hours for the previous day and seven days prior. This display allows for an immediate historical comparison of the respective metric. The day of reference can also be changed to allow historical comparison of previous days.

These dashboards provide two critical functions:

- ▶ An “at-a-glance” view of all the critical IBM Spectrum Virtualize monitoring metrics
- ▶ A historical comparison of the current metric profile with previous days that enables rapid detection of anomalous workloads and behaviors.



Figure 9-6 shows how to change the day of reference.

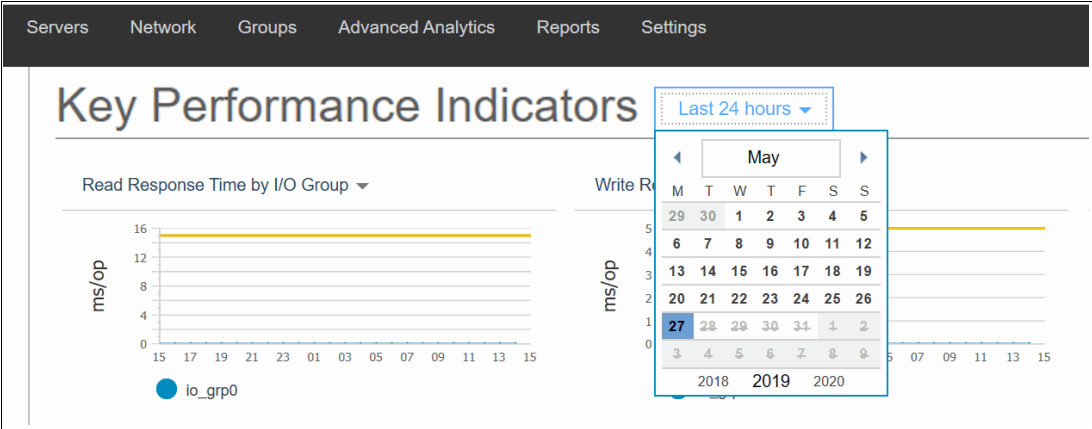


Figure 9-6 Change day of reference

Figure 9-7 shows a metric that is exceeding the best practice limit (orange line).

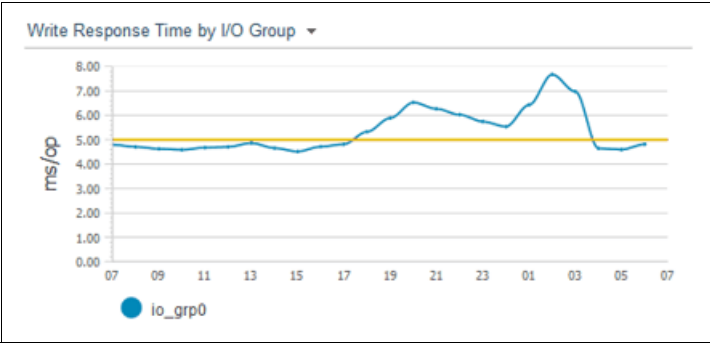


Figure 9-7 Metric exceeding best practice

Figure 9-8 shows the same chart as in Figure 9-7 with io\_grp0 selected, which overlays the previous day and 7 days prior.

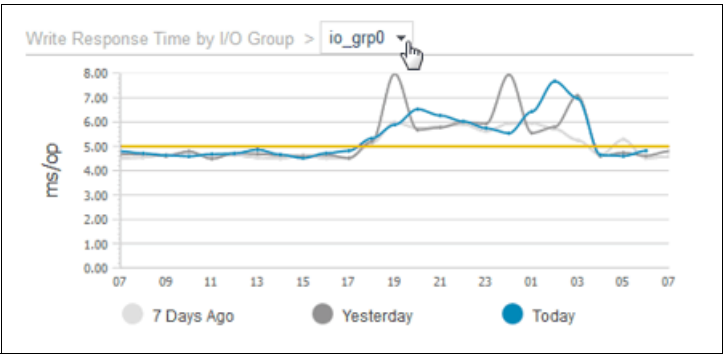


Figure 9-8 Changed chart due to iogrp selection

From this information, you can quickly conclude that this exception occurs every day at this same time, and is not a new phenomenon.

The line in yellow line indicates that if a component is constantly breaching the limit, then this component might be overly utilized; an occasional peak does not matter, but is more to help understand how the hardware is being used and is not a service indicator (response times > 10 ms are hardly acceptable these days).

**Note:** The panel has recently been renamed into *Key Performance Indicators* instead of *Best Practices Guidelines*.

## Key Performance Indicators

You can view the key metrics that are outside of a standard range for storage systems that run IBM Spectrum Virtualize by using the performance guidelines. The guidelines were established by a historical analysis of storage environments.

Most of the performance charts show an orange line that indicates the best practice value for the metric. These guidelines are established as the levels that allow for a diverse set of workload characteristics while maintaining a stable performance profile. The other lines on each chart represent the measured values for the metric for the resources on your storage system: I/O groups, ports, or nodes.

You can use the lines to compare how close to potentially becoming overloaded your resources are. If your storage system is responding poorly and the charts indicate overloaded resources, you might have to better balance the workload. You can balance the workload between the canisters of the cluster, potentially adding more canisters to the cluster, or move some workload to other storage systems.

Figure 9-9 shows the Key Performance Indicators in the dashboard.

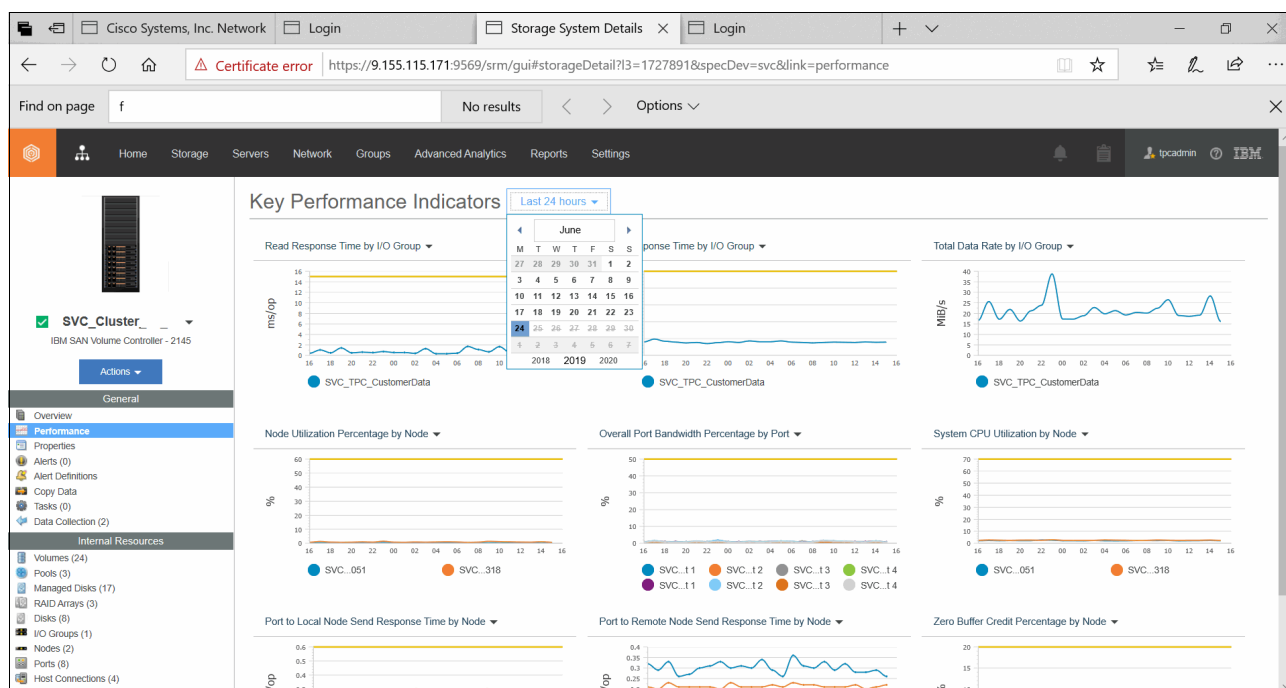


Figure 9-9 IBM Storage Insights

The charts show the hourly performance data measured for each resource on the selected day. Use the following charts to compare the workloads on your storage system with the best practice guidelines:

- ▶ **Node Utilization Percentage by Node:** Compare the guideline value for this metric, for example, 60% utilization, with the measured value from your system.  
The average of the bandwidth percentages of those ports in the node that are actively used for host and MDisk send and receive operations. The average is weighted by port speed and adjusted according to the technology limitations of the node hardware.  
Because for clusters without FC ports this chart is empty (or when no host IO is going on)
- ▶ **Overall Port Bandwidth Percentage by Port:** Compare the guideline value for this metric, for example, 50%, with the measured value from your system. Because a cluster can have many ports, the chart shows only the eight ports with the highest average bandwidth over the selected day.
- ▶ **Port-to-Local Node Send Response Time by Node:** Compare the guideline value for this metric, for example, 0.6 ms/op, with the measured value from your system.
- ▶ **Port-to-Remote Node Send Response Time by Node:** Because latencies for copy-services operations can vary widely, a guideline is not established for this metric. Use this chart to identify any discrepancies between the data rates of different nodes.
- ▶ **Read Response Time by I/O Group:** Compare the guideline value for this metric, for example, 15 ms/op, with the measured value from your system.  
It means, when you see this constantly being breached, then something might be wrong with the hardware.
- ▶ **System CPU Utilization by Node:** Compare the guideline value for this metric, for example, 70% utilization, with the measured value from your system.
- ▶ **Total Data Rate by I/O Group:** Because data rates can vary widely, a guideline is not established for this metric. Use this chart to identify any significant discrepancies between the data rates of different I/O groups because these discrepancies indicate that the workload is not balanced.
- ▶ **Write Response Time by I/O Group:** Compare the guideline value for this metric, for example, 5 ms/op, with the measured value from your system.
- ▶ **Zero Buffer Credit Percentage by Node:** Compare the guideline value for this metric, for example, 20%, with the measured value from your system.

**Note:** The guidelines are not thresholds, and they are not related to the alerting feature in IBM Spectrum Control. To create performance alerts that use the guidelines as thresholds, go to a resource detail window in the web-based GUI, click **Alerts** in the General section, and then click **Definitions**.

### 9.2.3 Performance monitoring with IBM Storage Insights

IBM Storage Insights (ISI) is an off-premises, IBM Cloud service that provides cognitive support capabilities, monitoring, and reporting for storage systems. Because it is an IBM Cloud service, getting started is simple and upgrades are handled automatically.

By using the IBM Cloud infrastructure, IBM Support can monitor your storage environment to help minimize the time to resolution of problems and collect diagnostic packages without requiring you to manually upload them. This wraparound support experience, from environment to instance, is unique to IBM Storage Insights and transforms how and when you get help.

IBM Storage Insights is a SaaS (Software as a Service) offering with its core running over IBM Cloud. IBM Storage Insights provides an unparalleled level of visibility across your storage environment to help you manage complex storage infrastructures and make cost-saving decisions. It combines proven IBM data management leadership with IBM analytics leadership from IBM Research® and a rich history of storage management expertise with a cloud delivery model, enabling you to take control of your storage environment.

As a cloud-based service, it enables you to deploy quickly and save storage administration time while optimizing your storage. It also helps automate aspects of the support process to enable faster resolution of issues. ISI optimizes storage infrastructure using cloud-based storage management and support platform with predictive analytics.

It allows you to optimize performance and to tier your data and storage systems for the right combination of speed, capacity and economy. IBM Storage Insights provides comprehensive storage management and helps to keep costs low, and might prevent downtime and loss of data or revenue. IBM Storage Insights Key features are:

- ▶ Rapid results when you need them
- ▶ Single pane view across your storage environment
- ▶ Performance analyses at your fingertips
- ▶ Valuable insight from predictive analytics
- ▶ Two editions that meet your needs
- ▶ Simplified, comprehensive and proactive product support

Figure 9-10 denotes an IBM Storage Insight® example window.

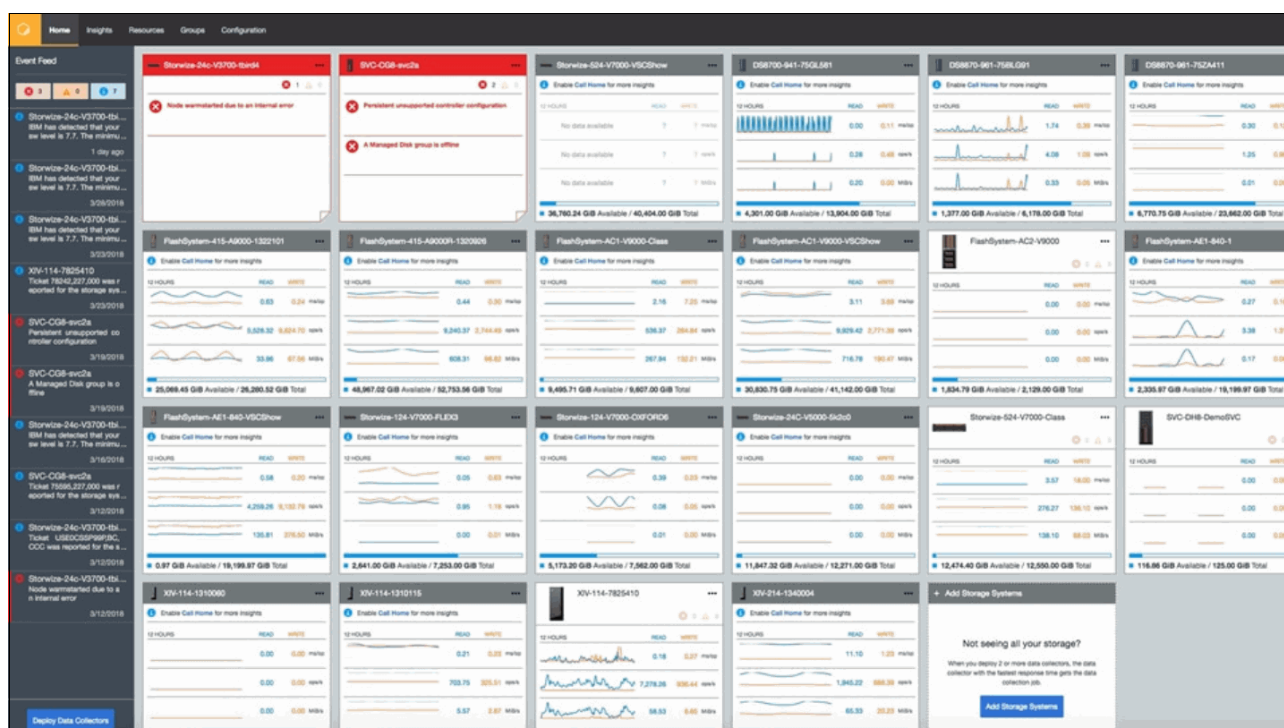


Figure 9-10 IBM Storage Insights

Understanding the security and data collection features of IBM Storage Insights Pro and IBM Storage Insights can help address the concerns of administrators and IT professionals who deploy the products in their environments and want to learn more about security and data collection. For more information, see [IBM Storage Insights Security](#).

**Note:** IBM strongly recommends the use of IBM Storage Insights or IBM Spectrum Control for a better use experience. IBM Storage Insights requires use of data collectors and the method of data collection changed recently to improve security and easy management. It is no longer required to have a user with admin privileges for data collectors; instead, a simple monitor user can get status information from the management node.

## Licensing and editions of IBM Storage Insights

Several editions of IBM Storage insights enable you to select the capabilities that serve your needs best. Licensing is implemented through different subscription levels.

- ▶ The no-charge version is called IBM Storage Insights and provides a unified view of a storage environment with a diagnostic events feed, an integrated support experience, and key capacity and performance metrics. IBM Storage Insights is available at no cost to IBM Storage Insights Pro subscribers and owners of IBM block storage systems who sign up. IBM Storage Insights provides an environment overview, integration in support processes, and shows you IBM analysis results.
- ▶ The capacity-based, subscription version is called IBM Storage Insights Pro and includes all the features of IBM Storage Insights plus a more comprehensive view of the performance, capacity, and health of storage resources. It also helps you reduce storage costs and optimize your data center by providing features like intelligent capacity planning, storage reclamation, storage tiering, and advanced performance metrics. The storage systems that you can monitor are expanded to include IBM file, object, software-defined storage (SDS) systems, and non-IBM block and file storage systems, such as EMC storage systems.

In both versions, when problems occur on your storage, you can get help to identify and resolve those problems and minimize potential downtime, where and when you need it.

Table 9-1 lists the different features of both versions.

*Table 9-1 Features in IBM Storage Insights and IBM Storage Insights Pro*

| Resource Management | Functions                                                       | IBM Storage Insights (free)                                                                                                 | IBM Storage Insights Pro (subscription)                             |
|---------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Monitoring          | Inventory management                                            | IBM block storage                                                                                                           | IBM and non-IBM block storage, file storage, and object storage     |
|                     | Logical configuration                                           | Basic                                                                                                                       | Advanced                                                            |
|                     | Health                                                          | Call Home events                                                                                                            | Call Home events                                                    |
|                     | Performance                                                     | Basic<br>(3 metrics: I/O rate, data rate, and response times aggregated for storage systems)                                | Advanced<br>(100+ metrics for storage systems and their components) |
|                     | Capacity                                                        | Basic<br>(4 metrics: allocated space, available space, total space, and compression savings aggregated for storage systems) | Advanced<br>(25+ metrics for storage systems and their components)  |
|                     | Drill down performance workflows to enable deep troubleshooting |                                                                                                                             | ✓                                                                   |
|                     | Explore virtualization relationships                            |                                                                                                                             | ✓                                                                   |
|                     | Explore replication relationships                               |                                                                                                                             | ✓                                                                   |
|                     | Retention of configuration and capacity data                    | Only the last 24 hours is shown                                                                                             | 2 years                                                             |
|                     | Retention of performance data                                   | Only the last 24 hours is shown                                                                                             | 1 year                                                              |
|                     | Reporting                                                       |                                                                                                                             | ✓                                                                   |
| Service             | Filter events to quickly isolate trouble spots                  | ✓*                                                                                                                          | ✓                                                                   |
|                     | Hassle-free log collection                                      | ✓                                                                                                                           | ✓                                                                   |
|                     | Simplified ticketing                                            | ✓                                                                                                                           | ✓                                                                   |
|                     | Show active PMRs and ticket history                             | ✓*                                                                                                                          | ✓                                                                   |

| Resource Management        | Functions                                                          | IBM Storage Insights (free) | IBM Storage Insights Pro (subscription) |
|----------------------------|--------------------------------------------------------------------|-----------------------------|-----------------------------------------|
| Analytics and optimization | Predictive Alerts                                                  | ✓                           | ✓                                       |
|                            | Customizable, multi-conditional alerting, including alert policies |                             | ✓                                       |
|                            | Performance planning                                               |                             | ✓                                       |
|                            | Capacity planning                                                  |                             | ✓                                       |
|                            | Business impact analysis (applications, departments, and groups)   |                             | ✓                                       |
|                            | Optimize data placement with tiering                               |                             | ✓                                       |
|                            | Optimize capacity with reclamation                                 |                             | ✓                                       |
| Security                   | ISO/IEC 27001 Information Security Management standards certified  | ✓                           | ✓                                       |
| Entitlements               |                                                                    | Free                        | Capacity-based subscription             |

**Restriction:** If you can access IBM Storage Insights but are not an IBM Storage Insights Pro subscriber, you must have a current warranty or maintenance agreement for an IBM block storage system to open tickets and send log packages.

The reporting feature is not available in IBM Storage Insights (free). To use the reporting feature, you must subscribe to IBM Storage Insights Pro or you can use IBM Spectrum Control.

For more information about how to purchase the IBM Storage Insights Pro version, see [IBM Support](#).

### ***IBM Storage Insights for IBM Spectrum Control***

IBM Storage Insights for IBM Spectrum Control is an IBM Cloud service that can help you predict and prevent storage problems before they impact your business. It is complementary to IBM Spectrum Control and is available at no additional cost if you have an active license with a current subscription and support agreement for IBM Virtual Storage Center, IBM Spectrum Storage Suite, or any edition of IBM Spectrum Control.

As an on-premises application, IBM Spectrum Control does not send the metadata about monitored devices off-site, which is ideal for dark shops and sites that do not want to open ports to the cloud. However, if your organization allows for communication between its network and the cloud, you can use IBM Storage Insights for IBM Spectrum Control to transform your support experience for IBM block storage.

IBM Storage Insights for IBM Spectrum Control and IBM Spectrum Control work hand in hand to monitor your storage environment. Here's how IBM Storage Insights for IBM Spectrum Control can transform your monitoring and support experience:

- ▶ Open, update, and track IBM Support tickets easily for your IBM block storage devices.
- ▶ Get hassle-free log collection by allowing IBM Support to collect diagnostic packages for devices so you do not have to.
- ▶ Use Call Home to monitor devices, get best practice recommendations, and filter events to quickly isolate trouble spots.
- ▶ Use IBM Support's ability to view the current and historical performance of your storage systems and help reduce the time-to-resolution of problems.

You can use IBM Storage Insights for IBM Spectrum Control for as long as you have an active license with a current subscription and support agreement for IBM Spectrum Control license. If your subscription and support lapses, you're no longer eligible for IBM Storage Insights for IBM Spectrum Control. To continue using IBM Storage Insights for IBM Spectrum Control, simply renew your IBM Spectrum Control license. You can also choose to subscribe to IBM Storage Insights Pro.

### **Feature comparison of IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control**

To understand the usability of IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control for your environment, we compare the features of IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control.



Table 9-2 shows the features in IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control.

*Table 9-2 IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control*

| Resource Management | Features                                                          | IBM Spectrum Control (Advanced edition)                                                     | IBM Storage Insights for IBM Spectrum Control                   |
|---------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Monitoring          | Inventory                                                         | IBM and non-IBM block storage, file storage, object storage, hypervisors, fabrics, switches | IBM and non-IBM block storage, file storage, and object storage |
|                     | Call Home events                                                  |                                                                                             | ✓                                                               |
|                     | Performance                                                       | ✓<br>(1-minute intervals)                                                                   | ✓<br>(5-minute intervals)                                       |
|                     | Capacity                                                          | ✓                                                                                           | ✓                                                               |
|                     | Drill down performance workflow to troubleshoot bottlenecks       | ✓                                                                                           | ✓                                                               |
|                     | Explore virtualization relationships                              |                                                                                             |                                                                 |
|                     | Explore replication relationships                                 | ✓                                                                                           | ✓                                                               |
|                     | Retain performance data                                           |                                                                                             |                                                                 |
| Service             | Deployment method                                                 |                                                                                             |                                                                 |
|                     | Filter Call Home events to quickly isolate trouble spots          |                                                                                             | ✓                                                               |
|                     | Hassle-free log collection                                        |                                                                                             | ✓                                                               |
|                     | Simplified ticketing                                              |                                                                                             | ✓                                                               |
|                     | Show active PMRs and ticket history                               |                                                                                             | ✓                                                               |
|                     | Active directory and LDAP integration for managing users          | ✓                                                                                           |                                                                 |
| Reporting           | Inventory, capacity, performance, and storage consumption reports | ✓                                                                                           | ✓                                                               |
|                     | Rollup reporting                                                  | ✓                                                                                           |                                                                 |
|                     | REST API                                                          | ✓                                                                                           |                                                                 |

| Resource Management | Features                                                           | IBM Spectrum Control (Advanced edition) | IBM Storage Insights for IBM Spectrum Control |
|---------------------|--------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------|
| Alerting            | Predictive Alerts                                                  | ✓                                       | ✓                                             |
|                     | Customizable, multi-conditional alerting, including alert policies | ✓                                       | ✓                                             |
| Analytics           | Performance planning                                               | ✓                                       | ✓                                             |
|                     | Capacity planning                                                  | ✓                                       | ✓                                             |
|                     | Business impact analysis (applications, departments, and groups)   | ✓                                       | ✓                                             |
|                     | Provisioning with service classes and capacity pools               | ✓                                       |                                               |
|                     | Balance workload across pools                                      | ✓                                       |                                               |
|                     | Optimize data placement with tiering                               | ✓                                       | ✓                                             |
|                     | Optimize capacity with reclamation                                 | ✓                                       | ✓                                             |
|                     | Transform and convert volumes                                      | ✓                                       |                                               |
| Pricing             |                                                                    | On-premises licensing                   | No charge for IBM Spectrum Control customers  |

You can upgrade IBM Storage Insights to IBM Storage Insights for IBM Spectrum Control, if you have an active license of IBM Spectrum Control. For more information, see [Storage Insights Registration](#), choose the option for IBM Spectrum Control, and follow the prompts.

IBM Storage Insights for IBM Spectrum Control does not include the service level agreement for IBM Storage Insights Pro. Terms and conditions for IBM Storage Insights for IBM Spectrum Control are available at [Cloud Services Terms](#).

IBM Storage Insights, IBM Storage Insights Pro, and IBM Storage Insights for IBM Spectrum Control show some similarities, but the following differences exist:

- ▶ IBM Storage Insights is an off-premises, IBM Cloud service that is available free of charge if you own IBM block storage systems. It provides a unified dashboard for IBM block storage systems with a diagnostic events feed, a streamlined support experience, and key capacity and performance information.
- ▶ IBM Storage Insights Pro is an off-premises, IBM Cloud service that is available on subscription and expands the capabilities of IBM Storage Insights. You can monitor IBM file, object, and software-defined storage (SDS) systems, and non-IBM block and file storage systems, such as Dell/EMC storage systems.

It also includes configurable alerts and predictive analytics that help you to reduce costs, plan capacity, and detect and investigate performance issues. You get recommendations for reclaiming unused storage, recommendations for optimizing the placement of tiered data, capacity planning analytics, and performance troubleshooting tools.

- ▶ IBM Storage Insights for IBM Spectrum Control is similar to IBM Storage Insights Pro in capability and is available for no additional cost if you have an active license with a current subscription and support agreement for IBM Virtual Storage Center, IBM Spectrum Storage Suite, or any edition of IBM Spectrum Control.

### ***IBM Spectrum Storage Suite***

IBM Spectrum Storage Suite gives you unlimited access to the IBM Spectrum Storage software family and IBM Cloud Object Storage software with licensing on a flat, cost-per-TB basis to make pricing easy to understand and predictable as capacity grows. Structured specifically to meet changing storage needs, the suite is ideal for organizations just starting out with software-defined storage, and for those with established infrastructures who need to expand their capabilities.

- ▶ IBM Spectrum Control: Analytics-driven hybrid cloud data management to reduce costs
- ▶ IBM Spectrum Protect: Optimized hybrid cloud data protection to reduce backup costs
- ▶ IBM Spectrum Protect Plus: Complete VM protection and availability that is easy to set up and manage yet scalable for the enterprise
- ▶ IBM Spectrum Archive: Fast data retention that reduces total cost of ownership for active archive data
- ▶ IBM Spectrum Virtualize: Virtualization of mixed block environments to increase data storage
- ▶ IBM Spectrum Accelerate: Enterprise block storage for hybrid cloud
- ▶ IBM Spectrum Scale: High-performance, highly scalable hybrid cloud storage for unstructured data driving cognitive applications
- ▶ IBM Cloud Object Storage: Flexible, scalable and simple object storage with geo-dispersed enterprise availability and security for hybrid cloud workloads

Because IBM Spectrum Storage Suite contains IBM Spectrum Control, you can deploy IBM Storage Insight for IBM Spectrum Control.

**Note:** Alerts are a good way to be notified of conditions and potential problems that are detected on your storage. If you use IBM Spectrum Control and IBM Storage Insights for IBM Spectrum Control together to enhance your monitoring capabilities, It is recommended that you define alerts in one of the offerings and not both.

By defining all your alerts in one offering, you can avoid receiving duplicate or conflicting notifications when alert conditions are detected.

## **Implementation and Setup of IBM Storage Insights**

To use IBM Storage Insights with the IBM Spectrum Virtualize, you must sign up. For more information, see the [Storage Insights Registration](#) web page.

### ***Sign-up process***

Consider the following points about the sign-up process:

- ▶ For the sign-up process, you need an IBM ID. If you do not have an IBM ID, create your IBM account and complete the short form.

- ▶ When you register, specify an owner for IBM Storage Insights. The owner manages access for other users and acts as the main contact.
- ▶ You receive a Welcome email when IBM Storage Insights is ready. The email contains a direct link to your dashboard.

Figure 9-11 shows the IBM Storage Insight registration window.

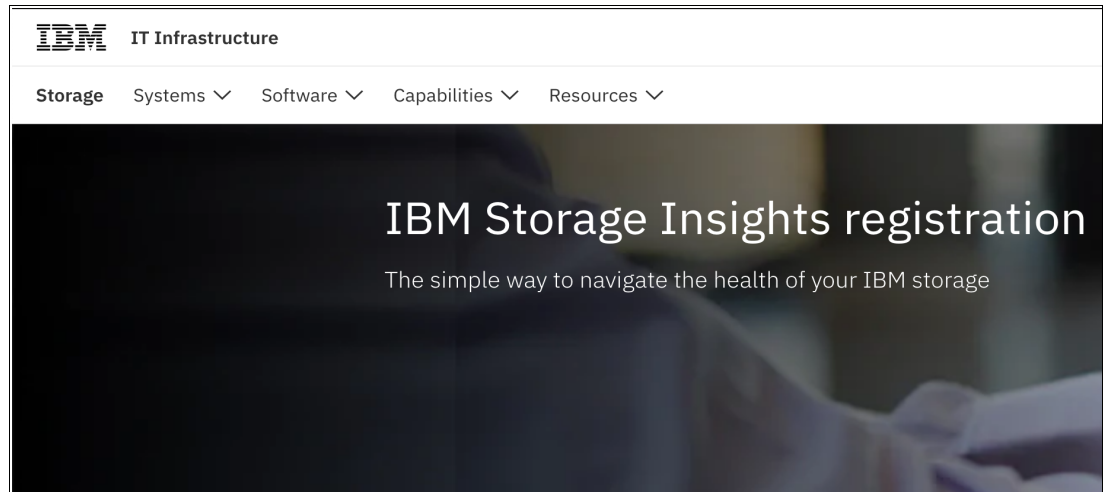


Figure 9-11 Storage Insights registration window

Figure 9-12 shows the registration website when you scroll down. You can select whether you want to register for IBM Storage Insights or IBM Storage Insights for Spectrum Control. For more information about the differences of the IBM Storage Insights software, see “Licensing and editions of IBM Storage Insights” on page 365.

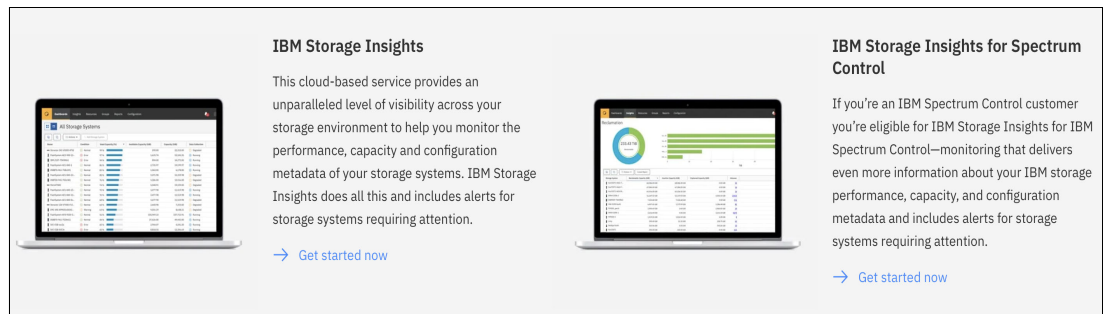


Figure 9-12 Choose IBM Storage Insights or IBM Storage Insights for Spectrum Control

Figure 9-13 shows the Log-in window in the registration process. If you have your credentials, enter your ID and proceed to the next window by clicking **Continue**. If you do not have an ID, click **Create an IBMid**.

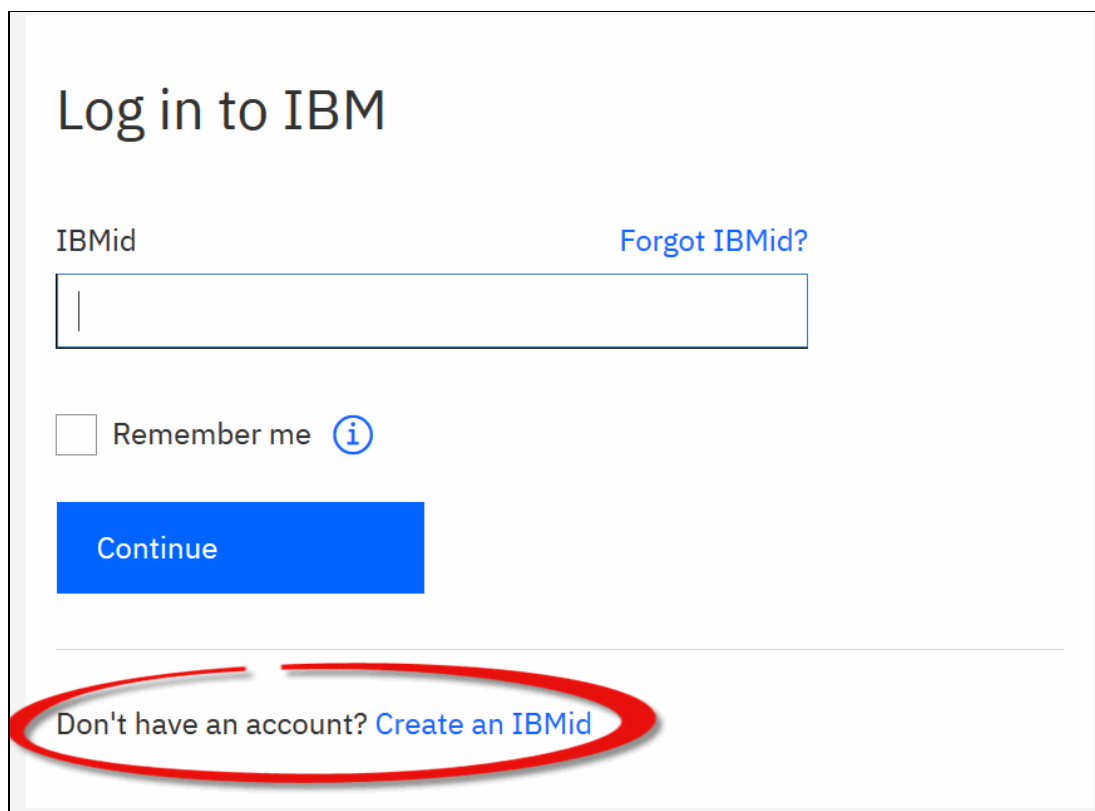


Figure 9-13 Registration login window

Figure 9-14 on page 374 shows the login window prompt for your ID and password.

If you want to create an IBMid, see Figure 9-14 on page 374 for reference and provide the following information:

- ▶ Email
- ▶ First name
- ▶ Last name
- ▶ Country or region
- ▶ Password

Select the option if you want to receive Information from IBM to keep you informed of products, services and offerings. You can withdraw your marketing consent at any time by sending an email to [net supp@us.ibm.com](mailto:net supp@us.ibm.com). Also, you can unsubscribe from receiving marketing emails by clicking the unsubscribe link in any email.

For more information about our processing, see the [IBM Privacy Statement](#).

## Create your IBM account

Access to trials, demos, starter kits, services and APIs



### Sign up for an IBMid

Already have an IBM account? [Log in](#)

Email \*

First name \*

Last name \*

Country or region \* (?) ✓

United States ▼

Set a password \*

- 8 characters minimum
- One uppercase character
- One lowercase character
- One number

IBM may use my contact data to keep me informed of products, services and offerings:

☒ by email.

You can withdraw your marketing consent at any time by sending an email to [netsupp@us.ibm.com](mailto:netsupp@us.ibm.com). Also you may unsubscribe from receiving marketing emails by clicking the unsubscribe link in each such email.

More information on our processing can be found in the [IBM Privacy Statement](#). By submitting this form, I acknowledge that I have read and understand the IBM Privacy Statement.

I accept the product [Terms and Conditions](#) of this registration form.

[Continue](#)

Figure 9-14 Create an IBM account

Figure 9-15 shows the log-in window prompt for your ID and password.

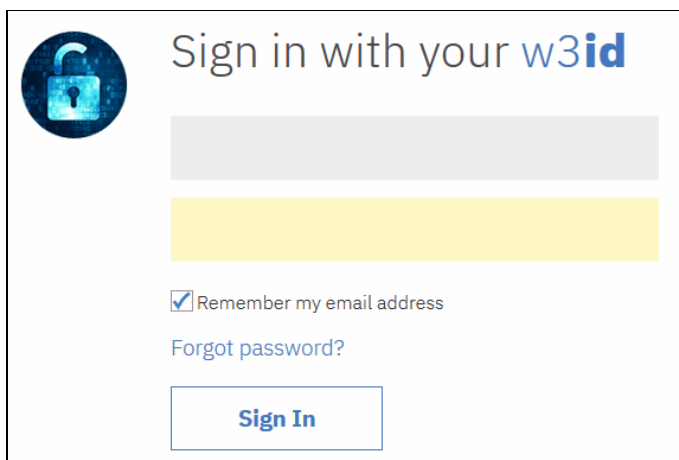
The image shows a sign-in window. On the left is a circular icon with a blue padlock. To the right of the icon, the text "Sign in with your w3id" is displayed. Below this text are two input fields: a light gray one for the ID and a yellow one for the password. Under the password field is a checkbox labeled "Remember my email address" which is checked. Below the checkbox is a link that says "Forgot password?". At the bottom is a blue button labeled "Sign In".

Figure 9-15 Registration - ID and password

Figure 9-16 shows the registration form. Complete the following information:

- ▶ Company name  
The name must be unique.
- ▶ You can consider other identifying features, such as a location or department:
  - Owner details
  - The person who registered for IBM Storage Insights
  - Access granted for storage trends, health of storage, and access to support
  - Email address or ID
  - First and last name

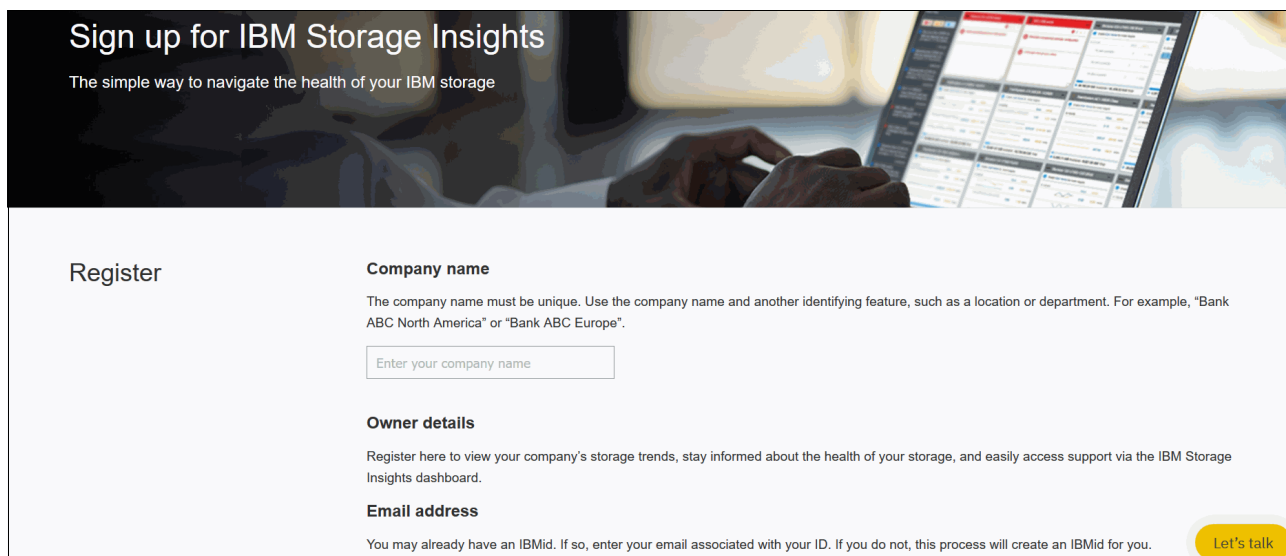
The image shows the registration form for IBM Storage Insights. At the top, it says "Sign up for IBM Storage Insights" and "The simple way to navigate the health of your IBM storage". Below this is a section titled "Register". The form has three main sections: "Company name" with a text input field and a note that the name must be unique; "Owner details" with a text input field and a note about registering to view storage trends; and "Email address" with a text input field and a note about entering an email associated with an ID. A yellow button labeled "Let's talk" is at the bottom right.

Figure 9-16 IBM storage insights registration form

After your registration for Storage Insights is complete, download and install the data collector for your system. Extract the data collector, run the data collector installer script, and ensure that your server (or virtual machine) can access the `host_name:port` that is specific to your instance of Storage Insights. After the data collector is installed on the system, you can add your storage devices to a Storage Insights dashboard.

**Note:** To connect to your instance of Storage Insights, you must configure your firewall to allow outbound communication on the default HTTPS port 443 using the Transmission Control Protocol (TCP). The User Datagram Protocol (UDP) is not supported.

### ***Deploying a data collector***

Complete the following steps to deploy a lightweight data collector in your data center to stream performance, capacity, and configuration metadata to IBM Storage Insights:

1. Log in to IBM Storage Insights (the link is in your Welcome email).
2. From the **Configuration** → **Data Collector** page, download the data collector for your operating system (Windows, Linux, or AIX).
3. Extract the contents of the data collector file on the virtual machine or physical server where you want it to run.
4. For Windows, run `installDataCollectorService.bat`.  
For Linux or AIX, run `installDataCollectorService.sh`.

After the data collector is deployed, it attempts to establish a connection to IBM Storage Insights. When the connection is complete, you're ready to start adding your storage systems for monitoring.

**Requirements:** The following requirements must be met:

- ▶ 1 GB RAM
- ▶ 1 GB disk space
- ▶ Windows, AIX, or Linux (x86-64 systems only)

For more information, see [this web page](#).

**Note:** To avoid potential problems, ensure that the operating system on the server or virtual machine where you install the data collector includes general or extended support for maintenance and security.

Storage system metadata is sent to IBM Storage Insights, such as the following information:

- ▶ The configuration of the storage system, such as name, firmware, and capacity.
- ▶ The internal resources of the storage system, such as volumes, pools, nodes, ports, and disks. This information includes the names and the configuration and capacity information for each internal resource.
- ▶ The performance of storage system resources and internal resources, such as pools and volumes.

For more information about the metadata that is collected and how it is used, see the following resources:

- ▶ [IBM Storage Insights Fact Sheet PDF Download](#)
- ▶ [IBM Storage Insights Security Guide PDF Download](#)

### ***Adding a storage system***

Complete the following steps to connect IBM Storage Insights to the storage systems that you want to monitor.

1. On the Operations dashboard in IBM Storage Insights, look for the button to add storage systems.



2. Click **Add Storage Systems** and follow the prompts. You can add one or more storage systems at a time.

For more information, see [this web page](#).

### ***View your dashboard***

On the Operations dashboard, view the following information:

- ▶ Storage systems that are being monitored.
- ▶ A dynamic diagnostic feed that tells you which storage systems require attention.
- ▶ Key capacity metrics so you know whether you have enough capacity to meet your storage demands.
- ▶ Key performance metrics so that you know whether the performance of your storage systems meets operational requirements.

For more information, see [this web page](#).

### ***Enable Call Home***

Get the most out of IBM Storage Insights by enabling Call Home on your IBM block storage systems. With Call Home, your dashboard includes a diagnostic feed of events and notifications about their health and status.

Stay informed so you can act quickly to resolve incidents before they affect critical storage operations.

For more information, see [this web page](#).

### ***Adding users to your dashboard***

Optional: Add users, such as other storage administrators, IBM Technical Advisors, and IBM Business Partners, at any time so that they can access your IBM Storage Insights dashboard.

1. In IBM Storage Insights, click your user name in the upper-right corner of the dashboard.
2. Click **Manage Users**.
3. On your MYIBM page, ensure that **IBM Storage Insights** is selected.
4. Click **Add new user**.

For more information, see [this web page](#).

## **9.3 Capacity metrics for block storage systems**

Effective and exact capacity management is based on fundamental knowledge of capacity metrics in the IBM SAN Volume Controller system. Data reduction pools, thin provisioning, compression, and deduplication add many metrics to the IBM SAN Volume Controller GUI, IBM Spectrum Control, IBM Storage Insights. The capacity metrics in this section are based on the IBM Spectrum Control, Version V5.3.3.

The Capacity section on the Dashboard provides an overall view of system capacity. This section displays physical capacity, volume capacity, and capacity savings.

Physical capacity indicates the total capacity in all storage on the system. Physical capacity includes all the storage that the system can virtualize and assign to pools. Physical capacity is displayed in a bar graph and divided into three categories: Stored Capacity, Available Capacity, and Total Physical. If the system supports self-compressing drives, certain system configurations make determining accurate physical capacity on the system difficult.

For example, if the system contains self-compressed drives and data reduction pools without compression enabled, the system cannot determine the accurate amount of physical capacity that is used on the system. In this case, over provisioning and losing access to write operations is possible. If this condition is detected by the system, the Physical Capacity section of the Dashboard page displays a message instead of capacity information.

To recover from this condition, you need to ensure that all thin-provisioned volumes and thin-provisioned volumes that are deduplicated are migrated to volumes with compression enabled in the data reduction pools. Alternatively, you can migrate the volumes to fully allocated volumes and use the drive compression to save capacity.

We discuss the different values and how they help to determine the capacity utilization, trends in capacity and space usage, and more importantly can prevent an out of space situation in the environment. To not run out of space you have to understand the different level of components, such as arrays, pools, system, and so on. You should understand which limits exist for each of them so that you understand although one pool might be fine, other pools could run out of storage. Simply monitoring at the system level is not appropriate if you have two or more pools.

Figure 9-17 shows how to interpret the capacity and savings in a storage environment.

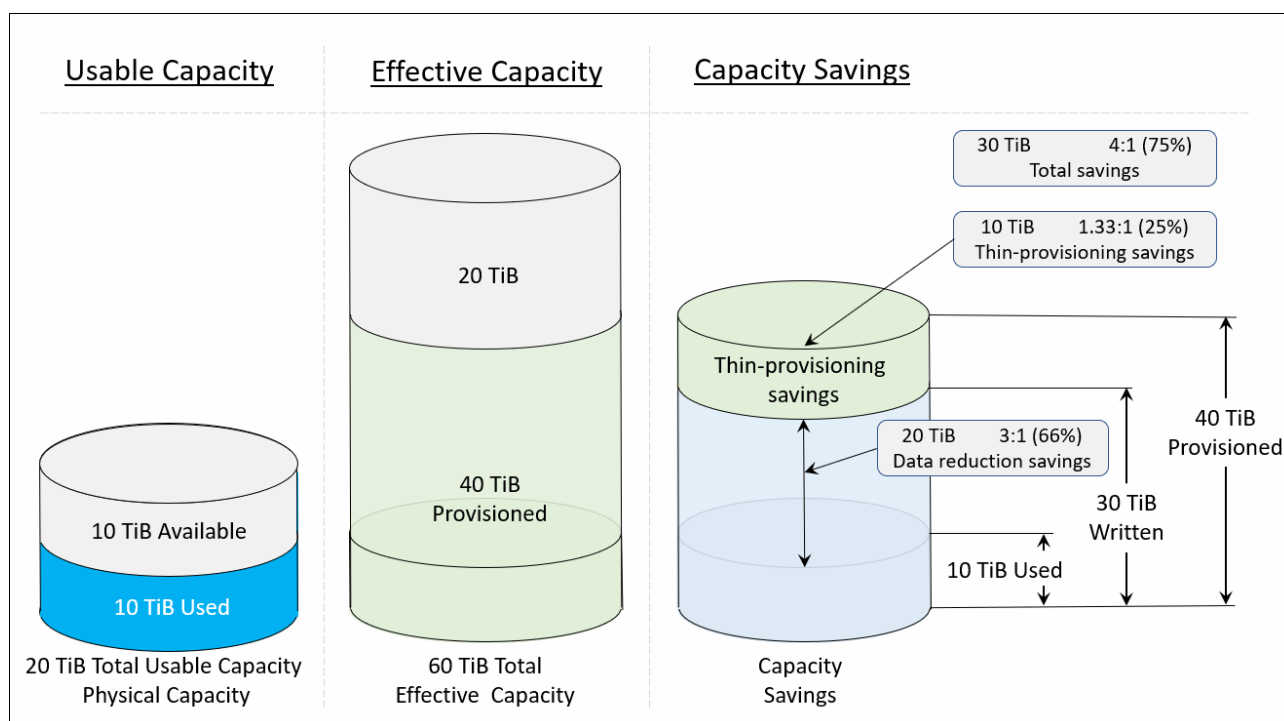


Figure 9-17 Understanding capacity information

The following metrics can be added to capacity charts for storage systems within capacity planning. Use the charts to detect capacity shortages and space usage trends.

Alphabetical lists of the capacity and space usage metrics that you can add to charts are provided in the following sections:

- Storage system capacity metrics
- Pool capacity metrics
- Volume capacity metrics

### 9.3.1 Storage system capacity metrics

In this section, we describe the system capacity metrics.

#### Allocated Space (GiB)

This metric is the amount of space that is allocated to the regular and thin-provisioned volumes in the pools. If the pool is a parent pool, the amount of space that is allocated to the volumes in the child pools is also included.

The space that is allocated for thin-provisioned volumes is less than their virtual capacity, which is shown in the Total Volume Capacity (GiB) column. If a pool does not have thin-provisioned volumes, the value for allocated space is the same as the value for total volume capacity.

Allocated space is the same as used space on all storage systems, with the exception of IBM SAN Volume Controller, Storwize, or IBM FlashSystem 9100 storage systems that are thin provisioned.

Figure 9-18 shows the Allocated Space (8,711 GiB) of an IBM SAN Volume Controller.

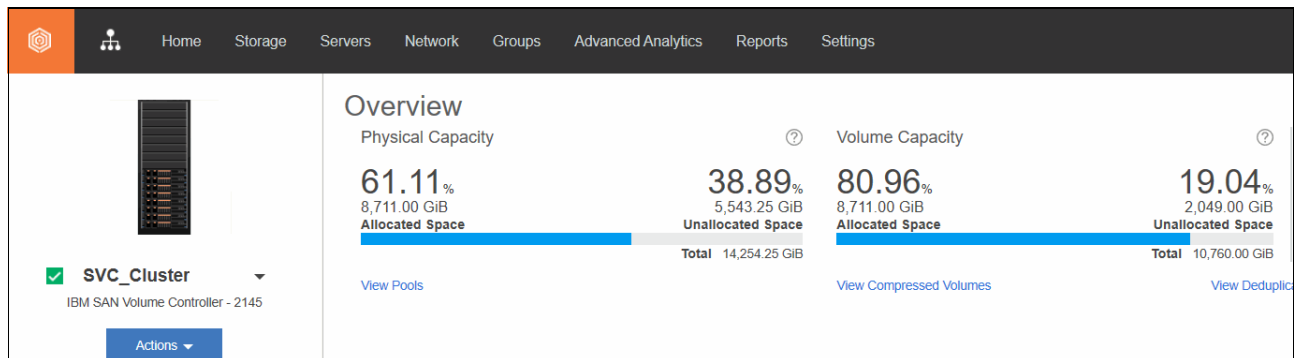


Figure 9-18 Allocated Space

#### Assigned Volume Space (GiB)

The total volume space in the storage system that is mapped or assigned to host systems, including child pool capacity. It is the sum of all volumes virtual size, so this value can exceed the physical capacity of your system when you over provision storage.

Go to IBM Spectrum Control and choose your Storage Volume Controller (for example SVC\_Cluster) then, select **Actions** → **View Capacity**, scroll down and verify if **Assigned Volume Space** is selected (right-click → **Assigned Volume Space** ensure that the option is selected).

Figure 9-19 on page 380 shows the Assigned Volume Space (GiB) in an IBM SAN Volume Controller system. The total volume space for this system is shown.

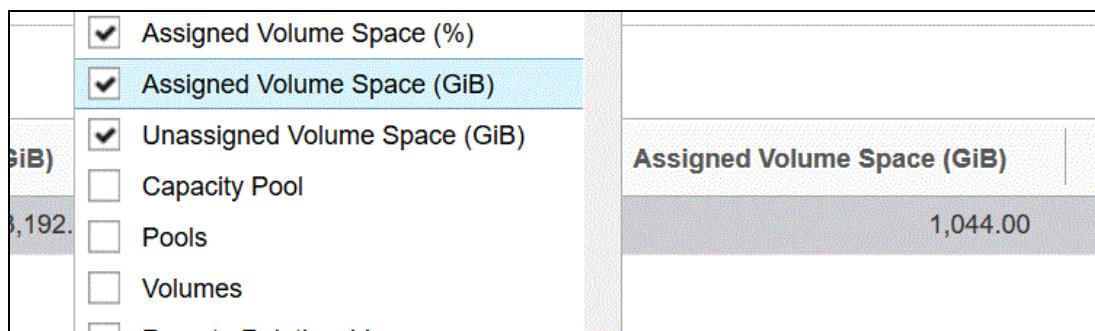


Figure 9-19 Assigned Volume Space (GiB)

Figure 9-20 shows a view of the actual and historic Assigned Volume Space in an IBM SAN Volume Controller.

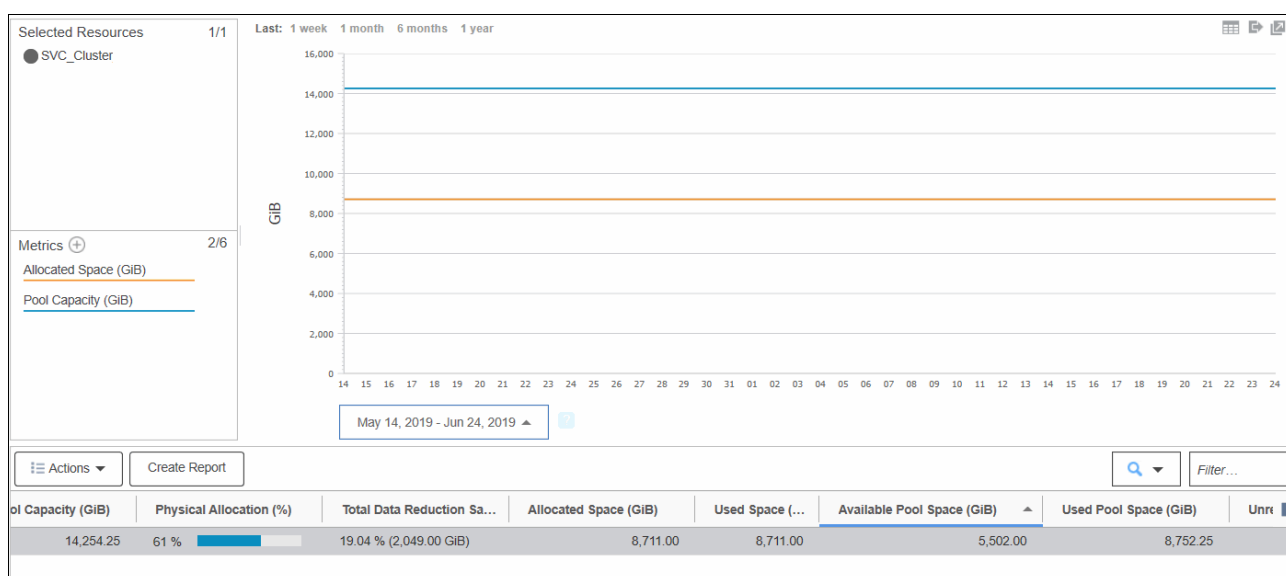


Figure 9-20 Assigned Volume Space - graph

To view other information, click the plus sign (+) that is next to metrics on the left.

### Available Pool Space (GiB)

This metric is the total amount of the space in the pools that is not allocated to the volumes in the pools. To calculate available space, the following formula is used:

$$(\text{pool capacity} - \text{allocated space})$$

For some storage systems, the pool space is limited by the physical capacity after data reduction, so the more you can compress data the more you can store in such a pool. For other systems there is a limit in the address space before compression, which means that even if you can compress the data extremely highly, you might not be able to use all of the physical space, because the address range before compression is exceeded.

### Compression Savings (%)

This metric is the estimated amount and percentage of capacity that is saved by using data compression, across all pools on the storage system. The percentage is calculated across all compressed volumes in the pools and does not include the capacity of non-compressed volumes.

For storage systems with drives that use inline data compression technology, the Compression Savings does not include the capacity savings that are achieved at the drive level. Drive level compression occurs after striping across MDisk and the RAID distribution on the disks/FCM level. There is no information available about which volume a block of data belongs to that was just compressed, so the information about compression within the drives is only available at the array level

The following formula is used to calculate the amount of storage space that is saved:

$(\text{written space} - \text{compressed size})$

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{written space} - \text{compressed size}) \div \text{written space}) \times 100$

For example, the written space, which is the amount of data that is written to the volumes before compression, is 40 GiB. The compressed size, which reflects the size of compressed data that is written to disk, is just 10 GiB. Therefore, the compression savings percentage across all compressed volumes is 75%.

### **Deduplication Savings (%)**

This metric is the estimated amount and percentage of capacity that is saved by using data deduplication, across all data reduction pools on the storage system. The percentage is calculated across all deduplicated volumes in the pools and does not include the capacity of volumes that are not deduplicated.

The following formula is used to calculate the amount of storage space that is saved:

$(\text{written space} - \text{deduplicated size})$

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{written space} - \text{deduplicated size}) \div \text{written space}) \times 100$

For example, the written space, which is the amount of data that is written to the volumes before deduplication, is 40 GiB. The deduplicated size, which reflects the size of deduplicated data that is written to disk, is just 10 GB. Therefore, data deduplication reduced the size of the data that is written by 75%.

### **Physical Allocation (%)**

This metric is the percentage of physical capacity in the pools that is allocated to the regular volumes, the thin-provisioned volumes, and the volumes in child pools. Check the value for physical allocation to see whether:

- ▶ The physical capacity of the pools is fully allocated; that is, the value for physical allocation is 100%.
- ▶ Sufficient capacity exists to:
  - Provision new volumes with storage
  - Allocate to the compressed and thin-provisioned volumes

Figure 9-21 shows a brief description of Physical Allocation (%).

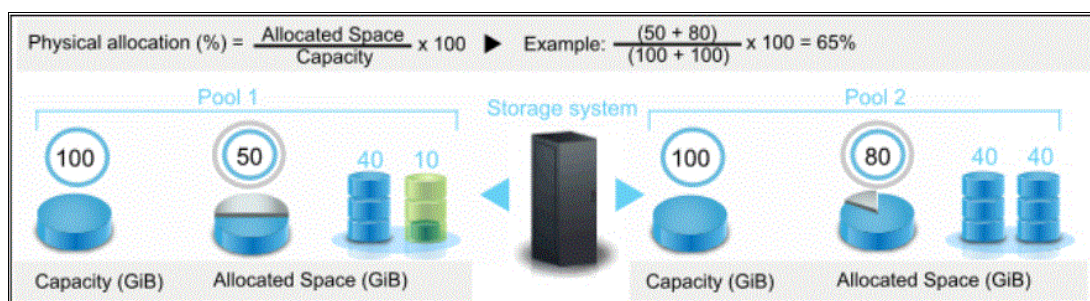


Figure 9-21 Physical Allocation (%)

### Pool Capacity (GiB)

This metric is total amount of storage space in the pools, which might include overhead space if the disks for the pools are not formatted.

For XIV systems and IBM Spectrum Accelerate, capacity represents the physical or “hard” capacity of the pool, not the virtual “soft” capacity. Pools that are allocated from other pools are not included in the total pool space.

### Pool Shortfall (%)

This metric is percentage of space that is over-committed to the pools with thin-provisioned volumes. For example, you commit 100 GiB of space to a thin-provisioned volume in a pool with a capacity of 50 GiB. As the space is allocated to the thin-provisioned volume in increments of 10 GiB, the space available for allocation decreases and the shortfall in capacity becomes more acute.

To calculate the shortfall, the following formula is used:

$$[(\text{unallocatable space} \div \text{committed but unallocated space}) \times 100]$$

A pool shortfall occurs when you commit more space to the volumes in the pools than is physically available to the pools. If the physical space available to the pools is less than the committed virtual space, then the pools do not have enough space to fulfill the commitment to the virtual space.

For example, the physical capacity of the pools is 70 GiB, but 150 GiB of virtual space was committed to the thin-provisioned volumes. If the volumes are using 50 GiB, then 100 GiB is still committed to those volumes (150 GiB - 50 GiB) with only 20 GiB of available pool space (70 GiB - 50 GiB). Because only 20 GiB of the pool space is available, 80 GiB of the committed space cannot be allocated (100 GiB - 20 GiB). In this case, the percentage of committed space that is unavailable is 80% [(80 GiB ÷ 100 GiB × 100)].

The advantage of using shortfall rather than a simple overprovisioning factor, is that shortfall always has values between 0 and 100% and therefore is well suited for simple alerting that can be applied to multiple pools.

Figure 9-22 explains the shortfall.

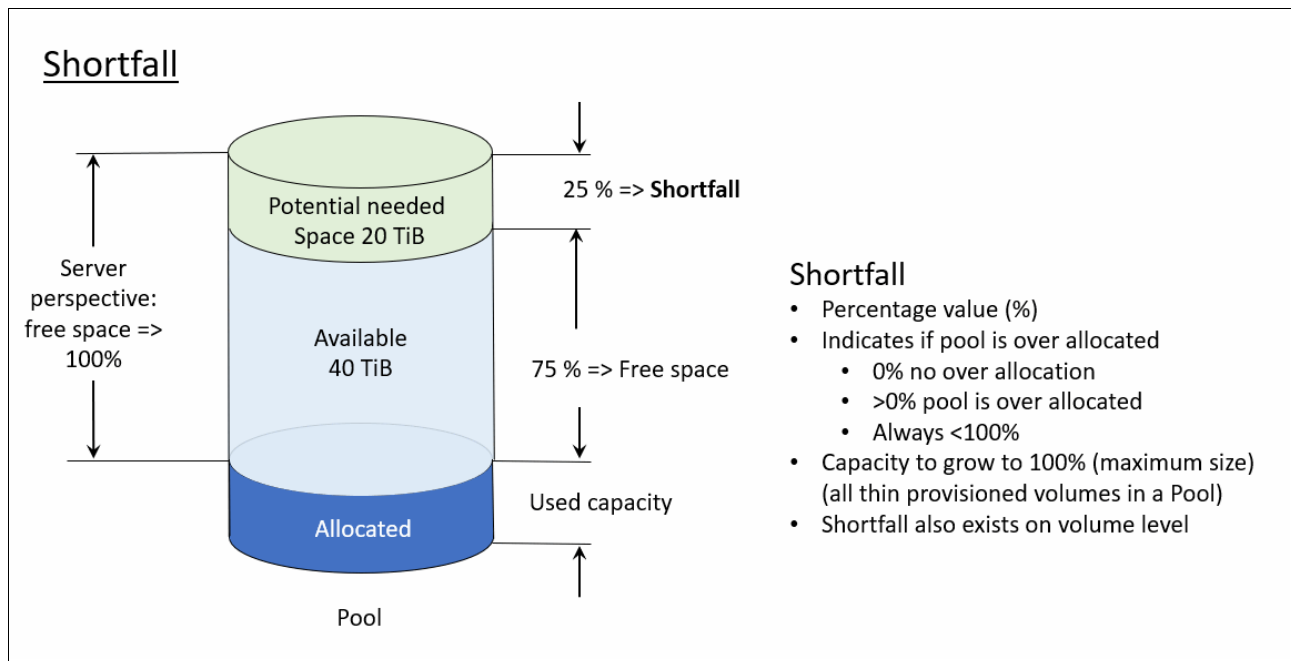


Figure 9-22 Shortfall

### Total Data Reduction Savings (%)

This metric is estimated amount and percentage of capacity that is saved by using data deduplication, data compression, and thin provisioning.

The following formula is used to calculate the amount of storage space that is saved:

(Total Volume Capacity - Allocated Space)

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{Total Volume Capacity} - \text{Allocated Space}) \div \text{Total Volume Capacity}) \times 100$

### Total Volume Capacity (GiB)

This metric is total amount of storage space that can be made available to the regular and thin-provisioned volumes in the pools. If the pool is a parent pool, it also includes the storage space that can be made available to the volumes in the child pools. In other words this is the capacity that a server will see.

### Unallocatable Volume Space (GiB)

This metric is amount of space that cannot be allocated to volumes because the physical capacity of the pools cannot meet the demands for virtual space. The following formula is used to calculate this value:

[Total Volume Capacity - Pool Capacity]

### Unallocated Volume Space (GiB)

This metric is total amount of remaining space that can be allocated to the volumes in the pools. The following formula is used to calculate this value:

[Total Volume Capacity - Allocated Space]

The space that is allocated for thin-provisioned volumes is typically less than their virtual capacity. Therefore, the unallocated space represents the difference between the virtual capacity and the allocated space for all the volumes in the pools.

### Virtual Allocation (%)

This metric is percentage of the physical capacity that is committed to the virtual capacity of the volumes in the pool. If the value exceeds 100%, the physical capacity does not meet the demands for virtual capacity. The following formula is used to calculate this value:

$$[(\text{Total Volume Capacity} \div \text{Pool Capacity}) \times 100]$$

Example: If the allocation percentage is 200% for a total storage pool size of 15 GiB, then the virtual capacity that is committed to the volumes in the pool is 30 GiB. This configuration means that twice as much space is committed than is physically contained in the pool. If the allocation percentage is 100% for the same pool, then the virtual capacity that is committed to the pool is 15 GiB. This configuration means that all the physical capacity of the pool is already allocated to volumes.

An allocation percentage that is higher than 100% is considered aggressive because insufficient physical capacity is available in the pool to satisfy the maximum allocation for all the thin-provisioned volumes in the pool. In such cases, you can use the value for Shortfall (%) to estimate how critical the shortage of space is for a pool.

## 9.3.2 Pool capacity metrics

If sufficient data is collected about the pools in your data center, you can view charts that compare the capacity of the pools with the space that is allocated to the pools and the space that is still available in the pools. In the **Zero Capacity** column on the Pools page, you can see the date, based on the space usage trends for the pool, when the pool will run out of available space.

**Tip:** To order the pools in the table by the amount of space available to the pools, click **Filter by column**, and then, click **Zero Capacity**.



Figure 9-23 shows an example of Zero Capacity trend.

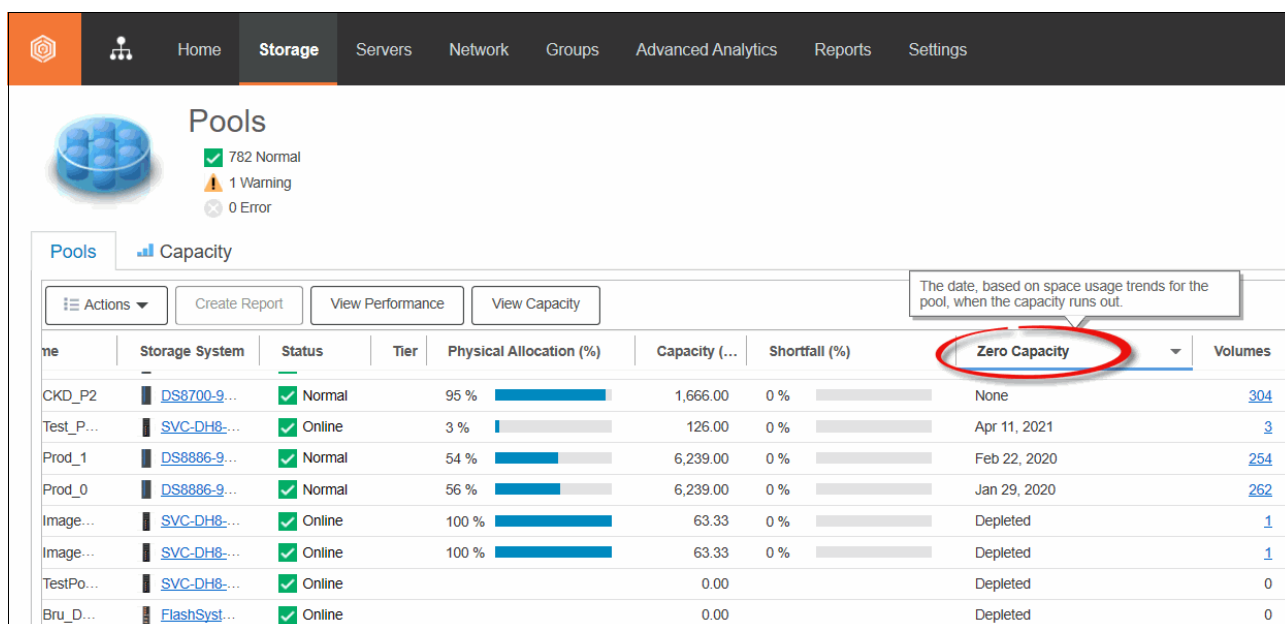


Figure 9-23 Zero Capacity trend

The following values can be shown in the Zero Capacity column:

- ▶ A date
 

The data that is based on space usage trends for the pool when the capacity runs out (projected).
- ▶ None
 

Based on the current trend, no date can be calculated for when the pool will be filled (for example, if the trend is negative) as data is moved out of the pool.
- ▶ Depleted
 

The pool is full.

The metrics that are described next can be added to capacity charts for storage systems within capacity planning. Use the charts to detect capacity shortages and space usage trends.

## Allocated Space (GiB)

The amount of space that is allocated to the regular and thin-provisioned volumes in the pool. If the pool is a parent pool, the amount of space that is allocated to the volumes in the child pools is also included.

The space that is allocated for thin-provisioned volumes is less than their virtual capacity, which is shown in the Total Volume Capacity column. If a pool does not contain thin-provisioned volumes, this value is the same as Total Volume Capacity.

Allocated space is the same as used space on all storage systems, except for resources that run IBM Spectrum Virtualize. These resources might have more allocated space than used space if the storage administrator preallocated some space for thin-provisioned volumes when the volumes were created.

### **Assigned Volume Space (GiB)**

The space on all of the volumes in a pool that are mapped or assigned to host systems. For a thin-provisioning pool, this value includes the virtual capacity of thin-provisioned volumes, which might exceed the total space in the pool.

### **Available Pool Space (GiB)**

The amount of space that is available to create new volumes in the pool. If the pool is a parent pool, the amount of space that is allocated to the volumes in the child pools is also included.

### **Available Soft Space (GiB)**

The amount of virtual storage space that is available to allocate to volumes in a storage pool.

### **Capacity (GiB)**

The total amount of storage space in the pool, which might include overhead space if the disks for the pool are not formatted. For XIV systems and IBM Spectrum Accelerate, capacity represents the physical or “hard” capacity of the pool, not the virtual “soft” capacity.

### **Compression Savings (%)**

The estimated amount and percentage of capacity that is saved by using data compression. The percentage is calculated across all compressed volumes in the pool and does not include the capacity of non-compressed volumes.

For storage systems with drives that use inline data compression technology, the Compression Savings does not include the capacity savings that are achieved at the drive level.

The following formula is used to calculate the amount of storage space that is saved:

(written space - compressed size)

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{written space} - \text{compressed size}) \div \text{written space}) \times 100$

For example, the written space, which is the amount of data that is written to the volumes before compression, is 40 GiB. The compressed size, which reflects the size of compressed data that is written to disk, is just 10 GiB. Therefore, the compression savings percentage across all compressed volumes is 75%.

### **Deduplication Savings (%)**

The estimated amount and percentage of capacity that is saved by using data deduplication. The percentage is calculated across all deduplicated volumes in the pool and does not include the capacity of volumes that are not deduplicated.

The following formula is used to calculate the amount of storage space that is saved:

(written space - deduplicated size)

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{written space} - \text{deduplicated size}) \div \text{written space}) \times 100$

For example, the written space, which is the amount of data that is written to the volumes before deduplication, is 40 GiB. The deduplicated size, which reflects the size of deduplicated data that is written to disk, is just 10 GB. Therefore, data deduplication reduced the size of the data that is written by 75%.

## Enterprise HDD Available Space (GiB)

The amount of storage space that is available on the Enterprise hard disk drives that can be used by IBM Easy Tier for re-tiering the volume extents in the pool.

## Enterprise HDD Capacity (GiB)

The total amount of storage space on the Enterprise hard disk drives that can be used by Easy Tier for re-tiering the volume extents in the pool

## Nearline HDD Available Space (GiB)

The amount of storage space that is available on the Nearline hard disk drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

## Nearline HDD Capacity (GiB)

The total amount of storage space on the Nearline hard disk drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

## Physical Allocation (%)

The percentage of physical capacity in the pool that is allocated to the regular volumes, the thin-provisioned volumes, and the volumes in child pools. This value is always less than or equal to 100% because you cannot allocate more physical space than is available in a pool. Check the value for physical allocation to see whether:

- ▶ The physical capacity of the pool is fully allocated; that is, the value for physical allocation is 100%.
- ▶ Whether you have sufficient capacity to:
  - Provision new volumes with storage
  - Allocate to the compressed and thin-provisioned volumes in the pool

Figure 9-24 denotes the Physical Allocation.

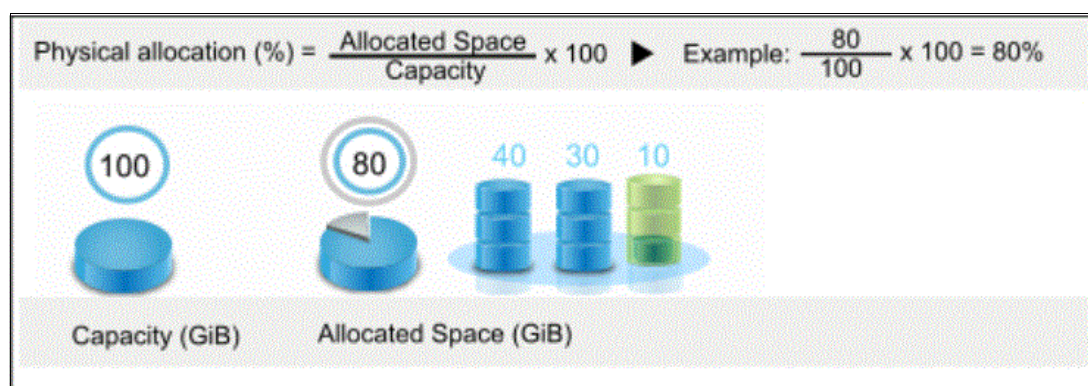


Figure 9-24 Physical Allocation

## Shortfall (%)

The percentage of space that is over committed to pools with thin-provisioned volumes. For example, you commit 100 GiB of space to a thin-provisioned volume in a pool with a capacity of 50 GiB. As the space is allocated to the thin-provisioned volume in increments of 10 GiB, the space available for allocation decreases and the shortfall in capacity becomes more acute.

If the pool is not thin-provisioned, the shortfall percentage equals zero. If shortfall percentage is not calculated for the storage system, the field is left blank.

To calculate shortfall, the following formula is used:

$$[(\text{Unallocatable Space} \div \text{Committed but Unallocated Space}) \times 100]$$

You can use this percentage to determine when the amount of over-committed space in a pool is at a critically high level. Specifically, if the physical space in a pool is less than the committed virtual space, then the pool does not have enough space to fulfill the commitment to virtual space. This value represents the percentage of the committed virtual space that is not available in a pool. As more space is used over time by volumes while the pool capacity remains the same, this percentage increases.

Example: The remaining physical capacity of a pool is 70 GiB, but 150 GiB of virtual space was committed to thin-provisioned volumes. If the volumes are using 50 GiB, then 100 GiB is still committed to the volumes (150 GiB - 50 GiB) with a shortfall of 30 GiB (70 GiB remaining pool space - 100 GiB remaining commitment of volume space to the volumes).

Because the volumes are overcommitted by 30 GiB based on the available space in the pool, the shortfall is 30% when the following calculation is used:

$$[(100 \text{ GiB unallocated volume space} - 70 \text{ GiB remaining pool space}) \div 100 \text{ GiB unallocated volume space}] \times 100$$

### **Soft Space (GiB)**

The amount of virtual storage space that is configured for the pool.

### **SSD Available Space (GiB)**

The amount of storage space that is available on the solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **SSD Capacity (GiB)**

The total amount of storage space on the solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Tier 0 Flash Available Space (GiB)**

The amount of storage space that is available on the Tier 0 flash solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Tier 0 Flash Capacity (GiB)**

The total amount of storage space on the Tier 0 flash solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Tier 1 Flash Available Space (GiB)**

The amount of storage space that is available on the Tier 1 flash, read-intensive solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Tier 1 Flash Capacity (GiB)**

The total amount of storage space on the Tier 1 flash, read-intensive solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Total Data Reduction Savings (%)**

The estimated amount and percentage of capacity that is saved by using data deduplication, data compression, and thin provisioning, across all volumes in the pool.

The following formula is used to calculate the amount of storage space that is saved:

$(\text{Total Volume Capacity} - \text{Allocated Space})$

The following formula is used to calculate the percentage of capacity that is saved:

$((\text{Total Volume Capacity} - \text{Allocated Space}) \div \text{Total Volume Capacity}) \times 100$

### **Total Volume Capacity (GiB)**

The total amount of storage space that can be made available to the regular and thin-provisioned volumes in the pool. If the pool is a parent pool, it also includes the storage space that can be made available to the volumes in the child pools.

### **Unallocatable Volume Space (GiB)**

The amount of space that cannot be allocated to volumes because the physical capacity of the pools cannot meet the demands for virtual space. The following formula is used to calculate this value:

$[\text{Total Volume Capacity} - \text{Pool Capacity}]$

### **Unallocated Volume Space (GiB)**

The total amount of remaining space that can be allocated to the volumes in the pools. The following formula is used to calculate this value:

$[\text{Total Volume Capacity} - \text{Allocated Space}]$

The space that is allocated for thin-provisioned volumes is typically less than their virtual capacity. Therefore, the unallocated space represents the difference between the virtual capacity and the allocated space for all the volumes in the pools.

### **Unassigned Volume Space (GiB)**

The total amount of space in the volumes that are not assigned to hosts.

### **Virtual Allocation (%)**

The percentage of the physical capacity that is committed to the virtual capacity of the volumes in the pool. If the value exceeds 100%, the physical capacity does not meet the demands for virtual capacity. The following formula is used to calculate this value:

$[(\text{Total Volume Capacity} \div \text{Pool Capacity}) \times 100]$

Example: If the allocation percentage is 200% for a total storage pool size of 15 GiB, then the virtual capacity that is committed to the volumes in the pool is 30 GiB. This configuration means that twice as much space is committed than is physically contained in the pool. If the allocation percentage is 100% for the same pool, then the virtual capacity that is committed to the pool is 15 GiB. This configuration means that all the physical capacity of the pool is already allocated to volumes.

An allocation percentage that is higher than 100% is considered aggressive because insufficient physical capacity is available in the pool to satisfy the maximum allocation for all the thin-provisioned volumes in the pool. In such cases, you can use the value for Shortfall (%) to estimate how critical the shortage of space is for a pool.

## **9.3.3 Volume capacity metrics**

You use the capacity chart to detect capacity shortages for the following types of volumes:

- ▶ Space-efficient volumes such as compressed volumes and thin-provisioned volumes
- ▶ Regular volumes that use Easy Tier to re-tier volume extents

You can review the allocation of space to space-efficient volumes to detect capacity shortfalls. You can also review the space usage of volumes that use Easy Tier to distribute volume extents across Enterprise HDD, Nearline HDD, and SSD storage.

The following metrics can be added to capacity charts for storage systems within capacity planning. Use the charts to detect capacity shortages and space usage trends.

### **Allocated Space (GiB)**

The amount of space that is allocated to the compressed, 5-provisioned, or the Easy Tier volume. Typically, the space that is allocated to the compressed or thin-provisioned volume is less than the capacity of the volume. For Easy Tier volumes, allocated space is the space that is allocated to the volume's extents on the Enterprise HDD, Nearline HDD, or SSD drives.

### **Capacity (GiB)**

The capacity of the compressed or the thin-provisioned volume, which comprises the sum of the allocated and the unallocated space. If the disks for the pool are not formatted, the capacity of the volume might include the overhead space. For thin-provisioned volumes in XIV systems pools or IBM Spectrum Accelerate pools, capacity is the physical or 'hard' capacity of the volume.

### **Compression Savings (%)**

The estimated amount and percentage of capacity that is saved by using data compression. the following formula is used to calculate the amount of storage space that is saved:

(written space - compressed size)

The following formula is used to calculate the percentage of capacity that is saved:

((written space - compressed size) ÷ written space) × 100

**Exception:** This column is blank for compressed volumes that are also deduplicated on storage systems that run IBM Spectrum Virtualize.

### **Enterprise HDD Capacity (GiB)**

The total amount of storage space on the Enterprise hard disk drive that the Easy Tier volume uses for re-tiering the volume extents.

### **Nearline HDD Capacity (GiB)**

The total amount of storage space on the Nearline hard disk drive that the Easy Tier volume uses for re-tiering the volume extents.

### **SSD Capacity (GiB)**

The total amount of storage space on the solid-state drive that the Easy Tier volume uses for re-tiering the volume extents.

### **Tier 0 Flash Capacity (GiB)**

The total amount of storage space on the Tier 0 flash solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### **Tier 1 Flash Capacity (GiB)**

The total amount of storage space on the Tier 1 flash, read-intensive solid-state drives that can be used by Easy Tier for re-tiering the volume extents in the pool.

### Used Space (GiB)

The amount of allocated space that is used by the compressed, thin-provisioned, or Easy Tier volume.

For compressed and thin-provisioned volumes, used space might not be the same as allocated space on the following types of storage systems:

- ▶ SAN Volume Controller
- ▶ Storwize family block storage systems
- ▶ FlashSystem storage systems

For thin-provisioned volumes, used space might not be the same as allocated space because you can preallocate space to the thin-provisioned volumes when the volumes are created. For compressed volumes, used space might not be the same as allocated space because more space is used to read the data than is used to write the data to the disk. For regular volumes that use Easy Tier on the storage systems that are listed, used space is the same as allocated space.

### Written Space (GiB)

The amount of data that is written from the assigned hosts to the volume before compression or data deduplication are used to reduce the size of the data. For example, the written space for a volume is 40 GiB. After compression, the volume used space, which reflects the size of compressed data that is written to disk, is just 10 GiB.

## 9.4 Creating alerts for IBM Spectrum Control and IBM Storage Insights

In this section we provide information about alerting with IBM Spectrum Control and IBM Storage Insights. Keep in mind that the free version of Storage Insights does not support alerting.

New data reduction technologies add more intelligence and capacity savings to your environment. If you use data reduction on different layers, such as hardware compression in the IBM FlashSystem 9100 Flash Core Modules (if a FS9100 is virtualized by the IBM SAN Volume Controller) and in the Data Reduction Pools you will have to pay more attention in preventing insufficient space remaining in the back-end storage device.

First, It is important to distinguish between thin provisioning and over-allocation (over-provisioning). Thin provisioning is a method for optimizing the use of available storage. It relies on allocation of blocks of data on-demand versus the traditional method of allocating all of the blocks up front. This method eliminates almost all white space, which helps avoid the poor usage rates (often as low as 10%) that occur in the traditional storage allocation method. Traditionally, large pools of storage capacity are allocated to individual servers, but remain unused (not written to).

Over provisioning means, that in total more space is being assigned and promised to the hosts. They can possibly try to store more data on the storage subsystem, as physical capacity is available. This will result in an out-of-space condition.

**Remember:** You must constantly monitor your environment to avoid over-provisioning situations that can be harmful to the environment and can cause data loss.

It is also important to keep at least 15% free space for Garbage Collection in the background. For more information, see “DRP internal details” on page 101.

Data reduction technologies return back some space. If the space that is used for the data can be reduced, the saved up space can be used for other data. But remember that, depending on the type of data, deleting might not result in freeing up much space.

Imagine if you have three identical or almost identical files on a file system that were deduplicated. This issue resulted in getting a good compression ratio (three files, but stored only once). If you now delete one file, you do not gain more space because the deduplicated data must stay on the storage (because two other versions refer to the data). Similar results can be seen when several FlashCopies of one source are used.

### 9.4.1 Alert examples

Table 9-3 shows an Alert for IBM SAN Volume Controller based on Pool level.

*Table 9-3 Event examples for IBM SAN Volume Controller*

| System | Entity | Resource type   | Event                      |
|--------|--------|-----------------|----------------------------|
| SVC    | Pool   | Usable Capacity | Physical allocation => nn% |

Other alerts are possible as well, but generally % alerts are best suited, as the alert definition applies to all pools in a storage system.

### 9.4.2 Alert to monitor back-end capacity: Physical allocation

In this section, we show how to deploy IBM Spectrum Control or IBM Storage Insights Pro to monitor storage capacity and set up thresholds to notify and prevent us from running out of space.

The following example shows how to create an alert to get status Information about the remaining physical space with a SAN Volume Controller.

First, assign a severity to an alert. Assigning a severity can help you more quickly identify and address the critical conditions that are detected on resources. The severity that you assign depends on the guidelines and procedures within your organization. Default assignments are provided for each alert.

Table 9-4 lists the possible alert severities.

*Table 9-4 Alert severities*

| Option   | Description                                                                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Critical | Alert is critical and needs to be resolved. For example, alerts that notify you when the amount of available space on a file system falls below a specified threshold. |
| Warning  | Alerts that are not critical, but represent potential problems. For example, alerts that notify you when the status of a data collection job is not normal.            |



| Option        | Description                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Informational | Alerts that might not require any action to resolve and are primarily for informational purposes. For example, alerts that are generated when a new pool is added to a storage system |

In this example, we created the following thresholds:

- ▶ Critical (15% space in the RAID Array left)
- ▶ Warning (20% space in the RAID Array left)
- ▶ Information (30% space in the RAID Array left)

Adjust the percentage levels to the required levels as needed. Keep in mind that the process to extend storage might take some time (ordering, installation, provisioning, and so on).

The advantage of this way to set up an Alert Policy is that you can add various IBM SAN Volume Controllers to this customized alert.

Figure 9-25 shows how to start creating a new Alert Policy that monitors the remaining free capacity at the Pool Level. The customized Alert is not tied to the Data Reduction Pool because it works regardless of the pool type.

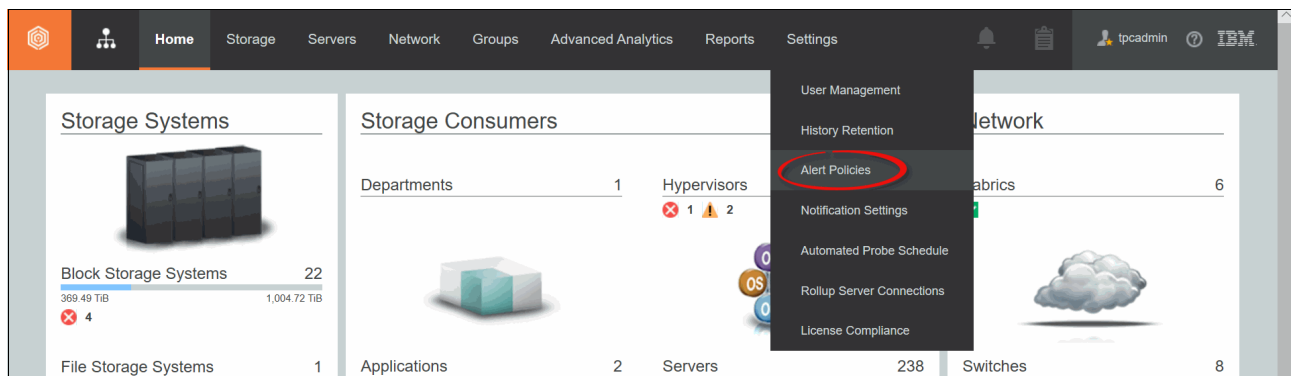
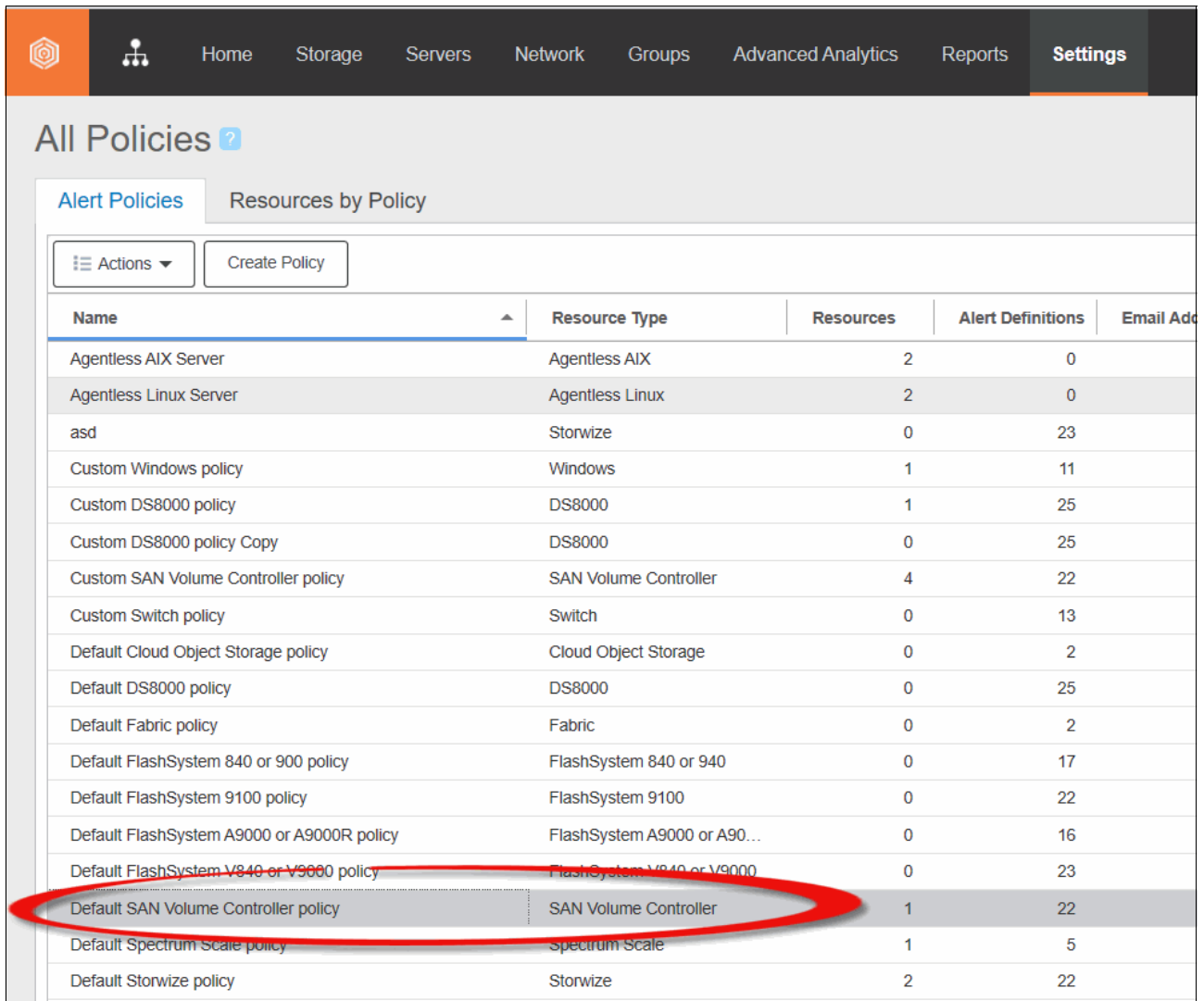


Figure 9-25 Create new Alert Policy

In the following example we create a new Alert Policy by copying the existing one. You might change an existing Alert Policy (in our example the Default Policy) as well. Consider that a storage subsystem can be active in only one Alert Policy.

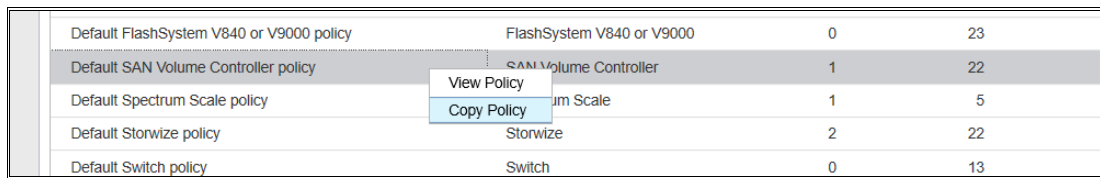
Figure 9-26 shows the Default Policy of IBM SAN Volume Controller.



| Name                                       | Resource Type               | Resources | Alert Definitions | Email Address |
|--------------------------------------------|-----------------------------|-----------|-------------------|---------------|
| Agentless AIX Server                       | Agentless AIX               | 2         | 0                 |               |
| Agentless Linux Server                     | Agentless Linux             | 2         | 0                 |               |
| asd                                        | Storwize                    | 0         | 23                |               |
| Custom Windows policy                      | Windows                     | 1         | 11                |               |
| Custom DS8000 policy                       | DS8000                      | 1         | 25                |               |
| Custom DS8000 policy Copy                  | DS8000                      | 0         | 25                |               |
| Custom SAN Volume Controller policy        | SAN Volume Controller       | 4         | 22                |               |
| Custom Switch policy                       | Switch                      | 0         | 13                |               |
| Default Cloud Object Storage policy        | Cloud Object Storage        | 0         | 2                 |               |
| Default DS8000 policy                      | DS8000                      | 0         | 25                |               |
| Default Fabric policy                      | Fabric                      | 0         | 2                 |               |
| Default FlashSystem 840 or 940 policy      | FlashSystem 840 or 940      | 0         | 17                |               |
| Default FlashSystem 9100 policy            | FlashSystem 9100            | 0         | 22                |               |
| Default FlashSystem A9000 or A9000R policy | FlashSystem A9000 or A9000R | 0         | 16                |               |
| Default FlashSystem V840 or V9000 policy   | FlashSystem V840 or V9000   | 0         | 23                |               |
| Default SAN Volume Controller policy       | SAN Volume Controller       | 1         | 22                |               |
| Default Spectrum Scale policy              | Spectrum Scale              | 1         | 5                 |               |
| Default Storwize policy                    | Storwize                    | 2         | 22                |               |

Figure 9-26 Default Alert Policy

Figure 9-27 describes how to copy a Policy to create one. Mouse over the Policy that you want to copy and then, click the left mouse button and choose **Copy Policy**.



|                                          |                           |   |    |
|------------------------------------------|---------------------------|---|----|
| Default FlashSystem V840 or V9000 policy | FlashSystem V840 or V9000 | 0 | 23 |
| Default SAN Volume Controller policy     | SAN Volume Controller     | 1 | 22 |
| Default Spectrum Scale policy            | Spectrum Scale            | 1 | 5  |
| Default Storwize policy                  | Storwize                  | 2 | 22 |
| Default Switch policy                    | Switch                    | 0 | 13 |

Figure 9-27 Copying a Policy and creating a Policy

Figure 9-28 shows how to rename the previously copied Policy. The new Policy is stored as another Policy. One SAN Volume Controller system can be added only to a single policy. You can add the system later on if you are not sure at this time (Optional: Select **Resource** and select the option).

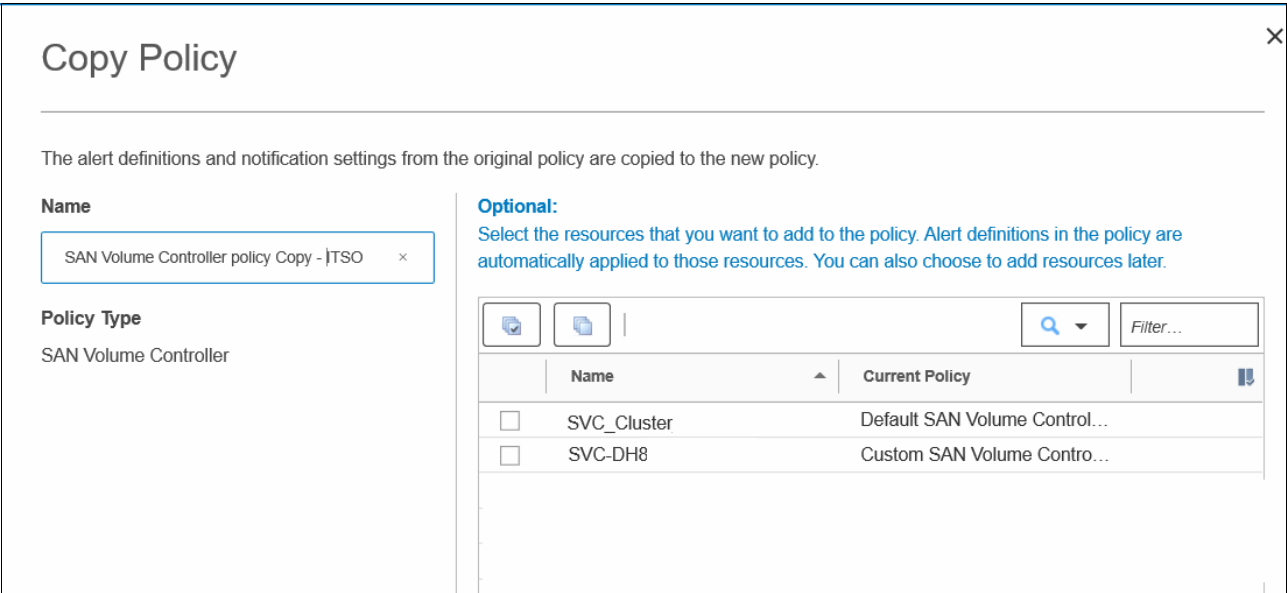


Figure 9-28 Create new Alert Policy

Figure 9-29 shows the newly created Alert Policy “SAN Volume Controller policy Copy - ITSO” with all the existing alerts inherited from the Default Policy.

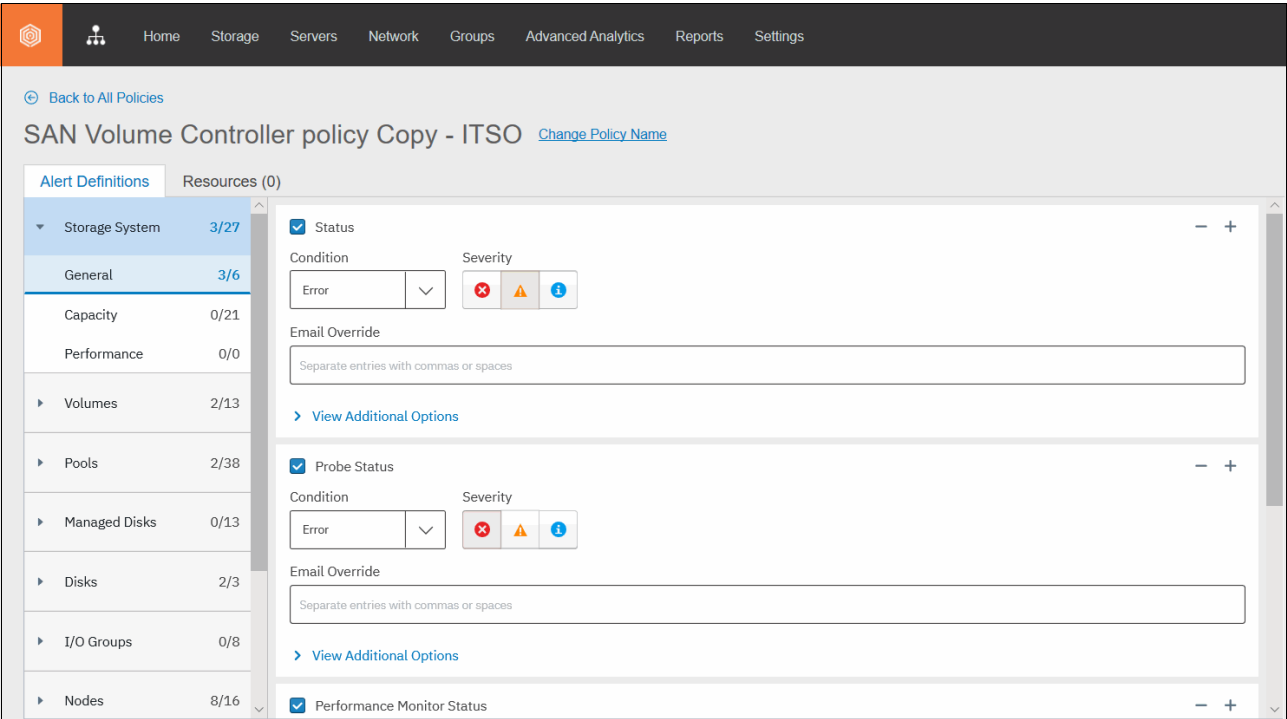


Figure 9-29 New Policy with inherited Alert Definitions

Figure 9-30 shows how to choose the required Alert Definitions by clicking **Pool** → **Capacity**.

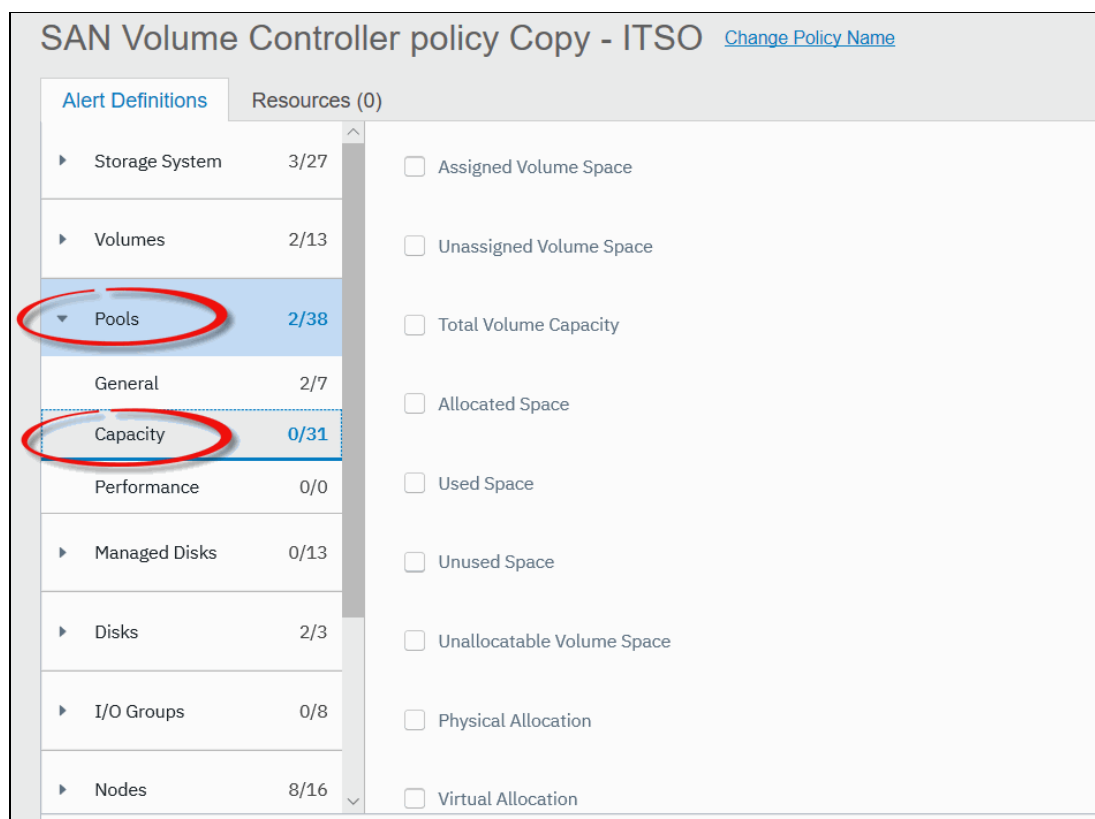


Figure 9-30 Alert Definition Pool -> Capacity

Figure 9-31 on page 397 denotes the tasks for setting up the Critical definition by monitoring the Available Physical Space (%) and releasing Policy Notifications at 15%. This example implies that when 85% or more physical space is taken a critical Notification will be sent using the predefined method.

Predefined methods can be one of the following options:

- ▶ Email Addresses
- ▶ SNMP
- ▶ IBM Netcool/OMNibus
- ▶ Windows Event Log or UNIX syslog

These methods have to be defined before you can choose them. If your environment does not have predefined methods, see Figure 9-35 on page 399.

**Note:** With IBM Storage Insights, you can only send emails.

Back to All Policies

SAN Volume Controller policy Copy - ITSO [Change Policy Name](#)

Alert Definitions Resources (0)

Storage System 3/27

Volumes 2/13

Pools 3/38

General 2/7

Capacity 1/31

Performance 0/0

☒ Physical Allocation

Operator Value Severity

<= 15 %

Email Override

Separate entries with commas or spaces

[View Additional Options](#)

☐ Virtual Allocation

Policy Notifications

Email Addresses ? No policy addresses

SNMP Not configured

Netcool / OMNIbus Not configured

Windows Log Disabled

Figure 9-31 Alert Definition 15% or less Available Physical Allocation- Critical

Figure 9-32 shows how to change the Frequency of the notification. You can choose here to get more frequent notification for the Critical Threshold “15% Available Physical Space”. In this example, we choose to set the frequency to “Send every 1 day”.

☒ Physical Allocation

Operator Value Severity

<= 15 %

Email Override

Separate entries with commas or spaces

[Hide Additional Options](#)

Notification Frequency

Send every 1 day(s)

How else do you want to be notified?

☐ Run script ☐ SNMP

☐ Netcool / OMNIbus ☐ Windows log

Figure 9-32 Alert Definition - Critical - change frequency

Figure 9-33 shows how to set up the Warning level at 20% or less Physical Allocation. To proceed, choose the plus sign at the previously defined Definition (Critical) and fill in the window as described in the picture (Operator: “<=”, Value: “20%”, and Severity “Warning”).

The screenshot displays the 'SAN Volume Controller policy Copy - ITSO' configuration window. On the left, a sidebar lists 'Alert Definitions' with categories like Storage System, Volumes, Pools, General, Capacity, Performance, and Managed Disks. The main area shows a list of existing alert definitions. A new definition is being added, indicated by a red circle around the '+' button. The new definition is for 'Physical Allocation' with the Operator set to '<=' and the Value set to '20'. The Severity is set to 'Warning' (indicated by a yellow triangle icon). The existing definition above it is for 'Physical Allocation' with the Operator set to '<=' and the Value set to '15', with a Severity of 'Critical' (indicated by a red 'X' icon).

Figure 9-33 Alert Definition 20% or less Physical Allocation - Warning

Figure 9-34 describes how to set up the Notification Threshold at 30%. Proceed as shown (Operator: “<=”, Value: “30%”, and Severity “Notification”).

The screenshot displays the 'SAN Volume Controller policy Copy - ITSO' configuration window. On the left, a sidebar lists 'Alert Definitions' with categories like Storage System, Volumes, Pools, General, Capacity, Performance, and Managed Disks. The main area shows a list of existing alert definitions. A new definition is being added, indicated by a red circle around the '+' button. The new definition is for 'Physical Allocation' with the Operator set to '<=' and the Value set to '30'. The Severity is set to 'Notification' (indicated by a blue 'i' icon). The existing definition above it is for 'Physical Allocation' with the Operator set to '<=' and the Value set to '20', with a Severity of 'Warning' (indicated by a yellow triangle icon).

Figure 9-34 Alert Definition 30% or less Available Physical Allocation- Notification

Figure 9-35 shows how to configure the Notification Settings in your monitoring environment.

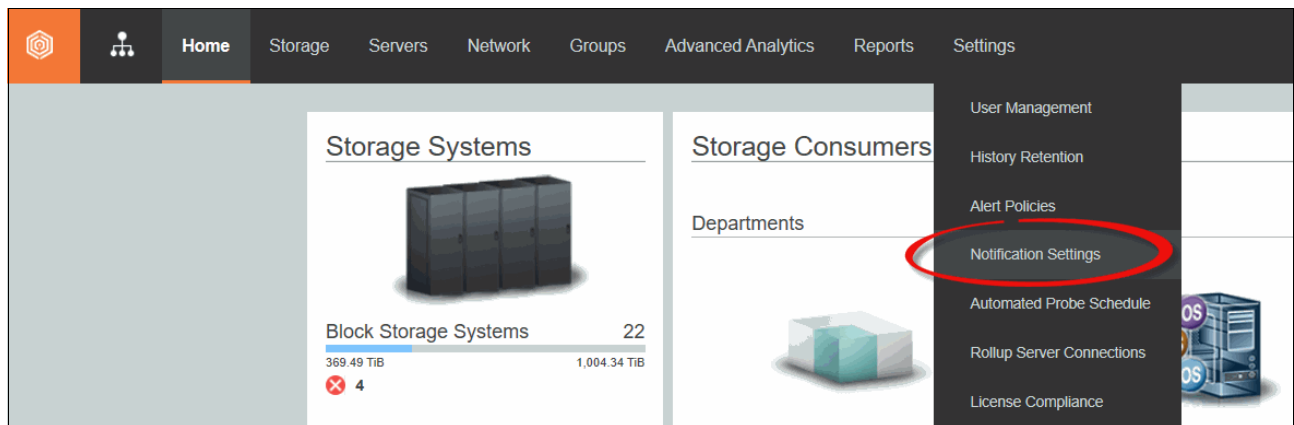


Figure 9-35 Change Notification Settings

Figure 9-36 denotes how to set up the Notification Settings.

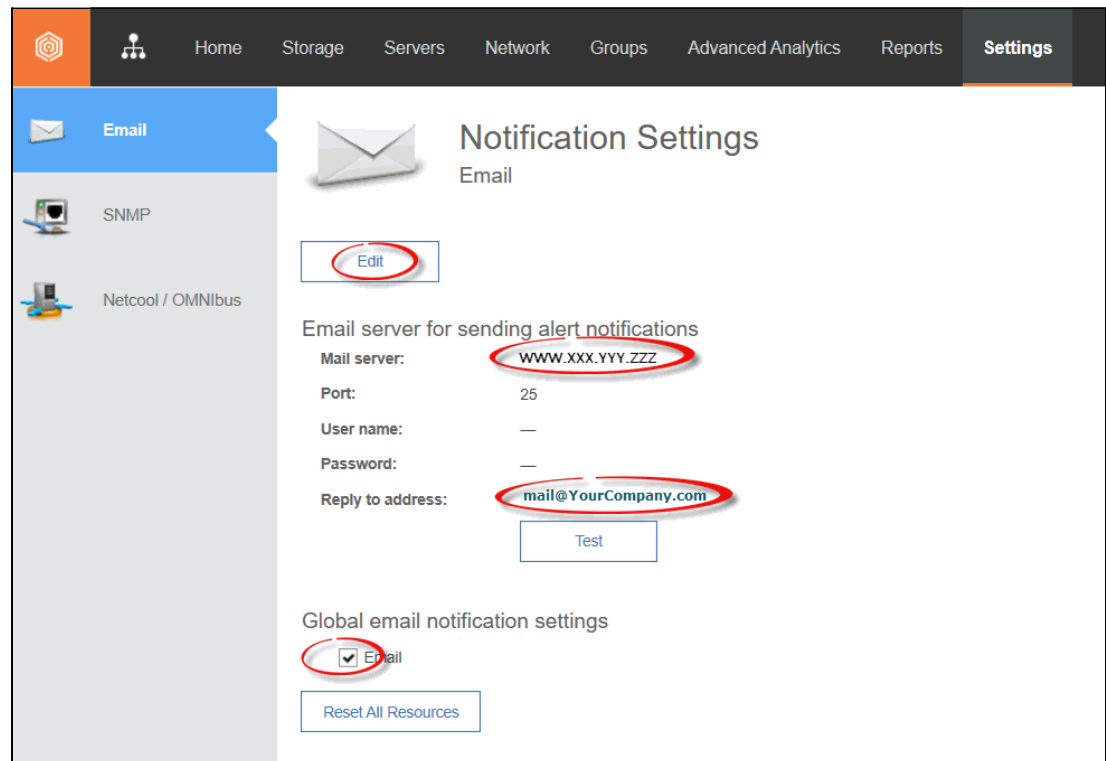


Figure 9-36 Notification Settings - Details

## 9.5 Important metrics

The following metrics are some of the most important metrics that need to be analyzed to understand a performance problem in IBM Spectrum Virtualize. Those metrics are valid to analyze the front end (by Node, by Host, or by volume) or the back-end (by MDisk or by Storage Pool).

**Note:** R/W stands for Read and Write operations.

- ▶ I/O Rate R/W: The term *I/O* is used to describe any program, operation, or device that transfers data to or from a computer, and to or from a peripheral device. Every transfer is an output from one device and an input into another. Typically measured in IOPS.
- ▶ Data Rate R/W: The data transfer rate (DTR) is the amount of digital data that is moved from one place to another in a specific time. In case of Disk or Storage Subsystem, this metric is the amount of data moved from a host to a specific storage device. Typically measured in MBps.
- ▶ Response time R/W: This is the time taken for a circuit or measuring device, when subjected to a change in input signal, to change its state by a specified fraction of its total response to that change. In case of Disk or Storage Subsystem, this is the time used to complete an I/O operation. Typically measured in ms.
- ▶ Cache Hit R/W: This is the percentage of times that read data or write data can be found in cache or can find cache free space that it can be written to.
- ▶ Average Data Block Size R/W: The block size is the unit of work for the file system. Every read and write is done in full multiples of the block size. The block size is also the smallest size on disk that a file can have.
- ▶ Port-to-Local Node Queue Time (Send): The average time in milliseconds that a send operation spends in the queue before the operation is processed. This value represents the queue time for send operations that are issued to other nodes that are in the local cluster. A good scenario has less than 1 ms on average.
- ▶ Port Protocol Errors (Zero Buffer Credit Percentage): The amount of time, as a percentage, that the port was not able to send frames between ports because of insufficient buffer-to-buffer credit. The amount of time value is measured from the last time that the node was reset. In Fibre Channel technology, buffer-to-buffer credit is used to control the flow of frames between ports. In our experience less is better than more. However, in a production environment this metric can be from 5% on average up to 20% peak without affecting performance.
- ▶ Port data rate (send and receive): The average amount of data in MBps for operations in which the port receives or sends data.
- ▶ Port Protocol Errors (Zero Buffer Credit Timer): The number of microseconds that the port is not able to send frames between ports because there is insufficient buffer-to-buffer credit. In Fibre Channel technology, buffer-to-buffer credit is used to control the flow of frames between ports. Buffer-to-buffer credit is measured from the last time that the node was reset. This value is related to the data collection sample interval.
- ▶ Port Congestion Index: The estimated degree to which frame transmission was delayed due to a lack of buffer credits. This value is generally 0 - 100. The value 0 means there was no congestion. The value can exceed 100 if the buffer credit exhaustion persisted for an extended amount of time. When you troubleshoot a SAN, use this metric to help identify port conditions that might slow the performance of the resources to which those ports are connected.
- ▶ Global Mirror (Overlapping Write Percentage): The percentage of overlapping write operations that are issued by the Global Mirror primary site. Some overlapping writes are processed in parallel, and so they are excluded from this value.
- ▶ Global Mirror (Write I/O Rate): The average number of write operations per second that are issued to the Global Mirror secondary site. Keep in mind that IBM SAN Volume Controller systems have limited number of GM I/Os that can be delivered.



- **Global Mirror (Secondary Write Lag):** The average number of extra milliseconds that it takes to service each secondary write operation for Global Mirror. This value does not include the time to service the primary write operations. Monitor the value of Global Mirror Secondary Write Lag to identify delays that occurred during the process of writing data to the secondary site.

**Note:** The host attributed response time is also an important metric that must be used with IBM Spectrum Control V5.3.3 or higher. Previous versions encountered a calculation error.

Also, IBM Spectrum Control V5.2.x is *not* supported as of September 30, 2019.

Many others metrics are supplied to IBM Spectrum Control from IBM SAN Volume Controller. For more information about all metrics, see [IBM Spectrum Control 5.4.2 Documentation - Performance metrics for resources that run IBM Spectrum Virtualize](#).

## 9.6 Performance support package

If you experience performance issues on your system at any level (Host, Volume, Nodes, Pools, and so on), consult IBM Support, who require detailed performance data about the IBM Spectrum Virtualize system to diagnose the problem. Generate a performance support package with detailed data by using IBM Spectrum Control.

In this scenario, you export performance data for a SAN Volume Controller to a compressed package. You then send the package to IBM Support, as shown in Figure 9-37.

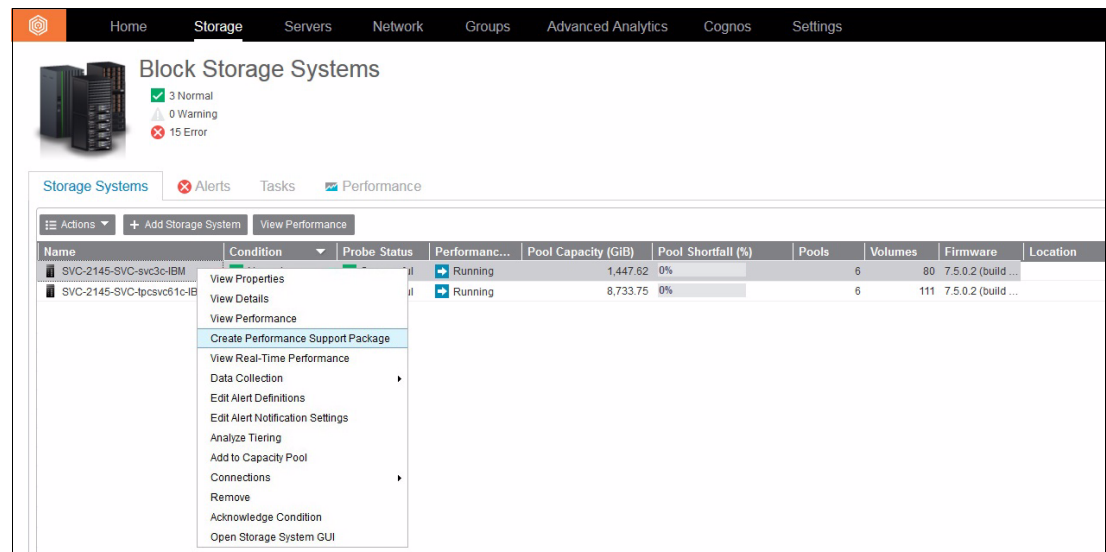


Figure 9-37 Performance support package creation

When the package has been created, you are requested to download it in .zip format. The package includes different reports in .csv format, as shown in Figure 9-38.

|                                                                        |               |       |
|------------------------------------------------------------------------|---------------|-------|
| log.txt                                                                | Text Document | 1 KB  |
| PerfReport_ITSO_SVC_ESC_Disks_20161017-233400_12hrs0mins.csv           | CSV File      | 1 KB  |
| PerfReport_ITSO_SVC_ESC_HostConnections_20161017-233400_12hrs0mins.csv | CSV File      | 1 KB  |
| PerfReport_ITSO_SVC_ESC_IOGroups_20161017-233400_12hrs0mins.csv        | CSV File      | 12 KB |
| PerfReport_ITSO_SVC_ESC_ManagedDisks_20161017-233400_12hrs0mins.csv    | CSV File      | 18 KB |
| PerfReport_ITSO_SVC_ESC_Nodes_20161017-233400_12hrs0mins.csv           | CSV File      | 18 KB |
| PerfReport_ITSO_SVC_ESC_Pools_20161017-233400_12hrs0mins.csv           | CSV File      | 16 KB |
| PerfReport_ITSO_SVC_ESC_StoragePorts_20161017-233400_12hrs0mins.csv    | CSV File      | 36 KB |
| PerfReport_ITSO_SVC_ESC_StorageSystem_20161017-233400_12hrs0mins.csv   | CSV File      | 12 KB |
| PerfReport_ITSO_SVC_ESC_Volumes_20161017-233400_12hrs0mins.csv         | CSV File      | 12 KB |

Figure 9-38 Package files example

For more information about how to create a performance support package, see [IBM Spectrum Control 5.4.2 Documentation - Exporting performance data for storage systems and fabrics](#).

**Note:** The performance data might be large, especially if the data is for storage systems that have many volumes, or the performance monitors are running with a 1-minute sampling frequency. If the time range for the data is greater than 12 hours, volume data and 1-minute sample data is automatically excluded from the performance data. To include volume data and 1-minute sample data, select the **Advanced package** option on the Create Performance Support Package wizard.

## 9.7 Metro and Global Mirror monitoring with IBM Copy Services Manager and scripts

Copy Services Manager is part of IBM Spectrum Control and controls copy services in storage environments. Copy services are features that are used by storage systems, such as IBM SAN Volume Controller, to configure, manage, and monitor data-copy functions. Copy services include IBM FlashCopy, Metro Mirror, Global Mirror, and Global Mirror Change Volumes.

You can use Copy Services Manager to complete the following data replication tasks and help reduce the downtime of critical applications:

- Plan for replication when you are provisioning storage
- Keep data on multiple related volumes consistent across storage systems if there is a planned or unplanned outage
- Monitor and track replication operations
- Automate the mapping of source volumes to target volumes

One of the most important events that needs to be monitored when IBM SAN Volume Controller systems are implemented in a Disaster Recovery (DR) solution with Metro Mirror (MM) or Global Mirror (GM) functions, is to check whether MM or GM was suspended because of a 1920 or 1720 error.

With IBM Spectrum Virtualize is able to suspend the MM or GM relationship to protect the performance on the primary site when MM or GM starts to affect write response time. That suspension can be caused by several factors.

IBM Spectrum Virtualize systems do not restart the MM or GM automatically. They must be restarted manually.

Setting IBM Spectrum Virtualize systems alert monitoring is explained in 9.1.1, “Monitoring with the GUI” on page 354. When MM or GM is managed by IBM CSM and if a 1920 error occurs, IBM CSM can automatically restart MM or GM sessions, and can set the delay time on the automatic restart option. This delay allows some time for the situation to correct itself.

Alternatively, if you have several sessions, you can stagger them so that they do not all restart at the same time, which can affect system performance. Choose the set delay time feature to define a time, in seconds, for the delay between when Copy Services Manager processes the 1720/1920 event and when the automatic restart is issued.

CSM is also able to automatically restart unexpected suspends. When you select this option, the Copy Services Manager server automatically restarts the session when it unexpectedly suspends due to reason code 1720 or 1920. An automatic restart is attempted for every suspend with reason code 1720 or 1920 up to a predefined number of times within a 30-minute time period.

The number of times that a restart is attempted is determined by the storage server **gmLinktolerance** value. If the number of allowable automatic restarts is exceeded within the time period, the session does not restart automatically on the next unexpected suspend. Issue a **Start** command to restart the session, clear the automatic restart counters, and enable automatic restarts.

**Warning:** When you enable this option, the session is automatically restarted by the server. When this situation occurs, the secondary site is not consistent until the relationships are fully resynched.

You can specify the amount of time (in seconds) that the copy services management server waits after an unexpected suspend before automatically restarting the session. The range of possible values is 0 - 43,200. The default is 0, which specifies that the session is restarted immediately following an unexpected suspend.

## 9.7.1 Monitoring MM and GM with scripts

IBM Spectrum Virtualize system provides a complete command-line interface (CLI), which allows you to interact with your systems by using scripts. Those scripts can run in the IBM Spectrum Virtualize shell, but with a limited script command set available, or they can run out of the shell using any scripting language that you prefer.

An example of script usage is one to check at a specific interval time whether MM or GM are still active, if any 1920 errors have occurred, or to react to an SNMP or email alert received. The script can then start some specific recovery action based on your recovery plan and environment.

Customers who do not use IBM Copy Service Manager have created their own scripts. These scripts are sometimes supported by IBM as part of ITS professional services or IBM System Lab services. Tell your IBM representative what kind of monitoring you want to implement with scripts, and together try to find if one exists in the IBM Intellectual Capital Management repository that can be reused.

## 9.8 Monitoring Tier1 SSD

Tier1 SSD requires that special attention is paid to the endurance events that can be triggered. For monitoring purposes, stay alert to the new fields listed in Table 9-5.

*Table 9-5 Field changes to drive and array devices*

| Field                      | Description                                                                                                                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| write_endurance_used       | Metric pulled from within drive (SAS spec) relating to the amount of data written across the life of the drive divided by the anticipated amount (2.42 PB for the 15.36 TB drive)<br><br>Starts at 0, and can continue > 100                                                     |
| write_endurance_usage_rate | Measuring / Low / Marginal / High<br>Takes 160 Days to get initial measurement;<br>Low: Approximately 5.5 Years or more<br>Marginal: Approximately 4.5 – 5.5 Years<br>High: Approximately < 4.5 years<br>High triggers event<br>SS_EID_VL_ER_SSD_WRITE_ENDURANCE_USAGE_RATE_HIGH |
| replacement_date           | The Current Date + Endurance Rate * Remaining Endurance<br>Triggers event<br>SS_EID_VL_ER_SSD_DRIVE_WRITE_ENDURANCE_LIMITED at 6 Months before limit                                                                                                                             |

If you see either of the following triggered events, contact your IBM service representative to put an action plan in place:

SS\_EID\_VL\_ER\_SSD\_WRITE\_ENDURANCE\_USAGE\_RATE\_HI4GH  
SS\_EID\_VL\_ER\_SSD\_DRIVE\_WRITE\_ENDURANCE\_LIMITED



# Maintenance

As an IT environment grows and is renewed, so must the storage infrastructure. Among the many benefits that the IBM SAN Volume Controller family software (IBM Spectrum Virtualize) provides is to greatly simplify the storage management tasks that system administrators must perform.

This chapter highlights guidance for the maintenance activities of storage administration by using the IBM SAN Volume Controller family software that is installed on the product. This guidance can help you to maintain your storage infrastructure with the levels of availability, reliability, and resiliency demanded by today's applications, and to keep up with storage growth needs.

This chapter concentrates on the most important topics to consider in IBM SAN Volume Controller administration so that you can use it as a checklist. It also provides best practice tips and guidance. To simplify the SAN storage administration tasks that you use often, such as adding users, storage allocation and removal, or adding and removing a host from the SAN, create step-by-step, standard procedures for them.

The discussion in this chapter focuses on the IBM SAN Volume Controller SV1 for the sake of simplicity by using screen shots and command outputs from this model. The recommendations and practices that are discussed in this chapter are applicable to the following models:

- ▶ IBM SAN Volume Controller SV2
- ▶ IBM SAN Volume Controller SA2
- ▶ IBM SAN Volume Controller DH8

**Note:** The practices that are described here are effective in many installations of different models of the IBM SAN Volume Controller family. These installations were performed in various business sectors for various international organizations. They all had one common need: to manage their storage environment easily, effectively, and reliably.

This chapter includes the following sections:

- ▶ 10.1, “User interfaces” on page 407
- ▶ 10.2, “Users and groups” on page 409
- ▶ 10.3, “Volumes” on page 411
- ▶ 10.4, “Hosts” on page 412
- ▶ 10.5, “Software updates” on page 413
- ▶ 10.6, “Drive firmware updates” on page 423
- ▶ 10.7, “SAN modifications” on page 425
- ▶ 10.8, “Server HBA replacement” on page 428
- ▶ 10.9, “Hardware upgrades” on page 429
- ▶ 10.10, “I/O Throttling” on page 443
- ▶ 10.11, “Automation” on page 449
- ▶ 10.12, “Documenting IBM SAN Volume Controller and SAN environment” on page 451

## 10.1 User interfaces

The IBM SAN Volume Controller family provides several user interfaces that you can use to maintain your system. The interfaces provide different sets of facilities to help resolve situations that you might encounter. The interfaces for servicing your system connect through the 1 Gbps Ethernet ports that are accessible from port 1 of each node.

Consider the following points:

- ▶ Use the management graphical user interface (GUI) to monitor and maintain the configuration of storage that is associated with your clustered systems.
- ▶ Use the service assistant tool GUI to complete service procedures.
- ▶ Use the command-line interface (CLI) to manage your system.

The best practice recommendation is to use the interface that is most suitable to the task you are attempting to complete. For example, a manual software update is best performed by using the service assistant GUI or the CLI. Running fix procedures to resolve problems or configuring expansion enclosures can only be performed by using the management GUI. Creating many volumes with customized names is best performed by way of the CLI by using a script. To ensure efficient storage administration, become familiar with all available user interfaces.

### 10.1.1 Management GUI

The management GUI is the primary tool that is used to service your system. Regularly monitor the status of the system by using the management GUI. If you suspect a problem, use the management GUI first to diagnose and resolve the problem. Use the views that are available in the management GUI to verify the status of the system, hardware devices, physical storage, and available volumes.

To access the Management GUI, start a supported web browser and point your web browser to [https://SVC\\_ip\\_address](https://SVC_ip_address) of your system where the *SVC\_ip\_address* is the management IP address set when the clustered system is created.

For more information about the task menus and functions of the Management GUI, see Chapter 4 of *Implementing the IBM System Storage SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8491.

### 10.1.2 Service Assistant Tool GUI

The service assistant interface is a browser-based GUI that can be used to service individual nodes.

**Important:** If used incorrectly, the service actions that are available through the service assistant can cause loss of access to data, or even data loss.

You connect to the service assistant on one node through the service IP address. If a working communications path exists between the nodes, you can view status information and perform service tasks on the other node by making the other node the current node. You do not have to reconnect to the other node. On the system, you can also access the service assistant interface by using the technician port.

The service assistant provides facilities only to help you service nodes. Always service the expansion enclosures by using the management GUI.

You can also complete the following actions by using the service assistant:

- ▶ Collect logs to create and download a package of files to send to support personnel.
- ▶ Provide detailed status and error summaries.
- ▶ Remove the data for the system from a node.
- ▶ Recover a system if it fails.
- ▶ Install a code package from the support site or rescue the code from another node.
- ▶ Update code on nodes manually.
- ▶ Configure a control enclosure chassis after replacement.
- ▶ Change the service IP address that is assigned to Ethernet port 1 for the current node.
- ▶ Install a temporary SSH key if a key is not installed and CLI access is required.
- ▶ Restart the services used by the system.

To access the Service Assistant Tool GUI, start a supported web browser and point your web browser to [https://SVC\\_ip\\_address/service](https://SVC_ip_address/service).

Where *SVC\_ip\_address* is the service IP address for the node or the management IP address for the system on which you want to work.

### 10.1.3 Command-line interface

The system CLI is intended for use by advanced users who are confident using a CLI. Up to 32 simultaneous interactive SSH sessions to the management IP address are supported.

Nearly all the functions that are offered by the CLI also are available through the management GUI. However, the CLI does not provide the fix procedures that are available in the management GUI. However, you can use the CLI when you require a configuration setting that is unavailable in the management GUI.

Entering `help` in a CLI displays a list of all available commands. You can access other UNIX commands in the restricted shell, such as `grep` and `more`, which are useful in formatting the output of the CLI commands. Reverse-i-search (Ctrl+R) is also available. Table 10-1 lists the available commands.

Table 10-1 UNIX commands available in the CLI

| UNIX command | Description                                 |
|--------------|---------------------------------------------|
| grep         | Filter output by keywords                   |
| more         | Moves through output one page at a time     |
| sed          | Filters output                              |
| sort         | Sorts output                                |
| cut          | Removes individual columns from output      |
| head         | Display only first lines                    |
| less         | Moves through the output one page at a time |
| tail         | Display only last lines                     |
| uniq         | Hides any duplicates in the output          |
| tr           | Translates characters                       |



| UNIX command | Description                                      |
|--------------|--------------------------------------------------|
| wc           | Counts lines, words and characters in the output |
| history      | Display command history                          |
| scp          | Secure copy protocol                             |

For more information about command references and syntax, see [IBM FlashSystem 9200 and 9100 documentation](#) and [IBM Spectrum Virtualize for SAN Volume Controller, FlashSystem, and Storwize Family Command-Line Interface User's Guide](#).

### Service command-line interface

You also can run service CLI commands on a specific node. Log in to the service IP address of the node that requires servicing. For more information about the use of the service command-line, see [this web page](#).

### USB command interface

When a USB flash drive is inserted into one of the USB ports on a node, the software searches for a control file on the USB flash drive and runs the command that is specified in the file. The use of the USB flash drive is required in the following situations:

- ▶ When you cannot connect to a node by using the service assistant and you want to see the status of the node.
- ▶ When you do not know, or cannot use, the service IP address for the node and must set the address.
- ▶ When you have forgotten the superuser password and must reset the password.

For more information about the use of the USB port, see [this web page](#).

### Technician port

The technician port is an Ethernet port on the back panel of the IBM SAN Volume Controller system. You can use this port to perform most of the system configuration operations, including the following tasks:

- ▶ Define a management IP address
- ▶ Initialize a new system
- ▶ Service the system

For more information about the use of the Technician port, see [IBM Documentation](#).

## 10.2 Users and groups

Almost all organizations use IT security policies that enforce the use of password-protected user IDs when their IT assets and tools are used. However, some storage administrators still use generic shared IDs (such as superuser, admin, or root) in their management consoles to perform their tasks. They might even use a factory-set default password. Their justification might be a lack of time, forgetfulness, or the fact that their SAN equipment does not support the organization's authentication tool.

SAN storage equipment management consoles often do not provide direct access to stored data, but a shared storage controller can be easily shut down (accidentally or deliberately) and any number of critical applications along with it.

Moreover, having individual user IDs set for your storage administrators allows changes to be better audited if logs must be analyzed.

IBM SAN Volume Controller supports the following authentication methods:

- ▶ Local:
  - By using a password
  - By using SSH keys
- ▶ Remote by using LDAP (Microsoft Active Directory or IBM Security Directory Server)

Local authentication is suitable for small, single enclosure environments whereas larger environments with multiple clusters and enclosures benefit from the ease of maintenance that is achieved by using single sign on (SSO) by way of remote authentication by using LDAP, for example.

By default, the following user groups are defined:

- ▶ Monitor

Users with this role can view objects but cannot manage the system or its resources. Support personnel can be assigned this role to monitor the system and to determine the cause of problems. This role is assigned to the IBM Storage Insights user. For more information about IBM Storage Insights, see Chapter 9, “Monitoring” on page 353.
- ▶ Copy Operator

Users with this role have monitor role privileges and can create, change, and manage all Copy Services functions.
- ▶ Service

Users can set the time and date on the system, delete dump files, add and delete nodes, apply service, and shut down the system. Users can also perform the same tasks as users in the monitor role.
- ▶ Administrator

Users with this role can access all functions on the system, except those that deal with managing users, user groups, and authentication.
- ▶ Security Administrator

Users with this role can access all functions on the system, including managing users, user groups, and user authentication.
- ▶ Restricted Administrator

Users with this role can complete some tasks, but are restricted from deleting specific objects. Support personnel can be assigned this role to solve problems.
- ▶ 3-Site Administrator

Users with this role can configure, manage, and monitor 3-site replication configurations through certain command operations that are available on the 3-Site Orchestrator only.
- ▶ VASA Provider

Users with this role can manage virtual volumes (VVOLs) that are used by VMware vSphere and managed through Spectrum Control software.

In addition to standard groups, you can configure ownership groups to manage access to resources on the system. An *ownership group* defines a subset of users and objects within the system. You can create ownership groups to further restrict access to specific resources that are defined in the ownership group.

Users within an ownership group can view or change only resources that are within the ownership group to which they belong. For example, you can create an ownership group for database administrators to provide monitor-role access to a single pool that is used by their databases. Their views and privileges in the management GUI are automatically restricted, as shown in Figure 10-1.

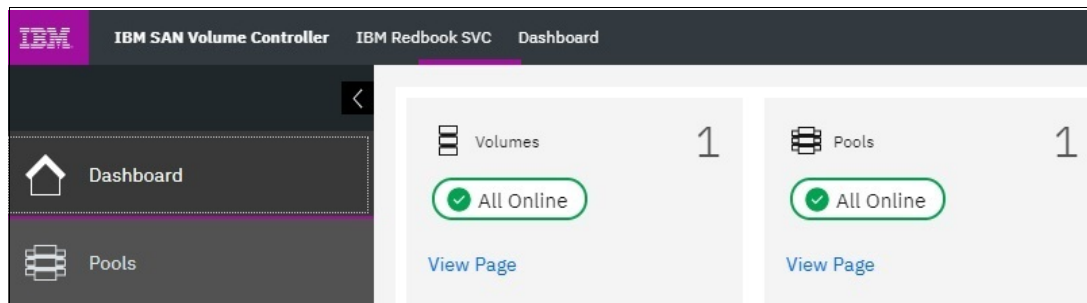


Figure 10-1 Restricted Dashboard view for a user in an ownership group

Regardless of the authentication method you choose, complete the following tasks:

- ▶ Create individual user IDs for your Storage Administration staff. Choose user IDs that easily identify the user and meet your organization's security standards.
- ▶ Include each individual user ID into the UserGroup with only enough privileges to perform the required tasks. For example, your first level support staff likely require only Monitor group access to perform their daily tasks whereas second level support might require Restricted Administrator access. Consider the use of Ownership groups to further restrict privileges.
- ▶ If required, create generic user IDs for your batch tasks, such as Copy Services or Monitoring. Include them in a Copy Operator or Monitor UserGroup. Never use generic user IDs with the SecurityAdmin privilege in batch tasks.
- ▶ Create unique SSH public and private keys for each administrator requiring local access.
- ▶ Store your superuser password in a safe location in accordance with your organization's security guidelines and use it only in emergencies.

## 10.3 Volumes

A *volume* is a logical disk that is presented to a host by an I/O group (pair of nodes). Within that group, a preferred node serves I/O requests to the volume.

When you allocate and deallocate volumes to hosts, consider the following guidelines:

- ▶ Before you allocate new volumes to a server with redundant disk paths, verify that these paths are working well, and that the multipath software is free of errors. Fix any disk path errors that you find in your server before you proceed.
- ▶ When you plan for future growth of space efficient volumes (VDisks), determine whether your server's operating system supports the specific volume to be extended online. For example, AIX V6.1 TL2 and lower do not support online expansion of rootvg LUNs. Test the procedure in a non-production server first.
- ▶ Always cross-check the host LUN ID information with the `vdisk_UID` of the IBM SAN Volume Controller. Do not assume that the operating system recognizes, creates, and numbers the disk devices in the same sequence or with the same numbers as you created them in the IBM SAN Volume Controller.

- ▶ Ensure that you delete any volume or LUN definition in the server *before* you unmap it in the IBM SAN Volume Controller. For example, in AIX, remove the HDisk from the volume group (reducevg) and delete the associated HDisk device (rmdev).
- ▶ Consider enabling volume protection by using **chsystem vdiskprotectionenabled yes -vdiskprotectiontime <value\_in\_minutes>**. Volume protection ensures that some CLI actions (most of those that explicitly or implicitly remove host-volume mappings or delete volumes) are policed to prevent the removal of mappings to volumes or deletion of volumes that are considered *active*; the system detected I/O activity to the volume from any host within a specified period (15 - 1440 minutes).

**Note:** Volume protection cannot be overridden by using the **-force** flag in the affected CLI commands. Volume protection must be disabled to continue an activity that is currently blocked.

- ▶ Ensure that you specifically remove a volume from any volume-to-host mappings and any copy services relationship to which it belongs *before* you delete it.

**Attention:** You must avoid the use of the **-force** parameter in **rmvdisk**.

- ▶ If you issue the **svctask rmvdisk** command and it still has pending mappings, the IBM SAN Volume Controller prompts you to confirm the action. This prompt is a hint that something might be incorrect.
- ▶ When you are deallocating volumes, plan for an interval between unmapping them to hosts (**rmvdiskhostmap**) and deleting them (**rmvdisk**). The IBM internal Storage Technical Quality Review Process (STQRP) asks for a minimum of a 48-hour period, which gives at least a one business day interval so that you can perform a quick back out if you later realize you still need some data on that volume.

For more information about volumes, see Chapter 5, “Volumes” on page 177.

## 10.4 Hosts

A *host* is a computer that is connected to the SAN switch through Fibre Channel (FC), iSCSI and other protocols.

When you add and remove hosts in the IBM SAN Volume Controller, consider the following guidelines:

- ▶ Before you map new servers to the IBM SAN Volume Controller, verify that they are all error free. Fix any errors that you find in your server and IBM SAN Volume Controller before you proceed. In the IBM SAN Volume Controller, pay special attention to anything that is inactive in the **lsfabric** command.
- ▶ Plan for an interval between updating the zoning in each of your redundant SAN fabrics, such as at least 30 minutes. This interval allows for failover to occur and stabilize, and for you to be notified if unexpected errors occur.
- ▶ After you perform the SAN zoning from one server’s host bus adapter (HBA) to the IBM SAN Volume Controller, you should list its WWPN by using the **lshbaportcandidate** command. Use the **lsfabric** command to certify that it was detected by the IBM SAN Volume Controller nodes and ports that you expected.

When you create the host definition in the IBM SAN Volume Controller (**mkhost**), try to avoid the **-force** parameter. If you do not see the host's WWPNs, it might be necessary to scan fabric from the host. For example, use the **cfgmgr** command in AIX.

For more information about hosts, see Chapter 8, "Hosts" on page 339.

## 10.5 Software updates

Because the IBM SAN Volume Controller might be at the core of your disk and SAN storage environment, its update requires planning, preparation, and verification. However, with the appropriate precautions, an update can be conducted easily and transparently to your servers and applications. This section highlights applicable guidelines for the IBM SAN Volume Controller update.

Most of the following sections explain how to prepare for the software update. These sections also present version-independent guidelines on how to update the IBM SAN Volume Controller family systems and flash drives.

Before you update the system, ensure that the following requirements are met:

- ▶ The latest update test utility was downloaded from IBM Fix Central to your management workstation.
- ▶ The latest system update package was downloaded from IBM Fix Central to your management workstation.
- ▶ All nodes are online.
- ▶ All errors in the system event log are addressed and marked as fixed.
- ▶ No volumes, MDisks, or storage systems exist with Degraded or Offline status.
- ▶ The service assistant IP is configured on every node in the system.
- ▶ The system superuser password is known.
- ▶ The current system configuration is backed up and saved (preferably off-site). Use the steps that are described in Example 10-11 on page 458.
- ▶ Physical access is available to the hardware.

Although the following actions are not required, they are suggestions to reduce unnecessary load on the system during the update:

- ▶ Stop all Metro Mirror, Global Mirror, or HyperSwap operations.
- ▶ Avoid running any FlashCopy operations.
- ▶ Avoid migrating or formatting volumes.
- ▶ Stop collecting IBM Spectrum Control performance data for the system.
- ▶ Stop any automated jobs that access the system.
- ▶ Ensure that no other processes are running on the system.
- ▶ If you want to update without host I/O, shut down all hosts.

**Note:** For customers who purchased the IBM SAN Volume Controller with a 3 year-warranty (2147 Models SV1, SV2, and SA2), Enterprise Class Support (ECS) is included, which entitles the customer to two code upgrades per year that are performed by IBM (total of six across the 3-year warranty). These upgrades are done by the IBM dedicated Remote Code Load (RCL) team or, where remote support is not allowed or enabled, by an on-site SSR.

For more information about ECS, see [this web page](#).

## 10.5.1 Determining the target software level

The first step is to determine your current and target IBM SAN Volume Controller software level.

By using the example of an IBM SAN Volume Controller, log in to the web-based GUI and find the current version. From the right side of the top menu drop-down menu, click the question mark symbol (?) and select **About IBM FlashSystem 9200** to display the current version or select **Settings** → **System** → **Update System** to display current and target levels.

Figure 10-2 shows the Update System output panel that displays the code levels. In this example, the current software level is 8.4.0.0.

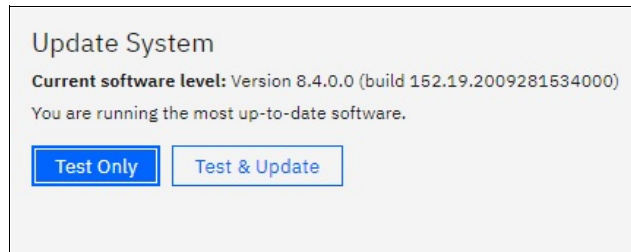


Figure 10-2 Update System output panel

Alternatively, if you use the CLI, run the `svcinfo lssystem` command. Example 10-1 shows the output of the `lssystem` CLI command and where the code level output can be found.

Example 10-1 `lssystem` command

---

```
IBM_2145:IBM Redbook SVC:superuser>lssystem|grep code
code_level 8.4.0.0 (build 152.19.2009101641000)
```

---

IBM SAN Volume Controller software levels are specified by four digits in the following format:

- ▶ In our example (V.R.M.F = 8.4.0.0):
  - V: Major version number
  - R: Release level
  - M: Modification level
  - F: Fix level

Use the latest IBM SAN Volume Controller release, unless you have a specific reason not to update, such as the following examples:

- ▶ The specific version of an application or other component of your SAN Storage environment has a known problem or limitation.
- ▶ The latest IBM SAN Volume Controller software release is not yet cross-certified as compatible with another key component of your SAN storage environment.
- ▶ Your organization has mitigating internal policies, such as the use of the “latest release minus 1” or requiring “seasoning” in the field before implementation in a production environment.

For more information, see this [IBM Support web page](#).

## 10.5.2 Obtaining software packages

To obtain a new release of software for a system update, see this [IBM Fix Central web page](#). Complete the following steps:

1. From the **Product selector** list, enter IBM SAN Volume Controller (or whatever model is suitable in your environment).
2. From the **Installed Version** list, select the current software version level that was determined as described in 10.5.1, “Determining the target software level” on page 414.
3. Select **Continue**.
4. In the **Product Software** section, select the three items that are shown in Figure 10-3.

|                          |   |                                                                                                                                                          |                                                                                                                               |
|--------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | 1 |  fix pack: → <a href="#">StorageSoftware-2145-SwUpgradeTestUtility</a>  | 2020/12/04                                                                                                                    |
|                          |   | Spectrum Virtualize Software Upgrade Test Utility V31.5                                                                                                  |                                                                                                                               |
|                          |   |  <a href="#">Software Upgrade Test Utility Documentation</a>            |  <a href="#">md5sums.txt</a>                 |
| <input type="checkbox"/> | 2 |  fix pack: → <a href="#">StorageSoftware-2145-DriveMicrocode-201027</a> | 2020/10/30                                                                                                                    |
|                          |   | SAN Volume Controller Drive Package 201027                                                                                                               |                                                                                                                               |
|                          |   |  <a href="#">Supported Drive Types and Firmware Levels</a>              |  <a href="#">Drive Release Note</a>          |
| <input type="checkbox"/> | 2 |  fix pack: → <a href="#">StorageSoftware-2145-8.4.0.0</a>              | 2020/11/12                                                                                                                    |
|                          |   | 8.4.0.0 - SAN Volume Controller Code                                                                                                                     |                                                                                                                               |
|                          |   |  <a href="#">Interoperability Information (SSIC)</a>                  |  <a href="#">Upgrade Planning Guidance</a> |
|                          |   |  <a href="#">Release Note</a>                                         |  <a href="#">md5sums</a>                   |

Figure 10-3 Fix Central software packages

5. Select **Continue**.
6. Click the option button for your preferred download options and then, click **Continue**.
7. Enter your machine type and serial number.
8. Select **Continue**.
9. Read the terms and conditions and then, select **I Agree**.
10. Select **Download Now** and save the three files onto your management computer.

## 10.5.3 Hardware considerations

Before you start the update process, always check whether your IBM SAN Volume Controller hardware and target code level are compatible.

If part or all your current hardware is not supported at the target code level that you want to update to, replace the unsupported hardware with newer models before you update to the target code level.

Conversely, if you plan to add or replace hardware with new models to an existing cluster, you might need to update your IBM SAN Volume Controller code first.

## 10.5.4 Update sequence

Check the compatibility of your target IBM SAN Volume Controller code level with all components of your SAN storage environment (SAN switches, storage controllers, server HBAs, on so on) and its attached servers (operating systems and eventually, applications).

Applications often certify only the operating system that they run under and leave to the operating system provider the task of certifying its compatibility with attached components (such as SAN storage). However, various applications might use special hardware features or raw devices and certify the attached SAN storage. If you have this situation, consult the compatibility matrix for your application to certify that your IBM SAN Volume Controller target code level is compatible.

The IBM SAN Volume Controller Supported Hardware List provides the complete information for using your IBM SAN Volume Controller SAN storage environment components with the current and target code level. For more information the supported hardware, device drivers, firmware, and recommended software levels for different products and code levels, see [this IBM Support web page](#).

By cross-checking the version of IBM SAN Volume Controller is compatible with the versions of your SAN environment components, you can determine which one to update first. By checking a component's update path, you can determine whether that component requires a multi-step update.

If you are not making major version or multi-step updates in any components, the following update order is less prone to eventual problems:

1. SAN switches or directors
2. Storage controllers
3. Servers HBAs microcode and multipath software
4. IBM SAN Volume Controller
5. IBM SAN Volume Controller internal drives
6. IBM SAN Volume Controller SAS attached SSD drives

**Attention:** Do *not* update two components of your IBM SAN Volume Controller SAN storage environment simultaneously, such as an IBM SAN Volume Controller model SV2 and one storage controller. This caution is true even if you intend to perform this update with your system offline. An update of this type can lead to unpredictable results, and an unexpected problem is much more difficult to debug.

## 10.5.5 SAN fabrics preparation

If you use symmetrical, redundant, independent SAN fabrics, preparing these fabrics for an IBM SAN Volume Controller update can be safer than hosts or storage controllers. This statement is true, assuming that you follow the guideline of a 30-minute minimum interval between the modifications that you perform in one fabric to the next.

Even if an unexpected error brings down your entire SAN fabric, the IBM SAN Volume Controller environment continues working through the other fabric and your applications remain unaffected.



Because you are updating your IBM SAN Volume Controller, also update your SAN switches code to the latest supported level. Start with your principal core switch or director, continue by updating the other core switches, and update the edge switches last. Update one entire fabric (all switches) before you move to the next one so that any problem you might encounter affects only the first fabric. Begin your other fabric update only after you verify that the first fabric update has no problems.

If you are not running symmetrical, redundant, independent SAN fabrics, fix this problem as a high priority because it represents a single point of failure.

## 10.5.6 Storage controllers preparation

As critical as with the attached hosts, the attached storage controllers must correctly handle the failover of MDisk paths. Therefore, they must be running supported microcode versions and their own SAN paths to IBM SAN Volume Controller must be free of errors.

## 10.5.7 Hosts preparation

If the suitable precautions are taken, the IBM SAN Volume Controller update is not apparent to the attached servers and their applications. The automated update procedure updates one IBM SAN Volume Controller node at a time, while the other node in the I/O group covers for its designated volumes.

However, to ensure that this feature works, the *failover capability* of your multipath software must be working properly. This capability can be mitigated by enabling NPIV if your current code level supports this function. For more information about NPIV, see Chapter 8, “Hosts” on page 339.

Before you start IBM SAN Volume Controller update preparation, check the following items for every server that is attached to IBM SAN Volume Controller that you update:

- ▶ Operating system type, version, and maintenance or fix level
- ▶ Make, model, and microcode version of the HBAs
- ▶ Multipath software type, version, and error log

For information about troubleshooting, see this [IBM Documentation web page](#).

Fix every problem or “suspect” that you find with the disk path failover capability. Because a typical IBM SAN Volume Controller environment can have hundreds of servers attached to it, a spreadsheet might help you with the Attached Hosts Preparation tracking process. If you have some host virtualization, such as VMware ESX, AIX LPARs, IBM VIOS, or Solaris containers in your environment, verify the redundancy and failover capability in these virtualization layers.

## 10.5.8 Copy services considerations

When you update an IBM SAN Volume Controller family product that participates in an inter-cluster Copy Services relationship, do *not* update both clusters in the relationship simultaneously. This situation is not verified or monitored by the automatic update process and might lead to a loss of synchronization and unavailability.

You must successfully finish the update in one cluster before you start the next one. Try to update the next cluster as soon as possible to the same code level as the first one. Avoid running them with different code levels for extended periods.

## 10.5.9 Running the Upgrade Test Utility

The latest IBM SAN Volume Controller Upgrade Test Utility must be installed and run before you update the IBM SAN Volume Controller software. For more information about the Upgrade Test Utility, see this [IBM Support web page](#).

This tool verifies the health of your IBM SAN Volume Controller storage array for the update process. It also checks for unfixed errors, degraded MDisks, inactive fabric connections, configuration conflicts, hardware compatibility, drive firmware, and many other issues that might otherwise require cross-checking a series of command outputs.

**Note:** The Upgrade Test Utility does not log in to storage controllers or SAN switches. Instead, it reports the status of the connections of the IBM SAN Volume Controller to these devices. It is the users' responsibility to check these components for internal errors.

You can use the management GUI or the CLI to install and run the Upgrade Test Utility.

### Using the management GUI

To test the software on the system, complete the following steps:

1. In the management GUI, select **Settings** → **System** → **Update System**.
2. Click **Test Only**.
3. Select the test utility that you downloaded from the IBM Fix Central support site.
4. Upload the Test utility file and enter the code level to which you are planning to update. Figure 10-4 shows the IBM SAN Volume Controller management GUI window that is used to install and run the Upgrade Test Utility.

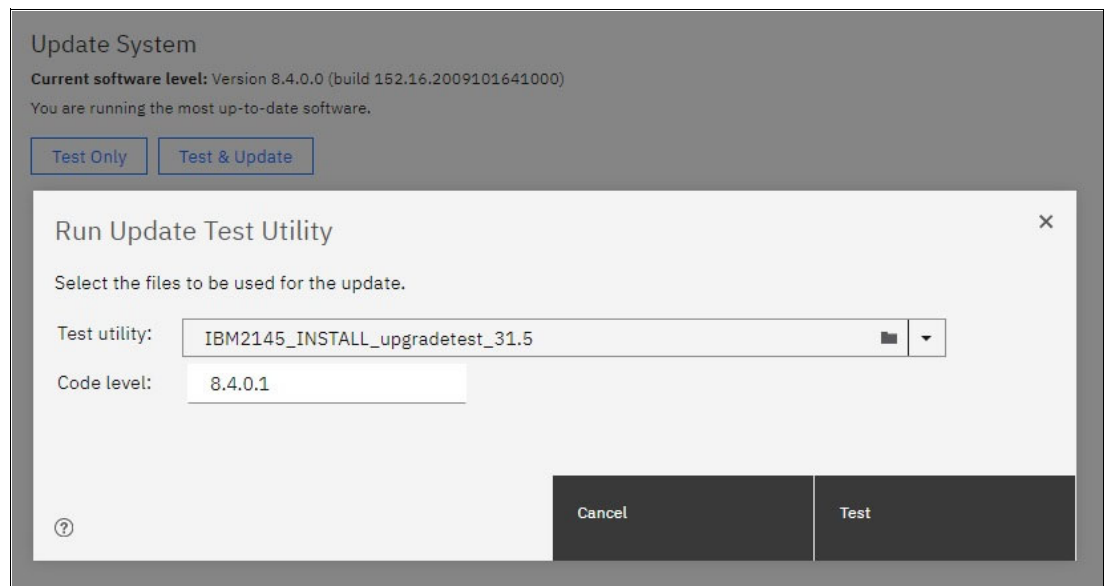


Figure 10-4 IBM SAN Volume Controller Upgrade Test Utility using the GUI

5. Click **Test**.

The test utility verifies that the system is ready to be updated. After the Update Test Utility completes, the results are shown. The results indicate that no warnings or problems were found, or direct you to more information about any known issues that were discovered on the system.

Figure 10-5 shows a successful completion of the update test utility.

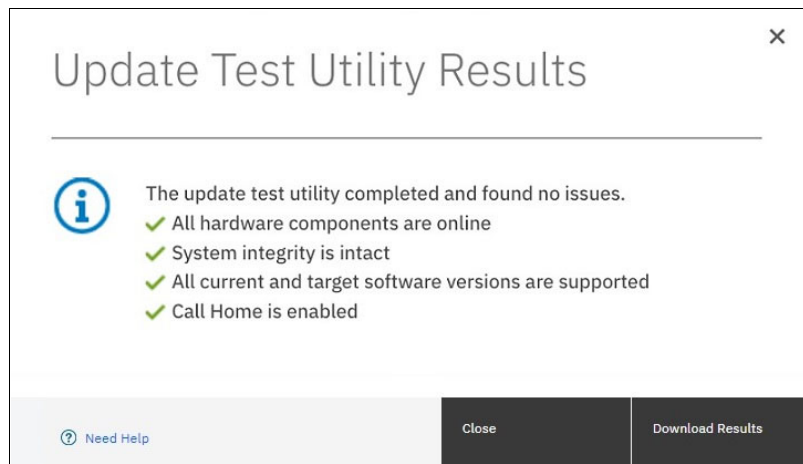


Figure 10-5 IBM SAN Volume Controller Upgrade Test Utility completion panel

6. Click **Download Results** to save the results to a file.
7. Click **Close**.

## Using the command-line

To test the software on the system, complete the following steps:

1. Using OpenSSH scp or PuTTY pscp, copy the software update file and the Software Update Test Utility package to the /home/admin/upgrade directory by using the management IP address of the IBM SAN Volume Controller. Some documentation and online help might refer to the /home/admin/update directory, which points to the same location on the system.

An example for the IBM SAN Volume Controller is shown in Example 10-2.

Example 10-2 Copying the upgrade test utility to IBM SAN Volume Controller

```
C:\>pscp -v -P 22 IBM2145_INSTALL_upgradetest_31.5
superuser@9.10.11.12:/home/admin/upgrade
Looking up host "9.10.11.12" for SSH connection
Connecting to 9.10.11.12 port 22
We claim version: SSH-2.0-PuTTY_Release_0.74
Remote version: SSH-2.0-OpenSSH_8.0
Using SSH protocol version 2
No GSSAPI security context available
Doing ECDH key exchange with curve Curve25519 and hash SHA-256 (unaccelerated)
Server also has ssh-rsa host key, but we don't know it
Host key fingerprint is:
ecdsa-sha2-nistp521 521 d3:00:2b:a0:24:cd:8c:df:3d:d5:d5:07:e5:e5:47:b9
Initialised AES-256 SDCTR (AES-NI accelerated) outbound encryption
Initialised HMAC-SHA-256 (unaccelerated) outbound MAC algorithm
Initialised AES-256 SDCTR (AES-NI accelerated) inbound encryption
Initialised HMAC-SHA-256 (unaccelerated) inbound MAC algorithm
Using username "superuser".
Attempting keyboard-interactive authentication
Keyboard-interactive authentication prompts from server:
| Password:
End of keyboard-interactive prompts from server
```

```
Access granted
Opening main session channel
Opened main channel
Primary command failed; attempting fallback
Started a shell/command
Using SCP1
Connected to 9.10.11.12
Sending file IBM2145_INSTALL_upgradetest_31.5, size=335904
Sink: C0644 335904 IBM2145_INSTALL_upgradetest_31.5
IBM2145_INSTALL_upgradete | 328 kB | 328.0 kB/s | ETA: 00:00:00 | 100%
Session sent command exit status 0
Main session channel closed
All channels closed
```

```
C:\>
```

- 
2. Ensure that the update file was successfully copied as indicated by the exit status 0 return code. You also can run the `lsdumps -prefix /home/admin/upgrade` command.

Example 10-3 shows how to install and run Upgrade Test Utility in the CLI. In this case, the Upgrade Test Utility found no errors and completed successfully.

---

*Example 10-3 Upgrade test using the CLI*

---

```
IBM_2145:IBM Redbook SVC:superuser>svctask applysoftware -file
IBM2145_INSTALL_upgradetest_31.5
```

```
CMMVC9001I The package installed successfully.
```

```
IBM_2145:IBM Redbook SVC:superuser>svcupgradetest -v 8.4.0.1
```

```
svcupgradetest version 31.5
```

Please wait, the test may take several minutes to complete.

Results of running svcupgradetest:

```
=====
```

The tool has found 0 errors and 0 warnings.

The tool has not found any problems with the cluster.

---

**Note:** The return code for the `applysoftware` command always is 1, whether the installation succeeded or failed. However, the message that is returned when the command completes reports the correct installation result.

---

Review the output to check whether any problems were found by the utility. The output from the command shows that no problems were found, or directs you to more information about any known issues that were discovered on the system.

## 10.5.10 Updating software

The SAN Volume Controller software is updated by using one of the following methods:

- ▶ During a standard update procedure in the management GUI, the system updates each of the nodes systematically. This method is recommended for updating the software that is on nodes.
- ▶ The command-line interface gives you more control over the automatic upgrade process. You can resolve multipathing issues when nodes go offline for updates. You also can override the default 30-minute mid-point delay, pause an update, and resume a stalled update.
- ▶ To provide even more flexibility in the update process, you also can manually update each node individually by using the Service Assistant Tool GUI.

When upgrading the software manually, you remove a node from the system, update the software on the node, and return the node to the system. You repeat this process for the remaining nodes until the last node is removed from the system. Then, the remaining nodes switch to running the new software.

When the last node is returned to the system, it updates and runs the new level of software. This action cannot be performed on an active node.

To update software manually, the nodes must be candidate nodes (a candidate node is a node that is not in use by the system and cannot process I/O) or in a service state. During this procedure, every node must be updated to the same software level and the node becomes unavailable during the update.

Whichever method (automatic or manual, GUI or CLI) that you choose to perform the update, ensure that you adhere to the following guidelines for your IBM SAN Volume Controller software update:

- ▶ Schedule the IBM SAN Volume Controller software update for a low I/O activity time. The update process puts one node at a time offline. It also disables the write cache in the I/O group that node belongs to until both nodes are updated. Therefore, with lower I/O, you are less likely to notice performance degradation during the update.
- ▶ Never power off, restart, or reset an IBM SAN Volume Controller node during software update unless you are instructed to do so by IBM Support. Typically, if the update process encounters a problem and fails, it backs out. The update process can take one hour per node with another optional, 30-minute mid-point delay.
- ▶ If you are planning for a major IBM SAN Volume Controller version update, update your current version to its latest fix level *before* you run the major update.
- ▶ Check whether you are running a web browser type and version that is supported by the IBM SAN Volume Controller target software level on every computer that you intend to use to manage your IBM SAN Volume Controller.

This section describes the required steps to update the software.

### Using the management GUI

To update the software on the system automatically, complete the following steps:

1. In the management GUI, select **Settings** → **System** → **Update System**.
2. Click **Test & Update**.
3. Select the test utility and the software package that you downloaded from the IBM Fix Central support site. The test utility verifies (again) that the system is ready to be updated.
4. Click **Next**. Select **Automatic update**.

5. Select whether you want to create intermittent pauses in the update to verify the process. Select one of the following options.
  - Fully automatic update without pauses (recommended)
  - Pausing the update after half of the nodes are updated
  - Pausing the update before each node updates
6. Click **Finish**. As the nodes on the system are updated, the management GUI displays the progress for each node.
7. Monitor the update information in the management GUI to determine when the process is complete.

## Using the command-line

To update the software on the system automatically, complete the following steps (you must run the latest version of the test utility to verify that no issues exist with the current system; see Example 10-3 on page 420):

1. Copy the software package to the IBM SAN Volume Controller by using the same method as described in Example 10-2 on page 419.

Before you begin the update, consider the following points:

- The installation process fails under the following conditions:
  - If the software that is installed on the remote system is not compatible with the new software or if an inter-system communication error does not allow the system to check that the code is compatible.
  - If any node in the system has a hardware type that is not supported by the new software.
  - If the system determines that one or more volumes in the system are taken offline by restarting the nodes as part of the update process. More information about which volumes are affected is available by running the **lsdependentvdisks** command. If you are prepared to lose access to data during the update, you can use the force flag to override this restriction.
- The update is distributed to all the nodes in the system by using internal connections between the nodes.
- Nodes are updated individually.
- Nodes run the new software concurrently with normal system activity.
- While the node is updated, it does not participate in I/O activity in the I/O group. As a result, all I/O activity for the volumes in the I/O group is directed to the other node in the I/O group by the host multipathing software.
- A 30-minute delay exists between node updates. The delay allows time for the host multipathing software to rediscover paths to the nodes that are updated. Access is not lost when another node in the I/O group is updated.
- The update is not committed until all nodes in the system are successfully updated to the new software level. If all nodes are successfully restarted with the new software level, the new level is committed. When the new level is committed, the system vital product data (VPD) is updated to reflect the new software level.
- Wait until all member nodes are updated and the update is committed before you start the new functions of the updated software.

- Because the update process takes time, the installation command completes when the software level is verified by the system. To determine when the update is completed, you must display the software level in the system VPD or look for the software update complete event in the error/event log. If any node fails to restart with the new software level or fails at any other time during the process, the software level is backed off.
  - During an update, the version number of each node is updated when the software is installed and the node is restarted. The system software version number is updated when the new software level is committed.
  - When the update starts, an entry is made in the error or event log and another entry is made when the update completes or fails.
2. Run the **applysoftware -file <software\_update\_file>** CLI command to start the update process.  
Where <software\_update\_file> is the file name of the software update file. If the system identifies any volumes that go offline as a result of restarting the nodes as part of the system update, the software update does not start. An optional force parameter can be used to indicate that the update continues regardless of the problem identified. If you use the force parameter, you are prompted to confirm that you want to continue.
  3. Issue the **lupdate** CLI command to check the status of the update process.  
This command displays a message that indicates that the process was successful when the update is complete.
  4. To verify that the update successfully completed, run the **lsnodecani stervpd** command for each node in the system. The `code_level` field displays the new code level for each node.

## 10.6 Drive firmware updates

Updating drive firmware is concurrent process that can be performed online while the drive is in use, whether it is any type of SSD drives in any SAS attached expansion enclosures.

When used on an array member drive, the update checks for volumes that are dependent on the drive and refuses to run if any such dependencies are found. Drive-dependent volumes often are caused by non-redundant or degraded RAID arrays.

Where possible, you should restore redundancy to the system by replacing any failed drives before upgrading drive firmware. When this restoration is not possible, you can add redundancy to the volume by adding a second copy in another pool or use the **-force** parameter to bypass the dependent volume check. Use the **-force** parameter only if you are willing to accept the risk of data loss on dependent volumes (if the drive fails during the firmware update).

**Note:** Because of some system constraints, it is not possible to produce a single NVMe firmware package that works on all NVMe drives on all Spectrum Virtualize code levels. Therefore, you find three different NVMe firmware files available for download, depending on the size of the drives you installed.

## Using the management GUI

To update the drive firmware automatically, complete the following steps:

1. Select **Pools** → **Internal Storage** → **Actions** → **Upgrade All**.
2. As shown in Figure 10-6, in the Upgrade Package window, browse to the drive firmware package that you downloaded as described in 10.5.2, “Obtaining software packages” on page 415.

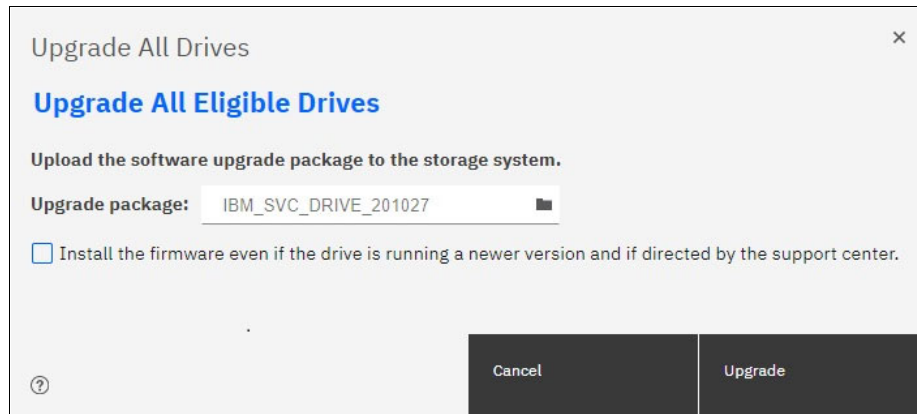


Figure 10-6 Drive firmware upgrade

3. Click **Upgrade**. Each drive upgrade takes approximately 6 minutes.  
You can also update individual drives by right-clicking a single drive and selecting **Upgrade**.
4. To monitor the progress of the upgrade, select **Monitoring** → **Background Tasks**.

## Using the command-line

To manually update the software on the system, complete the following steps:

1. Copy the drive firmware package to the IBM SAN Volume Controller by using the same method as described in Example 10-2 on page 419.
2. Issue the following CLI command to start the update process for all drives:

```
applydrivesoftware -file <software_update_file> -type firmware -all
```

Where <software\_update\_file> is the file name of the software update file. The use of the **-all** option updates firmware on all eligible drives including quorum drives, which is a slight risk. To avoid this risk, use the **-drive** option and make sure the quorum is moved by running the **lsquorum** and **chquorum** commands in between **applydrivesoftware** invocations.

**Note:** The maximum number of drive IDs that can be specified on a command line using the **-drive** option is 128. If you have more than 128 drives, use the **-all** option or run multiple invocations of **applydrivesoftware** to complete the update.

3. Issue the following CLI command to check the status of the update process:  
**lsdriveupgradeprogress**  
This command displays success when the update is complete.
4. To verify that the update successfully completed, run the **lsdrive** command for each drive in the system. The `firmware_level` field displays the new code level for each drive.



Example 10-4 shows how to list the firmware level for four specific drives.

*Example 10-4 List firmware level for drives 0,1, 2 and 3*

---

```
IBM_2145:IBM Redbook SVC:superuser>for i in 0 1 2 3; do echo "Drive $i = `lsdrive  
$i|grep firmware`; done  
Drive 0 = firmware_level 1_2_11  
Drive 1 = firmware_level 1_2_11  
Drive 2 = firmware_level 1_2_11  
Drive 3 = firmware_level 1_2_11
```

---

For more information, see this [IBM Documentation web page](#).

## 10.7 SAN modifications

When you administer shared storage environments, human error can occur when a failure is fixed, or a change is made that affects one or more servers or applications. That error can then affect other servers or applications because precautions were not taken.

Human error can include the following examples:

- ▶ Disrupting or disabling the working disk paths of a server while trying to fix failed ones.
- ▶ Disrupting a neighbor SAN switch port while inserting or removing an FC cable or SFP.
- ▶ Disabling or removing the working part in a redundant set instead of the failed one.
- ▶ Making modifications that affect both parts of a redundant set without an interval that allows for automatic failover during unexpected problems.

Adhere to the following guidelines to perform these actions with assurance:

- ▶ Uniquely and correctly identify the components of your SAN.
- ▶ Use the proper failover commands to disable only the failed parts.
- ▶ Understand which modifications are necessarily disruptive, and which can be performed online with little or no performance degradation.

### 10.7.1 Cross-referencing WWPN

With the WWPN of an HBA, you can uniquely identify one server in the SAN. If a server's name is changed at the operating system level and not at the IBM SAN Volume Controller host definitions, it continues to access its previously mapped volumes exactly because the WWPN of the HBA did not change.

Alternatively, if the HBA of a server is removed and installed in a second server and the first server's SAN zones and IBM SAN Volume Controller host definitions are not updated, the second server can access volumes that it likely should not access.

Complete the following steps to cross-reference HBA WWPNs:

1. In your server, verify the WWPNs of the HBAs that are used for disk access. Typically, you can complete this task by using the SAN disk multipath software of your server.

If you are using SDDPCM, run the **pcmpath query WWPN** command to see output similar to what is shown in Example 10-5.

*Example 10-5* Output of the `pcmpath query WWPN` command

```
[root@Server127]> pcmpath query wwpn
Adapter Name PortWWN
fscsi0       10000090FA021A13
fscsi1       10000090FA021A12
```

If you are using server virtualization, verify the WWPNs in the server that is attached to the SAN, such as AIX VIO or VMware ESX. Cross-reference with the output of the IBM SAN Volume Controller **lshost <hostname>** command, as shown in Example 10-6.

*Example 10-6* Output of the `lshost <hostname>` command

[illegible]

2. If necessary, cross-reference information with your SAN switches, as shown in Example 10-7. In Brocade switches use the **nodefind <WWPN>** command.

### Example 10-7 Cross-referencing information with SAN switches

```
b1g32sw1_B64:admin> nodefind 10:00:00:90:FA:02:1A:13
Local:
Type Pid      COS      PortName                                     NodeName                                     SCR
N    401000;    2,3;10:00:00:90:FA:02:1A:13;20:00:00:90:FA:02:1A:13; 3
Fabric Port Name: 20:10:00:05:1e:04:16:a9
Permanent Port Name: 10:00:00:90:FA:02:1A:13
Device type: Physical Unknown(initiator/target)
Port Index: 16
Share Area: No
Device Shared in Other AD: No
Redirect: No
Partial: No
Aliases: nybixtdb02_fcs0
b32sw1 B64:admin>
```

For storage allocation requests that are submitted by the server support team or application support team to the storage administration team, always include the server’s HBA WWPNs to which the new LUNs or volumes are supposed to be mapped. For example, a server might use separate HBAs for disk and tape access or distribute its mapped LUNs across different HBAs for performance. You cannot assume that any new volume is supposed to be mapped to every WWPN that server logged in the SAN.

If your organization uses a change management tracking tool, perform all your SAN storage allocations under approved change requests with the servers’ WWPNs listed in the Description and Implementation sections.

## 10.7.2 Cross-referencing LUN ID

Always cross-reference the IBM SAN Volume Controller `vdisk_UID` with the server LUN ID before you perform any modifications that involve IBM SAN Volume Controller volumes. Example 10-8 shows an AIX server that is running SDDPCM. The IBM SAN Volume Controller `vdisk_name` has no relation to the AIX device name. Also, the first SAN LUN mapped to the server (`SCSI_id=0`) shows up as `hdisk4` in the server because it had four internal disks (`hdisk0 - hdisk3`).

Example 10-8 Results of running the `lshostvdiskmap` command

```
IBM_2145:IBM Redbook SVC:superuser>lshostvdiskmap NYBIXTDB03
id name          SCSI_id vdisk_id vdisk_name      vdisk_UID
0  NYBIXTDB03  0          0      NYBIXTDB03_T01 60050768018205E12000000000000000

root@nybixtdb03:./> pcmpath query device
Total Dual Active and Active/Asymmetric Devices : 1
DEV#:  4  DEVICE NAME: hdisk4  TYPE: 2145  ALGORITHM:  Load Balance
SERIAL: 60050768018205E12000000000000000
=====
Path#      Adapter/Path Name      State   Mode      Select   Errors
  0*         fscsi0/path0           OPEN    NORMAL        7         0
    1         fscsi0/path1           OPEN    NORMAL    5597         0
  2*         fscsi2/path2           OPEN    NORMAL        8         0
    3         fscsi2/path3           OPEN    NORMAL   5890         0
```

If your organization uses a change management tracking tool, include the `vdisk_UID` and LUN ID information in every change request that performs SAN storage allocation or reclaim.

**Note:** Because a host can have many volumes with the same `scsi_id`, always cross-reference the IBM SAN Volume Controller volume UID with the host volume UID and record the `scsi_id` and LUN ID of that volume.

## 10.8 Server HBA replacement

Replacing a failed HBA in a server is a fairly trivial and safe operation if it is performed correctly. However, more precautions are required if your server has multiple, redundant HBAs on different SAN fabrics and the server hardware permits you to “hot” replace it (with the server still running).

Complete the following steps to replace a failed HBA and retain the working HBA:

1. In your server, that uses the multipath software, identify the failed HBA and record its WWPNs. For more information, see 10.7.1, “Cross-referencing WWPN” on page 425. Then, place this HBA and its associated paths offline, gracefully if possible. This approach is important so that the multipath software stops trying to recover it. Your server might even show a degraded performance while you perform this task.

Consider the following points:

- Some HBAs have an external label that shows the WWPNs. If you have this type of label, record the WWPNs before you install the new HBA in the server.
- If your server does not support HBA hot-swap, power off your system, replace the HBA, connect the used FC cable into the new HBA, and power on the system.
- If your server supports hot-swap, follow the suitable procedures to perform a “hot” replace of the HBA. Do *not* disable or disrupt the working HBA in the process.

2. Verify that the new HBA successfully logged in to the SAN switch. If it logged in successfully, you can see its WWPNs logged in to the SAN switch port. Otherwise, fix this issue before you continue to the next step.

Cross-check the WWPNs that you see in the SAN switch with the one you noted in step 1, and ensure that you did not record the incorrect WWNN.

3. In your SAN zoning configuration tool, replace the old HBA WWPNs for the new ones in every alias and zone to which they belong. Do *not* modify the other SAN fabric (the one with the working HBA) while you perform this task.

Only one alias should use each WWPN, and zones must reference this alias.

If you use SAN port zoning (although you should not be) and you did not move the new HBA FC cable to another SAN switch port, you do not need to reconfigure zoning.

4. Verify that the new HBA’s WWPNs appear in the IBM SAN Volume Controller by running the **lsfcportcandidate** command.

If the WWPNs of the new HBA do not appear, troubleshoot your SAN connections and zoning.

5. Add the WWPNs of this new HBA in the IBM SAN Volume Controller host definition by running the **addhostport** command. It is important that you do not remove the old one yet. Run the **lshost <servername>** command. Then, verify that the working HBA shows as active; the failed HBA should show as inactive or offline.

6. Use software to recognize the new HBA and its associated SAN disk paths. Certify that all SAN LUNs include redundant disk paths through the working HBA and the new HBA.

7. Return to the IBM SAN Volume Controller and verify again (by using the **lshost <servername>** command) that the working and the new HBA’s WWPNs are active. In this case, you can remove the old HBA WWPNs from the host definition by using the **rmhostport** command.

8. Do not remove any HBA WWPNs from the host definition until you ensure that you have at least two active ones that are working correctly.

By following these steps, you avoid removing your only working HBA by mistake.

## 10.9 Hardware upgrades

The IBM SAN Volume Controller scalability features allow significant flexibility in its configuration. The IBM SAN Volume Controller family features the following types of enclosures:

- ▶ **Control Enclosures**

These enclosures manage your storage systems, communicate with the host, and manage interfaces. Each “control enclosure” contains two nodes, which form an I/O group.

- ▶ **Expansion Enclosures**

These enclosures increase the available capacity of an IBM SAN Volume Controller cluster. They communicate with the control enclosure through a dual pair of 12 Gbps serial-attached SCSI (SAS) connections. These expansion enclosures can house many flash (solid-state drive [SSD]) SAS type drives, but are not available for all IBM SAN Volume Controller models.

Each SAN Volume Controller node is an individual server in a SAN Volume Controller clustered system. A basic configuration of an IBM SAN Volume Controller storage platform consists of two IBM SAN Volume Controller nodes, known as an *I/O group*. The nodes are always installed in pairs and all I/O operations that are managed by the nodes in an I/O group are cached on both nodes. For a balanced increase of performance and scale, up to four I/O groups can be clustered into a single storage system.

Similarly, to increase capacity, up to two chains (depending on IBM SAN Volume Controller model) of expansion enclosures can be added per node. Consequently, several scenarios are possible for growth. These processes are described next.

### 10.9.1 Adding nodes

You can add nodes to replace the existing nodes of your SAN Volume Controller cluster with newer ones, and the replacement procedure can be performed non-disruptively. The new node can assume the WWNN of the node you are replacing, which requires no changes in host configuration, SAN zoning, or multipath software. For more information about this procedure, see this [IBM Documentation web page](#).

Alternatively, you can add nodes to expand your system. If your IBM SAN Volume Controller cluster is below the maximum I/O groups limit for your specific product and you intend to upgrade it, you can install another I/O group.

It is also feasible that you might have a cluster of IBM Storwize V7000 nodes that you want to add the IBM SAN Volume Controller nodes to, because the IBM SAN Volume Controller nodes are more powerful than your existing nodes. Therefore, your cluster has different node models in different I/O groups.

To install these nodes, determine whether you need to upgrade your IBM SAN Volume Controller first (or Storwize V7000 code level if you are merging an Storwize V7000 Gen2 cluster with an IBM SAN Volume Controller, for example).

For more information, see 10.5.3, “Hardware considerations” on page 415.

**Note:** If two I/O groups are in a system, you must set up a quorum disk or application outside of the system. If the two I/O groups lose communication with each other, the quorum disk prevents both I/O groups from going offline.

For more information about adding a node to an IBM SAN Volume Controller cluster, see Chapter 3 of *Implementing the IBM System Storage SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8491.

**Note:** Use a consistent method (only the management GUI or only the CLI) when you add, remove, and re-add nodes. If a node is added by using the CLI and later re-added by using the GUI, it might get a different node name than it originally had.

After you install the newer nodes, you might need to redistribute your servers across the I/O groups. Consider the following points:

- Moving a server's volume to different I/O groups can be done online because of a feature called Non-Disruptive Volume Movement (NDVM). Although this process can be done without stopping the host, careful planning and preparation is advised. For more information about supported operating systems, see [this web page](#).

**Note:** You cannot move a volume that is in any type of Remote Copy relationship.

- If each of your servers is zoned to only one I/O group, modify your SAN zoning configuration as you move its volumes to another I/O group. As best you can, balance the distribution of your servers across I/O groups according to I/O workload.
- Use the **-iogrp** parameter in the **mkhost** command to define which I/O groups of the IBM SAN Volume Controller that the new servers will use. Otherwise, IBM SAN Volume Controller by default maps the host to all I/O groups, even if they do not exist and regardless of your zoning configuration. Example 10-9 shows this scenario and how to resolve it by using the **rmhostiogrp** and **addhostiogrp** commands.

*Example 10-9 Mapping the host to I/O groups*

```
IBM_2145:IBM Redbook SVC:superuser>lshost NYBIXTDB02
id 0
name NYBIXTDB02
port_count 2
type generic
mask 1111
iogrp_count 4
WWPN 10000000C9648274
node_logged_in_count 2
state active
WWPN 10000000C96470CE
node_logged_in_count 2
state active
IBM_2145:IBM Redbook SVC:superuser>lsiogrp
id name          node_count vdisk_count host_count
0 io_grp0        2          32          1
1 io_grp1        0          0           1
2 io_grp2        0          0           1
3 io_grp3        0          0           1
4 recovery_io_grp 0          0           0
IBM_2145:IBM Redbook SVC:superuser>lshostiogrp NYBIXTDB02
id name
```

```

0 io_grp0
1 io_grp1
2 io_grp2
3 io_grp3
IBM_2145:IBM Redbook SVC:superuser>rmhostiogrp -iogrp 1:2:3 NYBIXTDB02
IBM_2145:IBM Redbook SVC:superuser>lshostiogrp NYBIXTDB02
id name
0 io_grp0
IBM_2145:IBM Redbook SVC:superuser>lshiogrp
id name node_count vdisk_count host_count
0 io_grp0 2 32 1
1 io_grp1 0 0 0
2 io_grp2 0 0 0
3 io_grp3 0 0 0
4 recovery_io_grp 0 0 0
IBM_2145:IBM Redbook SVC:superuser>addhostiogrp -iogrp 3 NYBIXTDB02
IBM_2145:IBM Redbook SVC:superuser>lshostiogrp NYBIXTDB02
id name
0 io_grp0
3 io_grp3
IBM_2145:IBM Redbook SVC:superuser>lshiogrp
id name node_count vdisk_count host_count
0 io_grp0 2 32 1
1 io_grp1 0 0 0
2 io_grp2 0 0 0
3 io_grp3 0 0 1
4 recovery_io_grp 0 0 0

```

- 
- If possible, avoid setting a server to use volumes from different I/O groups that have different node types for extended periods of time. Otherwise, as this server's storage capacity grows, you might experience a performance difference between volumes from different I/O groups. This mismatch makes it difficult to identify and resolve eventual performance problems.

## Adding hot-spare nodes

To reduce the risk of a loss of redundancy or degraded system performance, hot-spare nodes can be added to the system. A hot-spare node has active system ports, but no host I/O ports, and is not part of any I/O group. If any node fails or is upgraded, this spare node automatically joins the system and assumes the place of the failed node, restoring redundancy.

The hot-spare node uses the same N\_Port ID Virtualization (NPIV) worldwide port names (WWPNs) for its Fibre Channel ports as the failed node, so host operations are not disrupted. After the failed node returns to the system, the hot-spare node returns to the **Spare** state, which indicates it can be automatically swapped for other failed nodes on the system.

The following restrictions apply to the use of hot-spare node on the system:

- Hot-spare nodes can be used with Fibre Channel-attached external storage only
- Hot-spare nodes *cannot* be used:
  - In systems that use RDMA-capable Ethernet ports for node-to-node communications
  - On enclosure-based systems
  - With SAS-attached storage
  - With iSCSI-attached storage
  - With storage that is directly attached to the system
- A maximum of four hot-spare nodes can be added to the system.

### ***Using the management GUI***

If your nodes are configured on your systems and you want to add hot-spare nodes, you must connect the extra nodes to the system. After hot-spare nodes are configured correctly on the system, you can add the spare node to the system configuration by completing the following steps:

1. In the management GUI, select **Monitoring** → **System Hardware**.
2. On the **System Hardware - Overview** window, click **Add Nodes**.
3. On the **Add Node** page, select the hot-spare node to add to the system.

If your system uses stretched or HyperSwap system topology, hot-spare nodes must be designated per site.

### ***Using the command-line interface***

To add a spare node to the system, run the following command:

```
addnode -panelname <panel_name> -spare
```

Where <panel\_name> is the name of the node that is displayed in the service assistant or in the output of the **lsnodecandidate** command.

For more information see, *IBM Spectrum Virtualize Hot-Spare Node and NPIV Target Ports*, REDP-5477.

## **10.9.2 Upgrading nodes in a cluster**

If you want to upgrade the nodes of your IBM SAN Volume Controller, the option is available to increase the cache memory size or the adapter cards in each node. This process can be done, one node at a time so as to be nondisruptive to the systems operations.

For more information, see [this IBM Documentation web page](#).

When evaluating cache memory upgrades, consider the following points:

- ▶ As your working set and total capacity increases, consider increasing your cache memory size. A *working set* is the most accessed workloads, excluding snapshots and backups. *Total capacity* implies more or larger workloads and a larger working set.
- ▶ If you are consolidating from multiple controllers, consider at least matching the amount of cache memory across those controllers.
- ▶ When externally virtualizing controllers (such as IBM SAN Volume Controller), a large cache can accelerate older controllers with smaller caches.
- ▶ If you use a Data Reduction Pool (DRP), maximize the cache size and consider adding SCM drives with Easy Tier for the best performance.
- ▶ If you are making heavy use of copy services, consider increasing the cache beyond just your working set requirements.
- ▶ A truly random working set might not benefit greatly from the cache.



**Important:** Do not power on a node that is:

- ▶ Shown as offline in the management GUI if you powered off the node to add memory to increase total memory. Before you increase memory, you must remove a node from the system so that it is not showing in the management GUI or in the output from the **lsnode** command.
- ▶ Still in the system and showing as offline with more memory than the node had when it powered off. Such a node can cause an immediate outage or an outage when you update the system software.

When evaluating adapter card upgrades, consider the following points:

- ▶ A single 32 Gb Fibre Channel port can deliver over 3 GBps (allowing for overheads).
- ▶ A 32 Gb FC card in each node with eight ports can deliver more than 24 GBps.
- ▶ An FCM NVMe device can perform at over 1 GBps.
- ▶ A single 32 Gb Fibre Channel port can deliver 80,000 - 125,000 IOPS with a 4 k block size.
- ▶ A 32 Gb FC card in each node with eight ports can deliver up to 1,000,000 IOPS.
- ▶ A FlashSystem 9200 can deliver 1,200,000 4 k read miss IOPS and up to 4,500,000 4 k read hit IOPS.
- ▶ If you have more than 12 NVMe devices, consider the use of two Fibre Channel cards per node. By using a third Fibre Channel card, up to 45 GBps can be achieved.
- ▶ If you want to achieve more than 800,000 IOPS, use at least two Fibre Channel cards per node.
- ▶ If the SAN Volume Controller is performing Remote Copy or clustering, consider the use of separate ports to ensure no conflict exists with host traffic.
- ▶ iSER by way of 25 GbE ports have similar capabilities as 16 Gb FC ports, but with less overall ports available. If you are planning to use 10 Gb iSCSI, ensure that it can service your expected workloads.

Real-time performance statistics are available in the management GUI from the **Monitoring** → **Performance** menu, as show in Figure 10-7.



Figure 10-7 IBM SAN Volume Controller performance statistics (IOPS)

## Memory options for an IBM SAN Volume Controller SV2 and SA2

The standard memory per node is 128 GB base memory with an option, by adding 32 GB memory modules, to support up to 768 GB (SA2) or 768 GB (SV2) of memory.

If you are adding memory to a node, you must remove that node from the system configuration before you start the following procedure. To do so, you can use the management GUI or the CLI.

To use the management GUI, select **Monitoring** → **System Hardware**. On the **System Hardware - Overview** page, select the directional arrow next to the node that you are removing to open the node details page. Select **Node Actions** → **Remove**.

To use the CLI, enter the following command, where `object_id` | `object_name` identifies the node that receives the added memory:

```
rmnodecanister object_id | object_name
```

A CPU processor has six memory channels, which are labeled A - F. Each memory channel has two DIMM slots, numbered 0 - 1. For example, DIMM slots A0 and A1 are in memory channel A.

On the system board, the DIMM slots are labeled according to their memory channel and slot. They are associated with the CPU nearest to their DIMM slots. You can install three distinct memory configurations in those 24 DIMM slots in each node.

The available memory configuration for each node is listed in Table 10-2. Each column shows the valid configuration for each total enclosure memory size. DIMM slots are listed in the same order that they appear in the node.

To ensure proper cooling and a steady flow of air from the fan modules in each node, blank DIMMs must be inserted in any slot that does not contain a memory module.

Table 10-2 Available memory configuration for one node

| DIMM Slot | Total node memory:<br>128 GB | Total node memory:<br>384 GB | Total node memory:<br>768 GB |
|-----------|------------------------------|------------------------------|------------------------------|
| F0 (CPU0) | Blank                        | 32 GB                        | 32 GB                        |
| F1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| E0 (CPU0) | Blank                        | 32 GB                        | 32 GB                        |
| E1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| D0 (CPU0) | 32 GB                        | 32 GB                        | 32 GB                        |
| D1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| A1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| A0 (CPU0) | 32 GB                        | 32 GB                        | 32 GB                        |
| B1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| B0 (CPU0) | Blank                        | 32 GB                        | 32 GB                        |
| C1 (CPU0) | Blank                        | Blank                        | 32 GB                        |
| C0 (CPU0) | Blank                        | 32 GB                        | 32 GB                        |
| C0 (CPU1) | Blank                        | 32 GB                        | 32 GB                        |

| DIMM Slot | Total node memory:<br>128 GB | Total node memory:<br>384 GB | Total node memory:<br>768 GB |
|-----------|------------------------------|------------------------------|------------------------------|
| C1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| B0 (CPU1) | Blank                        | 32 GB                        | 32 GB                        |
| B1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| A0 (CPU1) | 32 GB                        | 32 GB                        | 32 GB                        |
| A1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| D1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| D0 (CPU1) | 32 GB                        | 32 GB                        | 32 GB                        |
| E1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| E0 (CPU1) | Blank                        | 32 GB                        | 32 GB                        |
| F1 (CPU1) | Blank                        | Blank                        | 32 GB                        |
| F0 (CPU1) | Blank                        | 32 GB                        | 32 GB                        |

### Memory options for an IBM SAN Volume Controller SV1

The standard memory per node is 64 GB base memory with an option, by adding 32 GB memory modules, to support up to 256 GB of memory.

If you are adding memory to a node, you must remove that node from the system configuration before you start the following procedure. To do so, you can use the management GUI or the CLI.

To use the management GUI, select **Monitoring** → **System Hardware**. On the **System Hardware - Overview** page, select the directional arrow next to the node that you are removing to open the node details page. Select **Node Actions** → **Remove**.

To use the CLI, enter the following command, where `object_id` | `object_name` identifies the node that receives the added memory:

```
rmnodecanister object_id | object_name
```

### Memory options for an IBM SAN Volume Controller DH8

The standard memory per node is 32 GB base memory with an option, by adding a second CPU with 32 GB memory, to support up to 64 GB of memory.

If you are adding memory to a node, you must remove that node from the system configuration before you start the following procedure. To do so, you can use the management GUI or the CLI.

To use the management GUI, select **Monitoring** → **System Hardware**. On the **System Hardware - Overview** page, select the directional arrow next to the node that you are removing to open the node details page. Select **Node Actions** → **Remove**.

To use the CLI, enter the following command, where `object_id` | `object_name` identifies the node that receives the added memory:

```
rmnodecanister object_id | object_name
```

## Adapter card options for an IBM SAN Volume Controller SV2 and SA2

Four 10 Gb Ethernet ports for iSCSI connectivity are standard, but models SV2 and SA2 support the following optional host adapters for extra connectivity (feature codes in brackets):

- ▶ 4-port 16 Gbps Fibre Channel over NVMe adapter (AH14)
- ▶ 4-port 32 Gbps Fibre Channel over NVMe adapter (AH1D)
- ▶ 2-port 25 Gbps iSCSI/RoCE adapter (AH16)
- ▶ 2-port 25 Gbps iSCSI/iWARP adapter (AH17)

No more than three I/O adapter card features (AH14, AH16, AH17, and AH1D) can be used in a node. For more information about which specific slot supports which adapter type, see [this web page](#) and this [IBM Documentation web page](#).

Unlike in previous models, a Compression Accelerator is integrated directly with the processors for DRP compression workloads.

## Adapter card options for an IBM SAN Volume Controller SV1

Three 10 Gb Ethernet ports for iSCSI connectivity are standard, but model SV1 supports the following optional host adapters for additional connectivity (feature codes in brackets):

- ▶ 4-port 10 Gbps iSCSI/FC over Ethernet adapter (AH12)
- ▶ 4-port 16 Gbps Fibre Channel over NVMe adapter (AH14)
- ▶ 2-port 25 Gbps iSCSI/RoCE adapter (AH16)
- ▶ 2-port 25 Gbps iSCSI/iWARP adapter (AH17)

No more than four I/O adapter card features (AH12, AH14, AH16, or AH17) can be used in a node. It also can provide up to 16 16-Gb FC ports, up to four 10-Gb Ethernet (iSCSI/FCoE) ports, or up to eight 25 Gb Ethernet (iSCSI) ports.

**Note:** FCoE is no longer supported in Spectrum Virtualize 8.4. For more information about which specific slot supports which adapter type, see [this web page](#) and this [IBM Documentation web page](#).

The optional compression co-processor adapter increases the speed of I/O transfers to and from compressed volumes by using IBM Real-time Compression. You can optionally install one or two compression accelerator adapters in a SAN Volume Controller 2145-SV1 node. Each compression accelerator increases the speed of I/O transfers between nodes and compressed volumes. You must install at least one compression accelerator if you configured compressed volumes.

## Adapter card options for an IBM SAN Volume Controller DH8

Three 1 Gb Ethernet ports for iSCSI connectivity are standard, but model DH8 supports the following optional host adapters for more connectivity (feature codes in brackets):

- ▶ 4-port 8 Gbps Fibre Channel adapter (AH10)
- ▶ 2-port 16 Gbps Fibre Channel adapter (AH11)
- ▶ 4-port 16 Gbps Fibre Channel adapter (AH14)
- ▶ 2-port 10 Gbps iSCSI/FCoE adapter (AH12)

The maximum numbers and combinations of new adapters depends on the number of CPUs in the node and numbers and types of existing adapters. For more information about which specific slot supports which adapter type, see [this web page](#) and this [IBM Documentation web page](#).

The optional compression co-processor adapter increases the speed of I/O transfers to and from compressed volumes by using IBM Real-time Compression. You can optionally install one or two compression accelerator adapters in a SAN Volume Controller 2145-DH8 node. Each compression accelerator increases the speed of I/O transfers between nodes and compressed volumes. You must install at least one compression accelerator if you configured compressed volumes.

### 10.9.3 Moving to a new IBM SAN Volume Controller cluster

You might have a highly populated, intensively used IBM SAN Volume Controller cluster that you want to upgrade. You might also want to use the opportunity to refresh your IBM SAN Volume Controller and SAN storage environment.

Complete the following steps to replace your cluster with a newer, more powerful cluster:

1. Install your new IBM SAN Volume Controller cluster.
2. Create a replica of your data in your new cluster.
3. Migrate your servers to the new IBM SAN Volume Controller cluster when convenient.

If your servers can tolerate a brief, scheduled outage to switch from one IBM SAN Volume Controller cluster to another, you can use the IBM SAN Volume Controller Remote Copy services (Metro Mirror or Global Mirror) to create your data replicas, by completing the following steps:

1. Select a host that you want to move to the new IBM SAN Volume Controller cluster and find all the old volumes you must move.
2. Zone your host to the new IBM SAN Volume Controller cluster.
3. Create Remote Copy relationships from the old volumes in the old IBM SAN Volume Controller cluster to new volumes in the new IBM SAN Volume Controller cluster.
4. Map the new volumes from the new IBM SAN Volume Controller cluster to the host.
5. Discover new volumes on the host.
6. Stop all I/O from the host to the old volumes from the old IBM SAN Volume Controller cluster.
7. Disconnect and remove the old volumes on the host from the old IBM SAN Volume Controller cluster.
8. Unmap the old volumes from the old IBM SAN Volume Controller cluster to the host.
9. Ensure that the Remote Copy relationships between old and new volumes in the old and new IBM SAN Volume Controller cluster are synced.
10. Stop and remove Remote Copy relationships between old and new volumes so that the target volumes in the new IBM SAN Volume Controller cluster receive read/write access.
11. Import data from the new volumes and start your applications on the host.

If you must migrate a server online instead, you must use host-based mirroring by completing the following steps:

1. Select a host that you want to move to the new IBM SAN Volume Controller cluster and find all the old volumes that you must move.
2. Zone your host to the new IBM SAN Volume Controller cluster.
3. Create volumes in the new IBM SAN Volume Controller cluster of the same size as the old volumes in the old IBM SAN Volume Controller cluster.
4. Map the new volumes from the new IBM SAN Volume Controller cluster to the host.

5. Discover new volumes on the host.
6. For each old volume, use host-based mirroring (such as AIX **mirrorvg**) to move your data to the corresponding new volume.
7. For each old volume, after the mirroring is complete, remove the old volume from the mirroring group.
8. Disconnect and remove the old volumes on the host from the old IBM SAN Volume Controller cluster.
9. Unmap the old volumes from the old IBM SAN Volume Controller cluster to the host.

This approach uses the server's computing resources (CPU, memory, and I/O) to replicate the data. It can be done online if properly planned. Before you begin, ensure that it includes enough spare resources.

The biggest benefit to the use of either approach is that they easily accommodate (if necessary) the replacement of your SAN switches or your back-end storage controllers. You can upgrade the capacity of your back-end storage controllers or replace them entirely, as you can replace your SAN switches with bigger or faster ones. However, you do need to have spare resources, such as floor space, power, cables, and storage capacity, available during the migration.

#### 10.9.4 Splitting an IBM SAN Volume Controller cluster

Splitting an IBM SAN Volume Controller cluster might become a necessity if you have one or more of the following requirements:

- ▶ To grow the environment beyond the maximum number of I/O groups that a clustered system can support.
- ▶ To grow the environment beyond the maximum number of attachable subsystem storage controllers.
- ▶ To grow the environment beyond any other maximum system limit.
- ▶ To achieve new levels of data redundancy and availability.

By splitting the clustered system, you no longer have one IBM SAN Volume Controller that handles all I/O operations, hosts, and subsystem storage attachments. The goal is to create a second IBM SAN Volume Controller cluster so that you can equally distribute the workload over the two systems.

After safely removing enclosures from the existing cluster and creating a second IBM SAN Volume Controller cluster, choose from the following approaches to balance the two systems:

- ▶ Attach new storage subsystems and hosts to the new system and start adding only new workload on the new system.
- ▶ Migrate the workload onto the new system by using the approach that is described in 10.9.3, "Moving to a new IBM SAN Volume Controller cluster" on page 437.

#### 10.9.5 Adding expansion enclosures

As time passes and your environment grows, you must add more storage to your system. Depending on the IBM SAN Volume Controller family product and the code level that you installed, you can add different numbers of expansion enclosures to your system. Before you add an enclosure to a system, check that the licensed functions of the system support the extra enclosure.

Because all IBM SAN Volume Controller models were designed to make managing and maintaining them as simple as possible, adding an expansion enclosure is an easy task.

## **IBM SAN Volume Controller SV2 and SA2**

IBM SAN Volume Controller SV2 and SA2 models do *not* support any type of SAS expansion enclosures; all storage is external back-end storage. The IBM SAN Volume Controller can virtualize external storage that is presented to it from IBM and third-party storage systems. External back-end storage systems provide their logical volumes (LUs), which are detected by the IBM SAN Volume Controller as MDisks and can be used in a storage pool.

For more information, see Chapter 3, “Storage back-end” on page 67.

## **IBM SAN Volume Controller SV1**

The IBM SAN Volume Controller model SV1 can also support expansion enclosures with the following models:

- ▶ The IBM 2145 SVC LFF Expansion Enclosure Model 12F  
This model holds up to 12 3.5-inch SAS drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SVC SFF Expansion Enclosure Model 24F  
This model holds up to 24 2.5-inch SAS internal flash (solid state) drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SVC HD LFF Expansion Enclosure Model 92F  
This model holds up to 92 3.5-inch SAS internal flash (solid state) or HDD capacity drives in a 5U, 19-inch rack mount enclosure.

If Enterprise Class Support and three-year warranty is purchased, the model number changes from 2145 to 2147.

Each IBM SAN Volume Controller SV1 supports two chains of SAS expansion enclosures per node. Overall, the system supports up to four I/O groups (eight nodes) with a total of 80 expansion enclosures per system.

The best practice recommendation is to balance equally the expansion enclosures between chains. Therefore, if you have two more expansion enclosures, one should be installed on the first SAS chain and one on the second SAS chain. Also, when you add a single expansion enclosure to a system, it is preferable to add the enclosure directly below the nodes.

When you add a second expansion enclosure, it is preferable to add the enclosure directly above the nodes. As more expansion enclosures are added, alternate adding them above and below.

To limit contention for bandwidth on a chain of SAS enclosures, no more than 10 expansion enclosures can be chained to the SAS port of a node. On each SAS chain, the systems can support up to a SAS chain weight of 10 where:

- ▶ Each 2145-92F expansion enclosure adds a value of 2.5 to the SAS chain weight.
- ▶ Each 2145-12F or 2145-24F expansion enclosure adds a value of 1 to the SAS chain weight.

For example, each of the following expansion enclosure configurations has a total SAS weight of 10:

- ▶ Four 2145-92F enclosures per SAS chain
- ▶ Two 2145-92F enclosures and five 2145-12F enclosures per SAS chain

Figure 10-8 shows the cabling for adding four 2145-24F expansion enclosures (two at the top and two at the bottom of the figure).

For more information, see this [IBM Documentation web page](#).

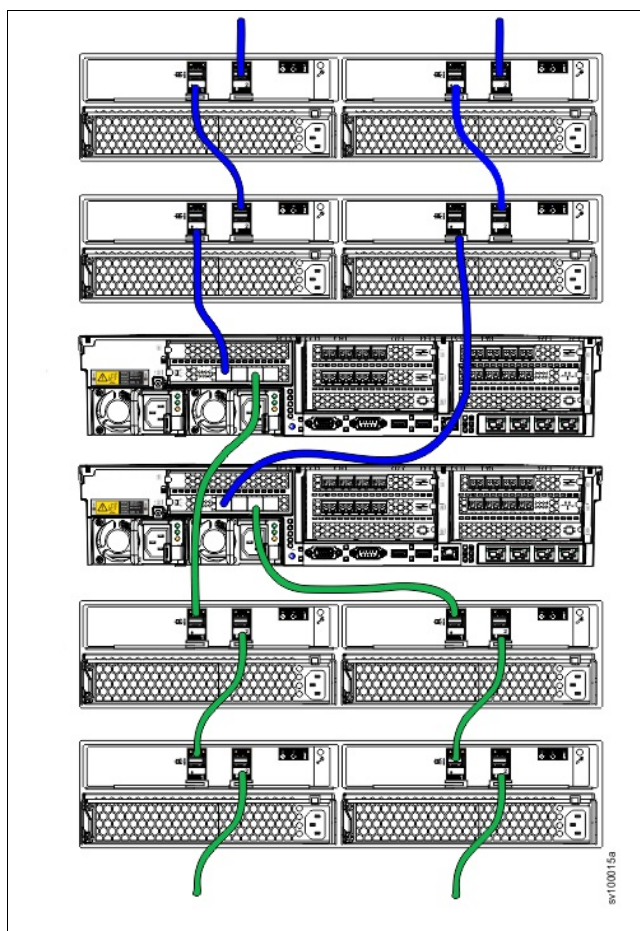


Figure 10-8 Cabling for adding four expansion enclosures in two SAS chains

### IBM SAN Volume Controller DH8

IBM SAN Volume Controller Model DH8 can also support expansion enclosures with the following models:

- ▶ The IBM 2145 SVC LFF Expansion Enclosure Model 12F  
This model holds up to 12 3.5-inch SAS drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SVC SFF Expansion Enclosure Model 24F  
This model holds up to 24 2.5-inch SAS internal flash (solid state) drives in a 2U, 19-inch rack mount enclosure.
- ▶ The IBM 2145 SVC HD LFF Expansion Enclosure Model 92F  
This model holds up to 92 3.5-inch SAS internal flash (solid state) drive capacity drives in a 5U, 19-inch rack mount enclosure.

If Enterprise Class Support and three-year warranty is purchased, the model number changes from 2145 to 2147.



Similar to the SV1 model, each model DH8 supports two chains of SAS expansion enclosures per node. Overall, the system supports up to four I/O groups (eight nodes) with a total of 80 expansion enclosures per system.

## 10.9.6 Removing expansion enclosures

As storage environments change and grow, it is sometimes necessary to move expansion enclosures between nodes. Removing an expansion enclosure is a straightforward task.

To remove an expansion enclosure from a node, complete the following steps:

**Note:** If the expansion enclosure that you want to move is not at the end of an SAS chain, you might need a longer pair of SAS cables to complete the procedure. In that case, ensure that you have two SAS cables of suitable length before you start this procedure.

1. Delete any volumes that are no longer needed and that depend on the enclosure that you plan to remove.
2. Delete any remaining arrays that are formed from drives in the expansion enclosure. Any data in those arrays is automatically migrated to other managed disks in the pool if there is enough capacity.
3. Wait for data migration to complete.
4. Mark all the drives (including any configured as spare or candidate drives) in the enclosures to be removed as unused.
5. Unmanage and remove the expansion enclosure by using the management GUI. Select **Monitoring** → **System Hardware**. On the **System Hardware - Overview** page, select the arrow next to the enclosure that you are removing to open the Enclosure Details page. Select **Enclosure Actions** → **Remove**.

**Important:** Do not proceed until the enclosure removal process completes successfully.

6. On the I/O group that contains the expansion enclosure that you want to remove, enter the following command to put the I/O group into maintenance mode:  

```
chlogr -maintenance yes <iogroup_name_or_id>
```
7. If the expansion enclosure that you want to move is at the end of a SAS chain, complete the following steps to remove the enclosure from the SAS chain:
  - a. Disconnect the SAS cable from port 1 of node 1 and node 2. The enclosure is now disconnected from the system.
  - b. Disconnect the other ends of the SAS cables from the previous enclosure in the SAS chain. The previous enclosure is now the end of the SAS chain. Proceed to step 10.
8. If the expansion enclosure is not at the end of a SAS chain, complete the following steps to remove the enclosure from the SAS chain.
  - a. Disconnect the SAS cable from port 2 of node 1 of the expansion enclosure that you want to move.
  - b. Disconnect the other end of the same SAS cable from port 1 of node 1 of the next expansion enclosure in the SAS chain.
  - c. Disconnect the SAS cable from port 1 of node 1 of the expansion enclosure that you want to move.

- d. Reroute the cable that was disconnected in the previous step and connect it to port 1 of node 1 of the next expansion enclosure in the SAS chain.

**Important:** Do not continue until you complete this cable connection step.

- e. Disconnect the SAS cable from port 2 of node 2 of the expansion enclosure that you want to move.
  - f. Disconnect the other end of the same SAS cable from port 1 of node 2 of the next expansion enclosure in the SAS chain.
  - g. Disconnect the SAS cable from port 1 of node 2 of the expansion enclosure that you want to move.
  - h. Reroute the cable that was disconnected in the previous step and connect it to port 1 of node 2 of the next expansion enclosure in the SAS chain.
9. Take the I/O group out of maintenance mode by entering the following command:  

```
chiogrp -maintenance no iogroup_name_or_id
```
  10. Check the event log for any errors and fix those errors as needed.
  11. Disconnect the power from the expansion enclosure that you want to remove.
  12. Remove the expansion enclosure from the rack along with its two power cables and two SAS cables.

**Note:** The IBM SAN Volume Controller products provide methods to securely erase data from a drive when an enclosure is decommissioned or before a drive is removed from the system during a repair activity. For more information about the CLI commands that are used to run this secure erase function, see this [IBM Documentation web page](#).

## 10.10 I/O Throttling

I/O throttling is a mechanism with which you can limit the volume of I/O processed by the storage controller at various levels to achieve quality of service (QoS). If a throttle is defined, the system processes the I/O or delays the processing of the I/O to free resources for more critical I/O. Throttling is a way to achieve a better distribution of storage controller resources.

IBM SAN Volume Controller V8.3 and later code brings the possibility to set the throttling at a volume level, host, host cluster, storage pool, and offload throttling by using the GUI. This section describes I/O throttling and shows how to configure the feature in your system.

### 10.10.1 I/O throttling overview

I/O throttling features the following characteristics:

- ▶ Both IOPS and bandwidth throttle limits can be set.
- ▶ It is an upper bound QoS mechanism.
- ▶ No minimum performance is guaranteed.
- ▶ Volumes, hosts, host clusters, and managed disk groups can be throttled.
- ▶ Queuing occurs at microsecond granularity.
- ▶ Internal I/O operations (such as FlashCopy and cluster traffic) are not throttled.
- ▶ Reduces I/O bursts and smooths the I/O flow with variable delay in throttled I/Os.
- ▶ Throttle limit is a per-node value.

### 10.10.2 I/O throttling on front-end I/O control

You can use throttling for a better front-end I/O control at the volume, host, host cluster, and offload levels:

- ▶ In a multi-tenant environment, hosts can have their own defined limits.  
You can use this to allow restricted I/Os from a data mining server and a higher limit for an application server.
- ▶ An aggressive host consuming bandwidth of the controller can be limited by a throttle.  
For example, a video streaming application can have a limit set to avoid consuming too much of the bandwidth.
- ▶ Restrict a group of hosts by their throttles.  
For example, Department A gets more bandwidth than Department B.
- ▶ Each volume can have a throttle defined.  
For example, a volume that is used for backups can be configured to use less bandwidth than a volume used for a production database.
- ▶ When performing migrations in a production environment consider, the use of host or volume level throttles.
- ▶ Offloaded I/Os.

Offload commands, such as **UNMAP** and **XCOPY**, free hosts and speed the copy process by offloading the operations of certain types of hosts to a storage system. These commands are used by hosts to format new file systems or copy volumes without the host needing to read and then write data.

Throttles can be used to delay processing for offloads to free bandwidth for other more critical operations, which can improve performance but limits the rate at which host features, such as VMware VMotion, can copy data.

### 10.10.3 I/O Throttling on back-end I/O control

You can also use throttling to control the back-end I/O by throttling the storage pools, which can be useful in the following scenarios:

- ▶ Each storage pool can have a throttle defined.
- ▶ Allows control of back-end I/Os from the IBM SAN Volume Controller.
- ▶ Avoids overwhelming any external back-end storage.
- ▶ Creates a VVOL in a child pool. A child pool (**mdiskgrp**) throttle can control I/Os coming from that VVOL.
- ▶ Only parent pools support throttles because only parent pools contain MDisk from internal or external back-end storage. For volumes in child pools, the throttle of the parent pool is applied.
- ▶ If more than one throttle applies to an I/O operation, the lowest and most stringent throttle is used. For example, if a throttle of 100 MBps is defined on a pool and a throttle of 200 MBps is defined on a volume of that pool, the I/O operations are limited to 100 MBps.

### 10.10.4 Overall benefits of using I/O throttling

The overall benefits of the use of I/O throttling is a better distribution all system resources:

- ▶ Avoids overwhelming the controller objects.
- ▶ Avoids starving the external entities, such as hosts, from their share of controller.
- ▶ A scheme of distribution of controller resources that, in turn, results in better use of external resources, such as host capacities.

With no throttling enabled, we have a scenario in which Host 1 dominates the bandwidth. After enabling the throttle, we see a much better distribution of the bandwidth among the hosts, as shown in Figure 10-9.

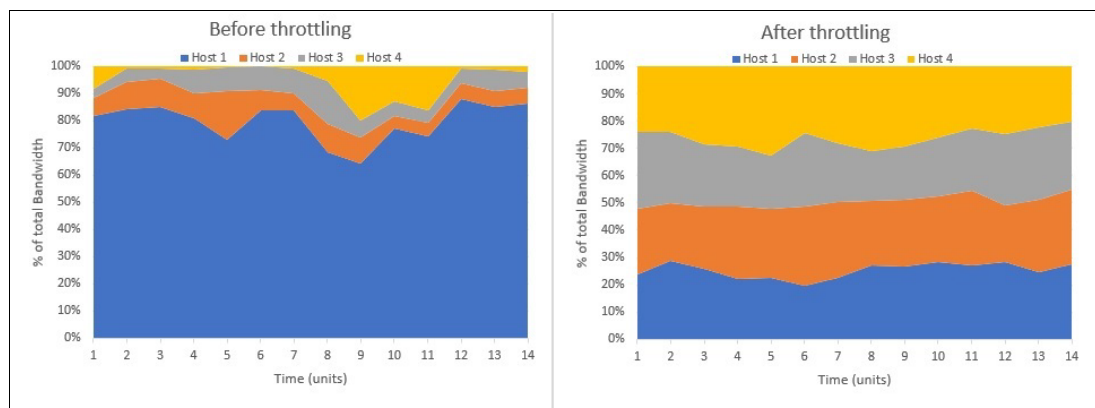


Figure 10-9 Distribution of controller resources before and after I/O throttling

### 10.10.5 I/O throttling considerations

When you are planning to use I/O throttling, consider the following points:

- ▶ The throttle cannot be defined for the host if it is part of a host cluster, which includes a host cluster throttle.

- ▶ If the host cluster does not have a throttle defined, its member hosts can have their individual host throttles defined.
- ▶ If a volume has multiple copies, throttling is done for the storage pool serving the primary copy. The throttling will not be applicable on the secondary pool for mirrored volumes and stretched cluster implementations.
- ▶ A host cannot be added to a host cluster if both have their individual throttles defined. If only one of the host/host cluster throttles is present, the command succeeds.
- ▶ A seeding host that is used for creating a host cluster cannot have a host throttle defined for it.

**Note:** Throttling is applicable only at the I/Os that an IBM SAN Volume Controller receives from hosts and host clusters. The I/Os generated internally, such as mirrored volume I/Os, cannot be throttled.

## 10.10.6 Configuring I/O throttling using the CLI

To create a throttle by using the CLI, you use the **mkthrottle** command (see Example 10-10). The bandwidth limit is the maximum amount of bandwidth the system can process before the system delays I/O processing. Similarly, the **iops\_limit** is the maximum amount of IOPS the system can process before the system delays I/O processing.

*Example 10-10 Creating a throttle using the mkthrottle command in the CLI*

Syntax:

```
mkthrottle -type [offload | vdisk | host | hostcluster | mdiskgrp]
              [-bandwidth bandwidth_limit_in_mb]
              [-iops iops_limit]
              [-name throttle_name]
              [-vdisk vdisk_id_or_name]
              [-host host_id_or_name]
              [-hostcluster hostcluster_id_or_name]
              [-mdiskgrp mdiskgrp_id_or_name]
```

Usage examples:

```
IBM_2145:IBM Redbook SVC:superuser>mkthrottle -type host -bandwidth 100 -host
ITS0_HOST3
```

```
IBM_2145:IBM Redbook SVC:superuser>mkthrottle -type hostcluster -iops 30000
-hostcluster ITS0_HOSTCLUSTER1
```

```
IBM_2145:IBM Redbook SVC:superuser>mkthrottle -type mdiskgrp -iops 40000 -mdiskgrp
0
```

```
IBM_2145:IBM Redbook SVC:superuser>mkthrottle -type offload -bandwidth 50
```

```
IBM_2145:IBM Redbook SVC:superuser>mkthrottle -type vdisk -bandwidth 25 -vdisk
volume1
```

```
IBM_2145:IBM Redbook SVC:superuser>lsthrottle
throttle_id throttle_name object_id object_name      throttle_type IOPs_limit
bandwidth_limit_MB
0          throttle0      2          ITS0_HOST3      host          100
1          throttle1      0          ITS0_HOSTCLUSTER1 hostcluster
30000
2          throttle2      0          Pool0          mdiskgrp
40000
```

|   |           |    |         |         |    |    |
|---|-----------|----|---------|---------|----|----|
| 3 | throttle3 |    |         | offload | 50 |    |
| 4 | throttle4 | 10 | volume1 | vdisk   |    | 25 |

**Note:** You can change a throttle parameter by using the **chthrottle** command.

### 10.10.7 Configuring I/O throttling using the GUI

The following pages shows how to configure the throttle by using the management GUI.

### 10.10.8 Creating a volume throttle

To create a volume throttle, go to **Volumes** → **Volumes**. Then, select the wanted volume, right-click it, and chose **Edit Throttle**, as shown in Figure 10-10. The bandwidth can be set from 1 MBps - 256 TBps and IOPS can be set from 1 - 33,254,432.

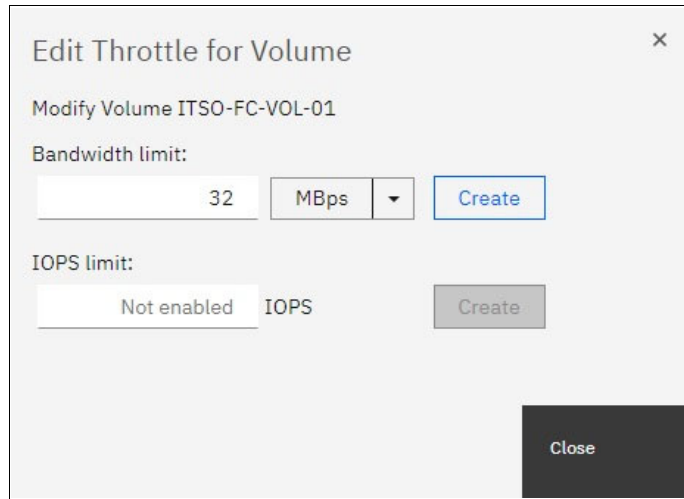


Figure 10-10 Creating a volume throttle in the GUI

If a throttle exists, the dialog box that is shown in Figure 10-10 also shows a **Remove** button that is used to delete the throttle.

### 10.10.9 Creating a host throttle

To create a host throttle, go to **Hosts** → **Hosts** and select the wanted host. Then, right-click it and chose **Edit Throttle**, as shown in Figure 10-11.

Dialog box titled "Edit Throttle for Host" (close button).

Modify Host NVMEHost1

Bandwidth limit:

10 MBps Create

IOPS limit:

2,000 IOPS Create

Close

Figure 10-11 Creating a host throttle in the GUI

### 10.10.10 Creating a host cluster throttle

To create a host cluster throttle, go to **Hosts** → **Host Clusters** and select the wanted host cluster. Then, right-click it and chose **Edit Throttle**, as shown in Figure 10-12.

Dialog box titled "Edit Throttle for Host Cluster" (close button).

Modify Host Cluster NVMECluster

Bandwidth limit:

200 MBps Create

IOPS limit:

Not enabled IOPS Create

Close

Figure 10-12 Creating a host cluster throttle in the GUI

### 10.10.11 Creating a storage pool throttle

To create a storage pool throttle, go to **Pools** → **Pools** and select the wanted storage pool. Then, right-click it and choose **Edit Throttle**, as shown in Figure 10-13.

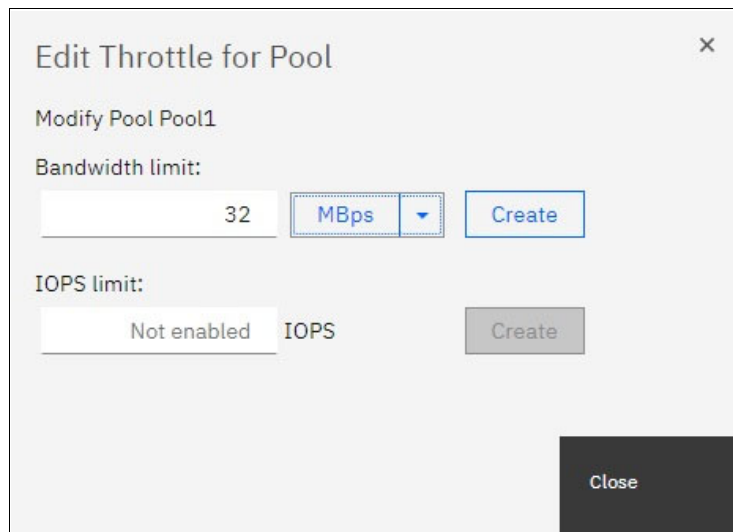


Figure 10-13 Creating a storage pool throttle in the GUI

### 10.10.12 Creating an offload throttle

To create an offload throttle, go to **Monitoring** → **System Hardware** → **System Actions**. Then, select **Edit System Offload Throttle**, as shown in Figure 10-14.

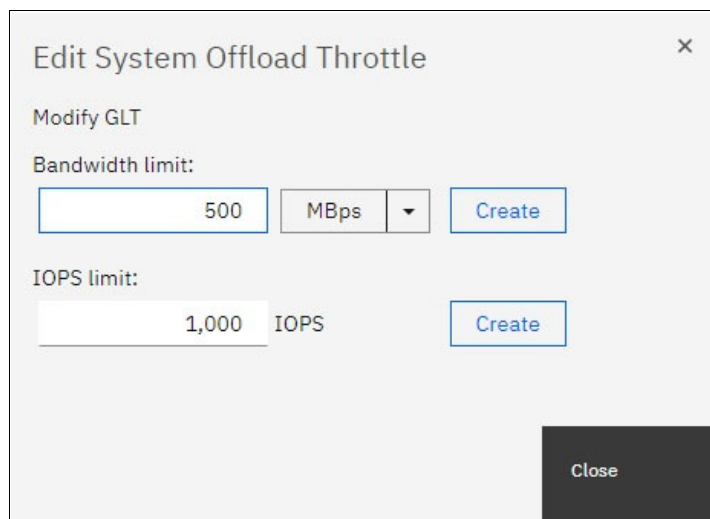


Figure 10-14 Creating system offload throttle in the GUI



## 10.11 Automation

Automation is a priority for maintaining today's busy storage environments. Automation software allows the creation of repeatable sets of instructions and processes to reduce the need for human interaction with computer systems. Red Hat Ansible and other third-party automation tools are becoming increasingly used across the enterprise IT environments and it is not unexpected that their use in storage environments is becoming more popular.

### 10.11.1 Red Hat Ansible

IBM SAN Volume Controller family includes integration with Red Hat Ansible Automation Platform, allowing IT to create an Ansible playbook that automates repetitive tasks across an organization in a consistent way, which helps improve outcomes and reduce errors.

Ansible is an agentless automation management tool that uses the SSH protocol. Currently, Ansible can be run from any machine with Python 2 (version 2.7) or Python 3 (versions 3.5 and higher) installed. This includes Red Hat, Debian, CentOS, macOS, any of the BSDs. Windows is not supported for the Ansible control node.

IBM is a Red Hat certified support module vendor, providing simple management for the following commands used in the IBM Spectrum Virtualize Ansible Collection:

- ▶ **Collect facts:** Collect basic information including hosts, host groups, snapshots, consistency groups, and volumes
- ▶ **Manage hosts:** Create, delete, or modify hosts
- ▶ **Manage volumes:** Create, delete, or extend the capacity of volumes
- ▶ **Manage MDisk:** Create or delete a managed disk
- ▶ **Manage pool:** Create or delete a pool (managed disk group)
- ▶ **Manage volume map:** Create or delete a volume map
- ▶ **Manage consistency group snapshot:** Create or delete consistency group snapshots
- ▶ **Manage snapshot:** Create or delete snapshots
- ▶ **Manage volume clones:** Create or delete volume clones

This collection provides a series of Ansible modules and plug-ins for interacting with the IBM Spectrum Virtualize Family storage systems. The modules in the IBM Spectrum Virtualize Ansible collection uses the Representational State Transfer (REST) application programming interface (API) to connect to the IBM Spectrum Virtualize storage system. These storage systems include the IBM SAN Volume Controller, IBM FlashSystem family including FlashSystem 5010, 5030, 5100, 7200, 9100, 9200, and 9200R and IBM Spectrum Virtualize for Public Cloud.

For more information, see *Automate and Orchestrate® Your IBM FlashSystem Hybrid Cloud with Red Hat Ansible*, REDP-5598.

For IBM Spectrum Virtualize modules, Ansible version 2.9 or higher is required. For more information about IBM Spectrum Virtualize modules, see [this web page](#).

## 10.11.2 RESTful API

The Spectrum Virtualize REST model API consists of command targets that are used to retrieve system information and to create, modify, and delete system resources. These command targets allow command parameters to pass through unedited to the Spectrum Virtualize command-line interface, which handles parsing parameter specifications for validity and error reporting. Hypertext Transfer Protocol Secure (HTTPS) is used to communicate with the RESTful API server.

To interact with the storage system by using the RESTful API, use the curl utility (see [this website](#)) to make an HTTPS command request with a valid configuration node URL destination. Open TCP port 7443 and include the keyword `rest` followed by the Spectrum Virtualize target command you want to run.

Each curl command takes the following form:

```
curl -k -X POST -H <header_1> -H <header_2> ... -d <JSON input>
https://SVC_ip_address:7443/rest/target
```

Where the following definitions apply:

- ▶ `POST` is the only HTTPS method that the Spectrum Virtualize RESTful API supports.
- ▶ Headers `<header_1>` and `<header_2>` are individually-specified HTTP headers (for example, `Content-Type` and `X-AuthUsername`).
- ▶ `-d` is followed by the JSON input; for example, `{“raid_level”: “raid5”}`.
- ▶ `<SVC_ip_address>` is the IP address of the IBM SAN Volume Controller to which you are sending requests.
- ▶ `<target>` is the target object of commands, which includes any object IDs, names, and parameters.

### Authentication

Aside from data encryption, the HTTPS server requires authentication of a valid user name and password for each API session. Use two authentication header fields to specify your credentials: `X-Auth-Username` and `X-Auth-Password`.

Initial authentication requires that you POST the authentication target (`/auth`) with the user name and password. The RESTful API server returns a hexadecimal token. A single session lasts a maximum of two active hours or 30 inactive minutes, whichever occurs first.

When your session ends because of inactivity, or if you reach the maximum time that is allotted, error code 403 indicates the loss of authorization. Use the `/auth` command target to reauthenticate with the user name and password.

The following example shows the correct procedure for authenticating. You authenticate by first producing an authentication token and then use that token in all future commands until the session ends.

For example, the following command passes the authentication command to IBM SAN Volume Controller node IP address 192.168.10.20 at port 7443:

```
curl -k -X POST -H 'Content-Type: application/json' -H 'X-Auth-Username:
superuser' -H 'X-Auth-Password: passwOrd' https://192.168.10.20:7443/rest/auth
```

**Note:** Make sure that you format the request correctly by using spaces after each colon in each header; otherwise, the command fails.

This request yields an authentication token, which can be used for all subsequent commands; for example:

```
{"token": "38823f60c758dca26f3eaac0ffee42aadc4664964905a6f058ae2ec92e0f0b63"}
```

### Example command

Most actions must be taken only after authentication. The following example of creating an array demonstrates the use of the previously generated token in place of the authentication headers that are used in the authentication process.

```
curl -k -X POST -H 'Content-Type: application/json' -H 'X-Auth-Token: 38823f60c758dca26f3eaac0ffee42aadc4664964905a6f058ae2ec92e0f0b63' -d '{"level": "raid5", "drive": "6:7:8:9:10", "raid6grp": "https://192.168.10.20:7443/rest/mkarray"}'
```

For more information about the use of the API, see this [IBM Documentation web page](#).

For more information about other examples, see the following web pages:

- ▶ [IBM Spectrum Virtualize Interfacing Using the RESTful API](#)
- ▶ [Implementing a RESTful API to the IBM Storwize Family](#)
- ▶ [Tips and tricks using the Spectrum Virtualize REST API](#)

## 10.12 Documenting IBM SAN Volume Controller and SAN environment

This section focuses on the challenge of automating the documentation that is needed for an IBM SAN Volume Controller solution. Consider the following points:

- ▶ Several methods and tools are available to automate the task of creating and updating the documentation. Therefore, the IT infrastructure might handle this task.
- ▶ Planning is key to maintaining sustained and organized growth. Accurate documentation of your storage environment is the blueprint with which you plan your approach to short-term and long-term storage growth.
- ▶ Your storage documentation must be conveniently available and easy to consult when needed. For example, you might need to determine how to replace your core SAN directors with newer ones, or how to fix the disk path problems of a single server. The relevant documentation might consist of a few spreadsheets and a diagram.
- ▶ Remember to include photographs in the documentation, where suitable.

**Storing documentation:** Avoid storing IBM SAN Volume Controller and SAN environment documentation only in the SAN. If your organization has a Disaster Recovery (DR) plan, include this storage documentation in it. Follow its guidelines about how to update and store this data. If no DR plan exists and you have the suitable security authorization, it might be helpful to store an updated copy offsite.

In theory, this IBM SAN Volume Controller and SAN environment documentation is written at a level that is sufficient for any system administrator who has average skills in the products to understand. Make a copy that includes all your configuration information.

Use the copy to create a functionally equivalent copy of the environment by using similar hardware without any configuration, off-the-shelf media, and configuration backup files. You might need the copy if you ever face a DR scenario, which is also why it is so important to run periodic DR tests.

Create the first version of this documentation (“as-built documentation”) as you install your solution. If you completed forms to help plan the installation of your IBM SAN Volume Controller solution, use these forms to help you document how your IBM SAN Volume Controller solution was first configured. Minimum documentation is needed for an IBM SAN Volume Controller solution. Because you might have more business requirements that require other data to be tracked, remember that the following sections do not address every situation.

## 10.12.1 Naming conventions

Whether you are creating your IBM SAN Volume Controller and SAN environment documentation, or you are updating what is in place, first evaluate whether you have a good naming convention in place. With a good naming convention, you can quickly and uniquely identify the components of your IBM SAN Volume Controller and SAN environment. System administrators can then determine whether a name belongs to a volume, storage pool, MDisk, host, or HBA by looking at it.

Because error messages often point to the device that generated an error, a good naming convention quickly highlights where to start investigating when an error occurs. Typical IBM SAN Volume Controller and SAN component names limit the number and type of characters you can use. For example, IBM SAN Volume Controller names are limited to 63 characters, which makes creating a naming convention a bit easier.

Many names in IBM SAN Volume Controller and SAN environment can be modified online. Therefore, you do not need to worry about planning outages to implement your new naming convention. The naming examples that are used in the following sections are effective in most cases, but might not be fully adequate for your environment or needs. The naming convention to use is your choice, but you must implement it in the entire environment.

### Enclosures, nodes and external storage controllers,

IBM SAN Volume Controller names its internal nodes as nodeX, with X being a sequential decimal number. These numbers range 2 - 8, in a four IBM SAN Volume Controller system cluster.

If multiple external controllers are attached to your IBM SAN Volume Controller solution, these controllers are detected as controllerX; therefore, you might need to change the name so that it includes, for example, the vendor name, the model, or its serial number. Therefore, if you receive an error message that points to controllerX, you do not need to log in to IBM SAN Volume Controller to know which storage controller to check.

**Note:** An IBM SAN Volume Controller detects external controllers that are based on their worldwide node name (WWNN). If you have an external storage controller that has one WWNN for each worldwide port name (WWPN), this configuration might lead to many controllerX names pointing to the same physical box. In this case, prepare a naming convention to cover this situation.

## MDisks and storage pools

When an IBM SAN Volume Controller detects new MDisks, it names them by default as `mdiskXX`, where `XX` is a sequential number. You should change the `XX` value to something more meaningful. MDisks are arrays (DRAID) from internal storage or volumes from an external storage system.

Ultimately, it comes down to personal preference and what works in your environment. The main “convention” that you must follow is to avoid the use of special characters in names, apart from the underscore, the hyphen, and the period (which are permitted), and spaces (which can make scripting difficult).

For example, you can change it to include the following information:

- ▶ For internal MDisks, refer to the IBM SAN Volume Controller system or cluster name.
- ▶ A reference to the external storage controller to which it belongs to (such as its serial number or last digits).
- ▶ The extpool, array, or RAID group that it belongs to in the storage controller.
- ▶ The LUN number or name it has in the storage controller.

Consider the following examples of MDisk names with this convention:

- ▶ `FS9200CL01-MD03`, where `FS9200CL01` is the system or cluster name, and `MD03` is the MDisk name.
- ▶ `23K45_A7V10`, where `23K45` is the serial number, `7` is the array, and `10` is the volume.
- ▶ `75VXYZ1_02_0206`, where `75VXYZ1` is the serial number, `02` is the extpool, and `0206` is the LUN.

Storage pools have several different possibilities. One possibility is to include the storage controller, type of back-end disks if external, RAID type, and sequential digits. If you use dedicated pools for specific applications or servers, another possibility is to use them instead. Consider the following examples:

- ▶ `FS9200-P00L01`, where `FS9200` is the system or cluster name, and `P00L01` is the pool.
- ▶ `P05XYZ1_3GR5`, where Pool 05 from serial `75VXYZ1`, LUNs with 300 GB FC DDMs, and RAID 5.
- ▶ `P16XYZ1_EX01`, where Pool 16 from serial `75VXYZ1`, and pool 01 dedicated to Exchange Mail servers.
- ▶ `XIV01_F9H02_ET`, where Pool with disks is from XIV named `XIV01` and FlashSystem 900 `F9H02`, which are both managed by Easy Tier.

## Volumes

Volume names should include the following information:

- ▶ The host or cluster to which the volume is mapped.
- ▶ A single letter that indicates its usage by the host, as shown in the following examples:
  - `B`: For a boot disk, or `R` for a rootvg disk (if the server boots from SAN)
  - `D`: For a regular data disk
  - `Q`: For a cluster quorum disk (do not confuse with IBM SAN Volume Controller quorum disks)
  - `L`: For a database log disk
  - `T`: For a database table disk

- ▶ A few sequential digits, for uniqueness.
- ▶ Sessions standard for VMware datastores:
  - esx01-sessions-001: For a datastore composed of a single volume
  - esx01-sessions-001a and esx01-sessions-001b: For a datastore composed of 2 volumes

For example, ERPNY01-T03 indicates a volume that is mapped to server ERPNY01 and database table disk 03.

## Hosts

In today's environment, administrators deal with large networks, the internet, and cloud computing. Use good server naming conventions so that they can quickly identify a server and determine the following information:

- ▶ Where it is (to know how to access it).
- ▶ What kind it is (to determine the vendor and support group in charge).
- ▶ What it does (to engage the proper application support and notify its owner).
- ▶ Its importance (to determine the severity if problems occur).

Changing a server's name in IBM SAN Volume Controller is as simple as changing any other IBM SAN Volume Controller object name. However, changing the name on the operating system of a server might have implications for application configuration or DNS and might require a server reboot. Therefore, you might want to prepare a detailed plan if you decide to rename several servers in your network.

The following example is for a server naming convention of LLAATRFFNN where:

- ▶ LL is the location, which might designate a city, data center, building floor, or room.
- ▶ AA is a major application; for example, billing, ERP, and Data Warehouse.
- ▶ T is the type; for example, UNIX, Windows, and VMware.
- ▶ R is the role; for example, Production, Test, Q&A, and Development.
- ▶ FF is the function; for example, DB server, application server, web server, and file server.
- ▶ NN is numeric.

## SAN aliases and zones

SAN aliases often must reflect only the device and port that is associated to it. Including information about where one specific device port is physically attached on the SAN might lead to inconsistencies if you make a change or perform maintenance and then forget to update the alias. Create one alias for each device port WWPN in your SAN and use these aliases in your zoning configuration.

Consider the following examples:

- ▶ AIX\_NYBIXTDB02\_FC2: Interface fcs2 of AIX server NYBIXTDB02.
- ▶ LIN-P0KBIXAP01-FC1: Interface fcs1 of Linux Server P0KBIXAP01.
- ▶ WIN\_EXCHSRV01\_HBA1: Interface HBA1 of physical Windows server EXCHSRV01.
- ▶ ESX\_NYVMCLUSTER01\_VMHBA2: Interface vmhba2 of ESX server NYVMCLUSTER01.
- ▶ IBM-NYFS9200-N1P1\_HOST: Port 1 of Node 1 from FS9200 Cluster NYFS9200 dedicated for hosts.
- ▶ IBM-NYFS9200-N1P5\_INTRA: Port 5 of Node 1 from FS9200 Cluster NYFS9200 dedicated to intracluster traffic.
- ▶ IBM-NYFS9200-N1P7\_REPL: Port 7 of Node 1 from FS9200 Cluster NYFS9200 dedicated to replication.

Be mindful of the IBM SAN Volume Controller port aliases. There are mappings between the last digits of the port WWPN and the node FC port.

- ▶ IBM\_D88870\_75XY131\_I0301: DS8870 serial number 75XY131, port I0301.
- ▶ TS4500-TD06: TS4500 tape library, tape drive 06.
- ▶ EMC\_VNX7500\_01\_SPA2: EMC VNX7500 hostname VNX7500\_01, SP A, port 2.

If your SAN does not support aliases, for example, in heterogeneous fabrics with switches in some interoperations modes, use WWPNs in your zones. However, remember to update every zone that uses a WWPN if you ever change it.

Have your SAN zone name reflect the devices in the SAN it includes (normally in a one-to-one relationship) as shown in the following examples:

- ▶ SERVERALIAS\_T1\_FS9200CLUSTERNAME (from a server to the IBM FlashSystem 9200, where you use T1 as an identifier to zones that uses, for example, node ports P1 on Fabric A, and P2 on Fabric B).
- ▶ SERVERALIAS\_T2\_FS9200CLUSTERNAME (from a server to the IBM FlashSystem 9200, where you use T2 as an identifier to zones that uses, for example, node ports P3 on Fabric A, and P4 on Fabric B).
- ▶ IBM\_DS8870\_75XY131\_FS9200CLUSTERNAME (zone between an external back-end storage and the IBM FlashSystem 9200).
- ▶ NYC\_FS9200\_P0K\_FS9200\_REPLICATION (for Remote Copy services).

## 10.12.2 SAN fabric documentation

The most basic piece of SAN documentation is a SAN diagram. It is likely to be one of the first pieces of information you need if you ever seek support from your SAN switches vendor. Also, a good spreadsheet with ports and zoning information eases the task of searching for detailed information, which, if included in the diagram, makes the diagram easier to use.

### Brocade SAN Health

The Brocade SAN Health Diagnostics Capture tool is a no-cost, automated tool that can help you retain this documentation. SAN Health consists of a data collection tool that logs in to the SAN switches that you indicate and collects data by using standard SAN switch commands. The tool then creates a compressed file with the data collection. This file is sent to a Brocade automated machine for processing by secure web or email.

After some time (typically a few hours), you receive an email with instructions about how to download the report. The report includes a Visio diagram of your SAN and an organized Microsoft Excel spreadsheet that contains all your SAN information. For more information and to download the tool, see [this web page](#).

The first time that you use the SAN Health Diagnostics Capture tool, explore the options that are provided to learn how to create a well-organized and useful diagram.

Figure 10-15 on page 456 shows an example of a poorly formatted diagram.

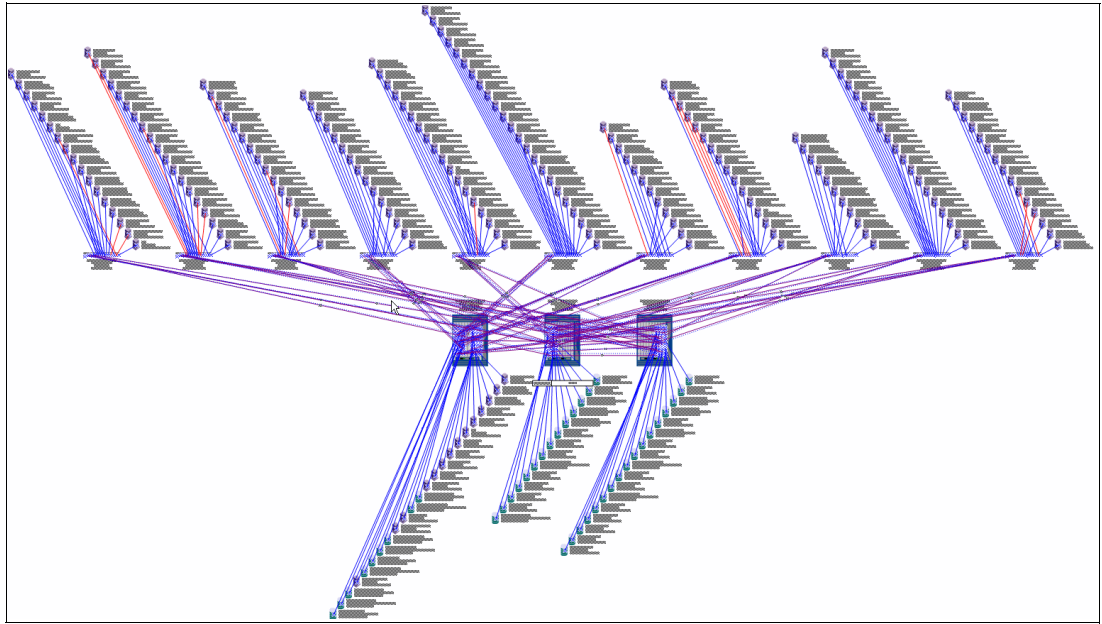


Figure 10-15 A poorly formatted SAN diagram

Figure 10-16 shows a tab of the SAN Health Options window in which you can choose the format of SAN diagram that best suits your needs. Depending on the topology and size of your SAN fabrics, you might want to manipulate the options in the Diagram Format or Report Format tabs.

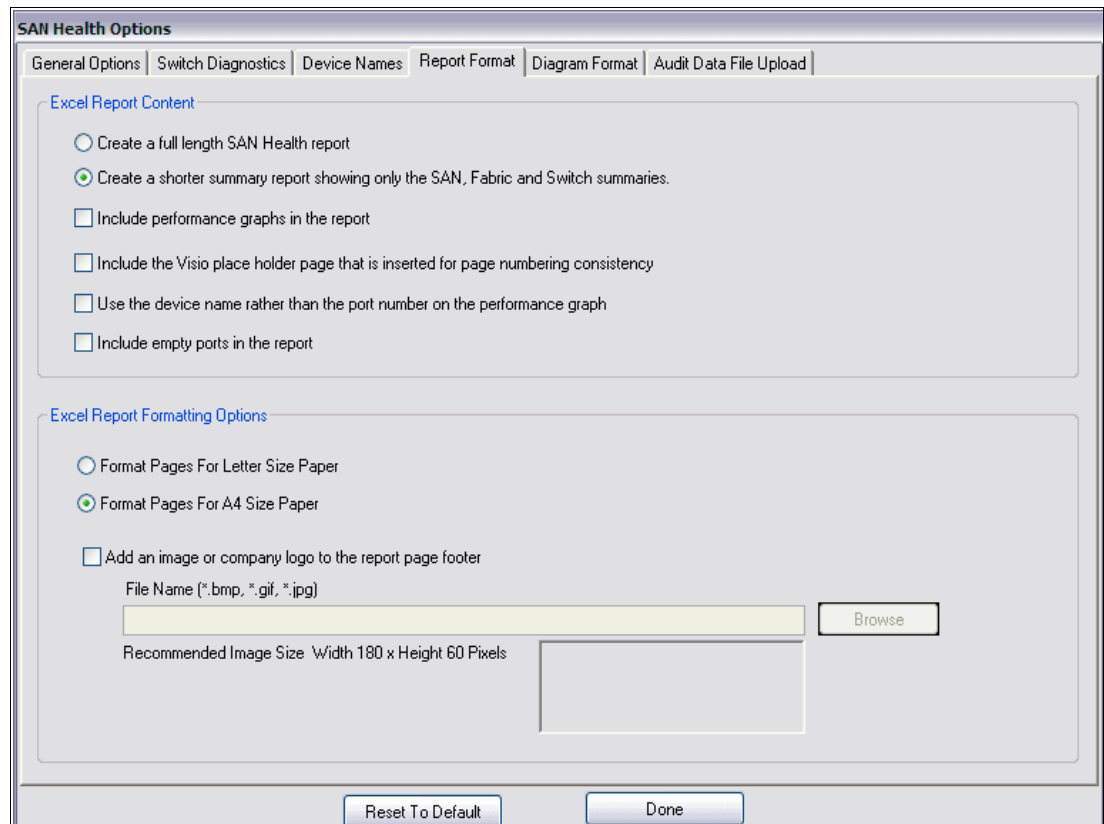


Figure 10-16 Brocade SAN Health Options window



SAN Health supports switches from manufacturers other than Brocade, such as Cisco. Both the data collection tool download and the processing of files are available at no cost. You can download Microsoft Visio and Excel viewers at no cost from the Microsoft website.

Another tool, which is known as SAN Health Professional, is also available for download at no cost. With this tool, you can audit the reports in detail by using advanced search functions and inventory tracking. You can configure the SAN Health Diagnostics Capture tool as a Windows scheduled task.

This tool is available for download at [this web page](#).

**Tip:** Regardless of the method that is used, generate a fresh report at least once a month or after any major changes are made. Keep previous versions so that you can track the evolution of your SAN.

### IBM Spectrum Control reporting

If you have IBM Spectrum Control running in your environment, you can use it to generate reports on your SAN. For more information about how to configure and schedule IBM Spectrum Control reports, see the [IBM Spectrum Control Documentation](#).

Also, see Chapter 9, “Monitoring” on page 353, for more information about how to configure and set up Spectrum Control.

Ensure that the reports that you generate include all of the information that you need. Schedule the reports with a period that you can use to backtrack any changes that you make.

## 10.12.3 IBM SAN Volume Controller documentation

You can back up the configuration data for an IBM SAN Volume Controller system after preliminary tasks are completed. Configuration data for the system provides information about your system and the objects that are defined in it. It also contains the configuration data of arrays, pools, volumes, and so on. The backup does not contain any data from the volumes themselves.

Before you back up your configuration data, the following prerequisites must be met:

- ▶ No independent operations that change the configuration for the system can be running while the **backup** command is running.
- ▶ No object name can begin with an underscore character (\_).

**Note:** The system automatically creates a backup of the configuration data each day at 1 AM. This backup is known as a *cron backup* and on the configuration node is copied to `/dumps/svc.config.cron.xml_<serial#>`.

Complete the following steps to generate a manual backup at anytime:

1. Run the **svcconfig backup** command to back up your configuration. The command displays messages similar to the messages that are shown in Example 10-11.

*Example 10-11 Sample svcconfig backup command output*

---

```
IBM_2145:IBM Redbook SVC:superuser>svcconfig backup
.....
...
.....
...
.....
CMMVC6155I SVCCONFIG processing completed successfully
```

---

The **svcconfig backup** command creates three files that provide information about the backup process and the configuration. These files are created in the /tmp directory and copied to/dumps directory of the configuration node. You can use the **lsdumps** command to list them. Table 10-3 lists the three files that are created by the backup process.

*Table 10-3 Files created by the backup process*

| File name                       | Description                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------|
| svc.config.backup.xml_<serial#> | Contains your configuration data.                                                       |
| svc.config.backup.sh_<serial#>  | Contains the names of the commands that were issued to create the backup of the system. |
| svc.config.backup.log_<serial#> | Contains details about the backup, including any reported errors or warnings.           |

2. Check that the **svcconfig backup** command completes successfully and examine the command output for any warnings or errors. The following output is an example of the message that is displayed when the backup process is successful:  
CMMVC6155I SVCCONFIG processing completed successfully
3. If the process fails, resolve the errors and run the command again.
4. Keep backup copies of the files outside the system to protect them against a system hardware failure. With Microsoft Windows, use the PuTTY **pscp** utility. With UNIX or Linux, you can use the standard **scp** utility. By using the **-unsafe** option, you can use a wild card to download all the svc.config.backup files by using a single command. Example 10-12 shows the output of the **pscp** command.

*Example 10-12 Saving the config backup files to your workstation*

---

```
C:\>
pscp -unsafe superuser@9.10.11.12:/dumps/svc.config.backup.* C:\
Using keyboard-interactive authentication.
Password:
svc.config.backup.log_78E | 33 kB | 33.6 kB/s | ETA: 00:00:00 | 100%
svc.config.backup.sh_78E0 | 13 kB | 13.9 kB/s | ETA: 00:00:00 | 100%
svc.config.backup.xml_78E | 312 kB | 62.5 kB/s | ETA: 00:00:00 | 100%
C:\>
```

---

The configuration backup file is in XML format and can be inserted as an object into your IBM SAN Volume Controller documentation spreadsheet. The configuration backup file might be quite large; for example, it contains information about each internal storage drive that is installed in the system.

**Note:** Directly importing the file into your IBM SAN Volume Controller documentation spreadsheet might make it unreadable.

Also consider collecting the output of specific commands. At a minimum, collect the output of the following commands:

- ▶ `svcinfo lsfabric`
- ▶ `svcinfo lssystem`
- ▶ `svcinfo lsmdisk`
- ▶ `svcinfo lsmdiskgrp`
- ▶ `svcinfo lsvdisk`
- ▶ `svcinfo lshost`
- ▶ `svcinfo lshostvdiskmap`

**Note:** Most of these CLI commands work without the `svcinfo` prefix; however, some commands do not work with only the short-name, and require the `svcinfo` prefix to be added.

Import the commands into the master spreadsheet, preferably with the output from each command on a separate sheet.

One way to automate either task is to first create a batch file (Windows), shell script (UNIX or Linux), or playbook (Ansible) that collects and stores this information. Then, use spreadsheet macros to import the collected data into your IBM SAN Volume Controller documentation spreadsheet.

When you are gathering IBM SAN Volume Controller information, consider the following preferred practices:

- ▶ If you are collecting the output of specific commands, use the `-delim` option of these commands to make their output delimited by a character other than tab, such as comma, colon, or exclamation mark. You can import the temporary files into your spreadsheet in comma-separated values (CSV) format, specifying the same delimiter.

**Note:** It is important to use a delimiter that is not part of the output of the command. Commas can be used if the output is a specific type of list. Colons might be used for special fields, such as IPv6 addresses, WWPNs, or iSCSI names.

- ▶ If you are collecting the output of specific commands, save the output to temporary files. To make your spreadsheet macros simpler, you might want to preprocess the temporary files and remove any “garbage” or unwanted lines or columns. With UNIX or Linux, you can use commands, such as **grep**, **sed**, and **awk**. Freeware software is available for Windows with the same commands, or you can use any batch text editor tool.

The objective is to fully automate this procedure so you can schedule it to run automatically on a regular basis. Make the resulting spreadsheet easy to consult and have it contain only the information that you use frequently. The automated collection and storage of configuration and support data (which is typically more extensive and difficult to use) is described in 10.12.7, “Automated support data collection” on page 462.

## 10.12.4 Storage documentation

You must generate documentation of your back-end storage controllers after configuration. Then, you can update the documentation when these controllers receive hardware or code updates. As such, there is little point to automating this back-end storage controller documentation. The same applies to the IBM SAN Volume Controller internal drives and enclosures.

Any portion of your external storage controllers that is used outside the IBM SAN Volume Controller solution might have its configuration changed frequently. In this case, see your back-end storage controller documentation for more information about how to gather and store the information that you need.

Fully allocate all of the available space in any of the optional external storage controllers that you might use as additional back-end to the IBM SAN Volume Controller solution. This way, you can perform all your disk storage management tasks by using the IBM SAN Volume Controller user interface.

## 10.12.5 Technical support information

If you must open a technical support incident for your storage and SAN components, create and keep available a spreadsheet with all relevant information for all storage administrators. This spreadsheet includes the following information:

- ▶ Hardware information:
  - Vendor, machine and model number, serial number (example: IBM 2145-SV2 S/N 7812345)
  - Configuration, if applicable
  - Current code level
- ▶ Physical location:
  - Data center, including the complete street address and phone number
  - Equipment physical location (room number, floor, tile location, and rack number)
  - Vendor's security access information or procedure, if applicable
  - Onsite person's contact name and phone or page number
- ▶ Support contract information:
  - Vendor contact phone numbers and website
  - Customer's contact name and phone or page number
  - User ID to the support website, if applicable
  - Do not store the password in the spreadsheet under any circumstances
  - Support contract number and expiration date

By keeping this data on a spreadsheet, storage administrators have all the information that they need to complete a web support request form or to provide to a vendor's call support representative. Typically, you are asked first for a brief description of the problem and then asked later for a detailed description and support data collection.

## 10.12.6 Tracking incident and change tickets

If your organization uses an incident and change management and tracking tool (such as IBM Tivoli® Service Request Manager®), you or the storage administration team might need to develop proficiency in its use for the following reasons:

- ▶ If your storage and SAN equipment are not configured to send SNMP traps to this incident management tool, manually open incidents whenever an error is detected.
- ▶ The IBM SAN Volume Controller can be managed by the IBM Storage Insights (SI) tool, which is available free of charge to owners of IBM storage systems. The SI tool allows you to monitor all the IBM storage devices' information that is on SI. For more information, see Chapter 9, "Monitoring" on page 353.
- ▶ Disk storage allocation and deallocation and SAN zoning configuration modifications should be handled under properly submitted and approved change requests.
- ▶ If you are handling a problem yourself, or calling your vendor's technical support, you might need to produce a list of the changes that you recently implemented in your SAN or that occurred since the documentation reports were last produced or updated.

When you use incident and change management tracking tools, adhere to the following guidelines for IBM SAN Volume Controller and SAN Storage Administration:

- ▶ Whenever possible, configure your storage and SAN equipment to send SNMP traps to the incident monitoring tool so that an incident ticket is automatically opened, and the suitable alert notifications are sent. If you do not use a monitoring tool in your environment, you might want to configure email alerts that are automatically sent to the mobile phones or pagers of the storage administrators on duty or on call.
- ▶ Discuss within your organization the risk classification that a storage allocation or deallocation change request is to have. These activities are typically safe and nondisruptive to other services and applications when properly handled.

However, they have the potential to cause collateral damage if a human error or an unexpected failure occurs during implementation. Your organization might decide to assume more costs with overtime and limit such activities to off-business hours, weekends, or maintenance windows if they assess that the risks to other critical applications are too high.

- ▶ Use templates for your most common change requests, such as storage allocation or SAN zoning modification, to facilitate and speed up their submission.
- ▶ Do not open change requests in advance to replace failed, redundant, hot-pluggable parts, such as disk drive modules (DDMs) in storage controllers with hot spares, or SFPs in SAN switches or servers with path redundancy. Typically, these fixes do not change anything in your SAN storage topology or configuration or t cause any more service disruption or degradation than you had when the part failed. Handle these fixes within the associated incident ticket because it might take longer to replace the part if you need to submit, schedule, and approve a non-emergency change request.

An exception is if you must interrupt more servers or applications to replace the part. In this case, you must schedule the activity and coordinate support groups. Use good judgment and avoid unnecessary exposure and delays.

- ▶ Keep handy the procedures to generate reports of the latest incidents and implemented changes in your SAN Storage environment. Typically, you do not need to periodically generate these reports because your organization probably already has a Problem and Change Management group that runs such reports for trend analysis purposes.

## 10.12.7 Automated support data collection

In addition to the easier-to-use documentation of your IBM SAN Volume Controller and SAN Storage environment, collect and store for some time the configuration files and technical support data collection for all your SAN equipment.

For IBM SAN Volume Controller, this information includes **snap** data. For other equipment, see the related documentation for more information about how to gather and store the support data that you might need.

You can create procedures that automatically create and store this data on scheduled dates, delete old data, or transfer the data to tape.

IBM Storage Insights also can be used to create support tickets and then attach the snap data to this record from within the SI GUI. For more information, see Chapter 11, “Troubleshooting and diagnostics” on page 463.

## 10.12.8 Subscribing to IBM SAN Volume Controller support

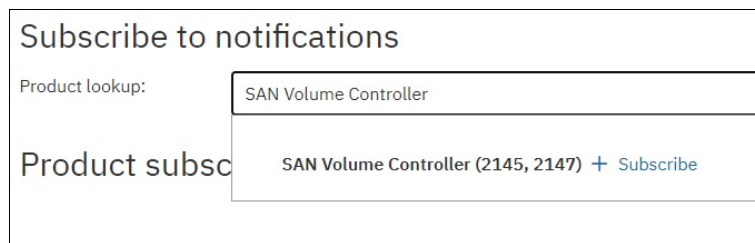
Subscribing to IBM SAN Volume Controller support is like the most overlooked practice in IT administration, and yet it is the most efficient way to stay ahead of problems. With this subscription, you can receive notifications about potential threats before they can reach you and cause severe service outages.

For more information about subscribing to this support and receiving support alerts and notifications for your products, see [this web page](#).

If you do not have an IBM ID, create an ID.

You can subscribe to receive information from each vendor of storage and SAN equipment from the IBM website. You can often quickly determine whether an alert or notification is applicable to your SAN storage. Therefore, open any alert or notification when you receive them and keep them in a folder of your mailbox.

Sign up and tailor the requests and alerts you want to receive. For example, enter IBM SAN Volume Controller in the Product lookup text box and then, click **Subscribe** to subscribe to SAN Volume Controller notifications, as shown in Figure 10-17.



The screenshot shows a web form titled "Subscribe to notifications". It contains a "Product lookup:" label and a text input field with "SAN Volume Controller" entered. Below the input field is a dropdown menu that is open, showing the selected item "SAN Volume Controller (2145, 2147)" followed by a blue plus icon and the word "Subscribe".

Figure 10-17 Creating a subscription to IBM SAN Volume Controller notifications



# Troubleshooting and diagnostics

This chapter provides information about troubleshooting common problems that can occur in IBM Spectrum Virtualize environment. It describes situations that are related to IBM SAN Volume Controller, the SAN environment, optional external storage subsystems, and hosts. It also explains how to collect the necessary problem determination data.

This chapter includes the following topics:

- ▶ 11.1, “Starting troubleshooting” on page 464
- ▶ 11.2, “Diagnostic data collection” on page 470
- ▶ 11.3, “Common problems and isolation techniques” on page 474
- ▶ 11.4, “Remote Support Assistance” on page 491
- ▶ 11.5, “Call Home Connect Cloud and Health Checker feature” on page 492
- ▶ 11.6, “IBM Storage Insights” on page 494

## 11.1 Starting troubleshooting

Troubleshooting is a systematic approach to solving a problem. The goal of troubleshooting or problem determination is to understand why something does not work as expected and find a resolution. Therefore, the first step is to describe the problem as accurately as possible, then perform log collection from all the involved products of the solution as soon as the problem is reported. Ideally, an effective problem report describes the expected behavior, the actual behavior, and, if possible, how to reproduce the behavior.

The following questions help define the problem for effective troubleshooting:

- ▶ What are the symptoms of the problem?
  - What is reporting the problem?
  - What are the error codes and messages?
  - What is the business impact of the problem?
  - Where does the problem occur?
  - Is the problem specific to one or multiple hosts, one or both nodes?
  - Is the current environment and configuration supported?
- ▶ When does the problem occur?
  - Does the problem occur only at a specific time of day or night?
  - How often does the problem occur?
  - What sequence of events leads up to the time that the problem is reported?
  - Does the problem occur after an environment change, such as upgrading or installing software or hardware?
- ▶ Under which conditions does the problem occur?
  - Does the problem always occur when the same task is being performed?
  - Does a certain sequence of events need to occur for the problem to surface?
  - Do any other applications fail at the same time?
- ▶ Can the problem be reproduced?
  - Can the problem be re-created on a test system?
  - Are multiple users or applications encountering the same type of problem?
  - Can the problem be re-created by running a single command, a set of commands, a specific application, or a stand-alone application?

Log file collection that is done close to the time of the incident and an accurate timeline are critical for effective troubleshooting.



### 11.1.1 Using the GUI

The graphical user interface (GUI) is a good starting point for your troubleshooting. It features two icons at the top that can be accessed from any window of the GUI.

As shown in Figure 11-1, the first icon shows IBM Spectrum Virtualize events, such as an error or a warning. The second icon shows suggested tasks and background tasks that are running, or that were recently completed.

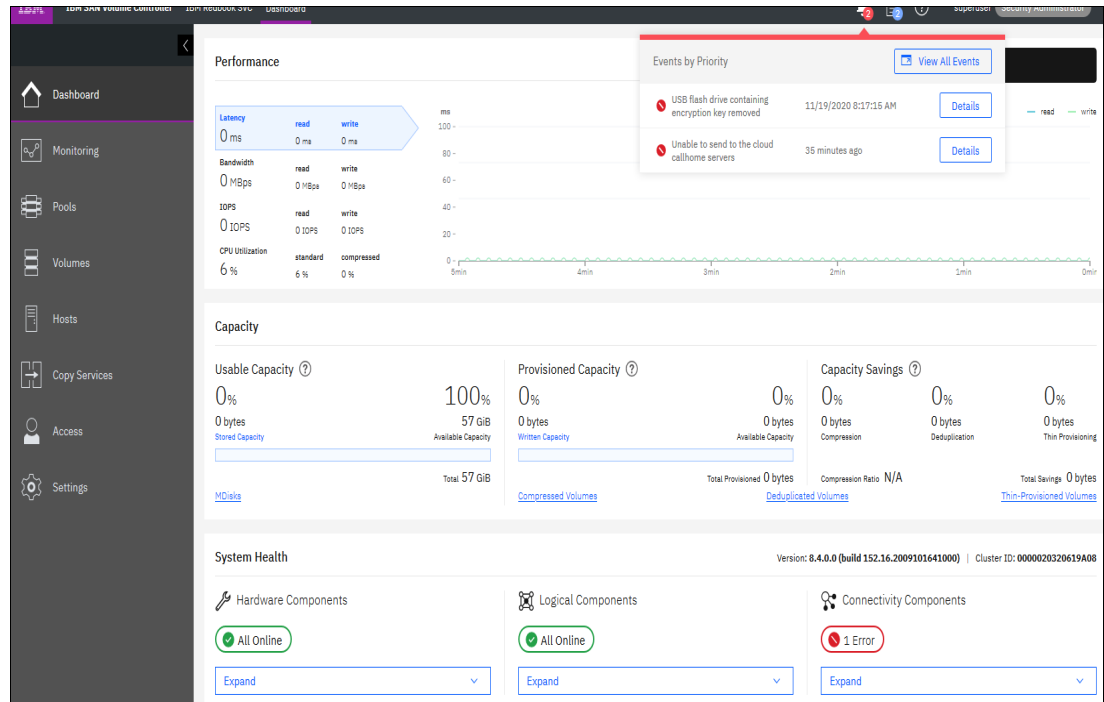


Figure 11-1 Events and tasks icons in GUI

The Dashboard window (see Figure 11-2) provides an at-a-glance look into the condition of the system and notification of any critical issues that require immediate action. It contains sections for performance, capacity, and system health that provide an overall understanding of what is happening on the system.

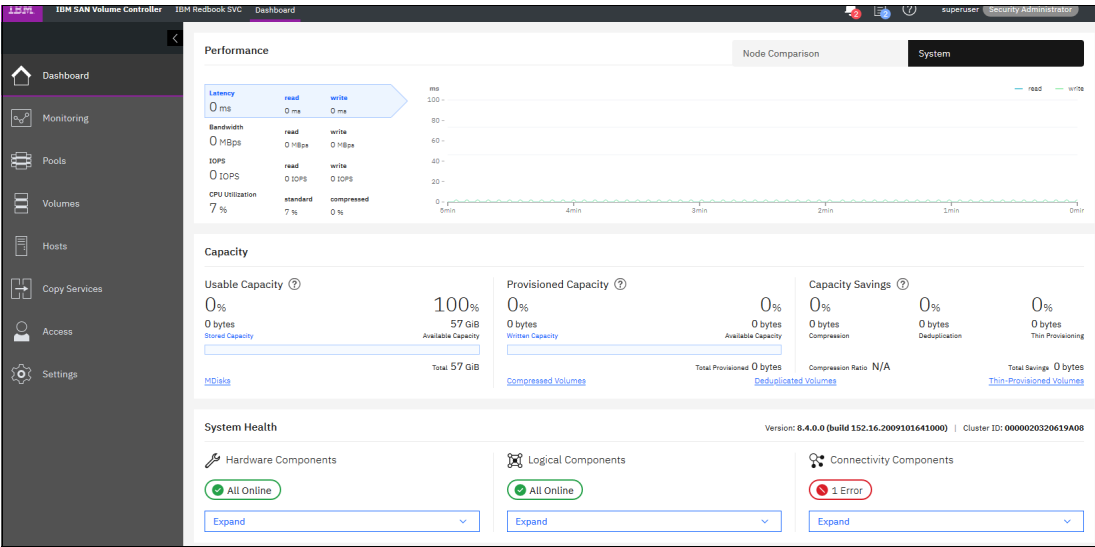


Figure 11-2 Dashboard

The System Health section in the bottom part of the Dashboard provides information about the health status of hardware, and logical and connectivity components. If you click **Expand** in each of these categories, the status of individual components is shown, as shown in the example in Figure 11-3. You can also click **More Details**, which takes you to the window that is related to that specific component, or is shows you more information about it.

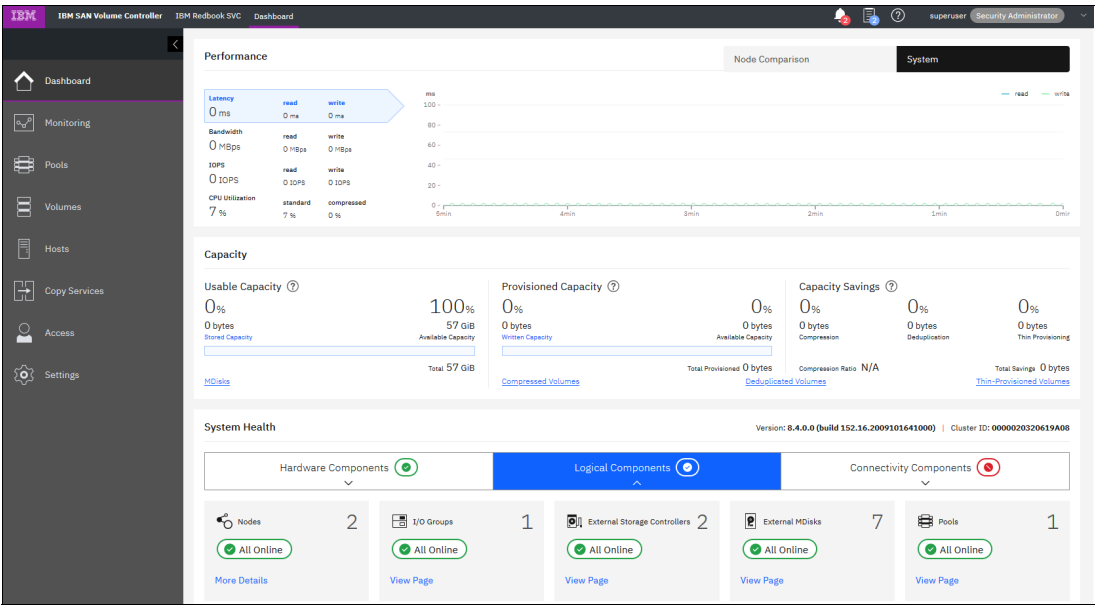


Figure 11-3 System Health section in Dashboard

The entire list of components in each category can be found in [IBM Documentation](#).

## 11.1.2 Recommended actions and fix procedure

The fix procedures were carefully designed to assist users to fix the problem without causing more issues. When multiple unfixed error codes are in the Event log, the management GUI provides a way to run the next recommended fix procedure. Therefore, the first step in troubleshooting is to run the fix procedures on the error codes in the Event log.

These messages and codes provide reference information about informational configuration events and error event codes when a service action is required. Cluster Error Code (CEC) is visible in the cluster event log; Node Error Code (NEC) is visible in node status in the service assistant GUI. A cluster can encounter the following types of failure recoveries because of various conditions:

- ▶ Node assert (warmstart or Tier1/T1 recovery) is reported as CEC 2030
- ▶ Cluster recovery(Tier2/T2 recovery) is reported as CEC 1001
- ▶ System recovery(Tier3/T3 recovery) is required when all nodes of the clustered system report NEC 550/578

Review the messages and codes section in [IBM Documentation](#) for full listing of the codes.

The **Monitoring** → **Events** window shows information messages, warnings, and issues on the IBM Spectrum Virtualize. So, this is a good place to check the current problems in the system.

Using the **Recommended Actions** filter, the most important events that need to be fixed are displayed.

If there is an important issue that needs to be fixed, the **Run Fix** button is available in the upper-left corner with an error message, indicating which event must be fixed as soon as possible. This fix procedure assists you to resolve problems in IBM Spectrum Virtualize. It analyzes the system, provides more information about the problem, suggest actions to be taken with steps to be followed, and finally checks to see whether the problem is resolved.

So, if any error is reported by the system, such as system configuration problems and hardware failures, always use the fix procedures to resolve it.

Figure 11-4 shows **Monitoring** → **Events** window with the Run Fix button.

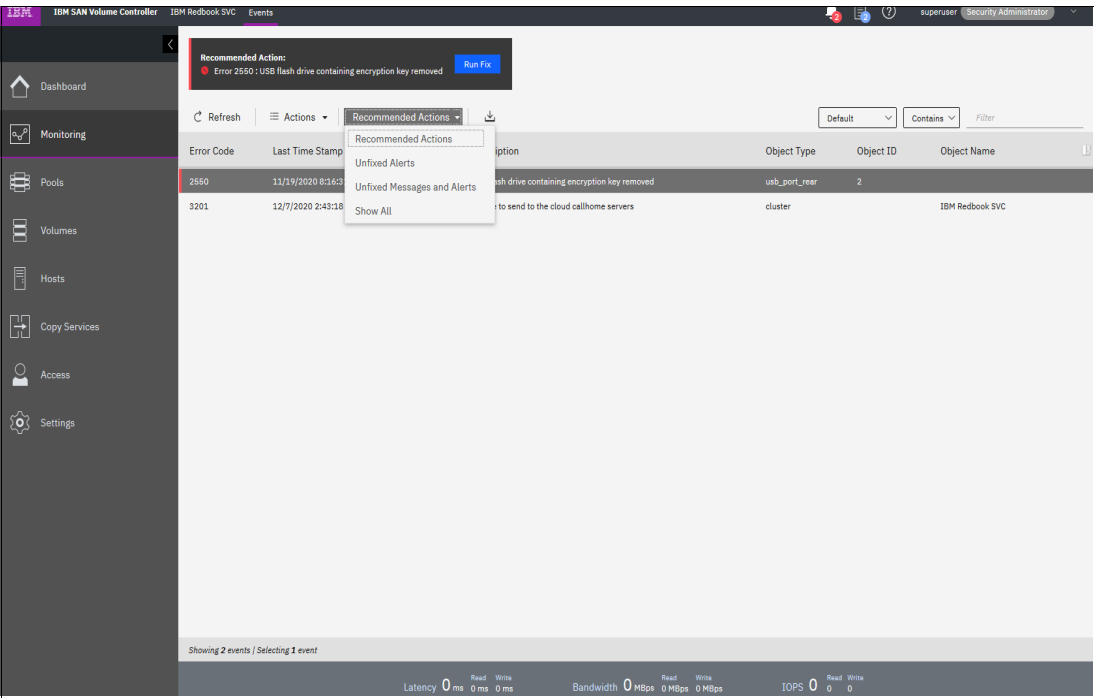


Figure 11-4 Monitoring > Events window

**Resolve alerts in a timely manner:** When an issue or a potential issue is reported, resolve it as quickly as possible to minimize its effect and potentially avoid more serious problems with your system.

For more information about any event, select an event in the table, and click **Properties** in the **Actions** menu. You can also access the Run Fix Procedure and properties by right-clicking an event.

In the Properties and Sense Data window for the specific event (see Figure 11-5 on page 469), more information about the event is displayed. You can review and also click **Run Fix** to run the fix procedure.

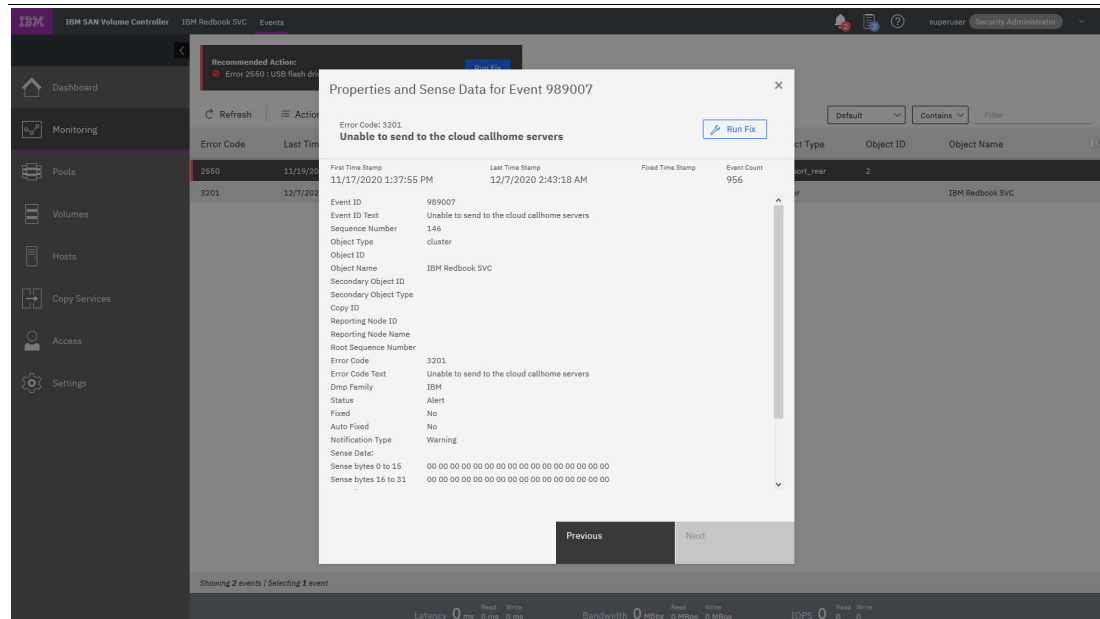


Figure 11-5 Properties and sense data for event window

**Tip:** From the Properties and Sense Data for Event Window, you can use the **Previous** and **Next** buttons to move between events.

Another common practice is to use the IBM Spectrum Virtualize CLI to find issues and resolve them. You can use the IBM Spectrum Virtualize CLI to perform common error recovery steps. Although the maintenance procedures perform these steps, it is sometimes faster to run these commands directly through the CLI.

Run these commands whenever you encounter the following issues:

- ▶ You experience a back-end storage issue (for example, error code 1370 or error code 1630).
- ▶ You performed maintenance on the back-end storage subsystems.

**Important:** Run the following commands when back-end storage is configured, a zoning change occurs, or any other type of changes that are related to the communication between IBM Spectrum Virtualize and back-end storage subsystem occurs. This process ensures that IBM Spectrum Virtualize recognized the changes.

Common error recovery involves the following IBM Spectrum Virtualize CLI commands:

- ▶ `lscontroller` and `lsmdisk`  
Provides current status of all controllers and MDisk.
- ▶ `detectmdisk`  
Discovers the changes in the back-end.
- ▶ `lscontroller <controller_id_or_name>`  
Checks the controller that was causing the issue and verifies that all the WWPNs are listed as you expect. It also checks that the `path_counts` are distributed evenly across the WWPNs.

► `lsmdisk`

Determines whether all MDisks are online.

**Note:** When an issue is resolved by using the CLI, check if the error was removed from the **Monitoring** → **Events** window. If not, ensure that the error was fixed, and if so, manually mark the error as fixed.

## 11.2 Diagnostic data collection

Data collection and problem isolation in an IT environment are sometimes difficult tasks. In the following section, the essential steps that are needed to collect debug data to find and isolate problems in an IBM Spectrum Virtualize environment are described.

### 11.2.1 Collecting data from IBM Spectrum Virtualize

When a problem exists with an IBM SAN Volume Controller and you must open a case with IBM support, you must provide the support packages for the device. To automatically collect and upload the support packages to IBM Support center, use IBM Spectrum Virtualize.

You also can download the package from the device and manually upload to IBM. The easiest way is to automatically upload the support packages from IBM Spectrum Virtualize by using the GUI or CLI.

#### Data collection by using the GUI

To collect data by using the GUI, complete the following steps:

1. Click **Settings** → **Support** → **Support Package**. Both options to collect and upload support packages are available.
2. To automatically upload them, click **Upload Support Package**.
3. In the pop-up window, enter the PMR number and the type of support package to upload to the IBM support center. The Snap Type 4 option can be used to collect standard logs and generate a new statesave on each node of the system.

The Upload Support Package window is shown in Figure 11-6.

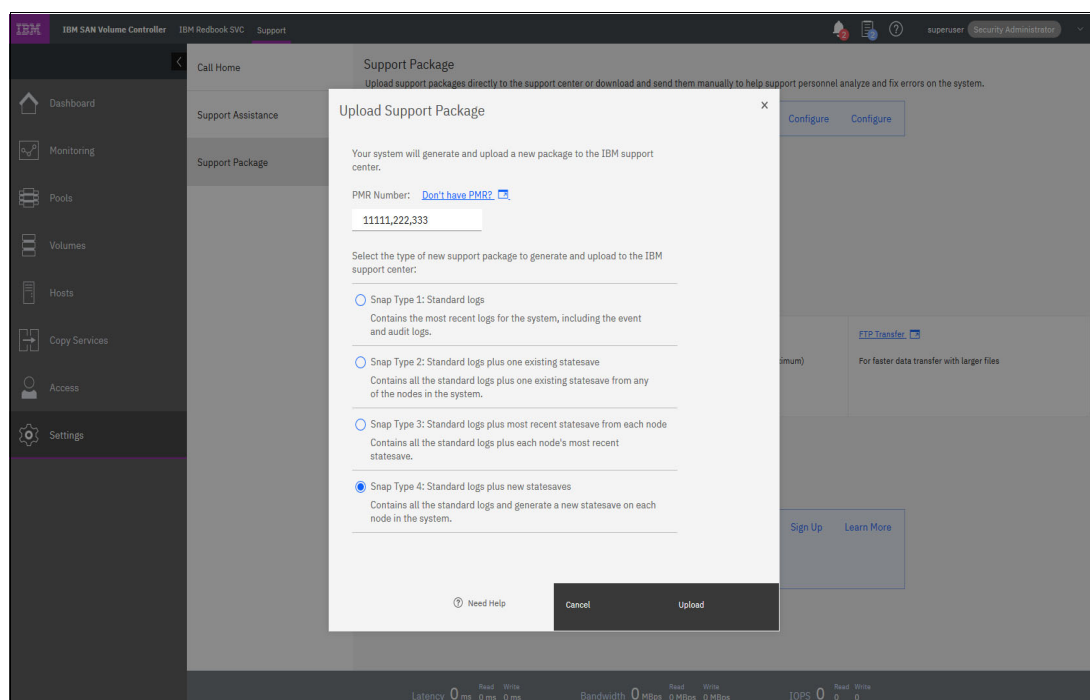


Figure 11-6 Upload Support Package window

For more information about the required support package that is most suitable to diagnose different type of issues, see this [IBM Support web page](#).

Consider the following points:

- ▶ For issues that are related to interoperability with hosts or storage, use Snap Type 4.
- ▶ For critical performance issues, collect Snap Type 1 and then, collect Snap Type 4.
- ▶ For general performance issues, collect Snap Type 4.
- ▶ For issues related to replication, including 1920 errors, collect Snap Type 4 from both systems.
- ▶ For issues related to compressed volumes, collect Snap Type 4.
- ▶ For 2030, 1196 or 1195 errors collect Snap Type 4.
- ▶ For all other issues, collect Snap Type 4.

## Data collection by using the CLI

To collect the same type of support packages by using the CLI, you must first generate a new livedump of the system by using the **svc\_livedump** command, and then upload the log files and new generated dumps by using the **svc\_snap** command, as shown in Example 11-1. To verify whether the support package was successfully uploaded, use the **sainfo lscmdstatus** command (XXXXX,YYY,ZZZ is the pmr number).

### Example 11-1 The **svc\_livedump** command

```
IBM_2145:ITS0_DH8_B:superuser>svc_livedump -nodes all -yes
Livedump - Fetching Node Configuration
Livedump - Checking for dependent vdisks
Livedump - Check Node status
```

```

Livedump - Prepare specified nodes - this may take some time...
Livedump - Prepare node 1
Livedump - Prepare node 2
Livedump - Trigger specified nodes
Livedump - Triggering livedump on node 1
Livedump - Triggering livedump on node 2
Livedump - Waiting for livedumps to complete dumping on nodes 1,2
Livedump - Waiting for livedumps to complete dumping on nodes 2
Livedump - Successfully captured livedumps on nodes 1,2
IBM_2145:ITS0_DH8_B:superuser>svc_snap upload pmr=XXXXX,YYY,ZZZ gui3
Collecting data
Packaging files
Snap data collected in /dumps/snap.ABCDEFG.171128.223133.tgz
IBM_2145:ITS0_DH8_B:superuser>sainfo lscmdstatus
last_command satask supportupload -pmr ppppp,bbb,ccc -filename
/dumps/snap.ABCDEFG.171128.223133.tgz
last_command_status CMMVC8044E Command completed successfully.
T3_status
T3_status_data
cpfiles_status Complete
cpfiles_status_data Copied 1 of 1
snap_status Complete
snap_filename /dumps/snap.ABCDEFG.171128.223133.tgz
installcanistersoftware_status
supportupload_status Complete
supportupload_status_data [PMR=ppppp,bbb,ccc] Upload complete
supportupload_progress_percent 0
supportupload_throughput_KBps 0
supportupload_filename /dumps/snap.ABCDEFG.171128.223133.tgz
downloadsoftware_status
downloadsoftware_status_data
downloadsoftware_progress_percent 0
downloadsoftware_throughput_KBps 0
downloadsoftware_size
IBM_2145:ITS0_DH8_B:superuser>

```

---

## 11.2.2 Host multipath software data collection

If a problem exists that is related to host communication with IBM SAN Volume Controller, collecting data from hosts and multipath software is useful.

For more information, see the operating system-specific multipathing documentation.

The Subsystem Device Driver Device Specific Module (SDDDSM) and the Subsystem Device Driver Path Control Module (SDDPCM) reached End of Service (EOS). Therefore, migrate SDDDSM to MSDSM on the Windows platform and SDDPCM to AIXPCM on the AIX/VIOS platforms, respectively. For more information, see the following IBM Support web pages:

- ▶ [Migrating from SDDDSM to Microsoft's MSDSM - SVC/Storwize](#)
- ▶ [How To Migrate SDDPCM to AIXPCM](#)



If the systems still use SDD multipathing modules, SDDPCM for AIX provides the **sddpcmgetdata** script to collect information that is used for problem determination. This script creates a .tar file in the current directory with the current date and time as a part of the file name. When you suspect an issue exists with SDDPCM, it is essential to run this script and send this .tar file to IBM Support.

SDDDSM for Windows hosts also includes a utility to collect information for problem determination. The **sddgetdata.bat** tool creates a CAB file in the installation directory with the current date and time as part of the file name. The CAB file includes the following information:

- ▶ SystemInfo
- ▶ HKLM \SYSTEM\CurrentControlSet, HKLM\HARDWARE\DEVICEMAP, and HKLM\Cluster output from the registry
- ▶ SDDDSM directory contents
- ▶ HBA details
- ▶ Datapath outputs
- ▶ Pathtest trace
- ▶ SDDSRV logs
- ▶ Cluster logs
- ▶ System disks and paths

The execution and output of the **sddgetdata.bat** tool is shown in Example 11-2.

*Example 11-2 The sddgetdata.bat tool*

---

```
C:\Program Files\IBM\SDDDSM>sddgetdata.bat
Collecting SDD trace Data

Flushing SDD kernel logs

SDD logs flushed

Collecting datapath command outputs

Collecting System Information

Collecting SDD and SDDsrv logs

Collecting Most current driver trace

Please wait for 30 secs... Writing DETAILED driver trace to trace.out

Generating a CAB file for all the Logs

sdddata_WIN-IWG6VLJN3U3_20171129_151423.cab file generated

C:\Program Files\IBM\SDDDSM>
```

---

For more information about diagnostics for IBM SDD, see this [IBM Support web page](#).

### 11.2.3 Additional data collection

Data collection methods vary by storage platform, SAN switch, and operating system.

When an issue exists in a SAN environment and it is not clear where the problem is occurring, you might need to collect data from several devices in the SAN.

The following basic information must be collected for each type of device:

- ▶ Hosts:
  - Operating system: Version and level
  - HBA: Driver and firmware level
  - Multipathing driver level
- ▶ SAN switches:
  - Hardware model
  - Software version
- ▶ Storage subsystems:
  - Hardware model
  - Software version

## 11.3 Common problems and isolation techniques

SANs, storage subsystems, and host systems can be complicated. They often consist of hundreds or thousands of disks, multiple redundant subsystem controllers, virtualization engines, and different types of SAN switches. All of these components must be configured, monitored, and managed correctly. If issues occur, administrators must know what to look for and where to look.

IBM Spectrum Virtualize features useful error logging mechanisms. It keeps track of its internal events and informs the user about issues in the SAN or storage subsystem. It also helps to isolate problems with the attached host systems. By using these functions, administrators can easily locate any issue areas and take the necessary steps to fix any events.

In many cases, IBM Spectrum Virtualize and its service and maintenance features guide administrators directly, provide help, and suggest remedial action. Furthermore, IBM Spectrum Virtualize determines whether the problem still persists.

Another feature that helps administrators to isolate and identify issues that might be related to IBM Spectrum Virtualize is the ability of their nodes to maintain a database of other devices that communicate with the IBM Spectrum Virtualize device. Devices, such as hosts and back-end storages, are added or removed from the database as they start or stop communicating to IBM Spectrum Virtualize.

Although IBM Spectrum Virtualize node hardware and software events can be verified in the GUI or CLI, external events, such as failures in the SAN zoning configuration, hosts, and back-end storages, are common. They also must have troubleshooting performed out of IBM Spectrum Virtualize. For example, a misconfiguration in the SAN zoning might lead to the IBM Spectrum Virtualize cluster not working properly.

This problem occurs because the IBM Spectrum Virtualize cluster nodes communicate with each other by using the Fibre Channel SAN fabrics.

- ▶ The attached hosts. For more information, see 11.3.1, “Host problems” on page 475.
- ▶ The SAN. For more information, see 11.3.2, “SAN problems” on page 479.
- ▶ The attached storage subsystem. For more information, see 11.3.3, “Storage subsystem problems” on page 480.
- ▶ The local FC port masking. For more information, see 8.1.3, “Port masking” on page 340.

From the host perspective, you can experience various situations that range from performance degradation to inaccessible disks. To diagnose any host-related issue, you can start checking the hosts configuration on IBM Spectrum Virtualize side. The Hosts window in the GUI or the following CLI commands are used to start a verification in any possible hosts-related issue:

- Checks the host's status. If the status is online, the host ports are online in both nodes of an I/O group. If the status is offline, the host ports are offline in both nodes of an I/O group. If the status is inactive, it means that the host features volumes that are mapped to it, but all of its ports received no SCSI commands in the last 5 minutes. Also, if status is degraded, it means at least one (but not all) of the host ports is not online in at least one node of an I/O group. Example 11-3 shows the **lshost** command output.

```
IBM_2145:ITS0_DH8_B:superuser>lshost
id name      port_count iogrp_count status  site_id site_name host_cluster_id
host_cluster_name
0 Win2K8      2          4        degraded
1 ESX_62_B 2          4        online
2 ESX_62_A 2          1        offline
```

- Shows more information about a specific host. It is often used when you must identify which host port is not online in IBM Spectrum Virtualize node.

*Example 11-4 The lshost <host\_id\_or\_name> command*

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```
WWPN 100000051E0F81CC
node_logged_in_count 0
state offline
```

---

► **lshostvdiskmap**

Check that all volumes are mapped to the correct hosts. If a volume is not mapped correctly, create the necessary host mapping.

► **lsfabric -host <host\_id\_or\_name>**

Use this command with parameter **-host <host\_id\_or\_name>** to display Fibre Channel (FC) connectivity between nodes and hosts. Example 11-5 shows the **lsfabric -host <host\_id\_or\_name>** command output.

*Example 11-5 The lsfabric -host <host\_id\_or\_name> command*

---

```
IBM_2145:ITS0_DH8_B:superuser>lsfabric -host Win2K8
remote_wwpn      remote_nportid id node_name local_wwpn      local_port
local_nportid state  name  cluster_name type
100000051E0F81CD 021800      1 node1    500507680C220416 2          020400
active Win2K8                                host
100000051E0F81CD 021800      2 node2    500507680C22041D 2          020000
active Win2K8                                host
```

---

To perform troubleshooting on the host side, check the following components:

- Any special software that you use.
- Any recent change in the operating system, such as patching the operating system and an upgrade.
- Operating system version and maintenance or service pack level
- Multipathing type and driver level
- Host bus adapter model, firmware, and driver level
- Host bus adapter connectivity issues

Based on this list, the host administrator must check and correct any problems.

Hosts with higher queue depth might overload shared storage ports. Therefore, it is recommended to verify that the sum total of the queue depth of all hosts sharing a single target Fibre Channel port is limited to 2048. If any of the hosts have a queue depth of more than 128, it must be reviewed because queue full conditions can lead to I/O errors and extended error recoveries.

For more information about managing hosts on IBM Spectrum Virtualize, see Chapter 8, “Hosts” on page 339.

Apart from hardware-related situations, problems can exist in such areas as the operating system or the software that is used on the host. These problems normally are handled by the host administrator or the service provider of the host system. However, the multipathing driver that is installed on the host and its features can help to determine possible issues.

For example, for a volume path issue reported by SDD output on the host by using the **datapath query adapter** and **datapath query device** commands. The adapter in degraded state means that the specific HBA on the server side cannot reach all of the nodes in the I/O group to which the volumes are associated.

**Note:** Subsystem Device Driver Device Specific Module (SDDDSM) and Subsystem Device Driver Path Control Module (SDDPCM) reached End of Service (EOS). Therefore, migrate SDDDSM to MSDSM on Windows platform and SDDPCM to AIXPCM on AIX/VIOS platforms.

For more information, see the following IBM Support web pages:

- ▶ [Migrating from SDDDSM to Microsoft's MSDSM - SVC/Storwize](#)
- ▶ [How To Migrate SDDPCM to AIXPCM](#)

Faulty paths can be caused by hardware and software problems, such as the following examples:

- ▶ Hardware:
  - Faulty Small Form-factor Pluggable transceiver (SFP) on the host or SAN switch
  - Faulty fiber optic cables
  - Faulty HBAs
- ▶ Software:
  - A back-level multipathing driver
  - Obsolete HBA firmware or driver
  - Wrong zoning
  - Incorrect host-to-VDisk mapping

Based on field experience, it is recommended that you complete the following hardware checks first:

- ▶ Whether any connection error indicators are lit on the host or SAN switch.
- ▶ Whether all of the parts are seated correctly. For example, cables are securely plugged in to the SFPs and the SFPs are plugged all the way into the switch port sockets.
- ▶ Ensure that no fiber optic cables are broken. If possible, swap the cables with cables that are known to work.

After the hardware check, continue to check the following aspects of software setup:

- ▶ Check that the HBA driver level and firmware level are at the preferred and supported levels.
- ▶ Check the multipathing driver level, and make sure that it is at the preferred and supported level.
- ▶ Check for link layer errors that are reported by the host or the SAN switch, which can indicate a cabling or SFP failure.
- ▶ Verify your SAN zoning configuration.
- ▶ Check the general SAN switch status and health for all switches in the fabric.

## iSCSI and iSER configuration and performance issues

In this section, we discuss the iSCSI and iSER configuration and performance issues.

### ***Links issues***

If the Ethernet port link does not come online, check whether the SFP or cables and check whether the port supports auto-negotiation with the switch. This check is especially true for SFPs that support 25 G and higher because a mismatch can exist in Forward Error Correction (FEC), which can prevent a port to auto-negotiate.

Longer cables are exposed to more noise and interference; that is, high Bit Error Ratio (BER); therefore, they require more powerful error correction codes.

Two IEEE 802.3 FEC specifications are available. If an auto-negotiation issue occurs, verify whether any compatibility issue exists with SFPs at both end points:

- ▶ Clause 74: Fire Code (FC-FEC) or BASE-R (BR-FEC) (16.4 dB loss specification).
- ▶ Clause 91: Reed-Solomon; that is, RS-FEC (22.4 dB loss specification).

### ***Priority flow control***

Priority flow control (PFC) is an Ethernet protocol that supports the ability to assign priorities to different types of traffic within the network. On most Data Center Bridging Capability Exchange protocol (DCBX) supported switches, verify that Link Layer Discovery Protocol (LLDP) is enabled. The presence of a VLAN is a prerequisite for configuring PFC. It is recommended to set the priority tag in the range 0 - 7.

A DCBX-enables switch and a storage adapter exchange parameters that describe traffic classes and PFC capabilities.

In the IBM FlashSystem, Ethernet traffic is divided into the following Classes of Service that are based on feature use case:

- ▶ Host attachment (iSCSI/iSER)
- ▶ Backend Storage (iSCSI)
- ▶ Node-to-node communication (RDMA clustering)

If challenges occur while configuring PFC, verify the following attributes to help determine the issue:

- ▶ Configure IP/VLAN by using `cfgportip`.
- ▶ Configure class of service (COS) by using `chsytsmethernet`.
- ▶ Ensure that the priority tag is enabled on the switch.
- ▶ Ensure that `lsportip` output shows: `dcbx_state`, `pfc_enabled_tags`.

Enhanced Transmission Selection (ETS) settings are recommended if a port is shared.

### ***Standard network connectivity check***

Verify that the required TCP/UDP ports are allowed in the network firewall. The following ports for various host attachments are available:

- ▶ Software iSCSI requires TCP Port 3260
- ▶ iSER/RoCE host requires 3260
- ▶ iSER/iWRAP host requires TCP Port 860

Verify that the IP addresses are reachable and the TCP ports are open.

### ***iSCSI performance issues***

In specific situations, the TCP/IP layer might try to combine several ACK responses together into a single response to improve performance, but that process can negatively affect iSCSI read performance as the storage target waits for the response to arrive. This issue is observed when the application is single-threaded and features a low queue depth.

It is recommended to disable the `TCPDe1ayedAck` parameter on the host platforms to improve overall storage I/O performance. If the host platform does not provide a mechanism to disable `TCPDe1ayedAck`, verify whether a smaller “Max I/O Transfer Size” with more concurrency (queue depth >16) improves overall latency and bandwidth usage for the specific host workload. In most Linux distributions, this size is controlled by the `max_sectors_kb` parameter with a suggested transfer size of 32 KB.

Also, review network switch diagnostic data to evaluate packet drop and retransmission in the network. It is advisable to enable flow control and PFC to enhance the reliability of the network delivery system to avoid packet loss, which enhances storage performance.

### 11.3.2 SAN problems

Introducing IBM Spectrum Virtualize into your SAN environment and the use of its virtualization functions are not difficult tasks. However, before you can use IBM Spectrum Virtualize in your environment, you must follow some basic rules. These rules are not complicated, but you can make mistakes that lead to accessibility issues or a reduction in the performance experienced.

Two types of SAN zones are needed to run IBM Spectrum Virtualize in your environment: a *host zone* and a *storage zone*. In addition, you must have an IBM Spectrum Virtualize zone that contains all of the IBM Spectrum Virtualize node ports of the IBM Spectrum Virtualize cluster. This IBM Spectrum Virtualize zone enables intracluster communication.

For more information about setting up IBM Spectrum Virtualize in a SAN fabric environment, see Chapter 2, “Storage area network” on page 19.

Because IBM FlashSystem is a major component of the SAN and connects the host to the storage subsystem, check and monitor the SAN fabrics.

Some situations of performance degradation and buffer-to-buffer credit exhaustion can be caused by incorrect local FC port masking and remote FC port masking. To ensure healthy operation of your IBM SAN Volume Controller, configure your local FC port masking and your remote FC port masking.

The ports that are intended to have only intracluster or node-to-node communication traffic must not have replication data or host or back-end data running on it. The ports that are intended to have only replication traffic must not have intracluster or node-to-node communication data or host or back-end data running on it.

Some situations can cause issues in the SAN fabric and SAN switches. Problems can be related to a hardware fault or to a software problem on the switch.

The following hardware defects often are the easiest problems to find:

- ▶ Switch power, fan, or cooling units
- ▶ Installed SFP modules
- ▶ Fiber optic cables

Software failures are more difficult to analyze. In most cases, you must collect data and involve IBM Support. However, before you take any other steps, check the installed code level for any known issues. Also, check whether a new code level is available that resolves the problem that you are experiencing.

The most common SAN issues often are related to zoning. For example, perhaps you chose the wrong WWPN for a host zone, such as when two IBM SAN Volume Controller node ports must be zoned to one HBA with one port from each IBM SAN Volume Controller node. As shown in Example 11-6, two ports are zoned that belong to the same node. Therefore, the result is that the host and its multipathing driver do not see all of the necessary paths.

*Example 11-6 Incorrect WWPN zoning*

---

```
zone: Senegal_Win2k3_itsosvcc11_iogrp0_Zone
      50:05:07:68:10:20:37:dc
      50:05:07:68:10:40:37:dc
      20:00:00:e0:8b:89:cc:c2
```

---

The correct zoning must look like the zoning that is shown in Example 11-7.

*Example 11-7 Correct WWPN zoning*

---

```
zone: Senegal_Win2k3_itsosvcc11_iogrp0_Zone
      50:05:07:68:10:40:37:e5
      50:05:07:68:10:40:37:dc
      20:00:00:e0:8b:89:cc:c2
```

---

The following IBM FlashSystem error codes are related to the SAN environment:

- ▶ Error 1060 - Fibre Channel ports are not operational
- ▶ Error 1220 - A remote port is excluded

A bottleneck is another common issue related to SAN switches. The bottleneck can be present in a port where a host, storage subsystem, or IBM Spectrum Virtualize device is connected, or in Inter-Switch Link (ISL) ports. The bottleneck can occur in some cases, such as when a device that is connected to the fabric is slow to process received frames or if a SAN switch port cannot transmit frames at a rate that is required by a device that is connected to the fabric.

These cases can slow down communication between devices in your SAN. To resolve this type of issue, refer to the SAN switch documentation or open a case with the vendor to investigate and identify what is causing the bottleneck and fix it.

If you cannot fix the issue with these actions, use the method that is described in 11.5, “Call Home Connect Cloud and Health Checker feature” on page 492, collect the SAN switch debugging data, and then, contact the vendor for assistance.

### 11.3.3 Storage subsystem problems

Today, various heterogeneous storage subsystems are available. These subsystems feature different management tools, setup strategies, and possible problem areas, depending on the manufacturer. To support a stable environment, all subsystems must be correctly configured, and follow the respective preferred practices with no existing issues.



Check the following areas if you experience a storage-subsystem-related issue:

- ▶ Storage subsystem configuration. Ensure that a valid configuration and preferred practices are applied to the subsystem.
- ▶ Storage subsystem controllers. Check the health and configurable settings on the controllers.
- ▶ Storage subsystem array. Check the state of the hardware, such as a disk drive module (DDM) failure or enclosure alerts.
- ▶ Storage volumes. Ensure that the logical unit number (LUN) masking is correct.
- ▶ Host attachment ports. Check the status, configuration, and connectivity to SAN switches.
- ▶ Layout and size of RAID arrays and LUNs. Performance and redundancy are contributing factors.

IBM Spectrum Virtualize has several CLI commands that you can use to check the status of the system and attached storage subsystem. Before you start a complete data collection or problem isolation on the SAN or subsystem level, use the following commands first and check the status from the IBM Spectrum Virtualize perspective:

▶ **lscontroller <controller\_id\_or\_name>**

Check that multiple worldwide port names (WWPNs) that match the back-end storage subsystem controller ports are available.

Check that the *path\_counts* are evenly distributed across each storage subsystem controller, or that they are distributed correctly based on the preferred controller. The total of all *path\_counts* must add up to the number of managed disks (MDisks) multiplied by the number of IBM Spectrum Virtualize nodes.

▶ **lsmdisk**

Check that all MDisks are online (not degraded or offline).

▶ **lsmdisk <MDisk\_id\_or\_name>**

Check several of the MDisks from each storage subsystem controller. Are they online? Do they all have *path\_count* = number of backend ports in the zone to IBM Spectrum Virtualize x number of nodes? An example of the output from this command is shown in Example 11-8.

*Example 11-8 Issuing an lsmdisk command*

---

```
IBM_2145:itsosvcc11:superuser>lsmdisk 0
id 0
name MDisk0
status online
mode array
MDisk_grp_id 0
MDisk_grp_name Pool0
capacity 198.2TB
quorum_index
block_size
controller_name
ctrl_type
ctrl_wwnn
controller_id
path_count
max_path_count
ctrl_LUN_#
UID
```

```

preferred_WWPN
active_WWPN
fast_write_state empty
raid_status online
raid_level raid6
redundancy 2
strip_size 256
spare_goal
spare_protection_min
balanced exact
tier tier0_flash
slow_write_priority latency
fabric_type
site_id
site_name
easy_tier_load
encrypt no
distributed yes
drive_class_id 0
drive_count 8
stripe_width 7
rebuild_areas_total 1
rebuild_areas_available 1
rebuild_areas_goal 1
dedupe no
preferred_iscsi_port_id
active_iscsi_port_id
replacement_date
over_provisioned yes
supports_unmap yes
provisioning_group_id 0
physical_capacity 85.87TB
physical_free_capacity 78.72TB
write_protected no
allocated_capacity 155.06TB
effective_used_capacity 16.58TB.

```

```

IBM_2145:itsosvcc11:superuser>lsmdisk 1

```

```

id 1
name flash9h01_itsosvcc11_0
status online
mode managed
mdisk_grp_id 0
mdisk_grp_name Pool0
capacity 1.6TB
quorum_index
block_size 512
controller_name itsoflash9h01
ctrl_type 4
ctrl_WWNN 500507605E852080
controller_id 1
path_count 32
max_path_count 32
ctrl_LUN_# 0000000000000000
UID 6005076441b53004400000000000000100000000000000000000000000000000

```

```
preferred_WWPN
active_WWPN many
.
lines removed for brevity
.
IBM_2145:itsosvcc11:superuser>
```

---

Example 11-8 on page 481 also shows that the external storage controller includes eight ports that are zoned to IBM Spectrum Virtualize, and IBM Spectrum Virtualize has four nodes. Therefore,  $8 \times 4 = 32$ .

► **lsvdisk**

Check that all volumes are online (not degraded or offline). If the volumes are degraded, are any stopped FlashCopy jobs present? Restart any stopped FlashCopy jobs or seek IBM Spectrum Virtualize support guidance.

► **lsfabric**

Use this command with the various options, such as **-controller controllerid**. Also, check different parts of the IBM Spectrum Virtualize configuration to ensure that multiple paths are available from each IBM Spectrum Virtualize node port to an attached host or controller. Confirm that all IBM Spectrum Virtualize node port WWPNs are connected to the back-end storage consistently.

## Determining the correct number of paths to a storage subsystem

By using IBM Spectrum Virtualize CLI commands, it is possible to determine the total number of paths to a storage subsystem. To determine the suitable value of the available paths, use the following formula:

Number of MDisks x Number of SVC nodes per Cluster = Number of paths  
mdisk\_link\_count x Number of SVC nodes per Cluster = Sum of path\_count

Example 11-9 shows how to obtain this information by using the **lscontroller <controllerid>** and **svcinfo lsnnode** commands.

*Example 11-9 Output of the svcinfo lscontroller command*

---

```
IBM_2145:itsosvcc11:superuser>lscontroller 1
id 1
controller_name itsof9h01
WWNN 500507605E852080
mdisk_link_count 16
max_mdisk_link_count 16
degraded no
vendor_id IBM
product_id_low FlashSys
product_id_high tem-9840
product_revision 1430
ctrl_s/n 01106d4c0110-0000-0
allow_quorum yes
fabric_type fc
site_id
site_name
WWPN 500507605E8520B1
path_count 64
max_path_count 64
WWPN 500507605E8520A1
path_count 64
```

```

max_path_count 64
WWPN 500507605E852081
path_count 64
max_path_count 64
WWPN 500507605E852091
path_count 64
max_path_count 64
WWPN 500507605E8520B2
path_count 64
max_path_count 64
WWPN 500507605E8520A2
path_count 64
max_path_count 64
WWPN 500507605E852082
path_count 64
max_path_count 64
WWPN 500507605E852092
path_count 64
max_path_count 64
IBM_2145:itsosvcc11:superuser>svcinfo lsnode
id name UPS_serial_number WWNN status IO_group_id IO_group_name
config_node UPS_unique_id hardware iscsi_name
iscsi_alias panel_name enclosure_id canister_id enclosure_serial_number site_id
site_name
1 node1 500507680C003AE1 online 0 io_grp0 yes
DH8 iqn.1986-03.com.ibm:2145.itsosvcc11.node1 78CBFEA0
2 node2 500507680C003ACA online 0 io_grp0 no
DH8 iqn.1986-03.com.ibm:2145.itsosvcc11.node2 78CBRB0
3 node3 500507680C003A9F online 1 io_grp1 no
DH8 iqn.1986-03.com.ibm:2145.itsosvcc11.node3 78CBLP0
4 node4 500507680C003DB6 online 1 io_grp1 no
DH8 iqn.1986-03.com.ibm:2145.itsosvcc11.node4 78CCAQ0
IBM_2145:itsosvcc11:superuser>

```

---

Example 11-9 on page 483 also shows that 16 MDisks are present for the storage subsystem controller with ID 1, and four IBM Spectrum Virtualize nodes are in the IBM Spectrum Virtualize cluster. In this example, the path\_count is  $16 \times 4 = 64$ .

IBM Spectrum Virtualize has useful tools for finding and analyzing back-end storage subsystem issues because it has a monitoring and logging mechanism.

Typical events for storage subsystem controllers include incorrect configuration, which results in a 1625 - Incorrect disk controller configuration error code. Other issues that are related to the storage subsystem include failures pointing to the managed disk I/O (error code 1310), disk media (error code 1320), and error recovery procedure (error code 1370).

However, all messages do not have only one specific reason for being issued. Therefore, you must check multiple areas for issues, not just the storage subsystem.

To determine the root cause of a problem, complete the following steps:

1. Check the Recommended Actions window by clicking **Monitoring** → **Events**.
2. Check the attached storage subsystem for misconfigurations or failures:
  - a. Independent of the type of storage subsystem, first check whether the system has any unfixed errors. Use the service or maintenance features that are provided with the storage subsystem to fix these issues.
  - b. Check whether volume mapping is correct. The storage subsystem LUNs must be mapped to a host object with IBM SAN Volume Controller ports. Also, observe the IBM Spectrum Virtualize restrictions for back-end storage subsystems, which can be found at this [IBM Support web page](#).

If you need to identify which of the attached MDisk has a corresponding LUN ID, run the IBM Spectrum Virtualize **lsmdisk** CLI command, as shown in Example 11-10. This command also shows to which storage subsystem a specific MDisk belongs (the controller ID).

*Example 11-10 Determining the ID for the MDisk*

---

```
IBM_2145:itsosvcc11:admin>lsmdisk
```

| id                                      | name           | status                           | mode              |            |
|-----------------------------------------|----------------|----------------------------------|-------------------|------------|
| mdisk_grp_id                            | mdisk_grp_name |                                  | capacity          | ctrl_LUN_# |
| controller_name                         |                | UID                              |                   |            |
| 0                                       | mdisk0         | online                           | managed           | 0          |
| MDG-1                                   |                | 600.0GB                          | 0000000000000000  |            |
| controller0                             |                |                                  |                   |            |
| <b>600a0b800017423300000059469cf845</b> |                | 00000000000000000000000000000000 |                   |            |
| 2                                       | mdisk2         | online                           | managed           | 0          |
| MDG-1                                   |                | 70.9GB                           | 00000000000000002 |            |
| controller0                             |                |                                  |                   |            |
| <b>600a0b800017443100000096469cf0e8</b> |                | 00000000000000000000000000000000 |                   |            |

---

3. Check the SAN environment for switch problems or zoning failures.  
 Make sure that the zones are properly configured, and the zone set is activated. The zones that allow communication between the storage subsystem and the IBM Spectrum Virtualize device must contain WWPNs of the storage subsystem and WWPNs of IBM SAN Volume Controller or Storwize V7000.
4. Collect all support data and contact IBM Support.

Collect the support data for the involved SAN, IBM Spectrum Virtualize, or storage systems as described in 11.5, “Call Home Connect Cloud and Health Checker feature” on page 492.

### 11.3.4 Native IP replication problems

The native IP replication feature uses the following TCP/IP ports for remote cluster path discovery and data transfer:

- ▶ IP Partnership management IP communication: TCP Port 3260
- ▶ IP Partnership data path connections: TCP Port 3265

If a connectivity issue exists between the cluster in the management communication path, a cluster reports error code 2021: Partner cluster IP address unreachable. However, when a connectivity issue exists in the data path, the cluster reports error code 2020: IP Remote Copy link unavailable.

If the IP addresses are reachable and TCP ports are open, verify whether the end-to-end network supports a Maximum Transmission Unit (MTU) of 1500 bytes without packet fragmentation. When an external host-based ping utility is used to validate end-to-end MTU support, use the “do not fragment” qualifier.

Fix the network path so that traffic can flow correctly. After the connection is made, the error corrects automatically.

Network quality of service largely influences the effective bandwidth usage of the dedicated link between the cluster. Bandwidth usage is inversely proportional to round-trip time (RTT) and rate of packet drop or retransmission in the network.

For standard block traffic, a packet drop or retransmission of 0.5% or more can lead to unacceptable use of the available bandwidth. Work with network team to investigate oversubscription or other quality of service of the link, with an objective to bring the packet drop percentage as low as possible (less than 0.1%).

### 11.3.5 Remote Direct Memory Access (RDMA) based Clustering

RDMA technology supports zero-copy networking, which makes it possible to read data directly from the main memory of one computer and write that data directly to the main memory of another computer. This technology bypasses the CPU intervention while processing the I/O leading to lower latency and a faster rate of data transfer.

IBM FlashSystem Cluster can be formed by using RDMA-capable NICs that use RoCE or iWARP technology. Consider the following points:

- ▶ Inter-node Ethernet connectivity can be done over identical ports only; such ports must be connected within the same switching fabric.
- ▶ If the cluster is to be created without any ISL (up to 300 meters), deploy Independent (isolated) switches.
- ▶ If the cluster is to be created on short-distance ISL (up to 10 km; that is, 6.2 miles), provision as many ISLs between switches as RDMA-capable cluster ports.
- ▶ For long-distance ISL (up to 100 km; that is 62 miles), DWDM and CWDM methods are applicable for L2 networks. Packet switched or VXLAN methods are deployed for L3 network as this equipment comes with deeper buffer “pockets”.

The following Ports must be opened in the firewall for IP-based RDMA clustering:

- ▶ TCP 4791, 21451, 21452, and 21455
- ▶ UDP 4791, 21451, 21452, and 21455

The first step to review if the node IP address is reachable and verify the required TCP/UDP ports are accessible in both directions. The following CLI output can be helpful to find the reason for connectivity error:

```
sainfo lsnodeipconnectivity
```

### 11.3.6 Advanced Copy services problems

The performance of a specific storage feature or overall storage subsystem are generally interlinked; that is, a bottleneck in one software or hardware layer might propagate to other layers. Therefore, problem isolation is critical part of performance analysis.

The first thing to check is if any unfixed events exist that require attention. After the fix procedure is followed to correct the alerts, the next step is to check the audit log to determine whether any activity exists that can trigger the performance issue. If that information correlates, more analysis can be done to check whether that specific feature is used.

The most common root causes for performance issues are SAN congestion, configuration changes, incorrect sizing/estimation of advanced copy services (replication, FlashCopy, and volume mirroring) or I/O load change, because of hardware component failure.

The following sections are a quick reference to common misconfigurations.

## Remote Copy

Any disturbances in the SAN/WAN can cause congestion and packet drop, which can affect Metro Mirror (MM) or Global Mirror (GM) traffic. Because host I/O latency depends on MM or GM I/O completion to the remote cluster, a host can experience high latency. Based on various parameters, replication can be operatively stopped to protect host. Therefore, the following conditions affect GM/MM:

- ▶ Network congestion or fluctuation. Fix the network. Also, verify that port masking is enabled so that the congestion in replication ports does not affect clustering or host or storage ports.
- ▶ Overload of secondary or primary cluster. Monitor and throttle the host, which causes the condition.
- ▶ High background copy rate, which leaves less bandwidth to replicate foreground host I/O. Adjust the background copy rate so that the link does not get oversubscribed.
- ▶ A large Global Mirror with Change Volumes (GMCV) consistency group might introduce hundreds of milliseconds of pause when the replication cycle starts. Reduce the number of relationships in a consistency group if the observed I/O pause is not acceptable.

## HyperSwap

Verify that the link between the sites is stable and has enough bandwidth to replicate the peak workload. Also, check if a volume must frequently change the replication direction from one site to other. This issue occurs when a specific volume is being written by hosts from both sites. Evaluate if this problem can be avoided to reduce frequent direction change. (Ignore it if the solution is designed to consider active/active access.)

If a single volume resynchronization between the sites takes too long, review the partnership `link_bandwidth_mbits` and `per_relationship_bandwidth_limit` parameters.

## FlashCopy

Consider the following points:

- ▶ Verify that the preferred node of FlashCopy source and target volumes is the same to avoid excessive internode communications.
- ▶ Verify the high background copy rates and clean rate of FlashCopy relations because these factors might cause back-end overload.
- ▶ Port saturation or node saturation. Review if the values are correctly sized.
- ▶ Check the number of FC relationships in any FlashCopy consistency group. The larger the number of relationships, the higher the I/O pause time (Peak I/O Latency) when the CG starts.
- ▶ If the host I/O pattern is small and random, evaluate if reducing the FlashCopy grain size to 64 KB provides any improvement in latency compared to the default grain size of 256 KB.

## Compression

Compress a volume if the data is compressible. No benefit is gained by compressing a volume where compression saving is less than 25% because that rate can reduce the overall performance of the RACE compression engine.

If the I/O access pattern is sequential, that might not be a suitable candidate for RACE. Use the Comprestimator/Data Reduction Estimation Tool to size the workload.

## Volume mirroring

Write performance of the mirrored volumes is dictated by the slowest copy. Reads are served from the Copy0 of the volume (in the case of a stretched cluster topology, both the copies can serve reads, which is dictated by the host site attribute). Therefore, size the solution accordingly.

**Note:** Because the mirroring layer maintains a bitmap copy on the quorum device, any unavailability of the quorum takes the mirroring volumes offline. Similarly, slow access to the quorum might also affect the performance of mirroring volumes.

## Data reduction pools

Data reduction pools (DRPs) internally implement a log structured array (LSA), which means that writes (new or over-writes or updates) always allocate newer storage blocks.

The older blocks (with invalid data) are marked for garbage collection later. The garbage collection process defers the work as much as possible because the more it is deferred, the higher the chance of having to move only some valid data from the block to make that block available to the free pool.

However, when the pool reaches more than 85% of its allocated capacity, garbage collection must speed up and move valid data more aggressively to make space available sooner. This process can lead to increased latency because of increased CPU usage and load on the back-end. Therefore, it is recommended to manage storage provisioning to avoid such scenarios.

Users also are encouraged to pay particular attention to any GUI notifications and use best practices for managing physical space. Use data reduction only at one layer (at the virtualization layer or the back-end storage or drives) because no benefit is gained to compress and deduplicate the same data twice.

Encrypted data cannot be compressed; therefore, data reduction must be done before the data is encrypted. Proper sizing is crucial to get the best of performance from data reduction; therefore, use data reduction estimation tools to evaluate system performance and space saving.

IBM FlashSystem makes use of the following types of data reduction techniques:

- ▶ IBM FlashSystem that uses the FCM NVMe drives feature built-in hardware compression.
- ▶ IBM FlashSystem that uses industry-standard NVMe drives relies on the Spectrum Virtualize software and DRPs to deliver data reduction.



### 11.3.7 Health status during upgrade

It is important to understand that during the software upgrade process, alerts that indicate the system is not healthy are reported. These alerts are normal behavior because the IBM FlashSystem node canisters go offline during this process; therefore, the system triggers these alerts.

While trying to upgrade an IBM FlashSystem, other messages might be issues, such as an error in verifying the signature of the update package.

This message does not mean that an issue exists in your system. At times, this issue occurs because not enough space exists on the system to copy the file, or the package is incomplete or contains errors. In this case, open a PMR with IBM Support and follow their instructions.

### 11.3.8 Managing physical capacity of over provisioned storage controllers

Drives and back-end controllers are available that include built-in hardware compression and other data reduction technologies that allow capacity to be provisioned above the available real physical capacity. Different data sets lead to different capacity savings and some data, such as encrypted data or compressed data, do not compress. When the physical capacity savings do not match the expected or provisioned capacity, the storage can run out of physical space, which leads to a write-protected drive or array.

To avoid running out of space on the system, the usable capacity must be carefully monitored on the GUI of the IBM FlashSystem. The IBM FlashSystem GUI is the only capacity dashboard that shows the physical capacity.

Monitoring is especially important when migrating substantial amounts of data onto the IBM FlashSystem. This migration typically occurs during the first part of the workload lifecycle because data is on-boarded or initially populated into the storage system. IBM strongly encourages users to configure Call Home on the IBM FlashSystem. Call Home monitors the physical free space on the system and automatically opens a service call for systems that reach 99% of their usable capacity.

IBM Storage Insights also can monitor and report on any potential out of space conditions. The new Advisor function also warns users when the IBM FlashSystem is nearing full capacity.

When IBM FlashSystem reaches an out of space condition, the device drops into a read-only state. An assessment of the data compression ratio and the replanned capacity estimation must be done to determine how much outstanding storage demand might exist. This extra capacity must be prepared and presented to the host so that recovery can begin.

The approaches that can be taken to reclaim space on the IBM FlashSystem in this scenario vary by the capabilities of the system, any optional external back-end controllers, and the system configuration and pre-planned capacity overhead needs.

The following options are available:

- ▶ Add capacity to the IBM FlashSystem. Users must a plan that allows them to add capacity to the system when needed.
- ▶ Reserve a set of space in the IBM FlashSystem that makes it “seem” fuller than it really is, and that you can free up in an emergency situation. IBM FlashSystem can create a volume that is not compressed, de-duped, or thin provisioned (a fully allocated volume).

Create some of these volumes to reserve an amount of physical spaces. You likely want to name them; for example, “emergency buffer space”. If you are reaching the limits for physical capacity, delete one or more of these volumes for a temporary reprieve.

**Important:** Running out of space can be a serious situation. Recovery can be complicated and time-consuming. For this reason, it is imperative that proper planning and monitoring are done to avoid reaching this condition.

The following sections describe the process for recovering from an out of space condition.

## Reclaiming and unlocking

After you assess and account for storage capacity, the first step is to contact IBM Support who can aid in unlocking the read-only mode and restoring operations. The reclamation task can take a long time to run, and larger flash arrays take longer to recover than smaller ones.

## Freeing up space

The amount of used space can be reduced by using several methods after the IBM FlashSystem is unlocked by IBM Support.

To recover from out of space conditions on Standard Pools, the following methods are available:

- ▶ Add storage to the system, if possible.
- ▶ Migrate extents from the write protected array to other non-write protected MDisk with enough extents, which can be an external back-end storage array.
- ▶ Migrate volumes with extents on the write protected array to another pool. If possible, moving volumes from the IBM FlashSystem pool to another external pool can free up space in the IBM FlashSystem pool to allow for space reclamation. Because this volume moves into the new pool, its previously occupied flash extents are freed up (by way of SCSI unmap), which then goes to provide more free space to the IBM FlashSystem enclosure to be configured to a proper provisioning to support the compression ratio.
- ▶ Deleting dispensable volumes to free space. If possible, within the pool (managed disk group) on the IBM FlashSystem, delete any unnecessary volumes. The IBM FlashSystem supports SCSI unmap so deleting volumes realizes space reclamation benefits by using this method.
- ▶ Bring the volumes in the pool back online by using a Directed Maintenance Procedure.

For more information about types of recovery, see can be found in this [IBM Support Technote](#).

## 11.4 Remote Support Assistance

Remonsite and remotely. Both methods use secure connections to protect data in the communication between support center and system. Also, you can audit all actions that support personnel conduct on the system.

Figure 11-7 shows how to set up the remote support options in the GUI by selecting **Settings** → **Support** → **Support Assistance** → **Reconfigure Settings**.

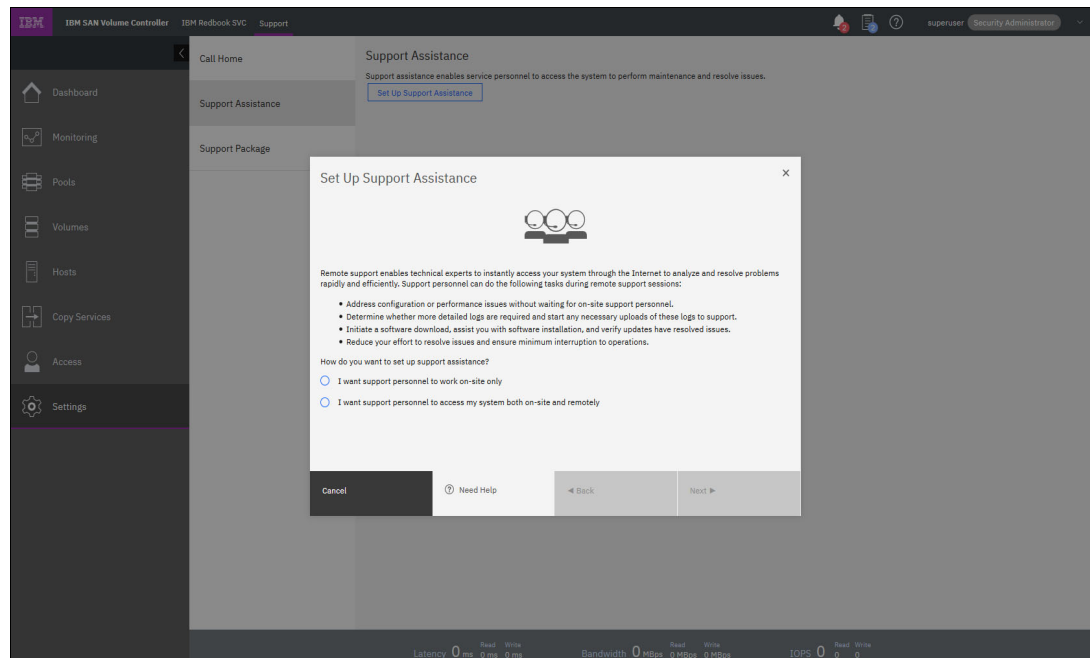


Figure 11-7 Reconfigure settings

You can use only local support assistance if you have security restrictions that do not allow support to connect remotely to your systems. With Remote Support Assistance, support personnel can work onsite and remotely through a secure connection from the support center. They can perform troubleshooting, upload support packages, and download software to the system with your permission. When you configure remote support assistance in the GUI, local support assistance also is enabled.

With the remote support assistance method, the following access types are available:

- ▶ **At any time**  
Support center can start remote support sessions at any time.
- ▶ **By permission only**  
Support center can start a remote support session only if permitted by an administrator. A time limit can be configured for the session.

**Note:** Systems that are purchased with a 3-year warranty and include Enterprise Class Support (ECS) are entitled to IBM support by using Remote Support Assistance to quickly connect and diagnose problems. However, IBM Support might choose to use this feature on non-ECS systems at their discretion; therefore, we recommend configuring and testing the connection on all systems.

To configure remote support assistance, the following prerequisites must be met:

- ▶ Call Home is configured with a valid email server.
- ▶ A valid service IP address is configured on each node on the system.
- ▶ A Remote Support Proxy server is configured if your system is behind a firewall or if you want to route traffic from multiple storage systems to the same place. Before you configure remote support assistance, the proxy server must be installed and configured separately. The IP address and the port number for the proxy server must be set up when remote support centers are enabled.

For more information about setting up the Remote Proxy Server, see [this web page](#).

- ▶ If you do not have firewall restrictions and the storage nodes are directly connected to the Internet, request your network administrator to allow connections to 129.33.206.139 and 204.146.30.139 on port 22.
- ▶ Both uploading support packages and downloading software require direct connections to the Internet. A DNS server must be defined on your system for both of these functions to work. The Remote Proxy Server cannot be used to download files.
- ▶ To ensure that support packages are uploaded correctly, configure the firewall to allow connections to the following IP addresses on port 443: 129.42.56.189, 129.42.54.189, and 129.42.60.189.
- ▶ To ensure that software is downloaded correctly, configure the firewall to allow connections to the following IP addresses on port 22: 170.225.15.105, 170.225.15.104, 170.225.15.107, 129.35.224.105, 129.35.224.104, and 129.35.224.107.

Remote support assistance can be configured by using GUI and CLI. For more information about configuring this assistance, see *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8467

## 11.5 Call Home Connect Cloud and Health Checker feature

Formerly known as Call Home Web, the new Call Home Connect Cloud is a cloud-based version with improved feature to view Call Home information on the web.

Call Home is a function that is available in several IBM systems, including IBM FlashSystem, which allows them to automatically report problems and status to IBM.

Call Home Connect Cloud provides the following information about IBM systems:

- ▶ Automated tickets
- ▶ Warranty and contract status
- ▶ Health check alerts and recommendations
- ▶ System connectivity heartbeat
- ▶ Recommended software levels
- ▶ Inventory
- ▶ Security bulletins

To access the Call Home Connect Cloud, see the [IBM Support home page](#).

In the IBM support website, Call Home Connect Cloud is available at **My support** → **Call Home Web**, as shown in Figure 11-8.

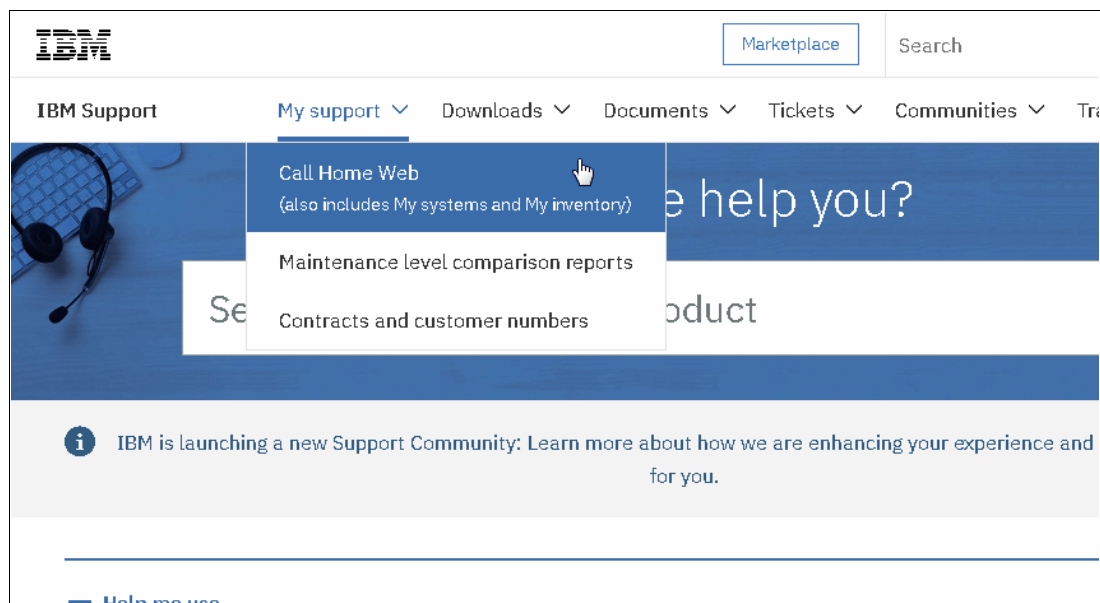


Figure 11-8 Call Home Connect Cloud (Call Home Web)

To allow Call Home Connect Cloud analyze data of IBM Spectrum Virtualize systems and provide useful information about them, the devices must be added to the tool. The machine type, model, and serial number are required to register the product in Call Home Connect Cloud. Also, it is required that IBM SAN Volume Controller have Call Home and inventory notification enabled and operating.

### 11.5.1 Health Checker

A new feature of Call Home Connect Cloud is the Health Checker, a tool that runs in the IBM Cloud.

It analyzes Call Home and inventory data of systems that are registered in Call Home Connect Cloud and validates their configuration. Then, it displays alerts and provide recommendations in the Call Home Connect Cloud tool.

**Note:** Call Home Connect Cloud is used because it provides useful information about your systems, and with the Health Checker feature, it helps you to monitor the system. It also operatively provides alerts and creates recommendations that are related to them.

Some of the functions of the IBM Call Home Connect Cloud and Health Checker were ported to IBM Storage Insights. For more information, see 11.6, “IBM Storage Insights” on page 494.

## 11.6 IBM Storage Insights

IBM Storage Insights is a part of the monitoring and ensuring continued availability of the IBM SAN Volume Controller.

Available at no charge, cloud-based IBM Storage Insights provides a single dashboard that gives you a clear view of all of your IBM block storage. You can make better decisions by seeing trends in performance and capacity. Storage health information enables you to focus on areas needing attention. In addition, when IBM support is needed, Storage Insights simplifies uploading logs, speeds resolution with online configuration data, and provides an overview of open tickets all in one place.

The following features are included:

- ▶ A unified view of IBM systems:
  - Provides a single window to see all of your system's characteristics.
  - See all of your IBM storage inventory.
  - Provides a live event feed so that you know, up to the second, what is going on with your storage and enables you to act fast.
- ▶ IBM Storage Insight collects telemetry data and Call Home data, and provides up-to-the-second system reporting of capacity and performance.
- ▶ Overall storage monitoring:
  - The overall health of the system.
  - Monitor the configuration to see whether it meets the best practices.
  - System resource management: determine whether the system is being overly taxed and provide proactive recommendations to fix it.
- ▶ Storage Insights provides advanced customer service with an event filter that enables the following functions:
  - The ability for you and support to view support tickets, open and close them, and track trends.
  - Auto log collection capability to enable you to collect the logs and send them to IBM before support starts looking into the problem. This can save as much as 50% of the time to resolve the case.

In addition to the no-charge Storage Insights, there is also the option of Storage Insights Pro, which is a subscription service that provides longer historical views of data, offers more reporting and optimization options, and supports IBM file and block storage together with EMC VNX and VMAX.

Figure 11-9 shows the comparison of Storage Insights and Storage Insights Pro.

| Product Comparison      |                                                                 |                             |                                         |
|-------------------------|-----------------------------------------------------------------|-----------------------------|-----------------------------------------|
|                         | Capability                                                      | IBM Storage Insights (Free) | IBM Storage Insights Pro (Subscription) |
| <b>Monitoring</b>       | Health, Performance and Capacity                                | ✓                           | ✓                                       |
|                         | Filter events to quickly isolate trouble spots                  | ✓                           | ✓                                       |
|                         | Drill down performance workflows to enable deep troubleshooting |                             | ✓                                       |
|                         | Application / server storage performance troubleshooting        |                             | ✓                                       |
|                         | Customizable multi-conditional alerting                         |                             | ✓                                       |
| <b>Support Services</b> | Simplified ticketing / log workflows and ticket history         | ✓                           | ✓                                       |
|                         | Proactive notification of risks (select systems)                | ✓                           | ✓                                       |
| <b>Device Analytics</b> | Part failure prediction                                         | ✓                           | ✓                                       |
|                         | Configuration best practice                                     | ✓                           | ✓                                       |
|                         | Customized upgrade recommendation                               | ✓                           | ✓                                       |
| <b>TCO Analytics</b>    | Capacity planning                                               |                             | ✓                                       |
|                         | Performance planning                                            |                             | ✓                                       |
|                         | Application / server storage consumption                        |                             | ✓                                       |
|                         | Capacity optimization with reclamation planning                 |                             | ✓                                       |
|                         | Data optimization with tier planning                            |                             | ✓                                       |

Figure 11-9 Storage Insights versus Storage Insights Pro comparison

Storage Insights provides a lightweight data collector that is deployed on a customer supplied server. This can be either a Linux, Windows, or AIX server, or a guest in a virtual machine (for example, a VMware guest).

The data collector streams performance, capacity, asset, and configuration metadata to your IBM Cloud instance.

The metadata flows in one direction: from your data center to IBM Cloud over HTTPS. In the IBM Cloud, your metadata is protected by physical, organizational, access, and security controls. IBM Storage Insights is ISO/IEC 27001 Information Security Management certified.

## Collected metadata

The following metadata about the configuration and operations of storage resources is collected:

- ▶ Name, model, firmware, and type of storage system.
- ▶ Inventory and configuration metadata for the storage system's resources, such as volumes, pools, disks, and ports.
- ▶ Capacity values, such as capacity, unassigned space, used space, and the compression ratio.
- ▶ Performance metrics, such as read and write data rates, I/O rates, and response times
- ▶ The application data that is stored on the storage systems cannot be accessed by the data collector.

## Metadata access

Access to the metadata that is collected is restricted to the following users:

- ▶ The customer who owns the dashboard.
- ▶ The administrators who are authorized to access the dashboard, such as the customer's operations team.
- ▶ The IBM Cloud team that is responsible for the day-to-day operation and maintenance of IBM Cloud instances.
- ▶ IBM Support for investigating and closing service tickets.

### 11.6.1 Storage Insights Customer Dashboard

Figure 11-10 shows a view of the Storage Insights (SI) main dashboard and the systems that it is monitoring.

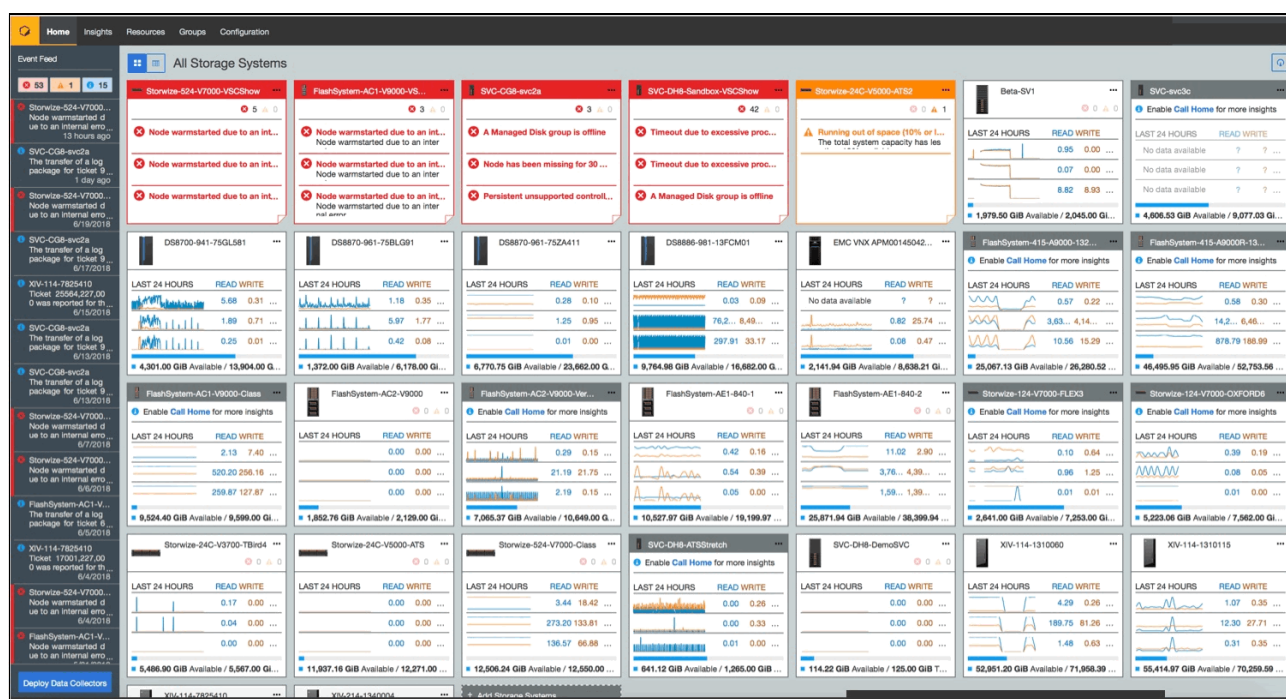


Figure 11-10 Storage Insights main dashboard

### 11.6.2 Customized dashboards to monitor your storage

With the latest release of IBM Storage Insights, you can customize the dashboard to show only a subset of the systems that are monitored. This feature is useful for customers that might be Cloud Service Providers (CSP) and want only a specific user to see those machines for which they are paying.

For more information about setting up the customized dashboard, see [this web page](#).



### 11.6.3 Creating support tickets

From the Dashboard GUI, IBM SI can create support tickets for one of the systems it reports about. Complete the following steps:

1. Go to the SI main dashboard and then, choose the system for which you want to raise the ticket. From this window, select **Action** → **Create/Update Ticket**.

Figure 11-11 shows how to create or update a support ticket from the SI dashboard.

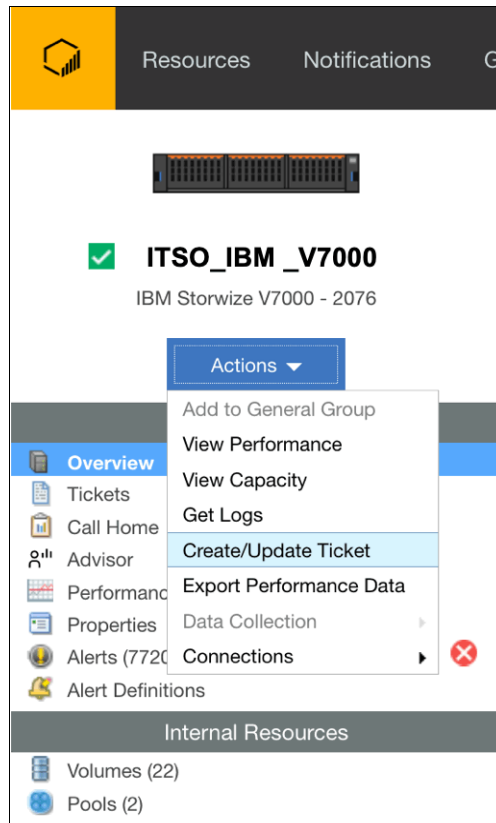


Figure 11-11 Creating or updating a support ticket

Figure 11-12 shows you the window where you can either create or update.

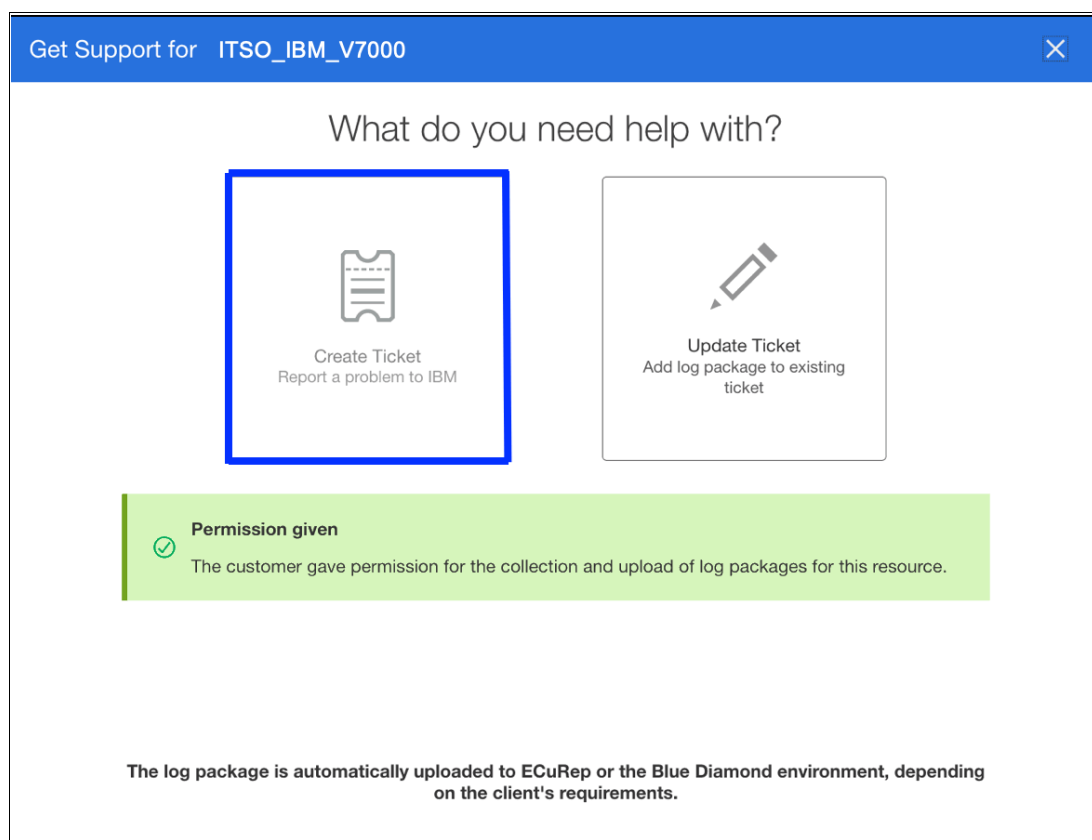


Figure 11-12 Create ticket

**Note:** The Permission given information box (see Figure 11-12) is an option that you must enable in the IBM Spectrum Virtualize GUI. For more information, see 11.4, “Remote Support Assistance” on page 491.

2. Select the **Create Ticket** option and you are presented with several windows with which you use to complete with the machine details, problem description, and the option to upload logs.

Figure 11-13 shows the ticket data collection that is done by the SI application.

The screenshot shows a window titled "Create Ticket for ITSO IBM V7000" with a close button (X) in the top right corner. The main heading is "Collecting information for ticket". Below this, a message states: "The following information for IBM is being added to the ticket:". The collected information is listed as follows:

- Customer number: 9717196
- Storage system: ITSO IBM V7000
- Type: IBM Storwize V7000
- Serial number: 0000020320208FCA
- Version: 8.1.3.6 (Build 148.13.....)

At the bottom of the window, there is a progress indicator consisting of four circles connected by a line. The first circle contains a checkmark, indicating the current step is complete. To the right of the progress indicator are two buttons: "Back" and "Next". The "Next" button is highlighted in blue, while the "Back" button is light gray.

Figure 11-13 Collecting information for ticket window

3. Add a problem description and attach other files, such as error logs or screen captures of error messages in the Add a note or attachment window (see Figure 11-14).

The screenshot shows a web-based interface for creating a ticket. At the top, a blue header bar contains the text "Create Ticket for ITSO\_IBM\_V7000" and a close button (X). Below the header, the main title "Add a note or attachment" is centered. The form consists of two text input fields. The first field contains the text "Device is showing a failure alert." and has a character count of "38" in the top right corner. Below this field is a hint: "Hint: Include what happened and the error code, if any." The second field is larger and contains the text "(Optional) Type a note to add to the ticket" and has a character count of "3000" in the top right corner. Below this field is a hint: "Hint: Include the time the problem or error occurred, the affected resources, and details of any maintenance or other activities that occurred before the problem." Below the text fields, there are two options for attaching files. On the left, under the heading "Attach Image or File:", there is a blue button labeled "Browse". To the right of this button is the word "OR". To the right of "OR" is a dashed blue box containing a blue upward-pointing arrow and the text "Drag file here". At the bottom of the form, there is a progress indicator consisting of four circles connected by a line; the first two circles are filled with blue, and the last two are empty. To the right of the progress indicator are two buttons: "Back" and "Next". The "Next" button is blue, while the "Back" button is white with a blue border.

Figure 11-14 Adding problem description and any other information

- Set a severity level for the ticket, ranging from a severity 1 for a system down or extreme business impact, to severity 4, which is for non-critical issues (see Figure 11-15).

Create Ticket for ITSO IBM V7000

### Select a severity level

Least critical      4      3      2      1      Most critical

Minor business impact - service is usable, not critical to operations

✓ ✓ ✓ ○

Back Next

Figure 11-15 Set severity level

A summary of the data that is used to create the ticket is shown in the Review the ticket window (see Figure 11-16).

Create Ticket for ITSO\_IBM\_V7000

Review the ticket

Problem summary: Device is showing a failure alert.

Severity level: 3 Minor business impact - service is usable, not critical to opera...

Type of problem: Hardware

Contact name: Jon Doe

Contact email: jdoe@us.ibm.com

Contact phone: 1234567890

Customer number: 666999

United States

Storage system: ITSO\_IBM\_V7000

Type: IBM Storwize V7000

Serial number: 0000020320208FCA

Version: 8.1.3.6 (Build 148.13.194231708000)

Did you know?

We can add a log package automatically for you if you monitor your storage system with a data collector and turn on Call Home.

[Learn more about monitoring resources with a data collector](#)

Figure 11-16 Review the ticket information

5. When completed, click the **Create Ticket** button to create the support ticket and send it to IBM (see Figure 11-17). The ticket number is created by the IBM Support system and sent back to your SI instance.

Create Ticket for ITSO\_IBM\_V7000

### Review the ticket

Problem summary: aaaa

Severity level: 3 Minor business impact - service is usable, not critical to opera...

Type of problem: Hardware

Contact name: Jon Doe

Contact email: jdoe@us.ibm.com

Contact phone: 1234567890

Customer number: 666555 United States

Storage system: ITSO\_IBM\_V7000

Type: IBM Storwize V7000

Serial number: 0000020320208FCA

Version: 8.1.3.6 (Build 148.13.194231708000)

Log package: Type 1: Standard logs

Progress: [✓] [✓] [✓] [✓]

Back Create Ticket

Figure 11-17 Final summary before ticket creation

6. Review the summary of the open and closed ticket numbers for the system that is selected by using the **Action** menu option (see Figure 11-18).

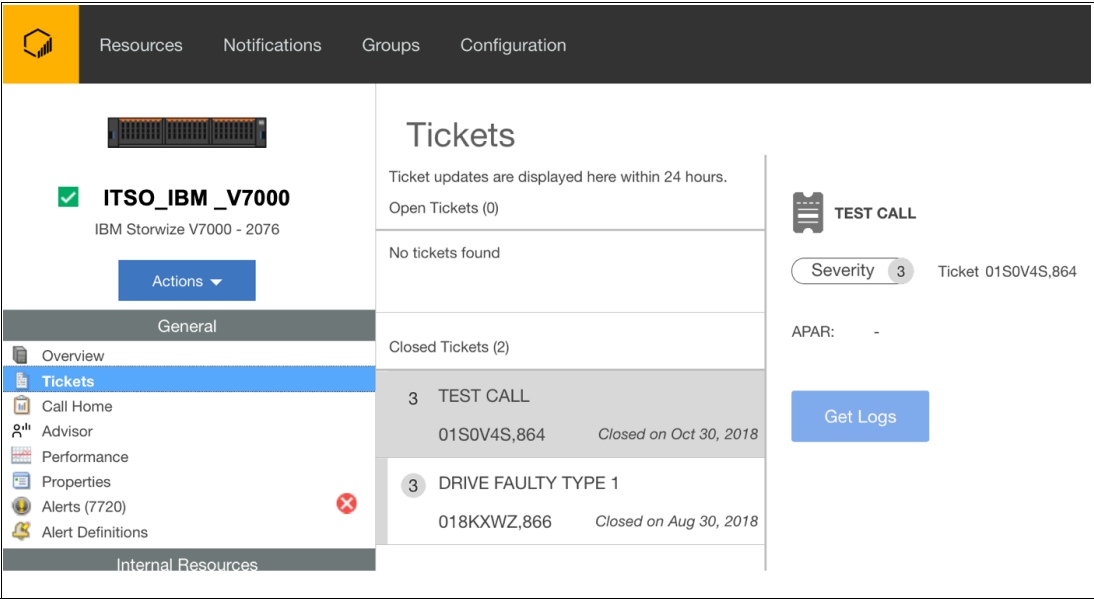


Figure 11-18 Ticket summary

### 11.6.4 Updating support tickets

IBM Storage Insights also can update support tickets from the Dashboard GUI for any of the systems it reports about.

Complete the following steps:

1. Go to the SI dashboard and then, choose the system for which you want to update the ticket. From this window, select **Actions** → **Create/Update Ticket**.



2. Figure 11-19 shows the initial menu to update a ticket. Select the **Update Ticket** option.

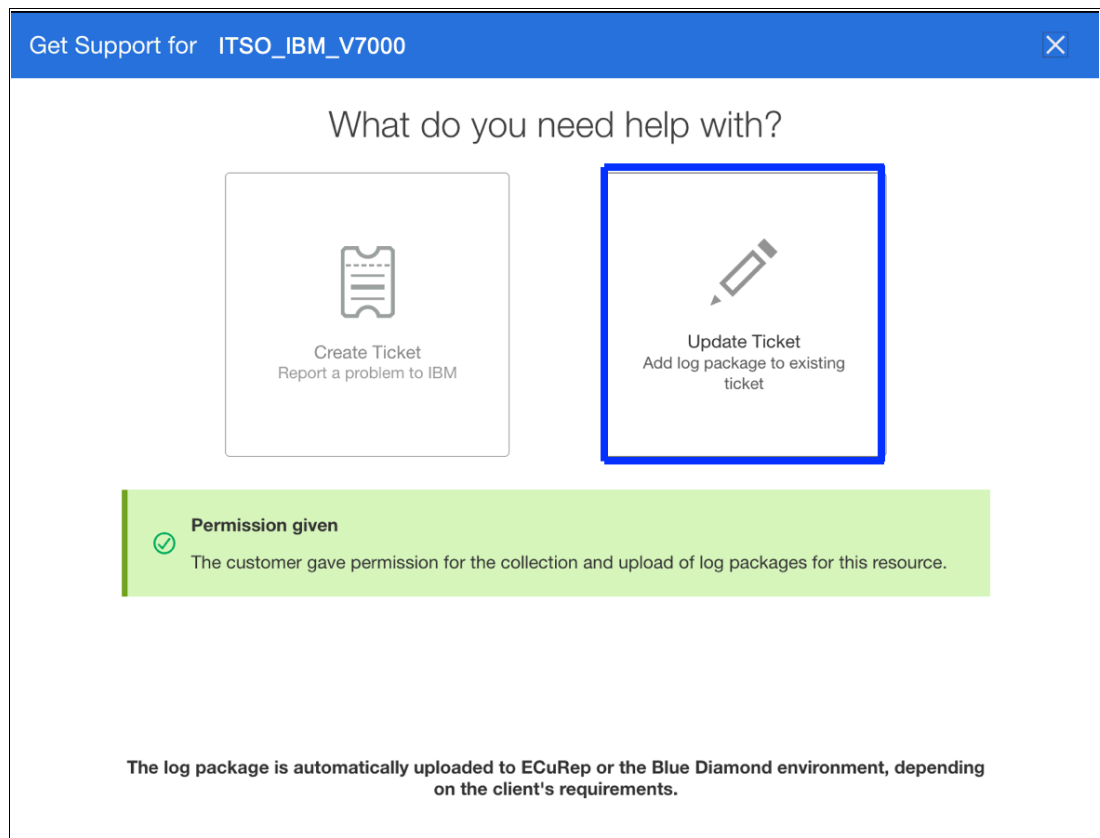


Figure 11-19 SI Update Ticket

3. Figure 11-20 shows the next window in which you must enter the PMR number and then, click **Next**. This PMR input is in the format XXXXX,YYY, ZZZ, where:

- XXXXX is the PMR record number
- YYY is the IBM Branch office number
- ZZZ in the IBM country number

This information was supplied when you created the ticket or by IBM Support in the event of the PMR being created by a problem Call Home event (assuming that Call Home is enabled).

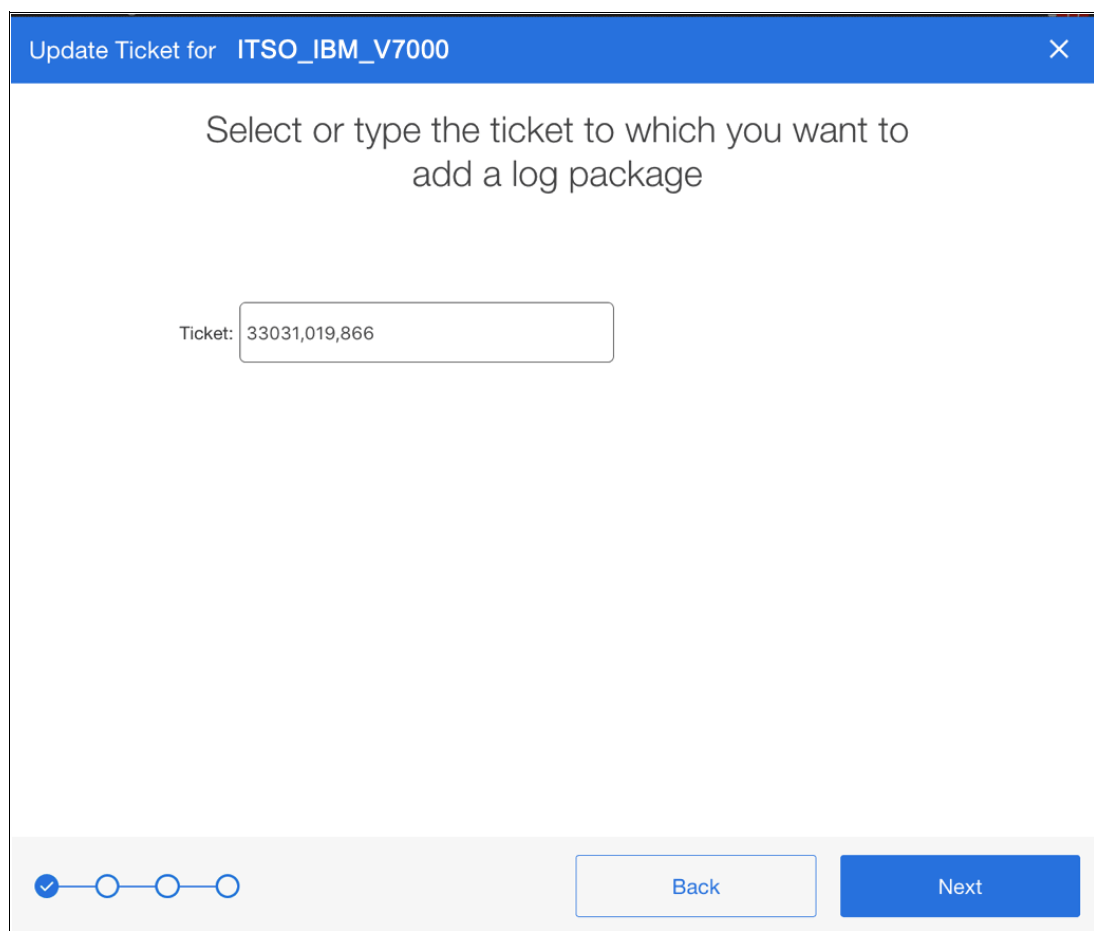


Figure 11-20 Entering the PMR ticket number

4. Click **Next**. A window opens in which you must choose the log type to upload. Figure 11-21 on page 507 shows the log selection window and the following available options:

- Type 1 - Standard logs, which is used For general problems, including simple hardware and simple performance problems.
- Type 2 - Standard logs and the most recent statesave log.
- Type 3 - Standard logs and the most recent statesave log from each node. Used for 1195 and 1196 node errors and 2030 software restart errors.
- Type 4 - Standard logs and new statesave logs. For complex performance problems, and problems with interoperability of hosts or storage systems, compressed volumes, and Remote Copy operations including 1920 errors.

Update Ticket 33031,019,866 for ITSO\_IBM\_V7000

Click the type of log package to add

Fastest ☒ Type 1 ☐ Type 2 ☐ Type 3 ☐ Type 4 Slowest

**Type 1: Standard logs**  
For general problems, including simple hardware and simple performance problems.

Language

✓ — — —

Back Next

Figure 11-21 Log type selection

If you are unsure which log type to upload, ask IBM Support for guidance. The most common type to use is Type 1; therefore, this type is the default. The other types are more detailed logs and for issues in order of complexity.

5. After selecting the type of logs, click **Next**. The log collection and upload starts. When completed, the log completion window opens.

### 11.6.5 SI Advisor

IBM Storage Insights continually evolves and the latest addition is a new option from the action menu called **Advisor**.

IBM Storage Insights analyzes your device data to identify violations of best practice guidelines and other risks, and to provide recommendations about how to address these potential problems. Select the system from the dashboard and then click the **Advisor** option to view these recommendations. To see details of a recommendation or to acknowledge it, double-click the recommendation.

Figure 11-22 shows the initial SI advisor menu.

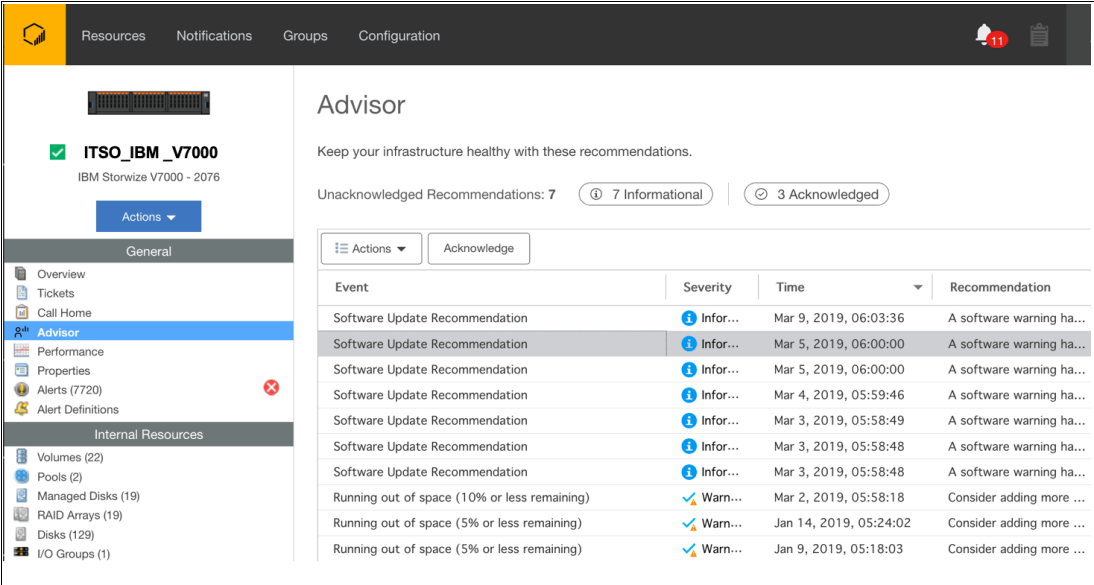


Figure 11-22 SI Advisor menu

Figure 11-23 shows an example of the detailed SI Advisor recommendations.

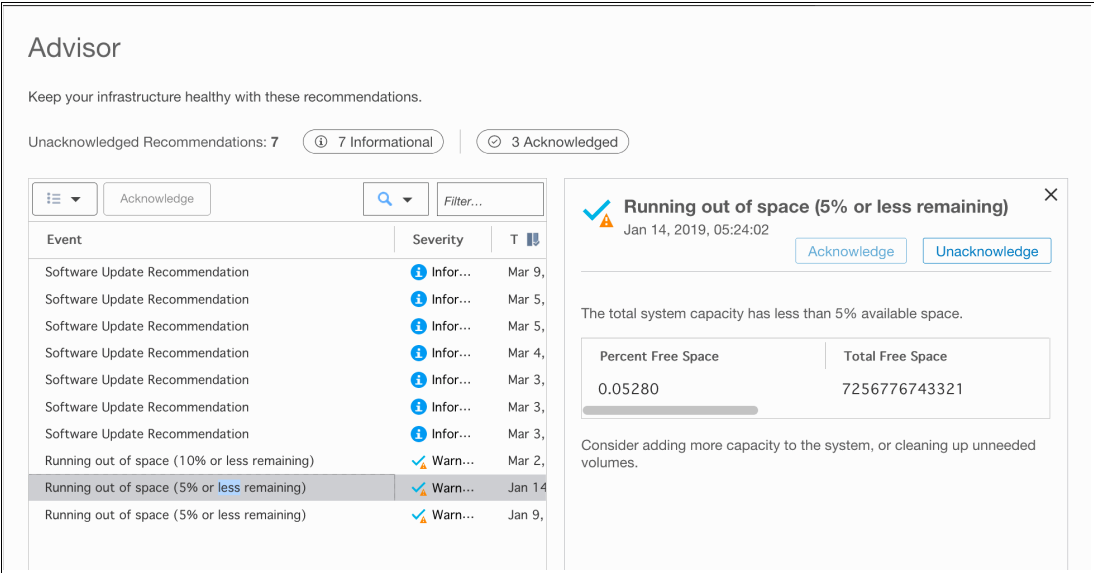


Figure 11-23 Advisor detailed summary of recommendations

The image shows the details of a “Running out of space” recommendation on the Advisor page. In this scenario, the user clicked the Warning tag to focus only on recommendations that have a severity of “Warning.”

For more information about setting and configuring the Advisor options, see [this web page](#).



# IBM Real-time Compression

This chapter highlights the preferred practices for IBM Real-time Compression (RtC) that uses IBM Spectrum Virtualize software that is installed on IBM SAN Volume Controller, IBM Storwize family, and IBM FlashSystem V9000. The main goal is to provide compression users with guidelines and factors to consider to achieve the best performance results and enjoy the compression savings that the RtC technology offers. The RtC technology is not to be confused with data compression implemented with Data Reduction Pools (DRP), which is covered in Chapter 4, “Storage pools” on page 91.

This chapter assumes that the reader is familiar with IBM Spectrum Virtualize RtC technology. For more information about this technology, see the following publications:

- ▶ *IBM Real-time Compression in IBM SAN Volume Controller and IBM Storwize V7000*, REDP-4859
- ▶ *Implementing IBM Real-time Compression in SAN Volume Controller and IBM Storwize V7000*, TIPS1083

**Important:** On 11 February 2020, IBM announced the arrival of SAN Volume Controller SA2 and SV2, and IBM FlashSystem 5X00, 7200, 9200 and 9200R to the family. These newer generations of hardware use improved hardware for the Quick Assist function.

RtC is no longer supported by the newer generation of hardware; therefore, volumes must be converted to decompressed or DRP compression before upgrading to the newer generation hardware.

This section was included for older versions of IBM SAN Volume Controller (DH8 and SV1) and Storwize V7000 Gen 2 or higher; however, some of the general principles also apply to DRP.

If you are in any doubt as to their applicability, work with your local IBM representative for clarification.

This book was comprehensively updated to include SAN Volume Controller SA2 and SV2 considerations. Guidance specific to the FlashSystem enclosure systems are provided in *IBM FlashSystem Best Practices and Performance Guidelines*, SG24-8503.

This chapter includes the following topics:

- ▶ 12.1, “Evaluating compression savings by using Comprestimator” on page 511
- ▶ 12.2, “Evaluating workload by using Disk Magic” on page 513
- ▶ 12.3, “Verifying available CPU resources” on page 513
- ▶ 12.4, “Configuring a balanced system” on page 514
- ▶ 12.5, “Standard benchmark tools” on page 515
- ▶ 12.6, “Compression with FlashCopy” on page 515
- ▶ 12.7, “Compression with Easy Tier” on page 515
- ▶ 12.8, “Compression on the backend” on page 516
- ▶ 12.9, “Migrating generic volumes” on page 516
- ▶ 12.10, “Converting RtC volumes to DRP compression” on page 517

## 12.1 Evaluating compression savings by using Comprestimator

Before you use RtC or DRP compression technology, it is important to understand the typical workloads you have in your environment. You need to determine whether these workloads are a good candidate for compression. Then, plan to implement workloads that are suitable for compression.

To determine the compression savings you are likely to achieve for the workload type, IBM developed an easy-to-use utility called *IBM Comprestimator*. The utility uses advanced mathematical and statistical algorithms to perform the sampling and analysis process in a short and efficient way. The utility also displays its accuracy level by showing the maximum error range of the results based on the internal formulas. The utility performs only read operations, so it has no effect on the data that is stored on the device.

From IBM Spectrum Virtualize V7.6, the Comprestimator utility can be used directly from the IBM Spectrum Virtualize shell. Example 12-1 shows the CLI commands to use the utility.

*Example 12-1 Estimating compression savings from the CLI*

---

```
IBM_2145:Spectrum_Virtualize_Cluster:user>analyzevdisk 0
IBM_2145:Spectrum_Virtualize_Cluster:user>lsvdiskanalysisprogress
vdisk_count pending_analysis estimated_completion_time
1           1                161014214700
IBM_2145:Spectrum_Virtualize_Cluster:user>lsvdiskanalysis -nohdr
0 vdisk0 sparse 161014214659 100.00GB 0.00MB 0.00MB 0 0.00MB 0.00MB 0 0.00MB 0 0
```

---

From IBM Spectrum Virtualize V7.7, the Comprestimator utility can be used directly from the IBM Spectrum Virtualize GUI. Figure 12-1 on page 512 shows how to start a system-wide analysis of compression estimates by clicking **Volumes** → **Actions** → **Space Savings** → **Estimate Compression Savings**.

**Note:** With the Spectrum Virtualize 8.4 code, the Comprestimator is now always on and constantly generating data. It provides instantaneous access to that information by way of the GUI, which eliminates the need to wait for sufficient sampling to provide meaningful data.

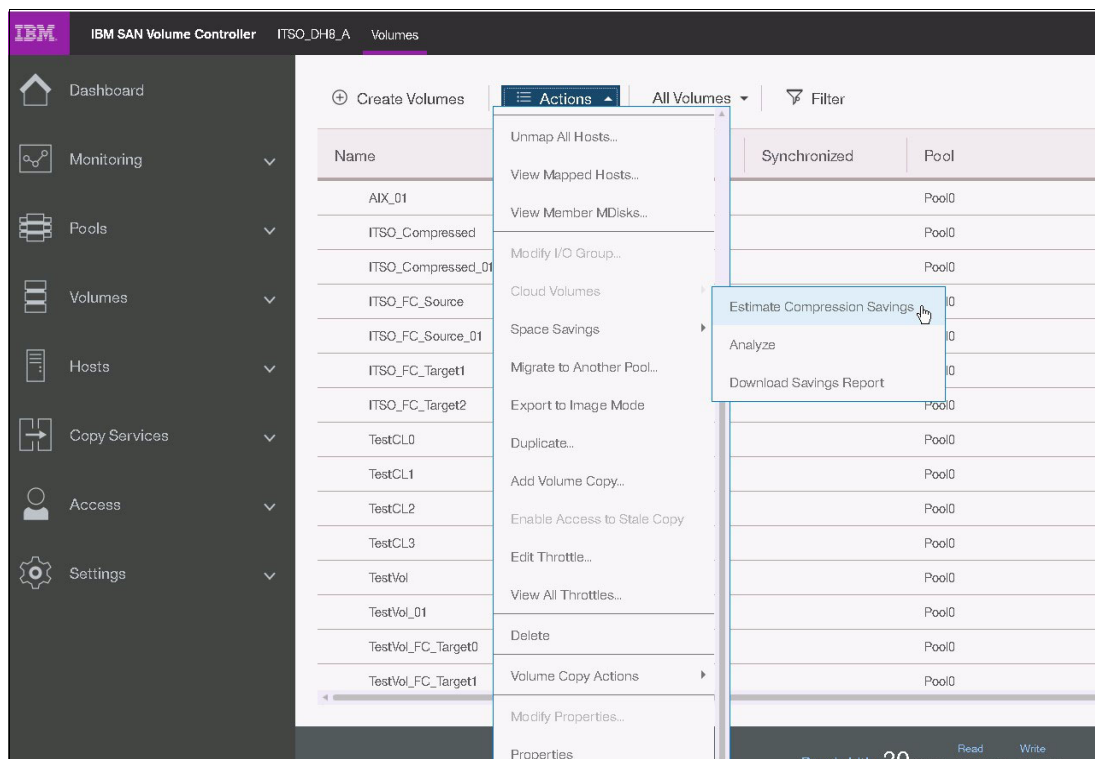


Figure 12-1 Estimating compression savings from the GUI

If an older IBM Spectrum Virtualize version is used or if you want to estimate the compression savings of a different storage system before changing to IBM Spectrum Virtualize, the Comprestimator utility can be installed on a host that has access to the devices that are to be analyzed.

The following preferred practices are suggested for the use of the Comprestimator:

- ▶ Run the Comprestimator utility before you implement an IBM Spectrum Virtualize solution and before you implement RtC technology.
- ▶ Download [latest version of the utility](#) from IBM if you are not using the version included with IBM Spectrum Virtualize.
- ▶ Use Comprestimator to analyze volumes that contain as much active data as possible rather than volumes that are mostly empty. This technique increases the accuracy level and reduces the risk of analyzing old data that is deleted but might still have traces on the device.

**Note:** Comprestimator can run for a long period (a few hours) when it is scanning a relatively empty device. The utility randomly selects and reads 256 KB samples from the device. If the sample is empty (that is, full of null values), it is skipped. A minimum number of samples with actual data are required to provide an accurate estimation.

When a device is mostly empty, many random samples are empty. As a result, the utility runs for a longer time as it tries to gather enough non-empty samples that are required for an accurate estimate. If the number of empty samples is over 95%, the scan is stopped.



- Use Table 12-1 thresholds for volume compressibility to determine whether to compress a volume.

Table 12-1 Thresholds for RtC implementation

|                                                                                                              | Data Compression Rate     | Recommendation    |
|--------------------------------------------------------------------------------------------------------------|---------------------------|-------------------|
| <b>On products that have Quick Assist compression acceleration cards installed and are on V7.4 and later</b> | > 40% compression savings | Use compression   |
|                                                                                                              | < 40% compression savings | Evaluate workload |
| <b>On all other products</b>                                                                                 | > 25% compression savings | Use compression   |
|                                                                                                              | < 25% compression savings | Evaluate workload |

## 12.2 Evaluating workload by using Disk Magic

Proper initial sizing greatly helps to avoid future sizing problems. Disk Magic is one such tool that is used for sizing and modeling storage subsystems for various open systems environments and various IBM platforms. It provides accurate performance and capacity analysis and planning for IBM Spectrum Virtualize products, other IBM storage solutions, and other vendors' storage subsystems. Disk Magic provides in-depth environment analysis, and is an excellent tool to estimate the performance of a system that is running RtC.

If you are an IBM Business Partner, for more information about Disk Magic, see [this web page](#).

If you are an IBM customer, ask an IBM representative to evaluate the workload of your storage environment when implementing an IBM Spectrum Virtualize RtC solution.

## 12.3 Verifying available CPU resources

Before compression is enabled on IBM Spectrum Virtualize systems, measure the current system utilization to ensure that the system has the CPU resources that are required for compression.

Compression is recommended for an I/O Group if the sustained CPU utilization is *below* the per-node values that are listed in Table 12-2. For node types for which the value listed is N/A, RtC can be implemented with no consideration regarding CPU utilization because these node types use dedicated CPU resources for RtC.

Table 12-2 CPU resources recommendations

| SAN Volume Controller |     | Storwize |            |                  | IBM Spectrum Virtualize Software |
|-----------------------|-----|----------|------------|------------------|----------------------------------|
| DH8 (Dual CPU)        | SV1 | V5030    | V7000 Gen3 | V7000 Gen2/Gen2+ |                                  |
| N/A                   | N/A | 30%      | N/A        | 50%              | 30%                              |

If any node in an I/O Group features sustained processor usage that is greater than the values that are listed Table 12-2 on page 513, do not create compressed volumes in this I/O Group. Doing so might affect existing non-compressed volumes that are owned by this I/O Group. If it is an option, add I/O groups. If you have any questions, speak to your IBM representative.

Customers who are planning to use RtC on V7000 Gen2/Gen2+ should install the extra Quick Assist compression acceleration card per node canister for better performance.

**Note:** To use the RtC feature on SAN Volume Controller DH8 and SV1 nodes, at least one Quick Assist compression acceleration card is required. To use the IBM RtC feature on the V9000 system, both Quick Assist compression acceleration cards are required.

## 12.4 Configuring a balanced system

In a system with more than one I/O group, it is important to balance the compression workload. Even in a single I/O group configuration, it is important to be aware of preferred node assignment of compressed volumes to ensure that one node is not overloaded compared to the other.

Consider a four-node (two I/O groups) IBM Spectrum Virtualize system with the following configuration:

- ▶ iogrp0: nodes 1 and 2 with 18 compressed volumes
- ▶ iogrp1: nodes 3 and 4 with two compressed volumes

This setup is not ideal because CPU and memory resources are dedicated for compression use in all four nodes. However, in nodes 3 and 4, this allocation is used only for serving two volumes out of a total of 20 compressed volumes. The following preferred practices in this scenario should be used:

- ▶ Alternative 1: Migrate all compressed volumes from iogrp1 to iogrp0 when only a few compressed volumes exist (that is, 10 - 20).
- ▶ Alternative 2: Migrate compressed volumes from iogrp0 to iogrp1 and balance the load across nodes when many compressed volumes exist (that is more than 20).

Table 12-3 lists the load distribution for each alternative.

*Table 12-3 Load distribution*

|                | node1 volumes                     | node2 volumes                     | node3 volumes                    | node4 volumes                    |
|----------------|-----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| Original setup | 9 compressed<br>X non-compressed  | 9 compressed<br>X non-compressed  | 1 compressed<br>X non-compressed | 1 compressed<br>X non-compressed |
| Alternative 1  | 10 compressed<br>X non-compressed | 10 compressed<br>X non-compressed | X non-compressed                 | X non-compressed                 |
| Alternative 2  | 5 compressed<br>X non-compressed  | 5 compressed<br>X non-compressed  | 5 compressed<br>X non-compressed | 5 compressed<br>X non-compressed |

## 12.5 Standard benchmark tools

Traditional block and file-based benchmark tools (such as IOmeter, IOzone, dbench, and fio) that generate truly random but not realistic I/O patterns do not run well with RtC.

These tools generate synthetic workloads that do not have any temporal locality. Data is not read back in the same (or similar) order in which it was written. Therefore, it is not useful to estimate what your performance looks like for an application with these tools.

Consider what data a benchmark application uses. If the data is compressed or it is all binary zero data, the differences that are measured are artificially bad or good, based on the compressibility of the data. The more compressible the data, the better the performance.

## 12.6 Compression with FlashCopy

By using the FlashCopy function of IBM Storage Systems, you can create a point-in-time copy of one or more volumes. You can use FlashCopy to solve critical and challenging business needs that require duplication of data on your source volume. Volumes can remain online and active while you create consistent copies of the data sets.

Follow these general guidelines:

- ▶ Consider configuring FlashCopy targets as non-compressed volumes. In some cases, the savings are not worth the other resources that are required because the FlashCopy target holds only the “split” grains that are backing the grains that were changed in the source. Therefore, total FlashCopy target capacity is a fraction of the source volume size.
- ▶ FlashCopy default grain size is 256 KB for non-compressed volumes and 64 KB for compressed volumes (new defaults from V6.4.1.5 and V7.1.0.1 and later). Use the default grain size for FlashCopy with compressed volumes (64 KB) because this size reduces the performance effect when compressed FlashCopy targets are used.
- ▶ Consider the use of the background copy method. There are two ways to use FlashCopy: With or without background copy. When it is used without background copy, the host I/O is pending until the split event is finished (copy-on-write process). For example, if the host sends a 4 KB write, this I/O waits until the corresponding grain (64 KB or 256 KB) is read and decompressed.

It is then written to FlashCopy target copy. This configuration adds latency to every I/O. When background copy is used, all the grains are copied to the FlashCopy target right after the FlashCopy mapping is created. Although the configuration adds latency during the copy, it eliminates latency after the copy is complete.

## 12.7 Compression with Easy Tier

IBM Easy Tier is a performance function that automatically and non-disruptively migrates frequently accessed data from magnetic media to solid-state drives (SSDs). In that way, the most frequently accessed data is stored on the fastest storage tier and the overall performance is improved.

Beginning with V7.1, Easy Tier supports compressed volumes. A new algorithm is implemented to monitor read operations on compressed volumes instead of reads and writes. The extents with the most read operations that are smaller than 64 KB are migrated to SSD MDisk.

As a result, frequently read areas of the compressed volumes are serviced from SSDs. Easy Tier on non-compressed volumes operates as before and it is based on read and write operations that are smaller than 64 KB.

For more information about implementing IBM Easy Tier with IBM RtC, see *Implementing IBM Easy Tier with IBM Real-time Compression*, TIPS1072.

## 12.8 Compression on the backend

If you have an IBM Spectrum Virtualize system setup with some back-end storage that supports compression (such as a Storwize product) and you plan to implement compression, configure compression volumes on the IBM Spectrum Virtualize system, not on the backend storage. This configuration minimizes I/O to the backend storage.

From V7.3, the existence of a lower-level write cache below the RtC component in the software stack allows for the coalescing of compressed writes. As a result, an even bigger reduction in back-end I/Os is achieved because of the ability to perform full-stride writes for compressed data.

**Note:** These recommendations do not apply with back-end controller with inherent compression mechanisms, such as IBM FlashSystem series systems that use FlashCore Modules with hardware compression. This configuration is designed to compress without any performance impact.

Do not enable compression on both IBM Spectrum Virtualize and back-end storage systems.

## 12.9 Migrating generic volumes

It is possible to migrate non-compressed volumes, both generic (fully allocated) or thin-provisioned, to compressed volumes by using volume mirroring. When migrating generic volumes that are created without initial zero formatting, extra considerations need to be taken into account. These volumes might contain traces of old data at the block device level. Such data is not accessible or viewable in the file system level. However, it might affect compression ratios and system resources during and after migration.

When the Comprestimator utility is used to analyze such volumes, the expected compression results reflect the compression rate for all the data in the block device level. This data includes the old data. This block device behavior is limited to generic volumes, and does not occur when using Comprestimator to analyze thin-provisioned volumes.

The second issue is that old data is also compressed. Therefore, system resources and system storage space are wasted on compression of old data that is effectively inaccessible to users and applications.

**Note:** Regardless of the type of block device that is analyzed or migrated, it is also important to understand a few characteristics of common file systems space management.

When data is deleted from a file system, the space that it occupied before it was deleted is freed and available to the file system. It is available even though the data at block device level was not deleted. When using Comprestimator to analyze a block device or when migrating a volume that is used by a file system, all underlying data in the device is analyzed or migrated regardless of whether this data belongs to files that were deleted from the file system. This process affects even thin-provisioned volumes.

There is no solution for existing generic volumes that were created without initial zero formatting. Migrating these volumes to compressed volumes might still be a good option and should not be discarded.

As a preferred practice, always format new volumes during creation. This process zeros all blocks in the disks and eliminates traces of old data. This is the default behavior from V7.7.

## 12.10 Converting RtC volumes to DRP compression

The following factors compel the conversion of RtC volumes to DRP technology instead:

- ▶ DRP compression was rewritten to scale much better than RtC.
- ▶ RtC is no longer supported on the newer generation of platforms that use improved hardware for the Quick Assist functions.
- ▶ DRP is the go-forward direction for Spectrum Virtualize capacity savings.
- ▶ RtC requires the dedication of CPU cores to the function of compression. However, since Spectrum Virtualize 8.1, when as all RtC volumes are migrated to decompressed or DRP compressed, the formerly dedicated cores can be used to service I/O.

Conversion is accomplished by way of volume mirroring, just like the process for converting non-compressed to compressed volumes. One obvious difference persists; that is, the conversion can occur in a single storage pool or **mdiskgrp**. Conversion from RtC necessarily requires that the destination DRP copy for the volume must be in a **mdiskgrp** that is a DRP.

**Note:** RtC *cannot* coexist in the same I/O group as DRP *deduplication*. If extra capacity savings through DRP deduplication are wanted, all RtC volumes in the I/O group must be converted before those volumes can then be converted again to use deduplication. Unfortunately, DRP volumes are not currently supported for Non-disruptive Volume Move to change I/O groups.





## Best practices for Spectrum Virtualize for Public Cloud in IBM Cloud

This chapter provides an overview of the Spectrum Virtualize for Public Cloud offering when hosted in the IBM Cloud. It includes following topics:

- ▶ 13.1, “Base architecture” on page 520
- ▶ 13.2, “Networking” on page 521
- ▶ 13.3, “Hosts” on page 522
- ▶ 13.4, “Storage” on page 522
- ▶ 13.5, “Replication” on page 524

## 13.1 Base architecture

The IBM Cloud Infrastructure as a Service (IaaS) offering provides a robust environment, as shown in Figure 13-1.

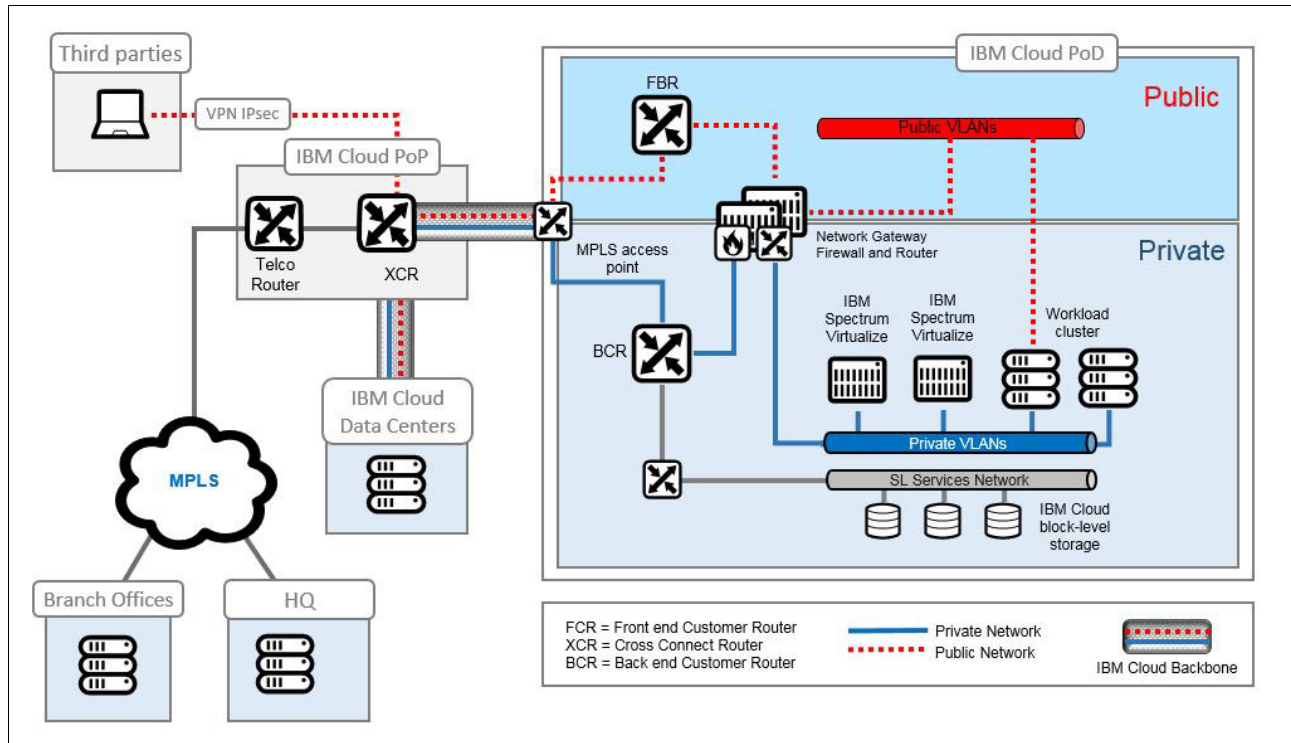


Figure 13-1 IBM Cloud networking

In this design, the Spectrum Virtualize nodes are bare metal servers from IBM Cloud's IaaS offering. It uses the private network for host access, storage access, and inter-node messaging.

For more information about Spectrum Virtualize for Public Cloud in the IBM Cloud, see *Implementing IBM Spectrum Virtualize for Public Cloud Version 8.3.1*, [REDP-5602](#).



## 13.2 Networking

Likely the most critical aspect of the Spectrum Virtualize for Public Cloud in IBM Cloud solution is the networking. The Spectrum Virtualize architecture heavily depends on network stability, reliability, and performance to provide optimal performance to the user.

When initially creating the system, the following basic methods can be used to create the system:

- ▶ Fully Automated
- ▶ Partially Automated
- ▶ Manual

When the Fully Automated installation procedure is used, a portable subnet is created on the private VLAN to which the bare metal servers are attached. This subnet is the network that is used for all Spectrum Virtualize services. At the time of this writing, the installer provisions a /26 network that contains 64 usable IP addresses and assigns five IP addresses per node

When performing the Partially Automated or Manual installation, you must have the portable private subnet created. This subnet must be large enough to hold all the IP addresses that are required for the Spectrum Virtualize cluster.

Portable subnets in the IBM Cloud are accessible to all resources on the same VLAN. If your environment spans beyond the addressable range of the subnet that is used for Spectrum Virtualize, or if the fully automated installation is used, it is suggested to define one portable private subnet for the Spectrum Virtualize cluster and extra portable private subnets on the same VLAN for hosts.

### 13.2.1 Node networking

The nodes that are provisioned in the IBM Cloud are provisioned by using dual port 10-Gb unbonded links. The Spectrum Virtualize software sees a total of two 10-Gb ports per node on the IBM Cloud Private Network, as shown in Figure 13-2.

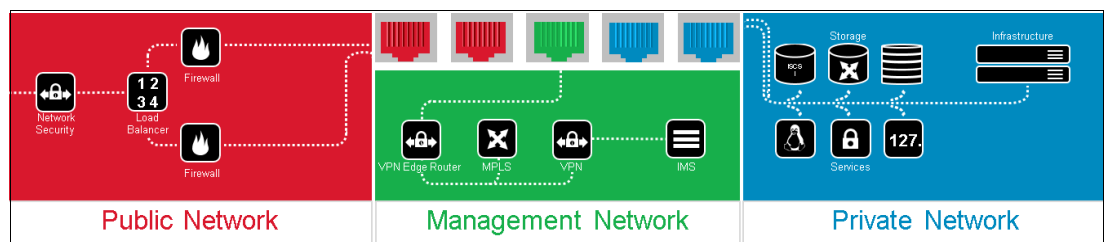


Figure 13-2 IBM Cloud Server Port Allocation

IBM Spectrum Virtualize does not use the public interfaces in any way. For security purposes, it is recommended to disable the public interfaces on the bare metal servers that are used as Spectrum Virtualize Nodes.

The management network is used by IBM Cloud infrastructure monitoring services. This interface is not visible to the user, IBM Spectrum Virtualize. It also is not externally facing.

The Ethernet ports that are connected to the Private Network on the nodes are shared for all uses and services in the cluster. Meaning management, inter-node messaging, hosts, storage, and replication all share the interfaces.

## 13.3 Hosts

In most cases, hosts in the IBM Cloud include one or two ports on the IBM Cloud Private Network. These ports connect to the Spectrum Virtualize Port IP addresses. As with most hosts, we recommend the use of a maximum of eight paths from a host to a volume. This configuration is easily achieved by applying the following calculation:

$$\text{paths} = \text{host\_port\_count} \times \text{storage\_port\_count}$$

In this solution, the storage port count almost always is 4 (2 per node). Therefore, to maintain a maximum of eighty paths, limit the number of Ethernet ports on the host that are used for storage access to two.

In an iSCSI configuration, we often recommend setting the MTU to 9000 to enable jumbo frames. However, the IBM Cloud Private network uses the default MTU of 1500. As such, it is not advisable to enable jumbo frames on the Spectrum Virtualize ports or the hosts.

For more information about best practice configurations for various host platforms, see the section for your specific operating system.

## 13.4 Storage

IBM Cloud provides IBM Spectrum Virtualize with flash-backed block storage on high-performance iSCSI targets. The storage is presented as a block-level device that customers can format to best fit their needs. The iSCSI storage is on the private network and does not count toward public and private bandwidth allotments.

The options that are available are Performance with granular I/O operations per second (IOPS) increments of 1,000 - 48,000 or Endurance with predefined IOPS per gigabyte.

Both options are available as volumes that are sized 20 GB - 12 TB. All volumes are encrypted by default with IBM Cloud-managed encryption.

**Note:** IBM Cloud Endurance and Performance volumes are no different from a technical perspective when used as back-end storage for IBM Spectrum Virtualize on IBM Cloud. After the IOPS profile fits the application requirements from an IBM Spectrum Virtualize perspective, the two solutions are identical. The only notable advantage is the IBM Cloud Performance storage granularity during the definition of the IOPS profile, which provides a more accurate capability estimation and minimizes waste.

### 13.4.1 Discovery and access

The Spectrum Virtualize for Public Cloud offering in the IBM Cloud uses standard iSCSI virtualization of a block storage device. The best practices for discovering the block storage to be virtualized was automated by using the Add Cloud Storage wizard, which is available in the GUI under the external storage section. This wizard guides you through the process of adding IBM Cloud Block Storage, as shown in Figure 13-3 on page 523.

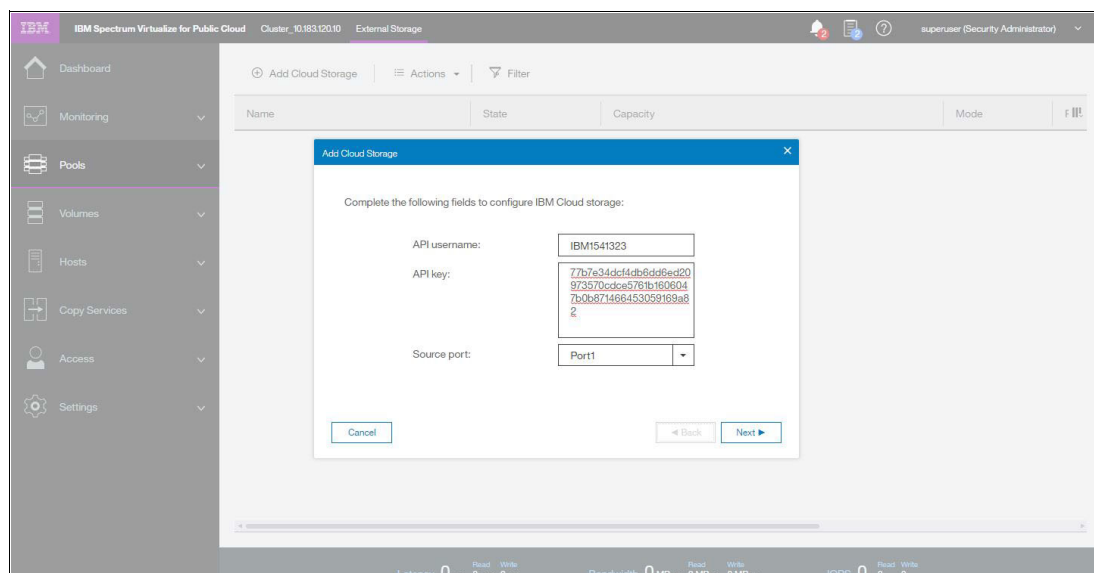


Figure 13-3 Add Cloud Storage

In this wizard, you must specify a source port in which the Spectrum Virtualize nodes communicate with the selected volumes. It is suggested to add MDisks to the cluster in a round-robin fashion to optimize port bandwidth usage and maximize performance.

## 13.4.2 EasyTier considerations

As with most Spectrum Virtualize solutions, it is possible to configure back-end storage devices with different performance characteristics within the same pool. However, all MDisks are marked as an Enterprise tier when discovered.

It is suggested to use define and use a standard set of capabilities and analogize them to the traditional tiers seen in typical IBM SAN Volume Controller environments. An example tiering scheme is provided in Table 13-1.

Table 13-1 An example profile of storage classes

| Class        | Capacity (TB) | I/O Density (IOPS/GB) | Maximum IOPS |
|--------------|---------------|-----------------------|--------------|
| Tier 0 Flash | 2             | 20                    | 40,000       |
| Tier 1 Flash | 2             | 10                    | 20,000       |
| Enterprise   | 2             | 5                     | 10,000       |
| Nearline     | 2             | 2                     | 4,000        |

Defining a sample scheme such as the one that is listed in Table 13-1 allows you to estimate and acquire block storage LUNs by using Performance or Endurance storage. It also allows you to assess which option offers the best price for the performance.

For more information about EasyTier behavior and best practices, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.

## 13.5 Replication

Because the solution makes use of the private network only, a connection to the IBM Cloud private network must be established to replicate between Spectrum Virtualize for Public Cloud in the IBM Cloud. The IBM Cloud offers the following options to connect external networks to the IBM Cloud private network:

- ▶ IBM Cloud Gateway Appliance
- ▶ IBM Cloud Direct Link

The IBM Cloud Direct Link is the preferred method of connecting to the IBM Cloud Private Network because it typically provides the best performance with lower costs compared to the Gateway Appliance. Carefully evaluate your bandwidth, budget, and usage requirements before purchasing or implementing the inter-site link.

For more information about the use of the IBM Cloud Direct Link, see [this web page](#).

For more information about the use of the IBM Cloud Gateway Appliance, see [this web page](#).

After the connectivity between the IBM Cloud Private Network and the remote network is established, the best practices for configuring the Spectrum Virtualize systems are consistent with the recommendations for Native IP Replication as described in 6.4, “Native IP replication” on page 303,

The only exception for the Spectrum Virtualize system configuration is that in the IBM Cloud, the number of ports is limited, therefore, the replication traffic is shared with host and storage traffic.



## Best practices for Spectrum Virtualize for Public Cloud in AWS

This chapter provides an overview of the Spectrum Virtualize for Public Cloud offering when hosted in Amazon Web Services (AWS). It includes the following topics:

- ▶ 14.1, “Base architecture” on page 526
- ▶ 14.2, “System resources” on page 526
- ▶ 14.3, “Back-end storage” on page 527
- ▶ 14.4, “Replication considerations” on page 528

## 14.1 Base architecture

Spectrum Virtualize for Public Cloud is deployed on Amazon EC2 compute instances with a base installation that is fully automated in a controlled topology. The base deployment uses an availability zone for the Spectrum Virtualize nodes and a second availability zone for the IP quorum host (which is also the bastion host), as shown in Figure 14-1.

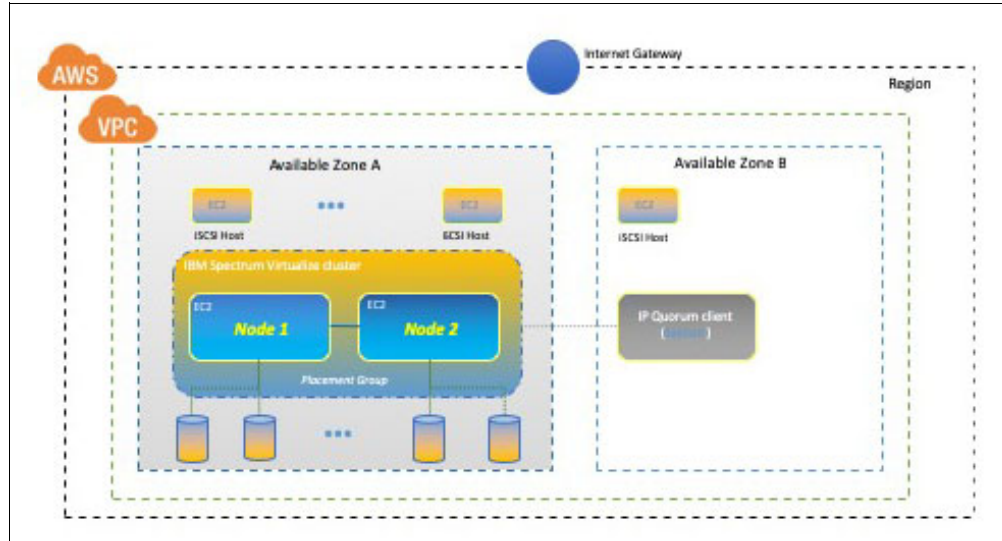


Figure 14-1 AWS base architecture

## 14.2 System resources

The network bandwidth, number of vCPUs, and amount of memory are determined by instance type. The AWS instances (C5.4xlarge, C5.9xlarge, and C5.18xlarge) are the available options at the time of this writing. The technical specifications are listed in Table 14-1.

Table 14-1 Amazon AWS EC2 on-demand resources

| EC2 instance | vCPU | Memory (GiB) | Dedicated EBS bandwidth (Mbps) | Network performance (Gbps) |
|--------------|------|--------------|--------------------------------|----------------------------|
| c5.4xlarge   | 16   | 32           | 3.500                          | Up to 10                   |
| c5.9xlarge   | 36   | 72           | 7.000                          | 10                         |
| c5.18xlarge  | 72   | 144          | 14.000                         | 25                         |

The selection of which EC2 instance to use must be done with care because no upgrade path is available after deployment (at the time of this writing). Because of this issue, it is suggested to deploy the largest configuration that the environment is expected to grow into.

If you must upgrade to a larger EC2 instance, you must deploy a new cluster and replicate the data to the new system.

## 14.3 Back-end storage

The Spectrum Virtualize nodes virtualize Amazon EBS storage to serve as MDisk. Because EBS storage can be presented only to a single EC2 instance at a time, the Spectrum Virtualize nodes perform MDisk forwards as required to access data on the back-end storage.

Different volume types are available for Amazon EBS. They differ in performance characteristics, as listed in Table 14-2.

Table 14-2 EBC volume types

| Item                             | SSDs                |                      | HDDs                     |          |
|----------------------------------|---------------------|----------------------|--------------------------|----------|
| Volume type                      | General-purpose SSD | Provisioned IOPS SSD | Throughput-optimized HDD | Cold HDD |
| API name                         | gp2                 | io1                  | st1                      | sc1      |
| Max IOPS / volume                | 16,000              | 64,000               | 500                      | 250      |
| Max throughput / volume in MiBps | 250                 | 1,000                | 500                      | 250      |

All volume types appear as “Enterprise Disks” tier in IBM Spectrum Virtualize for Public Cloud and the tier level is adapted afterward according to their capabilities. The assignment to the suitable Easy Tier level that is listed in Table 14-3 is a best practice and must be adjusted manually by using the `chmdisk -tier` command.

Table 14-3 AWS assignment to Easy Tier level

| Drive                    | AWS volume type | Easy Tier level | easytierload command flag |
|--------------------------|-----------------|-----------------|---------------------------|
| Provisioned IOPS SSD     | io1             | tier0_flash     | high                      |
| General-purpose SSD      | gp2             | tier1_flash     | low                       |
| Throughput-optimized HDD | st1             | tier_nearline   | low                       |

For more information about EasyTier behavior and best practices, see 4.6, “Easy Tier, tiered, and balanced storage pools” on page 148.

## 14.4 Replication considerations

Replication in the AWS infrastructure can be performed by creating an MPLS direct link to the VPC that contains the Spectrum Virtualize nodes or by using an AWS Site-to-Site VPN.

For more information about AWS Direct Links, see the [AWS documentation](#).

For more information about AWS Site-to-Site VPN usage, see the [AWS documentation](#).

After the connectivity is established, the best practices for configuring the Spectrum Virtualize systems are consistent with the recommendations for Native IP Replication as described in 2.5.5, “Native IP replication” on page 65.





## IBM i considerations

The IBM Spectrum Virtualize family of block storage systems, including the IBM SAN Volume Controller, IBM Flash System 5000 series, IBM FlashSystem 7200, and IBM FlashSystem 9200/9200R provides a broad range of flexible and scalable SAN storage solutions. These solutions can meet the demands of IBM i customers who are entering high-end storage infrastructure solutions.

All family members based on IBM Spectrum Virtualize software use a common management interface, and provide a comprehensive set of advanced functions and technologies, such as advanced Copy Services functions, encryption, compression, storage tiering, NVMe flash and storage class memory (SCM) devices, and external storage virtualization. Many of these advanced functions and technologies are also of interest to IBM i customers who are looking for a flexible, high performing, and highly available SAN storage solution.

This appendix provides important considerations and guidelines for successfully implementing the IBM Spectrum Virtualize family and its advanced functions with IBM i. Unless otherwise stated, the considerations also apply to previous generations of products, such as the IBM Storwize family, the IBM Flash System 9100 series, and IBM Flash System V9000.

This appendix includes the following topics:

- ▶ IBM i Storage management
- ▶ Single-level storage
- ▶ IBM i response time
- ▶ Planning for IBM i storage capacity
- ▶ Storage connection to IBM i
- ▶ Setting attributes in VIOS
- ▶ Guidelines for Virtual I/O Server resources
- ▶ Defining LUNs for IBM i
- ▶ Data layout
- ▶ Fibre Channel adapters in IBM i and VIOS
- ▶ Zoning SAN switches
- ▶ IBM i Multipath
- ▶ Booting from SAN
- ▶ IBM i mirroring
- ▶ Copy services considerations
- ▶ SAN Volume Controller stretched cluster
- ▶ DB2 mirroring for IBM i

## IBM i Storage management

Because of the unique IBM i storage architecture, special considerations for planning and implementing a SAN storage solution are required (also with IBM Spectrum Virtualize-based storage). This section describes how IBM i storage management manages its available disk storage. Many host systems require the user to take responsibility for how information is stored and retrieved from the disk units. An administrator must also manage the environment to balance disk usage, enable disk protection, and maintain balanced data to be spread for optimum performance.

The IBM i architecture is different in that the system takes over much of the storage management functions, which on other platforms are the responsibility of a system administrator.

IBM i, with its Technology Independent Machine Interface (TIMI), largely abstracts the underlying hardware layer from the IBM i operating system and its users and manages its system and user data in IBM i disk pools, which are also called *auxiliary storage pools* (ASPs).

When you create a file, you do not assign it to a storage location. Instead, the IBM i system places the file in the location that ensures the best performance from an IBM i perspective (see Figure A-1).

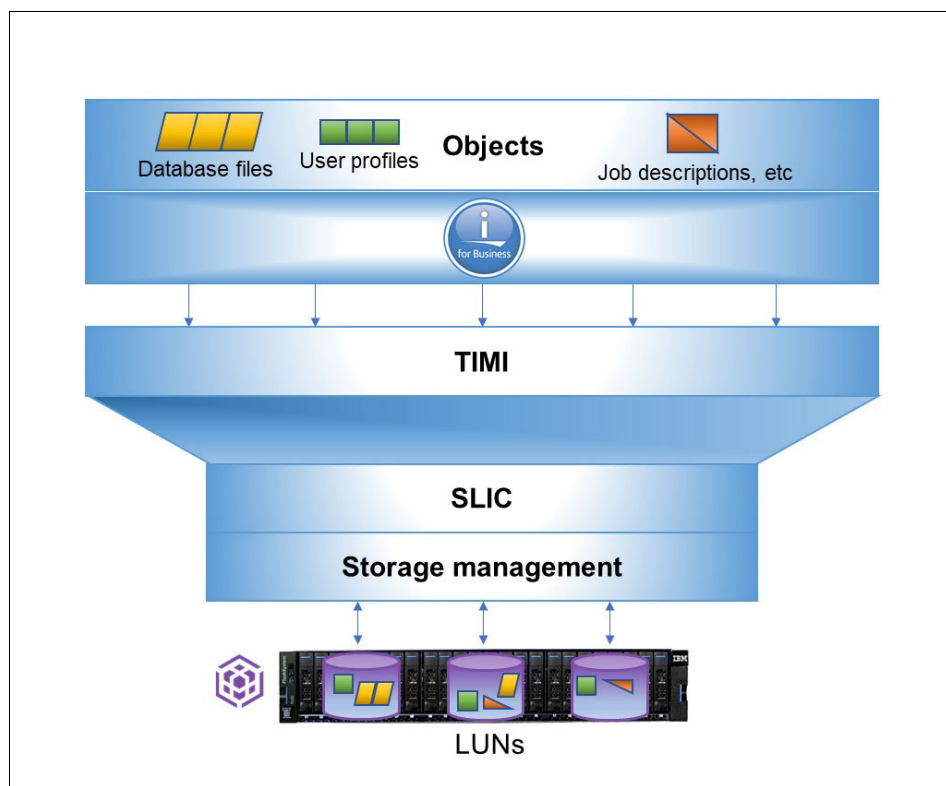


Figure A-1 IBM i storage management spreads objects across LUNs.

**Note:** When a program presents instructions to the machine interface for execution, the interface appears to the program as the system hardware, but it is not. The instructions that are presented to TIMI pass through a layer of microcode before they are understood by the hardware. Hence, TIMI and SLIC allow IBM Power Systems with IBM i to take technology in stride.

As a component of the IBM i System Licensed Internal Code (SLIC), IBM i storage management normally spreads the data in the file across multiple disk units (LUNs when external storage is used). When you add records to the file, the system automatically assigns more space on one or more disk units or LUNs.

## Single-level storage

IBM i uses a single-level storage, object-orientated architecture. It sees all disk space and the main memory or main storage as one address space. It also uses the same set of virtual addresses to cover main memory and disk space. Paging the objects in this virtual address space is performed in 4 KB pages, as shown in Figure A-2. After a page gets written to disk, it is stored with metadata, including its unique virtual address. For this purpose, IBM i originally used a proprietary 520 bytes per sector disk format.

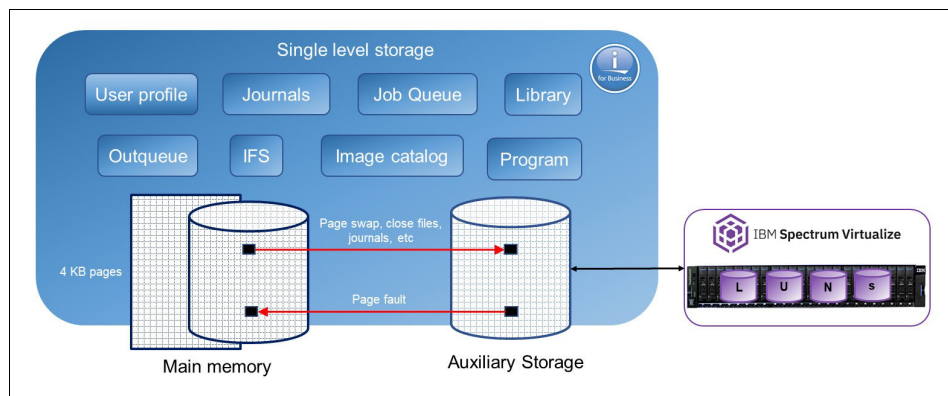


Figure A-2 Virtual address space.

**Note:** The system storage that is conformed with main storage or main memory and auxiliary storage is addressed in the same way. This single, device-independent addressing mechanism means that objects are referred to by name or name and library, *never* by disk location. The virtual addressing of IBM i is independent of the physical location of the object, type, capacity, and the number of disks units or LUNs on the system.

The IBM i disk storage space is managed by using auxiliary storage pools. Each IBM i system has a system ASP (ASP 1), which includes the load source (also known as *boot volume* on other systems) as disk unit 1, and optional user ASPs (ASP 2-33). The system ASP and the user ASPs are designated as SYSBAS and constitute the system database.

The single-level storage with its unique virtual addresses also implies that the disk storage that is configured in SYSBAS of an IBM i system must be available in its entirety for the system to remain operational. It cannot be shared for simultaneous access by other IBM i systems.

To allow for sharing of IBM i disk storage space between multiple IBM i systems in a cluster, switchable *independent auxiliary storage pools* (IASPs) can be configured. The IBM i auxiliary storage pools architecture is shown in Figure A-3.

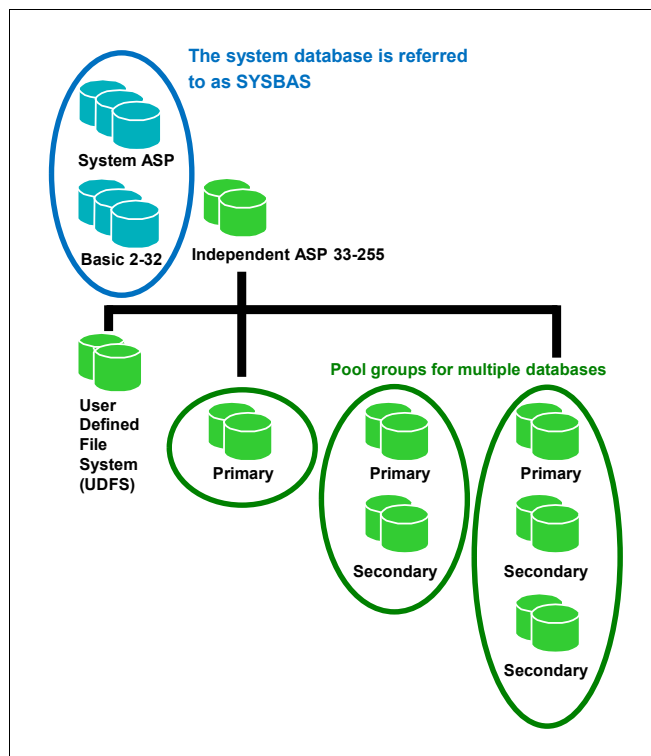


Figure A-3 IBM i auxiliary storage pools architecture

Single-level storage makes main memory work as a large cache. Reads are done from pages in main memory, and requests to disk are done only when the needed page is not there.

Writes are done to main memory or main storage, and write operations to disk are performed as a result of swap, file close, or forced write. Application response time depends on disk response time and many other factors.

Other storage-related factors include the IBM i storage pool configuration for the application, how frequently the application closes files, and whether it uses journaling. An example is shown in Figure A-4 on page 534.

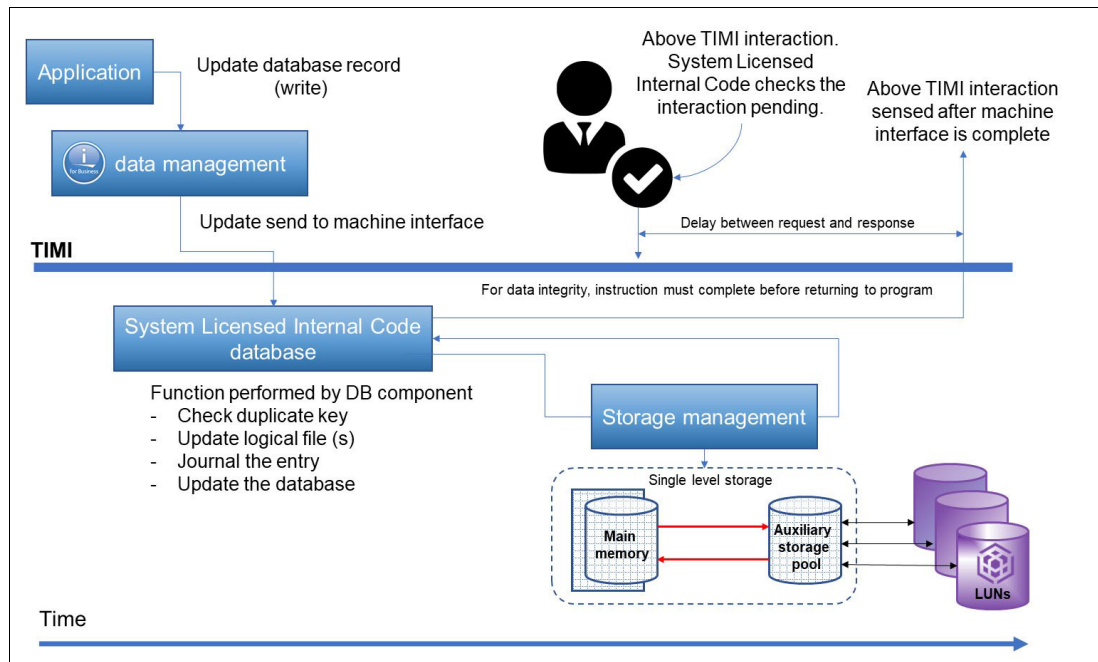


Figure A-4 TIMI atomicity

**Note:** In Figure A-5 on page 535, auxiliary storage pool (ASP) is conformed and assigned LUNs from IBM Spectrum Virtualize to the IBM i. It shows an application request and update to a database record. Throughout the time TIMI task is in progress, an interaction above TIMI can ensue. The interaction does not continue until the TIMI task concludes.

## IBM i response time

IBM i customers often are concerned about the following types of performance:

- ▶ **Application response time:** The response time of an application transaction. This time is usually critical for the customer.
- ▶ **Duration of batch job:** Batch jobs often are run during the night or other off-peak periods. The duration of a batch job is critical for the customer because it must be finished before regular daily transactions start.
- ▶ **Disk response time:** Disk response time is the time that is needed for a disk I/O operation to complete. It includes the service time for I/O processing and the wait time for potential I/O queuing on the IBM i host.

Disk response time can significantly influence application response time and the duration of a batch job. Because the performance of the disk subsystem has a significant effect on overall system performance, this issue is discussed next.

### Disk performance considerations

Disk subsystem performance has a significant effect on overall IBM i system performance, especially in a commercial data processing environment a large volume of data often must be processed. Disk drives or LUNs response times contribute to a major portion of the overall response time (OLTP) or runtime (batch).

Also, disk subsystem performance is affected by the type of protection (RAID, DRAID, or mirroring)

The amount of free space (GB) on the drives and the extent of fragmentation also has an effect. The reason is the need to find suitable contiguous space on the disks to create objects or extend objects. Disk space often is allocated in extents of 32 KB. If a 32 KB contiguous extent is not available, two extents of 16 KB are used.

The following disk performance considerations are discussed next:

- ▶ Disk I/O requests
- ▶ Disk subsystems
- ▶ Disk operation
- ▶ Asynchronous I/O wait
- ▶ Disk protection
- ▶ Logical Database I/O versus physical disk I/O

### Disk I/O requests

Greater sources of disk request are faults that arise from a request for information not being satisfied by what is in memory. Requests to bring information in to memory also result in disk I/O. Memory pages can also be purged from time to time, which results in disk I/O activity.

**Note:** The Set Object Access (**SETOBJACC**) command on IBM i temporarily changes the speed of access to an object by bringing the object into a main storage pool or purging it from all main storage pools. An object can be kept main storage resident by selecting a pool for the object that has available space and does not have jobs that are associated with it. For more information, see, [IBM Documentation](#).

### Disk subsystems

Typically, an external disk subsystem (storage system) connects a server through a SAN, as shown in Figure A-5.

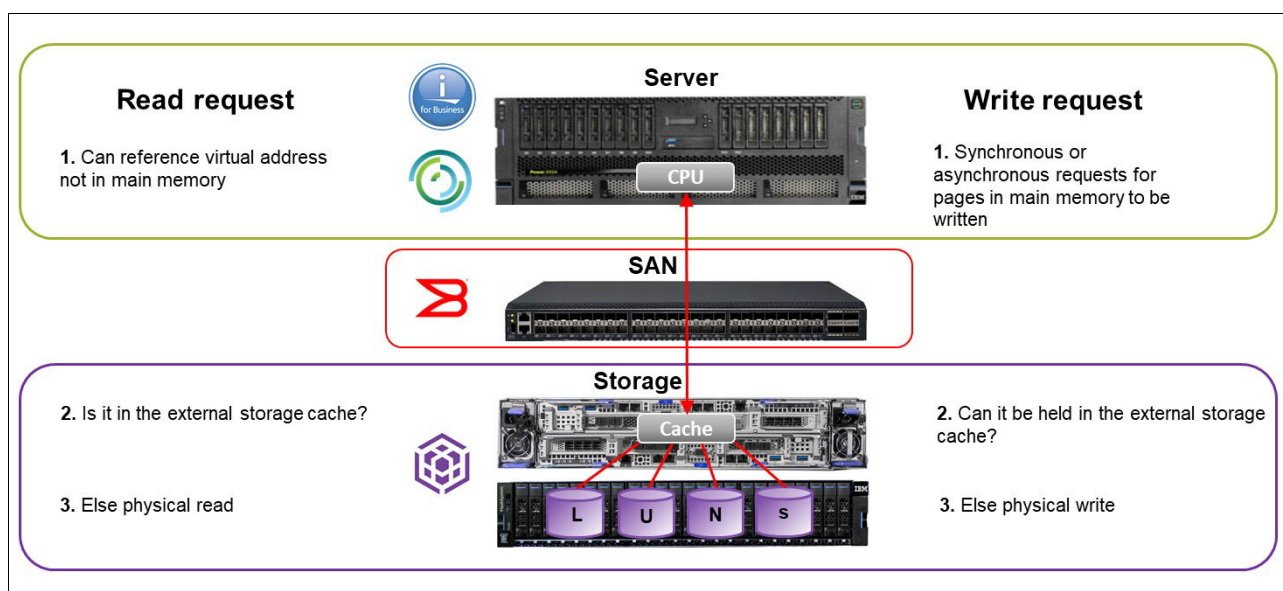


Figure A-5 Disk subsystem

A request information (data or instructions) from the CPU that is based on user interactions is submitted to the disk subsystem if it cannot be satisfied from the contents of memory.

Whether the request can be satisfied from the disk subsystem cache, it responds or forwards the request to the disk drives or LUNs.

Similarly, a write request is retained in memory, unless the operating system determines that it must be written to the disk subsystem. The operating system then attempts to satisfy the request by writing to the controller cache.

**Note:** The QAPMDISKRB from collections services data files in IBM i includes disk file response bucket entries. It also contains one record for each device resource name. It is intended to be used with the QAPMDISK file. For more information, see [this web page](#).

### Disk operation

On IBM i, physical disk I/O requests are categorized as database (physical or logical files) or non-database I/Os, as shown in Figure A-6.

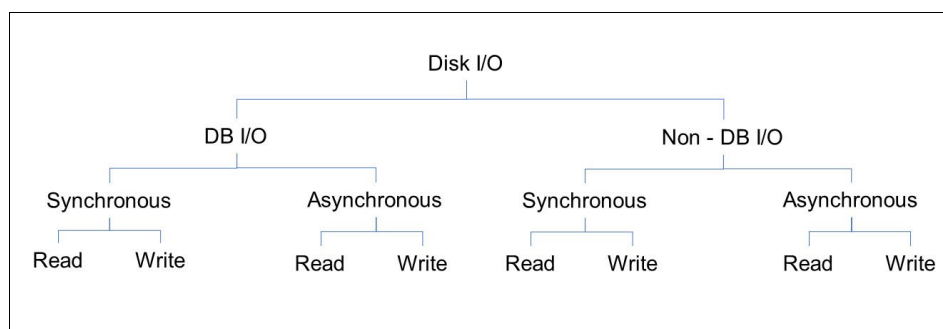


Figure A-6 Disk I/O on IBM i

It is the time that is taken to respond to synchronous disk I/Os that contribute to the OLTP response time or batch runtime. With asynchronous I/O, the progress of a request does not wait for the completion of I/O.

Usually, write requests are asynchronous, including journal deposits with commitment control. However, if journaling is active without commitment control, the writes become synchronous.

### Asynchronous I/O wait

On IBM i, might have to wait at times for asynchronous I/O requests to complete. The job issues a request but requires the data sooner than it can be made available by the disk subsystem. When a job waits for asynchronous, the I/O portion of the operation becomes synchronous. The time is recorded as asynchronous disk I/O wait in the QAPMJOBL file.

JBWIO is the number of times the process waited for outstanding asynchronous I/O operations to complete. For more information, see [this web page](#).

This issue might be caused by faster processors that are running with relatively poor disk subsystems performance. Disk subsystem performance can be affected by busy or slow disks, or small I/O cache.

### Disk protection

For more information about external storage consideration to set up your RAID protection, see Chapter 4, “Storage pools” on page 91.



**Note:** If you need high I/O performance on your IBM i workload, an option is to create a DRAID 1 on your supported storage system, such as IBM Flash System 7200 and 9200 with IBM Spectrum Virtualize 8.4. In this configuration, the rebuild area is distributed over all member drives. The minimum extent size for this type of DRAID is 1024 MB.

### ***Logical Database I/O versus physical disk I/O***

Information in partition buffer memory is available for use by any job or thread. Commonly, information is available in the partition buffer as a block of data rather than individual records. Data in a job buffer is available for use by the job only.

When an application program requests data, storage management checks whether they are available in memory. If so, it is moved to the open data path in the job buffer. If the data is not in memory, the request is submitted to the disk subsystem as a read command.

In that context, logical Database I/O information is moved between the open data path of user program and the partition buffer. This information is a count of the number of buffer movements and not a reflection of the records processed.

For more information, see the following web pages:

- ▶ [Sharing an Open Data Path](#)
- ▶ [Selecting the metrics](#)

Physical disk I/O occurs when information is read or written as a block of data to or from the disk. It involves the movement of data between the disk and the partition buffer in memory. For more information, see [this web page](#).

## **Planning for IBM i storage capacity**

To correctly plan the storage capacity provided by IBM Spectrum Virtualize family systems for IBM i, you must be aware of IBM i block translation for external storage that is formatted in 512-byte blocks. IBM i internal disks use a block size of either 520 or 4160 bytes.

IBM Spectrum Virtualize storage for hosts is formatted with a block size of 512 bytes, so a translation or mapping is required to attach it to IBM i. IBM i changes the data layout to support 512-byte blocks (sectors) in external storage by using an extra ninth sector to store the headers for every page.

The eight 8-byte headers from each 520-byte sectors of a page are stored in the ninth sector, which is different than 520-byte sector storage where the 8 bytes are stored continuous with the 512 bytes of data to form the 520-byte sector. The data that was previously stored in eight sectors is now stored by using nine sectors, so the required disk capacity on IBM Spectrum Virtualize based systems is 9/8 of the IBM i usable capacity. Similarly, the usable capacity in IBM i is 8/9 of the allocated capacity in these storage systems.

When attaching IBM Spectrum Virtualize family storage to IBM i, plan for the extra capacity on the storage system so that the 8/9ths of the effective storage capacity that is available to IBM i covers the capacity requirements for the IBM i workload.

The performance impact of block translation in IBM i is small or negligible.

In Figure A-7 shows the byte sectors for IBM i.

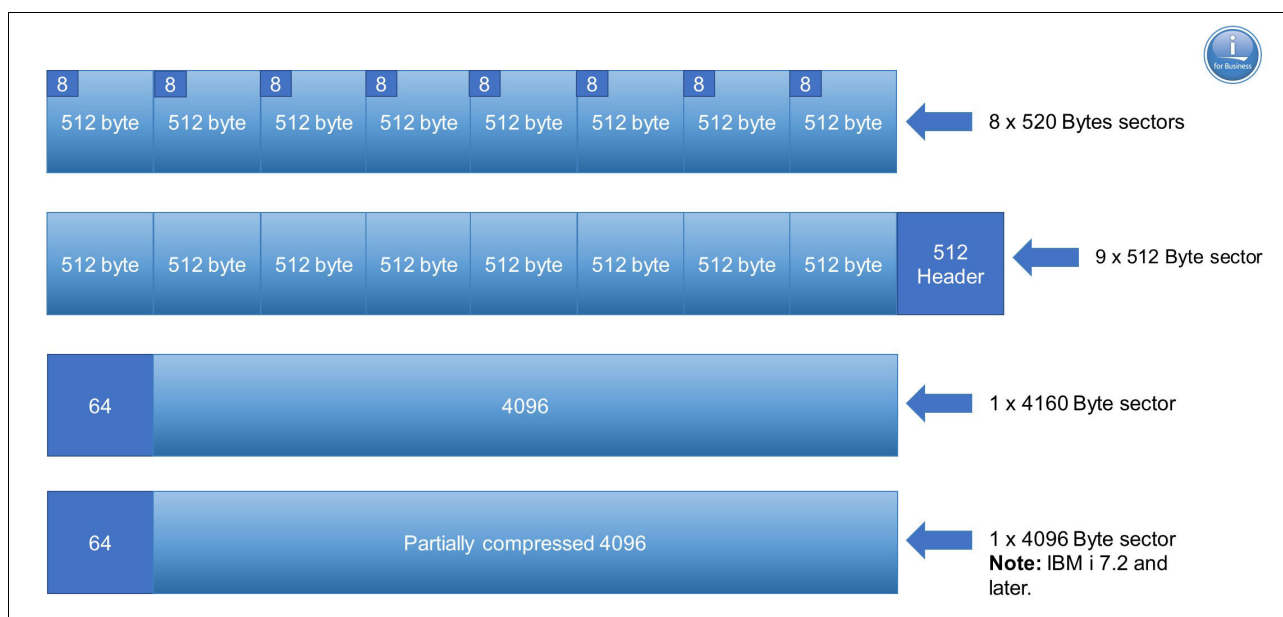


Figure A-7 IBM i with different sector sizes

## Storage connection to IBM i

IBM Spectrum Virtualize storage can be attached to IBM i in the following ways:

- ▶ Native connection without the use of the IBM PowerVM Virtual I/O Server (VIOS)
- ▶ Connection with VIOS in NPIV mode
- ▶ Connection with VIOS in virtual SCSI mode

The decision for IBM i native storage attachment or a VIOS attachment is based on the customer's requirements. Native attachment has its strength in terms of simplicity and can be a preferred option for static and smaller IBM i environments with only a few partitions. It does not require extra administration and configuration of a VIOS environment. However, it also provides the least flexibility and cannot be used with IBM PowerVM advanced functions, such as Live Partition Mobility or remote restart.

Table A-1 lists key criteria to help you with the decision for selecting an IBM i storage attachment method.

Table A-1 Comparing IBM i native and Virtual I/O Server attachment

| Criteria                                                                | Native attachment | VIOS attachment  |
|-------------------------------------------------------------------------|-------------------|------------------|
| <b>Simplicity</b><br>(Configuration, maintenance, and failure analysis) | ✓                 | More complex     |
| <b>Performance</b>                                                      | ✓                 | ✓<br>(with NPIV) |
| <b>Consolidation</b><br>(Storage and network adapters)                  | More limited      | ✓                |

| Criteria                                                                                                          | Native attachment | VIOS attachment |
|-------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|
| PowerVM advanced functions (partition mobility, suspend and resume, remote restart, and private cloud deployment) | Not available     | ✓               |
| Hardware support (storage and network adapters, and entry level servers)                                          | More limited      | ✓               |

The following sections describe the guidelines and preferred practices for each type of connection.

**Note:** For more information about the current requirements, see the following web pages:

- ▶ [IBM System Storage Interoperation Center \(SSIC\)](#)
- ▶ [IBM i POWER External Storage Support Matrix Summary](#)

## Native attachment

Native connection support for IBM i with IBM Spectrum Virtualize storage is available with IBM Power Systems POWER7® or later server technology. It requires IBM i 7.1, Technology Refresh (TR) 7 or later for POWER7, and IBM i 7.1 TR 8 or later for POWER8®.

Native connection *with* SAN switches can be done by using the following adapters:

- ▶ 32 Gb PCIe3 2-port FC adapters feature number #EN1A or #EN1B (IBM POWER9™ only)
- ▶ 16 Gb PCIe3 4-port FC adapters feature number #EN1C or #EN1D (POWER9 only)
- ▶ 16 Gb PCIe3 2-port FC adapters feature number #EN0A or #EN0B
- ▶ 8 Gb PCIe 2-port FC adapters feature number #5735 or #5273
- ▶ 4 Gb PCIe 2-port Fibre Channel (FC) adapters feature number #5774 or #5276

Direct native connection *without* SAN switches can be done by using the following adapters:

- ▶ 16 Gb adapters in IBM i connected to 16 Gb adapters in IBM Spectrum Virtualize V7.5 or later based storage with non-NPIV target ports
- ▶ 4 Gb FC adapters in IBM i connected to 8 Gb adapters in IBM Spectrum Virtualize based storage with non-NPIV target ports

For resiliency and performance reasons, connect IBM Spectrum Virtualize storage to IBM i with multipath using two or more FC adapters:

- ▶ You can define a maximum of 127 LUNs (up to 127 active + 127 passive paths) to a 16 or 32 Gb port in IBM i, with IBM i 7.2 Technology Refresh (TR) 7 or later, and with IBM i 7.3 TR3 or later.
- ▶ You can define a maximum of 64 LUNs (up to 64 active + 64 passive paths) to a 16 or 32 Gb port with IBM i release and TR lower than i 7.2 TR7 and i 7.3 TR3.
- ▶ You can define a maximum of 64 LUNs (up to 64 active + 64 passive paths) to a 4 or 8 Gb port, regardless of the IBM i level.

The LUNs report in IBM i as disk units with type 2145.

IBM i enables SCSI command tag queuing in the LUNs from natively connected IBM Spectrum Virtualize storage. The IBM i queue depth per LUN and path with this type of connection is 16.

## VIOS attachment

The following FC adapters are supported for VIOS attachment of IBM i to IBM Spectrum Virtualize storage:

- ▶ 32 Gb PCIe3 2-port FC adapter feature number #EN1A or #EN1B (POWER9 only)
- ▶ 16 Gb PCIe3 4-port FC adapter feature number #EN1C or #EN1D (POWER9 only)
- ▶ 16 Gb PCIe3 2-port FC adapter feature number #EN0A or #EN0B
- ▶ 8 Gb PCIe 2-port FC adapter feature number #5735 or #5273
- ▶ 8 Gb PCIe2 2-port FC adapter feature number #EN0G or #EN0F
- ▶ 8 Gb PCIe2 4-port FC adapter feature number #5729
- ▶ 8 Gb PCIe2 4-port FC adapter feature number #EN12 or #EN0Y

**Important:** For more information about the current requirements, see the following web pages:

- ▶ [IBM System Storage Interoperation Center \(SSIC\)](#)
- ▶ [IBM i POWER External Storage Support Matrix Summary](#)

## Connection with VIOS NPIV

IBM i storage attachment support that uses IBM PowerVM® Virtual I/O Server N\_Port ID Virtualization (NPIV) was introduced with POWER6 server technology. With NPIV, volumes (LUNs) from the IBM Spectrum Virtualize storage system are directly mapped to the IBM i server. VIOS does not see NPIV connected LUNs; instead, it is an FC pass-through.

The storage LUNs are presented to IBM i with their native device type of 2145 for IBM Spectrum Virtualize-based storage. NPIV attachment requires 8 Gb or newer generation FC adapter technology and SAN switches that must be NPIV enabled (see Figure A-8).

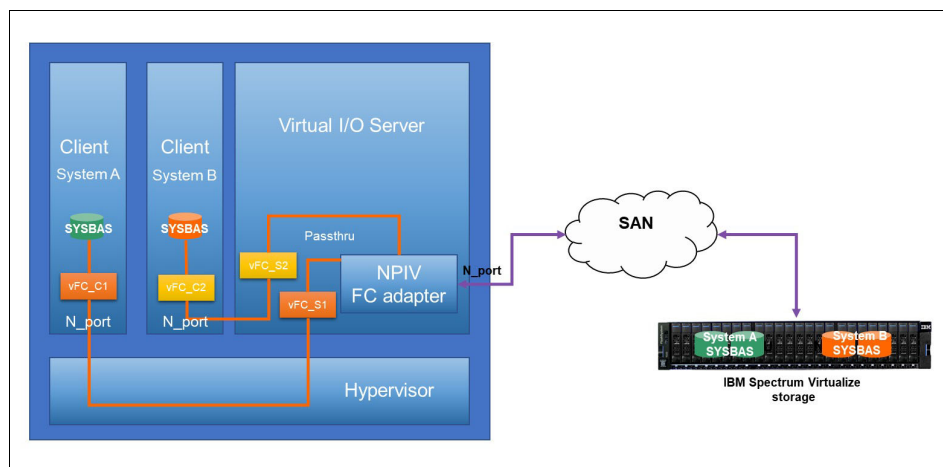


Figure A-8 IBM i SAN access using NPIV

## Redundant VIOS with NPIV

For resiliency and performance reasons, connect IBM Spectrum Virtualize storage to IBM i using multipathing across two or more VIOS servers.

Observe the following rules for mapping IBM i server virtual FC client adapters to the physical FC ports in VIOS when implementing NPIV connection:

- ▶ You can map up to 64 virtual FC adapters to the same physical FC adapter port in VIOS. With VIOS 3.1 and later this limit got increased for support of mapping of up to 255 virtual FC adapters to a 32 Gb physical FC adapter port.
- ▶ Mapping of more than one NPIV client virtual FC adapter from the *same* IBM i system to a VIOS physical FC adapter port is supported since IBM i 7.2 TR7 and i 7.3 TR3 – however when using PowerVM partition mobility only a single virtual FC adapter is allowed to be mapped from the *same* IBM i system to a VIOS physical FC adapter port.
- ▶ You can use the same port in VIOS for both NPIV mapping and connection with VIOS virtual SCSI (VSCSI).
- ▶ If PowerHA solutions with IBM i independent auxiliary storage pools (IASPs) are implemented, you need to use different virtual FC adapters for attaching the IASP LUNs, and not share an adapter between SYSBAS and IASP LUNs.

You can configure a maximum of 127 LUNs (up to 127 active + 127 passive paths) to a virtual FC adapter with IBM i 7.2 TR7 or later, and with IBM i 7.3 TR3 or later.

You can configure a maximum of 64 LUNs (up to 64 active + 64 passive paths) to a virtual FC adapter with IBM i release and TR lower than i 7.2 TR7 and i 7.3 TR3.

IBM i enables SCSI command tag queuing for LUNs from VIOS NPIV connected to IBM Spectrum Virtualize storage. The IBM i queue depth per LUN and path with this type of connection is 16.

**Note:** If you encounter issues with NPIV/Virtual FC of IBM i that is attached to an IBM Spectrum Virtualize, such as missing paths and missing disk units, consider the following suggestions:

- ▶ System Snapshot (SYSSNAP), for the SYSSNAP. Be sure to include LIC LOGs, QHST, and PALs. Change the date range to include the date range of the problem. For more information, see this [IBM Support web page](#).
- ▶ VIOS SNAPS can be collected from the VIOS partitions as part of the SYSSNAP or separately. For more information, see this [IBM Support web page](#).
- ▶ Collect switch logs as close as possible to the time of the problem.
- ▶ Collect the applicable State Save, SNAP, and so on, from the IBM Spectrum Virtualize at the time the problem is occurring. This information is needed by the storage support team.

If you experience a performance problem with poor disk response time and the IBM Spectrum Virtualize is connected with NPIV, see this [IBM Support web page](#).

### ***NPIV acceleration***

Virtual I/O Server version 3.1.2 or later strengthened FC N\_Port ID Virtualization (NPIV) to provide multiqueue support. This enhanced performance, including more throughput, reduced latency, and higher IOPS, spreads the I/O workload across multiple work queues.

The following FC adapter feature codes are supported:

- ▶ 32 Gb PCIe3 2-port FC adapters feature number #EN1A or #EN1B (POWER9 only)
- ▶ 16 Gb PCIe3 4-port FC adapters feature number #EN1C or #EN1D (POWER9 only)
- ▶ 16 Gb PCIe3 2-port FC adapters feature number #EN2A or #EN2B

**Note:** NPIV acceleration is supported by IBM i 7.2 or later, and by the firmware level for IBM Power Systems 9 is FW940 or later.

## Connection with VIOS virtual SCSI

IBM i storage attachment by using the IBM PowerVM Virtual I/O Server Connection that uses virtual SCSI was introduced with IBM Power Systems POWER6 technology.

When deciding on an IBM PowerVM Virtual I/O Server storage attachment for IBM i, NPIV attachment is often preferred over virtual SCSI attachment for the following reasons:

- ▶ With virtual SCSI, an emulation of generic SCSI devices is performed by VIOS for its client partitions, such as IBM i, which requires extra processing and adds a small delay to I/O response times.
- ▶ Virtual SCSI provides much lower scalability in terms of the maximum supported LUNs per virtual adapter than NPIV. It also requires more storage management, such as multipath configuration and customization at the VIOS layer, which adds complexity.
- ▶ Because of the virtual SCSI emulation unique device characteristics of the storage device, such as device type (or in the case of tape devices), media type and other device attributes are no longer presented to the IBM i client.
- ▶ Virtual SCSI attachment is not supported for PowerHA LUN level switching technology, which is required for IASP HyperSwap solutions with IBM Spectrum Virtualize.

Similar considerations as for NPIV apply with regards to the use of IBM i multipathing across two or more VIOS to improve resiliency and performance. However, because with virtual SCSI multipathing is also implemented at the VIOS layer, the following considerations apply:

- ▶ IBM i multipathing is performed with two or more VSCSI client adapters, each of them assigned to a VSCSI server adapter in different VIOS. With virtual SCSI, volumes (LUNs) from the IBM Spectrum Virtualize storage system are not mapped directly to an IBM i host but to the two or more VIOS servers. These LUNs that are detected as HDDs on each VIOS must be mapped as a virtual target device to the relevant VSCSI server adapters to be used by the IBM i client.
- ▶ In addition to IBM i multipathing across multiple VIOS servers, with virtual SCSI, multipathing is implemented at the VIOS server layer to provide further I/O parallelism and resiliency by using multiple physical FC adapters and SAN fabric paths from each VIOS server to its storage.
- ▶ The IBM recommended multipath driver for IBM Spectrum Virtualize-based storage running microcode V7.6.1 or later is the VIOS built-in AIXPCM multipath driver, which replaces the previously recommended SDDPCM multipath driver.

For more information, see this [IBM Support web page](#).

It is possible to connect up to 4095 LUNs per target, and up to 510 targets per port in a physical adapter in VIOS. With IBM i 7.2 and later, you can attach a maximum of 32 disk LUNs to a virtual SCSI adapter in IBM i. With IBM i releases before i 7.2, a maximum of 16 disk LUNs can be attached to a virtual SCSI adapter in IBM i. The LUNs are reported in IBM i as generic SCSI disk units of type 6B22.

IBM i enables SCSI command tag queuing in the LUNs from VIOS VSCSI connected to IBM Spectrum Virtualize storage. The queue depth on a LUN with this type of connection is 32.

# Setting attributes in VIOS

This section describes the values of certain device attributes in VIOS, which should be configured for resiliency and performance.

## FC adapter attributes

With either VIOS virtual SCSI connection or NPIV connection, use the VIOS **chdev** command to specify the following attributes for each SCSI I/O Controller Protocol Device (fscsi) device that connects an IBM Spectrum Virtualize storage LUN for IBM i:

- ▶ The attribute `fc_err_recov` should be set to `fast_fail`
- ▶ The attribute `dyntrk` should be set to `yes`

The specified values for the two attributes specify how the VIOS FC adapter driver or VIOS disk driver handle certain types of fabric-related failures and dynamic configuration changes. Without setting these values for the two attributes, the way these events are handled is different, and will cause unnecessary retries or manual actions.

**Note:** These attributes are also set to the recommended values when applying the default rules set that is available with VIOS 2.2.4.x or later.

## Disk device attributes

With VIOS virtual SCSI connection, use the VIOS **chdev** command to specify the following attributes for each hdisk device that represents an IBM Spectrum Virtualize storage LUN connected to IBM i:

- ▶ If IBM i multipathing across two or more VIOS servers is used, the attribute `reserve_policy` should be set to `no_reserve`.
- ▶ The attribute `queue_depth` should be set to 32.
- ▶ The attribute `algorithm` should be set to `shortest_queue`.

Setting `reserve_policy` to `no_reserve` is required to be set in each VIOS if multipath with two or more VIOS is implemented, to prevent SCSI reservations on the hdisk device.

Set `queue_depth` to 32 for performance reasons. Setting this value ensures that the maximum number of I/O requests that can be outstanding on an HDD in the VIOS at a time matches the maximum number of 32 I/O operations that IBM i operating system allows at a time to one VIOS VSCSI-connected LUN.

Set `algorithm` to `shortest_queue` for performance reasons. Setting this value allows the AIXPCM driver in VIOS to use a dynamic load balancing instead of the default path failover algorithm for distributing the I/O across the available paths to IBM Spectrum Virtualize storage.

Setting a physical volume identifier (PVID) for HHD devices that are used for virtual SCSI attachment of IBM i client partitions is not recommended because it makes those devices ineligible for a possible later migration to NPIV or native attachment.



**Important:** While working with SCSI and NPIV, you cannot mix both regarding the paths to the same LUN. However, VIOS supports NPIV and SCSI concomitantly; that is, some LUNs can be attached to the virtual WWPNs of the NPIV FC adapter. At the same time, the VIOS can also provide access to LUNs that are mapped to virtual target devices and exported as VSCSI devices.

You can have one or more Virtual I/O Servers providing the pass-through function for NPIV. Also, you can have one or more Virtual I/O Servers hosting VSCSI storage. Therefore, the physical HBA in the Virtual I/O Server supports NPIV and VSCSI traffic.

## Guidelines for Virtual I/O Server resources

Be aware of the memory requirements of the hypervisor when determining the overall memory of the system. Above and beyond the wanted memory for each partition, you must add memory for virtual resources (VSCSI and Virtual FC) and hardware page tables to support the maximum memory value for each partition.

The suggestion is to use the IBM Workload Estimator tool to estimate the needed Virtual I/O Server resources. However, as a starting point in context of CPU and memory for Virtual I/O Server, see this [IBM Support web page](#).

## Disk drives for IBM i

This section describes how to implement internal disk drives in IBM Spectrum Virtualize storage or externally virtualized back-end storage for an IBM i host. These suggestions are based on the characteristics of a typical IBM i workload, such as a relatively high write ratio, a relatively high access density, and a small degree of I/O skew due to the spreading of data by IBM i storage management.

Considering these characteristics and typical IBM i customer expectations for low I/O response times we expect that many SAN storage configurations for IBM i will be based on an all-flash storage configuration.

If for less demanding workloads, or for commercial reasons a multitier storage configuration with either using enterprise class (tier0\_flash) and high-capacity (tier1\_flash) flash drives or even enterprise hard disk drives (tier2\_HDD) is preferred, make sure that a sufficiently large part of disk capacity resides on flash drives. As a rule of thumb for a multitier configuration considering the typically low IBM i I/O skew, at least 20% of IBM i capacity should be based on the higher tier flash storage technology.

Even if specific parts of IBM i capacity are on flash drives, it is important that you provide enough HDDs with high rotation speed for a hybrid configuration with flash drives and HDDs. Preferably, use 15 K RPM HDDs of 300 GB or 600 GB capacity, along with flash technology.

IBM i transaction workload usually achieves the best performance when using disk capacity entirely from enterprise class flash (tier0\_flash) storage. High capacity or read-intensive flash drives are typically not the best choice for IBM i performance critical workload, especially for the top storage tier, considering a usually high IBM i write percentage of often 50% and higher, the disk write amplification by using RAID 6 and the significant lower random write performance of tier1 compared to tier0 flash drives.



The use of a multitier storage configuration by IBM Spectrum Virtualize storage is achieved through Easy Tier. For more information, see *Implementing the IBM System Storage SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8491.

Even if you do not plan to install multitier storage configuration, or currently have no multitier storage configuration that is installed, you can still use Easy Tier for intra-tier rebalancing. You also can evaluate your workload with its I/O skew, which provides information about the benefit you might gain by adding flash technology in the future.

## Compression considerations

If compression is wanted, the preferred choice for using compression at the IBM Spectrum Virtualize storage system layer for performance critical IBM i workload is by using IBM FlashCore® module (FCM) hardware compression technology at the disk drive level within IBM Spectrum Virtualize standard pools or data reduction pools (DRPs) with fully allocated volumes. These configuration options do not affect performance compared to other compression technologies, such as DRP compressed volumes or Real-time Compression at the storage subsystem level.

**Important:** Data reduction or deduplication can be used with IBM i, which affects performance positively.

Nevertheless, the performance is tremendously affected and different whenever something is touched, such as 30 minutes taking 3 - 18 hours. The data is affected whenever something is created, changed, or used. The integrity of the objects is okay.

However, if a physical page on disk is corrupted, potentially have hundreds or thousands of objects become corrupted instead of only one. Another consideration is the amount of wear that occurs on the drives from so much read/write activity.

If you plan to use deduplication for archival or test purposes, deduplication might be a viable solution for saving huge amounts of storage. If the deduplication solution is planned for a production or development environment, we strongly recommend that you test thoroughly before committing.

## Storage sizing and performance modeling

IBM provides tools, such as IBM Storage Modeller (StorM) and IntelliMagic Disk Magic for IBM representatives and Business Partners, which are recommended to be used for performance modelling and sizing before implementing a wanted IBM Spectrum Virtualize storage configuration for IBM i. These tools allow the user to enter the performance data of the current IBM i workload manually or by using file import from IBM i (5770-PT1 reports or PDI data) or from IBM Spectrum Control performance data. Enter the current storage configuration and model the wanted configuration.

When modeling Easy Tier, specify the lowest skew level for IBM i workload or import an existing I/O skew curve from available Easy Tier reports. The steps that are taken for sizing and modeling IBM i are shown in Figure A-9 on page 546.

The modeling helps assure an adequate solution sizing by providing predictions for the modeled IBM Spectrum Virtualize storage resource of system usage, the predicted disk response time for IBM i, and the usage and response times at workload growth.

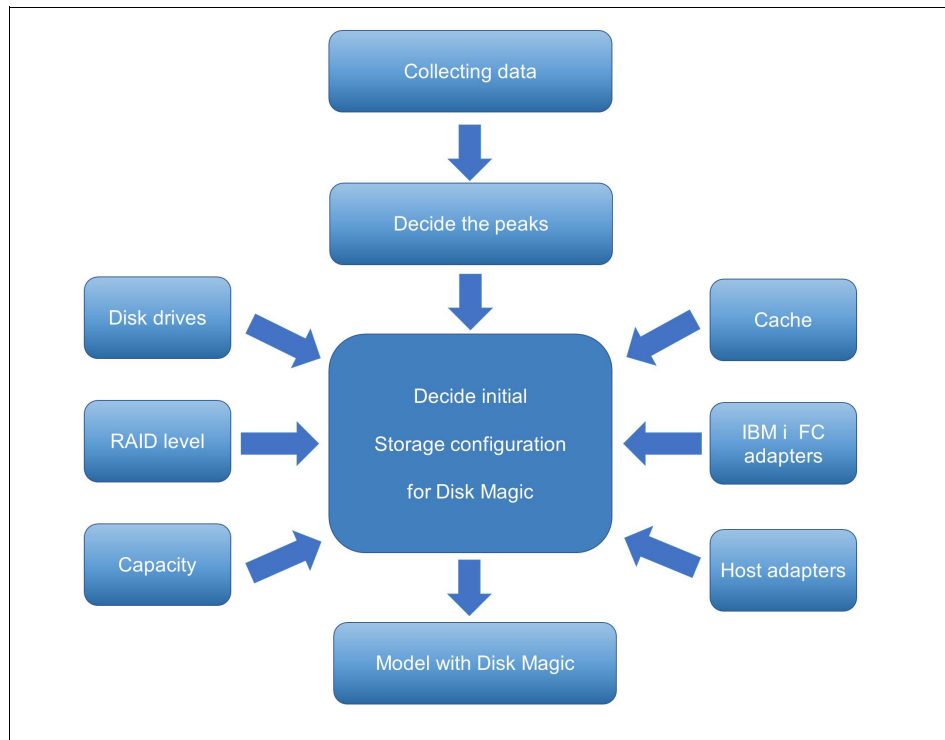


Figure A-9 Diagram of sizing and modelling for IBM i using Disk Magic.

**Note:** Contact your IBM representative or IBM Business Partner to discuss a performance modeling and sizing for a planned IBM Spectrum Virtualize storage solution for IBM i.

## IBM i Unmap support

To better use IBM Spectrum Virtualize storage flash technology with an efficient storage space allocation and deallocation, IBM i supports the space of storage system unmap capabilities by corresponding host unmap functions.

Initially, IBM i unmap support that is implemented by way of the SCSI Write Same command was introduced with i 7.2 TR8 and i 7.3 TR4 for LUN initialization only; that is, for the add disk units to ASP function.

With i 7.3 TR9 and i 7.3 TR5, runtime support was added, which also supports synchronous unmap for scenarios, such as object deletion and journal clearance. The runtime unmap algorithm was further enhanced supported by i 7.3 TR7 and i 7.4 TR1, which implements an asynchronous periodic free-space cleaning.

IBM Spectrum Virtualize V8.1.1 and later storage systems can use the unmap function for efficiently deallocate space, such as for volume deletion, on their back-end storage by sending SCSI unmap commands to specific supported internal SSDs and FCMs, and selected virtualized external flash storage. Space reclamation that is triggered by host unmap commands is supported with IBM Spectrum Virtualize V8.1.2 and later for DRP thinly-thinly provisioned volumes, which can increase the free capacity in the storage pool so that it becomes available also for use by other volumes in the pool.

For more information about IBM Spectrum Virtualize storage SCSI unmap support, see 4.1.2, “Data reduction pools” on page 97, and this [IBM Support web page](#).

## Defining LUNs for IBM i

LUNs for an IBM i host are defined from IBM Spectrum Virtualize block-based storage. They are created from available extents within a storage pool, the same way as for open system hosts.

Even though IBM i supports a usable LUN large-size of up to 2 TB - 1 byte for IBM Spectrum Virtualize storage, using only a few large size LUNs for IBM i is not recommended for performance reasons.

In general, the more LUNs that are available to IBM i, the better the performance. The following are the reasons for this:

- ▶ If more LUNs are attached to IBM i, storage management uses more threads and therefore enables better performance.
- ▶ More LUNs provide a higher I/O concurrency, which reduces the likelihood of I/O queuing and therefore the wait time component of the disk response time resulting in lower latency of disk I/O operations.

For planning purposes, consider that a higher number of LUNs might also require more physical or virtual FC adapters on IBM i based on the maximum number of LUNs supported by IBM i per FC adapter port.

The sizing process helps to determine a reasonable number of LUNs required to access the needed capacity while meeting performance objectives. Regarding both these aspects and the preferred practices, our guidelines are as follows:

- ▶ For any IBM i disk pool (ASP), define all the LUNs as the same size.
- ▶ 40 GB is the preferred minimum LUN size.
- ▶ You should not define LUNs larger than about 200 GB.

**Note:** This rule is not a fixed rule because it is important that enough LUNs are configured, with which this guideline helps. Selecting a larger LUN size should not lead to configurations, such as storage migrations, with a significantly fewer number of LUNs being configured with possibly detrimental effects on performance.

- ▶ A minimum of 8 LUNs for each ASP is preferred for small IBM i partitions and typically a couple of dozen LUNs for medium and up to a few hundreds for large systems.

When defining LUNs for IBM i, consider the following required minimum capacities for the load source (boot disk) LUN:

- ▶ With IBM i release 7.1, the minimum capacity is 20 GB
- ▶ With IBM i release 7.2 before TR1, the minimum capacity is 80 GB in IBM i
- ▶ With IBM i release 7.2 TR1 and later, the minimum capacity is 40 GB in IBM i

IBM Spectrum Virtualize dynamic volume expansion is supported for IBM i with IBM i 7.3 TR4 and later. An IBM i IPL is required to use the extra volume capacity.

**Tip:** For more information about cross-referencing IBM i disks units with IBM Spectrum Virtualize LUNs by using N-Port ID Virtualization (NPIV), see this [IBM Support web page](#).

## Disk arms and maximum LUN size

Selected limits that are related to disk arms and LUNs sizes were increased in IBM i 7.4, as listed in Table 14-4.

Table 14-4 Limits increased for maximum disk arms and LUN sizes

| System limits                                                                                   | IBM i 7.2 | IBM i 7.3 | IBM i 7.4 |
|-------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| Disk arms in all basic auxiliary storage pools (ASPs 1 - 32), per LPAR                          | 2047      | 2047      | 3999      |
| Disk arms in all independent auxiliary storage pools (IASPs 33 - 255) in all nodes in a cluster | 2047      | 2047      | 5999      |
| Maximum combined number of disk arms and redundant connections to disk units                    | 35.600    | 35.600    | 35.600    |
| 512-byte block size LUNs <sup>a</sup>                                                           | 2 TB      | 2 TB      | 2 TB      |
| 4096-byte block size LUNs <sup>b</sup>                                                          | 2 TB      | 2 TB      | 16 TB     |

a. Actual limit is one block short of the maximum that is listed in Table 14-4. For all 512 block LUNs, the maximum is still up to 2 TB, including IBM Storwize LUNs and SAN Volume Controller LUNs.

b. This size includes IBM FlashSystems LUNs, and 4 K block SAS disks (VSCSI attached).

**Note:** For more information about these limits, and others, see [this web page](#).

## Data layout

Spreading workloads across all IBM Spectrum Virtualize storage components maximizes the use of the hardware resources in the storage subsystem. I/O activity should be balanced between the two nodes or controllers of the IBM Spectrum Virtualize storage system I/O group, which is usually taken care of by the alternating preferred node volume assignments at LUN creation.

However, especially with improper sizing or unanticipated workload increases, it is possible when sharing resources that performance problems might arise due to resource contention. Some isolation of workloads, at least regarding a shared back-end storage, can be accomplished by a configuration where each IBM i ASP or LPAR has its own managed storage pool. Such a configuration with dedicated storage pools results in a tradeoff between accomplishing savings from storage consolidation and isolating workloads for performance protection. This is because a dedicated storage pool configuration likely requires more back-end storage hardware resources because it cannot use the averaging effect of multiple workloads typically showing their peaks at different time intervals.

Consider the following data layout:

- For all-flash storage configurations, assuming a correctly sized storage backend, there is typically no reason for not sharing the disk pool among multiple IBM i workloads.
- For hybrid configurations with Easy Tier on mixed HDD and flash disks, the storage pool also might be shared among IBM i workloads. Only large performance critical workloads should be configured in isolated disk pools.

- For HDD only pools, make sure that you isolate performance critical IBM i workloads in separate storage pools.
- Avoid mixing IBM i LUNs and non-IBM i LUNs in the same disk pool.

Apart from using Easy Tier on IBM Spectrum Virtualize for managing a multitier storage pool, there is also an option to create a separate storage pool for different storage tiers on IBM Spectrum Virtualize storage and create different IBM i ASPs for each tier. IBM i applications that have their data in an ASP of a higher storage tier experience a performance boost compared to those using an ASP with a lower storage tier.

IBM i internal data relocation methods, such as the ASP balancer hierarchical storage management function and IBM Db2 media preference, are not available to use with IBM Spectrum Virtualize flash storage.

## Fibre Channel adapters in IBM i and VIOS

When you size the number of FC adapters for an IBM i workload for native or VIOS attachment, take into account the maximum I/O rate (IOPS) and data rate (MBps) that a port in a particular adapter can sustain at 70% utilization. Also, consider the I/O rate and data rate of the IBM i workload.

If multiple IBM i partitions connect through the same FC port in VIOS, take into account the maximum rate of the port at 70% utilization and the sum of I/O rates and data rates of all connected LPARs.

For sizing, you might consider the throughput that is specified in Table A-2 that shows the throughput of a port in a particular adapter at 70% utilization.

*Table A-2 Throughput of FC adapters*

| Maximal I/O rate per port       | 16 Gb 2-port adapter | 8 Gb 2-port adapter |
|---------------------------------|----------------------|---------------------|
| IOPS per port                   | 52,500 IOPS          | 23,100 IOPS         |
| Sequential throughput per port  | 1,330 MBps           | 770 MBps            |
| Transaction throughput per port | 840 MBps             | 371 MBps            |

Make sure to plan for using separate FC adapters for IBM i disk and tape attachment. This separation is recommended due to the required IBM i virtual I/O processor (IOP) reset for tape configuration changes and for workload performance isolation.

## Zoning SAN switches

With IBM i native attachment, or VIOS NPIV attachment, zone the SAN switches so that one IBM i FC initiator port is in a zone with two FC ports from the IBM Spectrum Virtualize storage target, each port from one node canister of the I/O group, as shown in Figure A-10. This provides resiliency for the I/O to and from a LUN assigned to the IBM i FC initiator port. If the preferred node for that LUN fails, the I/O continues using the non-preferred node.

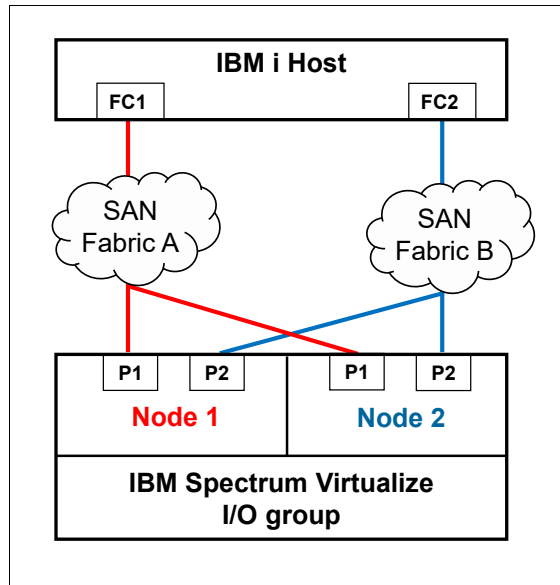


Figure A-10 SAN switch zoning for IBM i with IBM Spectrum Virtualize storage

For VIOS virtual SCSI attachment, zone one physical port in VIOS with one or more available FC ports from each of both node canisters of the IBM Spectrum Virtualize storage I/O group. IBM SAN Volume Controller or Storwize ports that are zoned with one VIOS port should be evenly spread between both node canisters. Keep in mind that a maximum of eight host paths is supported from VIOS to IBM Spectrum Virtualize storage.

## IBM i Multipath

Multipath provides greater resiliency for SAN-attached storage, and can improve performance as well. IBM i supports up to eight active paths and up to eight passive paths to each LUN. In addition to the availability considerations, lab performance testing shows that two or more paths provide performance improvements when compared to a single path.

Typically, two active paths to a LUN is a good balance of price and performance. The scenario that is shown in Figure A-10 results in two active and two passive paths to each LUN for IBM i. However, you can implement more than two active paths for workloads where high I/O rates are expected to the LUNs where a high I/O access density is expected.

It is important to understand that IBM i multipath for a LUN is achieved by connecting the LUN to two or more FC ports that belong to different adapters in an IBM i partition. Adding more than one FC port from the same IBM Spectrum Virtualize storage node canister to a SAN switch zone with an IBM i FC initiator port does not provide more active paths because an IBM i FC initiator port, by design, logs in into one target port of a node only.

With IBM i native attachment, the ports for multipath must be from different physical FC adapters in IBM i. With VIOS NPIV, the virtual FC adapters for multipath must be assigned to different VIOS for redundancy. However, if more than two active paths are used, you can use two VIOS and split the paths among them. With VIOS virtual SCSI attachment, the virtual SCSI adapters for IBM i multipath must be assigned to different VIOS.

IBM Spectrum Virtualize storage uses a redundant dual active controller design that implements SCSI asymmetrical logical unit access (ALUA). That is, some of the paths to a LUN are presented to the host as optimized and others as non-optimized.

With an ALUA aware host, such as IBM i, the I/O traffic to and from a specific LUN normally goes through only the optimized paths, which often are associated with a specific LUN of preferred node. The non-optimized paths, which often are associated with the non-preferred node, are not actively used.

In the case of an IBM Spectrum Virtualize storage topology, such as HyperSwap or IBM SAN Volume Controller Enhanced Stretched Cluster that implements host site awareness, the optimized paths are not necessarily associated with a preferred node of a LUN but with the node of the I/O group that includes the same site attribute as the host.

If the node with the optimized paths fails, the other node of the I/O group takes over the I/O processing. With IBM i multipath, all of the optimized paths to a LUN are reported as *active* on IBM i, while the non-optimized paths are reported as *passive*. IBM i multipath employs its load balancing among the active paths to a LUN and starts using the passive paths if all active paths failed.

## Booting from SAN

All IBM i storage attachment options that is native, VIOS NPIV, and VIOS virtual SCSI, support IBM i boot from SAN. The IBM i load source is on an IBM Spectrum Virtualize storage LUN that is connected the same way as the other LUNs. Apart from the required minimum size there are not any special requirements for the load source LUN. The FC or SCSI I/O adapter for the load source needs to be *tagged* that is to say specified, by the user in the IBM i partition profile on the IBM Power Systems Hardware Management Console (HMC). When installing the IBM i System Licensed Internal Code (SLIC) with disk capacity on IBM Spectrum Virtualize storage, the installation prompts you to select one of the available LUNs for the load source.

## IBM i mirroring

Some clients prefer to use IBM i mirroring functions for resiliency. For example, they use IBM i mirroring between two IBM Spectrum Virtualize storage systems, each connected with one VIOS.

When setting up IBM i mirroring with VIOS connected IBM Spectrum Virtualize storage, complete the following steps add the LUNs to the mirrored ASP:

1. Add the LUNs from two virtual adapters with each adapter connecting one to-be mirrored half of the LUNs.
2. After mirroring is started for those LUNs, add the LUNs from another two new virtual adapters, each adapter connecting one to-be mirrored half, and so on. This way, you ensure that IBM i mirroring is started between the two IBM Spectrum Virtualize storage systems and not among the LUNs from the same storage system.

## Copy services considerations

This section covers IBM Spectrum Virtualize Copy Services considerations for usage with IBM i.

### Remote replication

The IBM Spectrum Virtualize family products support both Metro Mirror synchronous remote replication and Global Mirror asynchronous remote replication. For Global Mirror there are two options: *Standard* Global Mirror, and Global Mirror with *change volumes*, which allows for a flexible and configurable recovery point objective (RPO) that allows data replication to be maintained during peak periods of bandwidth constraints, and data consistency at the remote site to be maintained and also during resynchronization.

Regarding the use of IBM Spectrum Virtualize Copy Services functions, the IBM i single-level storage architecture requires that the disk storage of an IBM i system must be treated as a single entity; that is, the scope of copying or replicating an IBM i disk space must include SYSBAS (referred to as *full system replication*) or an IASP (referred to as *IASP replication*).

Full system replication is used for Disaster Recovery (DR) purposes where an IBM i standby server is used at the DR site, as shown in Figure A-11. When a planned or unplanned outage occurs for the IBM i production server, the IBM i standby server can be started (IPLed) from the replicated SYSBAS volumes after they are turned on IBM Spectrum Virtualize to a primary role to become accessible for the IBM i standby host.

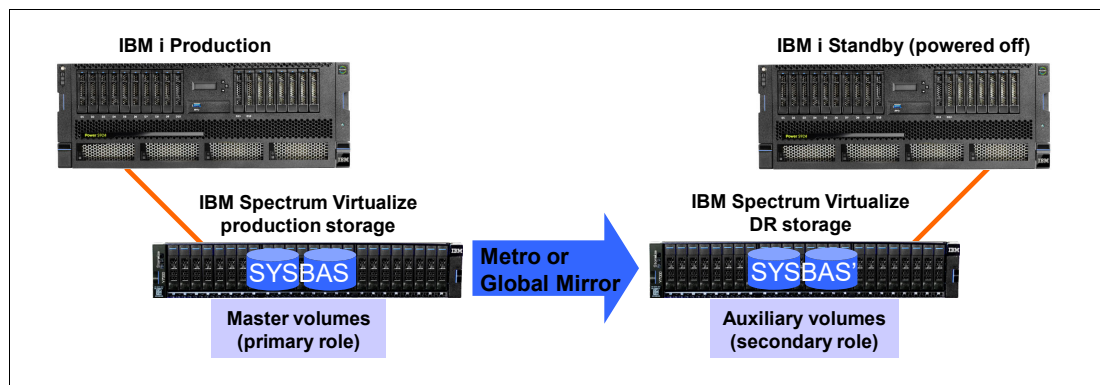


Figure A-11 IBM i full system replication with IBM Spectrum Virtualize



IASP IASP-based replication for IBM i is used for a high availability (HA) solution where an IBM i production and an IBM i backup node are configured in an IBM i cluster and the IASP that is replicated by IBM Spectrum Virtualize remote replication is switchable between the two cluster nodes as shown in Figure A-12.

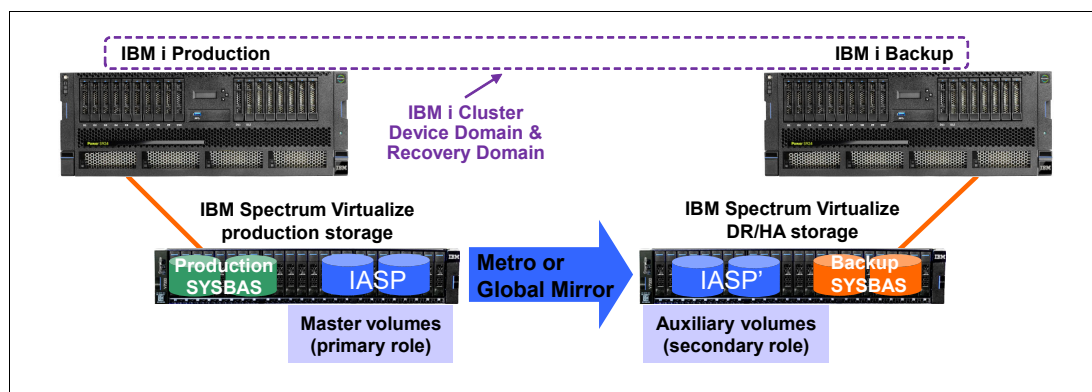


Figure A-12 IBM i IASP replication with IBM Spectrum Virtualize

In this scenario, the IBM i production system and the IBM i backup system each have their own non-replicated SYSBAS volumes and only the IASP volumes are replicated. This solution requires IBM PowerHA SystemMirror for i Enterprise Edition (5770-HAS \*BASE and option 1) for managing the IBM i cluster node switch and fail overs and the IBM Spectrum Virtualize storage remote replication switching.

For more information about IBM i high availability solutions with IBM Spectrum Virtualize Copy Services, see *PowerHA SystemMirror for IBM i Cookbook*, SG24-7994.

The sizing of the required replication link bandwidth for Metro Mirror or Global Mirror must be based on the peak write data rate of the IBM i workload to avoid affecting production performance. For more information, see 6.3.3, “Remote copy network planning” on page 262.

For more information about current IBM Spectrum Virtualize storage zoning guidelines, see 2.3.2, “Port naming and distribution” on page 30.

For environments that use remote replication, a minimum of two FC ports is suggested on each IBM Spectrum Virtualize storage node that is used for remote mirroring. The remaining ports on the node should not have any visibility to any other IBM Spectrum Virtualize cluster. Following these zoning guidelines helps to avoid configuration-related performance issues.

## FlashCopy

When planning for FlashCopy with IBM i, make sure that enough disk drives are available to the FlashCopy target LUNs to maintain a good performance of the IBM i production workload while FlashCopy relationships are active. This guideline is valid for both FlashCopy with background copy and without background copy.

When using FlashCopy with thinly provisioned target LUNs, make sure that there is sufficient capacity available in the storage pool to be dynamically allocated when needed for the copy-on-write operations. The required thin target LUN capacity depends on the amount of write operations to both the source and target LUNs, the locality of the writes, and the duration of the FlashCopy relationship.

## **FlashCopy temperature and considerations for IBM i**

FlashCopy temperature indicates the amount of disruption to source system and quality of the FlashCopy target. FlashCopy copies what was sent to disk. Updates that are sitting in memory on the IBM i are not known to the storage system.

### ***FlashCopy cold***

The following considerations apply to FlashCopy cold:

- ▶ All memory is flushed to disk.
- ▶ Source IASP must be varied off before performing a FlashCopy.
- ▶ This method is only method to ensure all write are sent out to disk and included.

### ***FlashCopy warm***

The following considerations apply to FlashCopy warm:

- ▶ No memory is flushed to disk.
- ▶ Writes in memory are excluded from the FlashCopy target.
- ▶ Zero disruption to IBM i source system.

### ***FlashCopy quiesced***

IBM i provides a quiesce function that can suspend database transactions and database and integrated file system (IFS) file change operations for the system and configured basic auxiliary storage pools (ASPs) or independent ASPs (IASPs).

The following considerations apply to FlashCopy quiesced:

- ▶ Some memory flushed to disk.
- ▶ Attempt to flush writes to disk and suspend DB I/O and to reach commitment control boundaries.
- ▶ Minimal disruption to source, is the preferred practice, and better quality than warm.

## **HyperSwap**

IBM Spectrum Virtualize storage HyperSwap as an active-active remote replication solution is supported for IBM i full system replication with IBM i 7.2 TR3 or later. It is supported for native and for VIOS NPIV attachment.

HyperSwap for IBM i IASP replication is supported by IBM i 7.2 TR5 or later and by IBM i 7.3 TR1 or later. With this solution, you need to install IBM PowerHA SystemMirror® for i Standard Edition (5770-HAS \*BASE and option 2) that enables LUN level switching to site 2. It is supported for native and VIOS NPIV attachment.

IBM Spectrum Virtualize HyperSwap relies on the SCSI ALUA aware IBM i host multipath driver to manage the paths to the local and remote IBM Spectrum Virtualize storage systems, which are logically configured as a single clustered system. From a SAN switch zoning perspective, HyperSwap requires that the IBM i host is zoned with both IBM Spectrum Virtualize nodes of the I/O group on each site. For a balanced configuration, the SAN switches from a dual fabric configuration should be evenly used.

An example of the SAN fabric connections for IBM i HyperSwap with VIOS NPIV attachment is shown in Figure A-13. This configuration example results in four active paths and 12 passive paths that are presented on IBM i for each HyperSwap LUN.

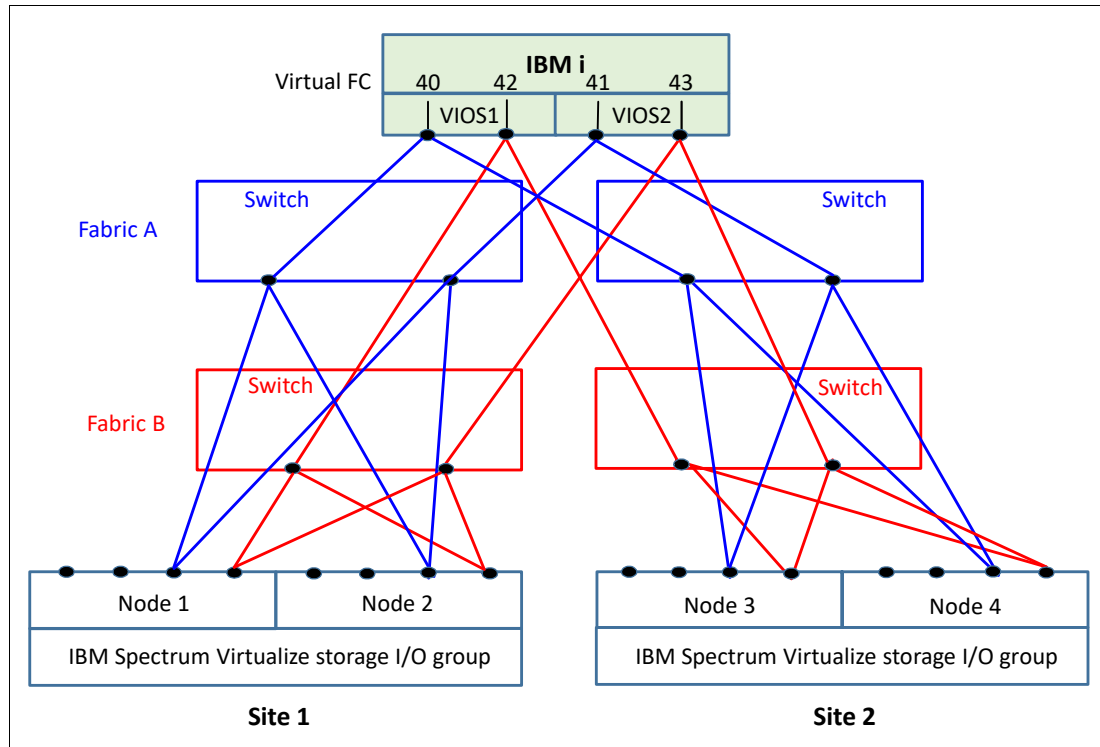


Figure A-13 IBM i HyperSwap SAN fabric connection example

Next, we briefly describe some high availability scenarios that use HyperSwap for IBM i.

### Outage of Spectrum Virtualize I/O group at site 1

In this scenario, the entire IBM i storage capacity is on HyperSwap LUNs.

After the outage of I/O group at site 1 occurs the I/O rate automatically transfers to the IBM Spectrum Virtualize nodes at site 2. The IBM i workload keeps running, and there are no relevant messages in IBM i message queues.

When the outage completes, the IBM i I/O rate automatically transfers to nodes on site 1. The IBM i workload keeps running without interruption.

### Disaster at site 1 with full system HyperSwap

In this scenario, we use a prepared IBM i standby system at site 2. The entire IBM i storage capacity is on HyperSwap LUNs. Two hosts are defined in the IBM Spectrum Virtualize storage cluster: one host with the WWPNs of IBM i at site 1, and one with WWPNs of site 2.

After a failure of site 1, including a failure of the IBM i production system and the storage at site 1, the IBM i LUNs are still available from the IBM Spectrum Virtualize nodes at site 2. In the HyperSwap cluster, we manually unmap the HyperSwap LUNs from the IBM i production host at site 1, map the LUNs to the IBM i standby host at site 2, and IPL the IBM i standby host at site 2. After the IPL is finished, we can resume the workload on site 2.

After the outage of site 1 completes, we power-down IBM i at site 2, unmap the IBM i LUNs from the host at site 2 and map them to the host at site 1. IPL IBM i at site 1 and resume the workload. The I/O rate is transferred to the IBM Spectrum Virtualize storage nodes at site 1.

## Disaster at site 1 with IASP HyperSwap

This scenario requires IBM PowerHA SystemMirror for i software to be installed, and the corresponding IBM i setup, which consists of two IBM i partitions in a cluster and a switchable IASP on IBM i at site 1, a PowerHA cluster resource group, and PowerHA copy description. The workload is running in the IASP.

For more information about the PowerHA for i setup, see *IBM PowerHA SystemMirror for i: Preparation (Volume 1 of 4)*, SG24-8400.

In this scenario, ensure that all IBM i LUNs (not only the IASP LUNs) are HyperSwap volumes.

If a disaster occurs at site 1, PowerHA automatically switches the IASP to the system at site 2, and the workload can be resumed at site 2.

After the failure at site 1 is fixed, use PowerHA to switch the IASP back to site 1 and resume the workload at this site.

## Planned outage with Live Partition Mobility

IBM PowerVM Live Partition Mobility (LPM) allows you to move a running logical partition, including its operating system and running applications, from one system to another without any shutdown or without disrupting the operation of that logical partition.

In this scenario, we combine Live Partition Mobility with HyperSwap to transfer the workload onto site 2 during a planned outage of site 1. This combination requires VIOS NPIV attachment and all IBM i LUNs configured as HyperSwap LUNs.

For more information about LPM and its requirements, see *IBM PowerVM Virtualization Introduction and Configuration*, SG24-7940.

To use LPM, you must define the IBM i host in IBM Spectrum Virtualize with the WWPNs of the second port of the virtual FC adapters. We recommend creating a separate host object definition for the secondary ports to specify site 2 for this host object. Then, you enable the I/O rate to be transferred to the nodes at site 2 after migrating the IBM i partition with LPM.

After the outage is completed, you can use LPM again to transfer the IBM i partition back to site 1. After the migration, the I/O rate automatically moves to the nodes at site 1.

**Important:** Live Partition Mobility now supports multiple client virtual FC (vFC) adapter ports being mapped to a single physical FC port. Each client virtual FC must be mapped to a separate physical port in advance, whether LPM with FC N\_Port ID Virtualization is used. That restriction was removed for the use of Virtual I/O Server version 3.1.2.10 or later and IBM i 7.2 or later. Therefore, the same physical port can be double-mapped to the same IBM i client partition. This configuration allows for better adapter use.

## IBM SAN Volume Controller stretched cluster

IBM SAN Volume Controller is a hardware and software storage solution that implements IBM Spectrum Virtualize. IBM SAN Volume Controller appliances map physical volumes in the storage device to virtualize volumes, which makes them visible to host systems (for this example IBM i). IBM SAN Volume Controller provides Copy Services functions that can be used to improve availability and support DR, including Metro Mirror, Global Mirror, and FlashCopy.

Therefore, IBM PowerHA SystemMirror for i interfaces is compatible with IBM SAN Volume Controller. After the basic IBM SAN Volume Controller environment is configured, PowerHA can create a copy session with the volumes.

The use of PowerHA with IBM SAN Volume Controller management creates an automated highly available DR solution with minimal extra configurations. PowerHA and IBM SAN Volume Controller interfaces are compatible with hardware that is running IBM Spectrum Virtualize and IBM Storwize series.

### Full system replication in a stretched cluster

This high availability storage solution with IBM SAN Volume Controller uses stretched cluster topology and volume mirroring. For more information about stretched clusters, see Chapter 7, “Business continuity” on page 329.

A scenario that uses full system replication with IBM Spectrum Virtualize in SAN Volume Controller present a full system replication by using volume mirroring is shown in Figure A-14.

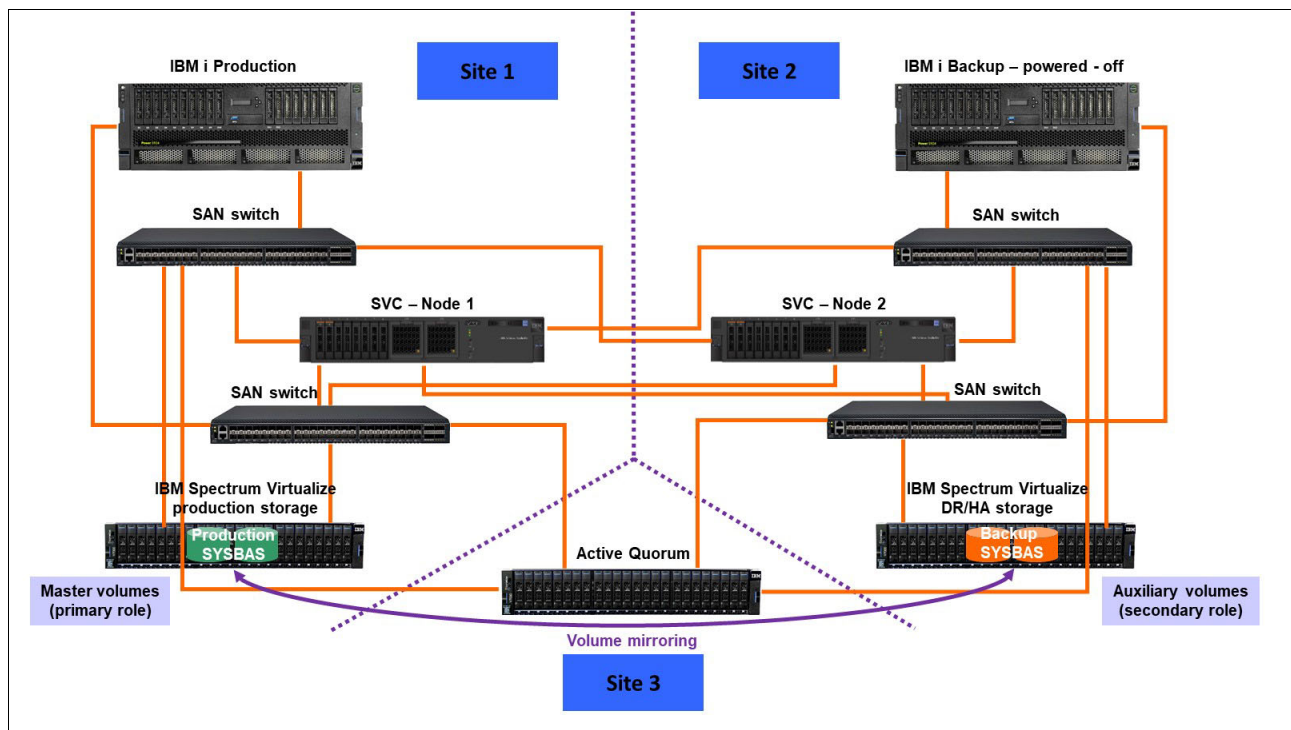


Figure A-14 Full system replication using IBM SAN Volume Controller volume mirroring

The scenario that is shown in Figure A-14 shows an IBM i production at site 1, a prepared IBM i backup system at site 2 that is powered off, and a third site that is the active quorum.

Two nodes of IBM SAN Volume Controller are in a stretched cluster topology, named split-cluster.

The LUNs of production SYSBAS are in an IBM Spectrum Virtualize production storage system at site 1. Those LUNs include a copy at site 2 in a second IBM Spectrum Virtualize DR/HA storage system, and this copy is done by using volume mirroring. Therefore, the IBM SAN Volume Controller stretch cluster configuration provides a continuous availability platform in which IBM i access is maintained, whether any single failure domain is lost (in this example, three domains exist).

Simultaneous IBM i access to both copies must be prevented; that is, the IBM i backup must be powered off whether IBM i production is active, and vice versa.

In our example, after a failure of site 1 (including a failure of the IBM i production system and the storage at site 1), the IBM i LUNs are still available because of the two data copies (the second at site 2).

An abnormal IPL is done at IBM i backup systems. Later, the IPL ended and we can resume the workload at site 2.

After the outage of site 1 is finished, we power down IBM i backup at site 2, and the resynchronization between both copies is incremental and started by the IBM SAN Volume Controller automatically. Volume mirroring is below the cache and copy services. Then, we restart the workload of IBM i production at site 1.

**Note:** For this example, HA testing and configuration changes are more challenging than with Remote Copy. For example, manual assignment is needed for the preferred node to enable local reads. Therefore, Enhanced Stretched Cluster, which was introduced with IBM SAN Volume Controller V7.2, adds a site awareness feature (reads always locally) and DR capability if simultaneous site and active quorum failures occur.

## LUN-level switching

This solution uses a single copy of an IASP group that can be switched between two IBM i systems. Likewise, LUN-level switching is supported for NPIV attachment and native attachment for storage that is based on IBM Spectrum Virtualize or IBM Storwize series.

It also is supported for an IBM SAN Volume Controller. This solution is engaging in heterogeneous environments where IBM SAN Volume Controller Stretched Cluster is used as the basis for a cross-platform, two-site HA solution.

Consider the following points regarding LUN-level switching features:

- ▶ LUN-level switching uses a single copy of an IASP on a single storage system.
- ▶ The IASP LUNs are assigned to host object definitions for the primary IBM i system.
- ▶ A second set of host object is defined for the secondary IBM i partition, but no LUNs are presented to the secondary system.
- ▶ The primary and secondary partitions are nodes in a cluster device domain.
- ▶ A device description for the IASP must be created on the secondary node before IBM PowerHA is configured.
- ▶ LUN-level switching can be used alone or together with Metro Mirror or Global Mirror in a three-node cluster. In a three-node cluster, LUN-level switching provides local HA. Whether the whole local site goes down, Metro Mirror or Global Mirror provides a DR option to a remote site.

**Note:** LUN-level switching plus Metro Mirror or Global Mirror (IASP) in the three-site solutions is not available for IBM Spectrum Virtualize as of this writing.

- If you want to add LUN-level switching to a Metro or Global Mirror, you do not need to create the cluster, IASP, or change the cluster administrative domain. You must create the IASP device description on the backup system from the LUN switching perspective.

The following IBM i license program products are required:

- IBM PowerHA SystemMirror for i (5770-HAS).
- Option 41 (HA Switchable Resources), and installed in all IBM i systems pertaining.
- Option 33 (5770-SS1 - Portable Application Solutions Environment), and installed in all IBM i systems pertaining.
- IBM Portable Utilities for i and OpenSSH, Open SSI, zlib (5733-SC1 base and option 1), and installed in all IBM i systems pertaining.

The LUN-level switching IBM PowerHA SystemMirror for i editions is listed in Table A-3.

*Table A-3 LUN-level switching IBM PowerHA SystemMirror for i editions*

| IBM i HA/DR clustering       | Express Edition | Standard Edition | Enterprise Edition |
|------------------------------|-----------------|------------------|--------------------|
| Cluster admin domain         | No              | Yes              | Yes                |
| Cluster device domain        | No              | Yes              | Yes                |
| Integrated heartbeat         | No              | Yes              | Yes                |
| Application monitoring       | No              | Yes              | Yes                |
| IBM i event/error management | No              | Yes              | Yes                |
| Automated planned failover   | No              | Yes              | Yes                |
| Managed unplanned failover   | No              | Yes              | Yes                |
| Centralized FlashCopy        | No              | Yes              | Yes                |
| LUN-level switching          | No              | Yes              | Yes                |
| Geomirror Sync mode          | No              | Yes              | Yes                |
| Geomirror Async mode         | No              | No               | Yes                |
| Multi-Site HA/DR management  | No              | No               | Yes                |
| Metro Mirror                 | No              | No               | Yes                |
| Global Mirror                | No              | No               | Yes                |
| IBM HyperSwap                | Yes             | Yes              | Yes                |

## IBM PowerHA LUN-level switching for IBM SAN Volume Controller in stretched cluster

This HA solution (see Figure A-15) supports two sites by using a combination of IBM i PowerHA LUN-level switching for server redundancy and IBM SAN Volume Controller in a stretched cluster that uses volume mirroring for storage redundancy.

The following requirements must be met:

- ▶ IBM i 7.1 TR6 or later
- ▶ NPIV or native attachment of IBM SAN Volume Controller

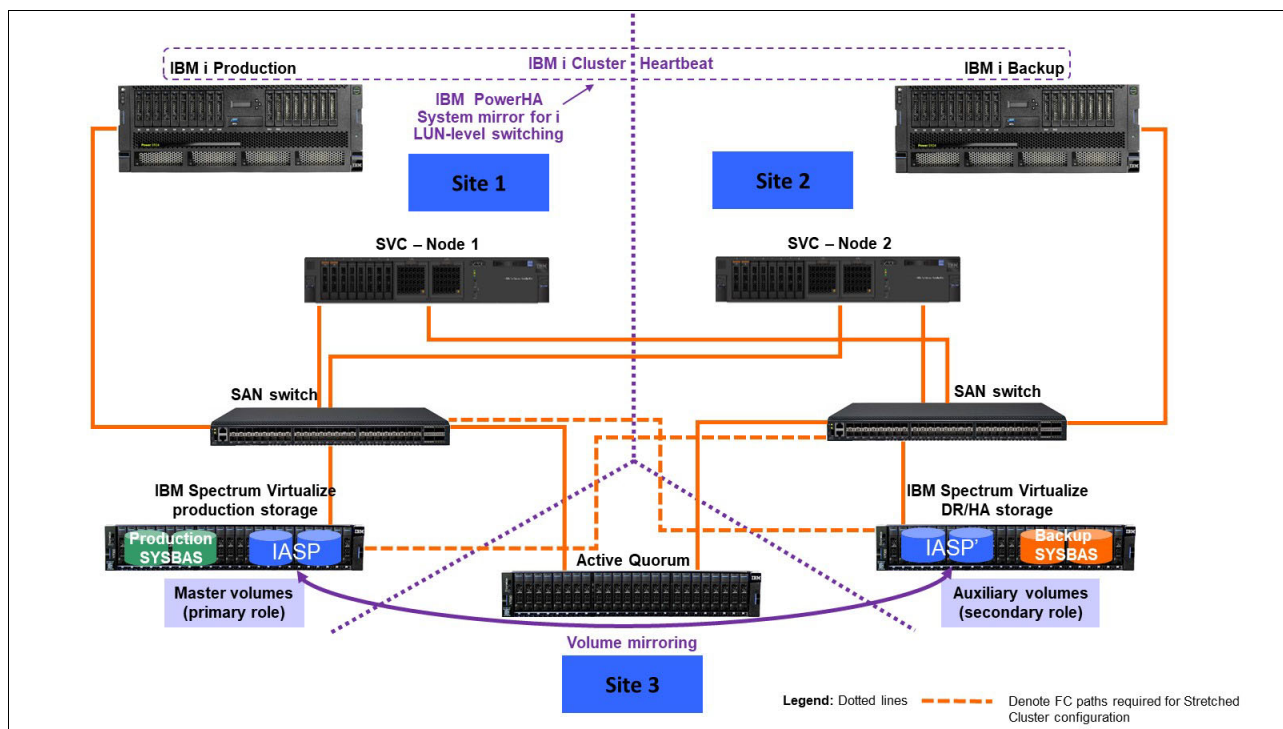


Figure A-15 IBM PowerHA System Mirror for i LUN-level switching with IBM SAN Volume Controller stretched cluster

The scenario that uses IBM PowerHA System Mirror for i LUN-level switching with IBM SAN Volume Controller Stretched Cluster provides the following benefits over IBM i full system replication with IBM SAN Volume Controller stretched cluster:

- ▶ High degree of automation for planned and unplanned site switches and fail overs.
- ▶ Shorter recovery times by using IASP.
- ▶ Reduced mirroring bandwidth requirements by using IASP (temporary writes in SYSBAS, such as for index builds are not mirrored).

As shown in Figure A-15, the availability is accomplished through the inherent active architecture of IBM SAN Volume Controller with the use of volume mirroring.

During a failure, the IBM SAN Volume Controller nodes and associated mirror copy of the data remain online and available to service all host I/O. The two data copies are placed in different MDisk groups or IBM Spectrum Virtualize storage systems. The resynchronization between both copies of IASP is incremental. Mirrored volumes feature the same functions and behavior as a standard volume.



**Note:** Since HyperSwap was introduced with IBM Spectrum Virtualize V7.5 on Storwize and IBM SAN Volume Controller, the scenario that uses the topology of HyperSwap with IBM SAN Volume Controller also is valid for IBM i.

Even with HyperSwap, we can use consistency groups that are enabled by using IBM i multipath driver, but not in stretched cluster scenarios. Note that remote mirroring license is required for using HyperSwap with IBM i.

Even with HyperSwap, we can use consistency groups that are enabled by using IBM i multipath driver, but not in stretched cluster scenarios. Remote mirroring license is required in this HyperSwap scenario.

In IBM Spectrum Virtualize 8.4., the maximum number of stretched volumes that is used per system is 5000; the maximum in HyperSwap per system is 2000.

For more information see *IBM Storwize HyperSwap with IBM i*, REDP-5490.

For more information about limits and restrictions for IBM System Storage SAN Volume Controller, see this [IBM Support web page](#).

## DB2 mirroring for IBM i

The DB2 Mirror base configuration consists of two systems that are in the same data center. This configuration does not span locations, because it is active-active read/write, which means that by definition all write operations are synchronous (by using Remote DirectMemory Access (RDMA) over Converged Ethernet (RoCE) network) to the application state. All write operations between two systems necessitates that the distance between the systems is limited to not affect performance.

The following broad approaches can be used to deploy active-active solutions:

- ▶ Use distributed lock management where multiple application servers can access the common or shared database. The servers are prevented from stepping on each other by the distributed lock management, which locks the other users out while you perform an update.
- ▶ The replication approach is used when each update of any type is totally synchronous to the application state. Therefore, when an application performs an update, it does not proceed to the next application step until the current write operations complete on the primary and secondary objects; that is, a two-phase commit exists between the two systems.

**Note:** Applications can be deployed in an active-active manner whereby each application server has simultaneous access to the database on both systems in the two-node active-active complex. If one of the database servers fails, the application servers continue performing I/O operations to the other system in the mirrored pair. This configuration features the added benefit of enabling workload balancing.

However, applications can also be deployed in an active-passive manner whereby application servers conduct write operations to one of the two systems in the two-system complex. If the primary system goes away, the application groups are switched to the secondary system. The active-active case necessitates that the application servers be hosted separately from the database servers and be connected through a client/server construct, such as JDBC.

**Note:** IBM i JDBC drivers now contain alternative server fail-over support to automatically transition the JDBC request between systems when one connection is no longer available. For many IBM i application workloads, deployment is completed through the traditional 5250 emulation window and contained in the same LPAR as the operating system and database. In this case, if the primary fails, the database is continuously replicated to the secondary system synchronously and is immediately available. The application must be restarted on the secondary system before the workload processing is resumed.

When one of the systems in the Db2® Mirror configuration is not available, Db2 Mirror tracks all update, change, and delete operations to the database table and all other mirror-eligible objects. When the pair is reconnected, all changes are synchronized between the systems. This process includes databases that are in an IASP or as part of the base system storage.

Db2 Mirror is compatible with IASPs and uses IASPs for IFS support within the Db2 Mirror configuration. For non-IFS objects, IASPs can be used, but are not required.

Also, Db2 Mirror supports applications that use traditional record-level access or SQL-based database access. Support for IFS and IFS journals is accomplished through deployment into an IASP, which can be configured as a switchable LUN or in a mirrored pair of IASPs through storage replication.

This solution requires IBM Power Systems POWER8 or later and IBM i 7.4 with IBM Db2 Mirror for i V7.4 (5770-DBM); Option 48, Db2 Data Mirroring, is required for Db2 Mirror for i; therefore, entitlement for Option 48 is automatically included with Db2 Mirror for i orders.

Ensure that IBM i Option 48 is installed and a key is applied along with the Db2 Mirror for i Licensed Program Product. For more information about software requirements for DB2 Mirror, see [IBM Documentation](#).

DR can be achieved by using various options, such as the IBM PowerHA SystemMirror for i Enterprise Edition, full system replication, or logical replication.

**Important:** DB2 Mirror local continuous availability can be combined with HA/DR replication technologies. Consider the following points:

- ▶ Remote replication for DR can be implemented by storage-based replication; that is, by using the Copy Services of IBM Spectrum Virtualize software.
- ▶ Any IFS IASP must remain switchable between both local DB2 Mirror nodes by choosing a DR topology that is supported by IBM PowerHA SystemMirror for i.
- ▶ Any DB IASP is available on both local nodes (no switch between local nodes).  
A DB IASP is not required for local DB2 Mirror database replication, but might be preferred for implementing a remote replication solution with shorter recovery times compared to SYSBAS replication.
- ▶ For a complete business continuity solution at the DR site, a remote DB2 Mirror node pair can be configured for a 4-node DB2 Mirror PowerHA Cluster configuration. Both IFS IASPs and DB IASPs must be registered with the remote DB2 Mirror pair (by using the SHADOW option for the DB IASP to maintain its Db2 Mirror configuration data such as default inclusion state and RCL).

For more information, see *IBM DB2 Mirror for i Getting Started*, REDP-5575.

## Set up process overview

During the setup and configuration of DB2 Mirror, the following nodes are referred to:

- ▶ Managing node
- ▶ Setup source node
- ▶ Setup copy node

For more information about these nodes, the setup process, and configuration, see [IBM Documentation](#).

Db2 Mirror is initially configured on a single partition that is called the *setup source node*. During the setup and configuration process, the setup source node is cloned to create the second node of the Db2 Mirror pair called the *setup copy node*. The setup copy node is configured and initialized automatically by Db2 Mirror during its first IPL.

Db2 Mirror configuration and setup process supports external and internal storage. External storage systems are used during the cloning process and IBM storage systems are recommended rather than non-IBM external storage because the cloning process is automated; that is, DB2 Mirror automates the cloning for IBM Spectrum Virtualize family.

The cloning technologies that are used for IBM storage systems are FlashCopy (cold and warm) and Remote Copy.

FlashCopy is used when both Db2 Mirror nodes connect to the same IBM Spectrum Virtualize storage system; cold clone requires the setup source node to be shut down during the cloning portion of the setup process. A warm clone allows the setup source node to remain active during the entire DB2 Mirror setup and configuration process.

Remote Copy is used when the Db2 Mirror nodes are connected to different IBM Spectrum Virtualize storage.

However, a manual copy also is available. For more information, see [IBM Documentation](#).

**Note:** Volume mirroring that is supported in IBM FlashSystem 9200 and IBM SAN Volume Controller is a valid cloning method for DB2 Mirror for a manual copy category. It is not automated as is it by using FlashCopy, Metro Mirror, or Global Mirror.

## IBM Spectrum Virtualize and DB2 Mirror

IBM Spectrum Virtualize storage systems establish communication with DB2 Mirror by using Secure Shell (SSH) to manage Copy Services functions. IBM Spectrum Virtualize user IDs must have the user role of administrator. For a managing node, the following products are required:

- ▶ 5733SC1 \*BASE IBM Portable Utilities for i
- ▶ 5733SC1 Option 1 OpenSSH, OpenSSL, zlib
- ▶ 5770SS1 Option 33 Portable Application Solutions Environment

**Note:** For more information about creating an SSH key pair, see [this web page](#). After an SSH key pair is created, attach the SSH public key to a use on the IBM Spectrum Virtualize storage system. The corresponding private key file must be uploaded to the managing node so that it can be used during the DB2 Mirror setup process.

### ***Virtual I/O Server and native attachment***

The DB2 Mirror storage cloning process for IBM Spectrum Virtualize requires FC adapters with native attachment or attachment with Virtual I/O Server N\_Port ID Virtualization.

### ***Host object definition and volume planning***

Before you start to set up Db2 Mirror, you must define the host object and assign volumes to the hosts to be used by the setup copy node. The same number of host objects, volumes, and the same size volumes must be defined for the setup source node and setup copy node. Later, the DB2 Mirror cloning process pairs storage volumes between the setup source node and setup copy node, which applies for SYSBAS and IASPs. Consider the following points:

- ▶ The set-up source node and set up copy node must have the same number and sizes of LUNs or disks in SYSBAS.
- ▶ The host object and volumes for any database IASPs must be predefined for the set-up copy node before a database IASP is added to DB2 Mirror.

### ***Remote Copy cloning***

In this case, DB2 Mirror Remote Copy cloning uses the following IBM Spectrum Virtualize Copy Services operations to copy the setup source node volumes to the setup copy nodes volumes:

- ▶ Global Mirror for cold clone
- ▶ Global Mirror with change volumes for warm clone

Whether you plan to perform the Remote Copy during a planned outage window, you must ensure that your bandwidth between storage systems is sufficient to complete the Remote Copy during that period. The Db2 Mirror cloning process does not provide the capability to pause the cloning and then resume it later. Therefore, you must plan enough time for the Remote Copy to complete.

**Important:** For IBM Spectrum Virtualize, the Copy Services partnership between storage systems must be manually created before Db2 Mirror is configured.

## **Architectures and considerations for DB2 Mirror**

Because of the synchronous design of Db2 Mirror, the distance between the nodes is limited to within a data center for most cases. Multiple configurations are supported for a data center Db2 Mirror implementation and the addition of a DR solution.

Several options are described in this section as examples with IBM Spectrum Virtualize storage systems. A specific implementation depends on your business resilience requirement.

**Note:** DB2 Mirror supports IBM SAN Volume Controller topologies, such as Enhanced Stretched Cluster and HyperSwap.

### ***DB2 Mirror environment with one IBM Spectrum Virtualize storage***

In this example, one IBM Spectrum Virtualize storage is used as a basic configuration for using DB2 Mirror. This configuration features some key advantages.

By using one storage system, you can take advantage of FlashCopy to set up your configuration rapidly. Regarding a DR strategy to provide storage resiliency, this solution might be considered.

Also, as shown in Figure A-16, two IBM Power System servers are used (at least one RoCE adapter per server). However, you can reduce this scenario in terms of cost of decreased resiliency by implementing DB2 Mirror across two IBM i LPARs on the same IBM Power Systems. For this example, a SYSBAS is cloned; however, IASP also can be added by using another set of volumes.

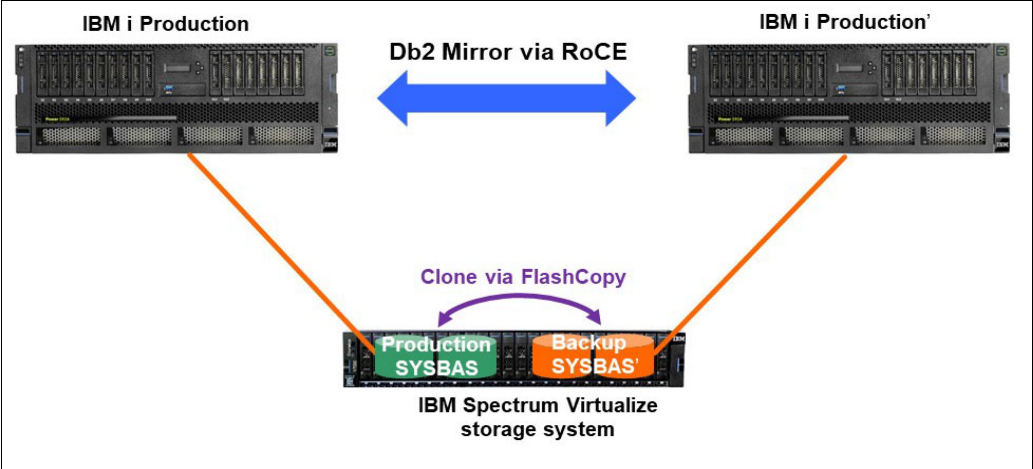


Figure A-16 DB2 Mirror environment with one IBM Spectrum Virtualize storage

### DB2 Mirror environment with two IBM Spectrum Virtualize storages

The use of two IBM Spectrum Virtualize storage provides further redundancy by helping to ensure that the active node remains running and available during a storage outage. In this example, two IBM Power Systems servers and IBM Spectrum Virtualize storages are used. Also, Remote Copy is used to set up DB2 Mirror.

As shown in Figure A-17, the set of volumes for SYSBAS and the set of volumes for IASP are replicated. Global Mirror also can be use.

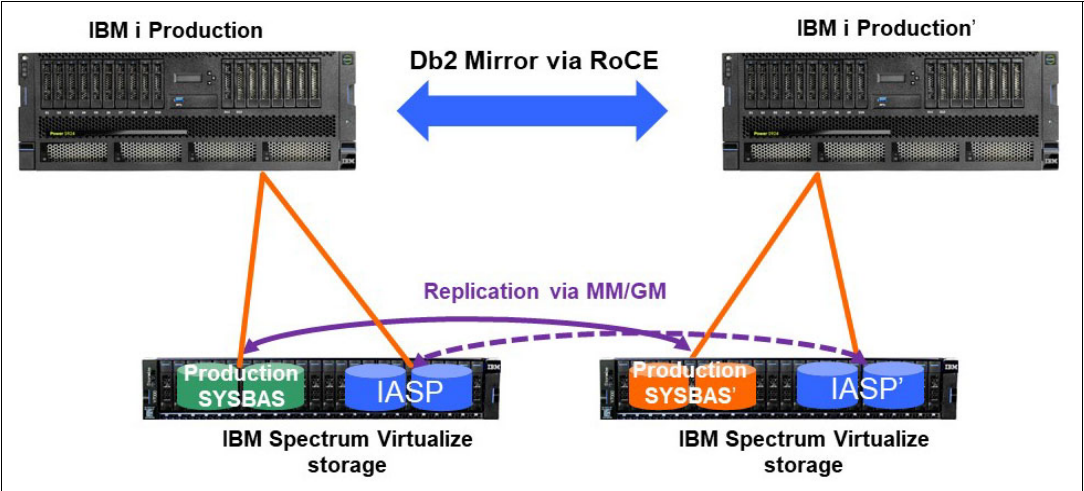


Figure A-17 DB2 Mirror environment with two IBM Spectrum Virtualize storages

### DB2 Mirror and DR considerations

Db2 Mirror is a continuous availability solution; therefore, it is *not* considered a DR solution. DB2 Mirror can be used within your DR strategy to improve your availability, even within a disaster situation.

The DB2 Mirror nodes must be close to each other because the maximum distance between IBM Power Systems servers is 200 meters (656 feet). At site 1, DB2 Mirror nodes are used, and at site 2 (the DR location), we can have a single server or multiple servers with DB2 Mirror nodes, and a unique or multiple IBM Spectrum Virtualize storages

The communication between the continuous availability at site 1 and the DR at site 2 can be achieved by using technologies, such as IBM PowerHA SystemMirror for use with Metro Mirror or Global Mirror with IASPs, full system replication, and logical replication from a third-party vendor.

### ***DB2 Mirror and full system replication***

The use of a mirrored pair within the disaster site provides extra protection if you are required to role swap to the DR location. With this scenario, a continuously available environment exists, even in DR.

A topology with multiple IBM Spectrum Virtualize storages and multiple IBM Power Systems servers is shown in Figure A-18.

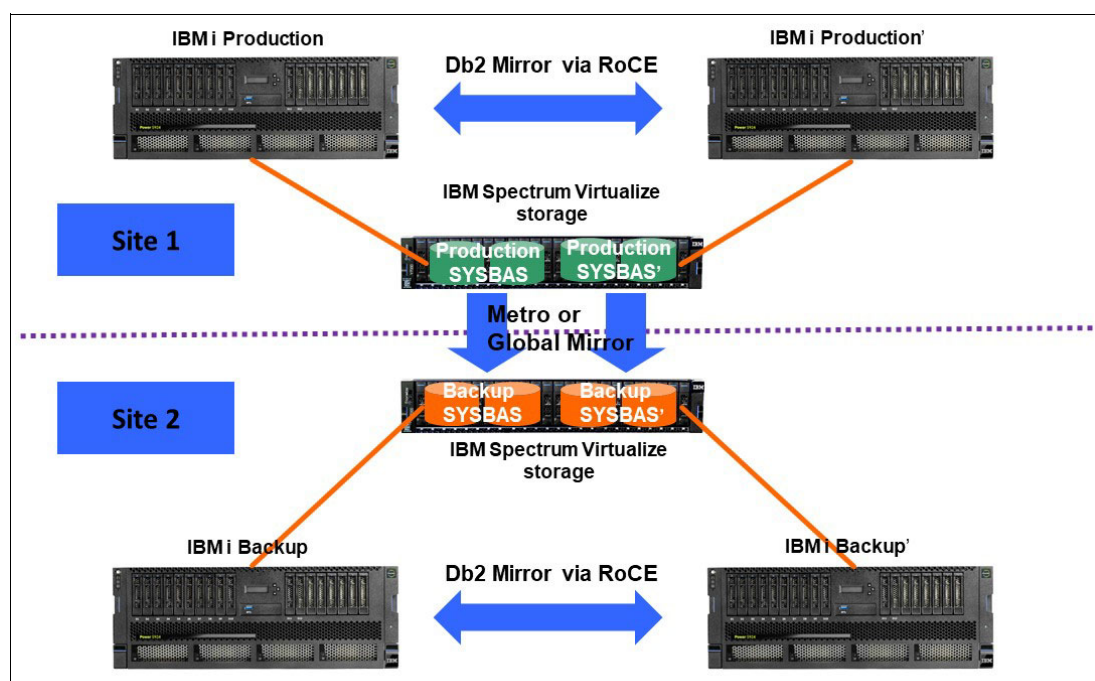


Figure A-18 DB2 Mirror and full system replication

Full system replication is fully supported. If you are not using IASP, this type of replication can be done for IBM i at IBM Spectrum Virtualize storage level.

At site 1, an active side exists because of full system replication; However, at site 2, the IBM i systems are powered off, and the replication is active across sites.

Two copies are at a DR location because if one side fails, the other side must continue replicating. If only three nodes are replicating, you cannot predict which side fails and does not have a valid copy of storage to switch.



# Related publications

The publications that are listed in this section are considered particularly suitable for a more detailed discussion of the topics that are covered in this book.

## IBM Redbooks

The following IBM Redbooks publications provide more information about the topic in this document. Note that some publications that are referenced in this list might be available in softcopy only:

- ▶ *IBM System Storage b-type Multiprotocol Routing: An Introduction and Implementation*, SG24-7544
- ▶ *IBM/Cisco Multiprotocol Routing: An Introduction and Implementation*, SG24-7543
- ▶ *IBM FlashSystem Best Practices and Performance Guidelines*, SG24-8503
- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.4*, SG24-8467
- ▶ *Implementing the IBM FlashSystem with IBM Spectrum Virtualize V8.4*, SG24-8465
- ▶ *PowerHA SystemMirror for IBM i Cookbook*, SG24-7994
- ▶ *IBM Real-time Compression in IBM SAN Volume Controller and IBM Storwize V7000*, REDP-4859
- ▶ *IBM Spectrum Virtualize HyperSwap SAN Implementation and Design Best Practices*, REDP-5597
- ▶ *Implementing IBM Real-time Compression in SAN Volume Controller and IBM Storwize V7000*, TIPS1083

You can search for, view, download, or order these documents and other Redbooks, Redpapers, Web Docs, draft, and additional materials, at the following website:

[ibm.com/redbooks](http://ibm.com/redbooks)

## Online resources

The following websites are also relevant as further information sources:

- ▶ IBM Documentation:  
<http://www.ibm.com/support/knowledgecenter/>
- ▶ Comprestimator download site:  
<https://www.ibm.com/support/pages/ibm-flashsystem-comprestimator>
- ▶ IBM SAN Volume Controller documentation:  
[https://www.ibm.com/support/knowledgecenter/STPVGU\\_8.4.0/com.ibm.storage.svc.console.840.doc/svc\\_ichome.html](https://www.ibm.com/support/knowledgecenter/STPVGU_8.4.0/com.ibm.storage.svc.console.840.doc/svc_ichome.html)

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SG24-8502-00  
ISBN 0738459593



(1.0" spine)  
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460 <-> 788 pages







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ISBN 0738459593

Printed in U.S.A.

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